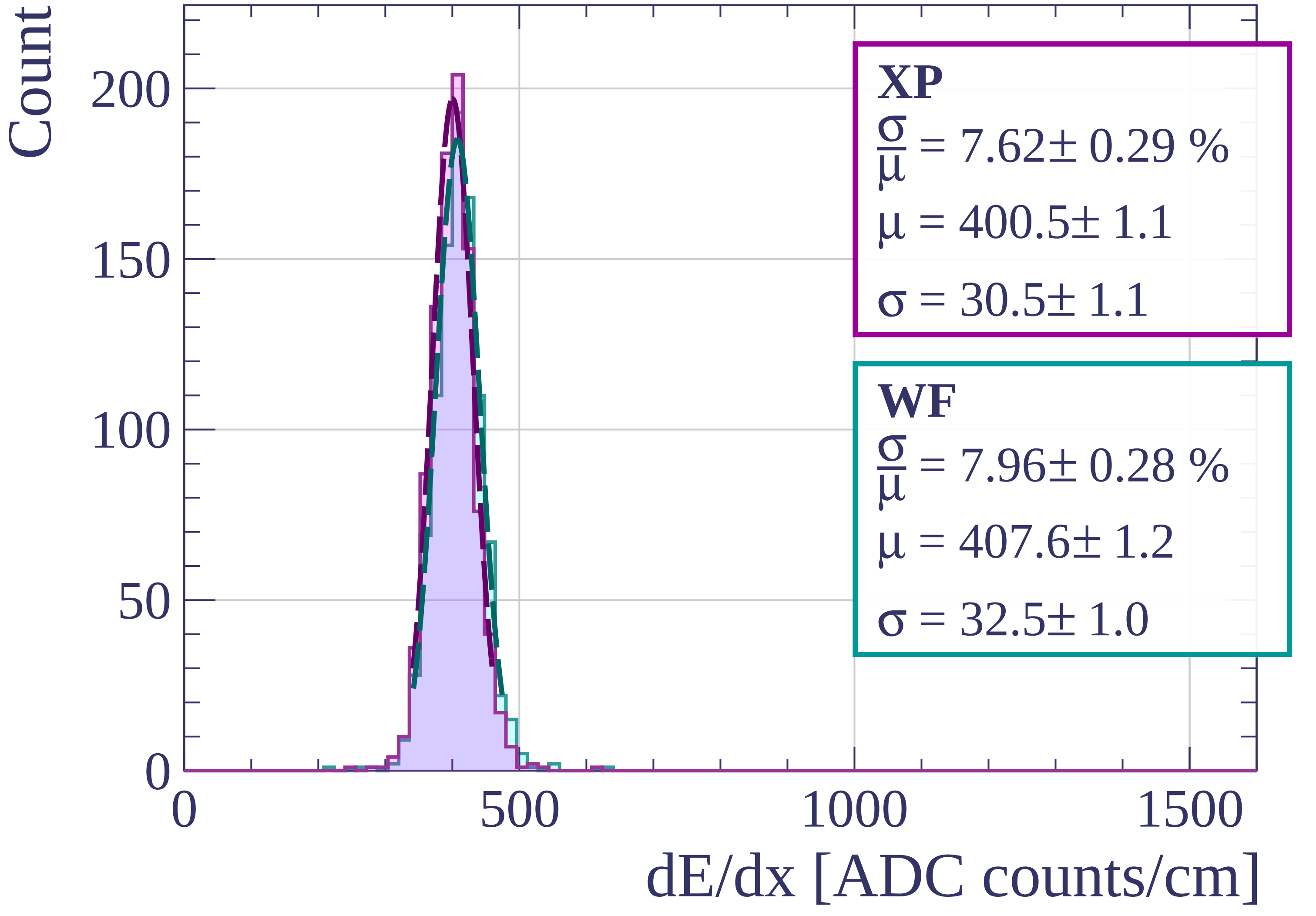
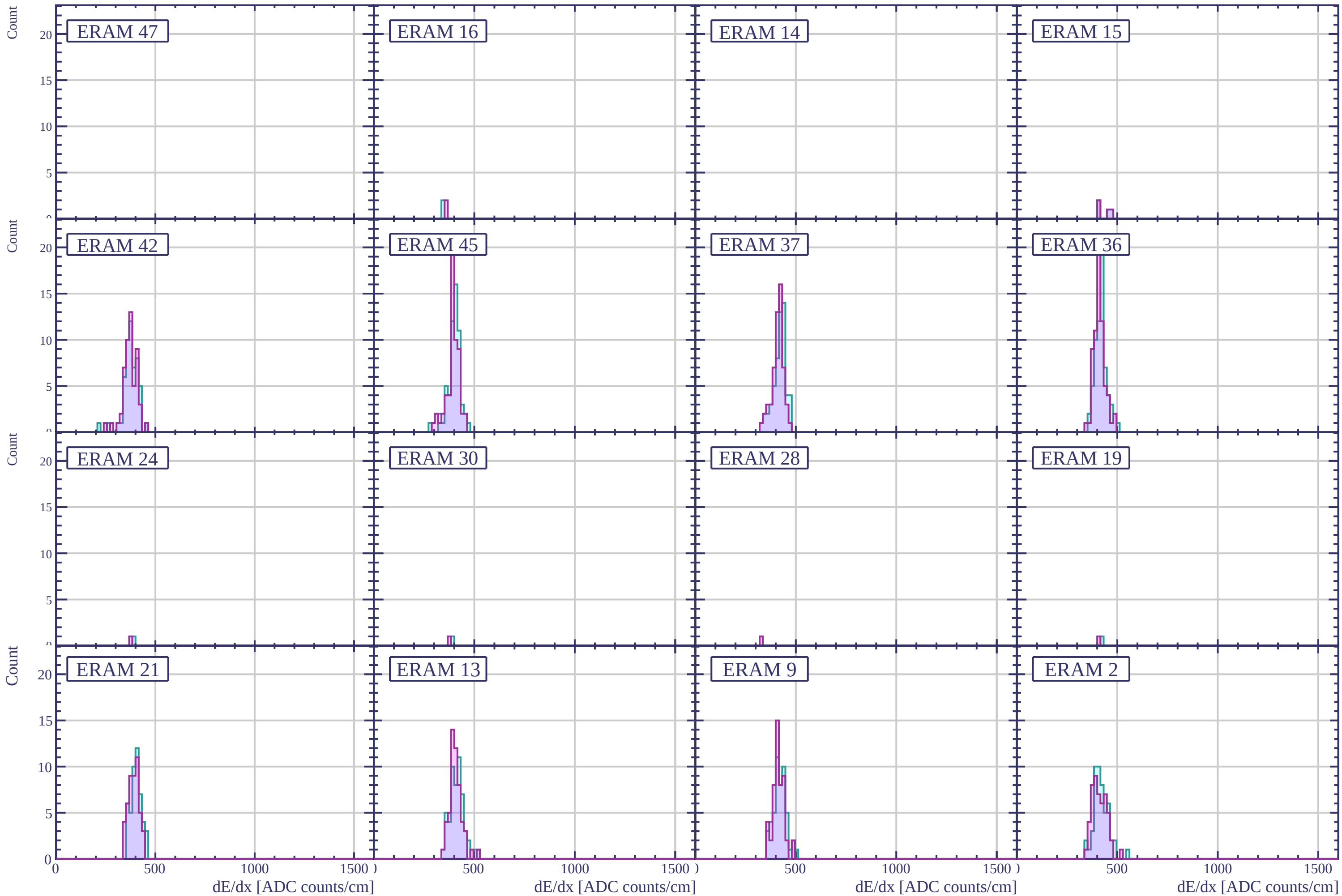
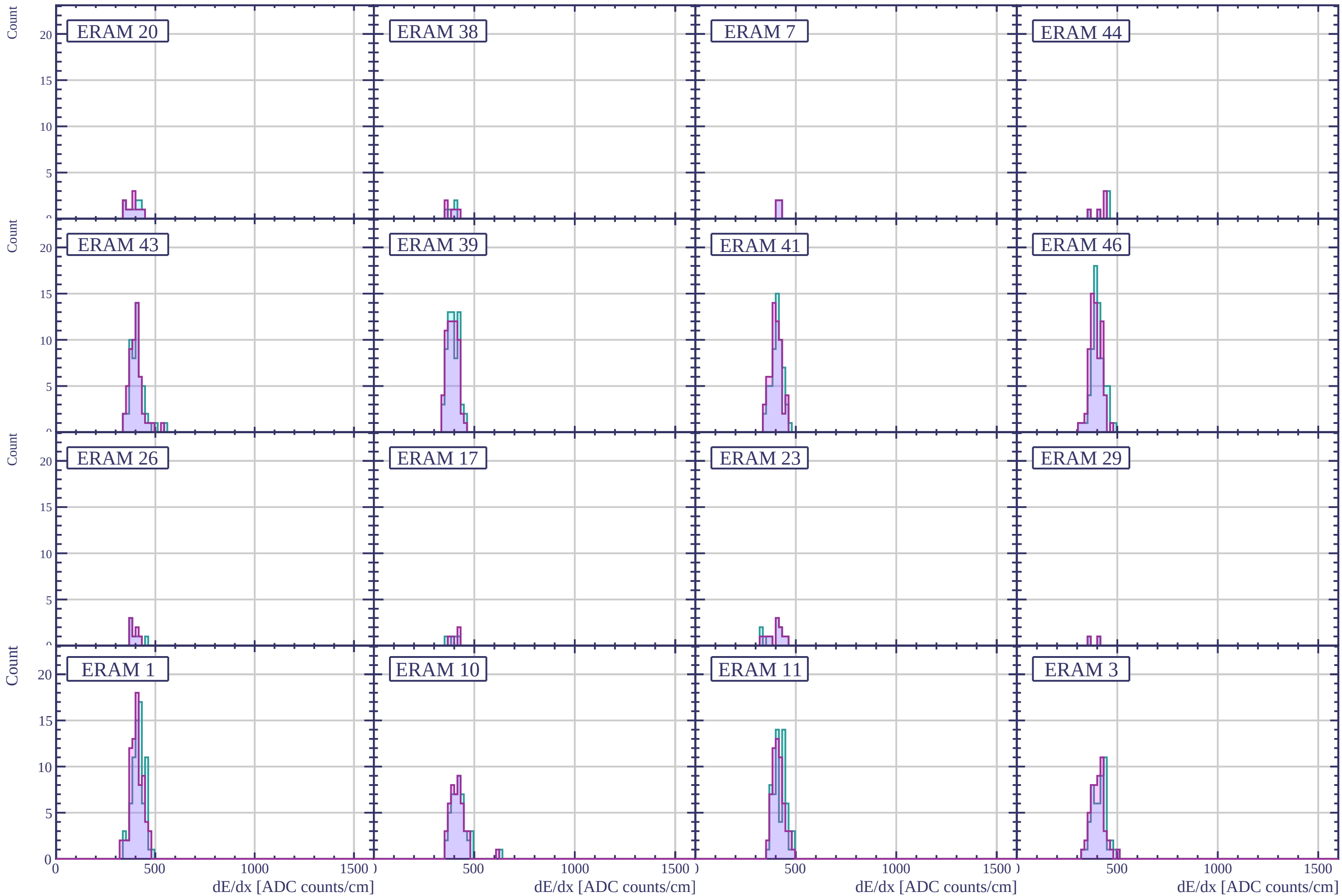
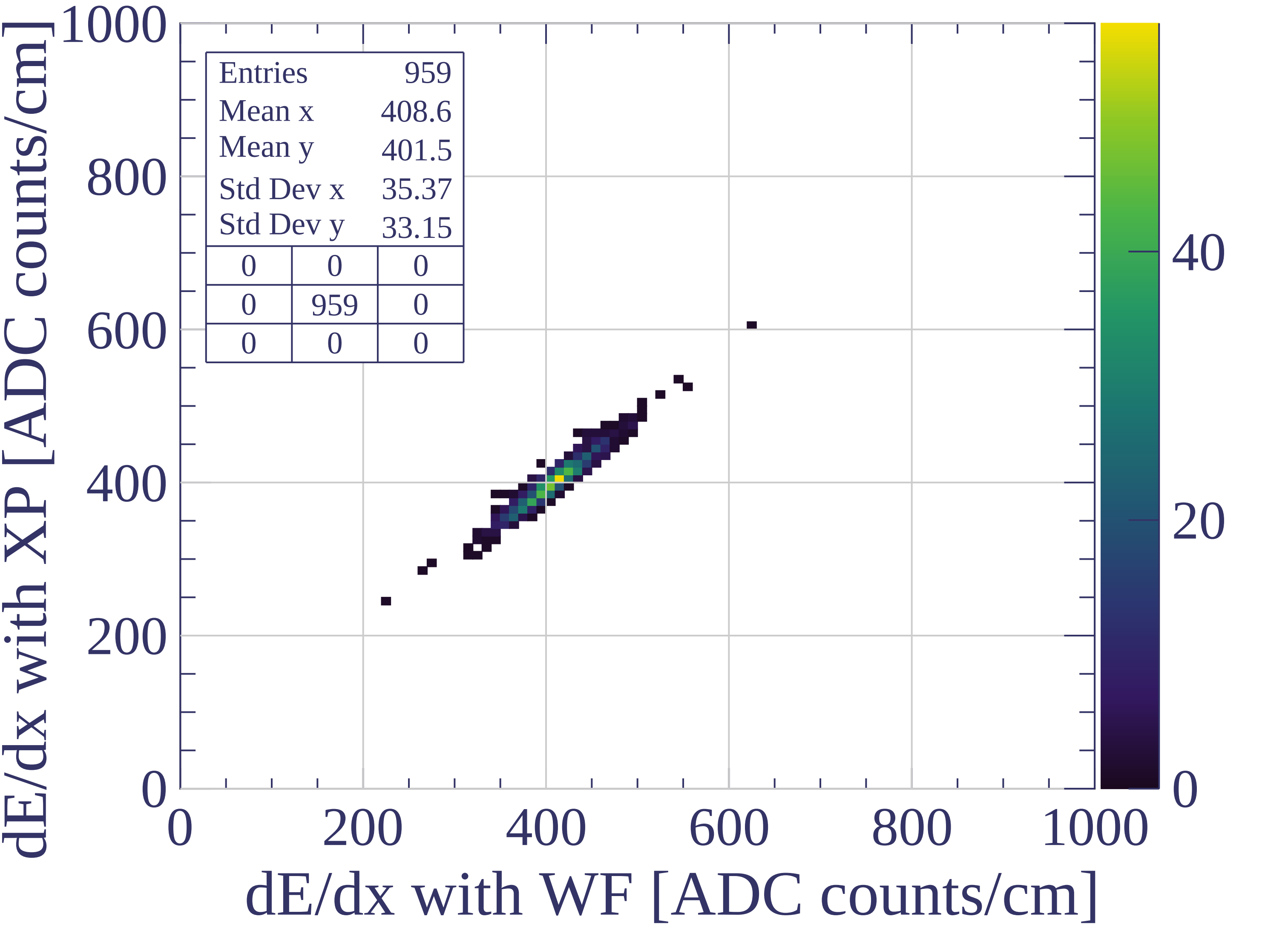
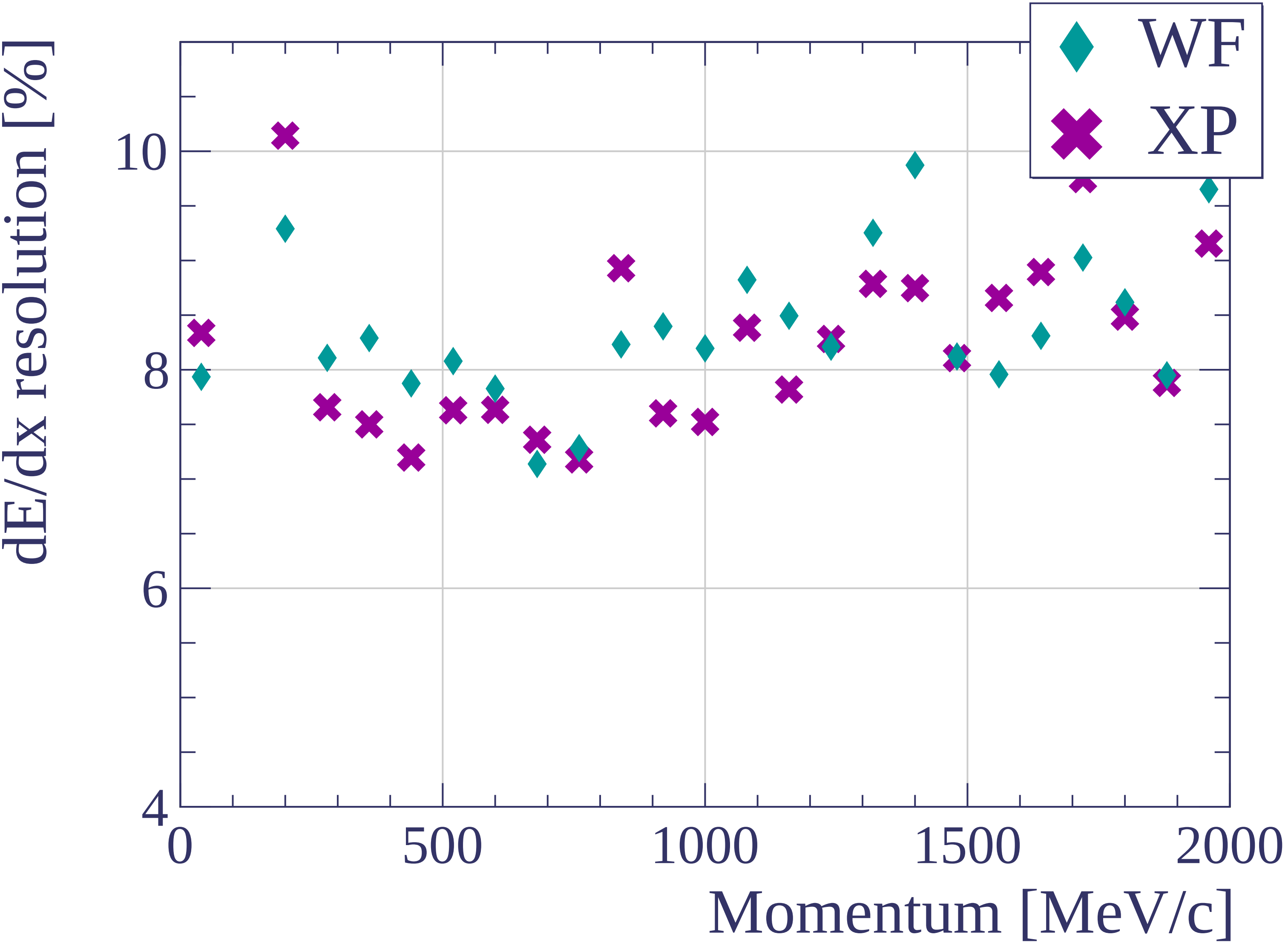

Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.62± 0.29 %
μ = 400.5± 1.1
σ = 30.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 7.96± 0.28 %
μ = 407.6± 1.2
σ = 32.5± 1.0


ERAM 47
ERAM 16
ERAM 14
ERAM 15
ERAM 42
ERAM 45
ERAM 37
ERAM 36
ERAM 24
ERAM 30
ERAM 28
ERAM 19
ERAM 21
ERAM 13
ERAM 9
ERAM 2
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
20

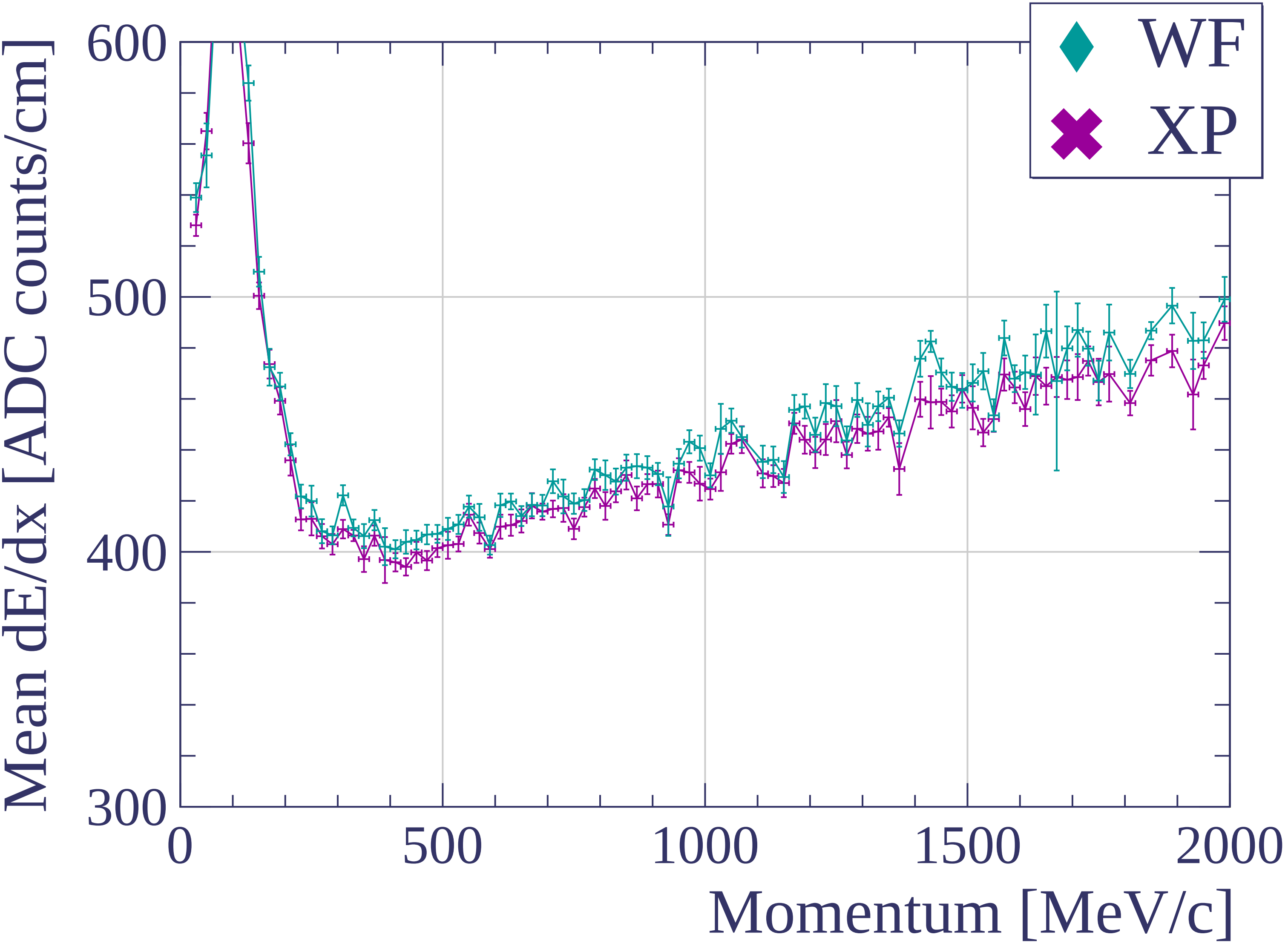

ERAM 20
ERAM 38
ERAM 7
ERAM 44
ERAM 43
ERAM 39
ERAM 41
ERAM 46
ERAM 26
ERAM 17
ERAM 23
ERAM 29
ERAM 1
ERAM 10
ERAM 11
ERAM 3
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
20



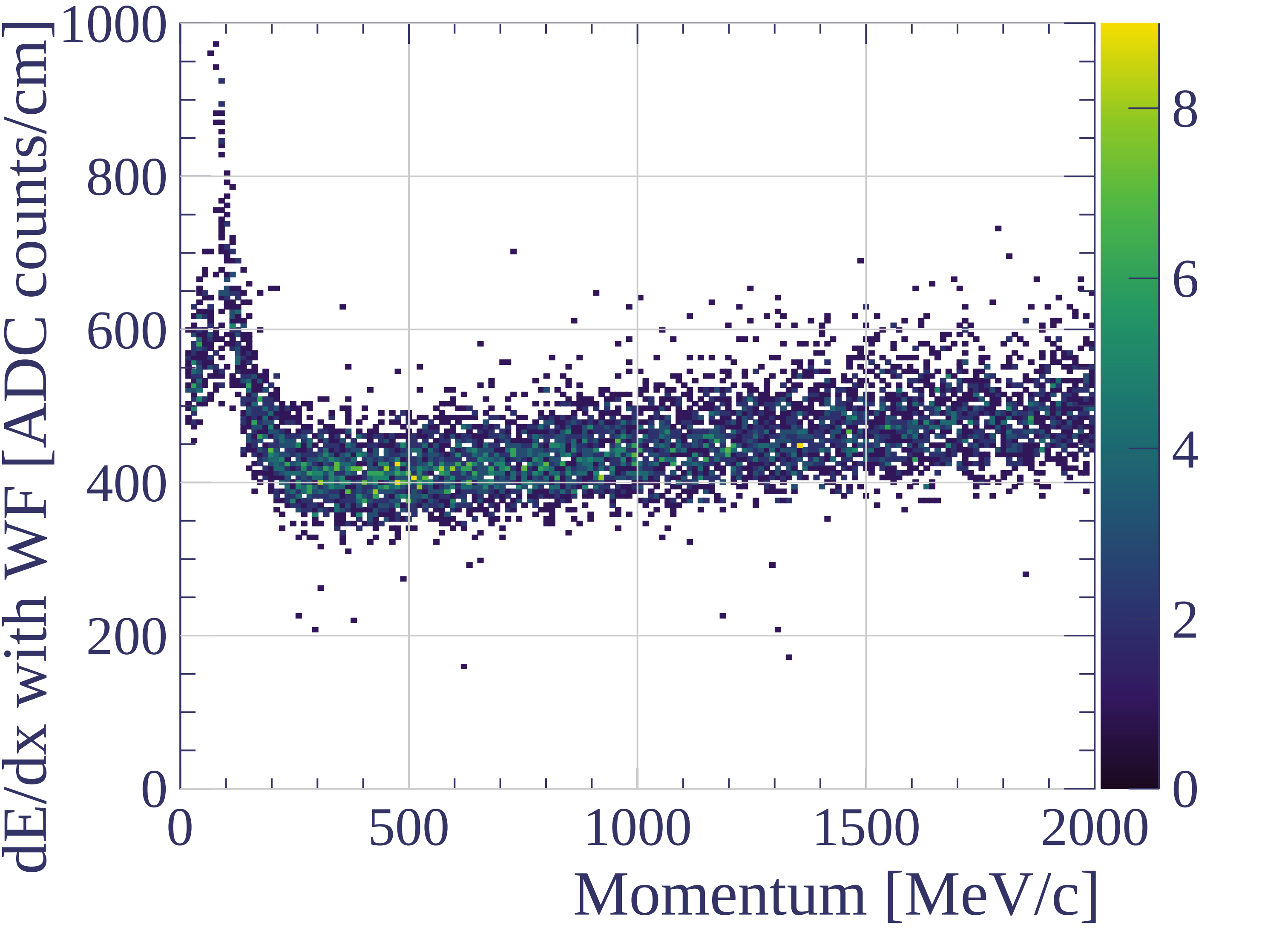
Entries 959
Mean x 408.6
Mean y 401.5
Std Dev x 35.37
Std Dev y 33.15
0 0 0
0 959 0
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
20
40

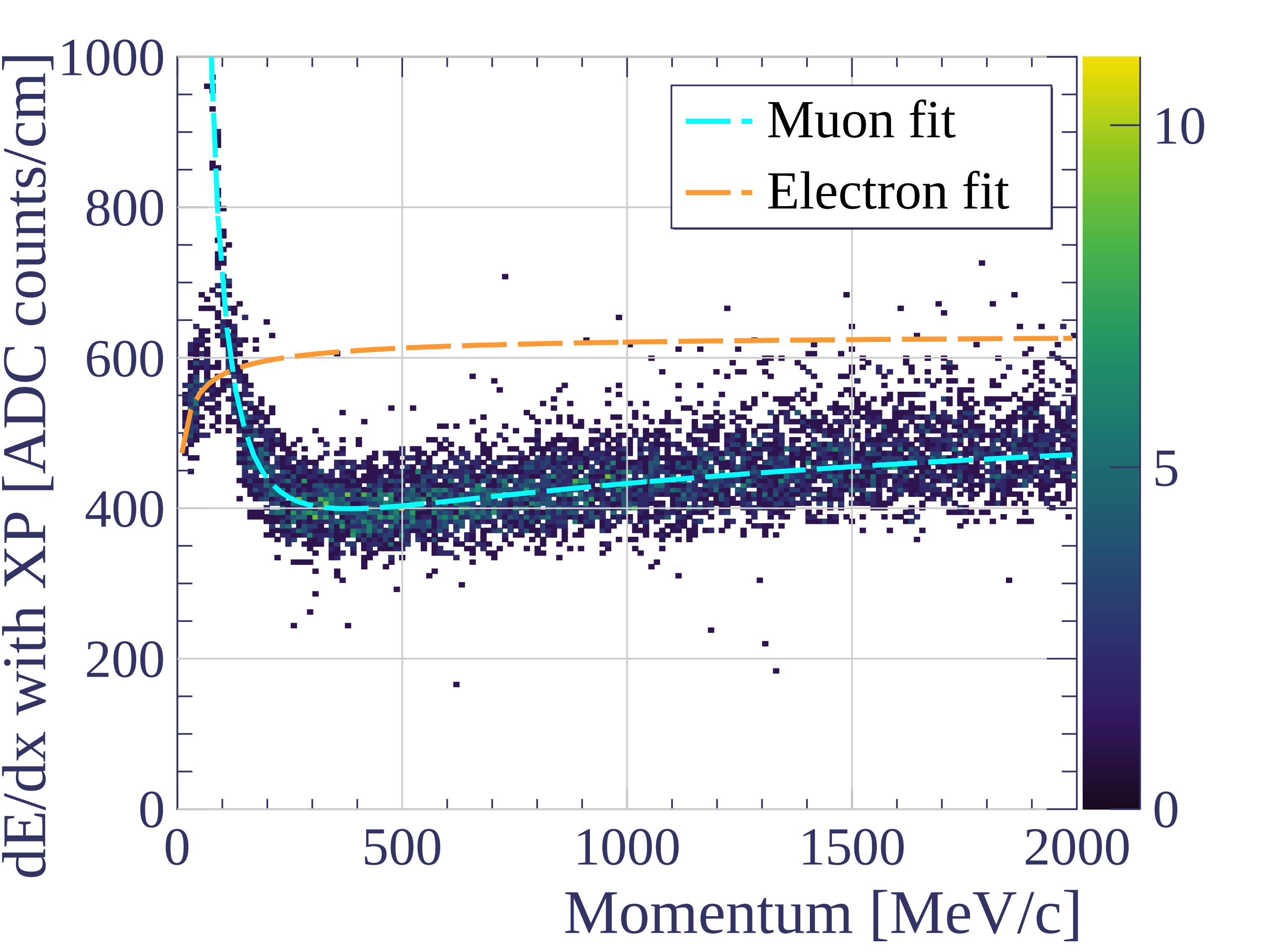


WF
XP
dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
2000
Momentum [MeV/c]

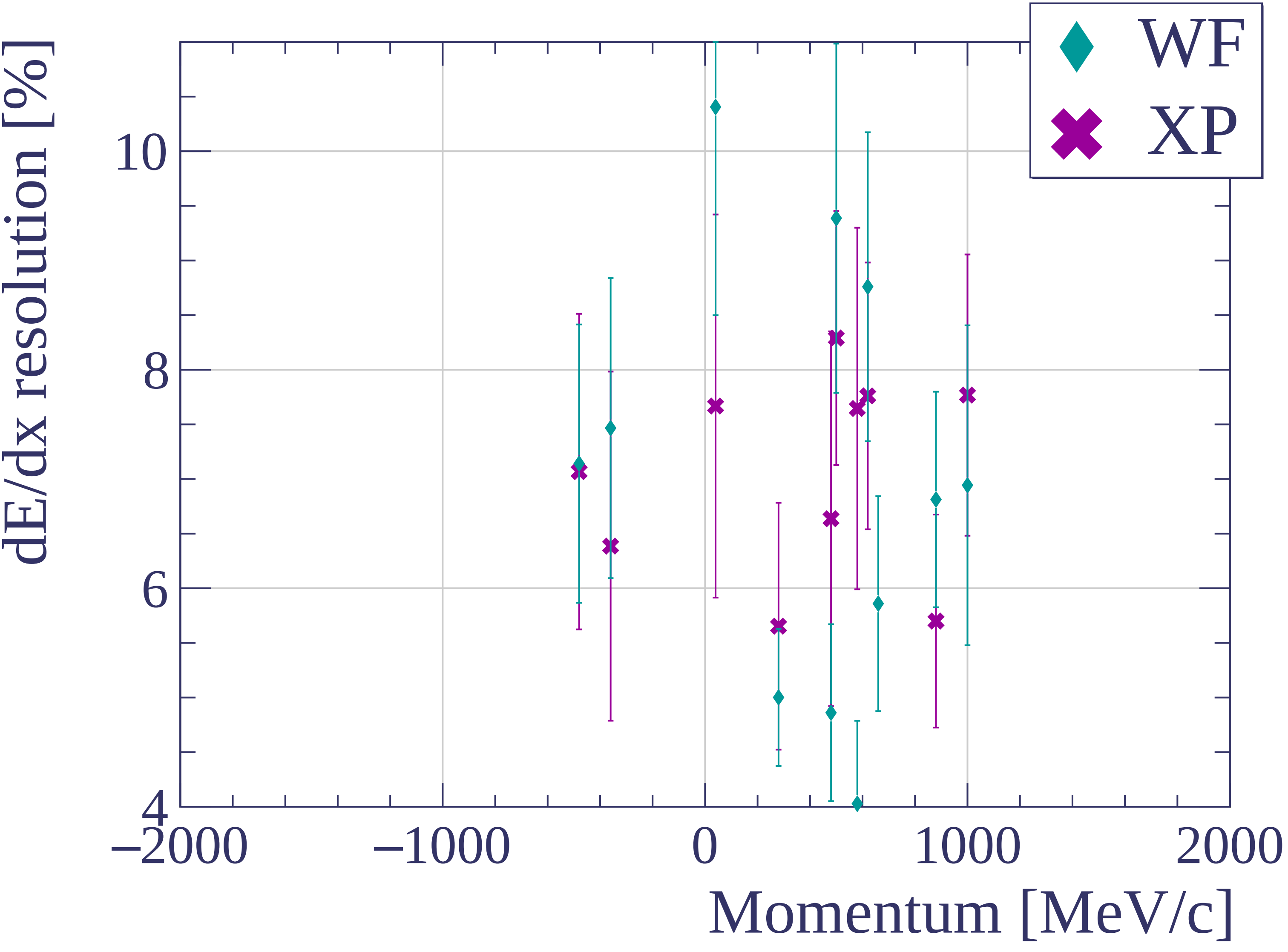


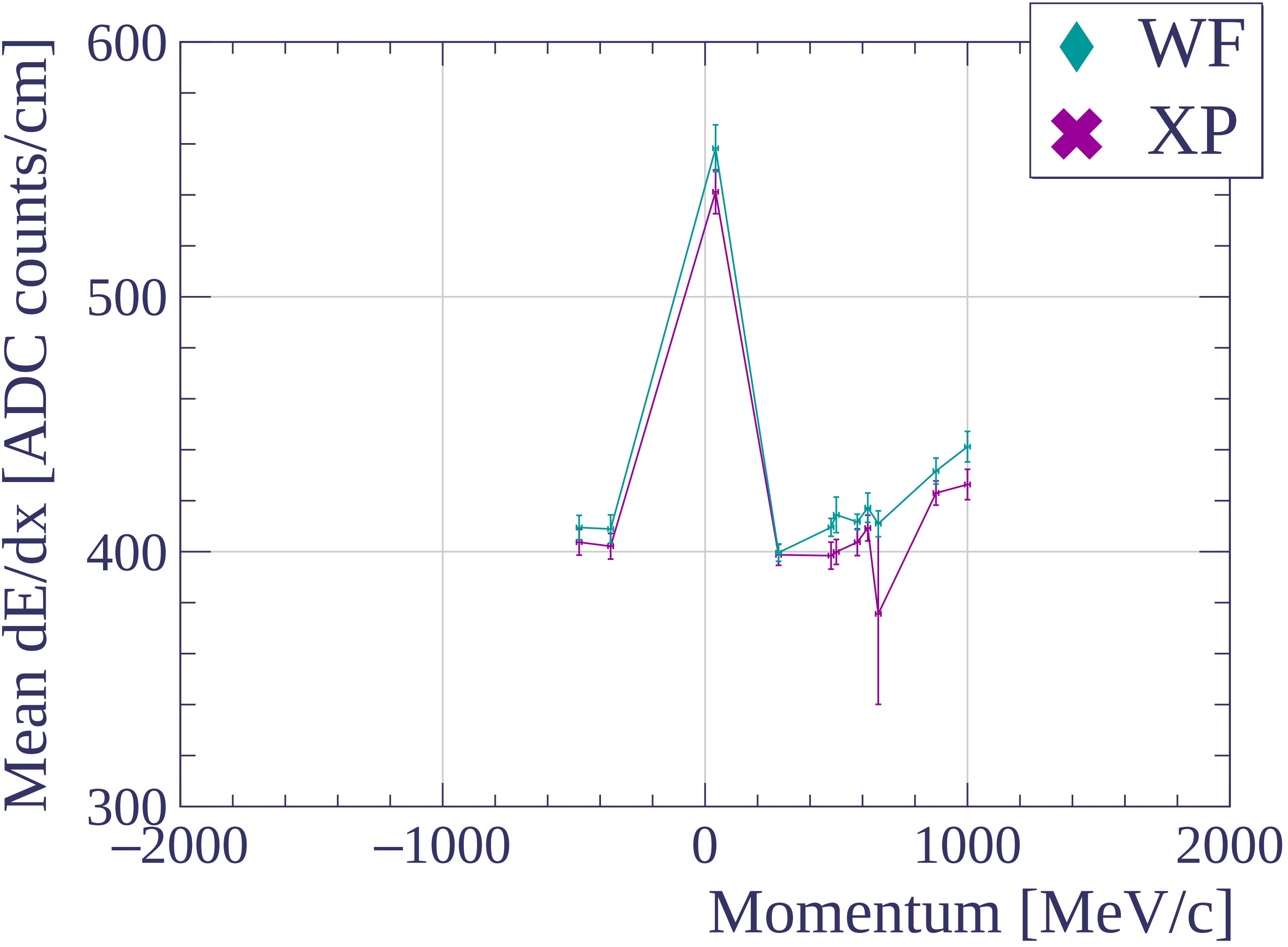
WF
XP
600
500
400
300
0
500
1000
1500
2000
Mean dE/dx [ADC counts/cm]
Momentum [MeV/c]


dE/dx with WF [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
0
500
1000
1500
2000
0
2
4
6
8

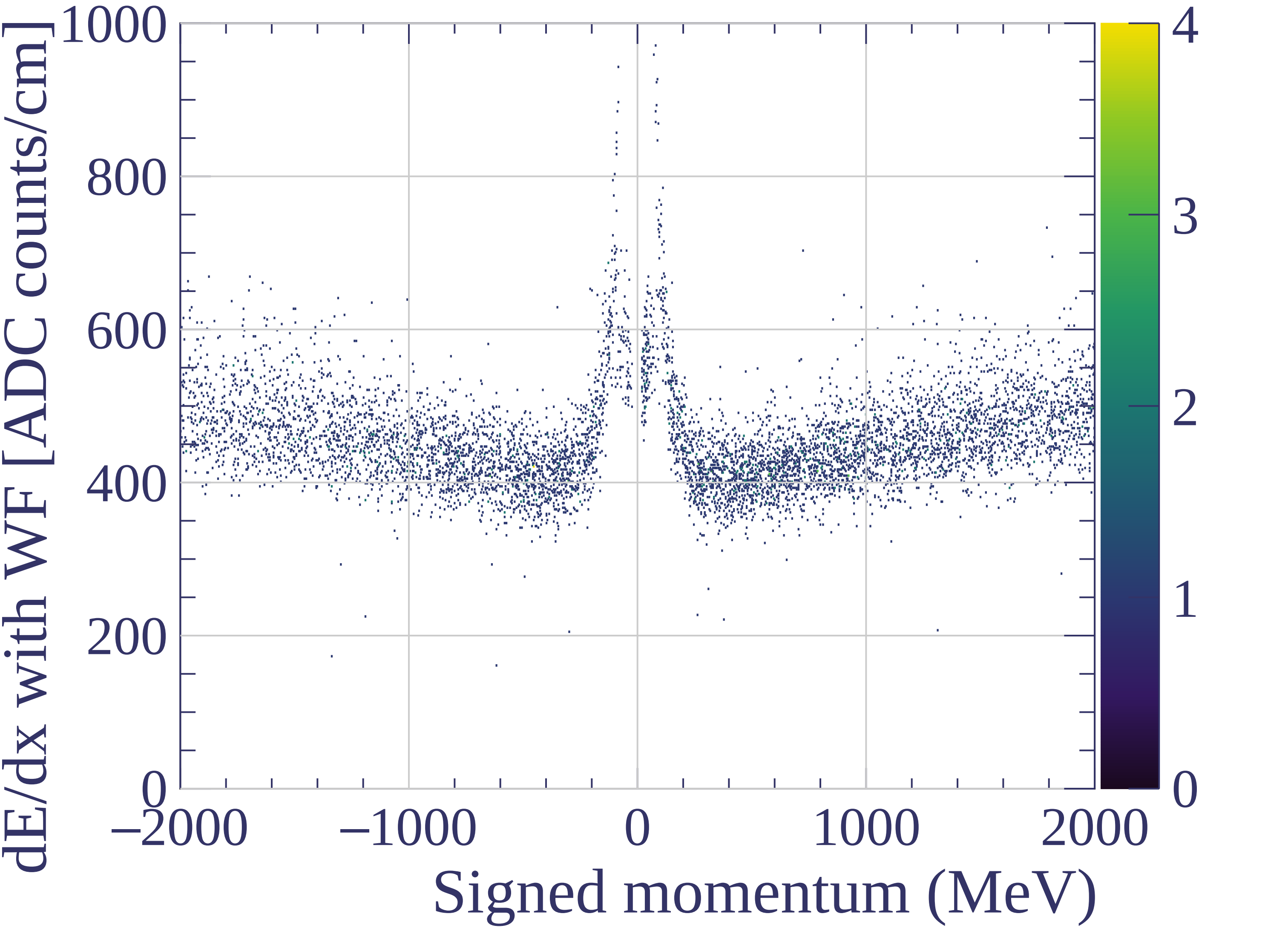


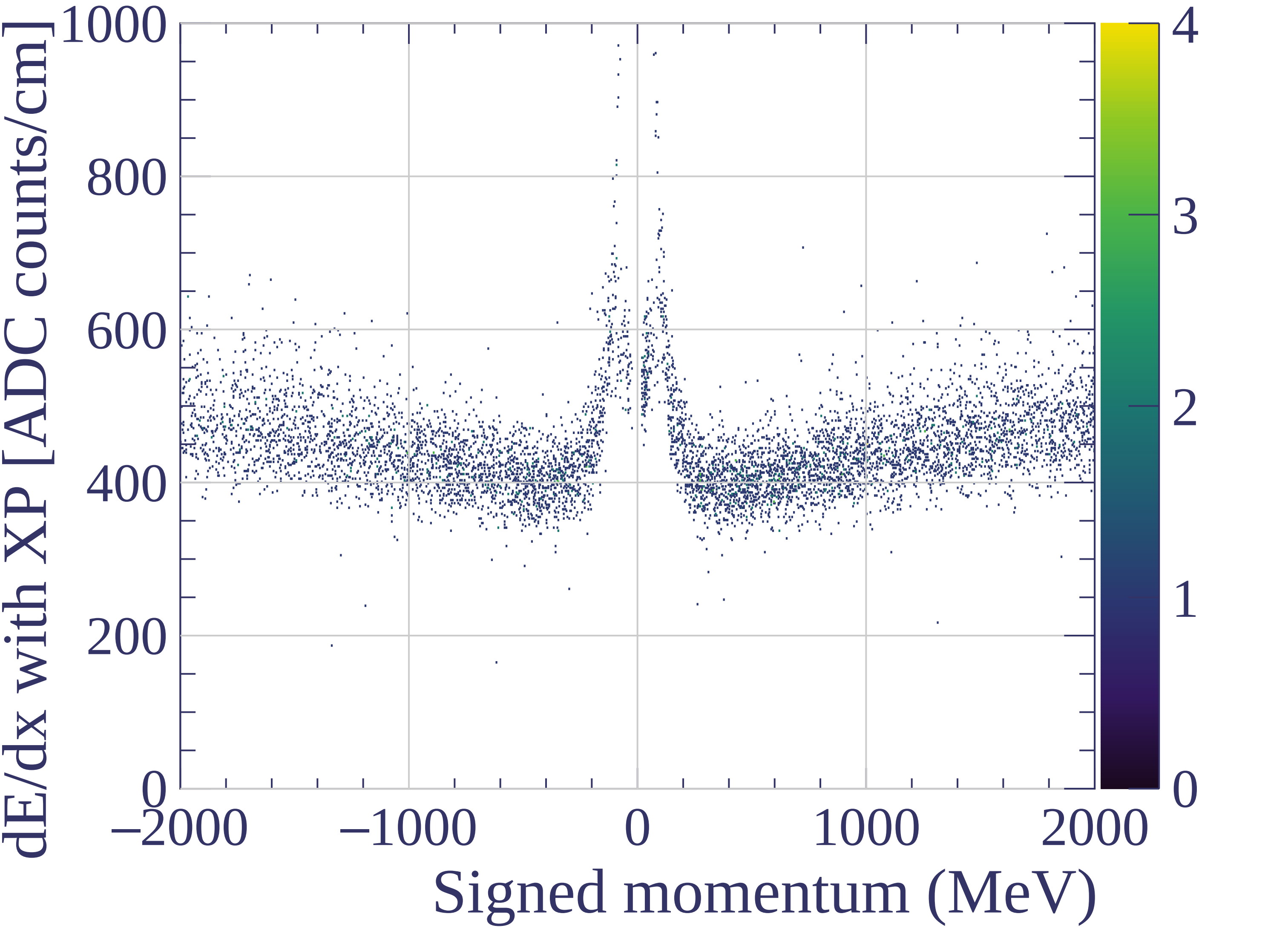
Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1500
2000
5
10


WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

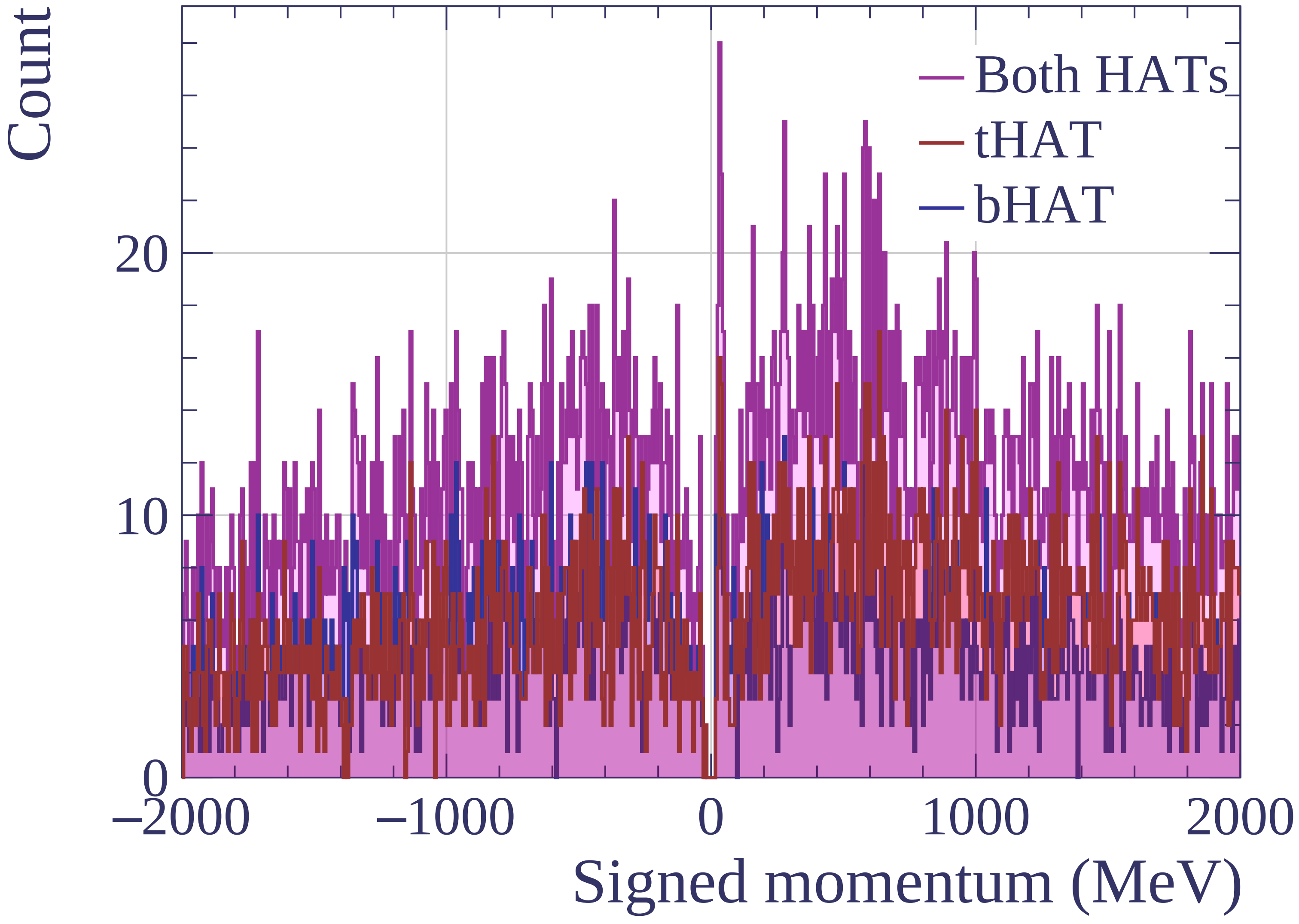


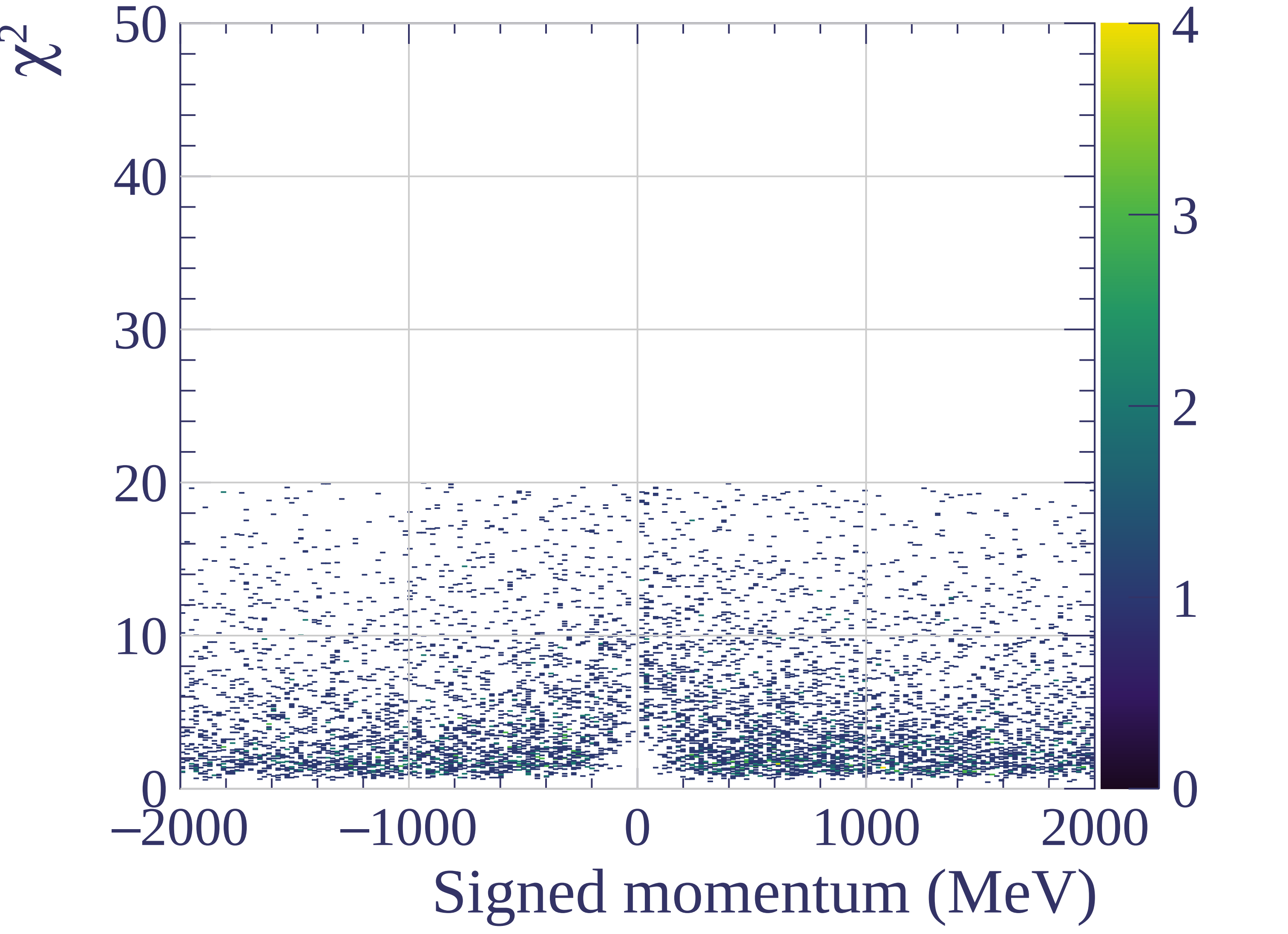
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−2000
−1000
0
1000
2000
Momentum [MeV/c]

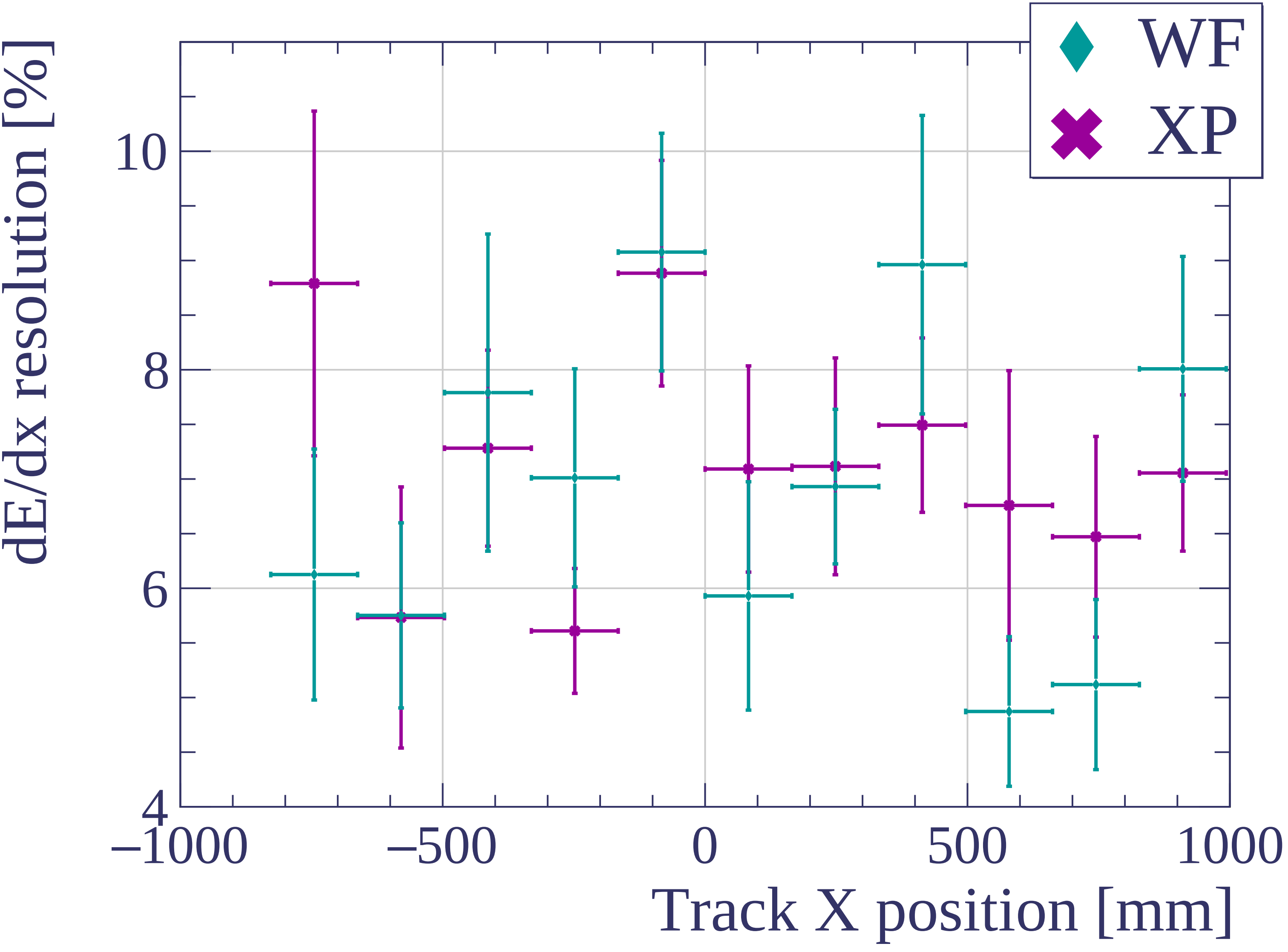

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
4
3
2
1
0



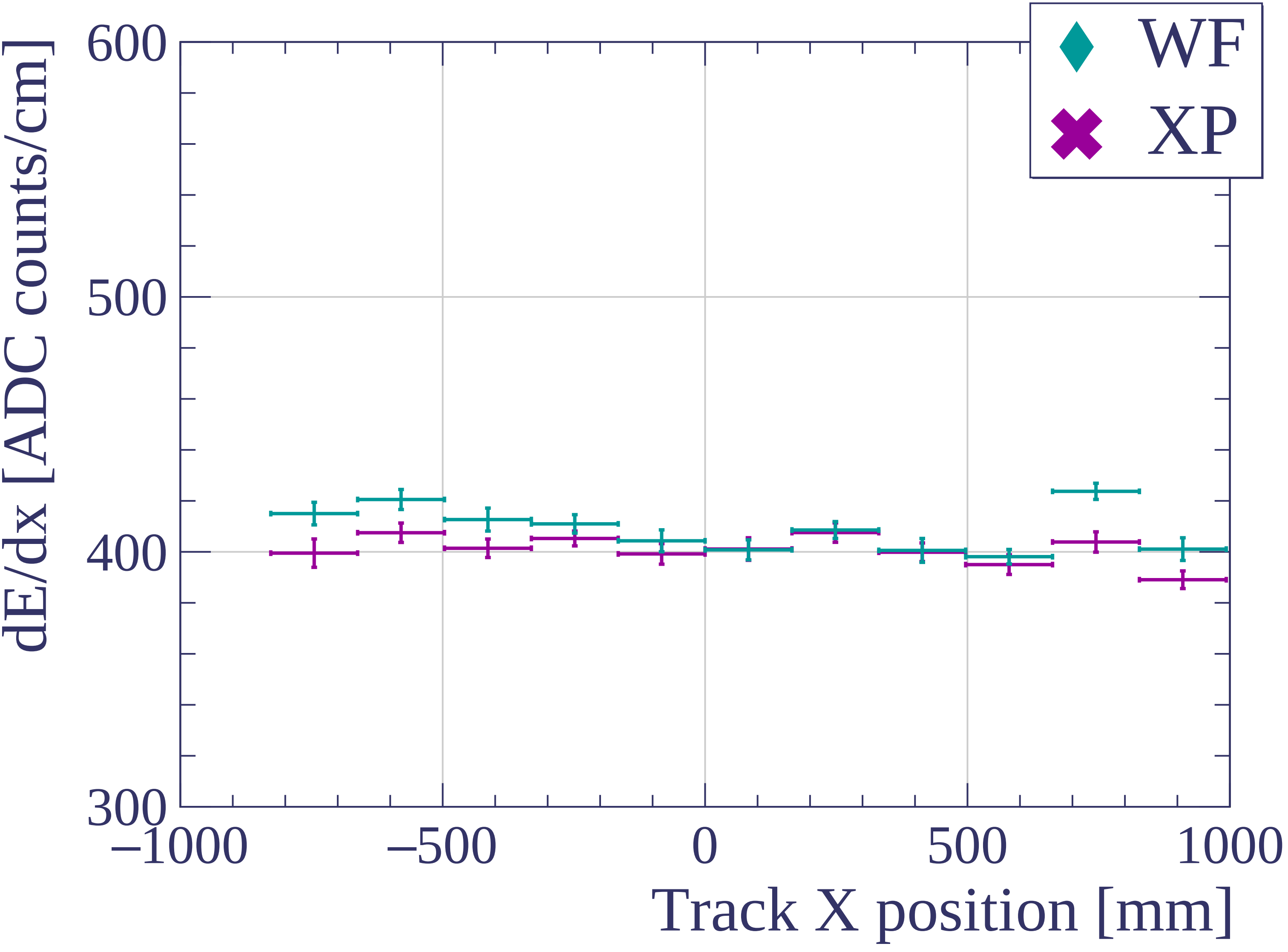
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
4
3
2
1
0


Count
Both HATs
tHAT
bHAT
0
10
20
−2000
−1000
0
1000
2000
Signed momentum (MeV)

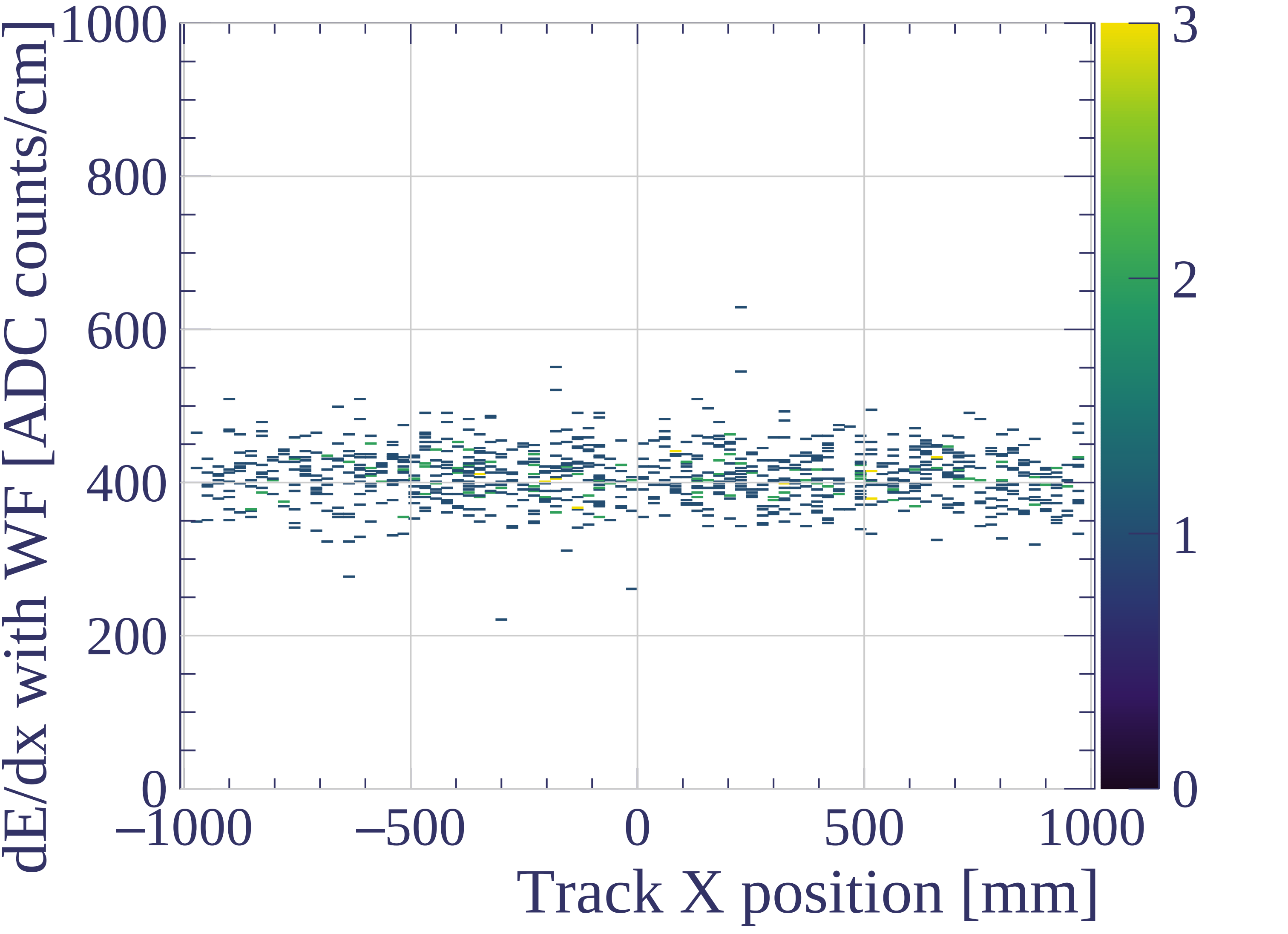

χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
4
3
2
1
0



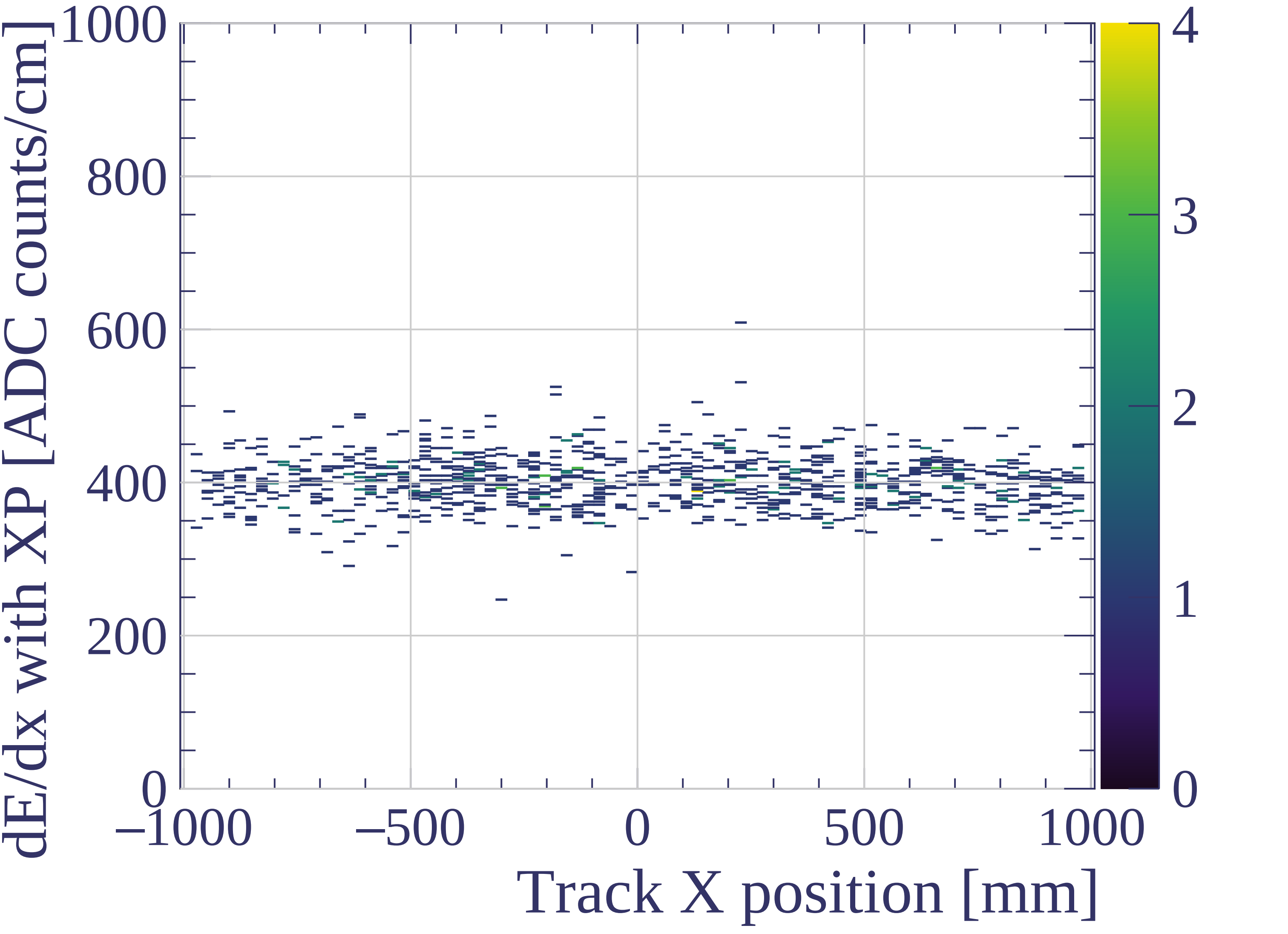
WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

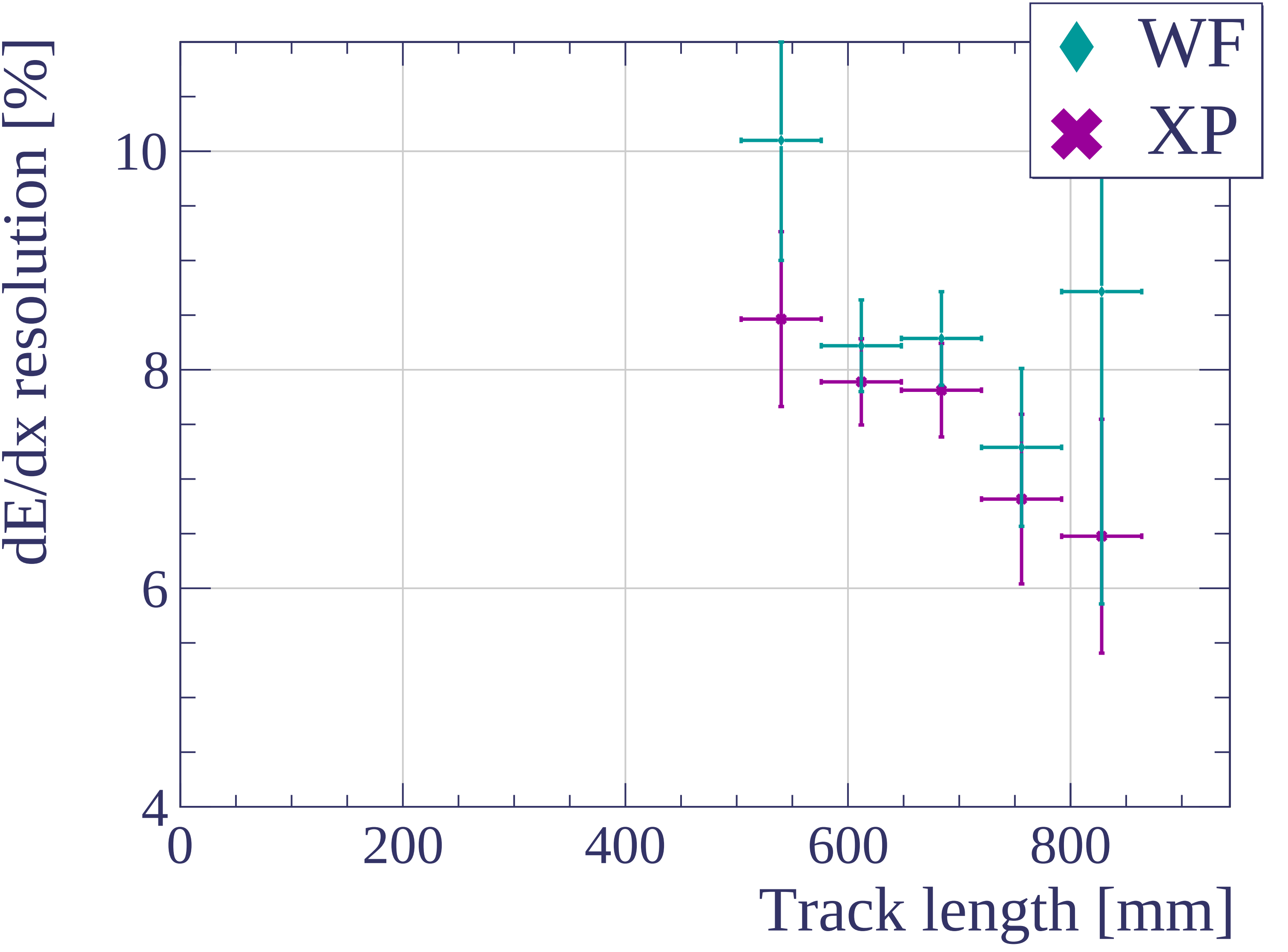


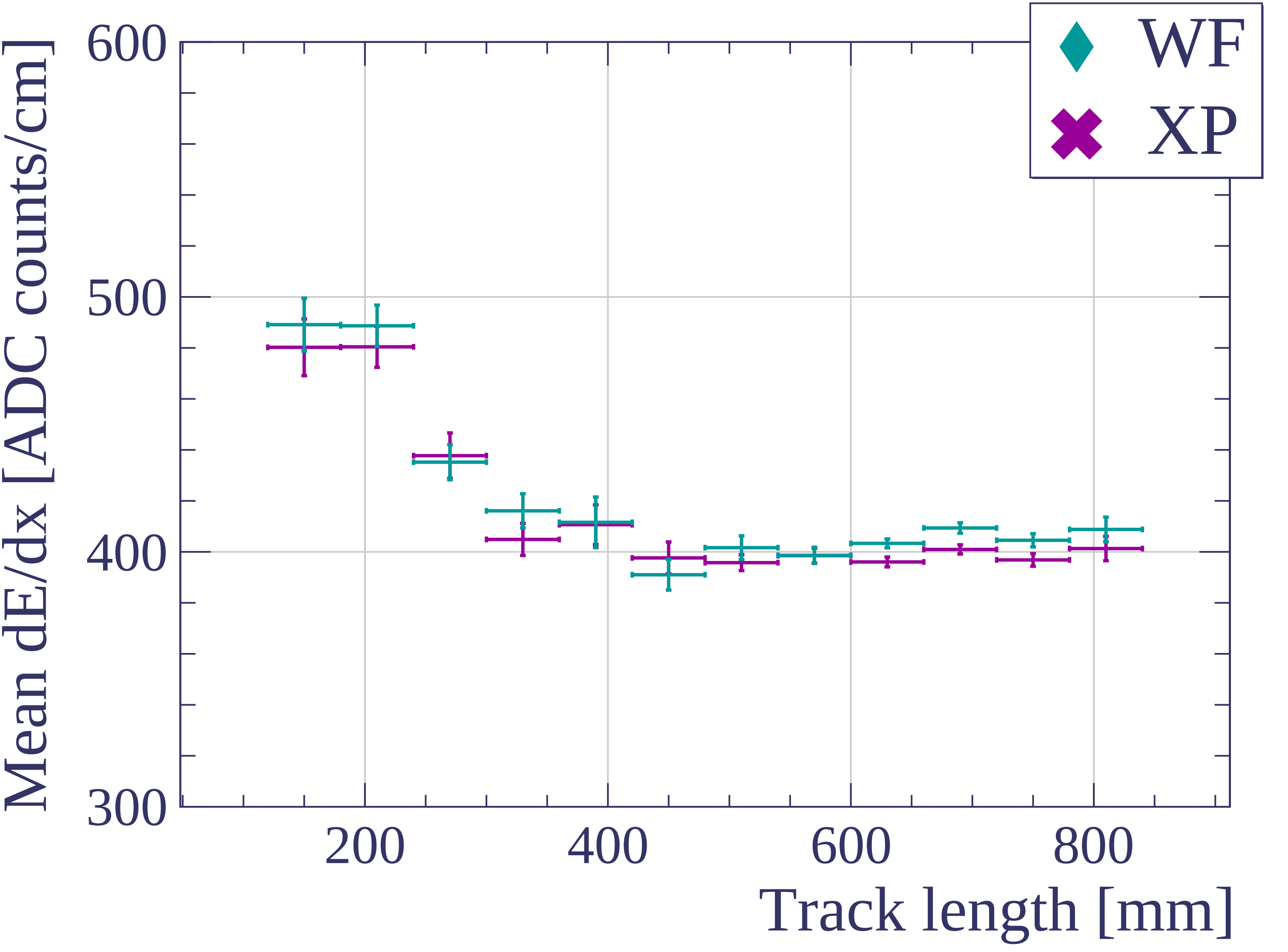
WF
XP
600
500
400
300
dE/dx [ADC counts/cm]
−1000
−500
0
500
1000
Track X position [mm]



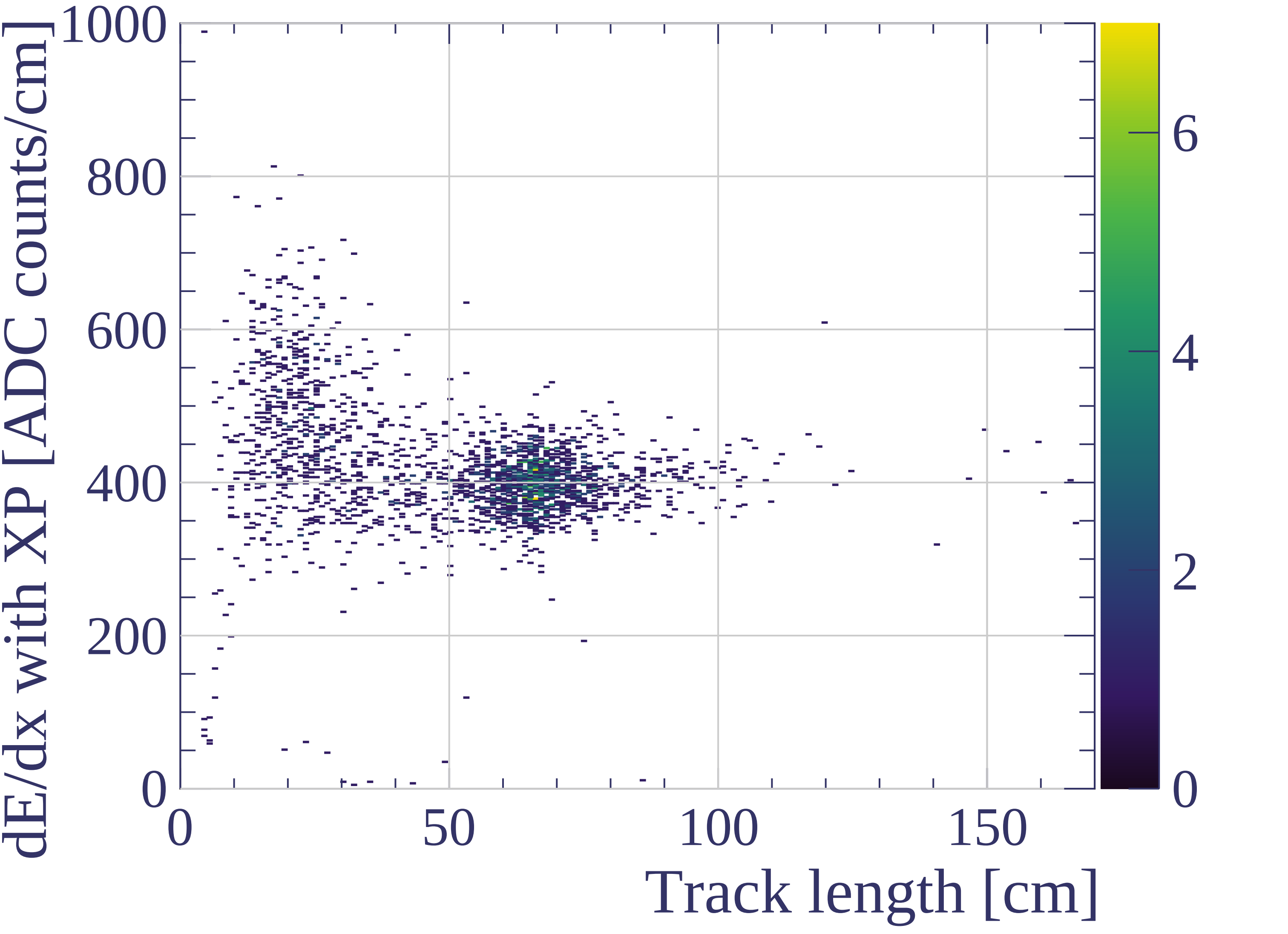
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
3
2
1
0


dE/dx with XP [ADC counts/cm]
Track X position [mm]
0
200
400
600
800
1000
−1000
−500
0
500
1000
0
1
2
3
4

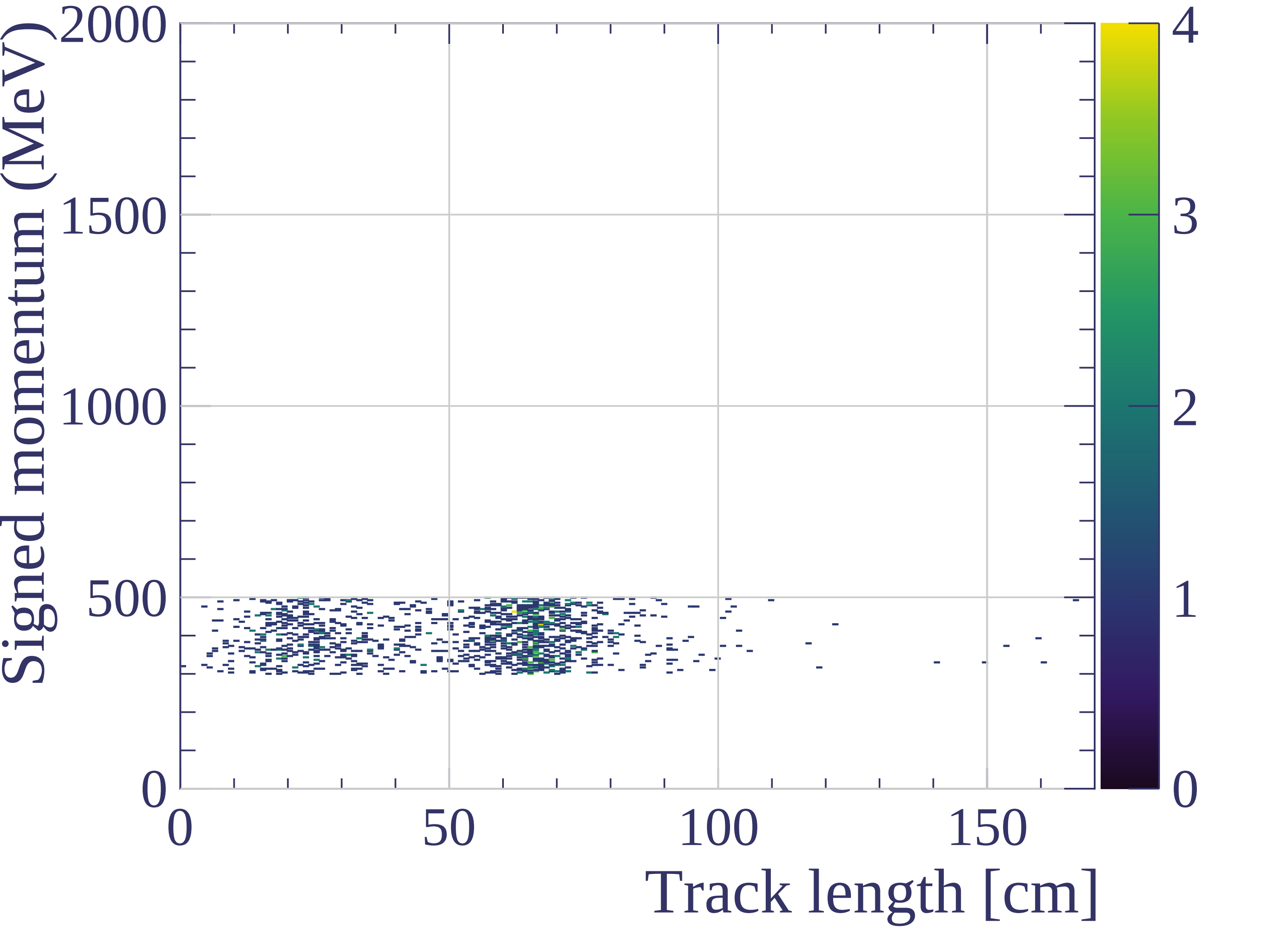

WF
XP
dE/dx resolution [%]
10
8
6
4
0
200
400
600
800
Track length [mm]



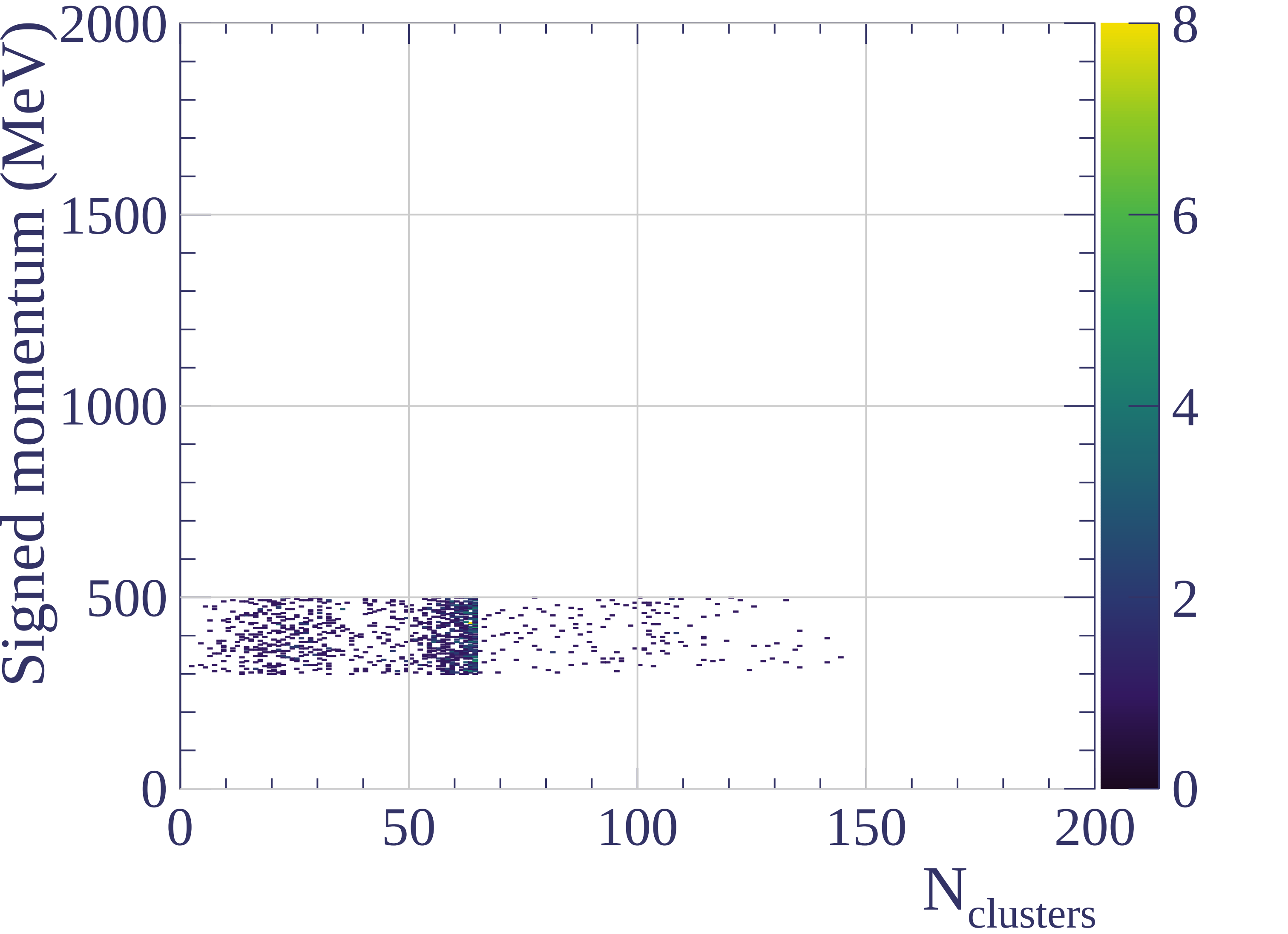
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
200
400
600
800
Track length [mm]

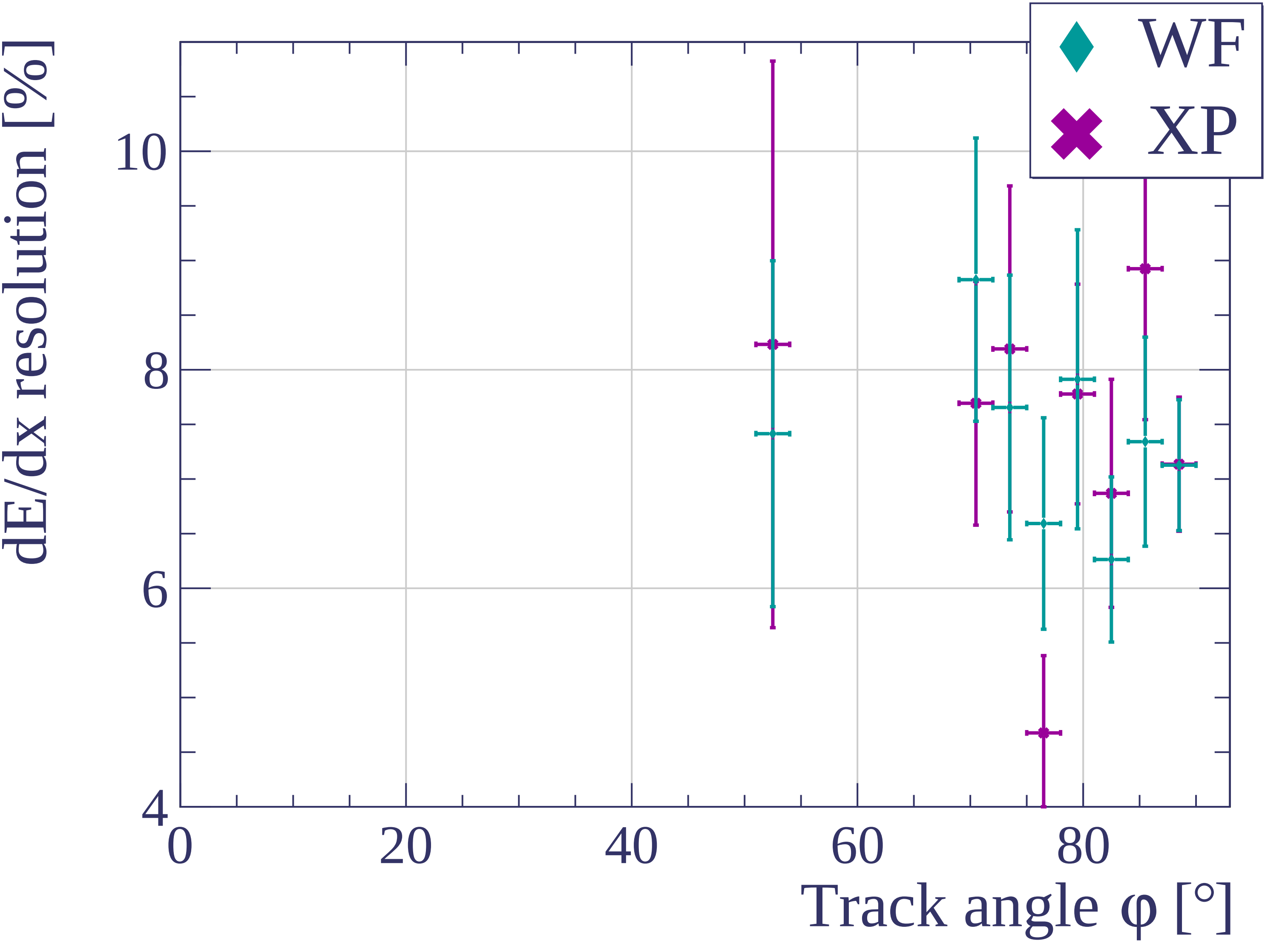


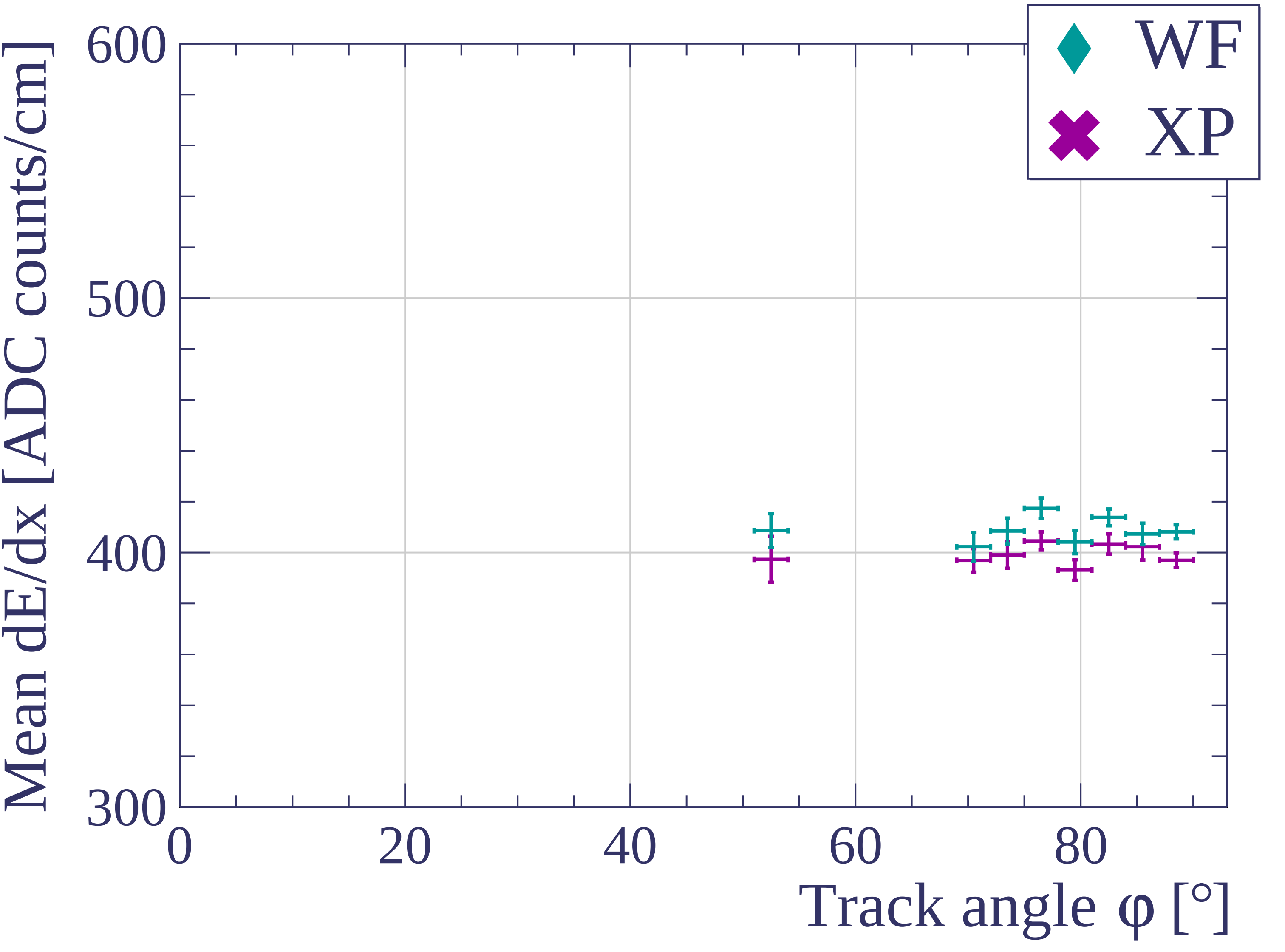
dE/dx with XP [ADC counts/cm]
Track length [cm]
0
200
400
600
800
1000
0
50
100
150
0
2
4
6

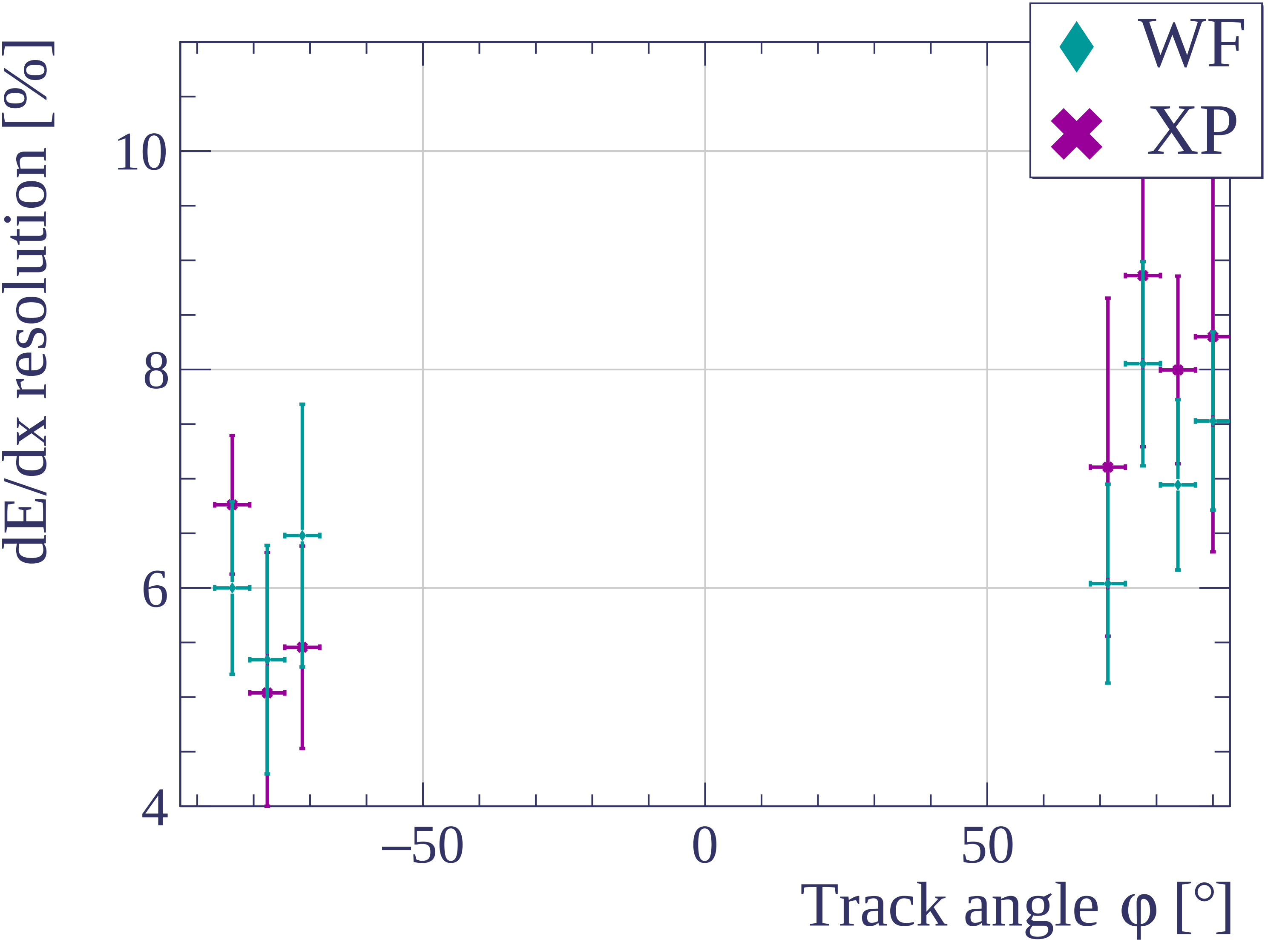


Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
4
3
2
1
0

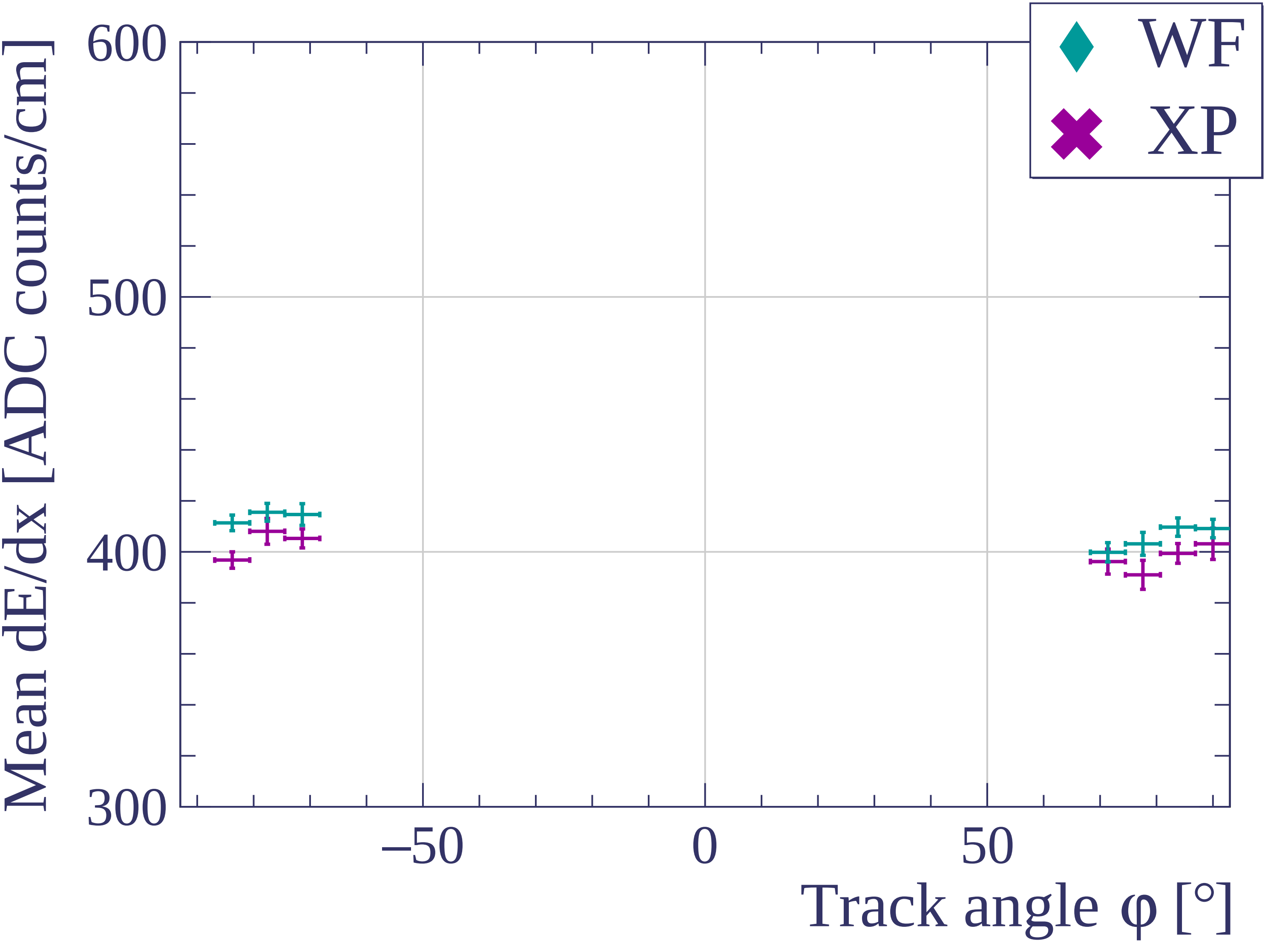

Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
8
6
4
2
0


WF
XP
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle φ [°]

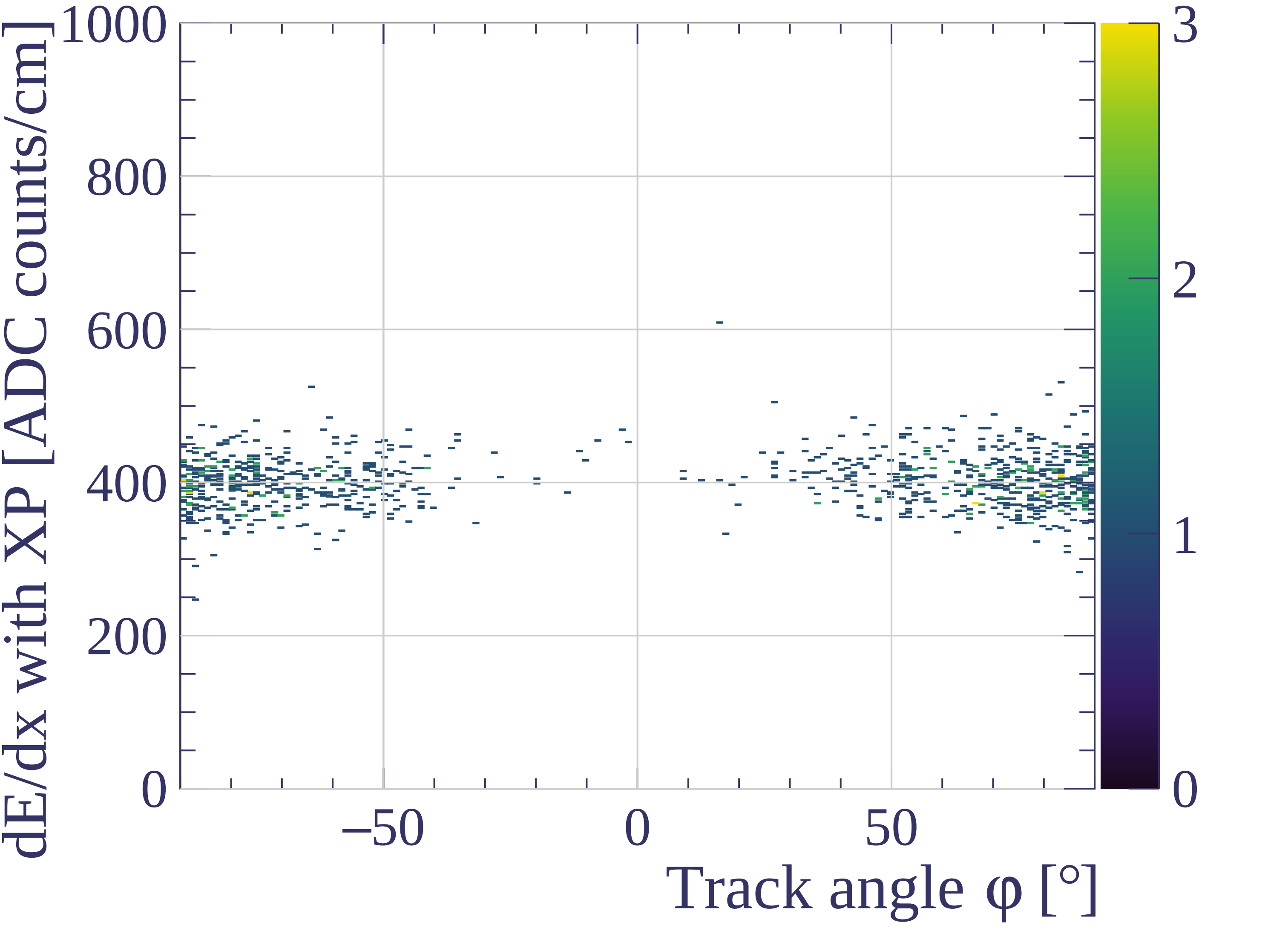

WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
0
20
40
60
80
Track angle φ [°]



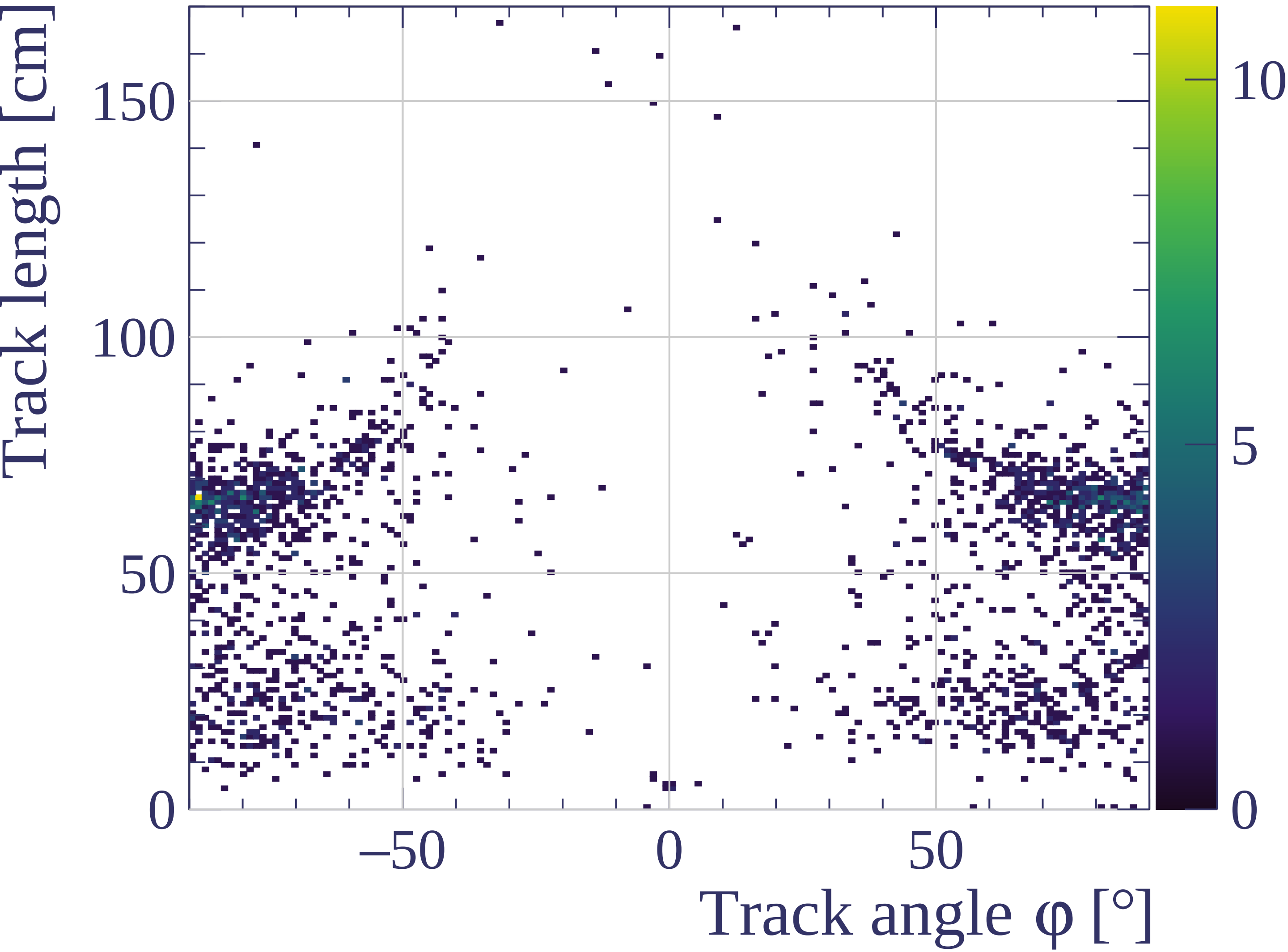
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

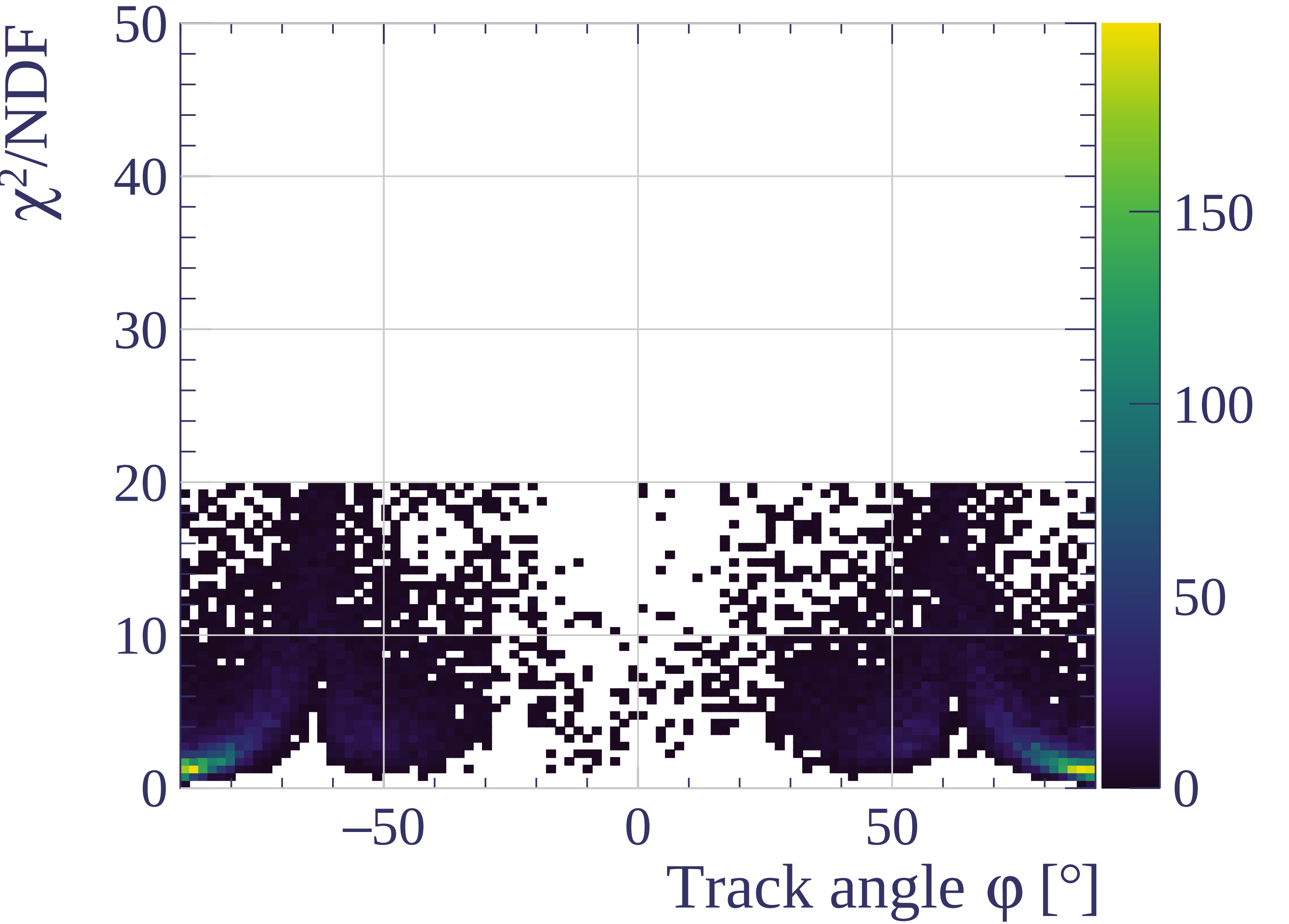


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle φ [°]

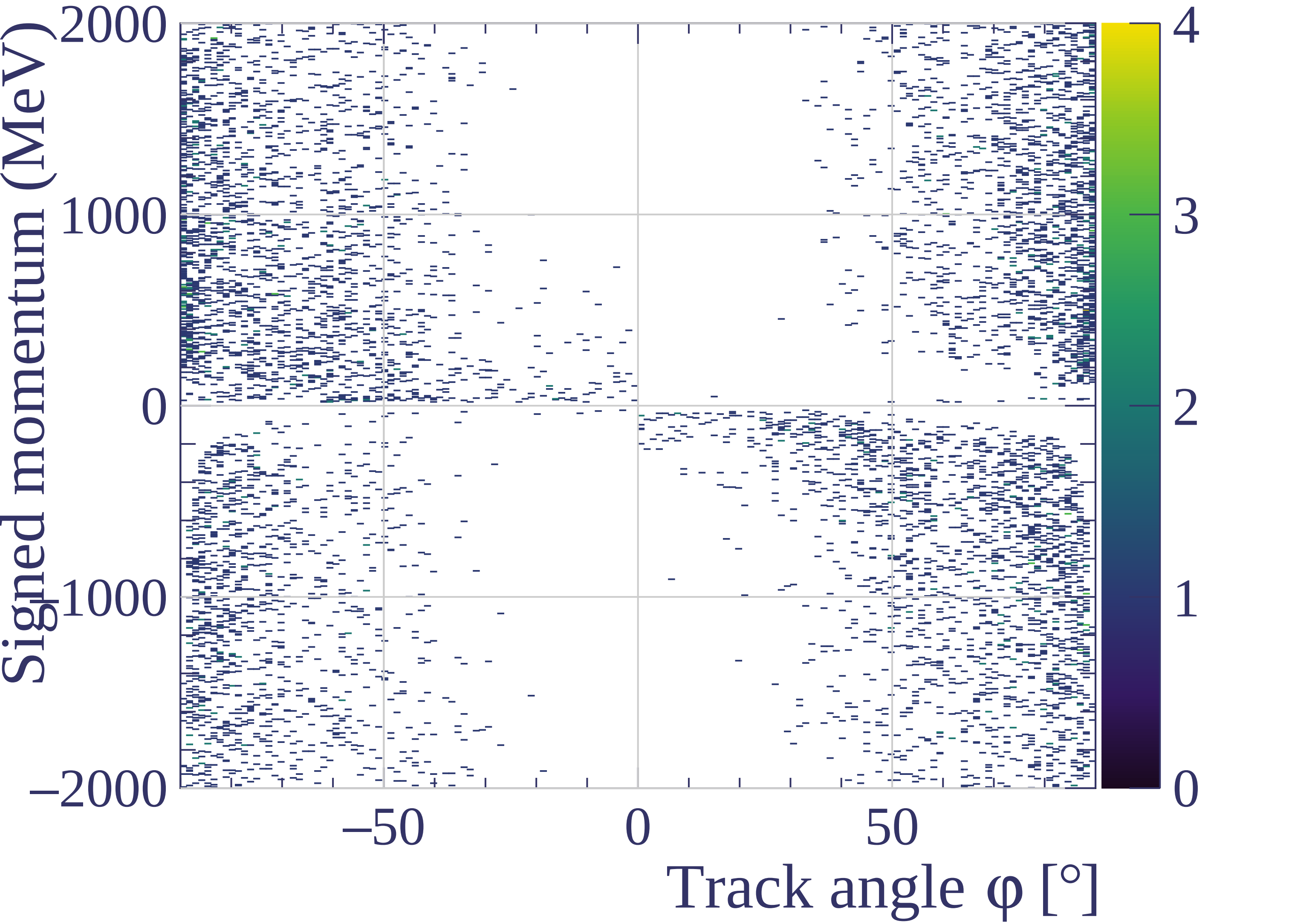


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
3
2
1
0

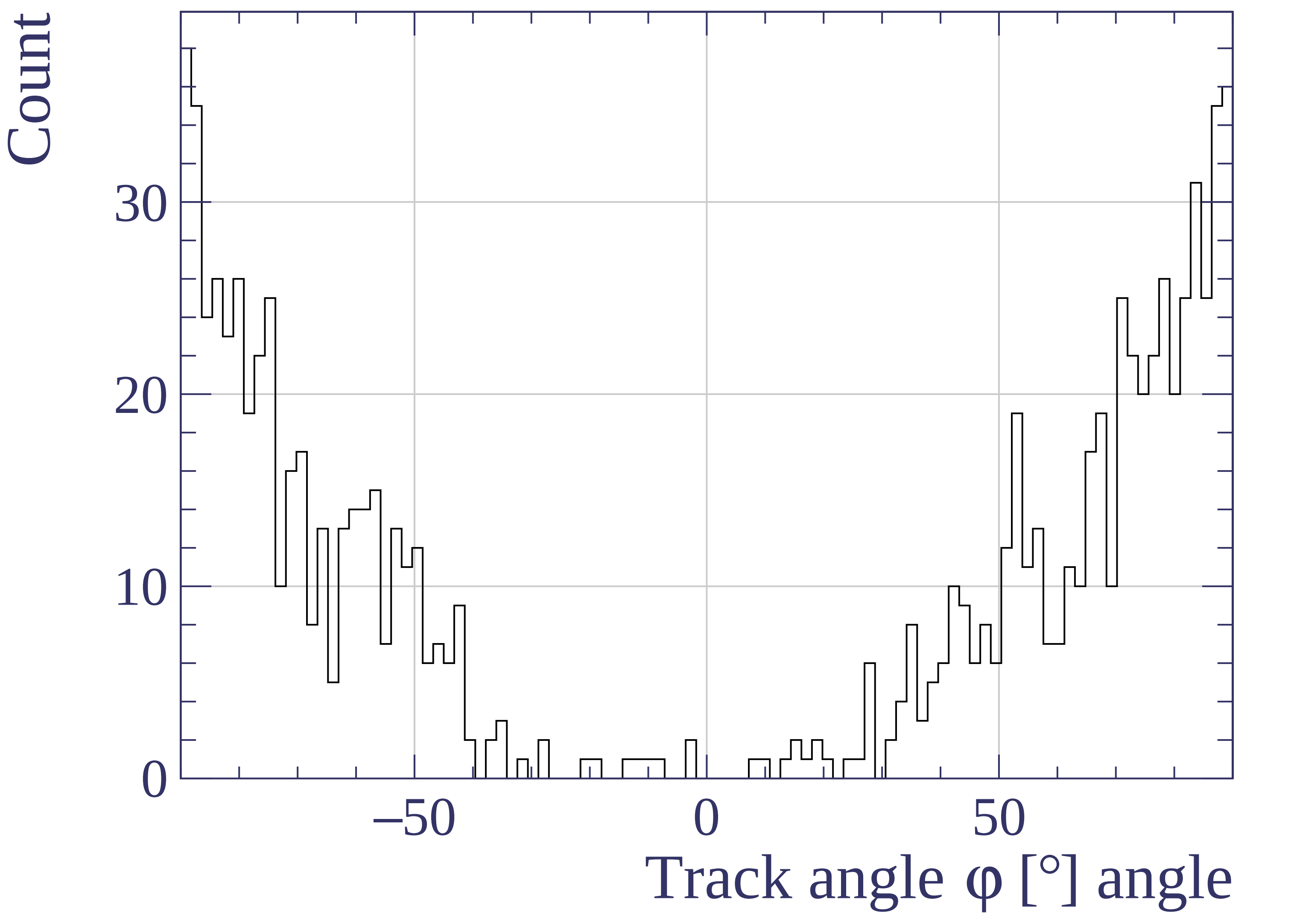

Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
10
5
0



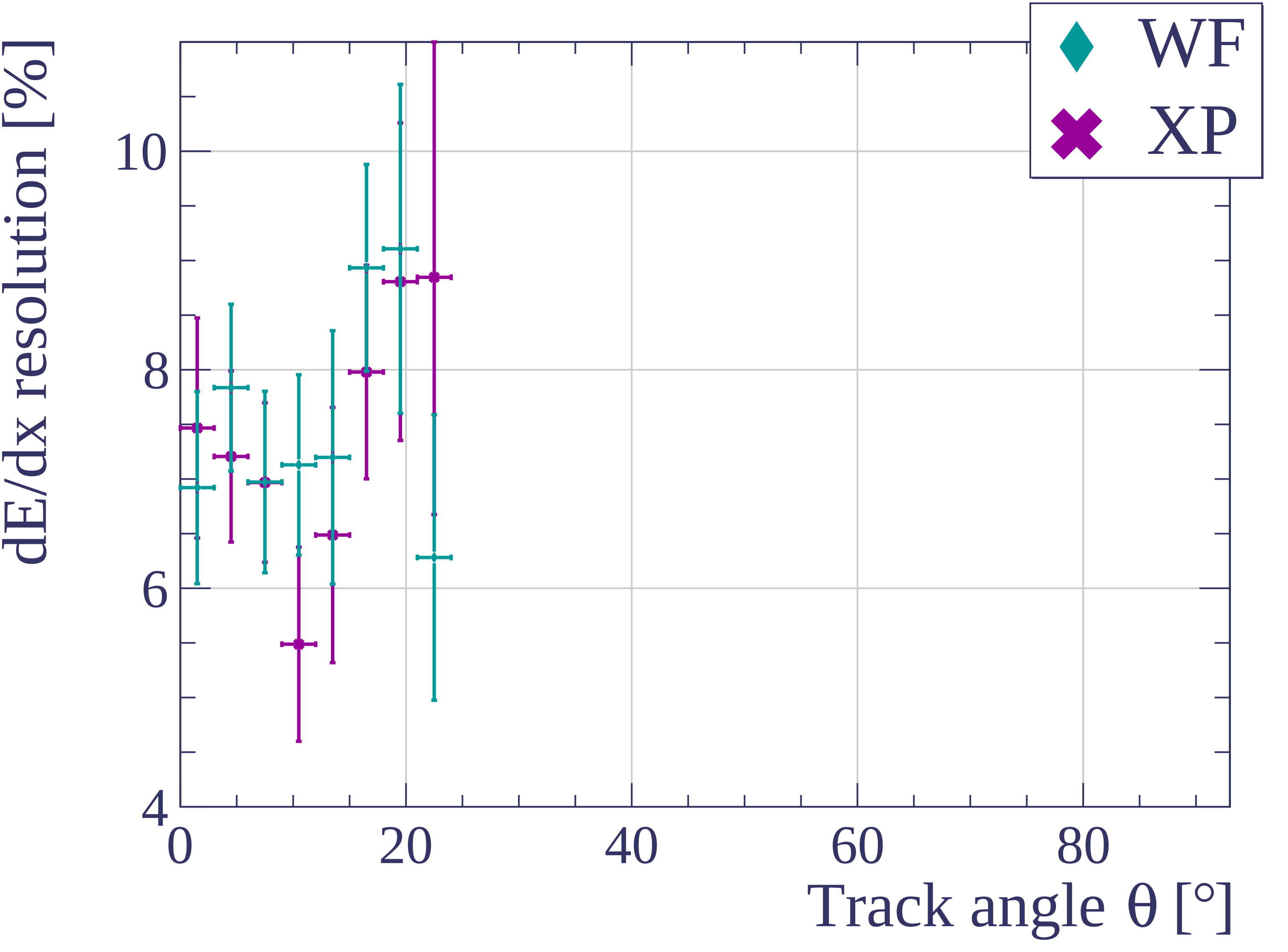
χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
150
100
50
0



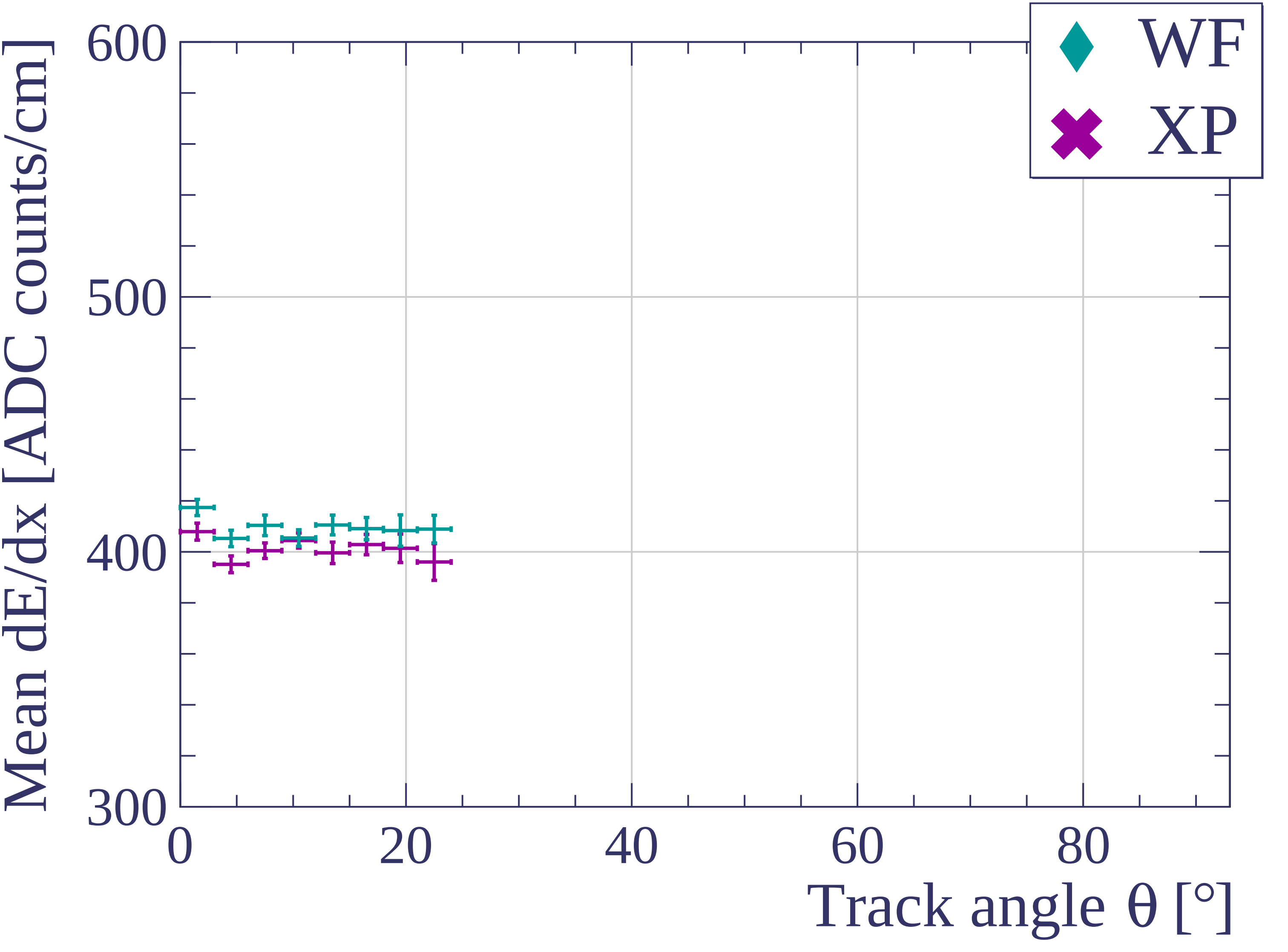
2000
1000
0
−1000
−2000
Signed momentum (MeV)
−50
0
50
Track angle φ [°]
4
3
2
1
0



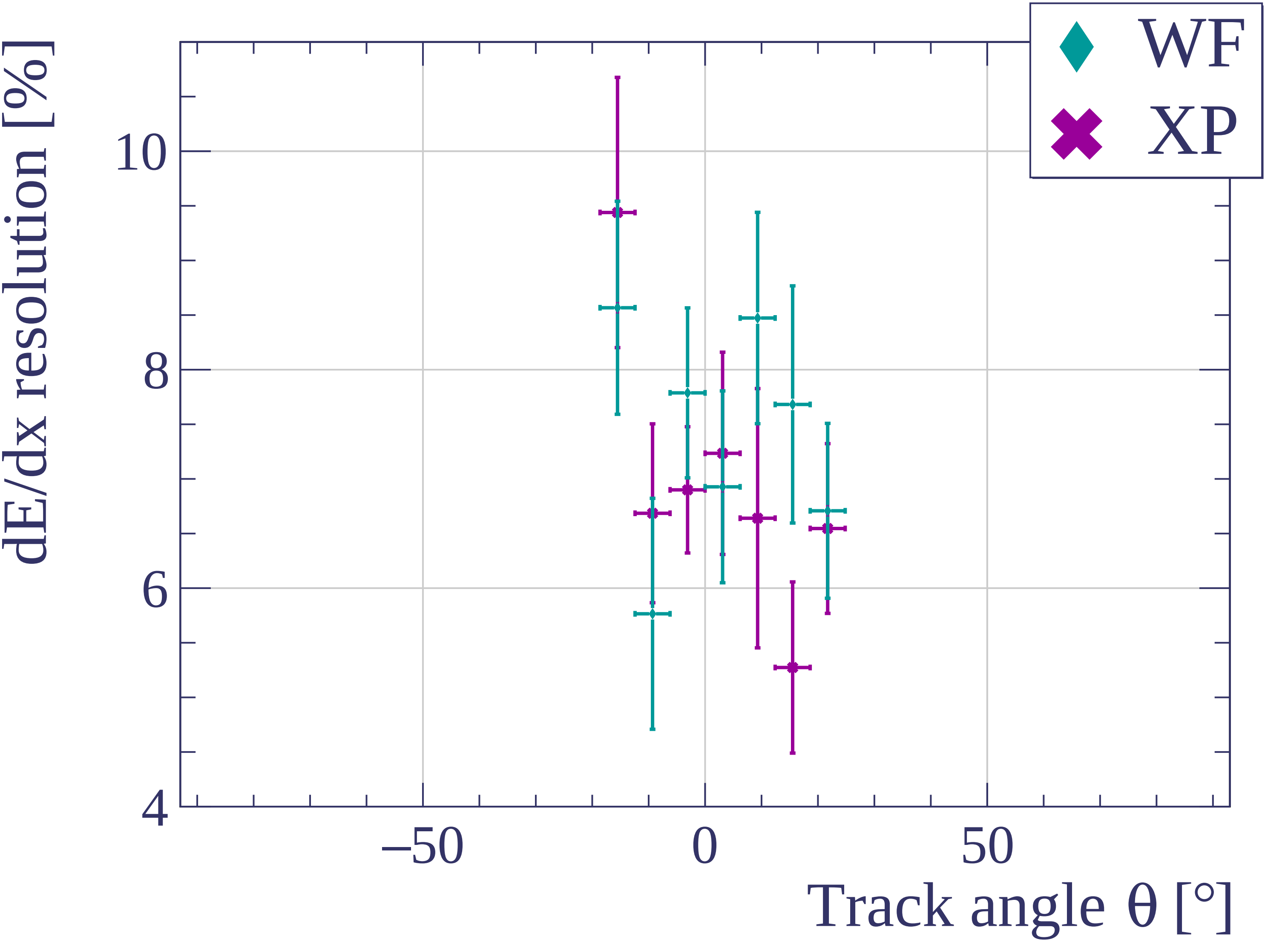
Count
30
20
10
0
−50
0
50
Track angle φ [°] angle



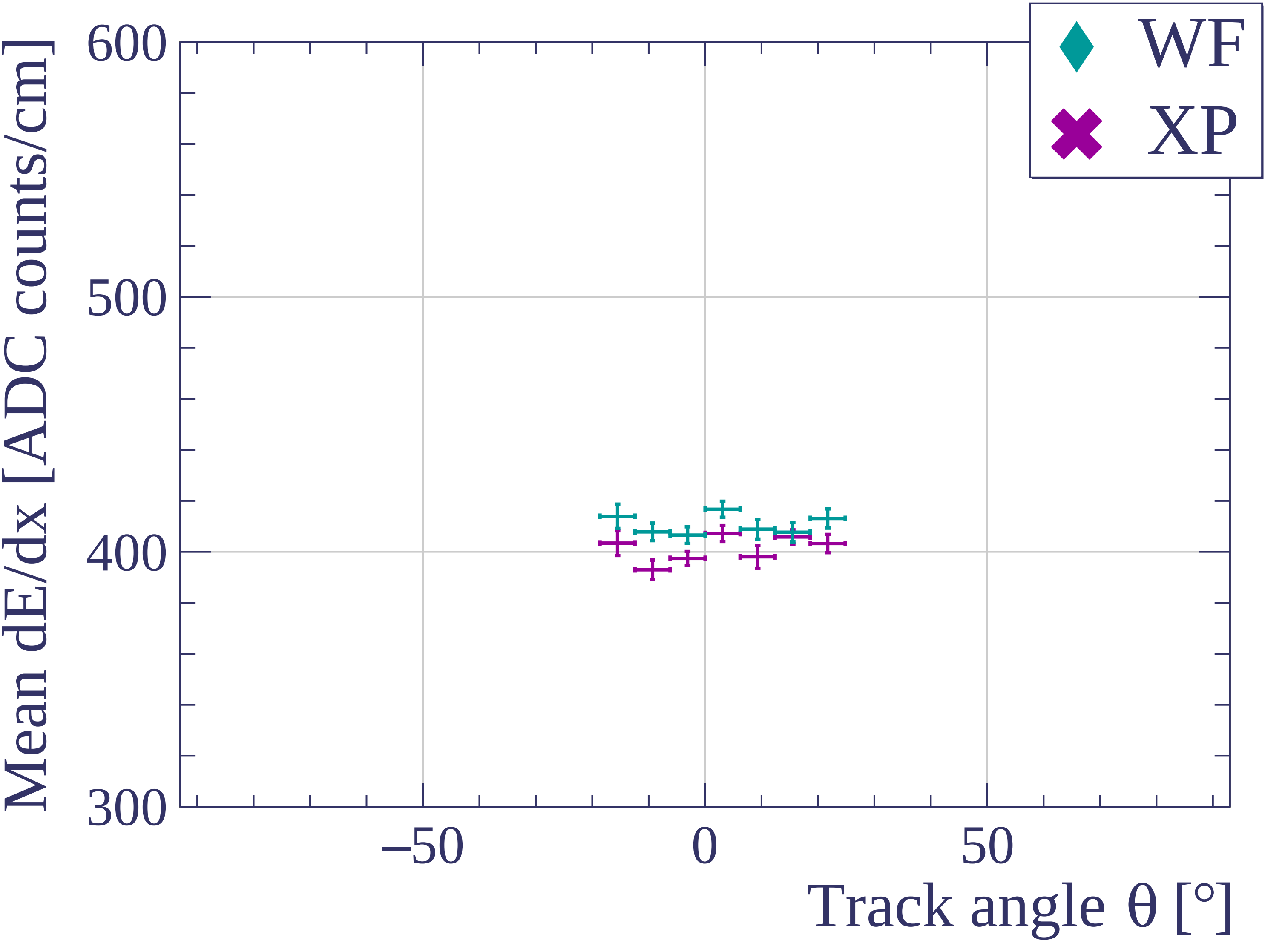
dE/dx resolution [%]
4
6
8
10
0
20
40
60
80
Track angle θ [°]
WF
XP



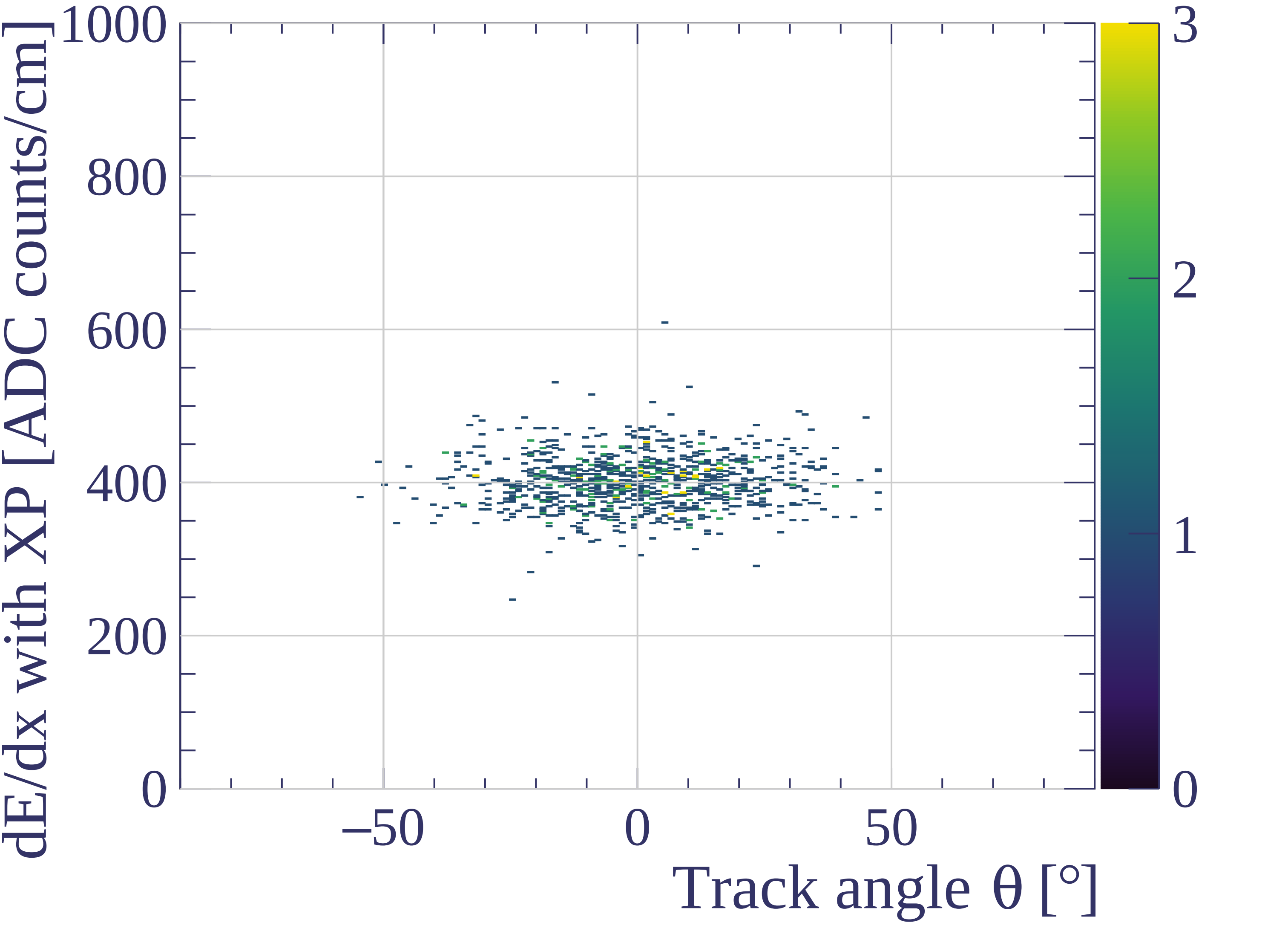
WF
XP
600
500
400
300
0
20
40
60
80
Mean dE/dx [ADC counts/cm]
Track angle θ [°]



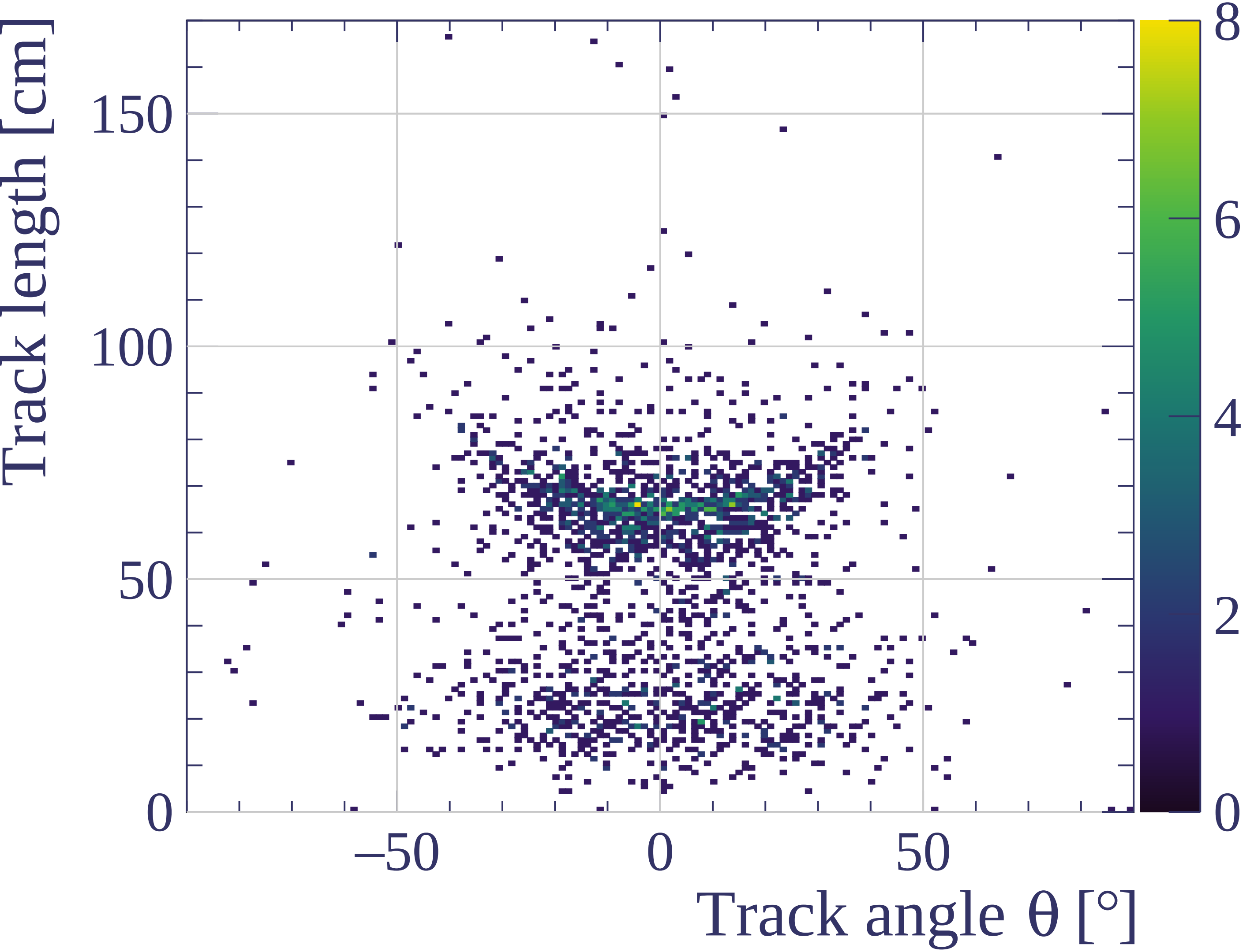
WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]



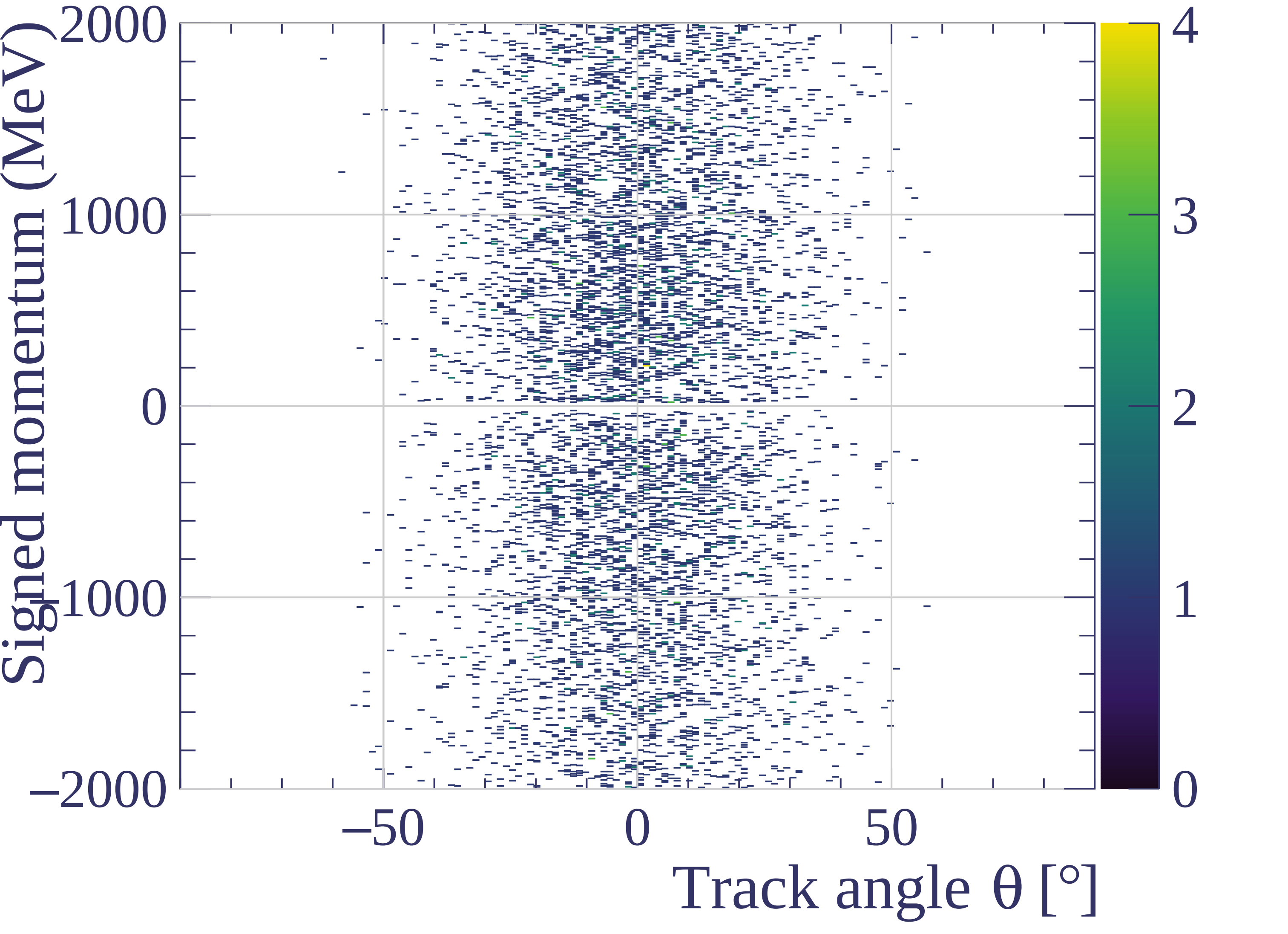
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]



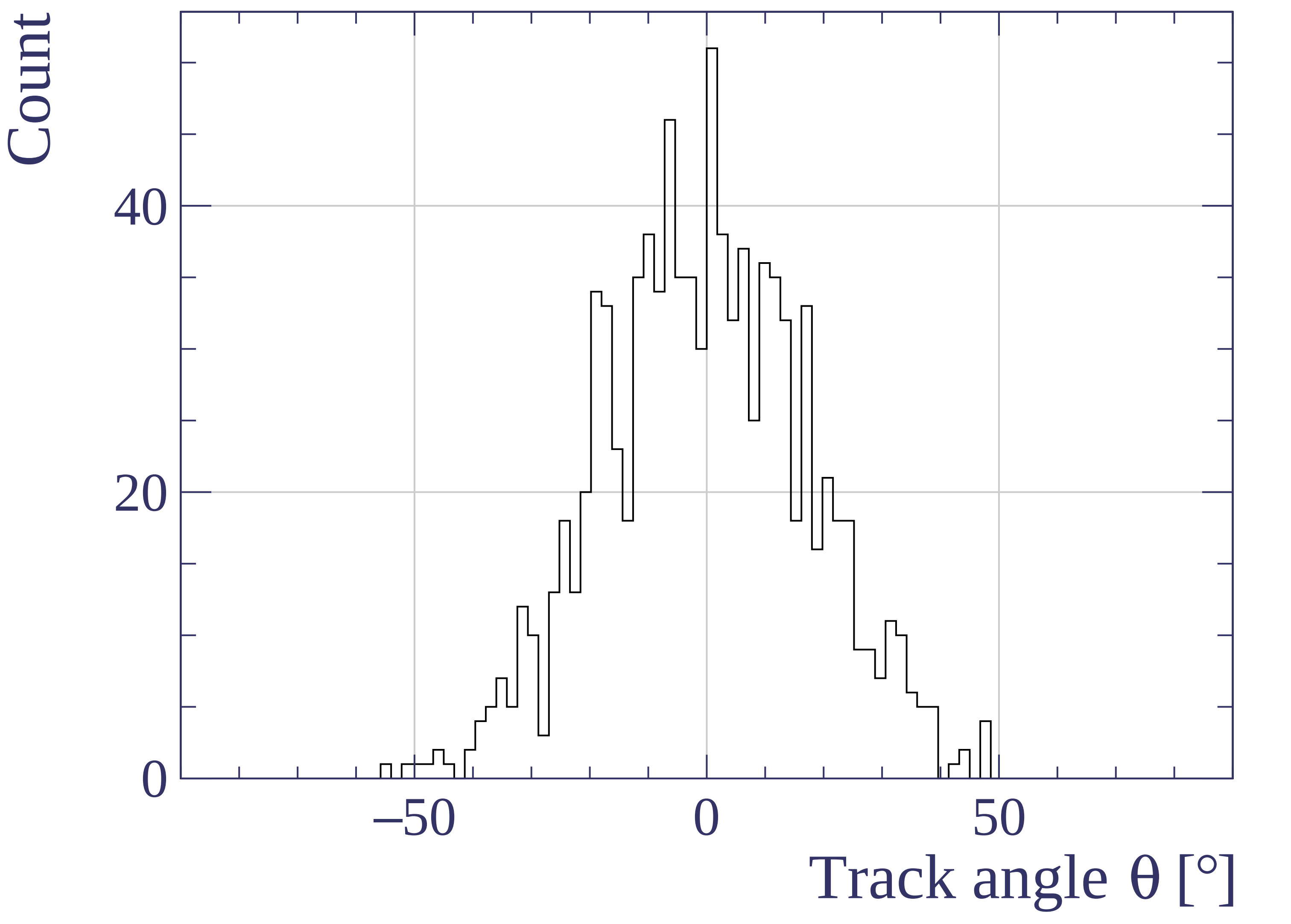
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
3
2
1
0



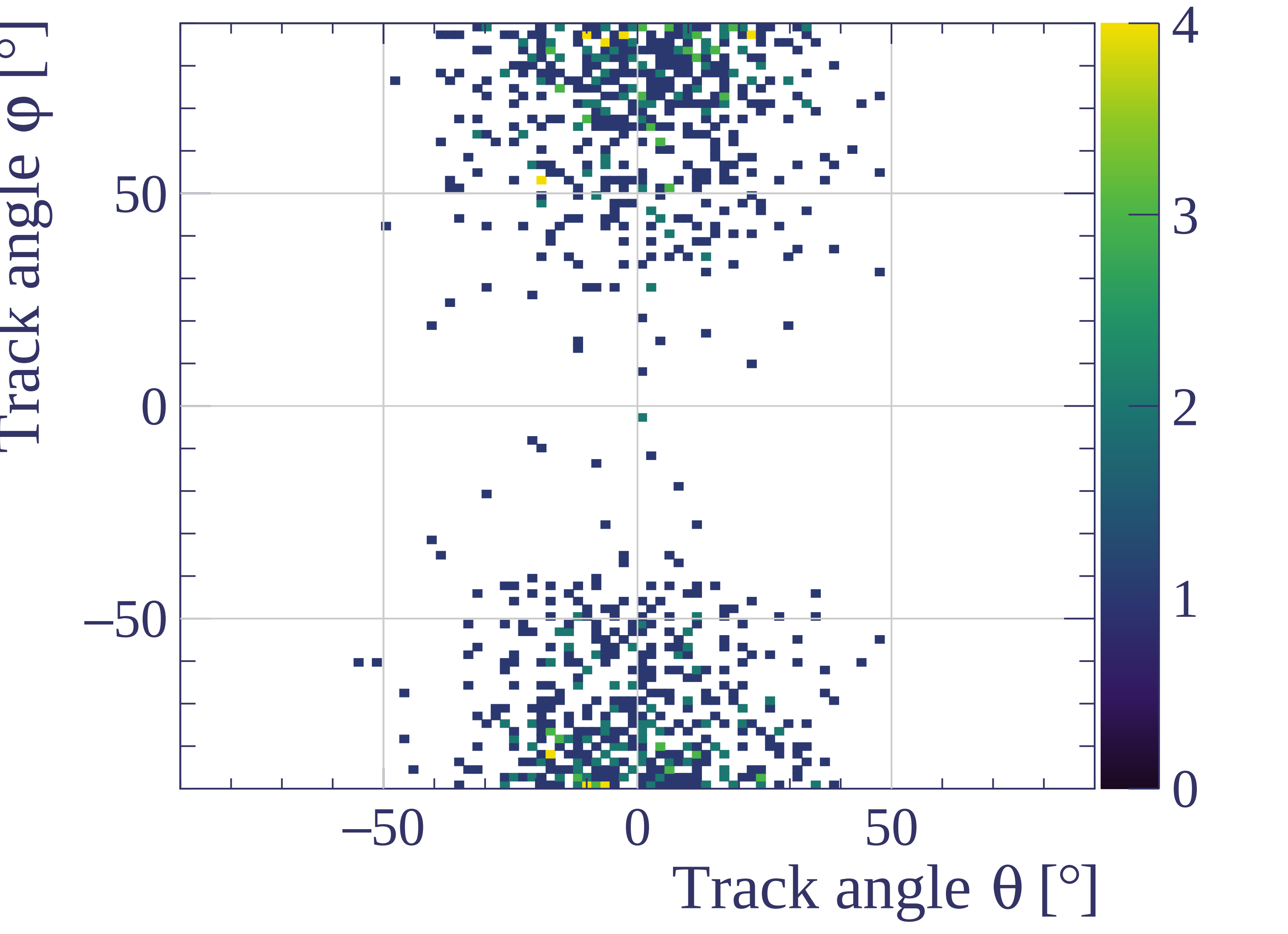
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
8
6
4
2
0



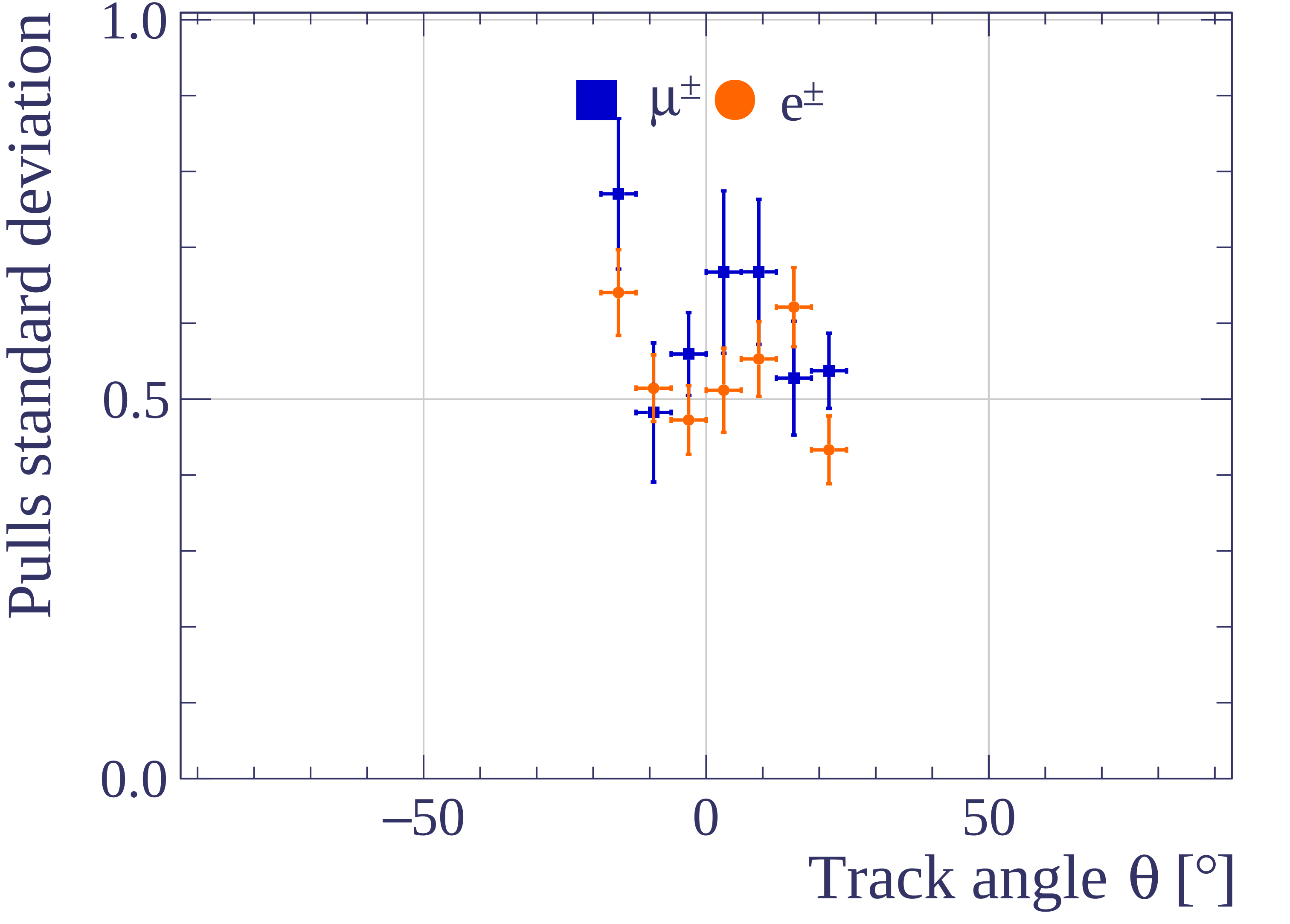
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
0
1
2
3
4



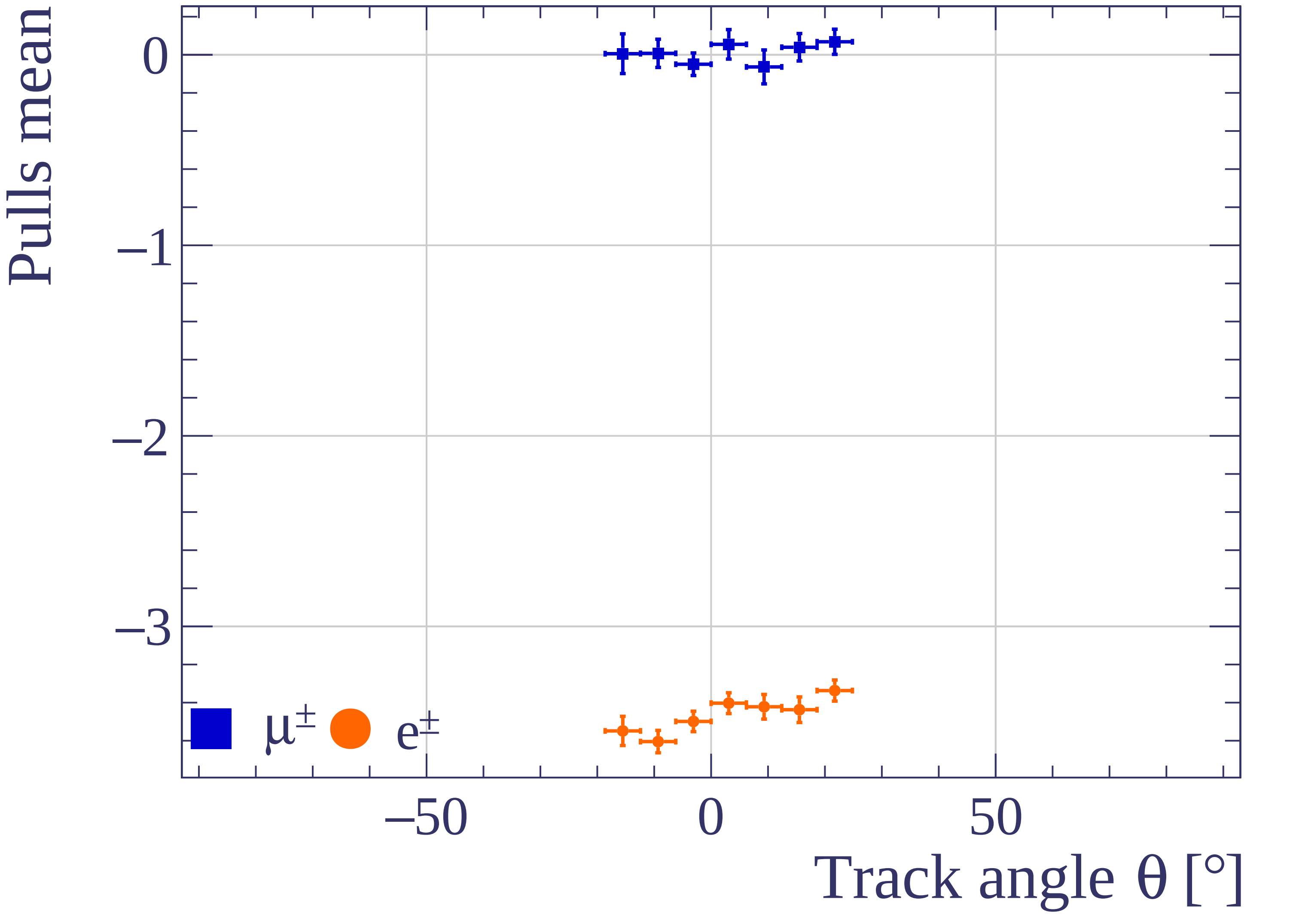
Count
40
20
0
−50
0
50
Track angle θ [°]



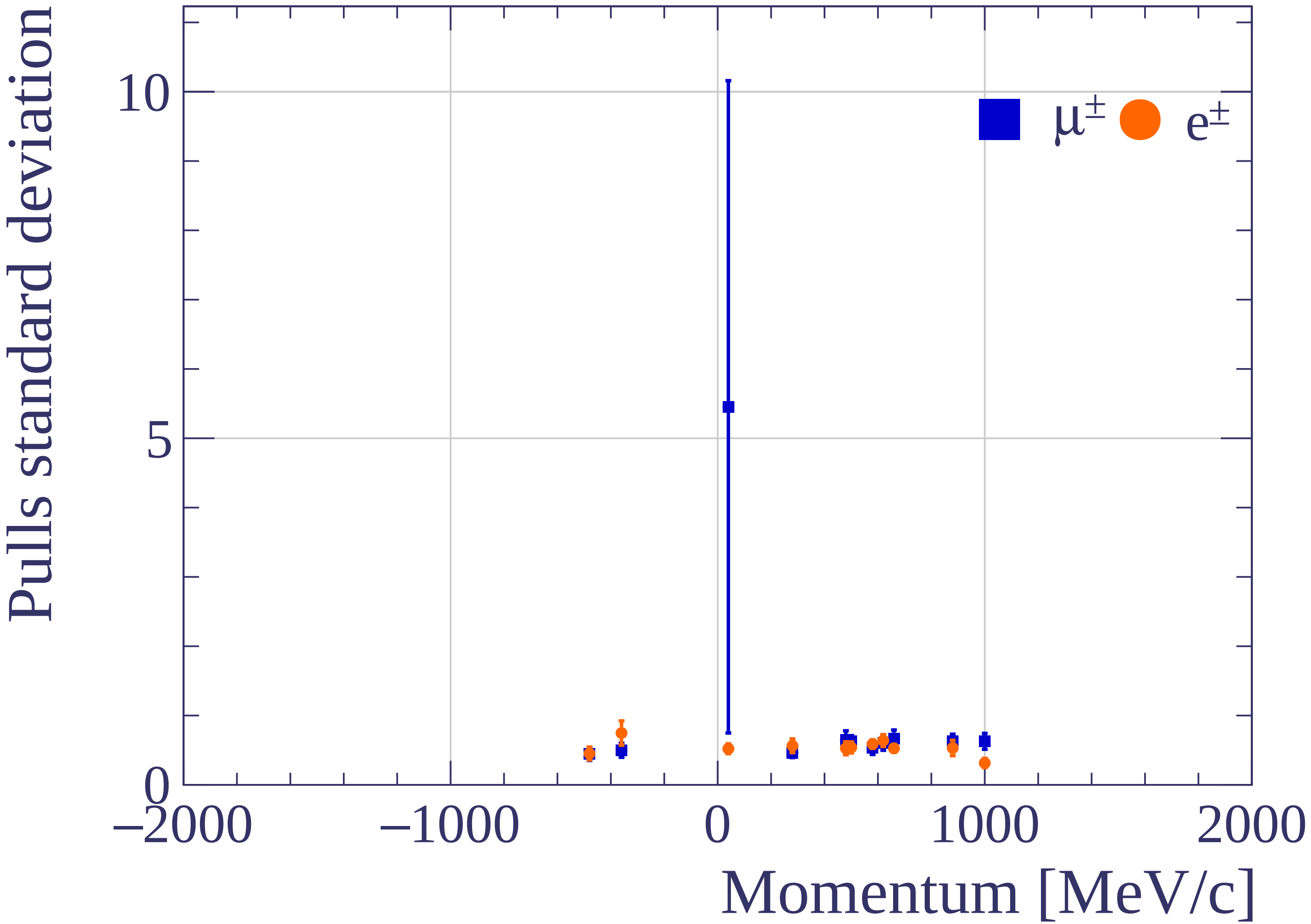
Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
4
3
2
1
0



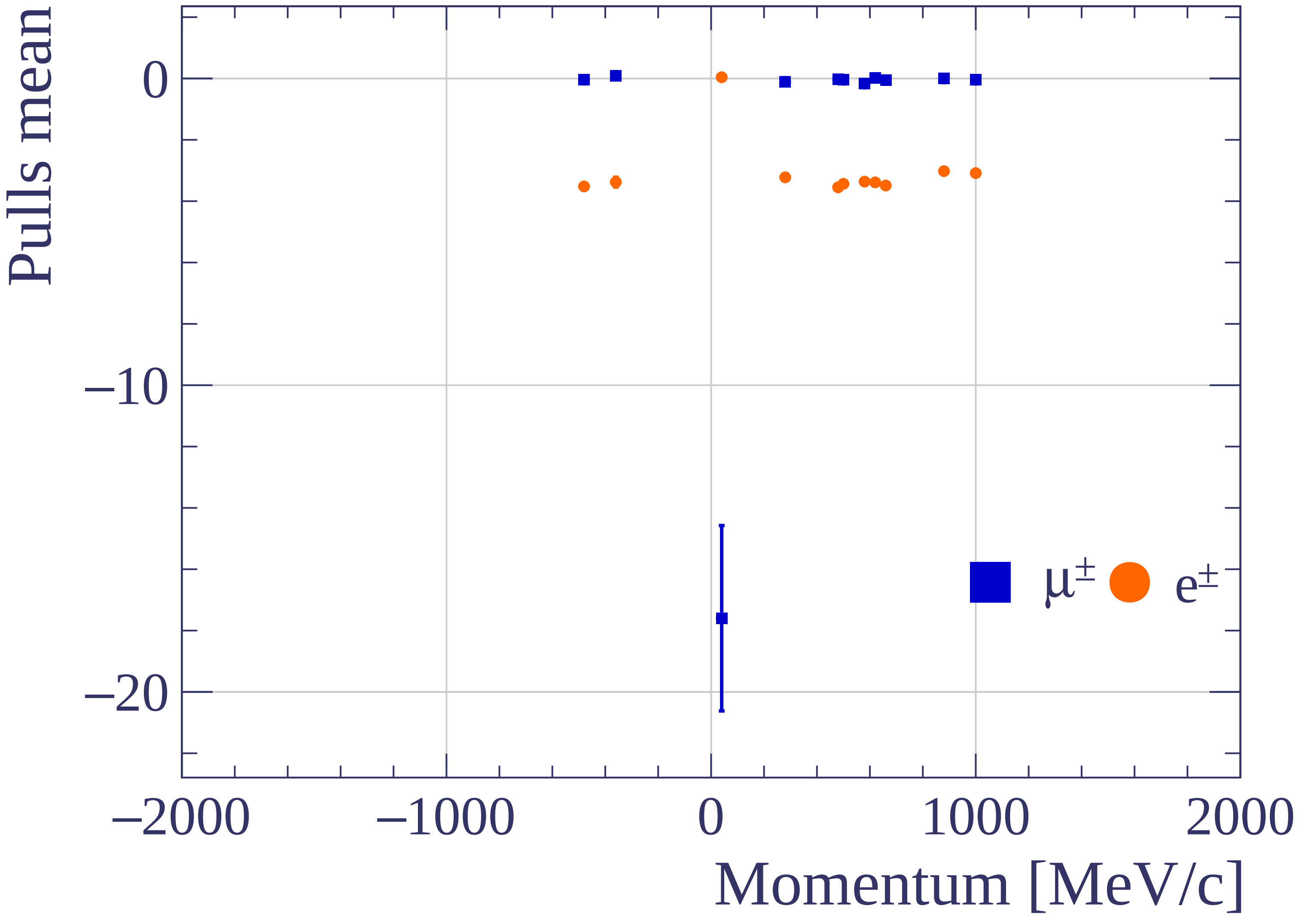
Pulls standard deviation
1.0
0.5
0.0
−50
0
50
Track angle θ [°]
μ±
e±



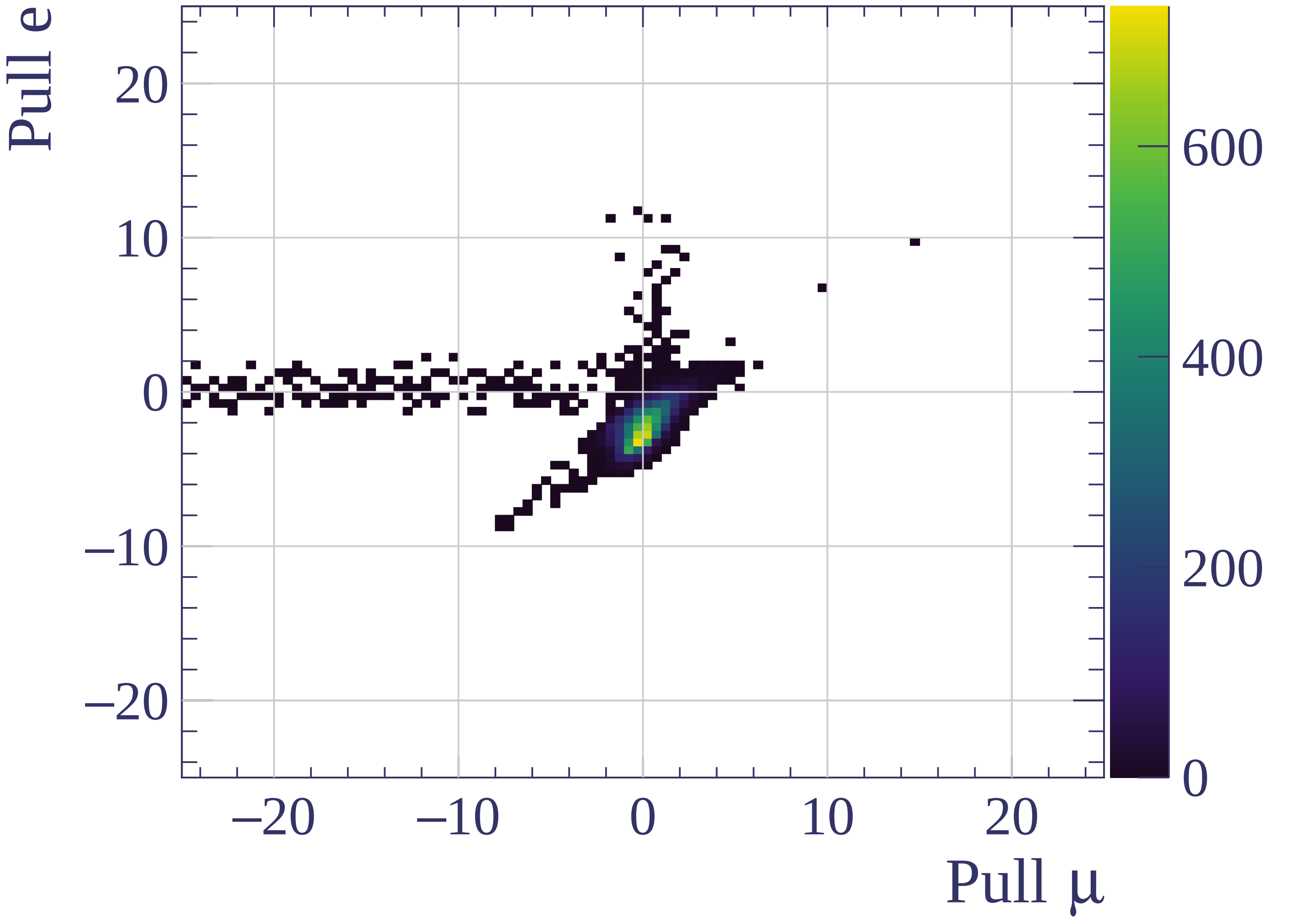
Pulls mean
0
−1
−2
−3
μ±
e±
−50
0
50
Track angle θ [°]



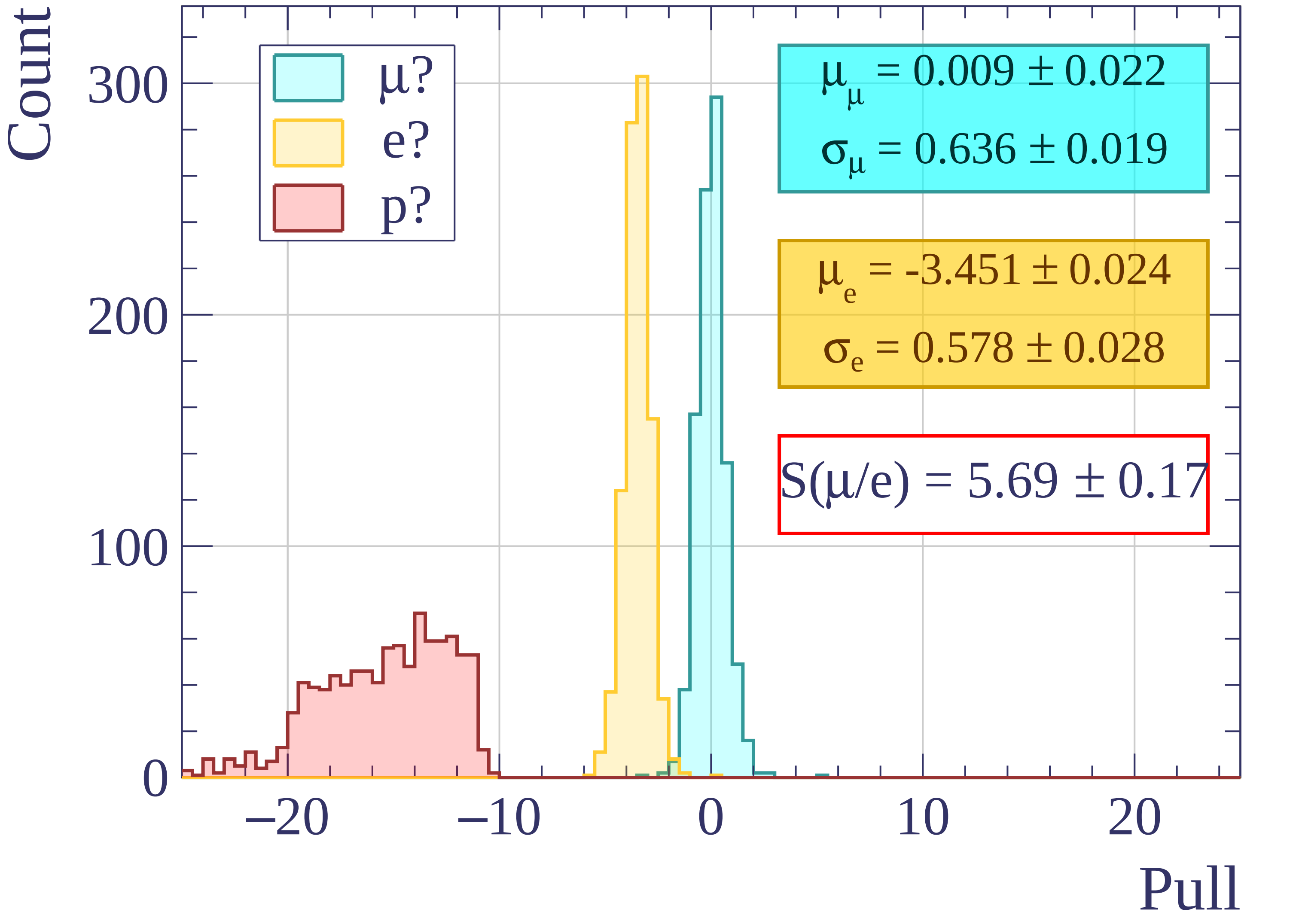
Pulls standard deviation
10
5
0
−2000
−1000
0
1000
2000
Momentum [MeV/c]
μ±
e±



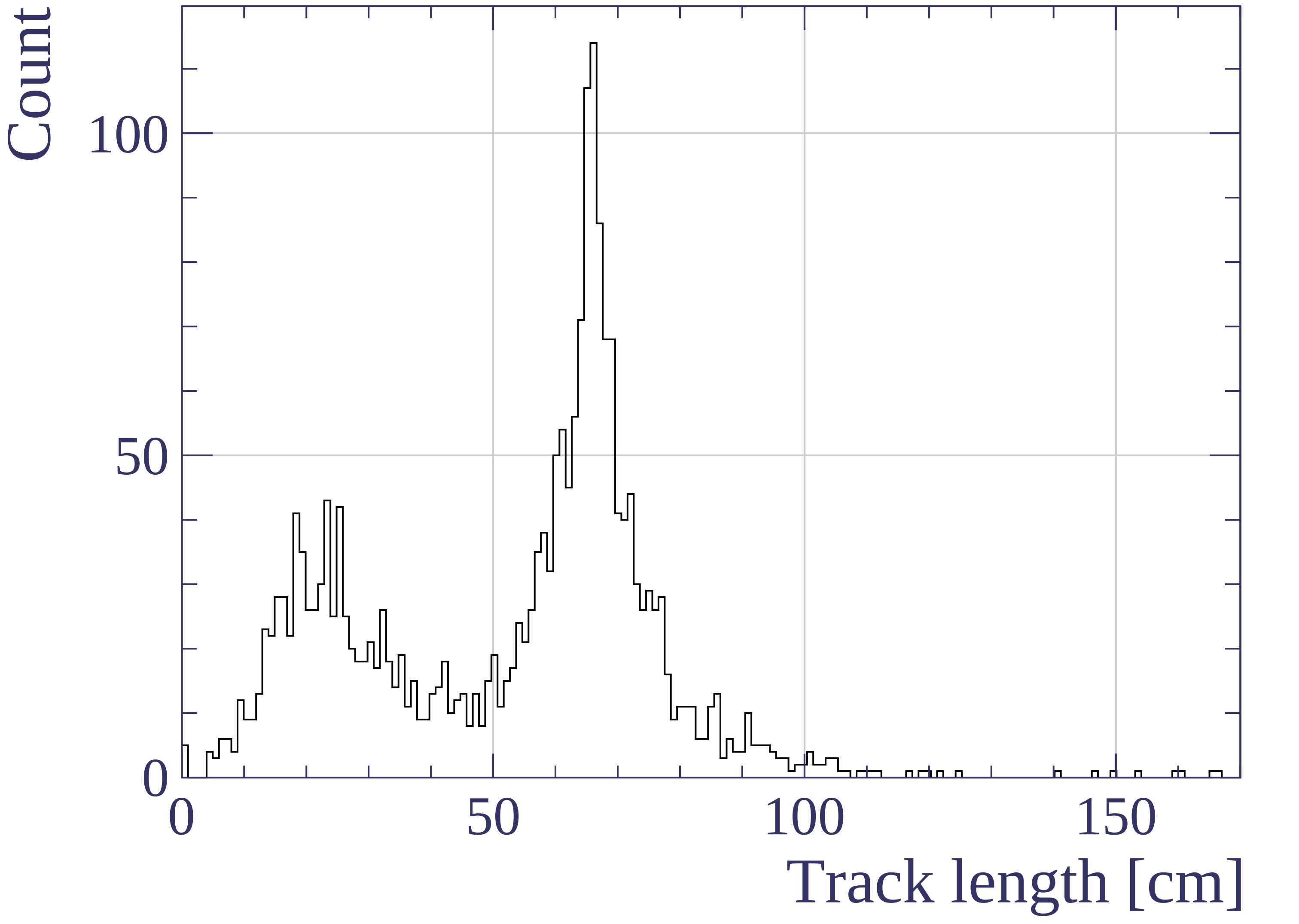
Pulls mean
0
−10
−20
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



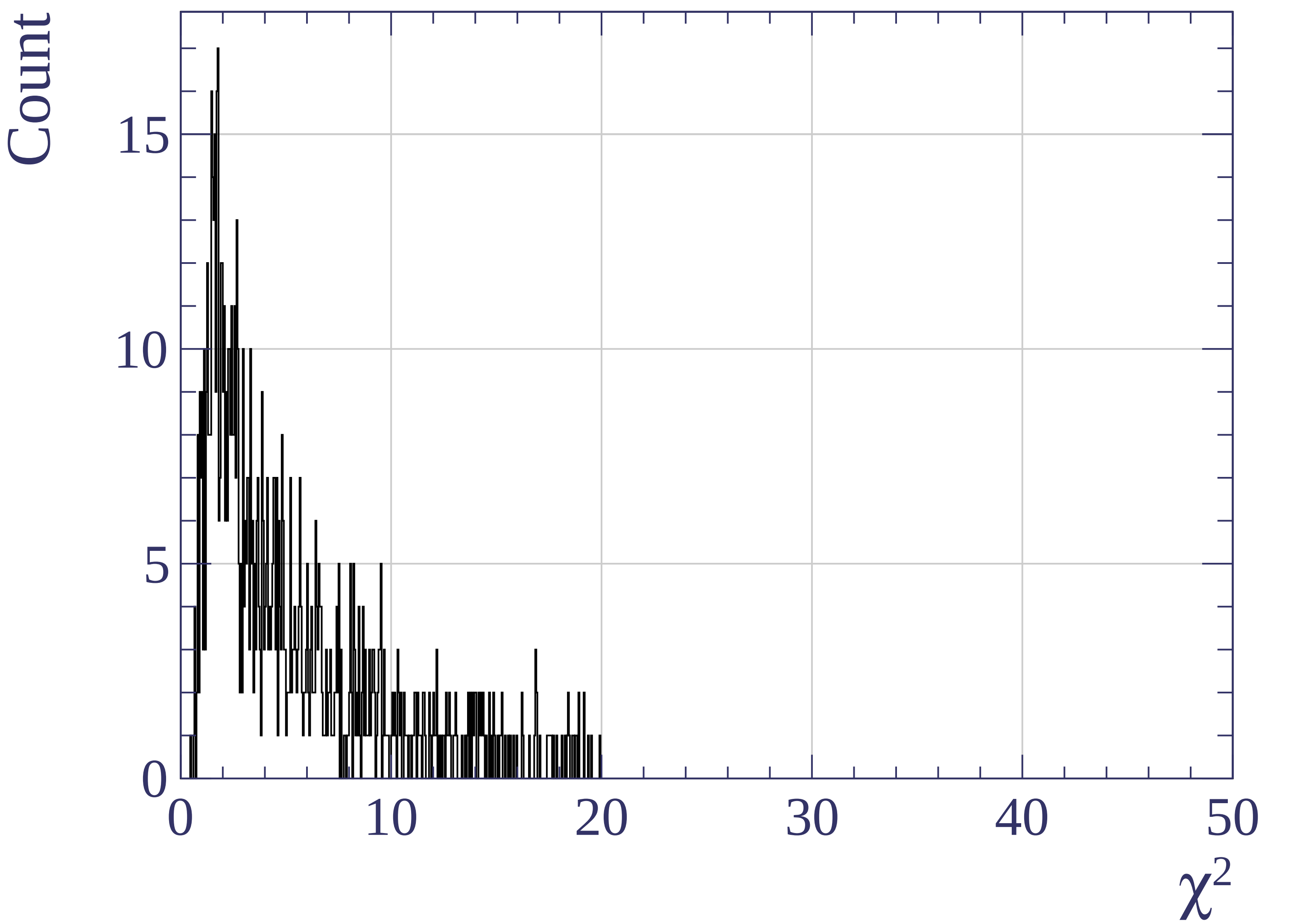
Pull e
20
10
0
−10
−20
−20
−10
0
10
20
Pull μ
600
400
200
0

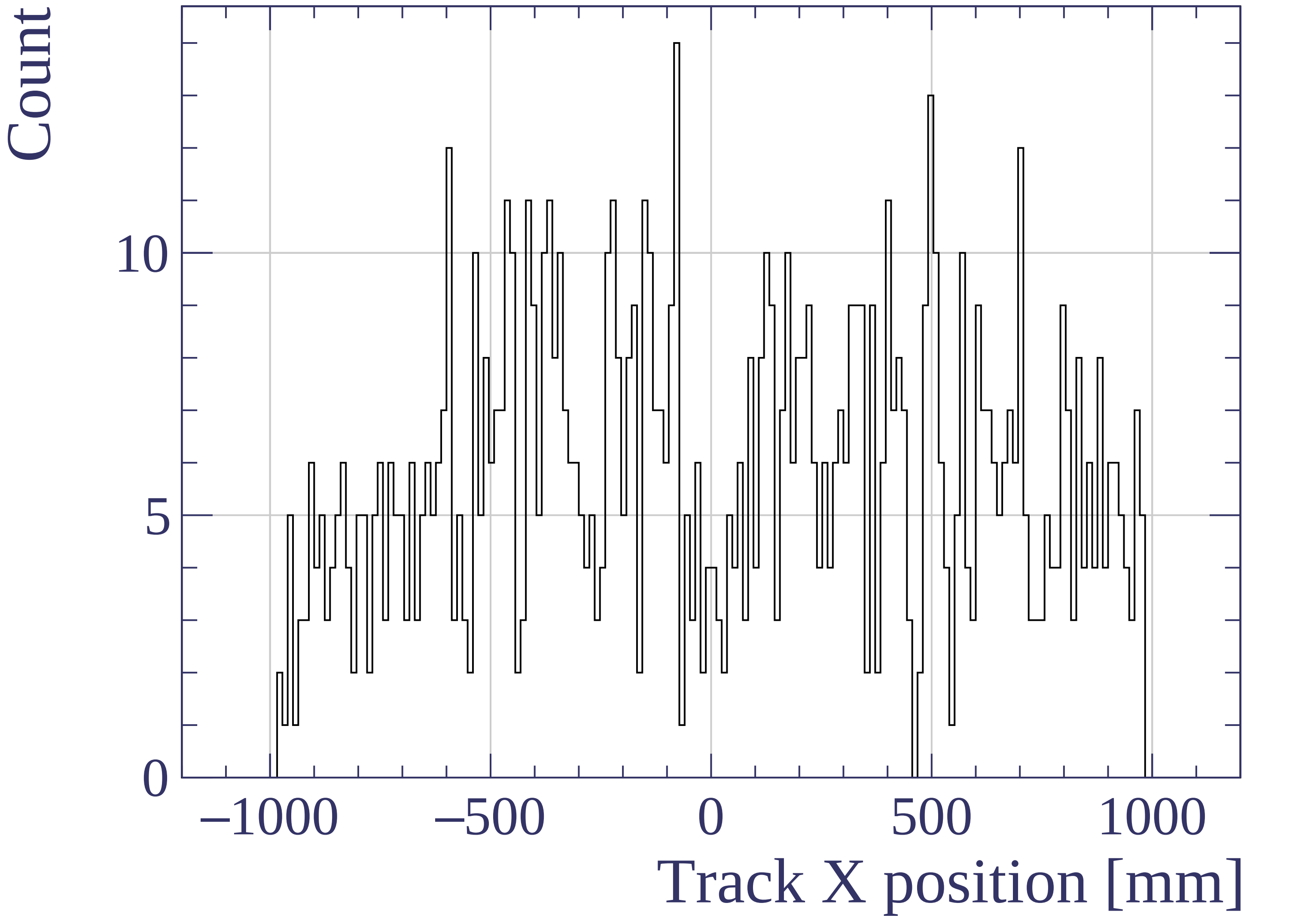


Count
300
200
100
0
−20
−10
0
10
20
Pull
μ?
e?
p?
μ_μ = 0.009 ± 0.022
σ_μ = 0.636 ± 0.019
μ_e = -3.451 ± 0.024
σ_e = 0.578 ± 0.028
S(μ/e) = 5.69 ± 0.17

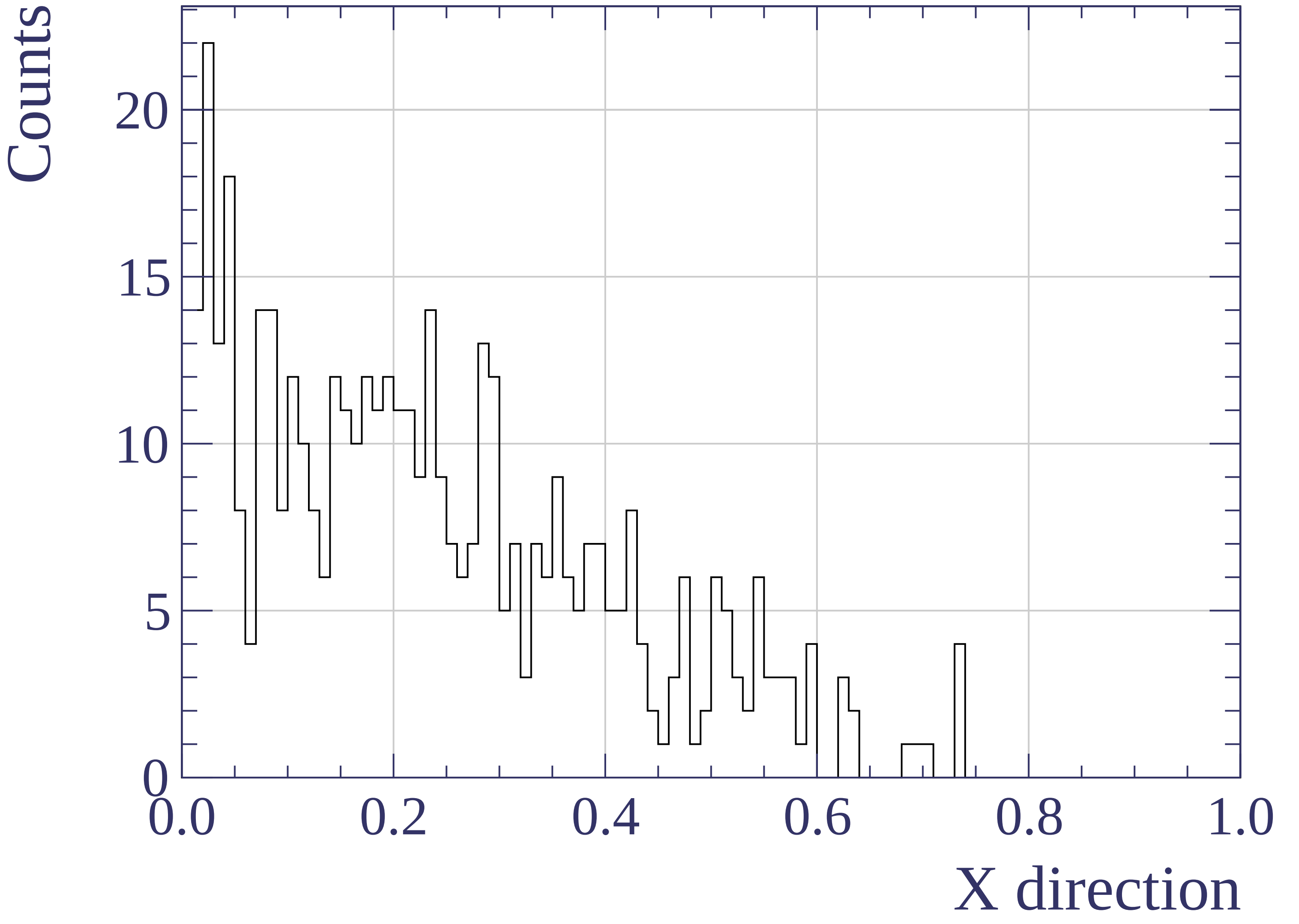


Count
100
50
0
0
50
100
150
Track length [cm]

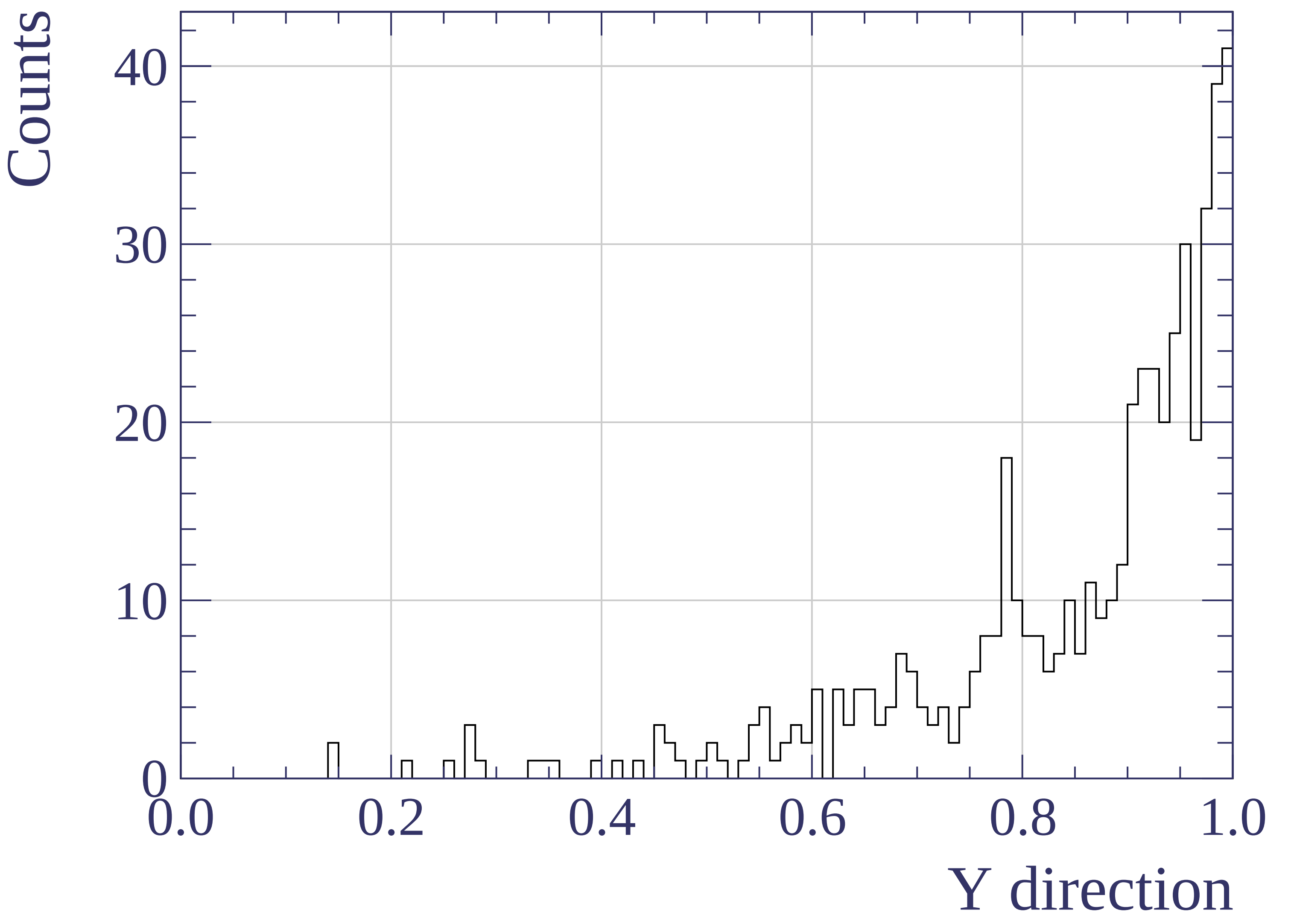

Count
0
5
10
15
0
10
20
30
40
50
χ^2



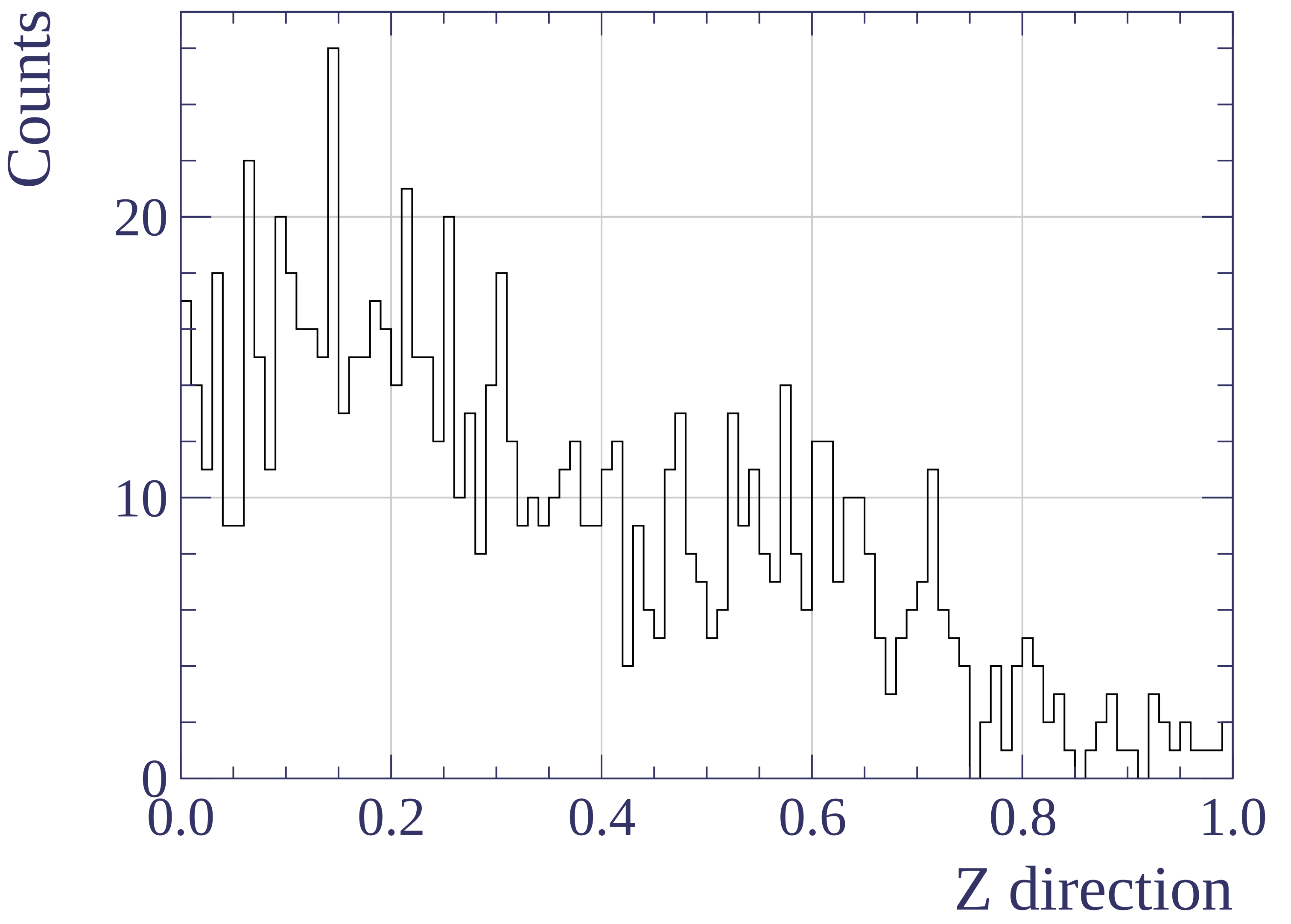
Count
0
5
10
−1000
−500
0
500
1000
Track X position [mm]

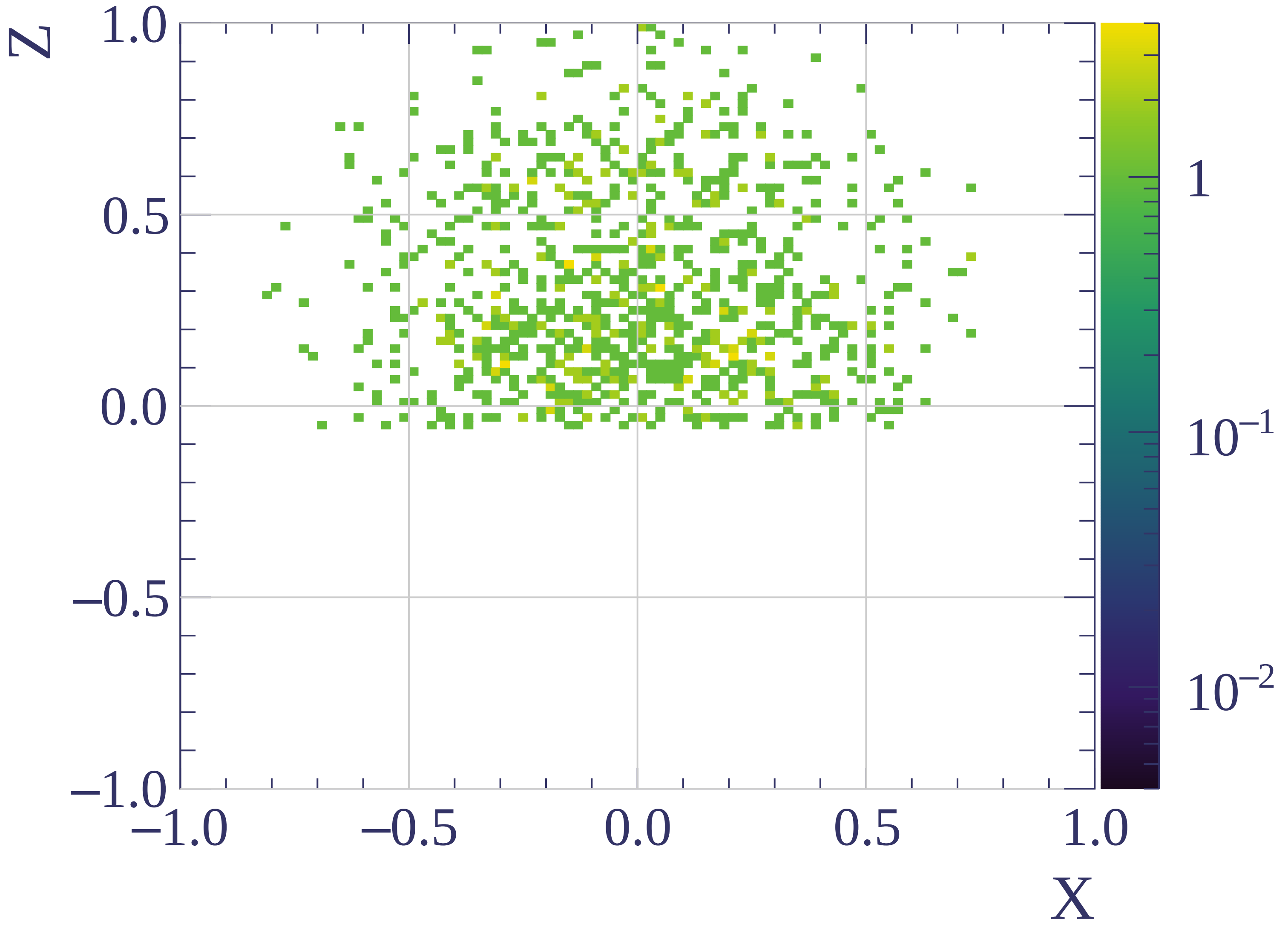


Counts
0
5
10
15
20
0.0
0.2
0.4
0.6
0.8
1.0
X direction

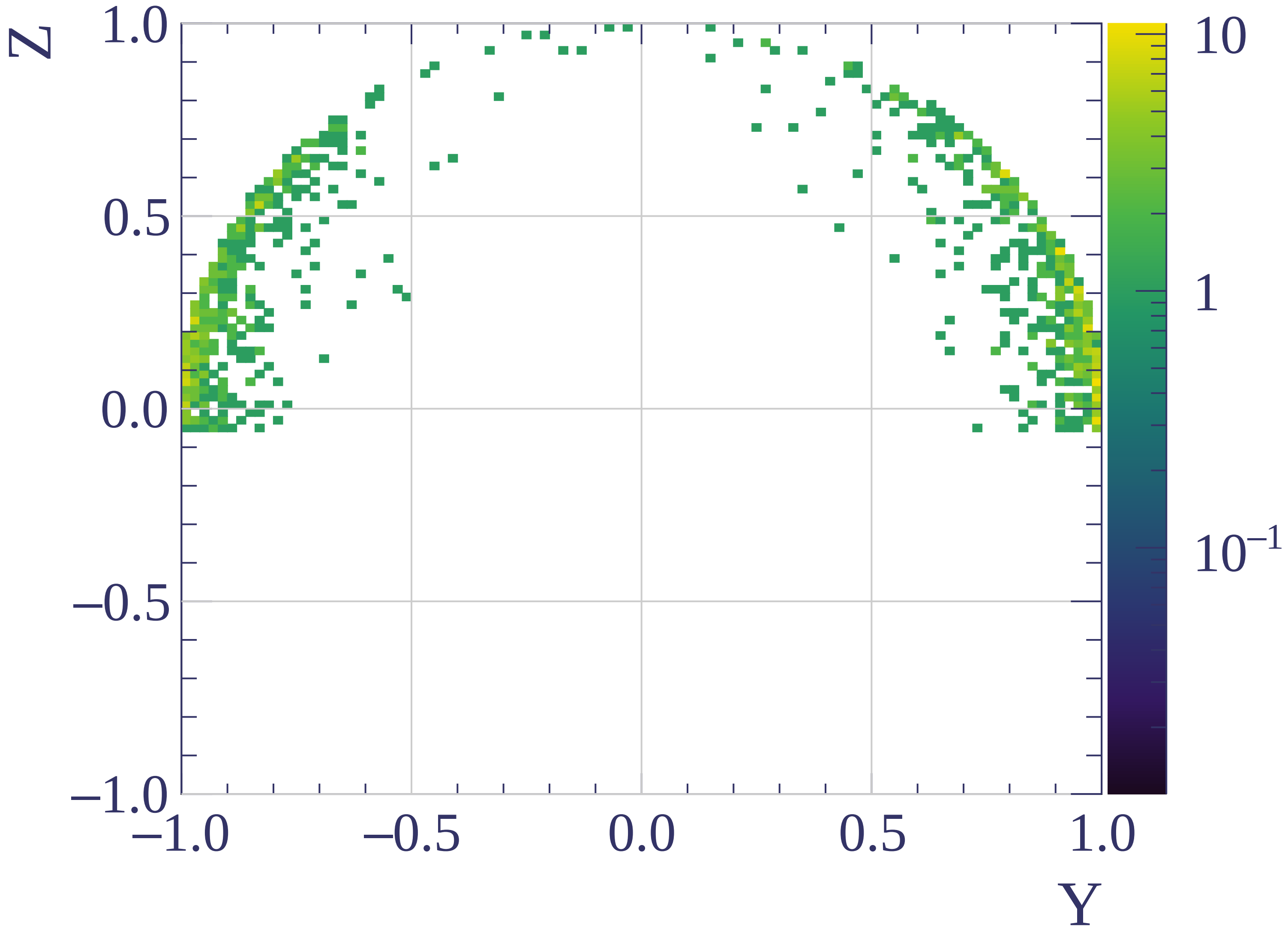


Counts
40
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

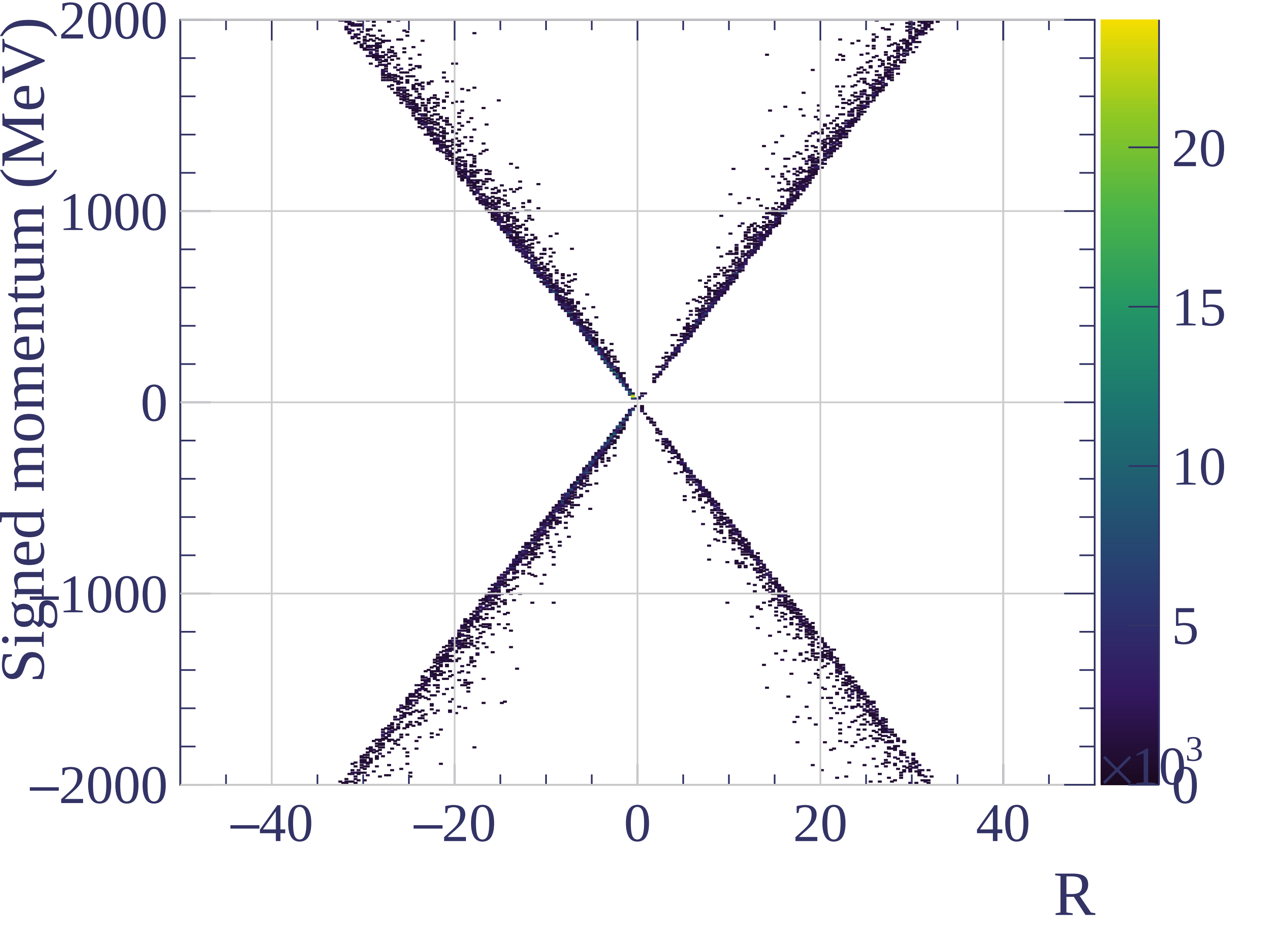

Counts
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



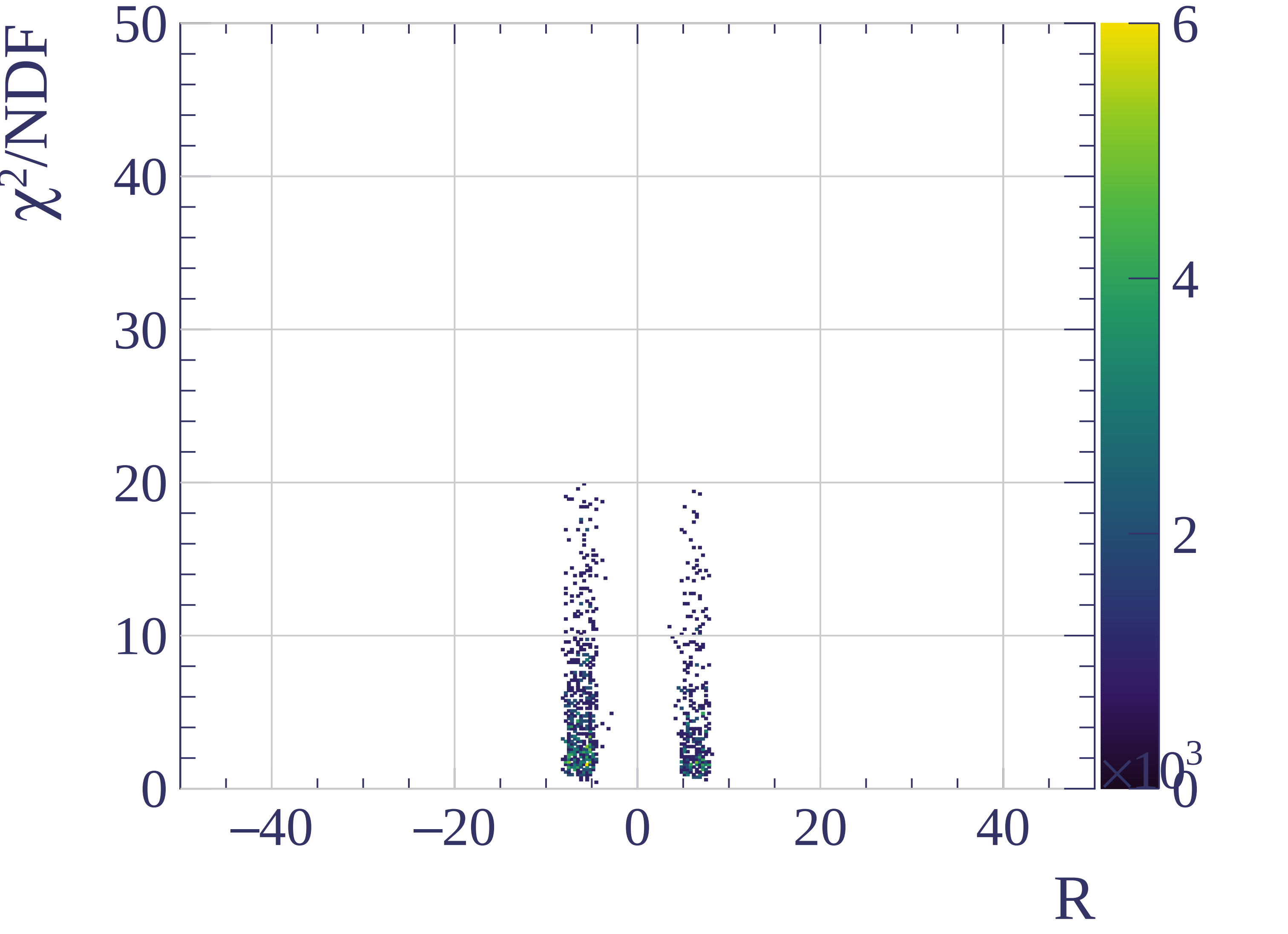
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
1
10^{-1}
10^{-2}



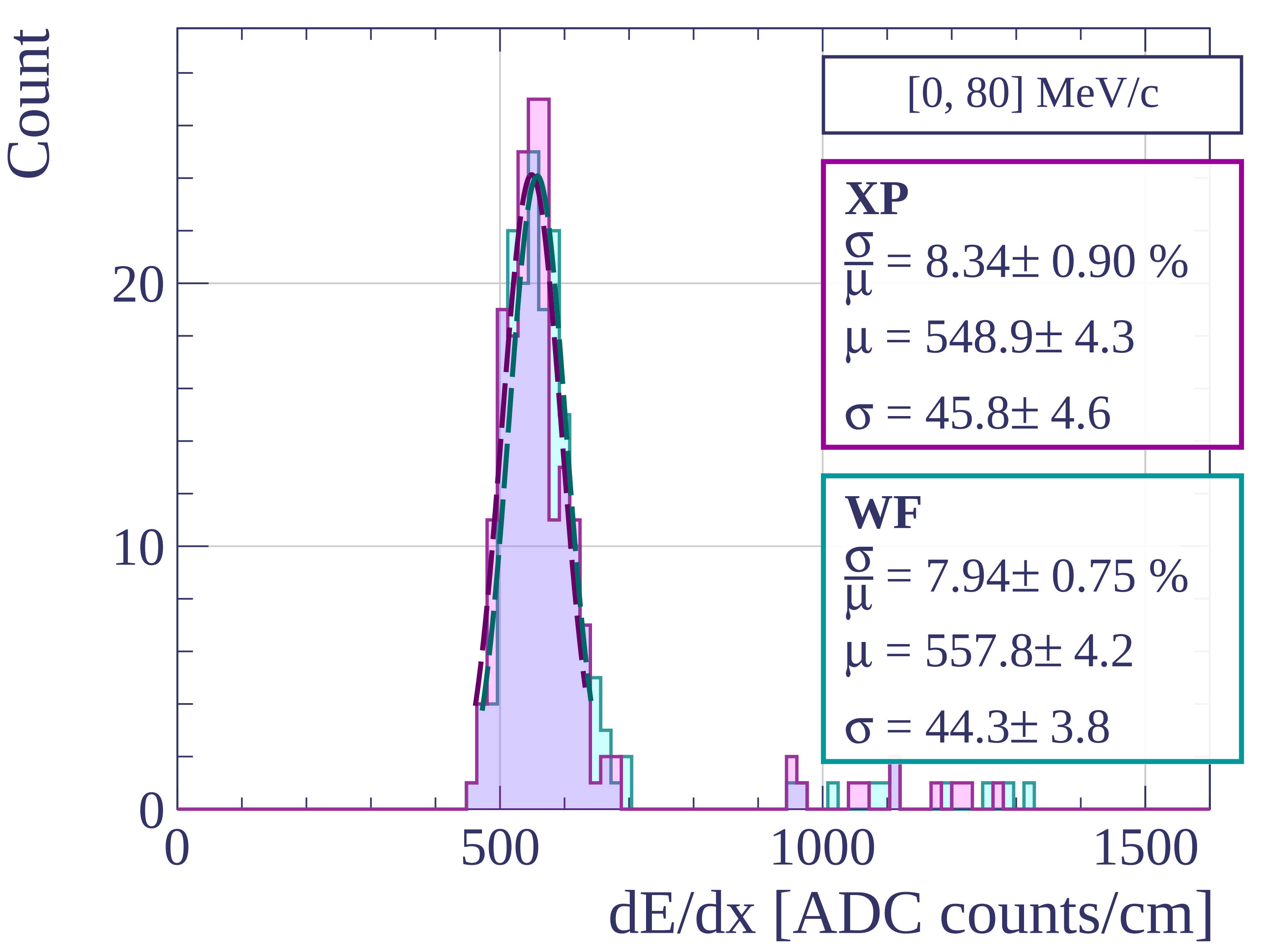
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10
1
10^{-1}



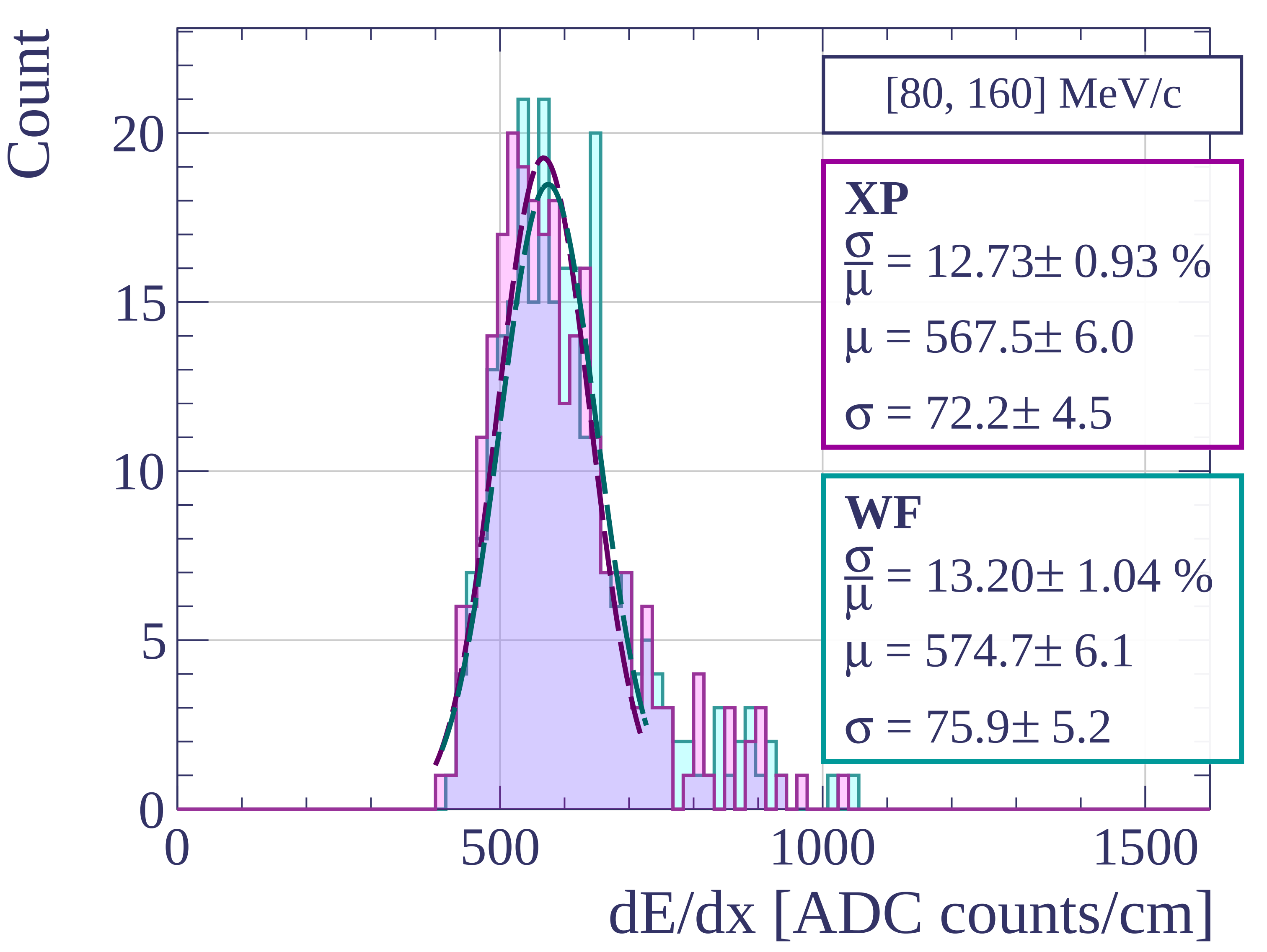
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
×10^3
20
15
10
5
0



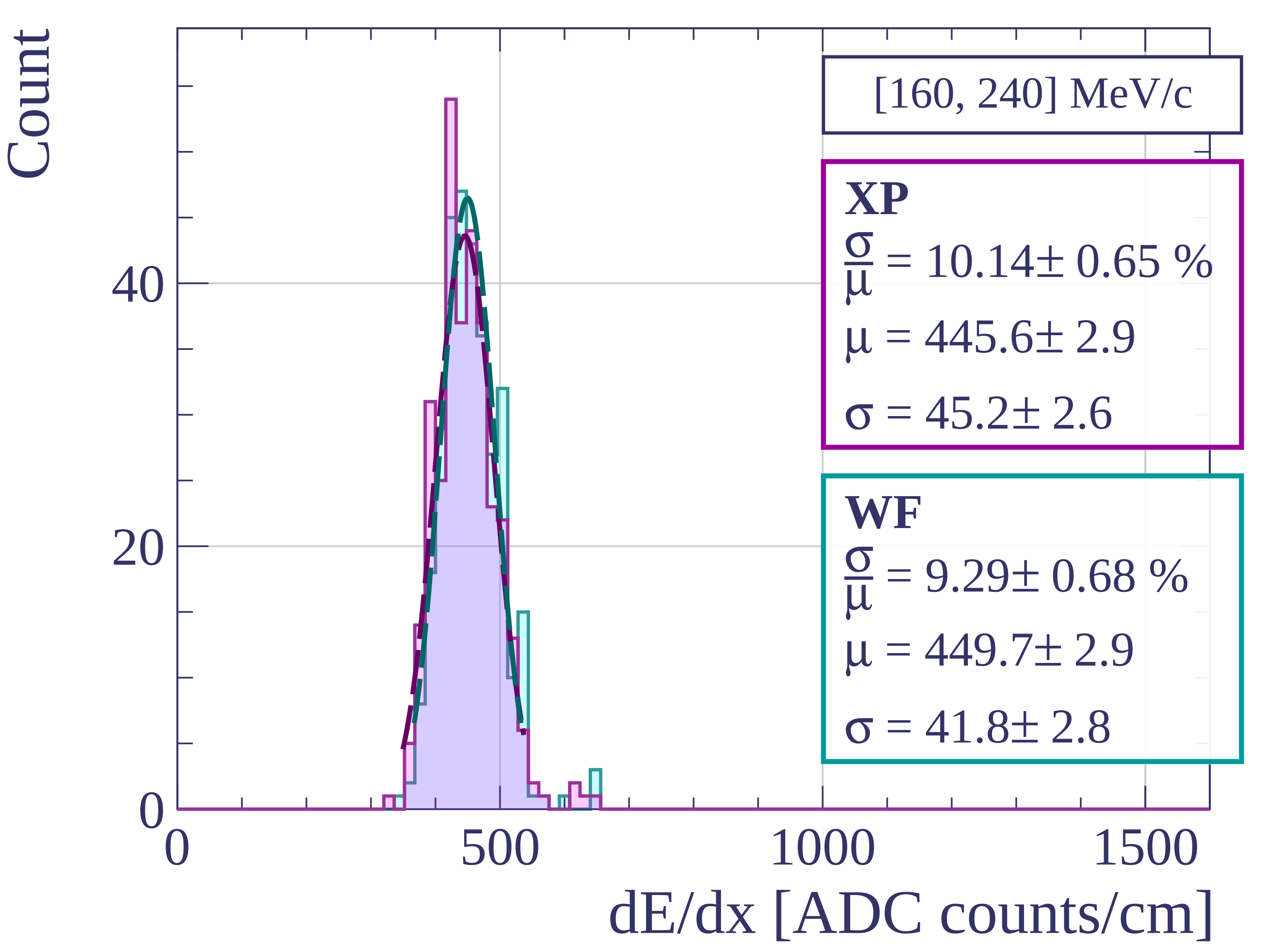
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
6
4
2
0
$\times10^3$



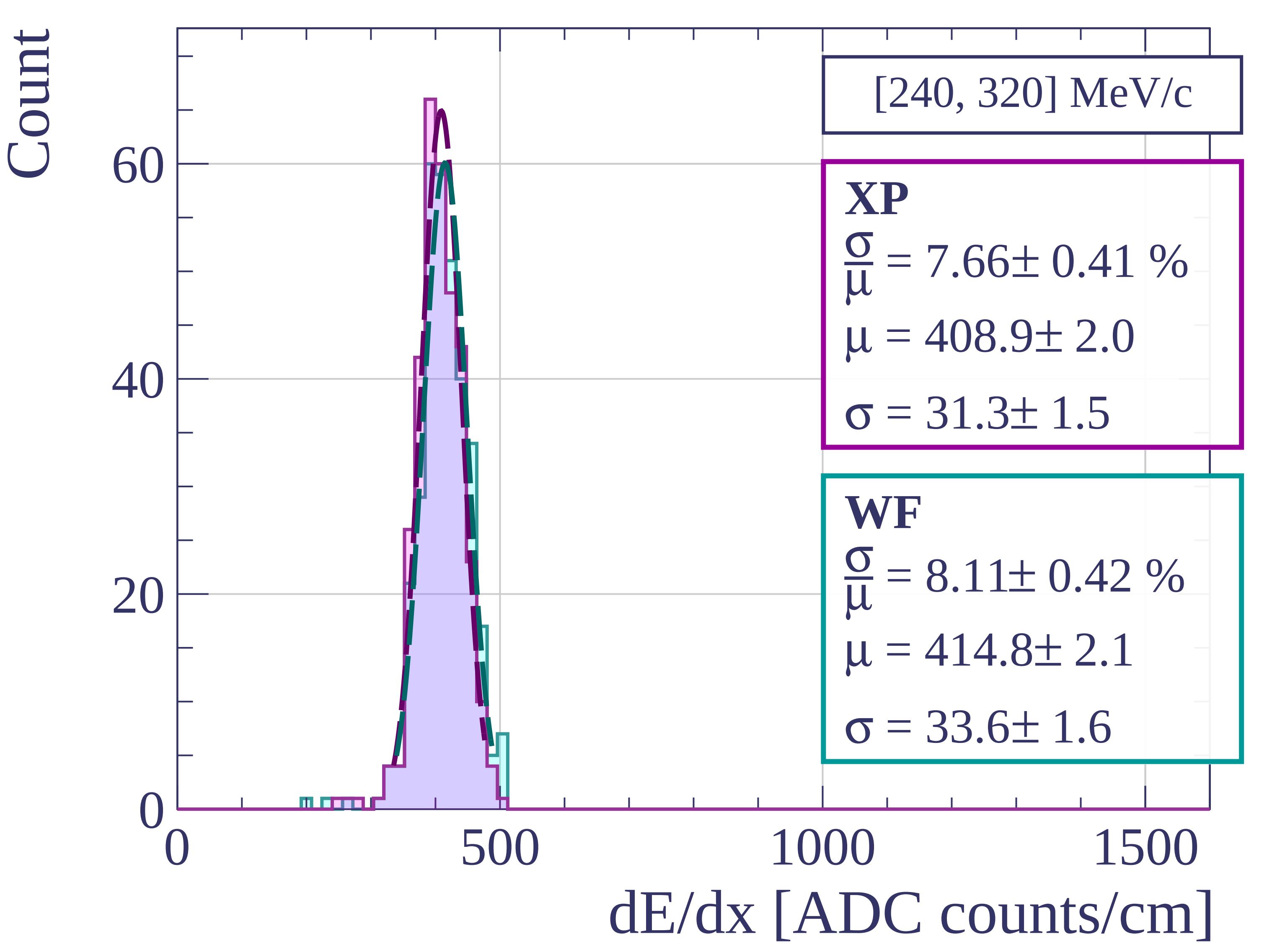
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
[0, 80] MeV/c
XP
σ/μ = 8.34± 0.90 %
μ = 548.9± 4.3
σ = 45.8± 4.6
WF
σ/μ = 7.94± 0.75 %
μ = 557.8± 4.2
σ = 44.3± 3.8



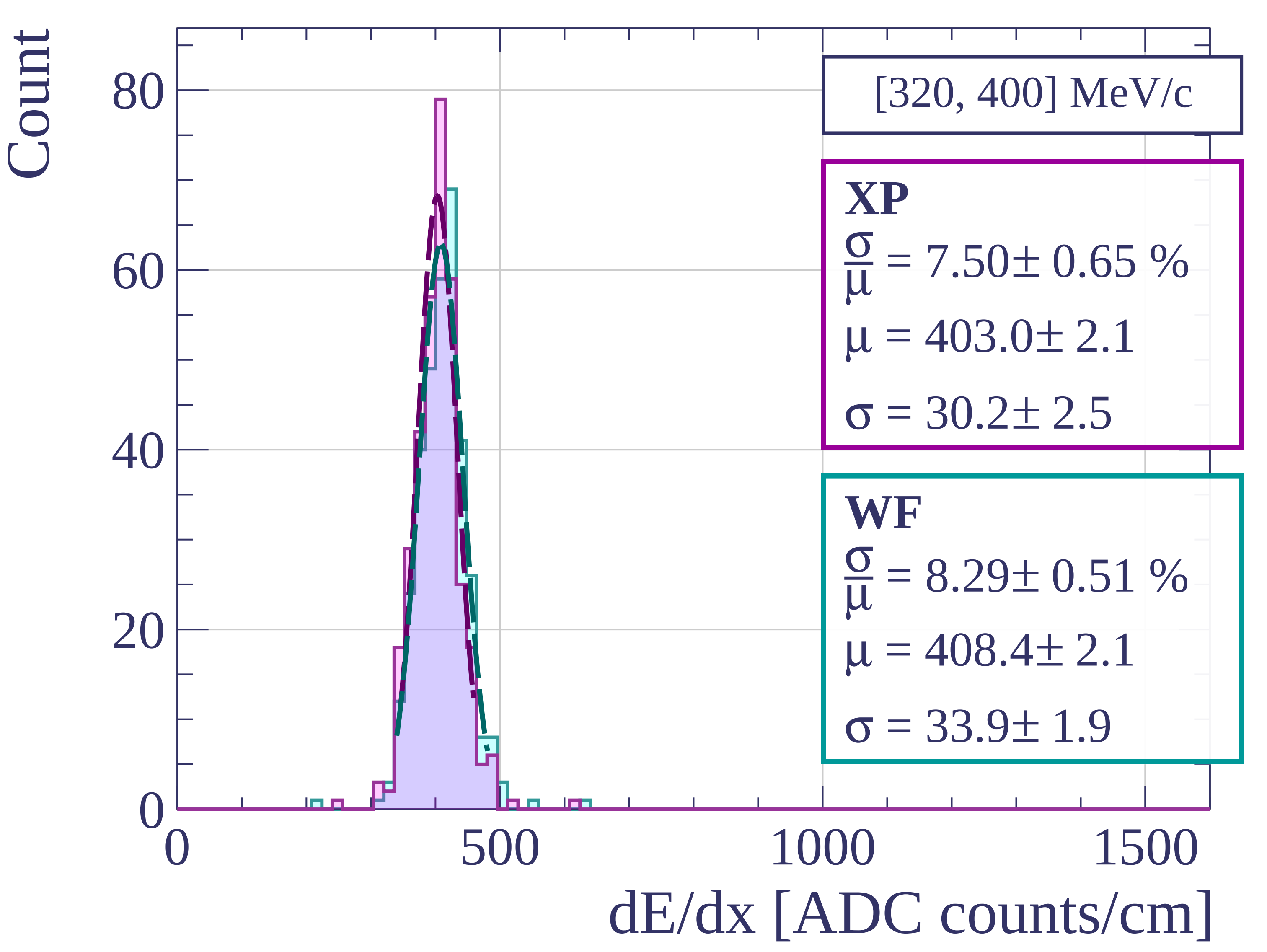
Count
0
5
10
15
20
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
σ/μ = 12.73± 0.93 %
μ = 567.5± 6.0
σ = 72.2± 4.5
WF
σ/μ = 13.20± 1.04 %
μ = 574.7± 6.1
σ = 75.9± 5.2



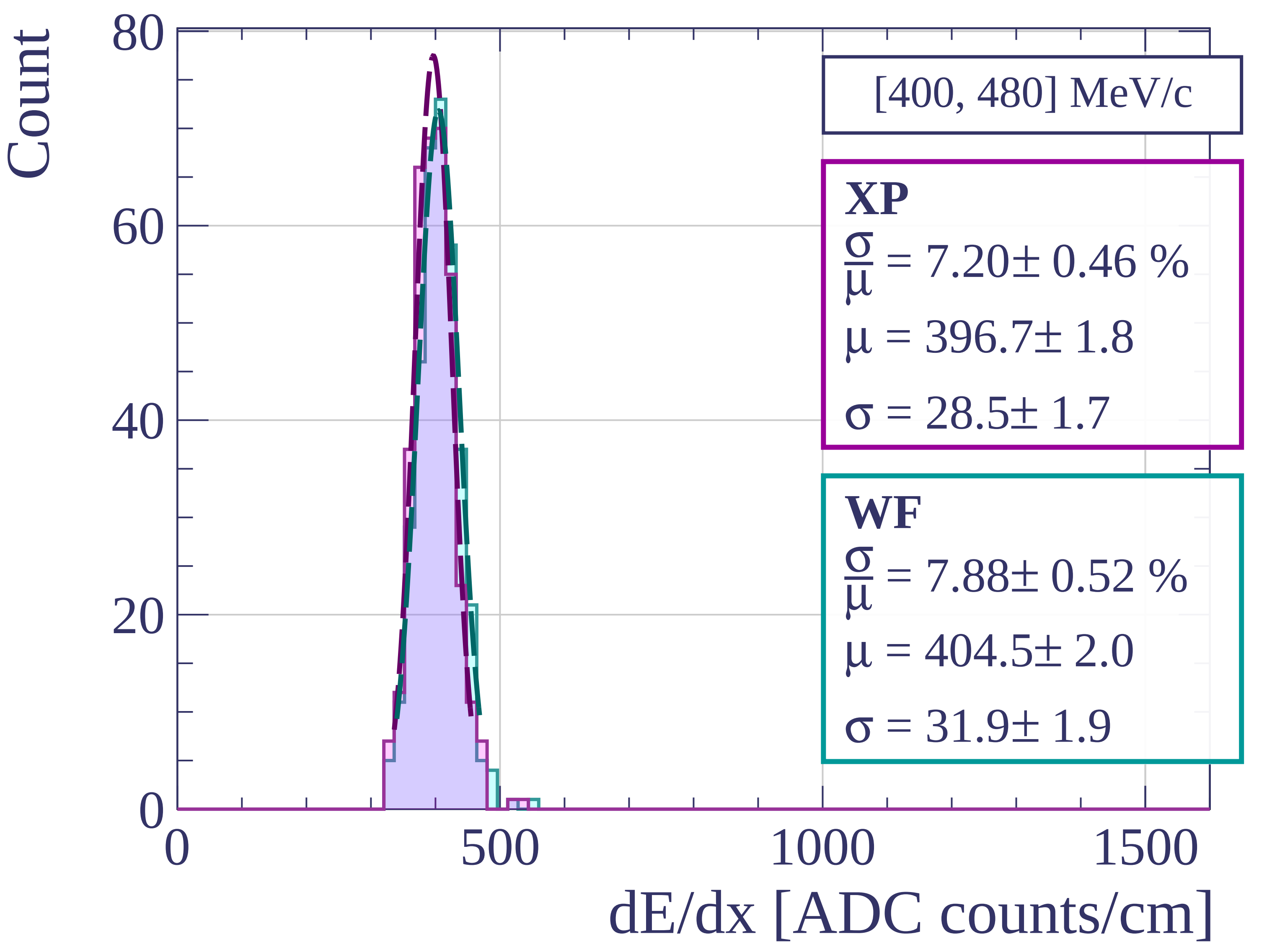
Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[160, 240] MeV/c
XP
σ/μ = 10.14± 0.65 %
μ = 445.6± 2.9
σ = 45.2± 2.6
WF
σ/μ = 9.29± 0.68 %
μ = 449.7± 2.9
σ = 41.8± 2.8



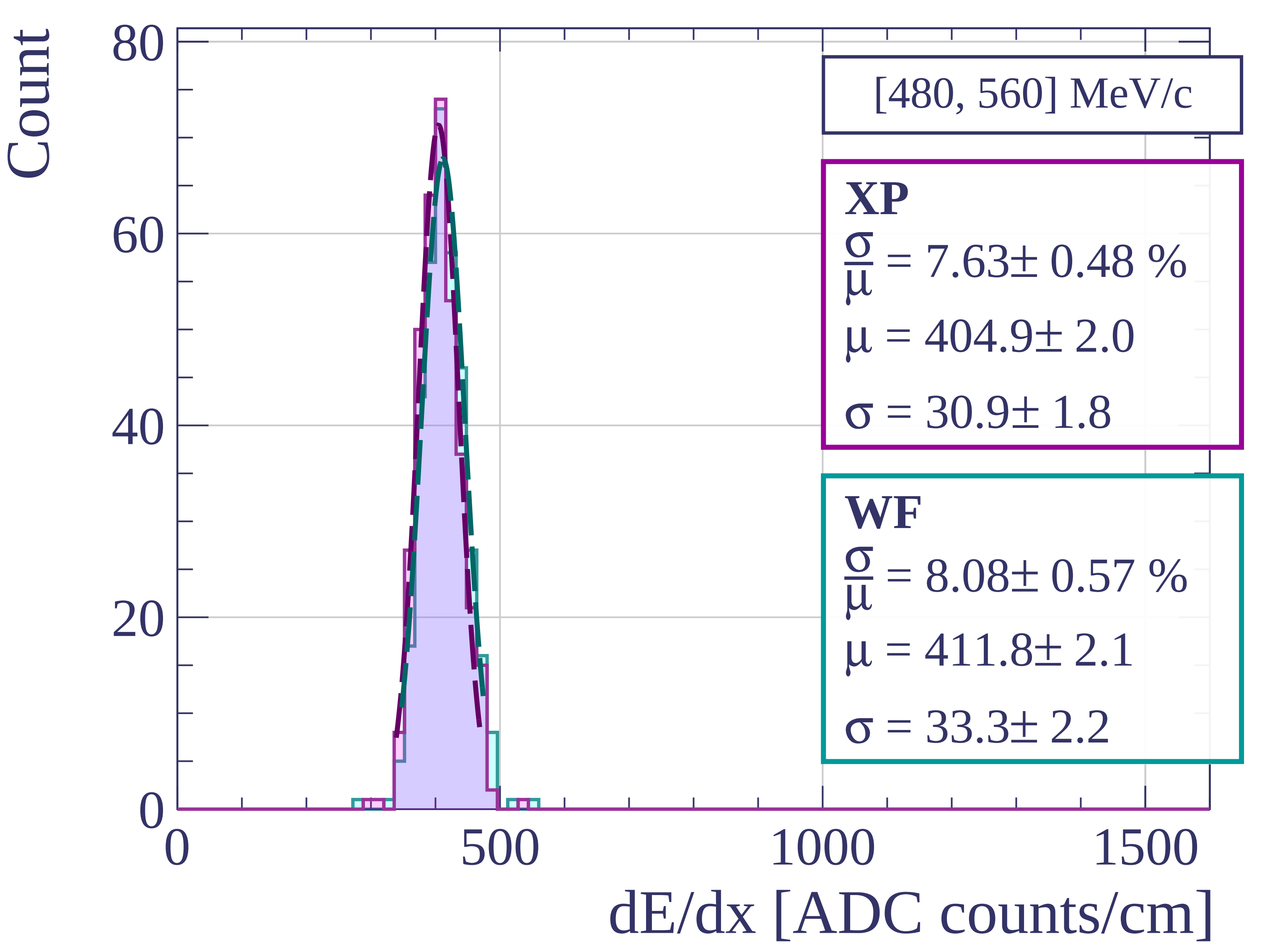
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.66± 0.41 %
μ = 408.9± 2.0
σ = 31.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.11± 0.42 %
μ = 414.8± 2.1
σ = 33.6± 1.6



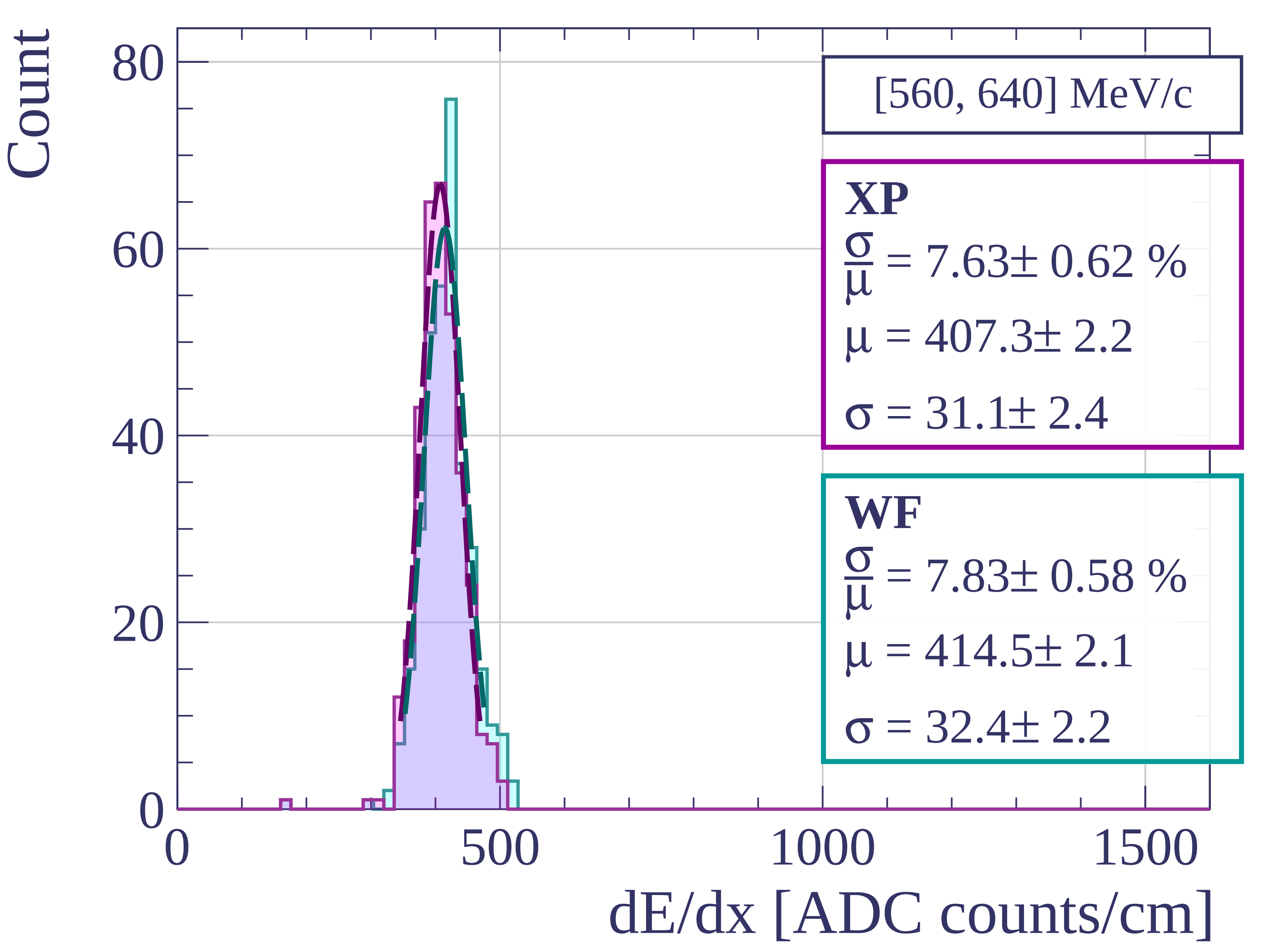
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[320, 400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.50± 0.65 %
μ = 403.0± 2.1
σ = 30.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 8.29± 0.51 %
μ = 408.4± 2.1
σ = 33.9± 1.9



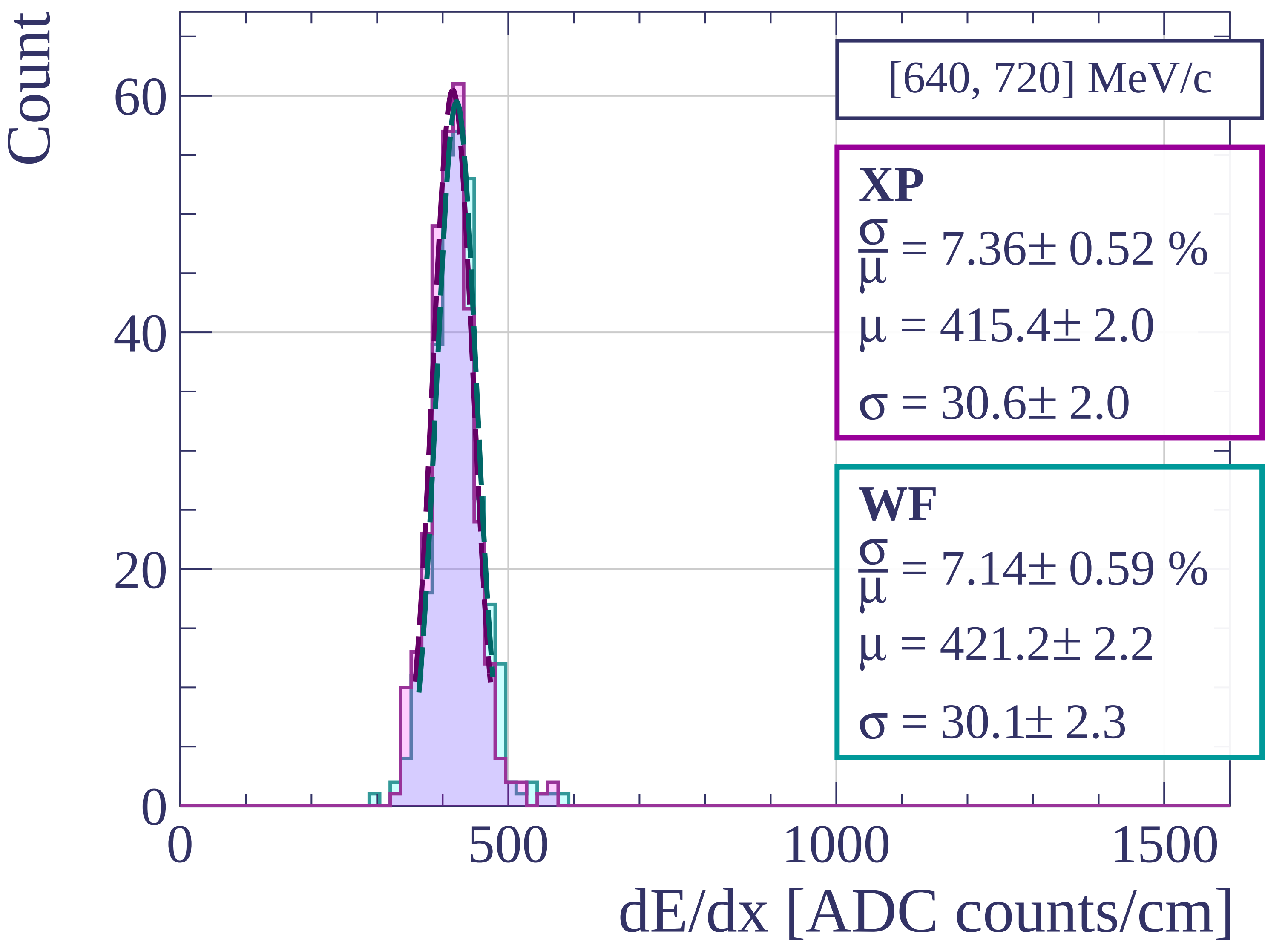
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
σ/μ = 7.20± 0.46 %
μ = 396.7± 1.8
σ = 28.5± 1.7
WF
σ/μ = 7.88± 0.52 %
μ = 404.5± 2.0
σ = 31.9± 1.9



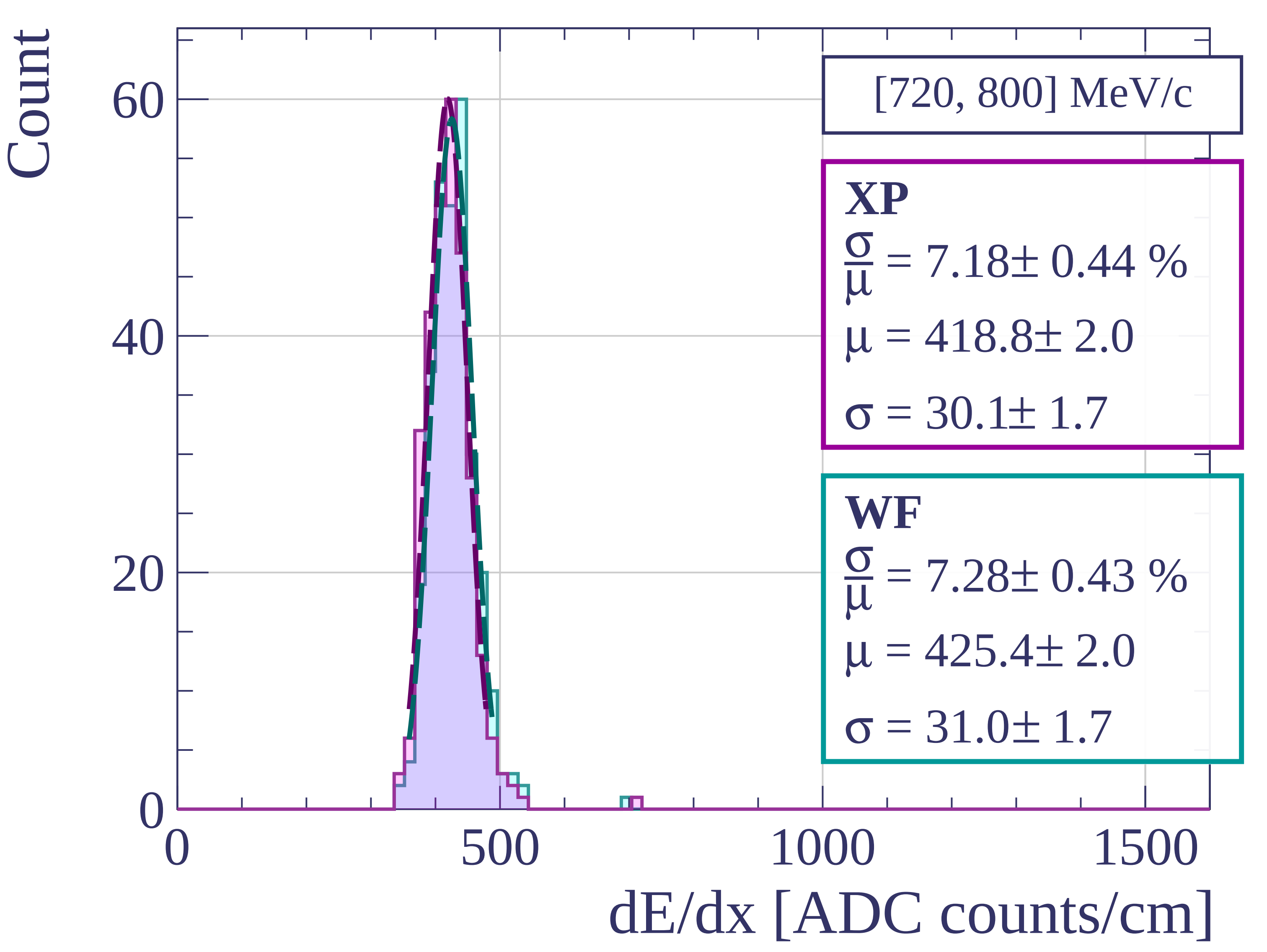
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.63± 0.48 %
μ = 404.9± 2.0
σ = 30.9± 1.8
WF
$\frac{\sigma}{\mu}$ = 8.08± 0.57 %
μ = 411.8± 2.1
σ = 33.3± 2.2



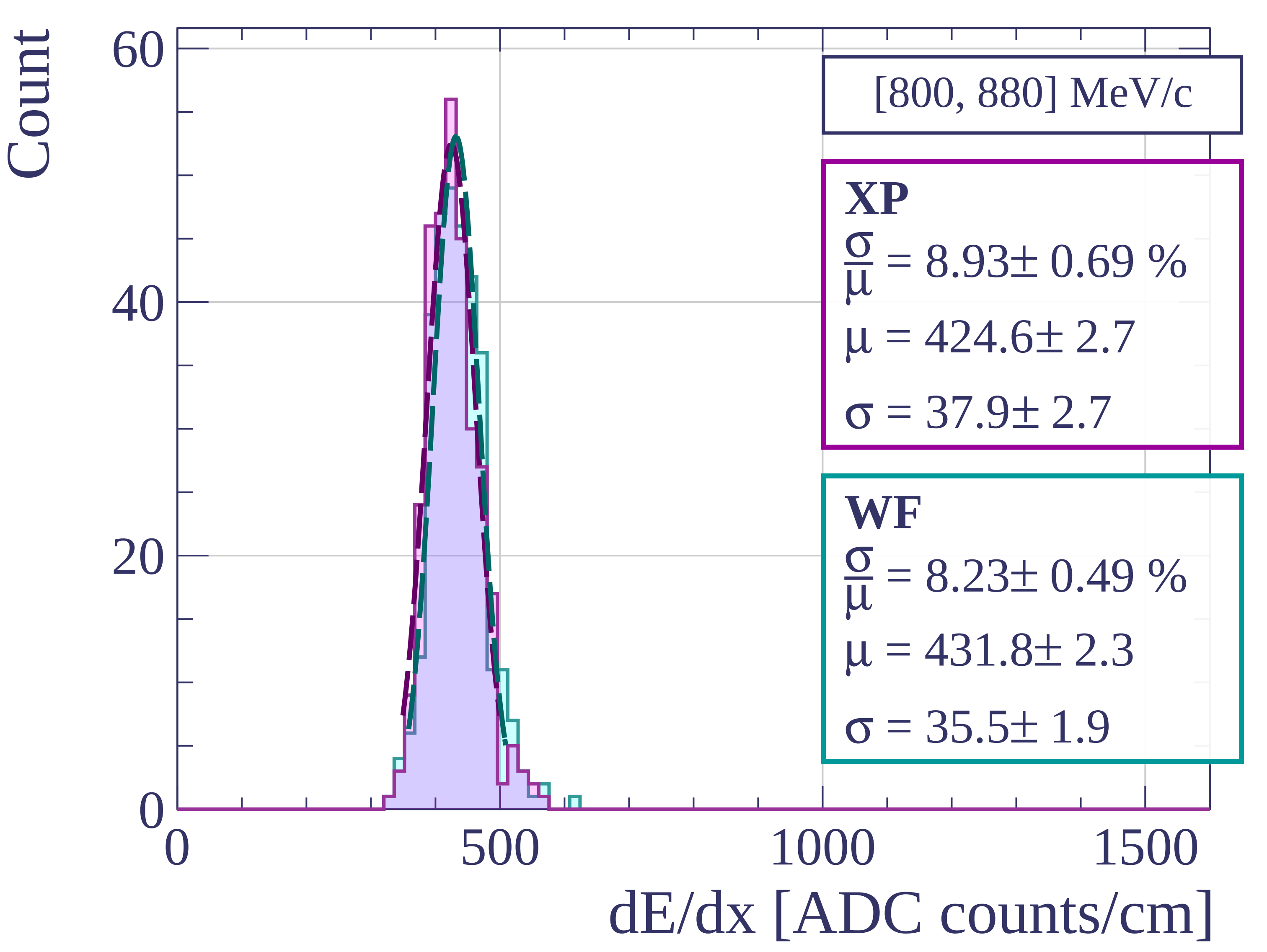
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.63± 0.62 %
μ = 407.3± 2.2
σ = 31.1± 2.4
WF
$\frac{\sigma}{\mu}$ = 7.83± 0.58 %
μ = 414.5± 2.1
σ = 32.4± 2.2

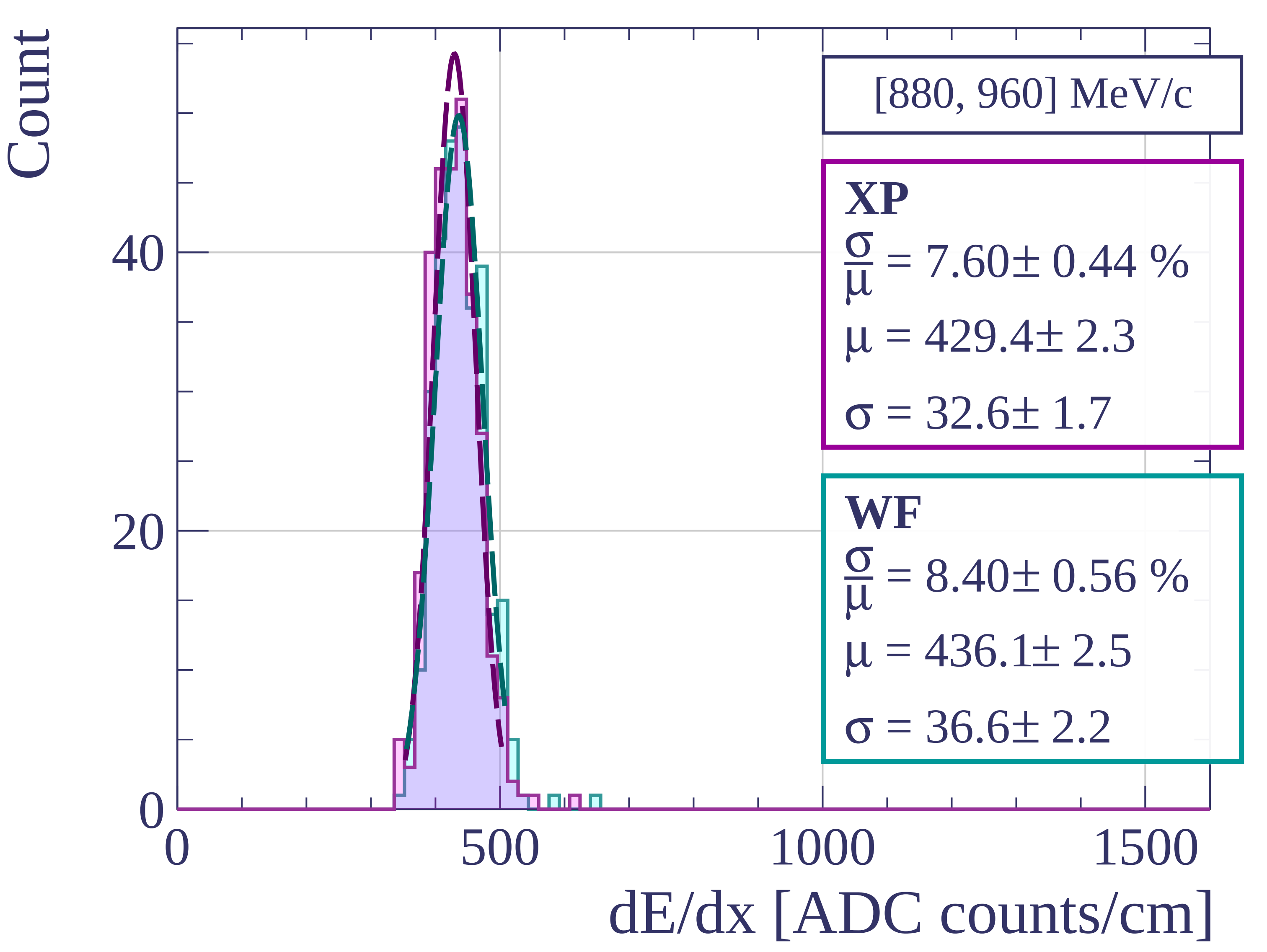


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[640, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.36± 0.52 %
μ = 415.4± 2.0
σ = 30.6± 2.0
WF
$\frac{\sigma}{\mu}$ = 7.14± 0.59 %
μ = 421.2± 2.2
σ = 30.1± 2.3

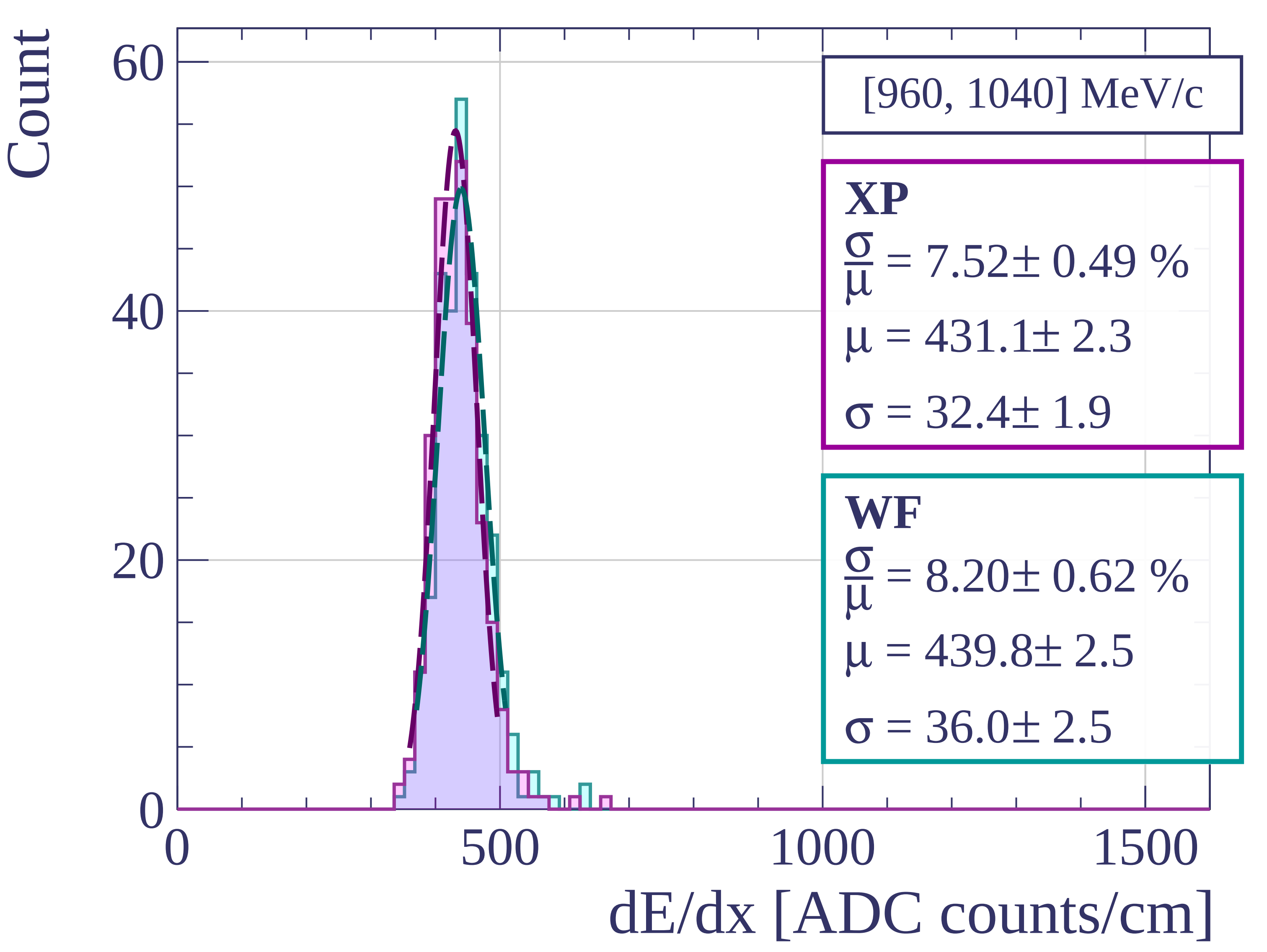


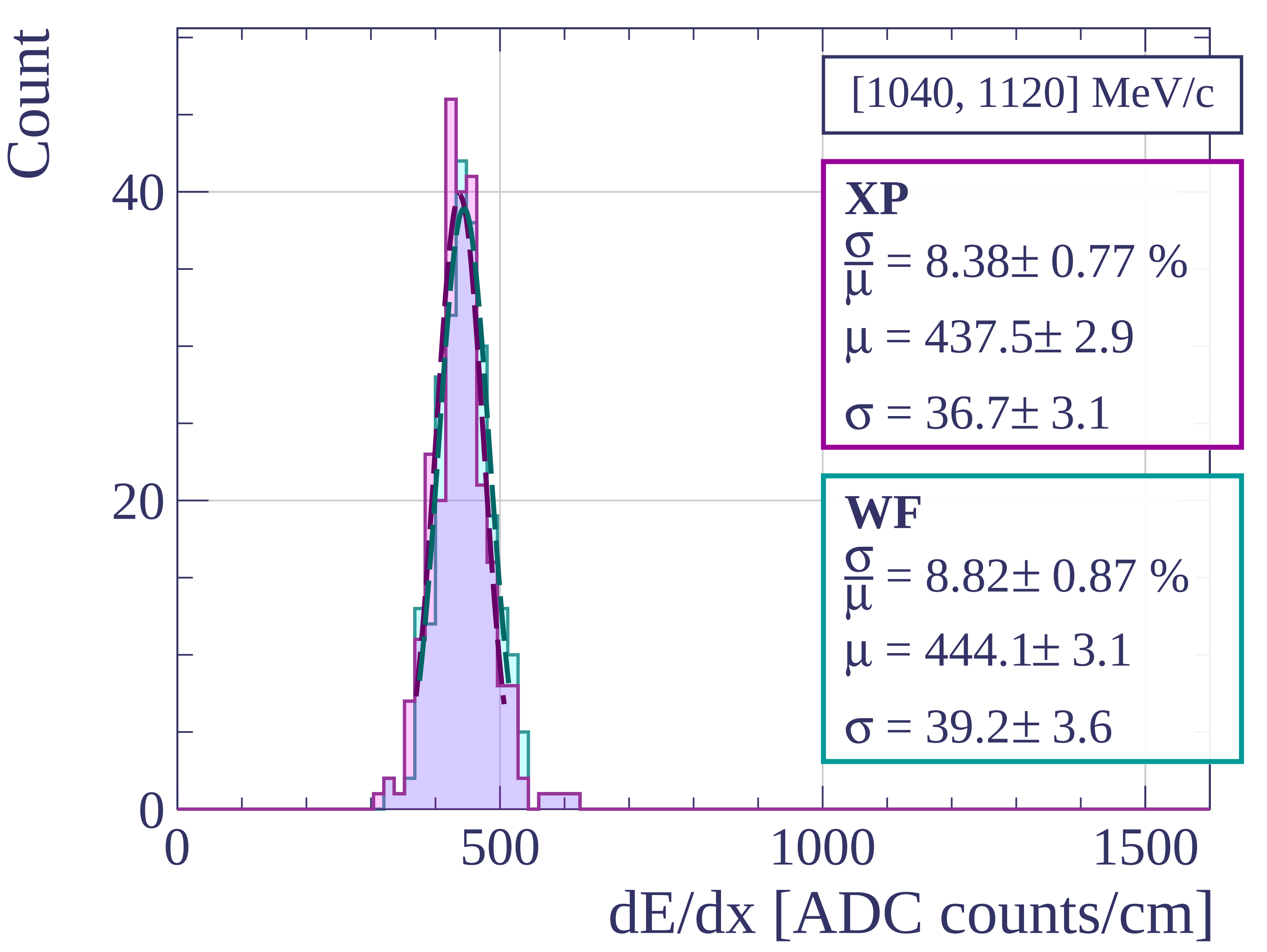
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
[720, 800] MeV/c
XP
σ/μ = 7.18± 0.44 %
μ = 418.8± 2.0
σ = 30.1± 1.7
WF
σ/μ = 7.28± 0.43 %
μ = 425.4± 2.0
σ = 31.0± 1.7


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
σ/μ = 8.93± 0.69 %
μ = 424.6± 2.7
σ = 37.9± 2.7
WF
σ/μ = 8.23± 0.49 %
μ = 431.8± 2.3
σ = 35.5± 1.9

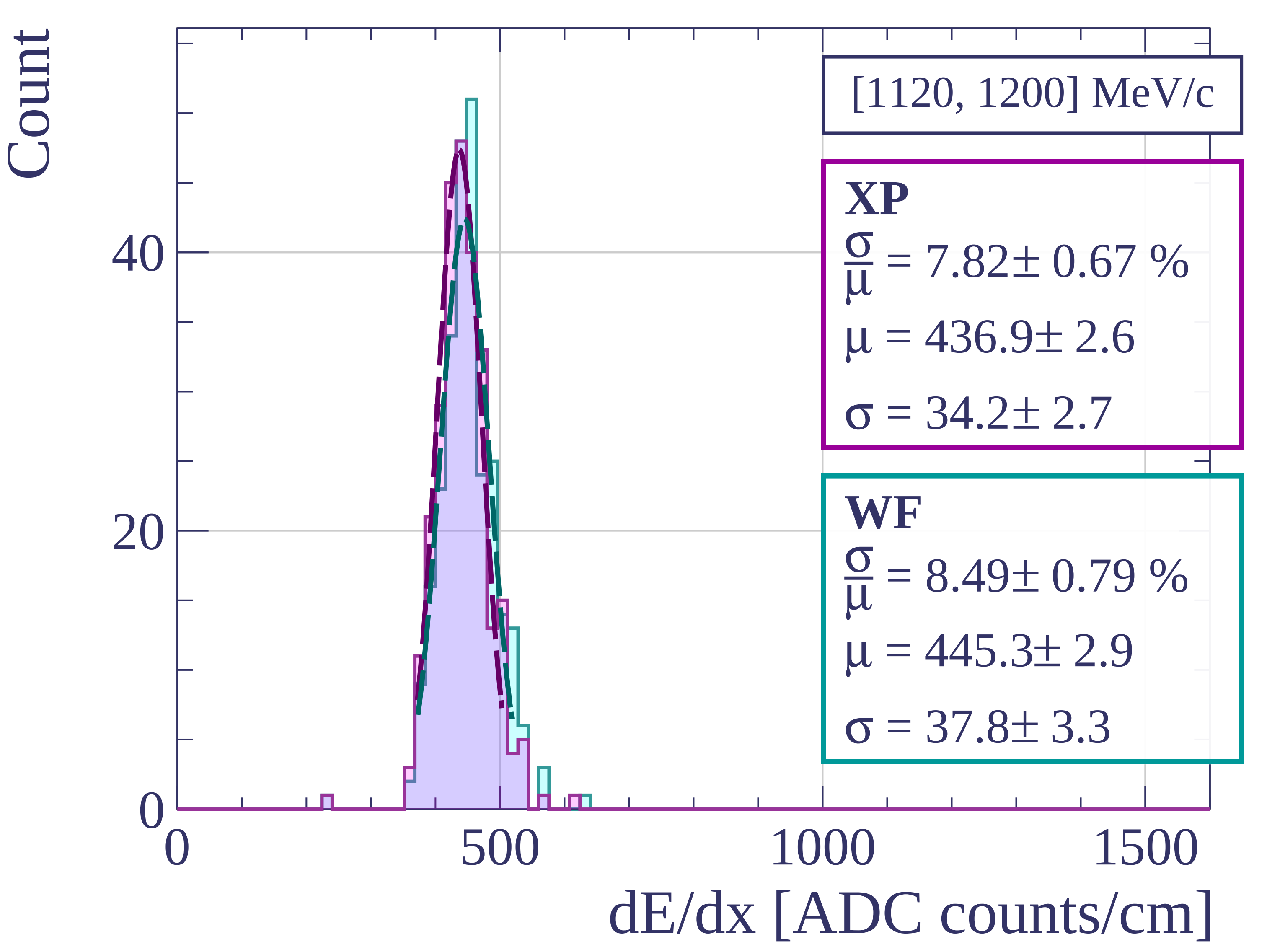


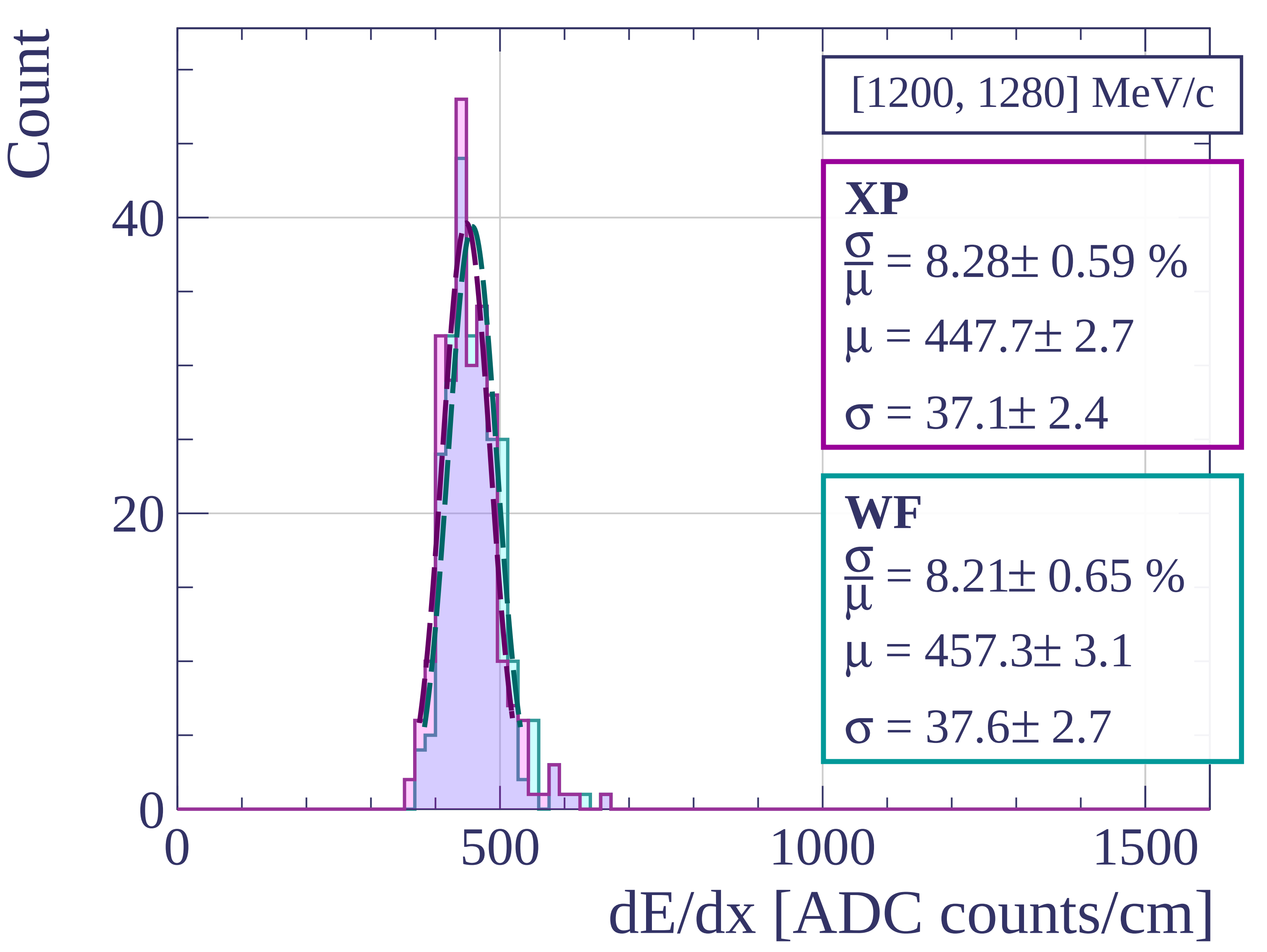
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
σ/μ = 7.60± 0.44 %
μ = 429.4± 2.3
σ = 32.6± 1.7
WF
σ/μ = 8.40± 0.56 %
μ = 436.1± 2.5
σ = 36.6± 2.2


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.52± 0.49 %
μ = 431.1± 2.3
σ = 32.4± 1.9
WF
$\frac{\sigma}{\mu}$ = 8.20± 0.62 %
μ = 439.8± 2.5
σ = 36.0± 2.5

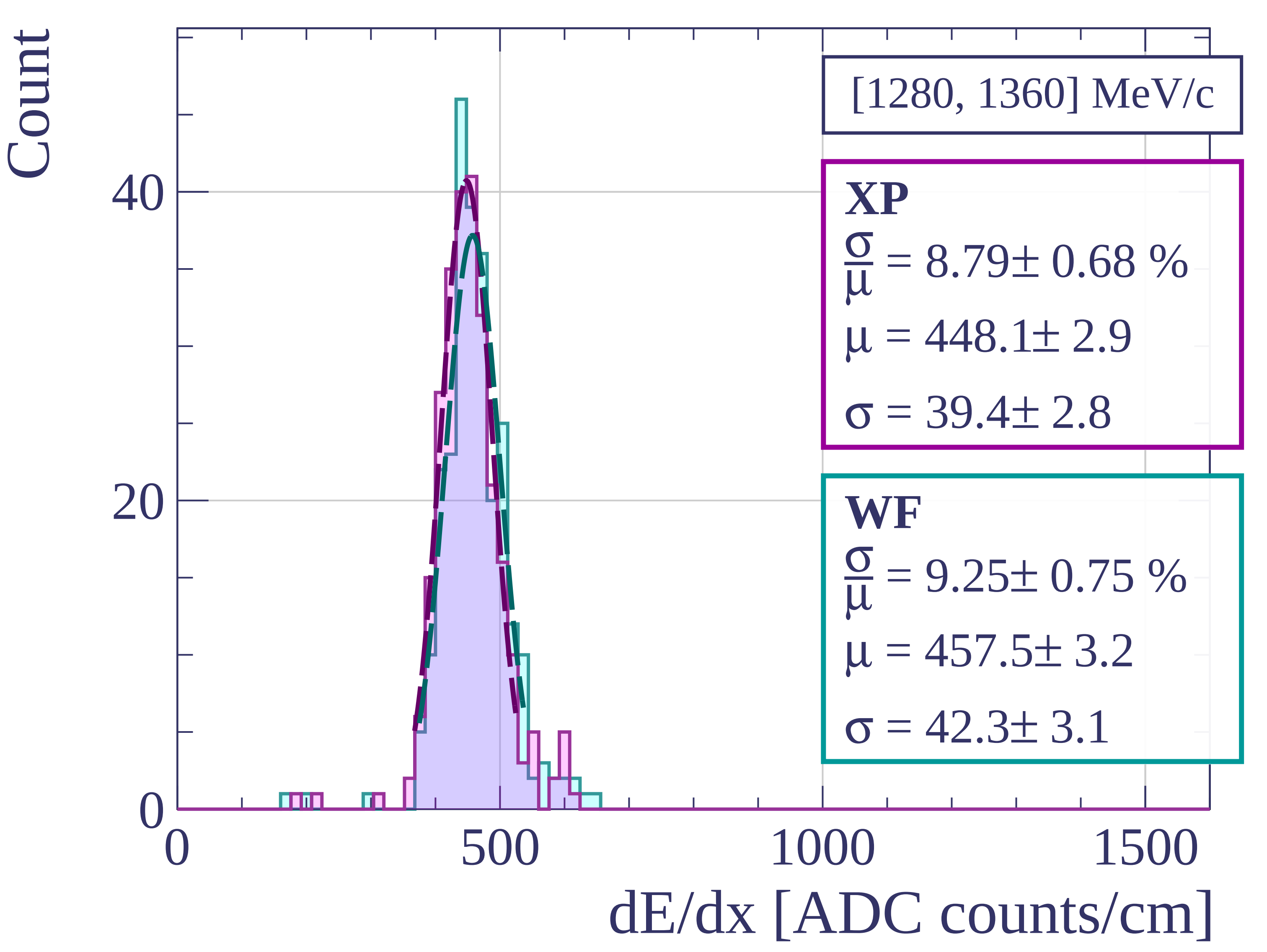


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.77 %
μ = 437.5± 2.9
σ = 36.7± 3.1
WF
$\frac{\sigma}{\mu}$ = 8.82± 0.87 %
μ = 444.1± 3.1
σ = 39.2± 3.6

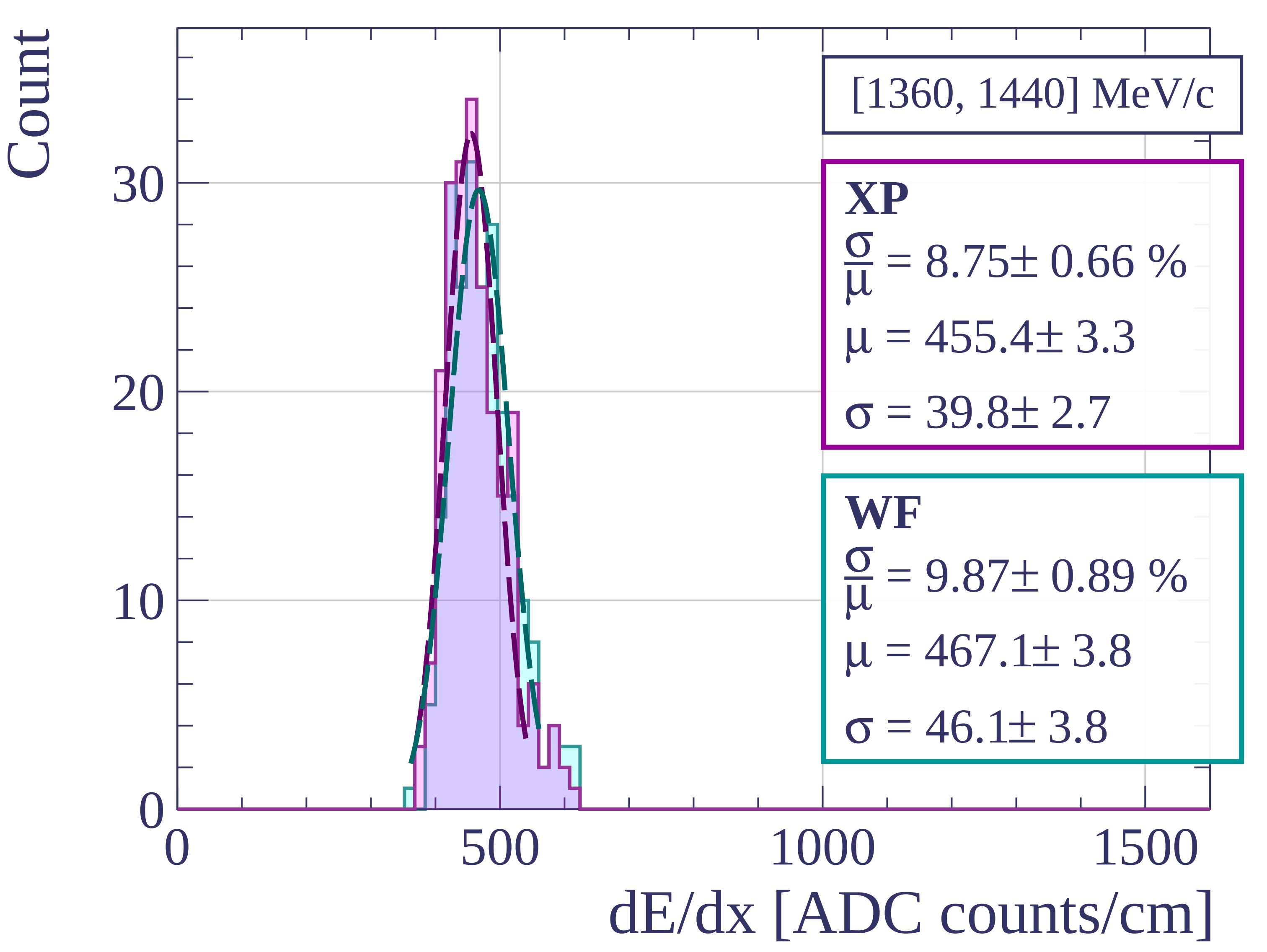

Count
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
σ/μ = 7.82± 0.67 %
μ = 436.9± 2.6
σ = 34.2± 2.7
WF
σ/μ = 8.49± 0.79 %
μ = 445.3± 2.9
σ = 37.8± 3.3
0
20
40
0
500
1000
1500



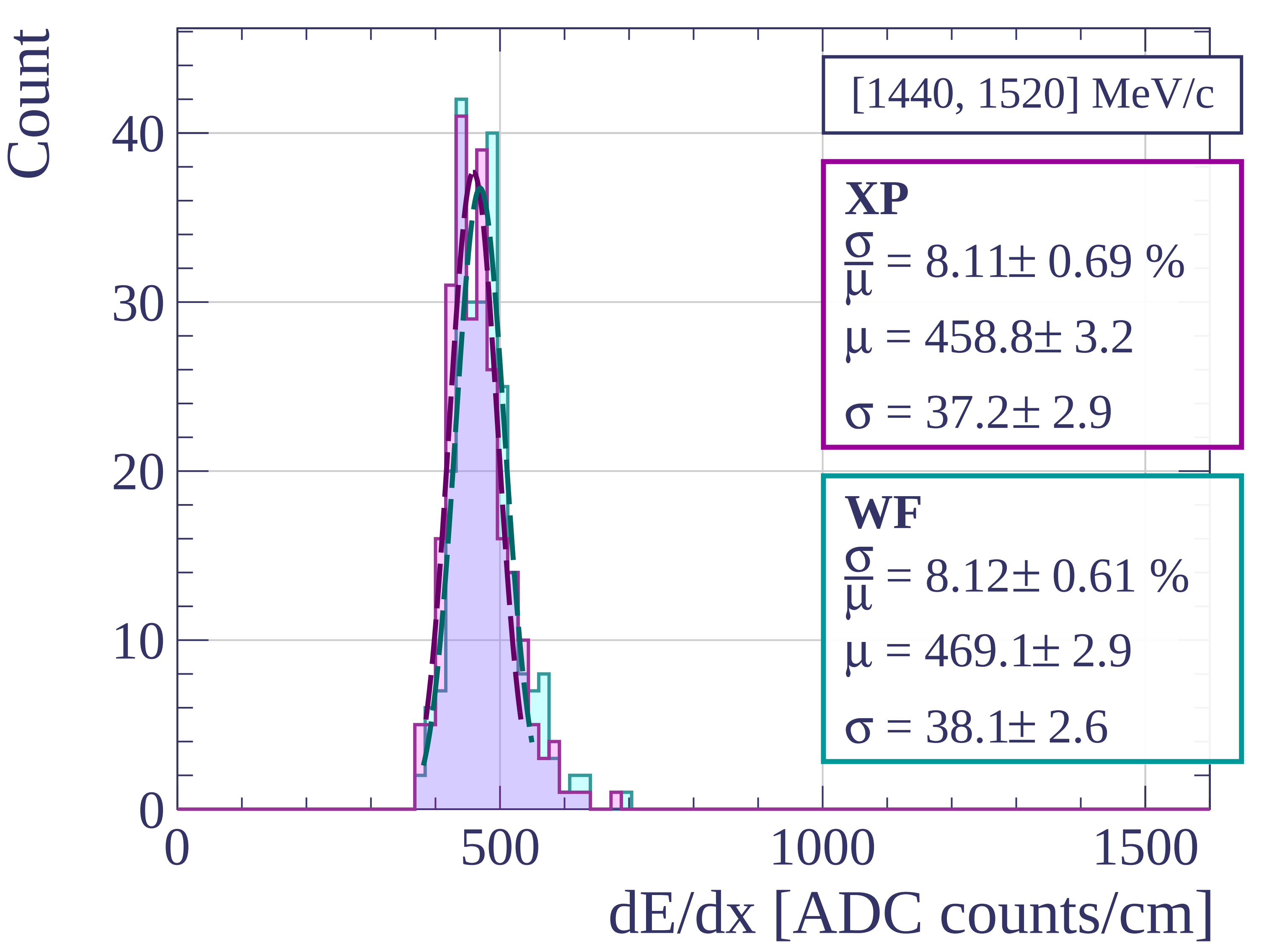
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.28± 0.59 %
μ = 447.7± 2.7
σ = 37.1± 2.4
WF
$\frac{\sigma}{\mu}$ = 8.21± 0.65 %
μ = 457.3± 3.1
σ = 37.6± 2.7

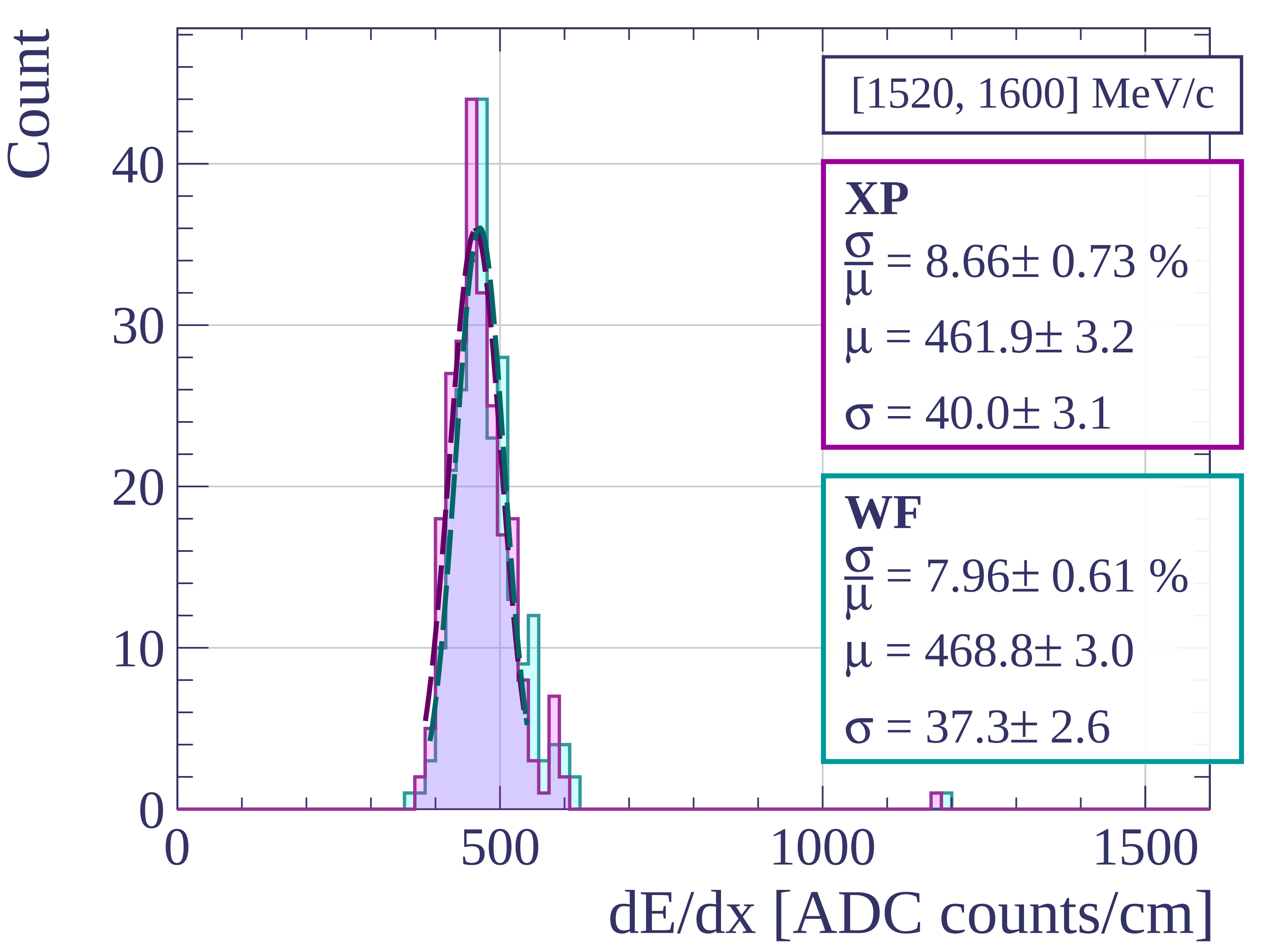


Count
dE/dx [ADC counts/cm]
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.79± 0.68 %
μ = 448.1± 2.9
σ = 39.4± 2.8
WF
$\frac{\sigma}{\mu}$ = 9.25± 0.75 %
μ = 457.5± 3.2
σ = 42.3± 3.1

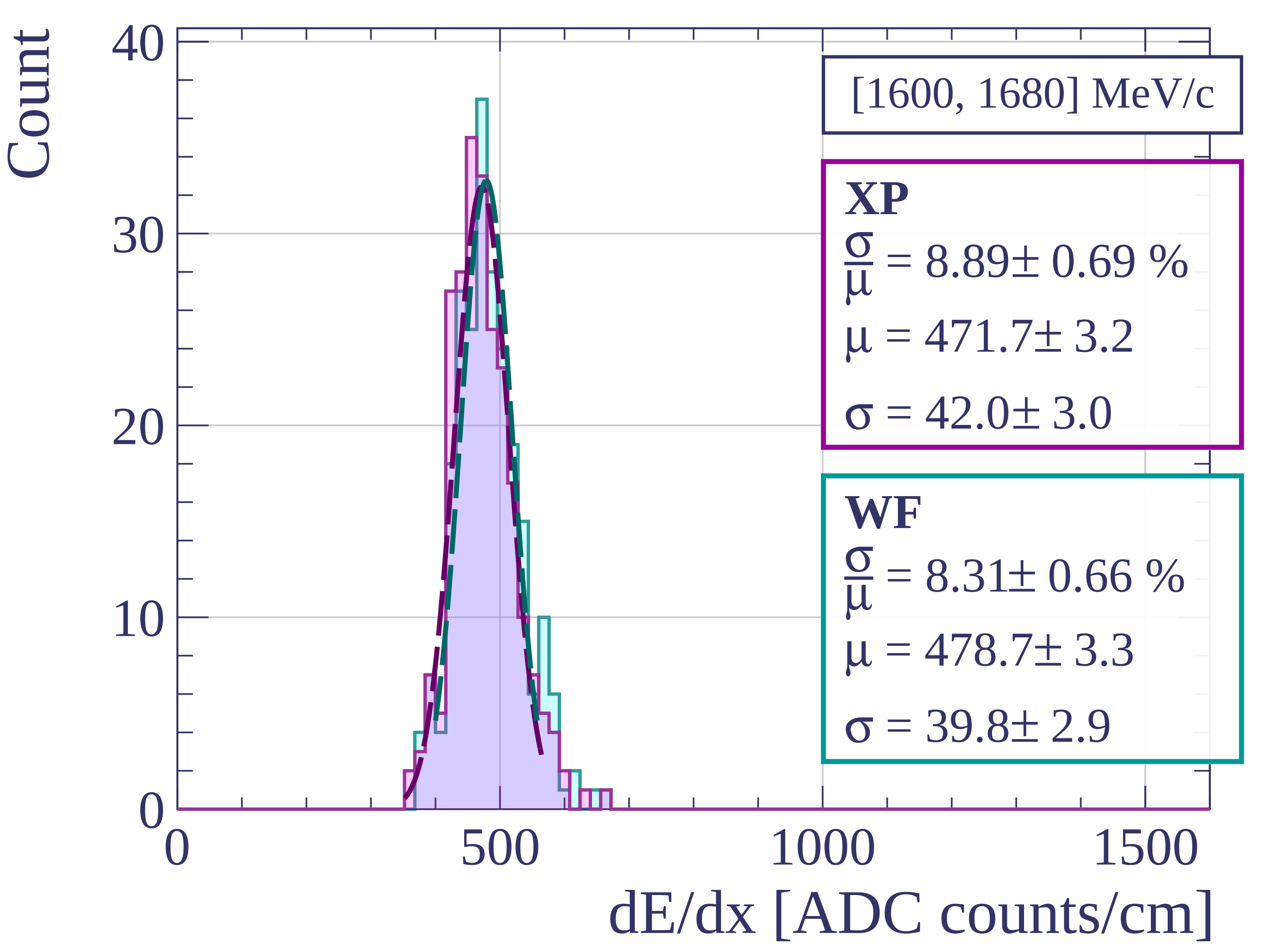


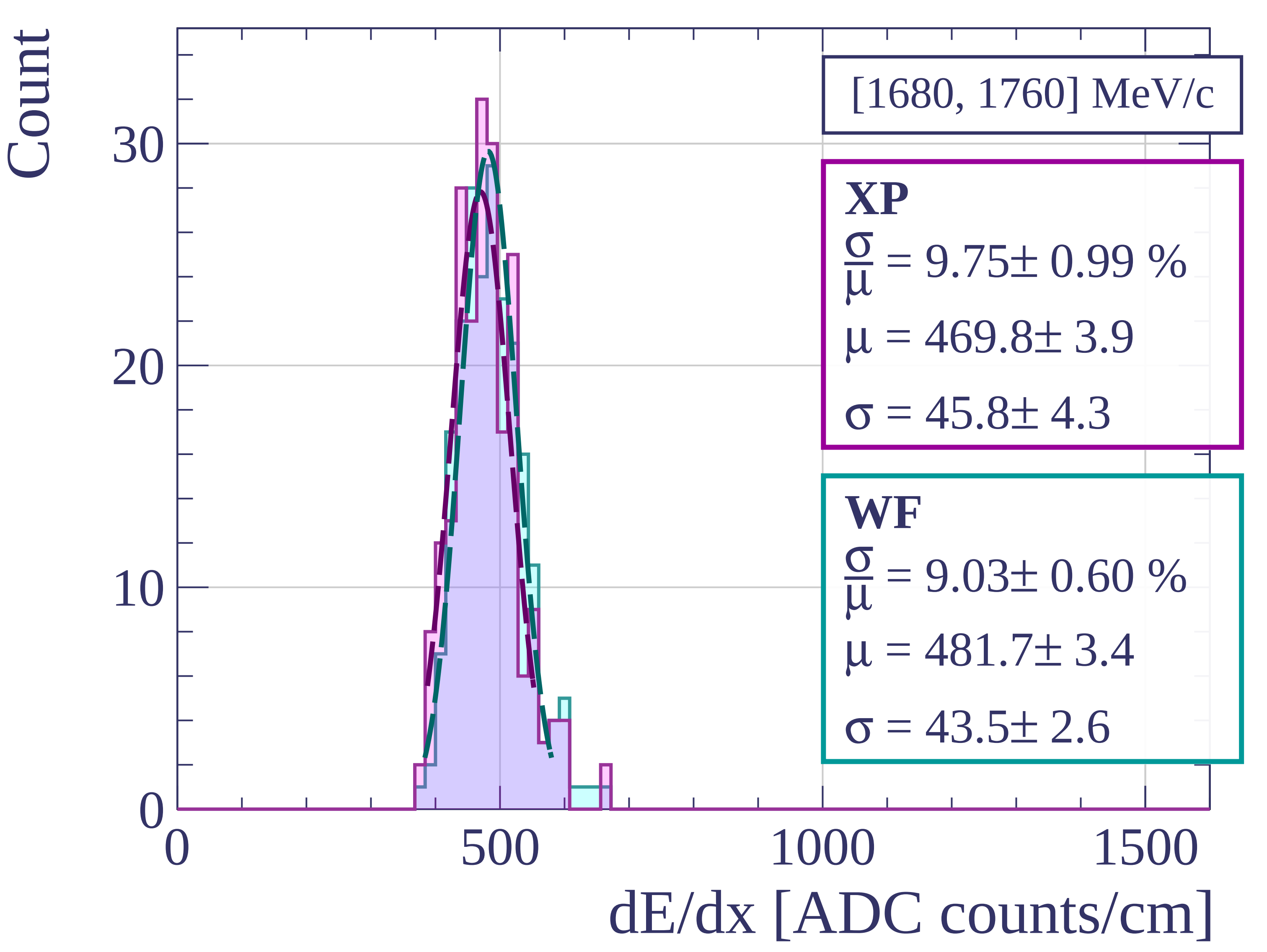
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.75± 0.66 %
μ = 455.4± 3.3
σ = 39.8± 2.7
WF
$\frac{\sigma}{\mu}$ = 9.87± 0.89 %
μ = 467.1± 3.8
σ = 46.1± 3.8


Count
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.11± 0.69 %
μ = 458.8± 3.2
σ = 37.2± 2.9
WF
$\frac{\sigma}{\mu}$ = 8.12± 0.61 %
μ = 469.1± 2.9
σ = 38.1± 2.6
0
10
20
30
40
0
500
1000
1500

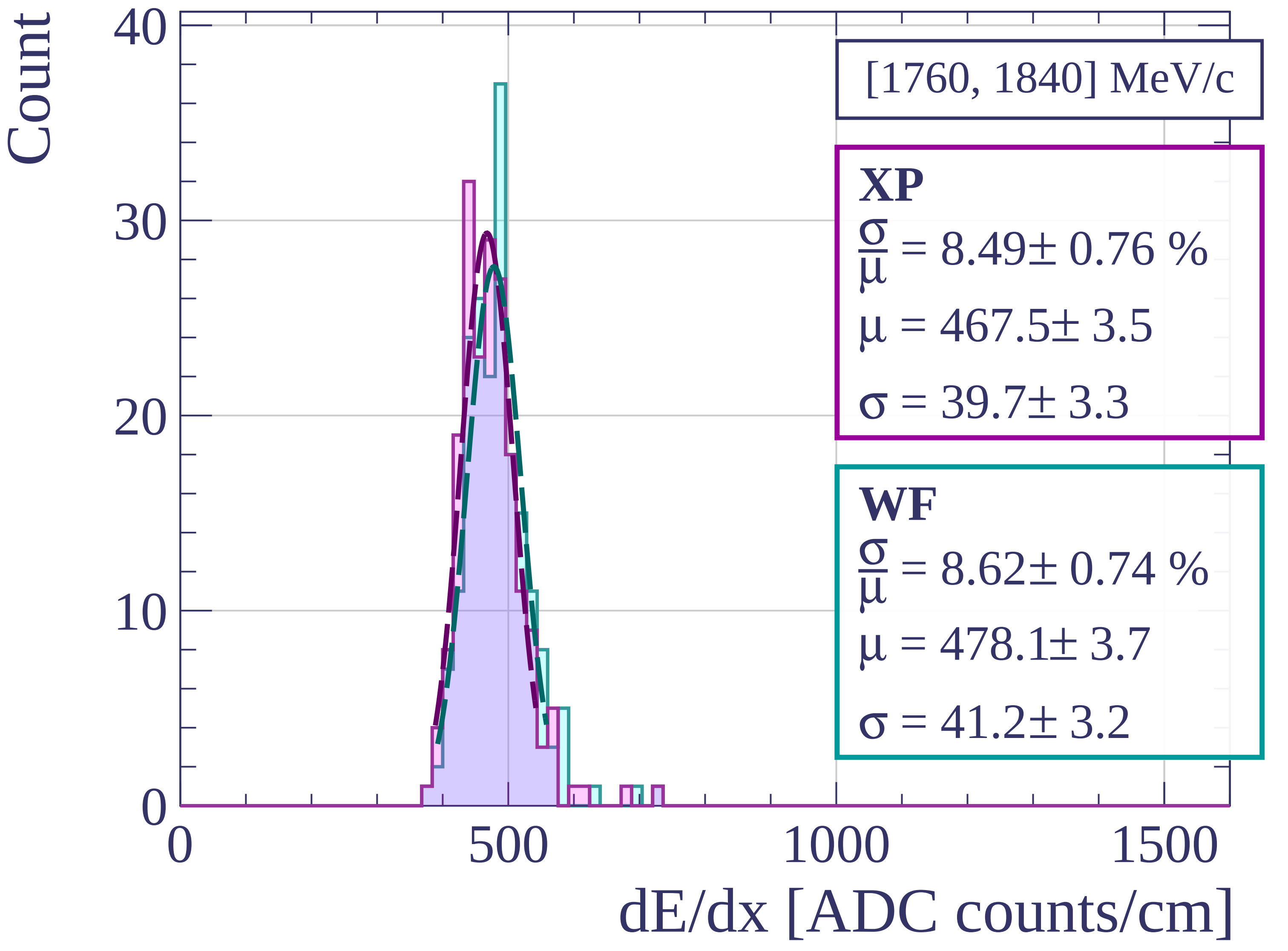


Count
0
10
20
30
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1600] MeV/c
XP
σ/μ = 8.66± 0.73 %
μ = 461.9± 3.2
σ = 40.0± 3.1
WF
σ/μ = 7.96± 0.61 %
μ = 468.8± 3.0
σ = 37.3± 2.6

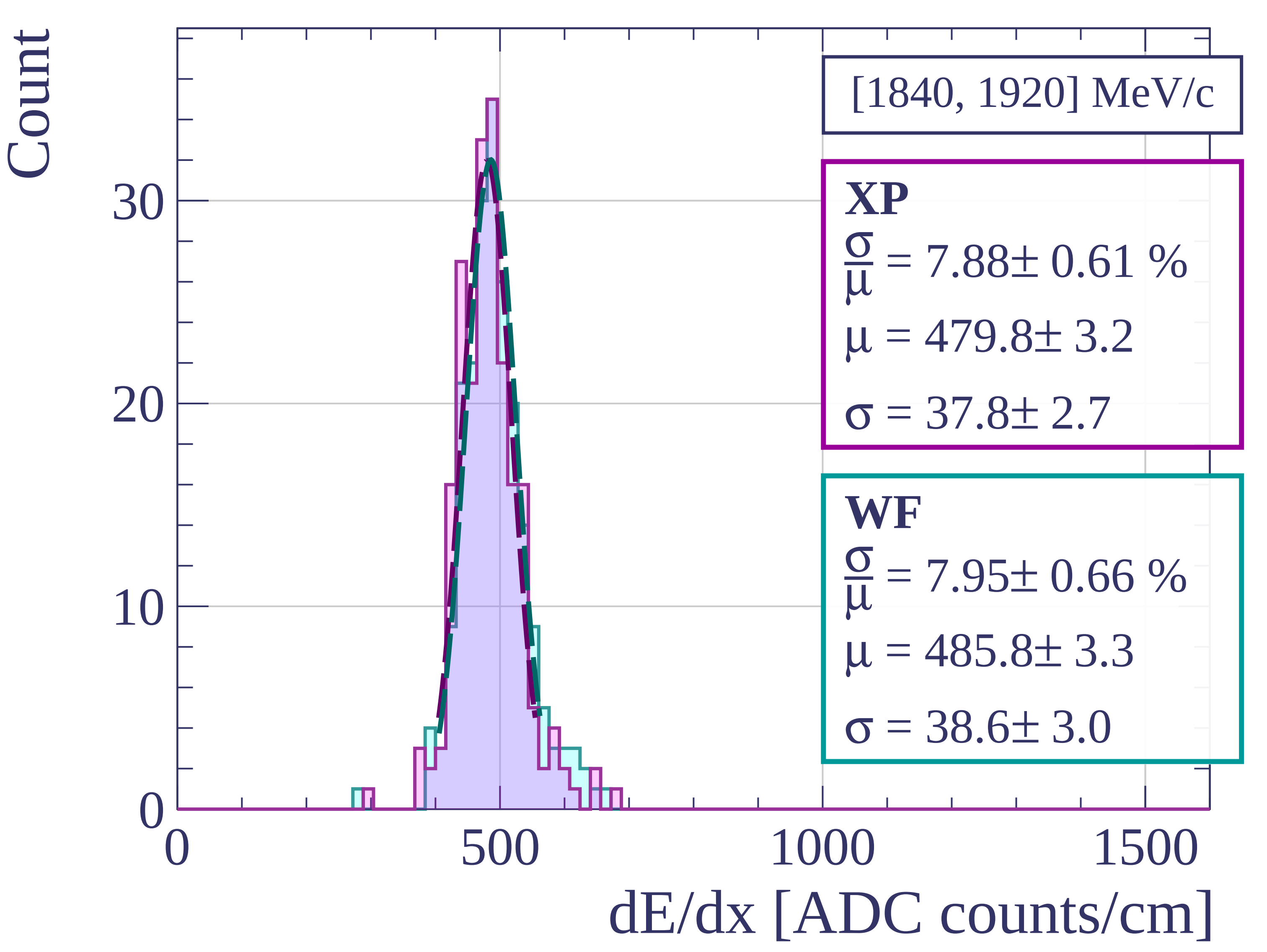

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
10
20
30
40
[1600, 1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.89± 0.69 %
μ = 471.7± 3.2
σ = 42.0± 3.0
WF
$\frac{\sigma}{\mu}$ = 8.31± 0.66 %
μ = 478.7± 3.3
σ = 39.8± 2.9



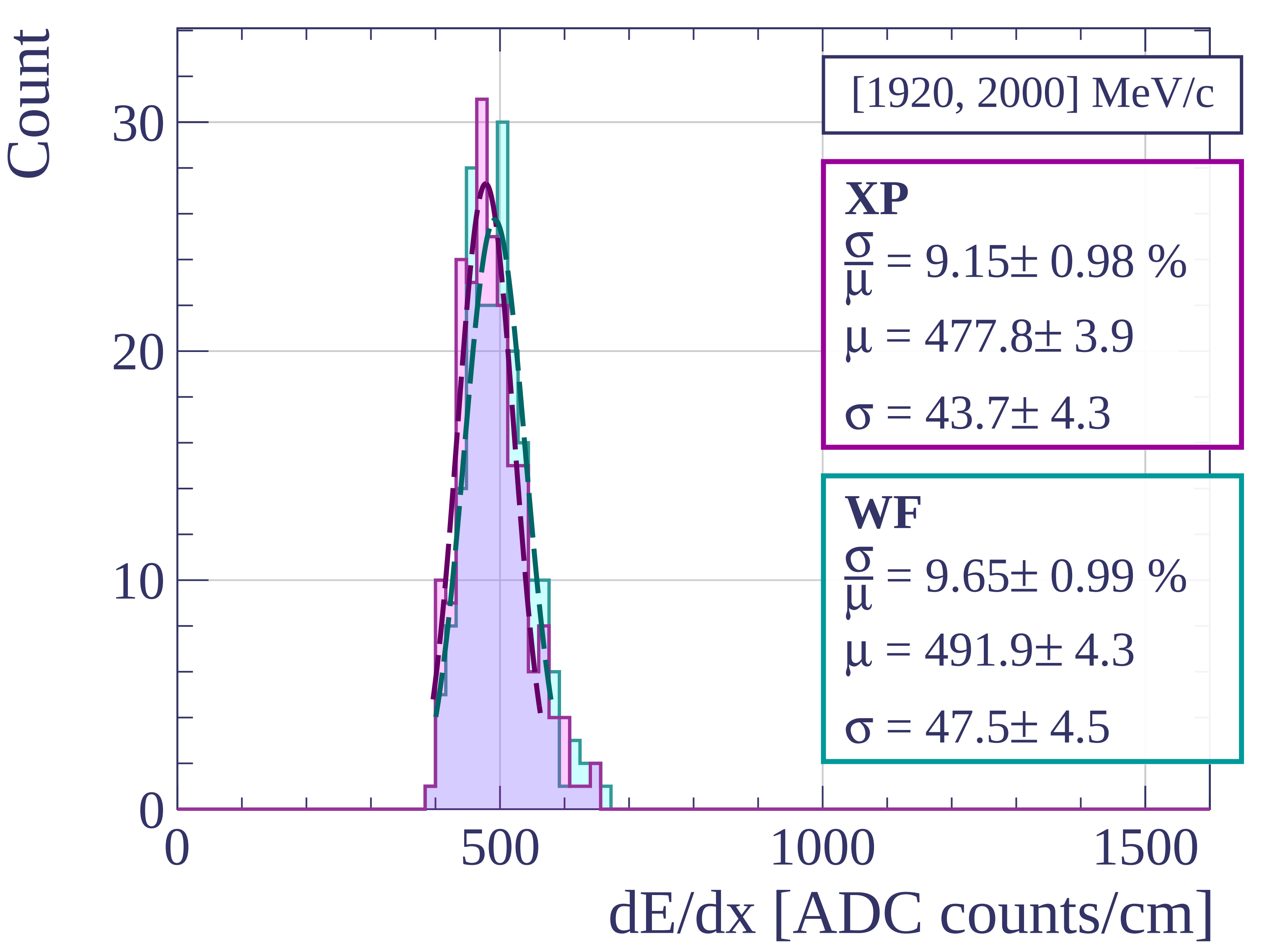
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
[1680, 1760] MeV/c
XP
σ/μ = 9.75± 0.99 %
μ = 469.8± 3.9
σ = 45.8± 4.3
WF
σ/μ = 9.03± 0.60 %
μ = 481.7± 3.4
σ = 43.5± 2.6



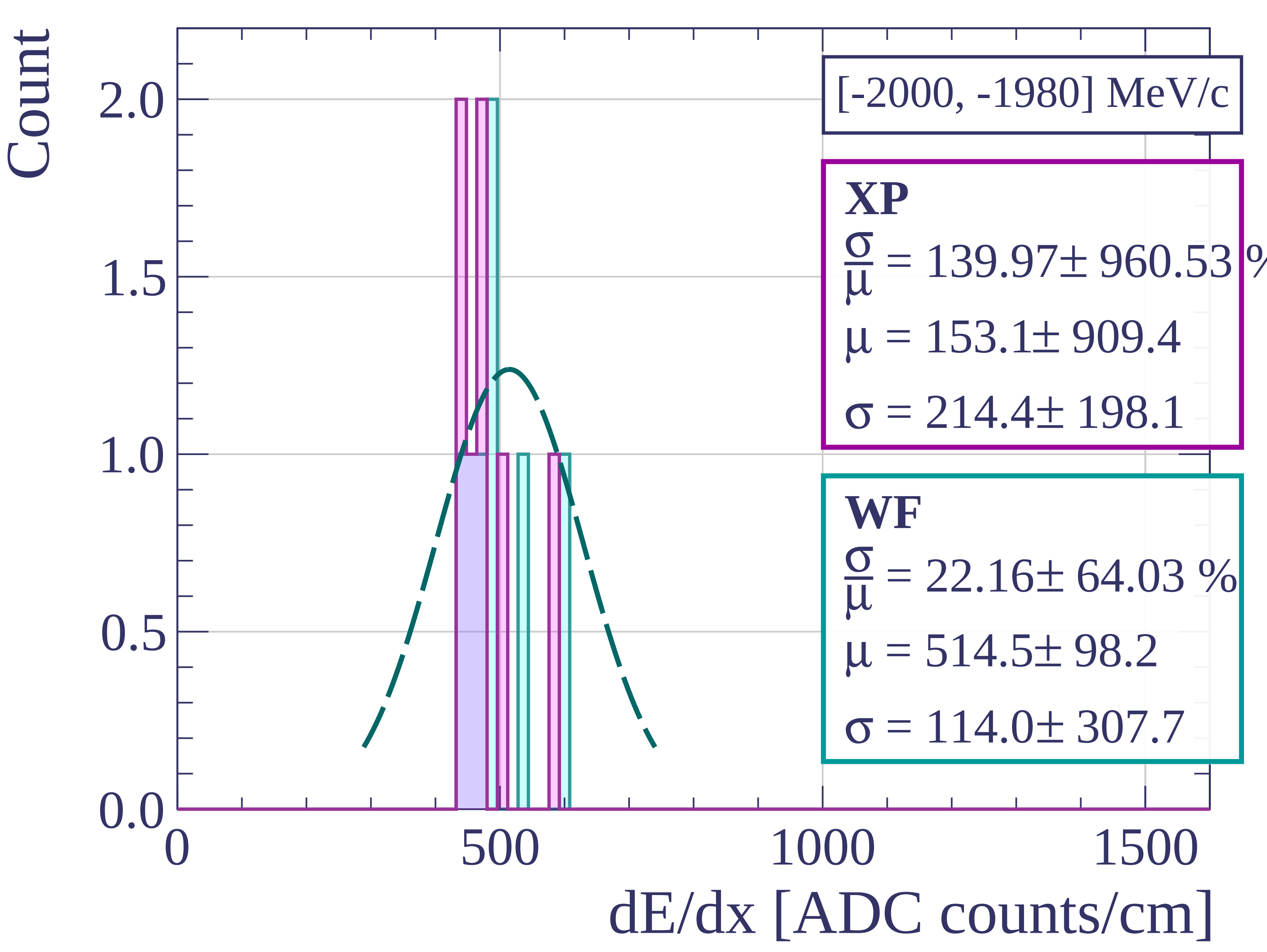
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.49± 0.76 %
μ = 467.5± 3.5
σ = 39.7± 3.3
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.74 %
μ = 478.1± 3.7
σ = 41.2± 3.2

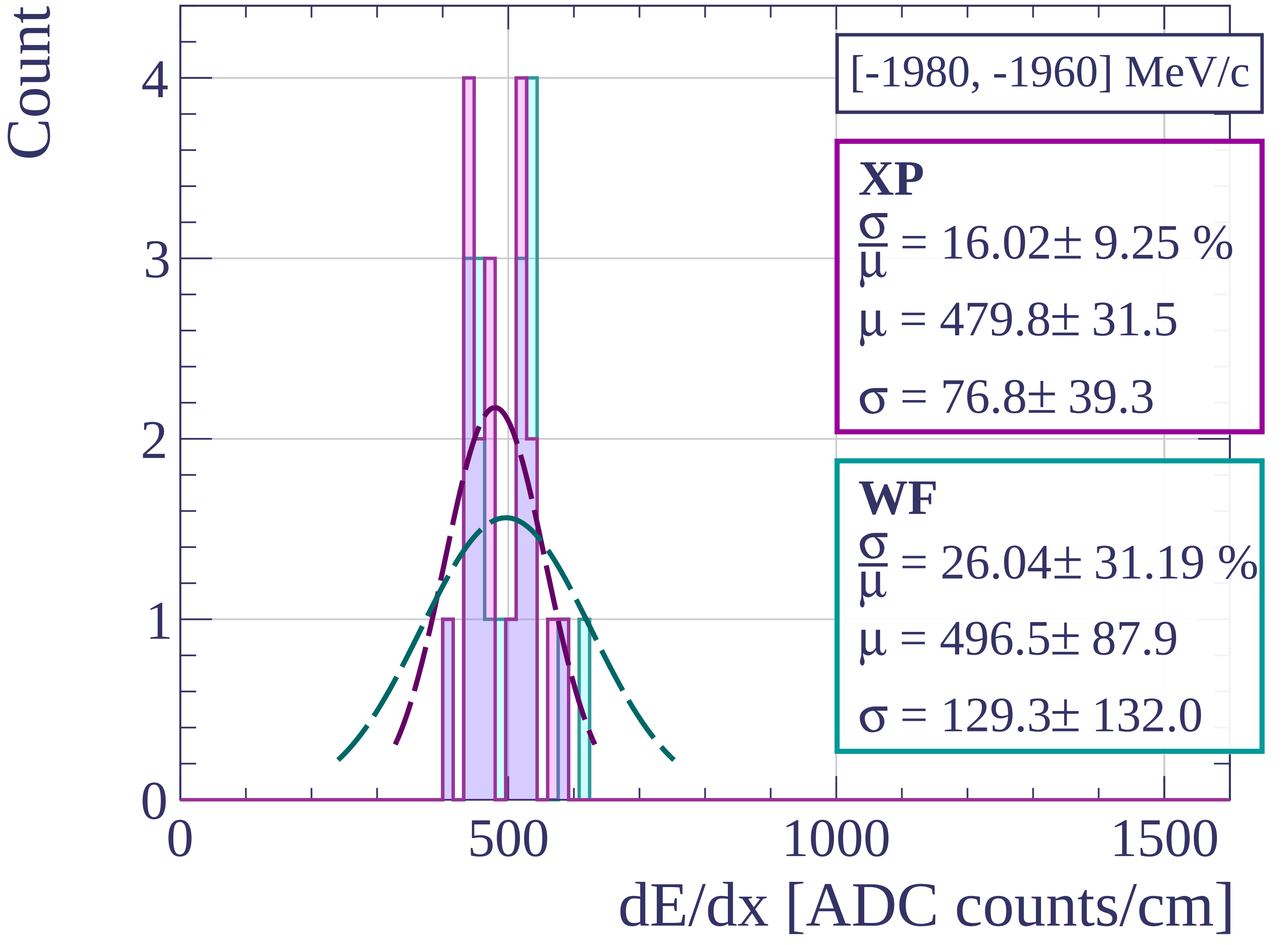


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.88± 0.61 %
μ = 479.8± 3.2
σ = 37.8± 2.7
WF
$\frac{\sigma}{\mu}$ = 7.95± 0.66 %
μ = 485.8± 3.3
σ = 38.6± 3.0

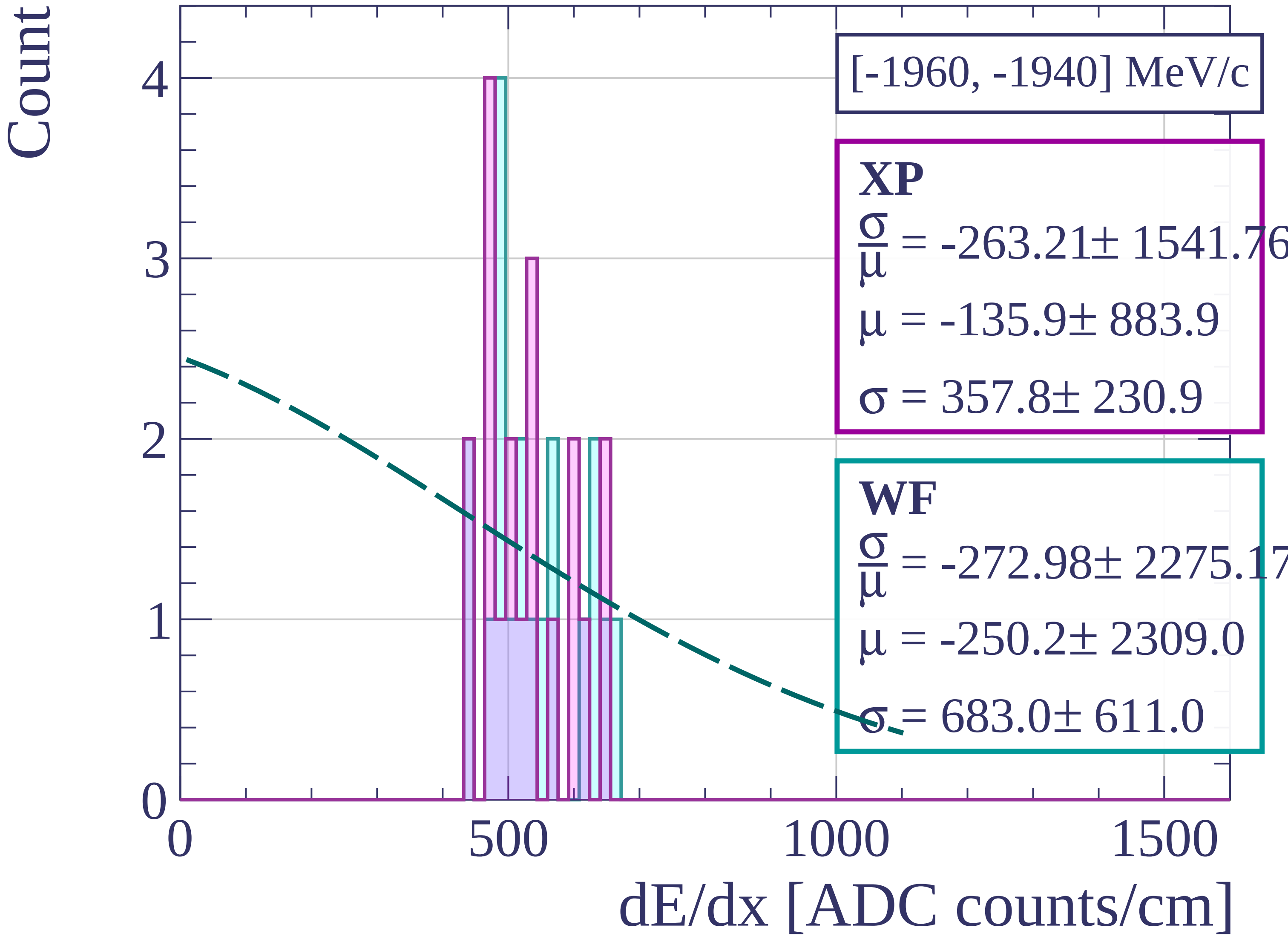


Count
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.15± 0.98 %
μ = 477.8± 3.9
σ = 43.7± 4.3
WF
$\frac{\sigma}{\mu}$ = 9.65± 0.99 %
μ = 491.9± 4.3
σ = 47.5± 4.5
0
10
20
30
0
500
1000
1500

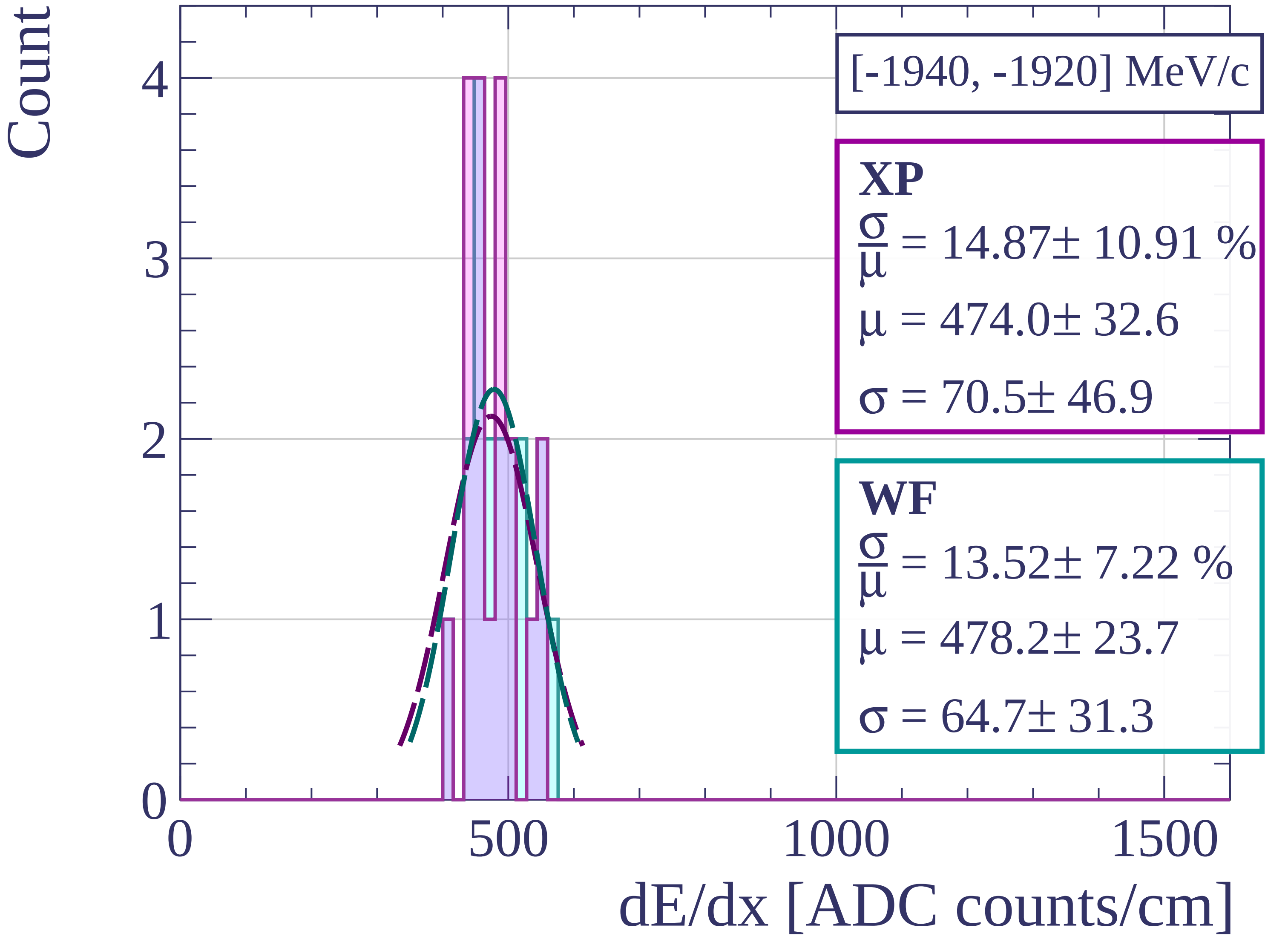

Count
dE/dx [ADC counts/cm]
[-2000, -1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 139.97± 960.53 %
μ = 153.1± 909.4
σ = 214.4± 198.1
WF
$\frac{\sigma}{\mu}$ = 22.16± 64.03 %
μ = 514.5± 98.2
σ = 114.0± 307.7



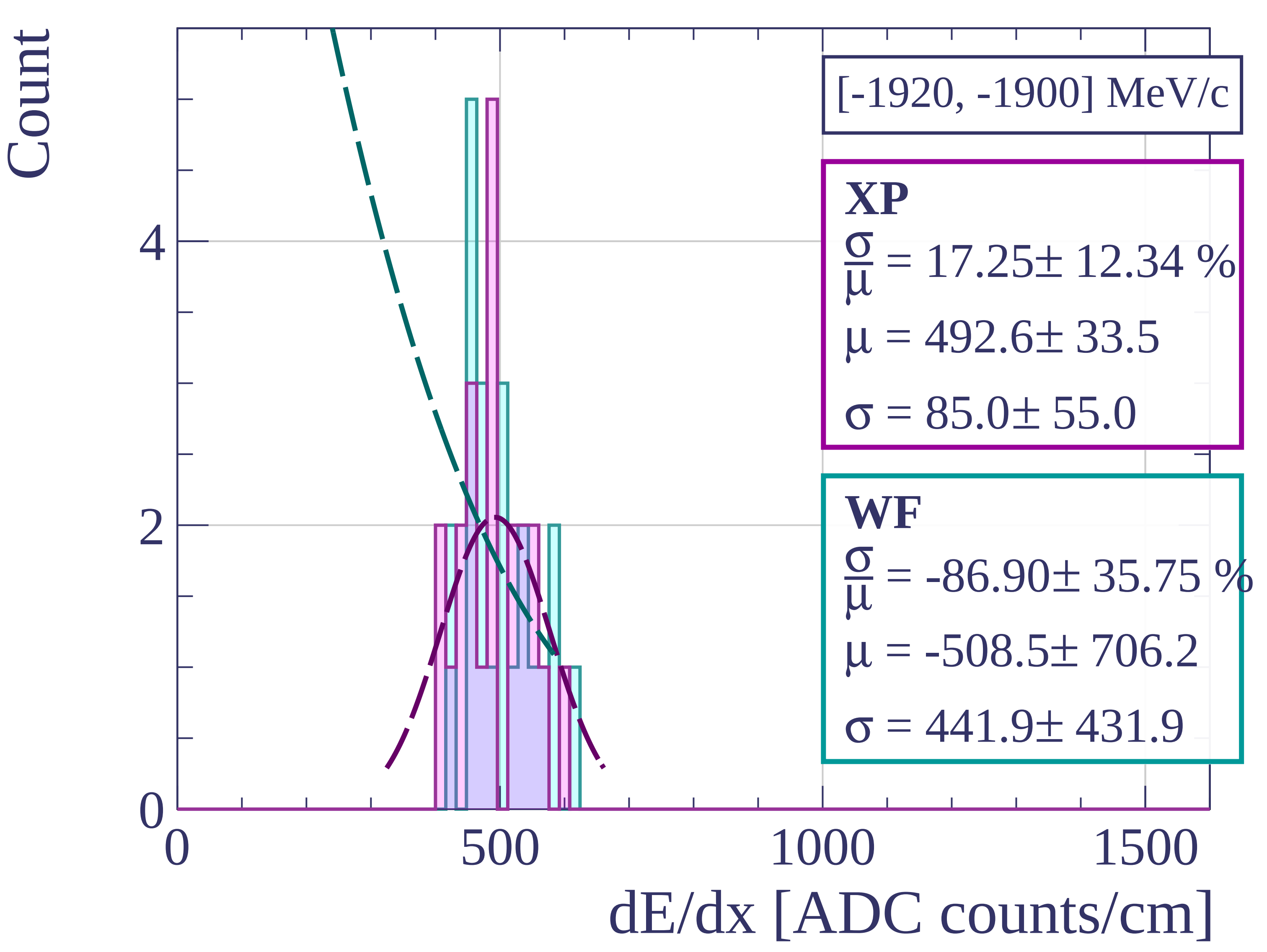
Count
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
σ/μ = 16.02± 9.25 %
μ = 479.8± 31.5
σ = 76.8± 39.3
WF
σ/μ = 26.04± 31.19 %
μ = 496.5± 87.9
σ = 129.3± 132.0
0
1
2
3
4
0
500
1000
1500



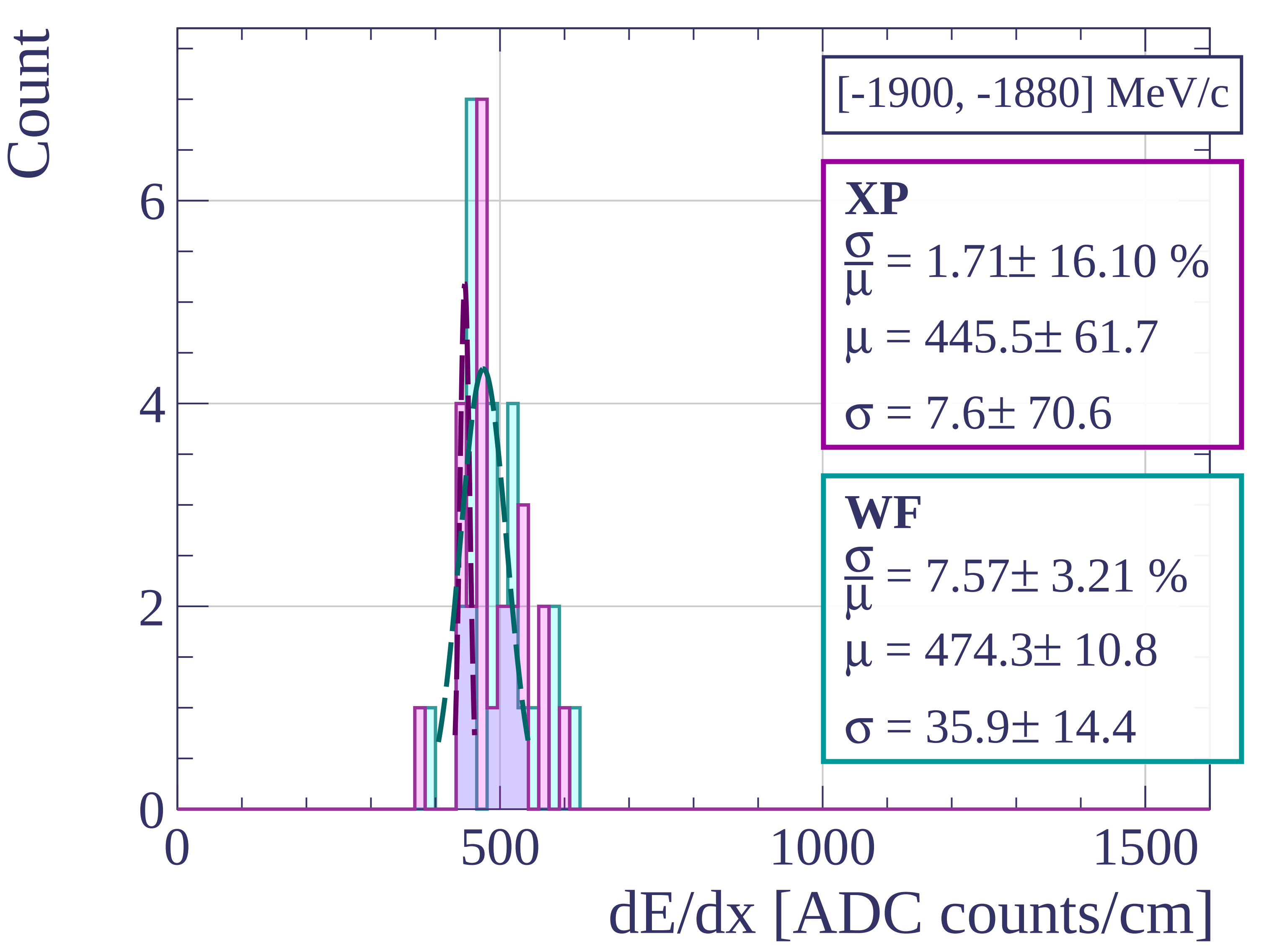
Count
dE/dx [ADC counts/cm]
[-1960, -1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = -263.21± 1541.76
μ = -135.9± 883.9
σ = 357.8± 230.9
WF
$\frac{\sigma}{\mu}$ = -272.98± 2275.17
μ = -250.2± 2309.0
σ = 683.0± 611.0



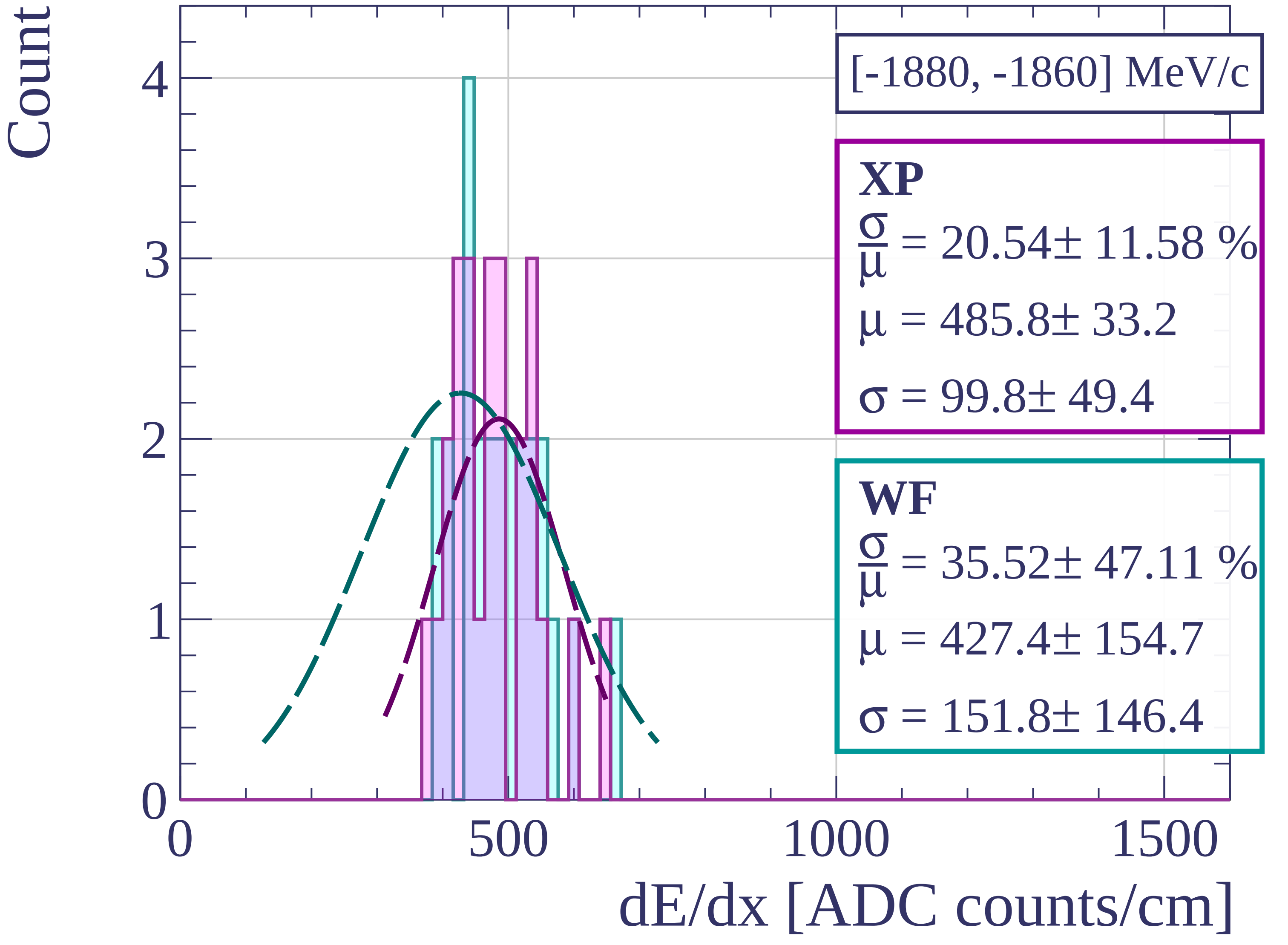
Count
[-1940, -1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.87± 10.91 %
μ = 474.0± 32.6
σ = 70.5± 46.9
WF
$\frac{\sigma}{\mu}$ = 13.52± 7.22 %
μ = 478.2± 23.7
σ = 64.7± 31.3
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]



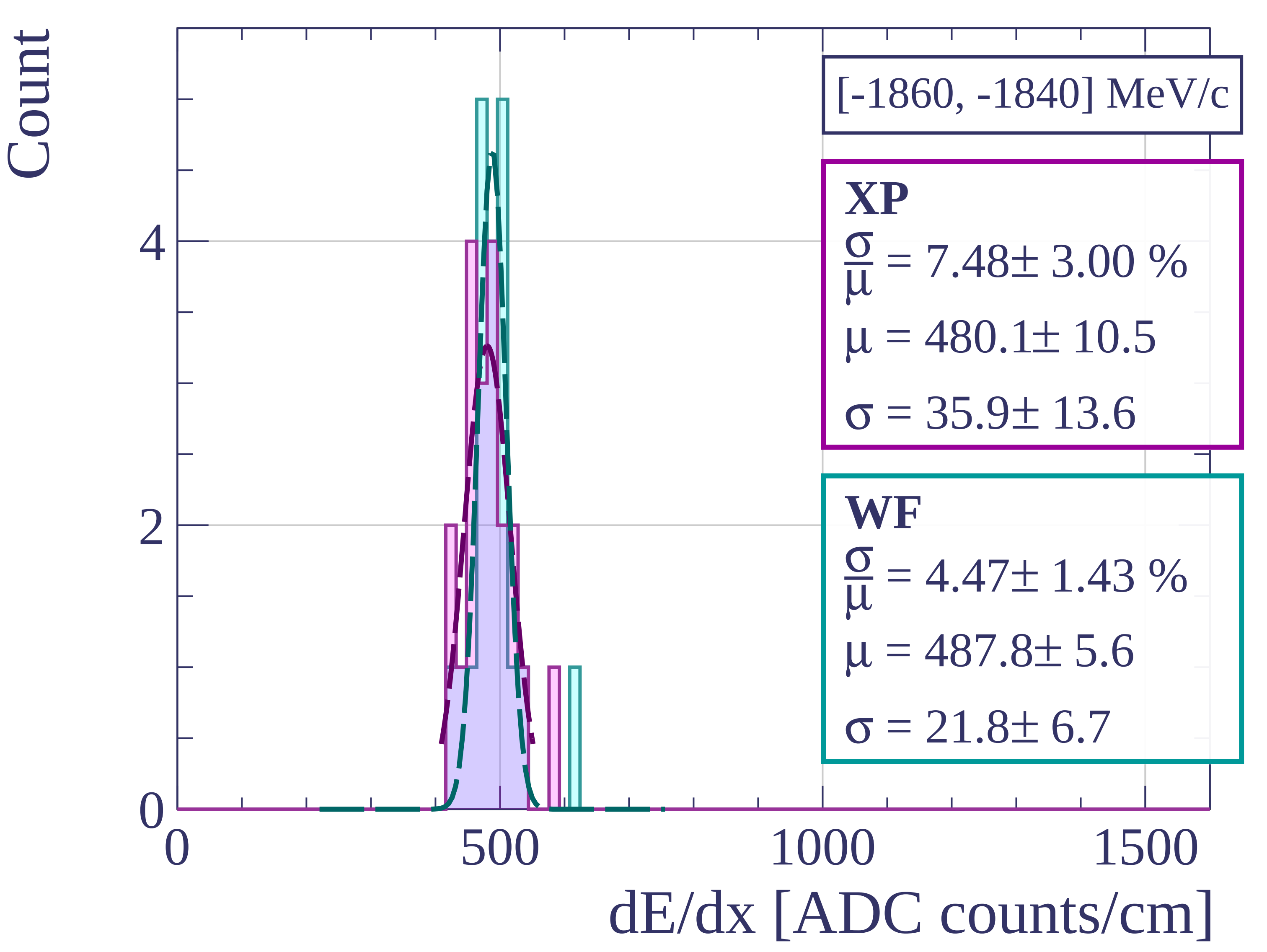
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1920, -1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 17.25± 12.34 %
μ = 492.6± 33.5
σ = 85.0± 55.0
WF
$\frac{\sigma}{\mu}$ = -86.90± 35.75 %
μ = -508.5± 706.2
σ = 441.9± 431.9

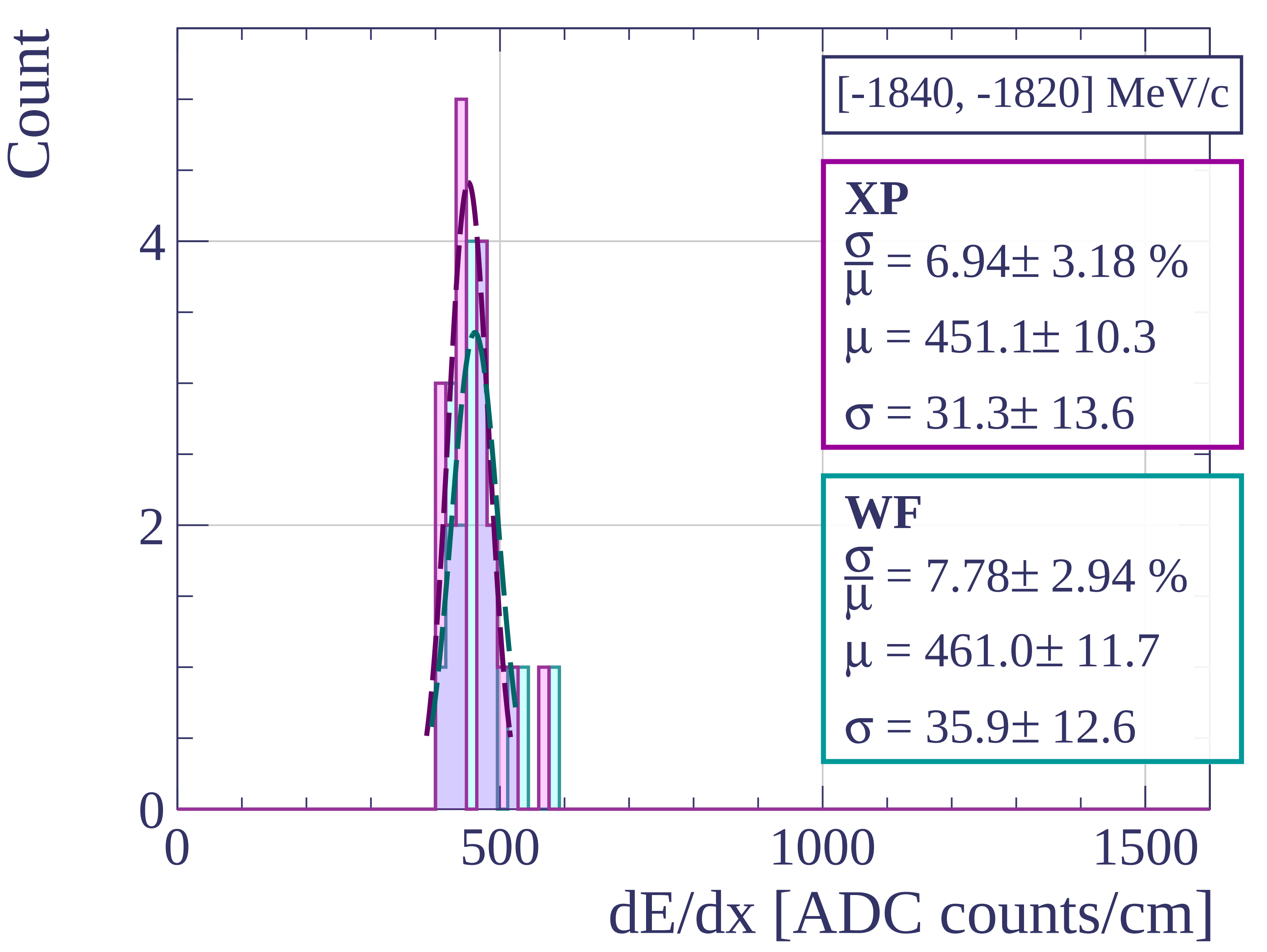


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1900, -1880] MeV/c
XP
σ/μ = 1.71± 16.10 %
μ = 445.5± 61.7
σ = 7.6± 70.6
WF
σ/μ = 7.57± 3.21 %
μ = 474.3± 10.8
σ = 35.9± 14.4

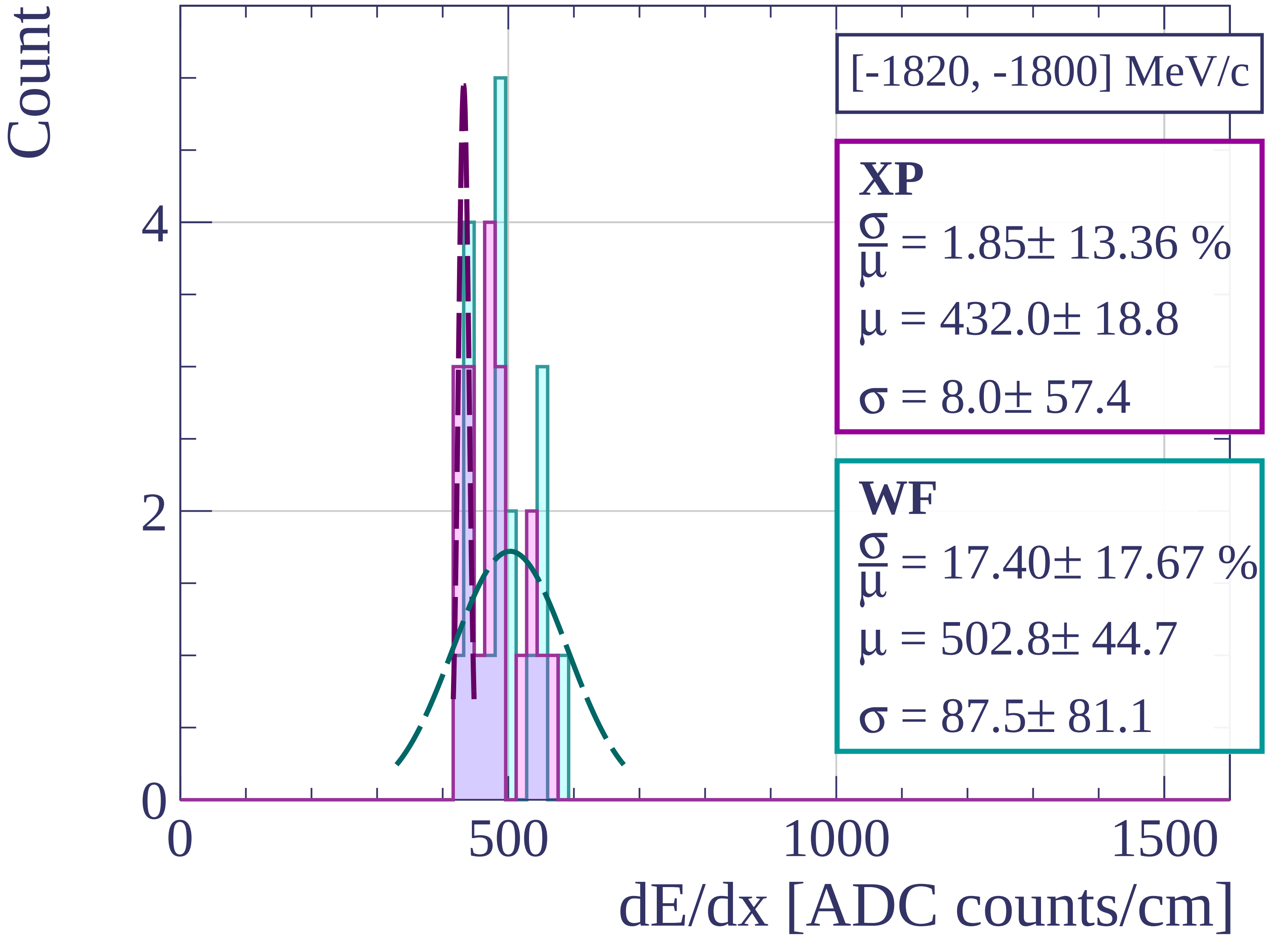


Count
dE/dx [ADC counts/cm]
0
1
2
3
4
0
500
1000
1500
[-1880, -1860] MeV/c
XP
σ/μ = 20.54± 11.58 %
μ = 485.8± 33.2
σ = 99.8± 49.4
WF
σ/μ = 35.52± 47.11 %
μ = 427.4± 154.7
σ = 151.8± 146.4

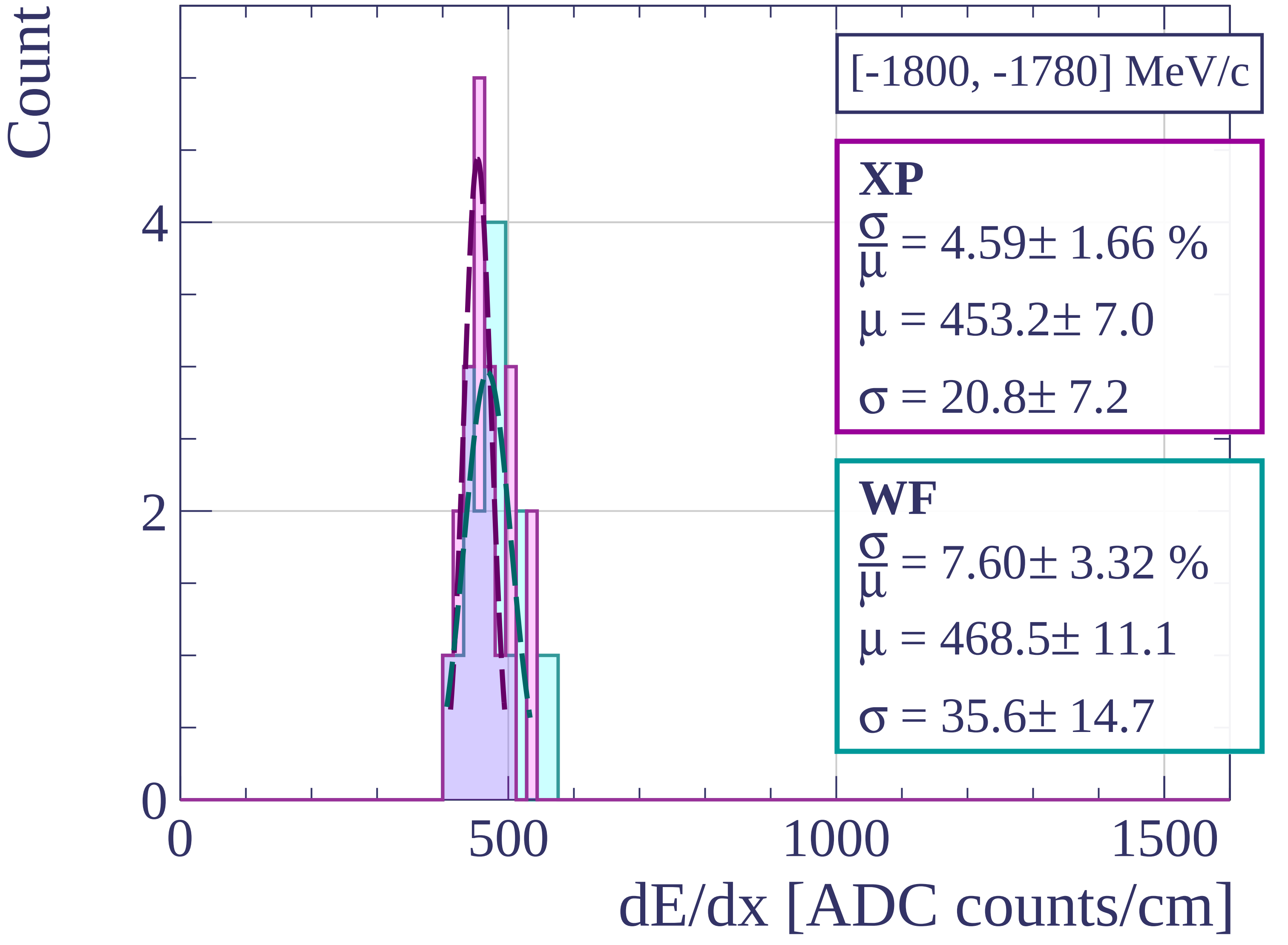

Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1860, -1840] MeV/c
XP
σ/μ = 7.48± 3.00 %
μ = 480.1± 10.5
σ = 35.9± 13.6
WF
σ/μ = 4.47± 1.43 %
μ = 487.8± 5.6
σ = 21.8± 6.7



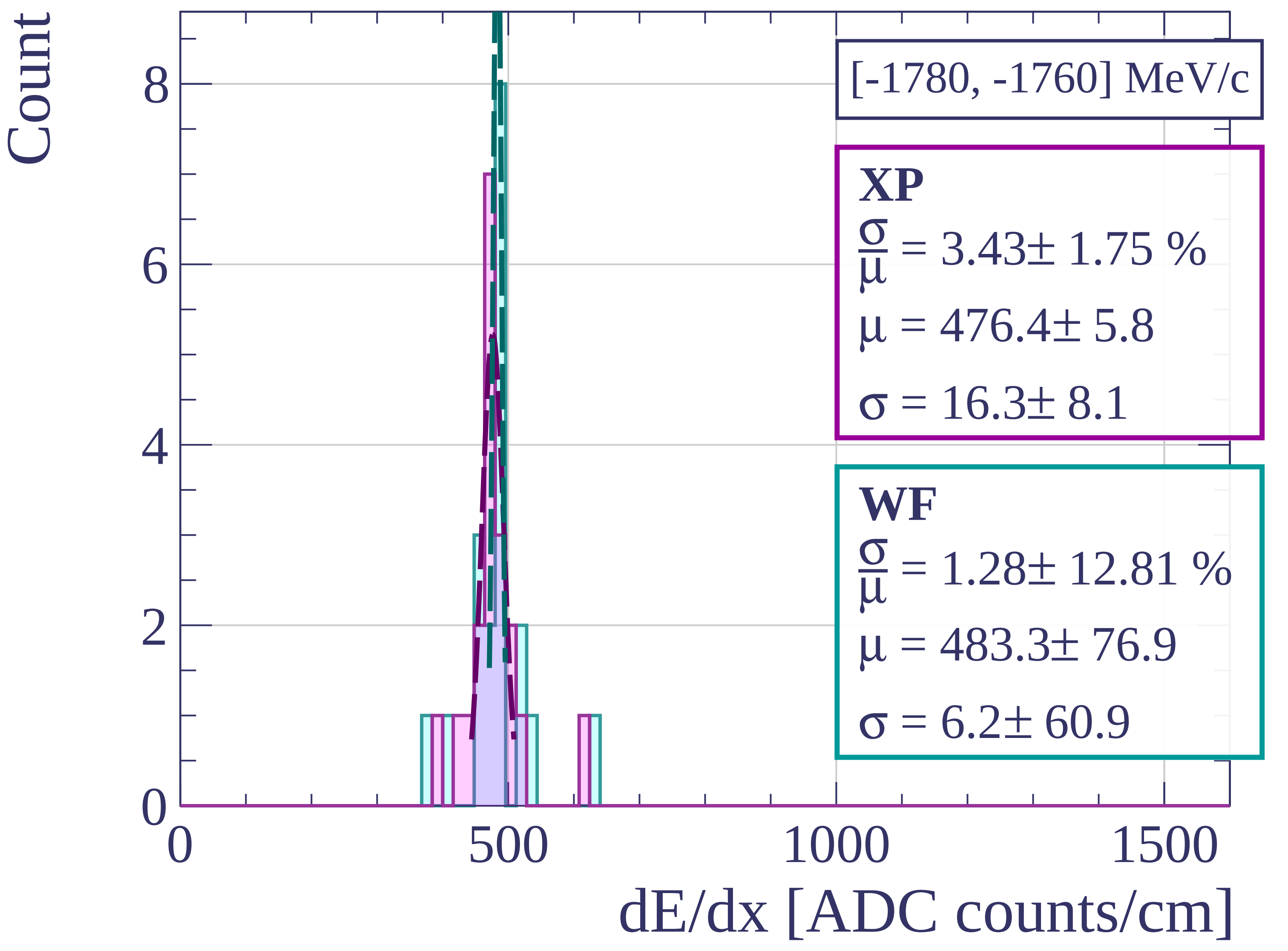
Count
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.94± 3.18 %
μ = 451.1± 10.3
σ = 31.3± 13.6
WF
$\frac{\sigma}{\mu}$ = 7.78± 2.94 %
μ = 461.0± 11.7
σ = 35.9± 12.6
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]



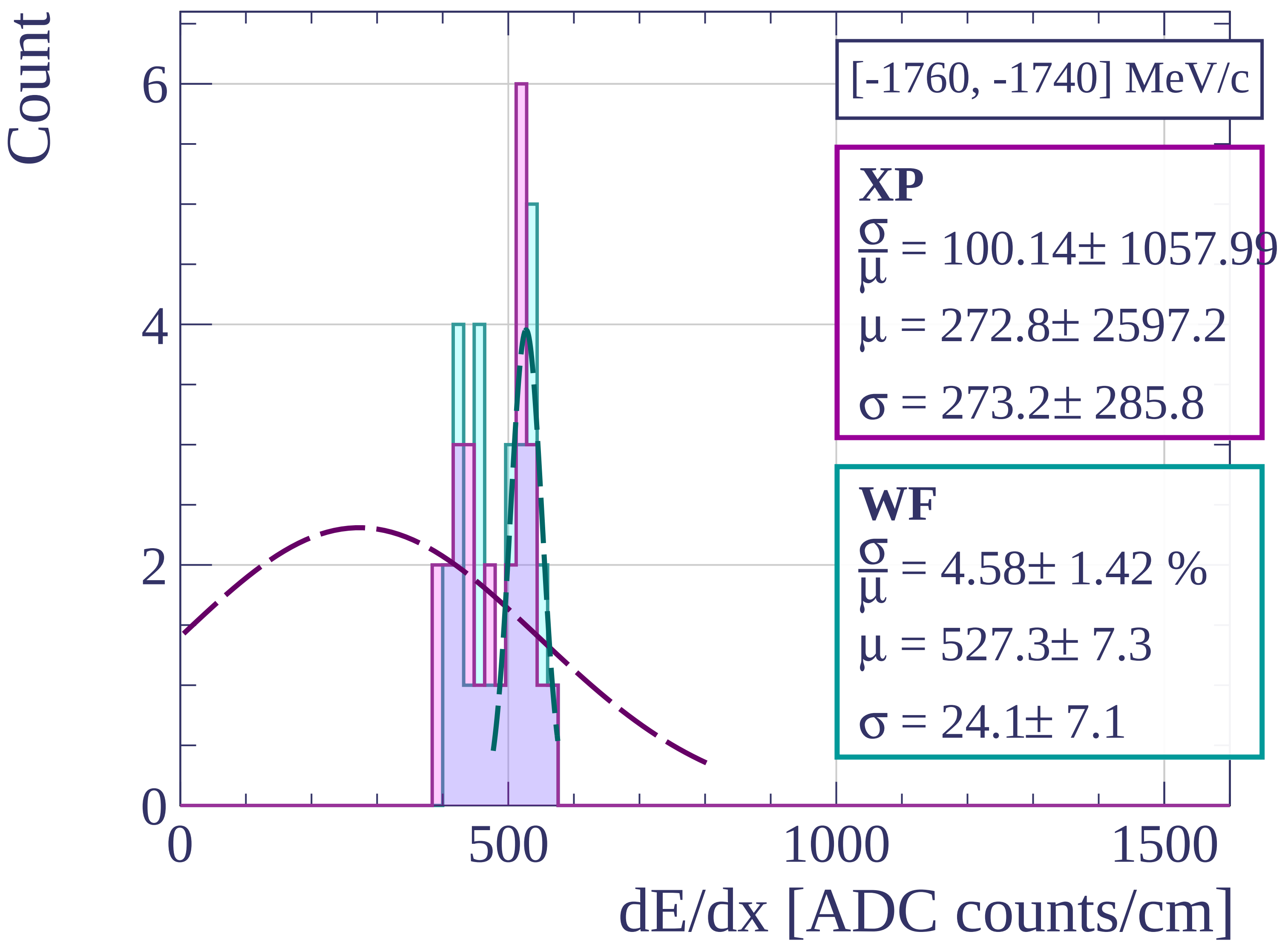
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1820, -1800] MeV/c
XP
σ/μ = 1.85± 13.36 %
μ = 432.0± 18.8
σ = 8.0± 57.4
WF
σ/μ = 17.40± 17.67 %
μ = 502.8± 44.7
σ = 87.5± 81.1



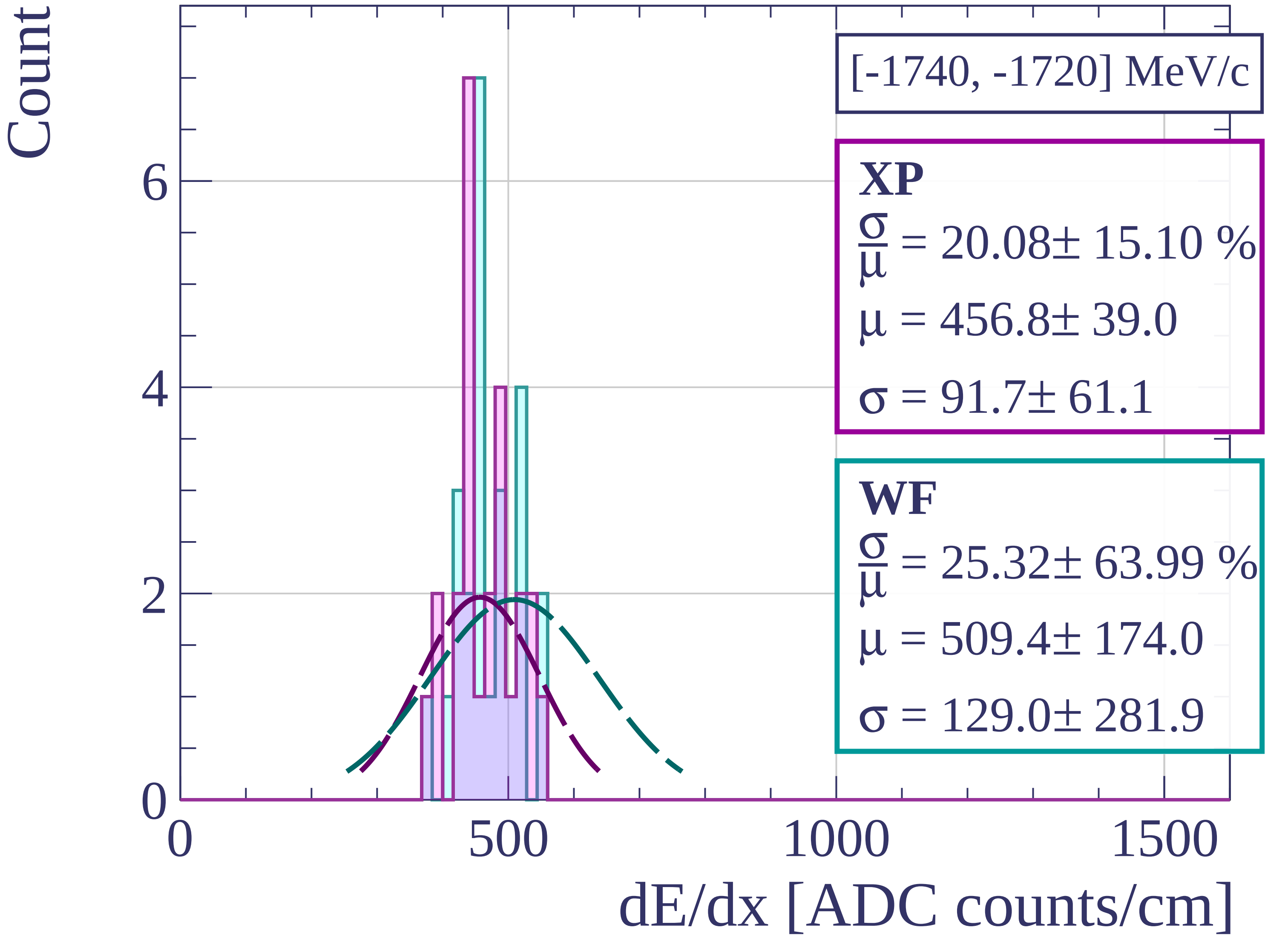
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1800, -1780] MeV/c
XP
σ/μ = 4.59± 1.66 %
μ = 453.2± 7.0
σ = 20.8± 7.2
WF
σ/μ = 7.60± 3.32 %
μ = 468.5± 11.1
σ = 35.6± 14.7

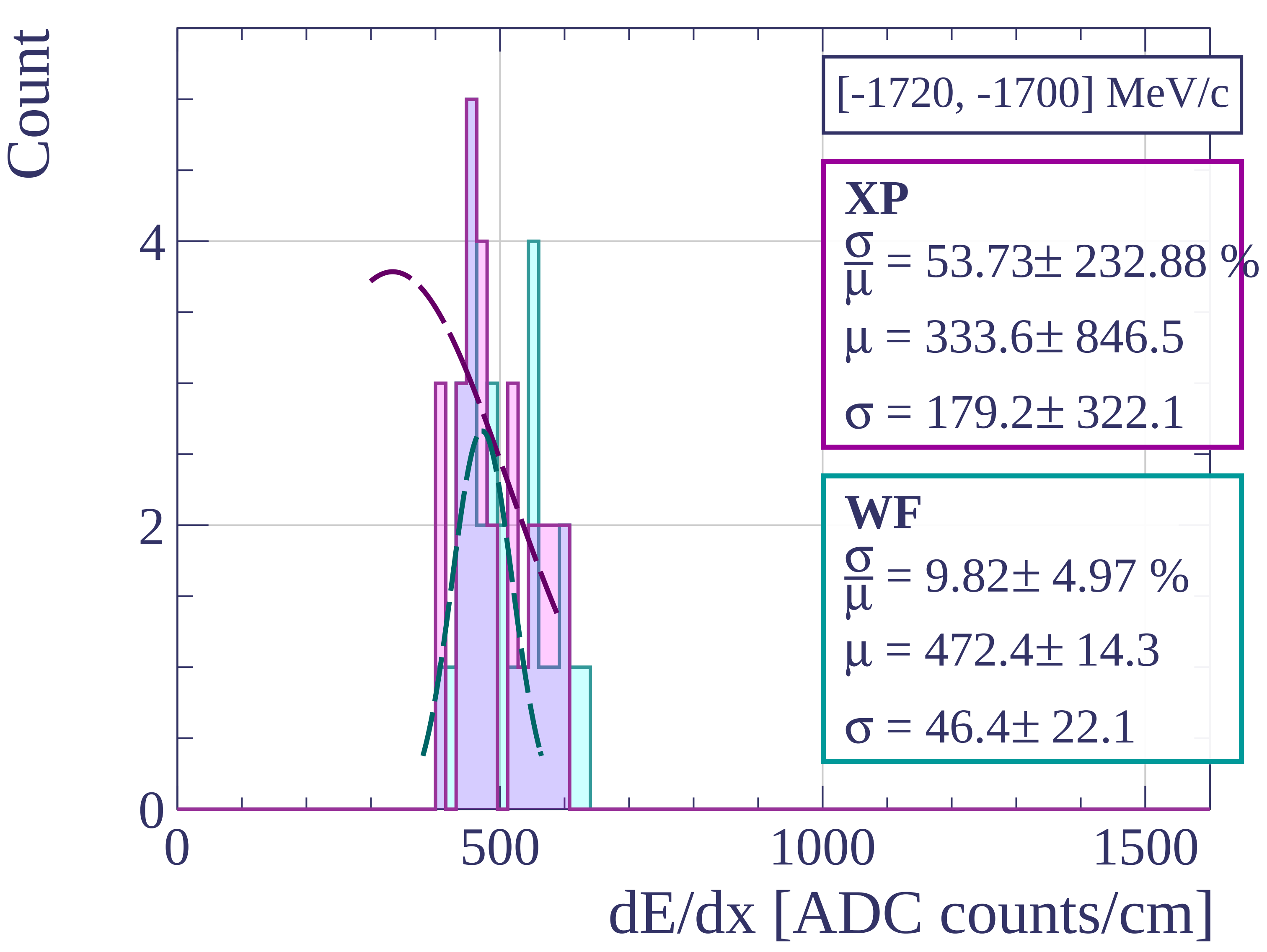


Count
[-1780, -1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.43± 1.75 %
μ = 476.4± 5.8
σ = 16.3± 8.1
WF
$\frac{\sigma}{\mu}$ = 1.28± 12.81 %
μ = 483.3± 76.9
σ = 6.2± 60.9
dE/dx [ADC counts/cm]

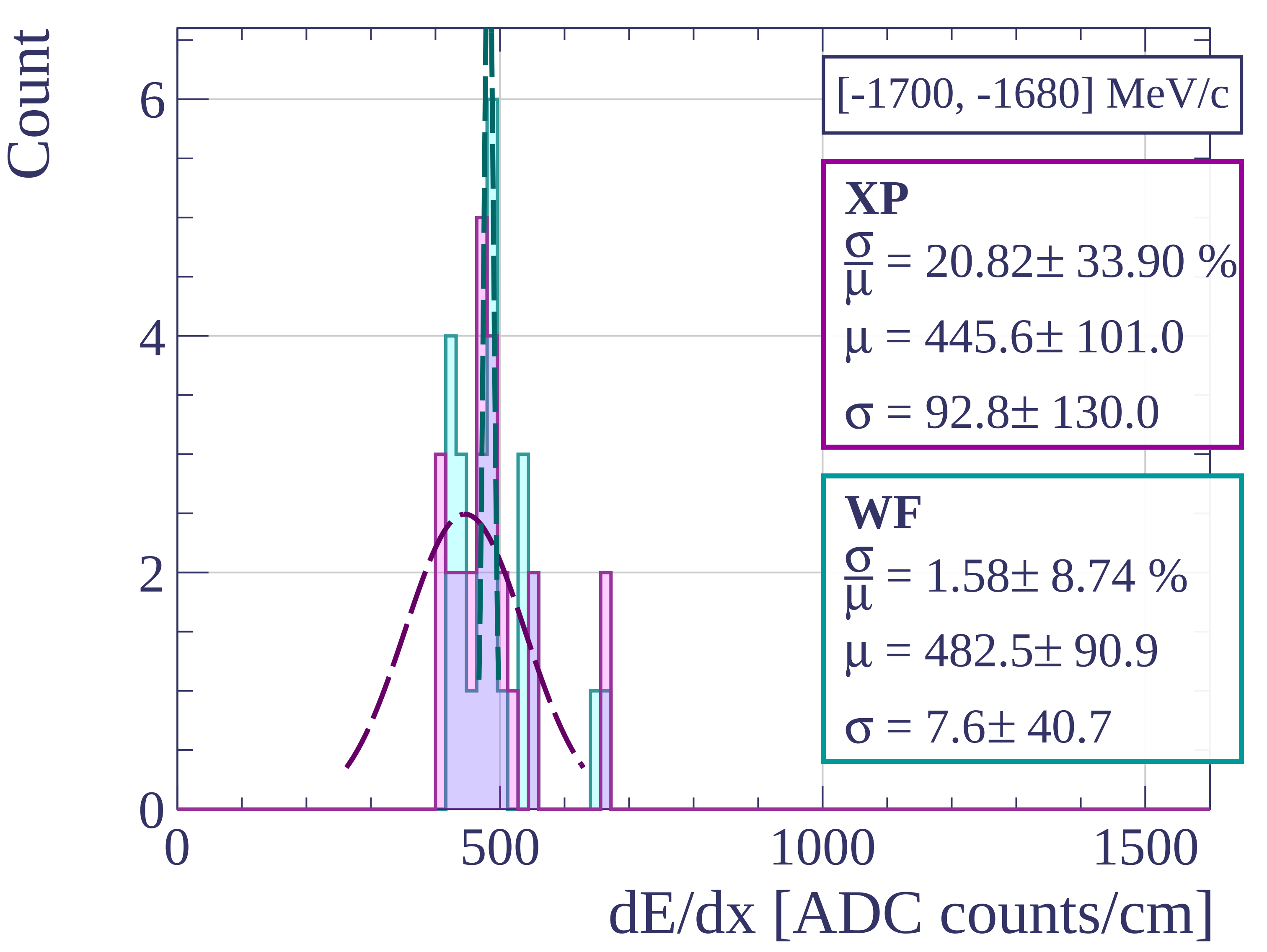


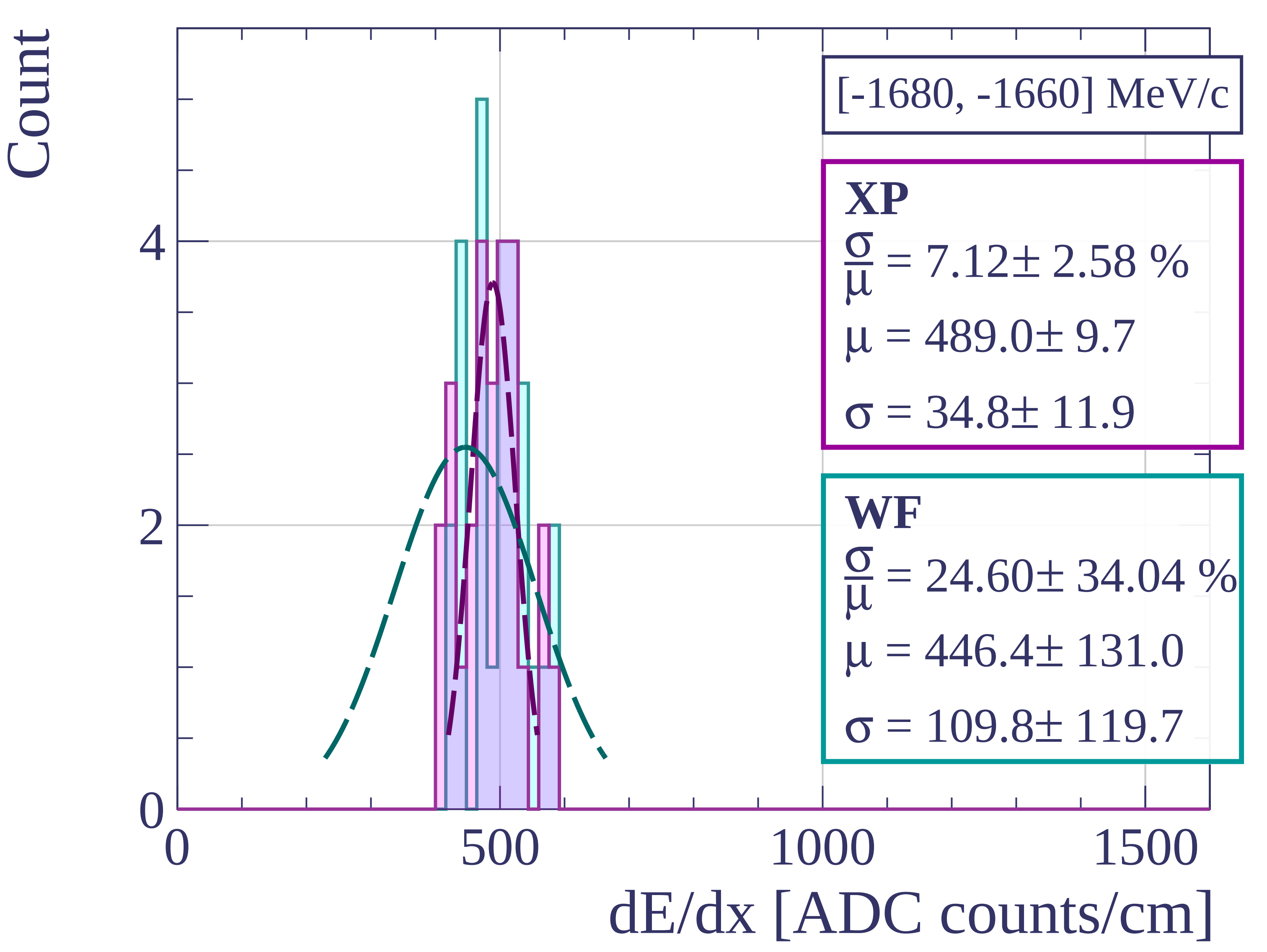
Count
[-1760, -1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 100.14± 1057.99
μ = 272.8± 2597.2
σ = 273.2± 285.8
WF
$\frac{\sigma}{\mu}$ = 4.58± 1.42 %
μ = 527.3± 7.3
σ = 24.1± 7.1
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]


[-1740, -1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.08± 15.10 %
μ = 456.8± 39.0
σ = 91.7± 61.1
WF
$\frac{\sigma}{\mu}$ = 25.32± 63.99 %
μ = 509.4± 174.0
σ = 129.0± 281.9
Count
dE/dx [ADC counts/cm]
0
2
4
6
500
1000
1500

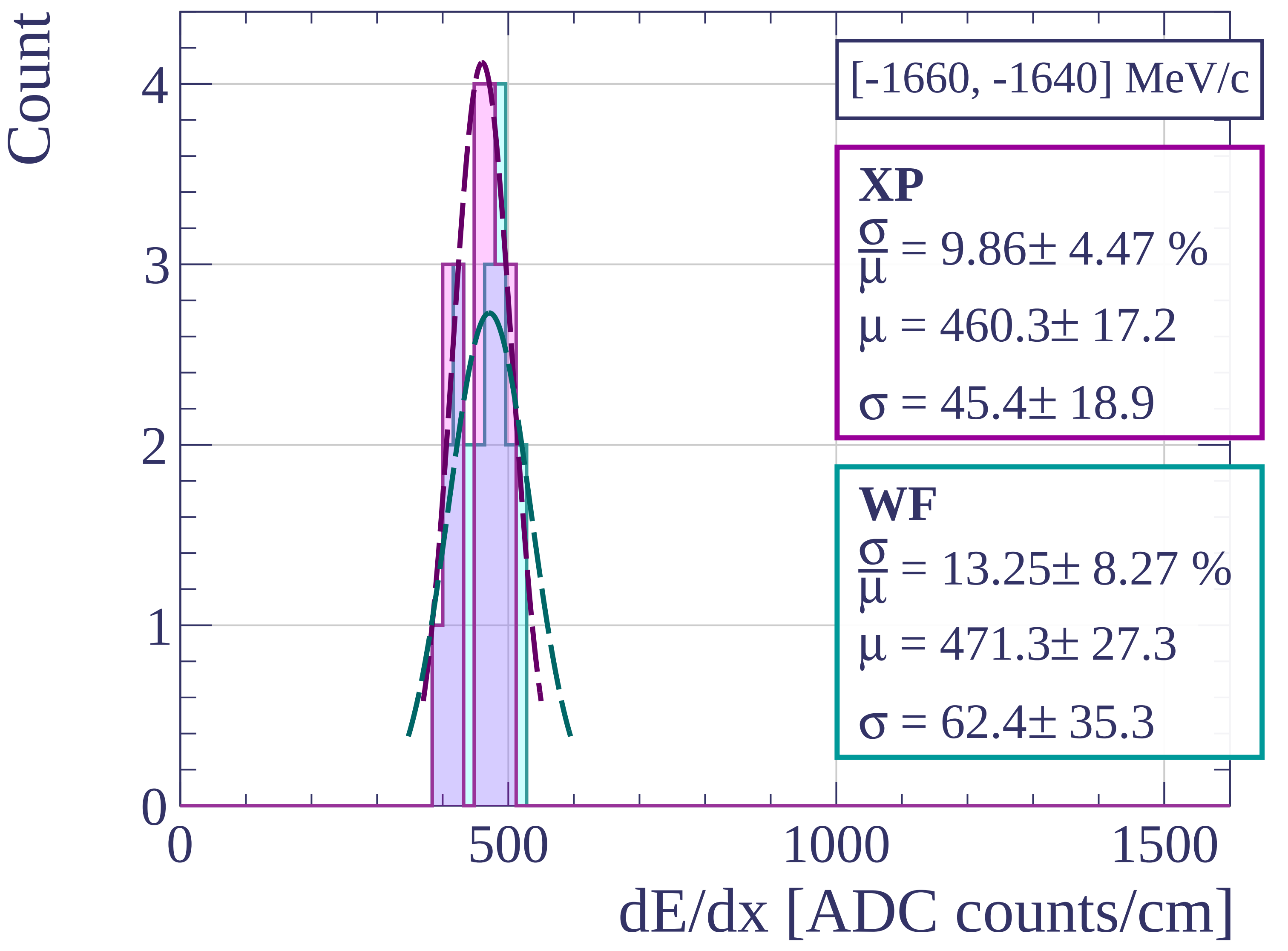


Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1720, -1700] MeV/c
XP
σ/μ = 53.73± 232.88 %
μ = 333.6± 846.5
σ = 179.2± 322.1
WF
σ/μ = 9.82± 4.97 %
μ = 472.4± 14.3
σ = 46.4± 22.1

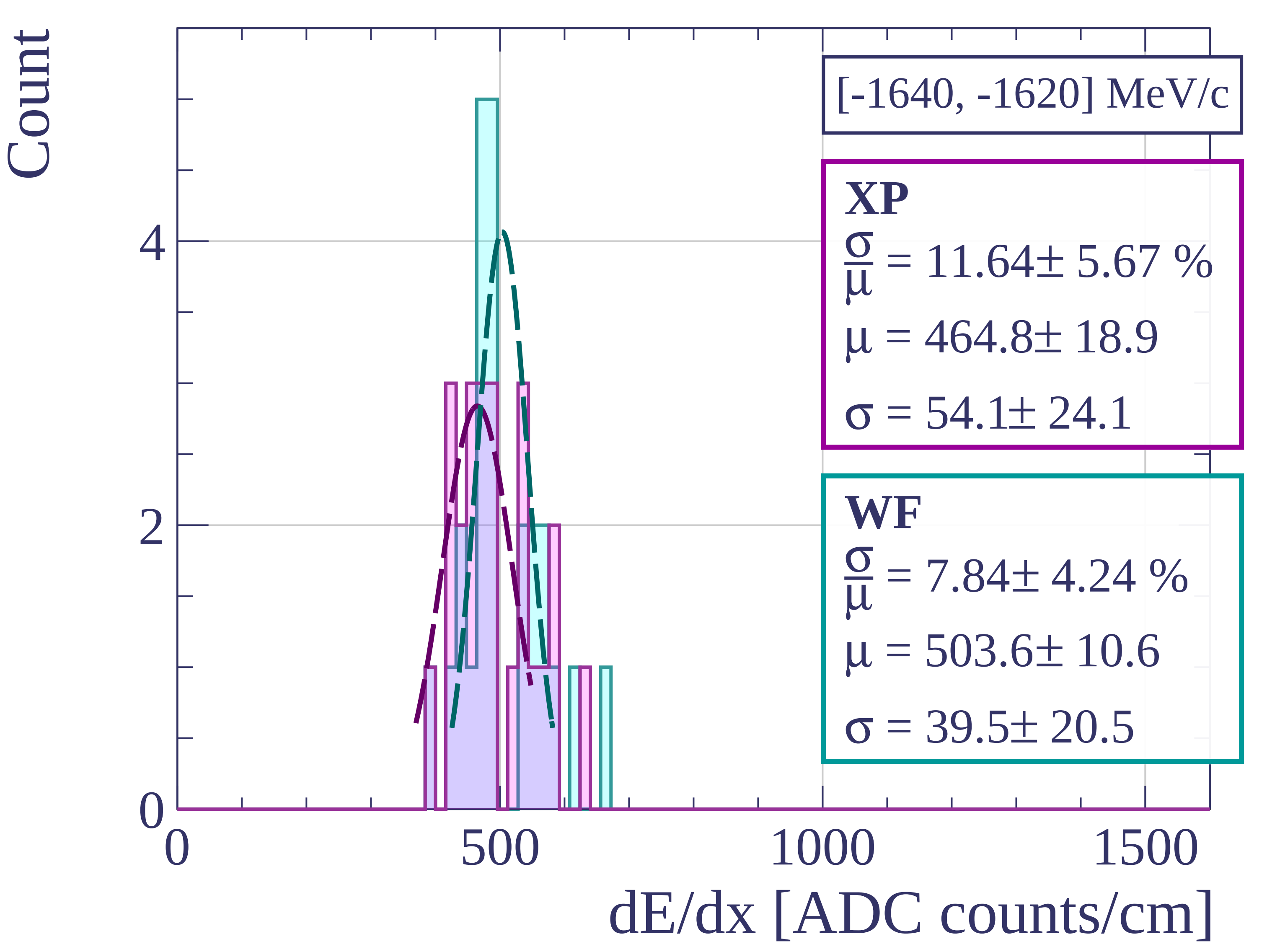

Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1700, -1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.82± 33.90 %
μ = 445.6± 101.0
σ = 92.8± 130.0
WF
$\frac{\sigma}{\mu}$ = 1.58± 8.74 %
μ = 482.5± 90.9
σ = 7.6± 40.7



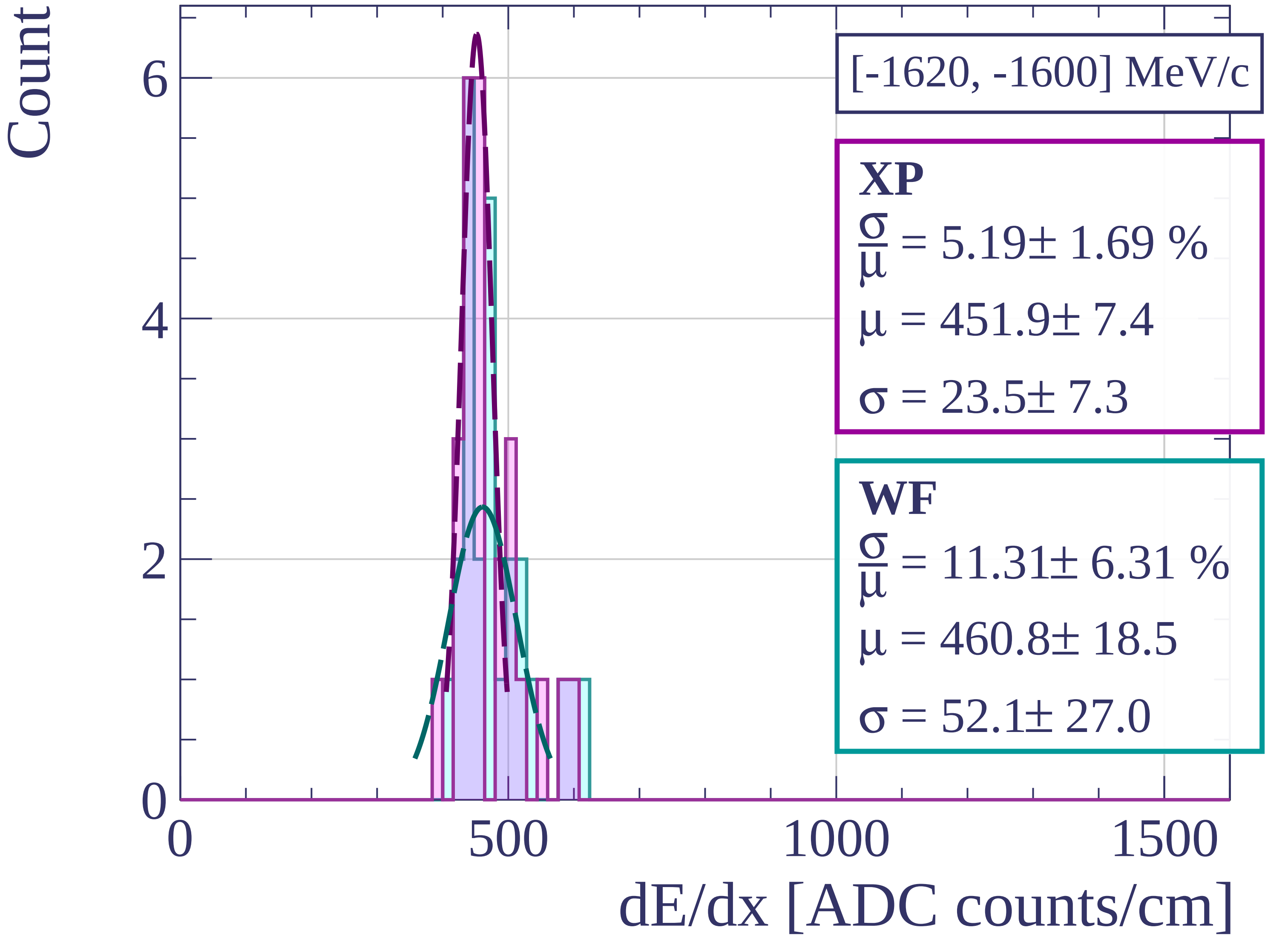
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1680, -1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.12± 2.58 %
μ = 489.0± 9.7
σ = 34.8± 11.9
WF
$\frac{\sigma}{\mu}$ = 24.60± 34.04 %
μ = 446.4± 131.0
σ = 109.8± 119.7



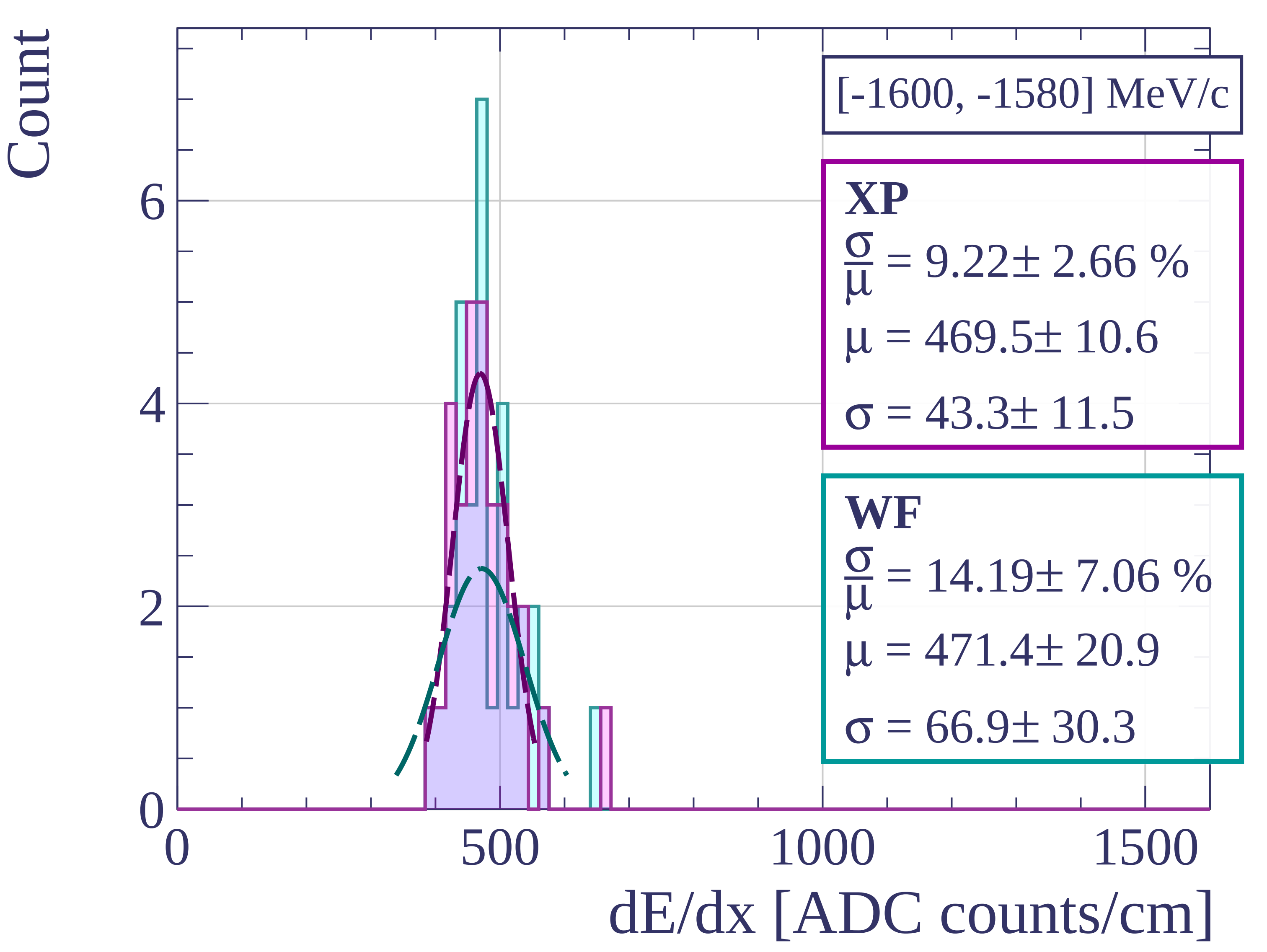
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
1
2
3
4
[-1660, -1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.86± 4.47 %
μ = 460.3± 17.2
σ = 45.4± 18.9
WF
$\frac{\sigma}{\mu}$ = 13.25± 8.27 %
μ = 471.3± 27.3
σ = 62.4± 35.3



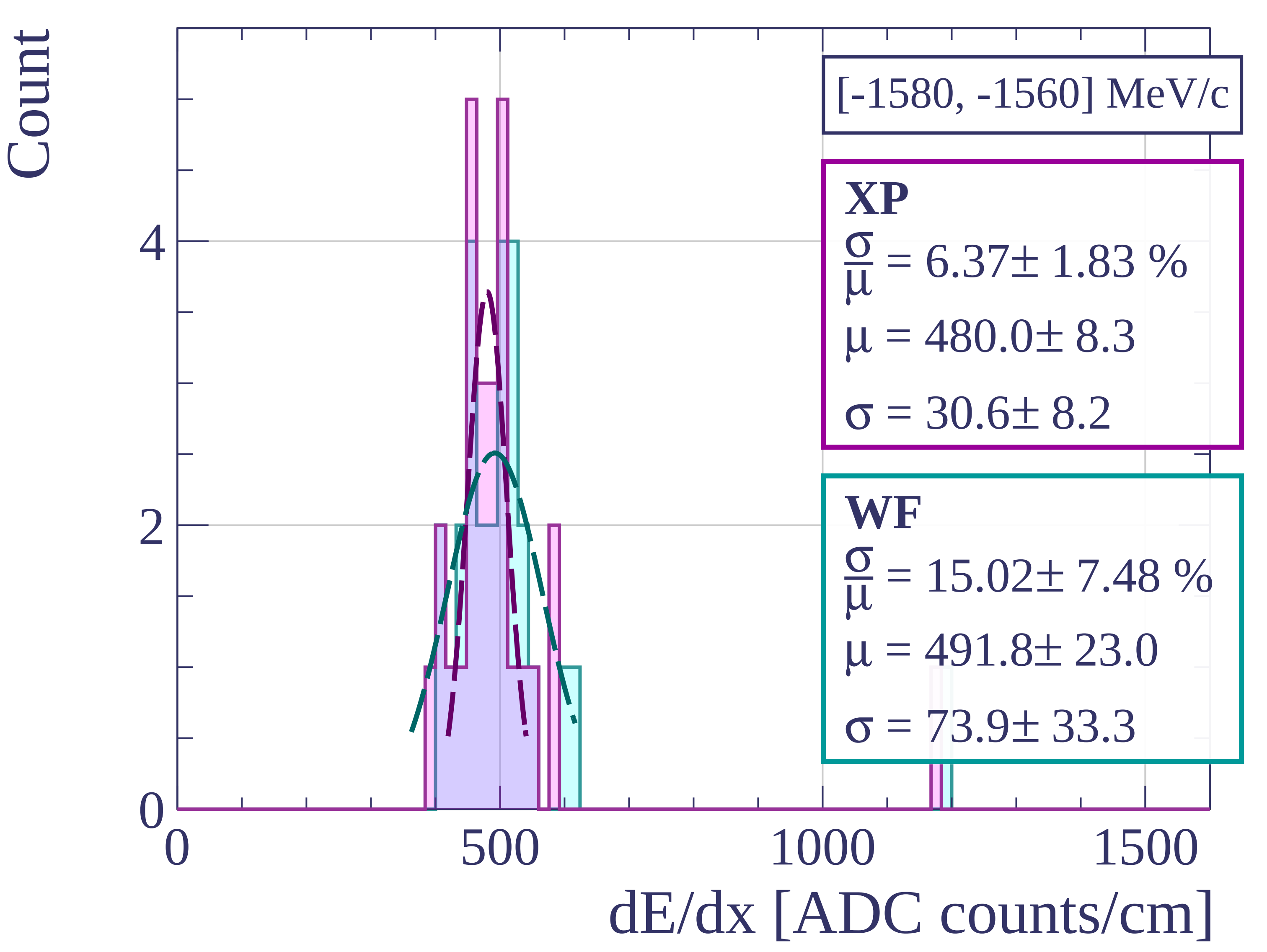
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
[-1640, -1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.64± 5.67 %
μ = 464.8± 18.9
σ = 54.1± 24.1
WF
$\frac{\sigma}{\mu}$ = 7.84± 4.24 %
μ = 503.6± 10.6
σ = 39.5± 20.5



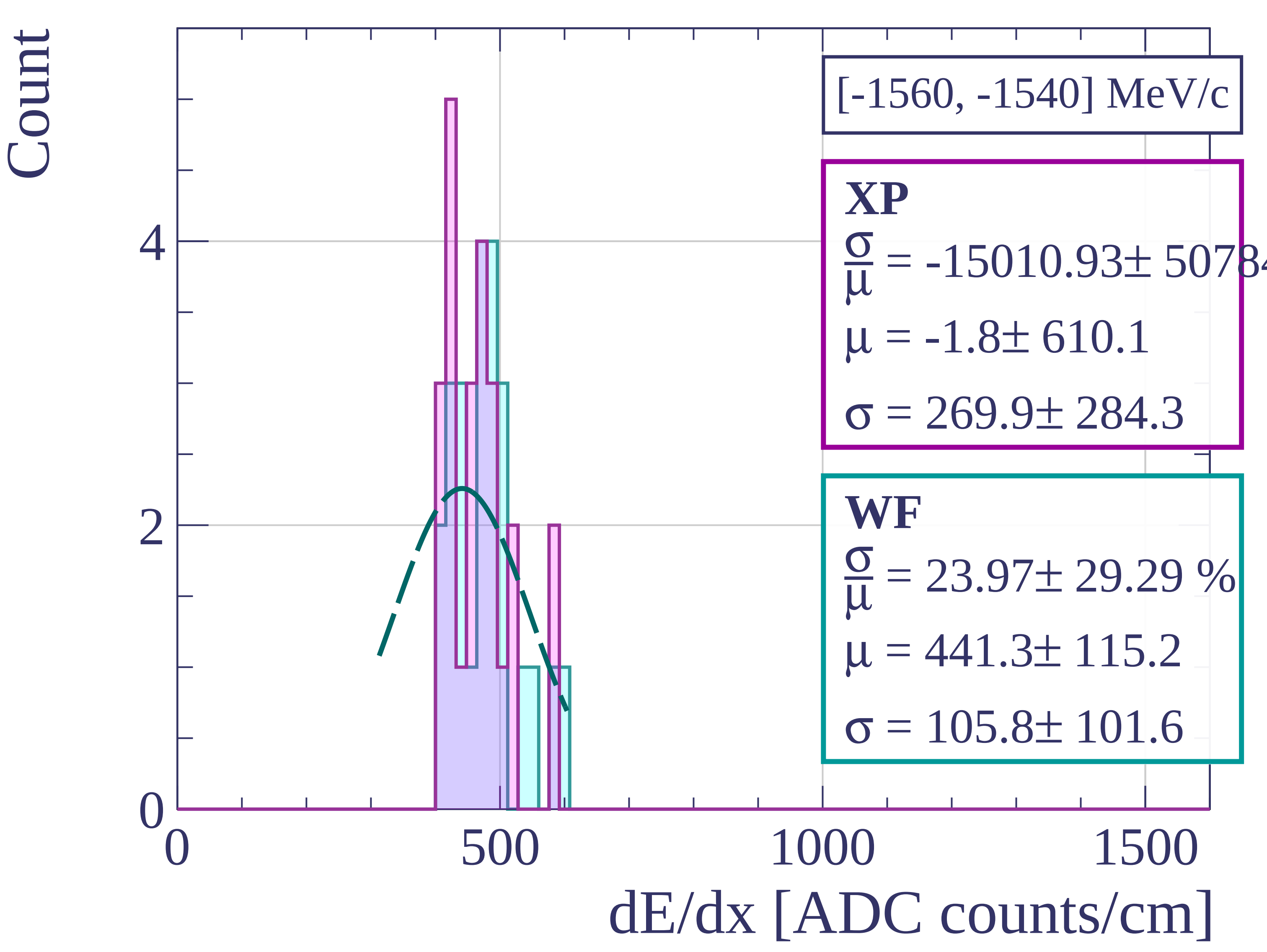
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1620, -1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.19± 1.69 %
μ = 451.9± 7.4
σ = 23.5± 7.3
WF
$\frac{\sigma}{\mu}$ = 11.31± 6.31 %
μ = 460.8± 18.5
σ = 52.1± 27.0



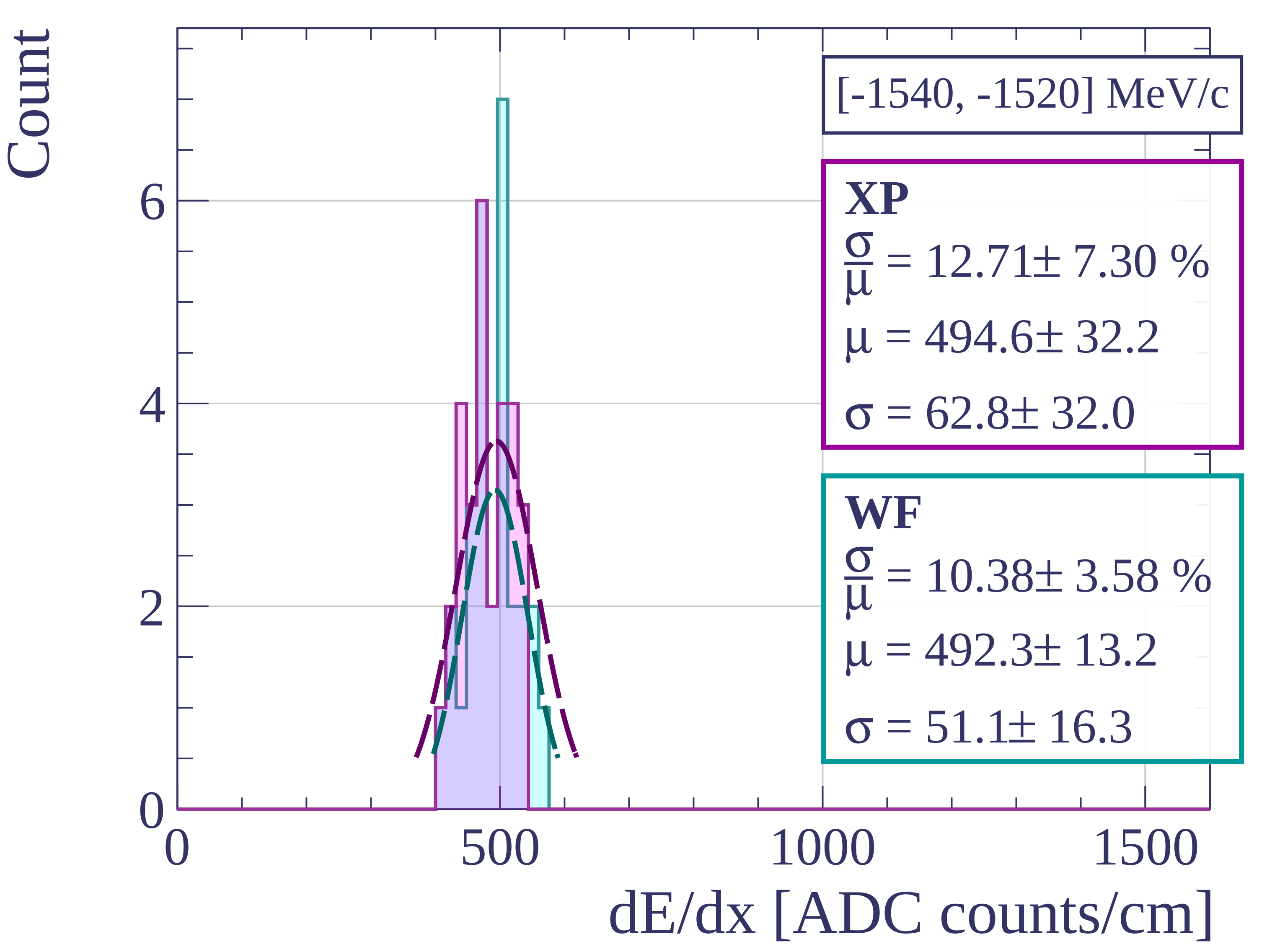
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1600, -1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.22± 2.66 %
μ = 469.5± 10.6
σ = 43.3± 11.5
WF
$\frac{\sigma}{\mu}$ = 14.19± 7.06 %
μ = 471.4± 20.9
σ = 66.9± 30.3



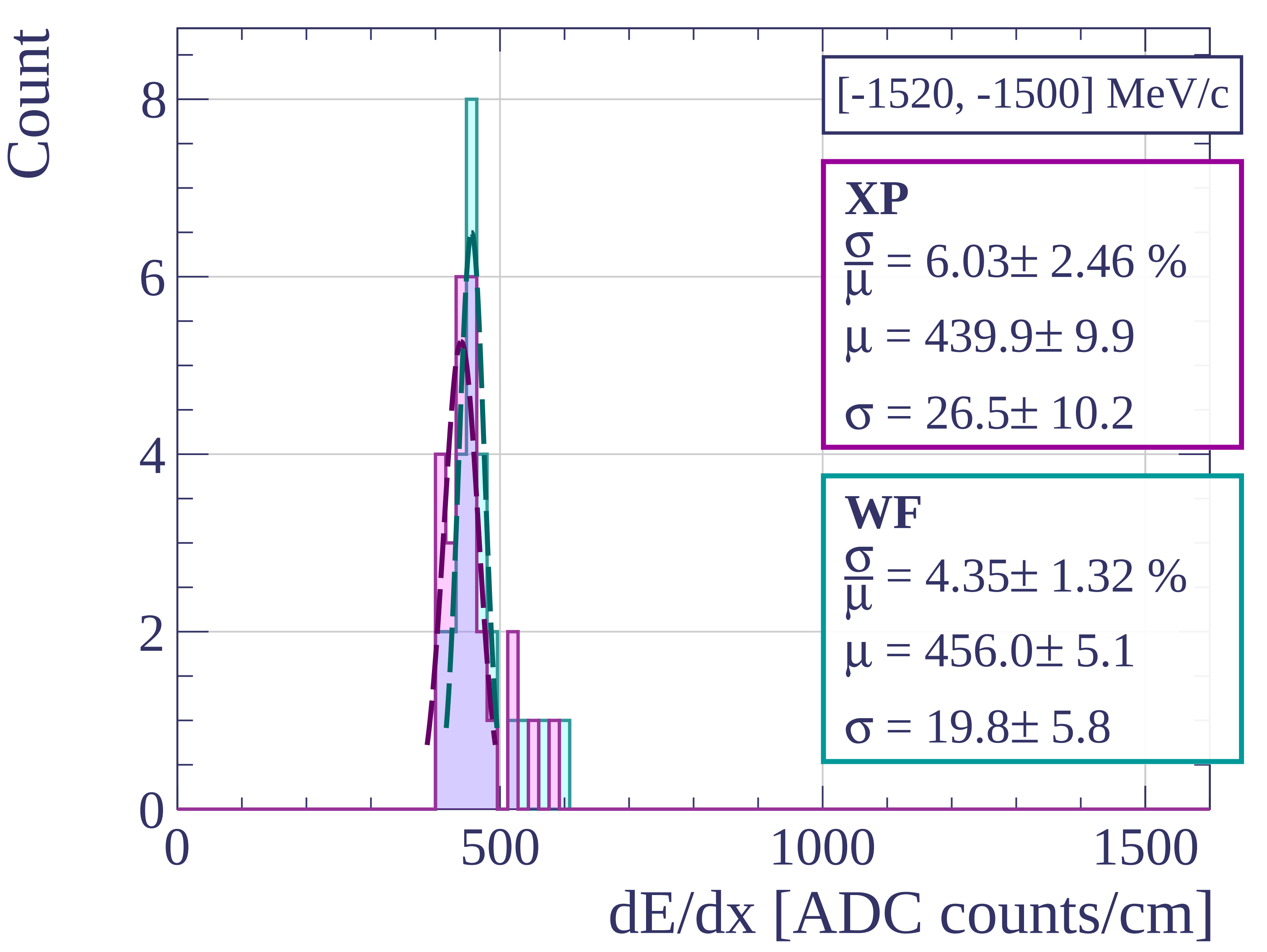
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1580, -1560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.37± 1.83 %
μ = 480.0± 8.3
σ = 30.6± 8.2
WF
$\frac{\sigma}{\mu}$ = 15.02± 7.48 %
μ = 491.8± 23.0
σ = 73.9± 33.3



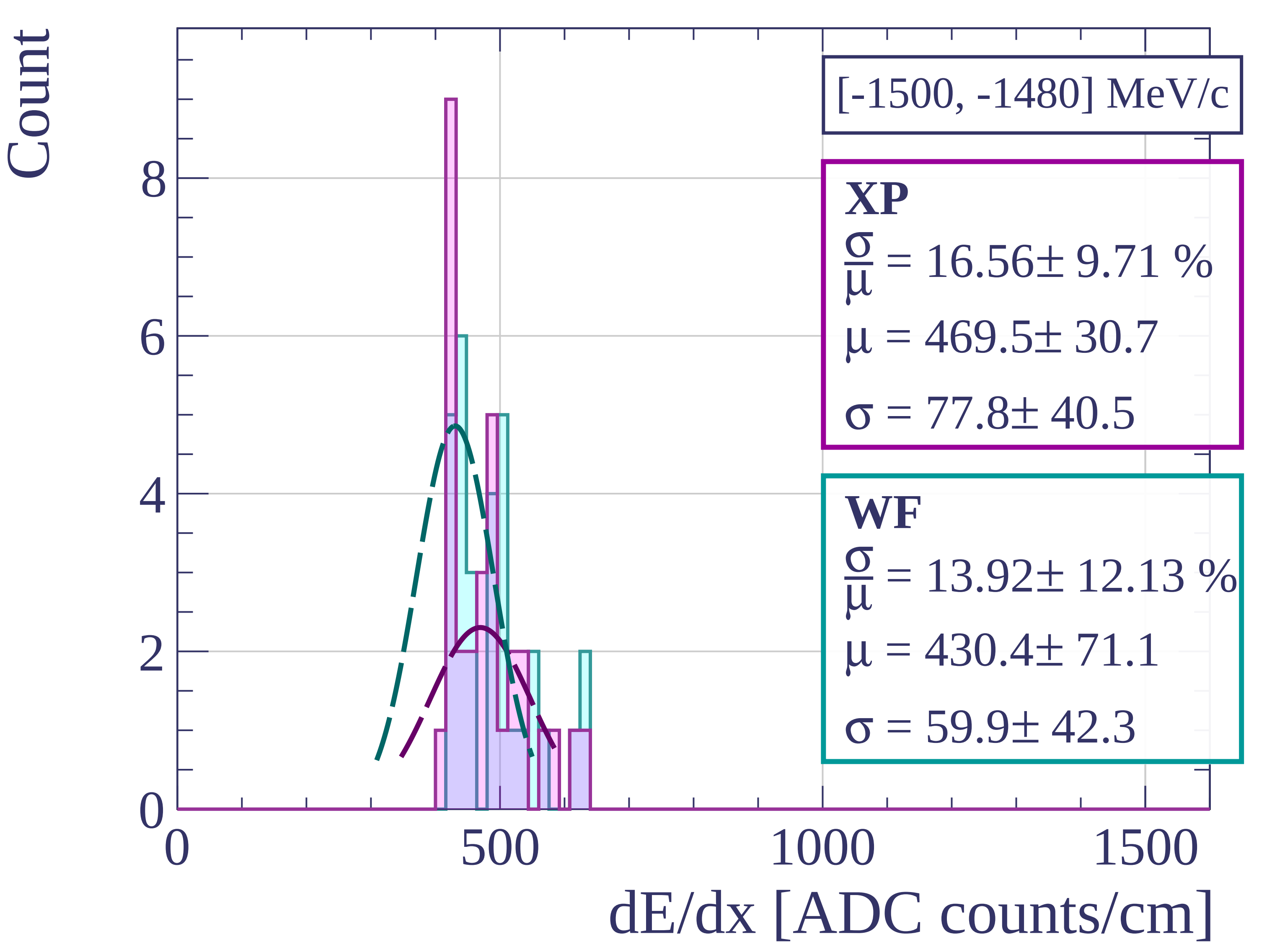
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1560, -1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = -15010.93± 5078
μ = -1.8± 610.1
σ = 269.9± 284.3
WF
$\frac{\sigma}{\mu}$ = 23.97± 29.29 %
μ = 441.3± 115.2
σ = 105.8± 101.6

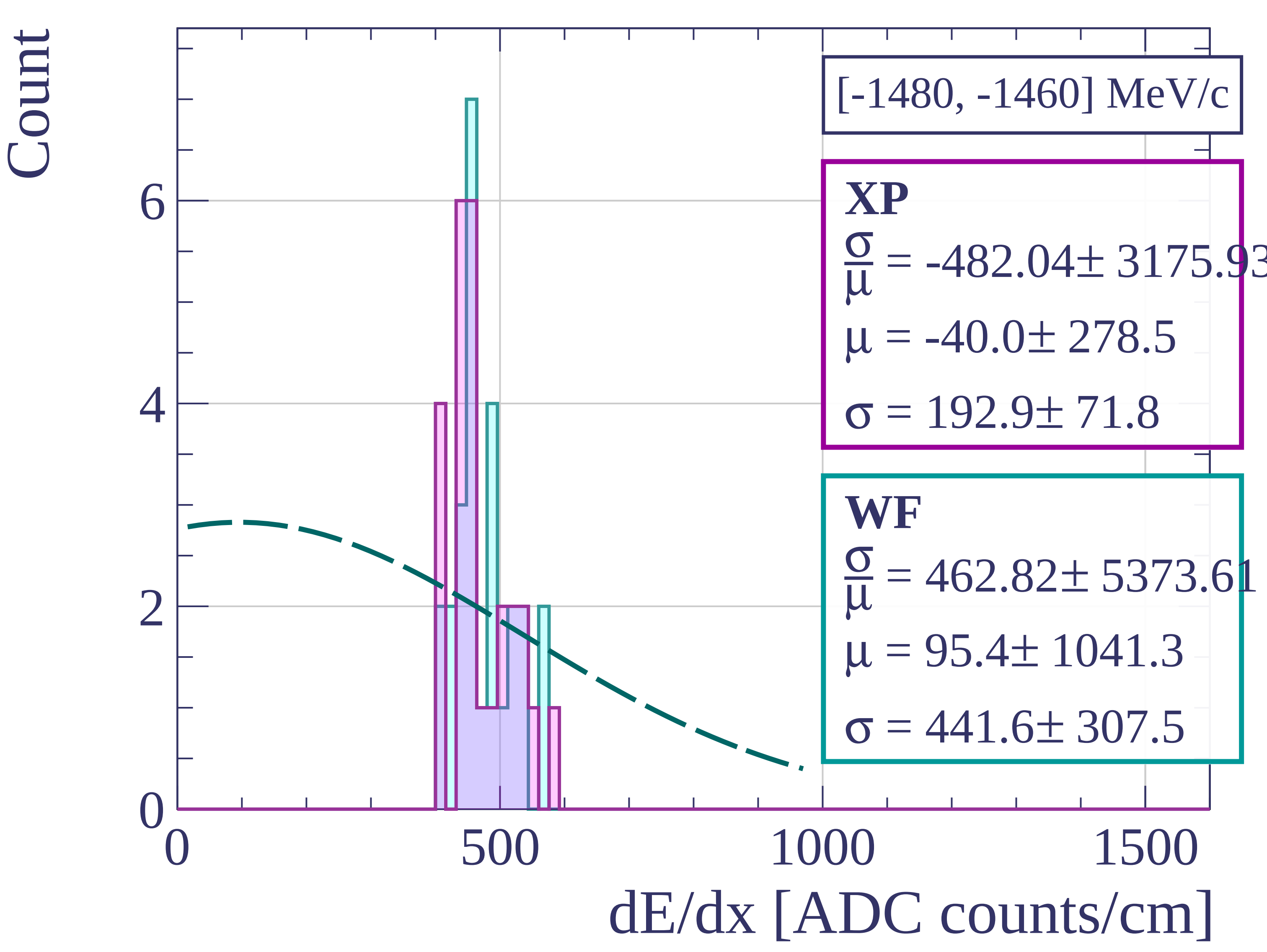


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.71± 7.30 %
μ = 494.6± 32.2
σ = 62.8± 32.0
WF
$\frac{\sigma}{\mu}$ = 10.38± 3.58 %
μ = 492.3± 13.2
σ = 51.1± 16.3

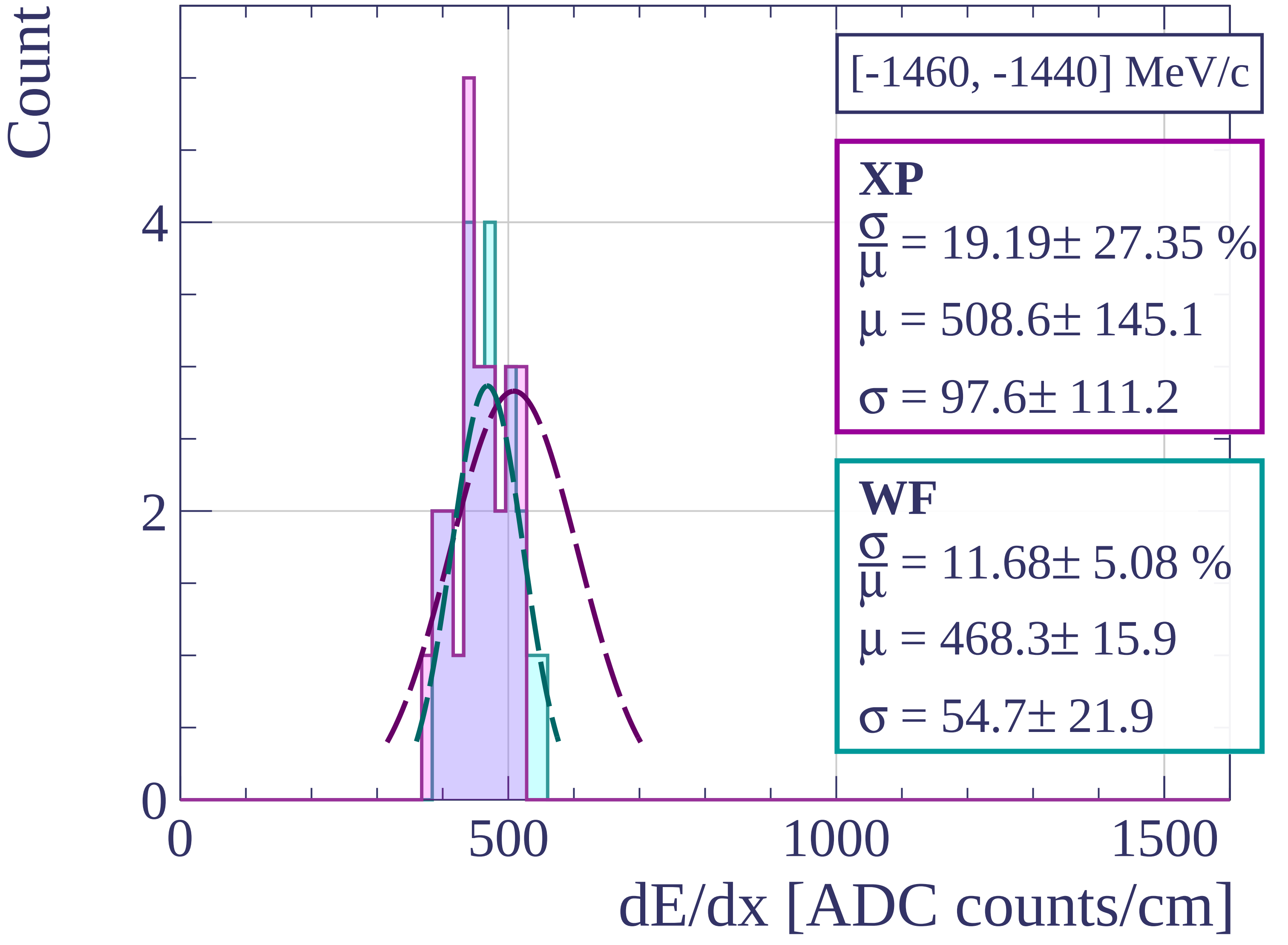


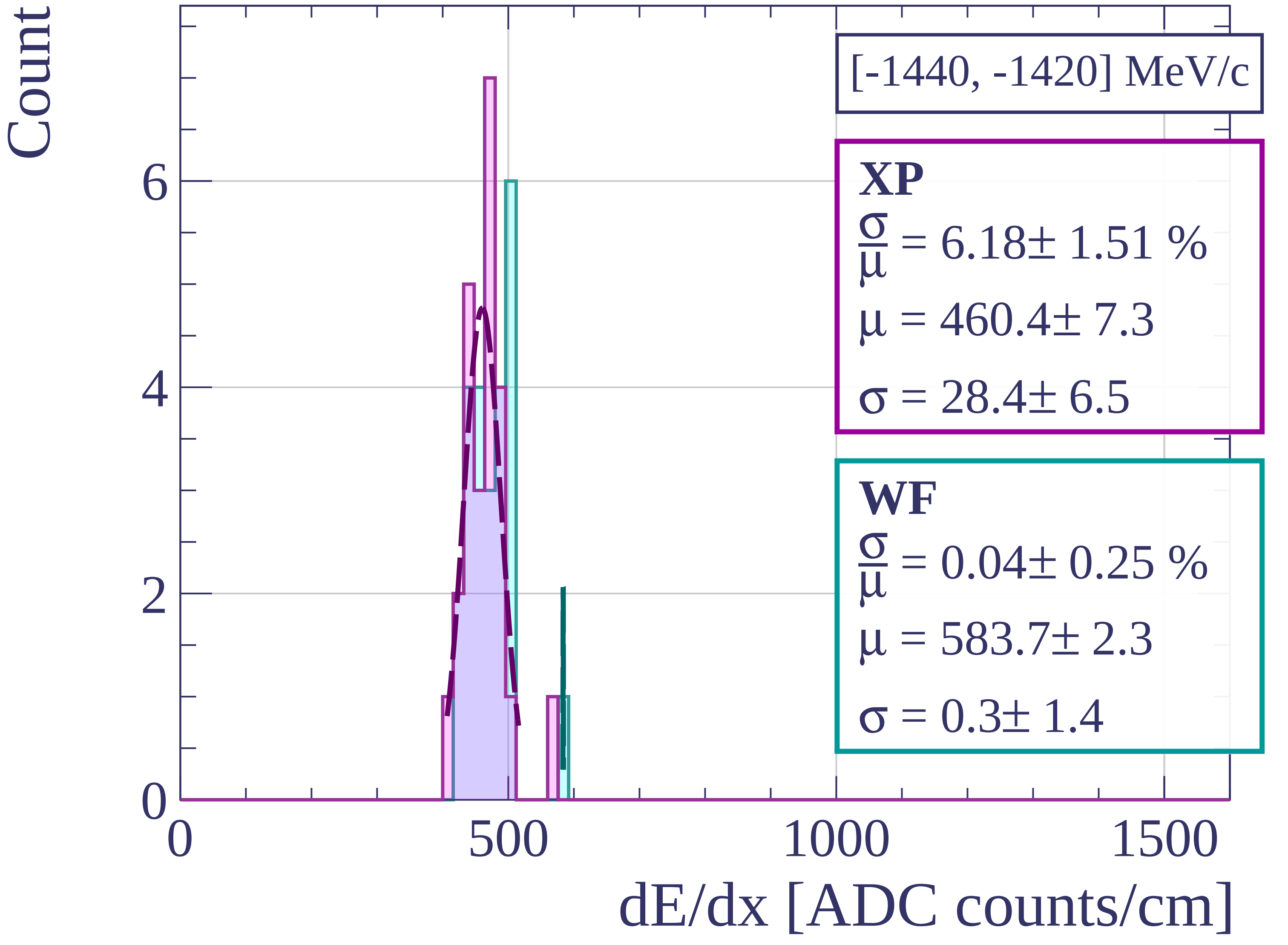
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.03± 2.46 %
μ = 439.9± 9.9
σ = 26.5± 10.2
WF
$\frac{\sigma}{\mu}$ = 4.35± 1.32 %
μ = 456.0± 5.1
σ = 19.8± 5.8


Count
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 16.56± 9.71 %
μ = 469.5± 30.7
σ = 77.8± 40.5
WF
$\frac{\sigma}{\mu}$ = 13.92± 12.13 %
μ = 430.4± 71.1
σ = 59.9± 42.3
0
2
4
6
8
0
500
1000
1500

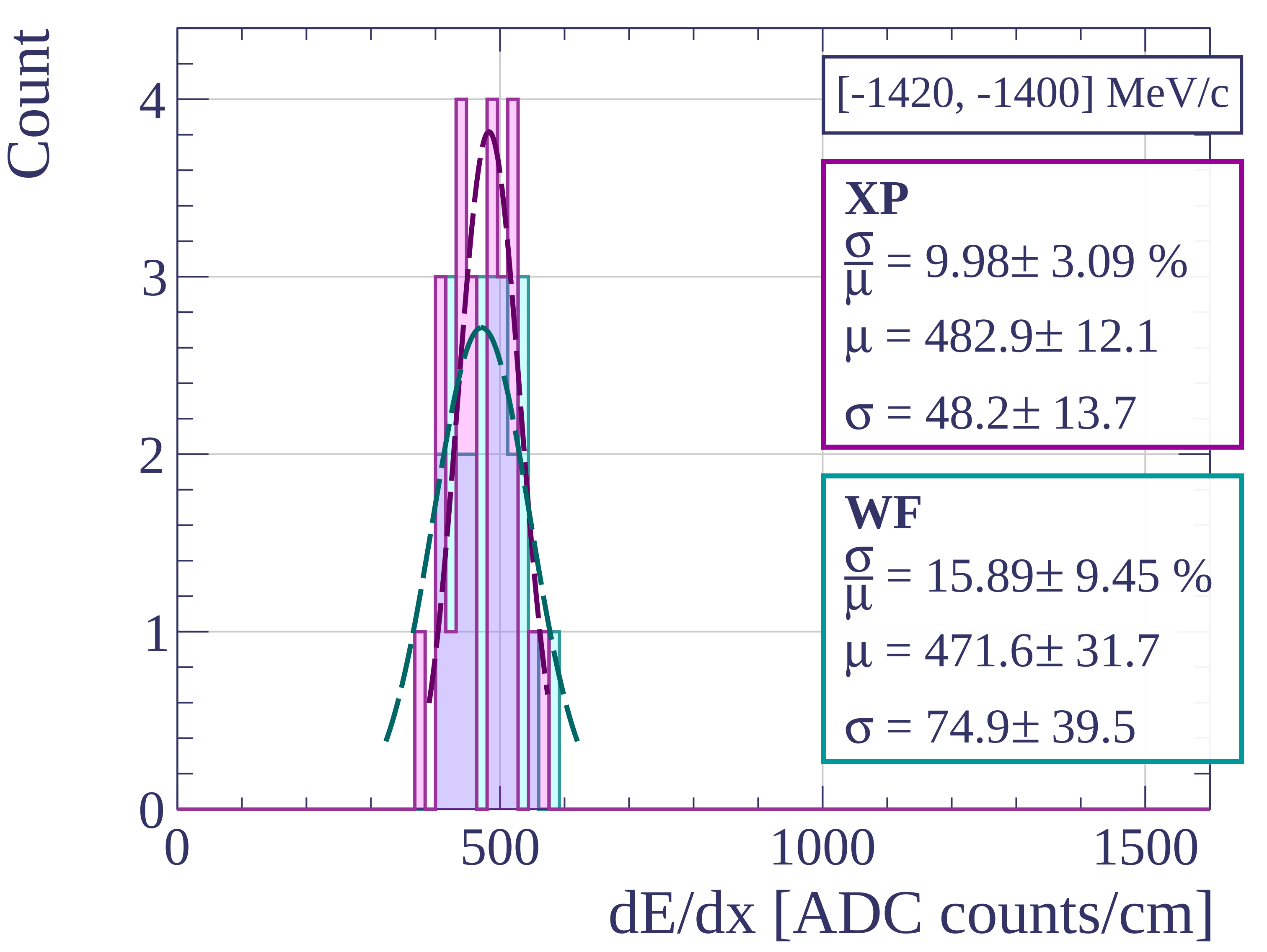


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1480, -1460] MeV/c
XP
σ/μ = -482.04± 3175.93
μ = -40.0± 278.5
σ = 192.9± 71.8
WF
σ/μ = 462.82± 5373.61
μ = 95.4± 1041.3
σ = 441.6± 307.5

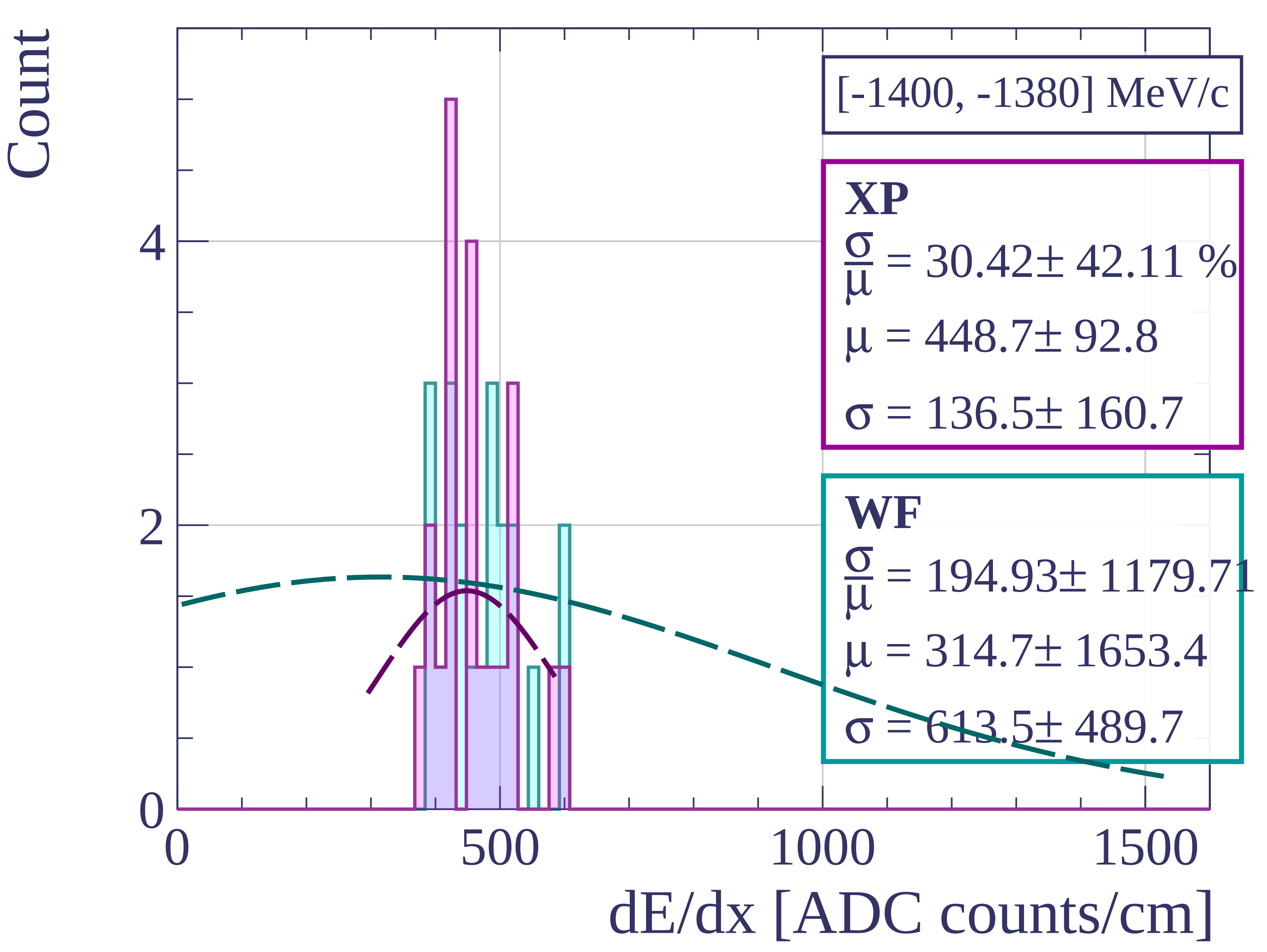

Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1460, -1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 19.19± 27.35 %
μ = 508.6± 145.1
σ = 97.6± 111.2
WF
$\frac{\sigma}{\mu}$ = 11.68± 5.08 %
μ = 468.3± 15.9
σ = 54.7± 21.9



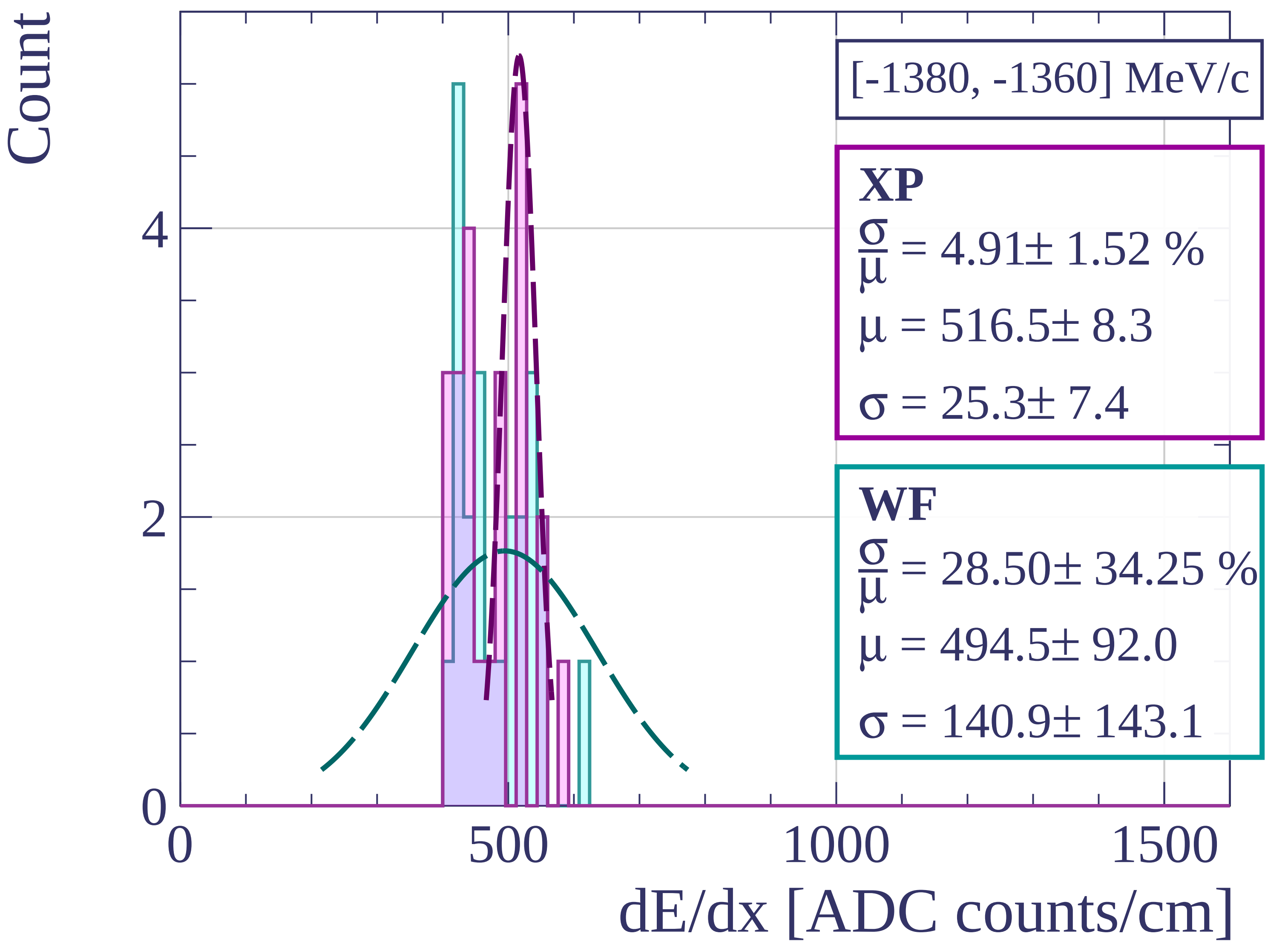
Count
[-1440, -1420] MeV/c
XP
σ/μ = 6.18± 1.51 %
μ = 460.4± 7.3
σ = 28.4± 6.5
WF
σ/μ = 0.04± 0.25 %
μ = 583.7± 2.3
σ = 0.3± 1.4
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



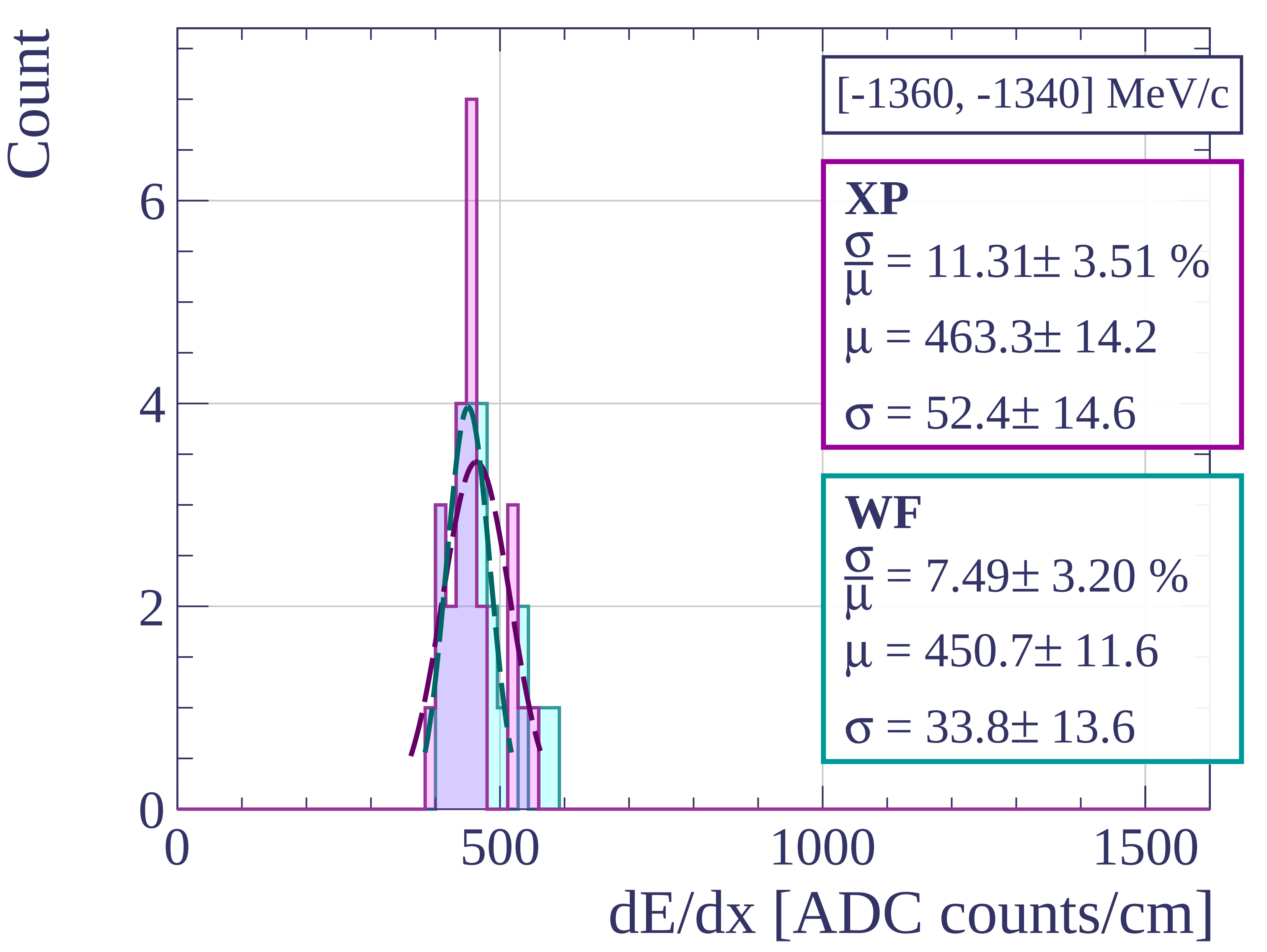
Count
4
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1420, -1400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.98± 3.09 %
μ = 482.9± 12.1
σ = 48.2± 13.7
WF
$\frac{\sigma}{\mu}$ = 15.89± 9.45 %
μ = 471.6± 31.7
σ = 74.9± 39.5



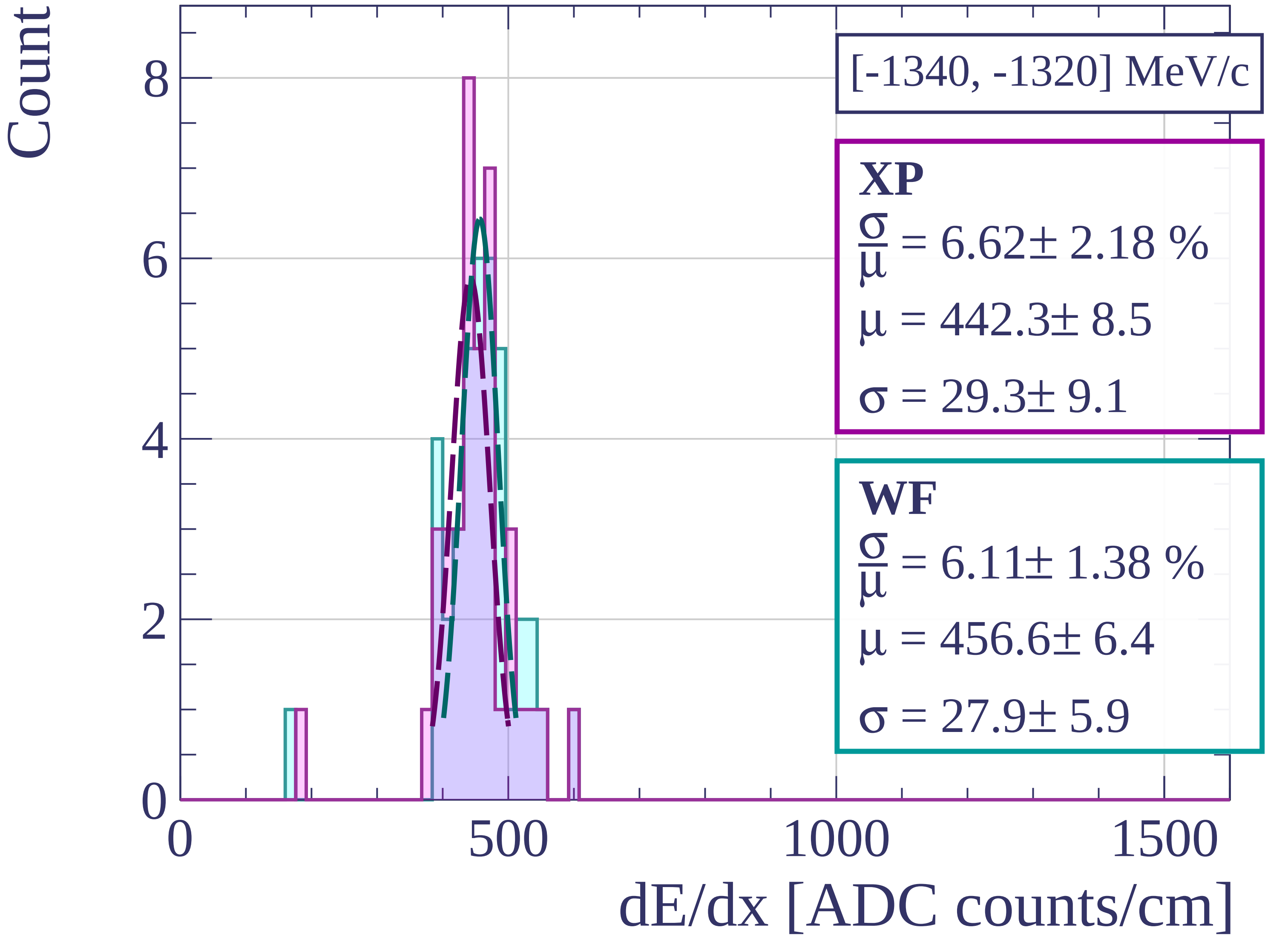
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
σ/μ = 30.42± 42.11 %
μ = 448.7± 92.8
σ = 136.5± 160.7
WF
σ/μ = 194.93± 1179.71
μ = 314.7± 1653.4
σ = 613.5± 489.7



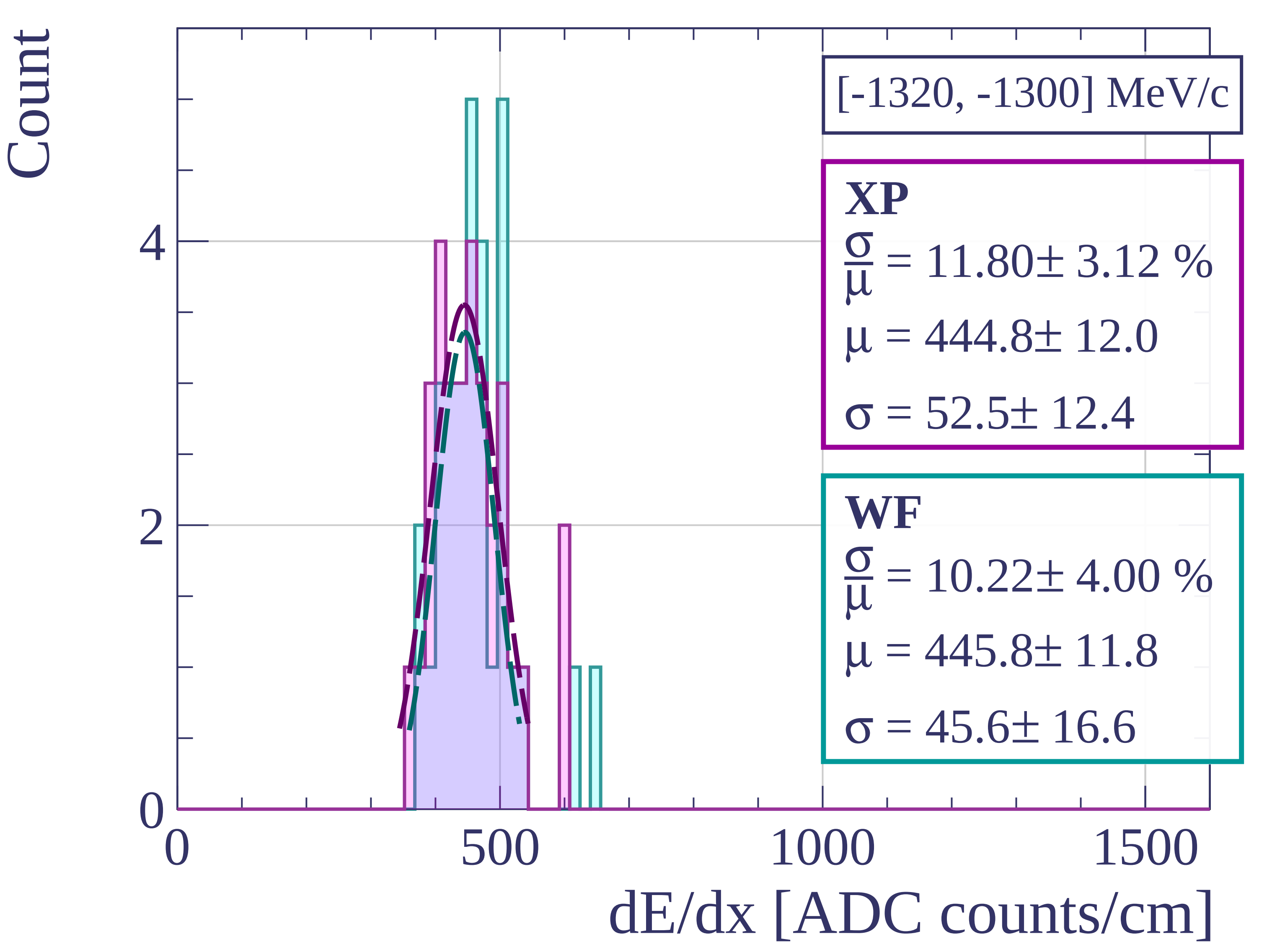
Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1380, -1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.91± 1.52 %
μ = 516.5± 8.3
σ = 25.3± 7.4
WF
$\frac{\sigma}{\mu}$ = 28.50± 34.25 %
μ = 494.5± 92.0
σ = 140.9± 143.1



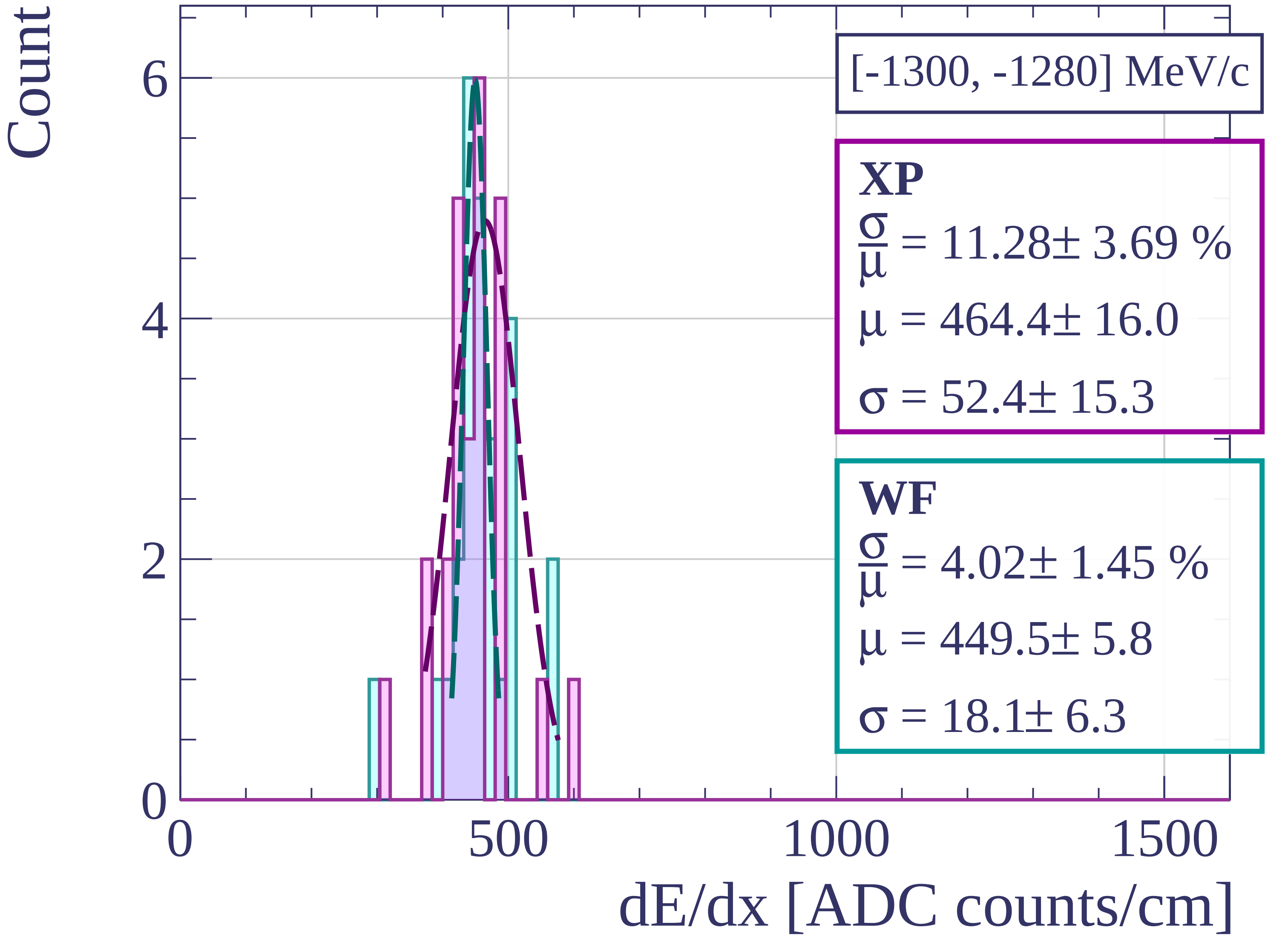
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1360, -1340] MeV/c
XP
σ/μ = 11.31± 3.51 %
μ = 463.3± 14.2
σ = 52.4± 14.6
WF
σ/μ = 7.49± 3.20 %
μ = 450.7± 11.6
σ = 33.8± 13.6



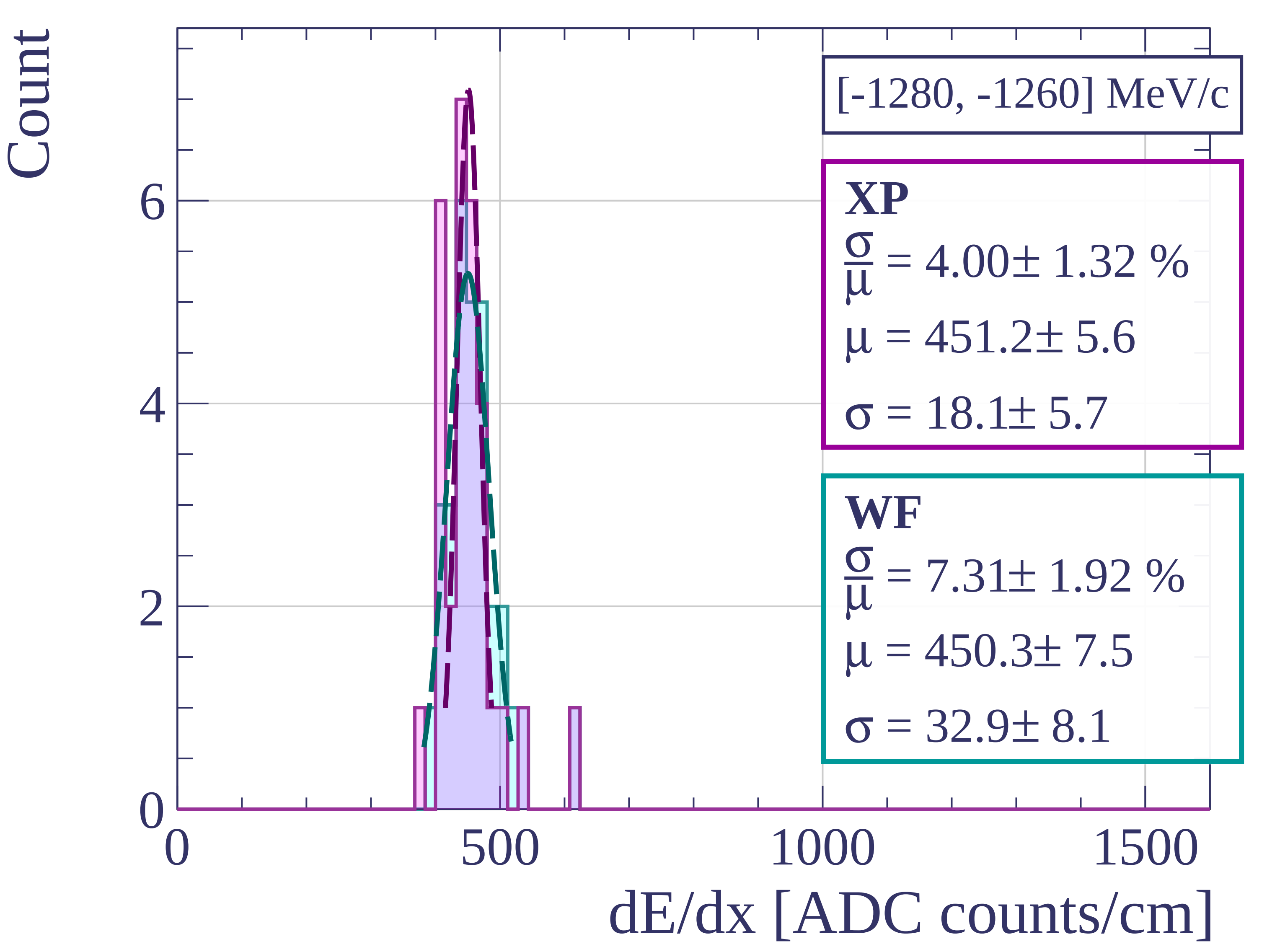
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
σ/μ = 6.62± 2.18 %
μ = 442.3± 8.5
σ = 29.3± 9.1
WF
σ/μ = 6.11± 1.38 %
μ = 456.6± 6.4
σ = 27.9± 5.9



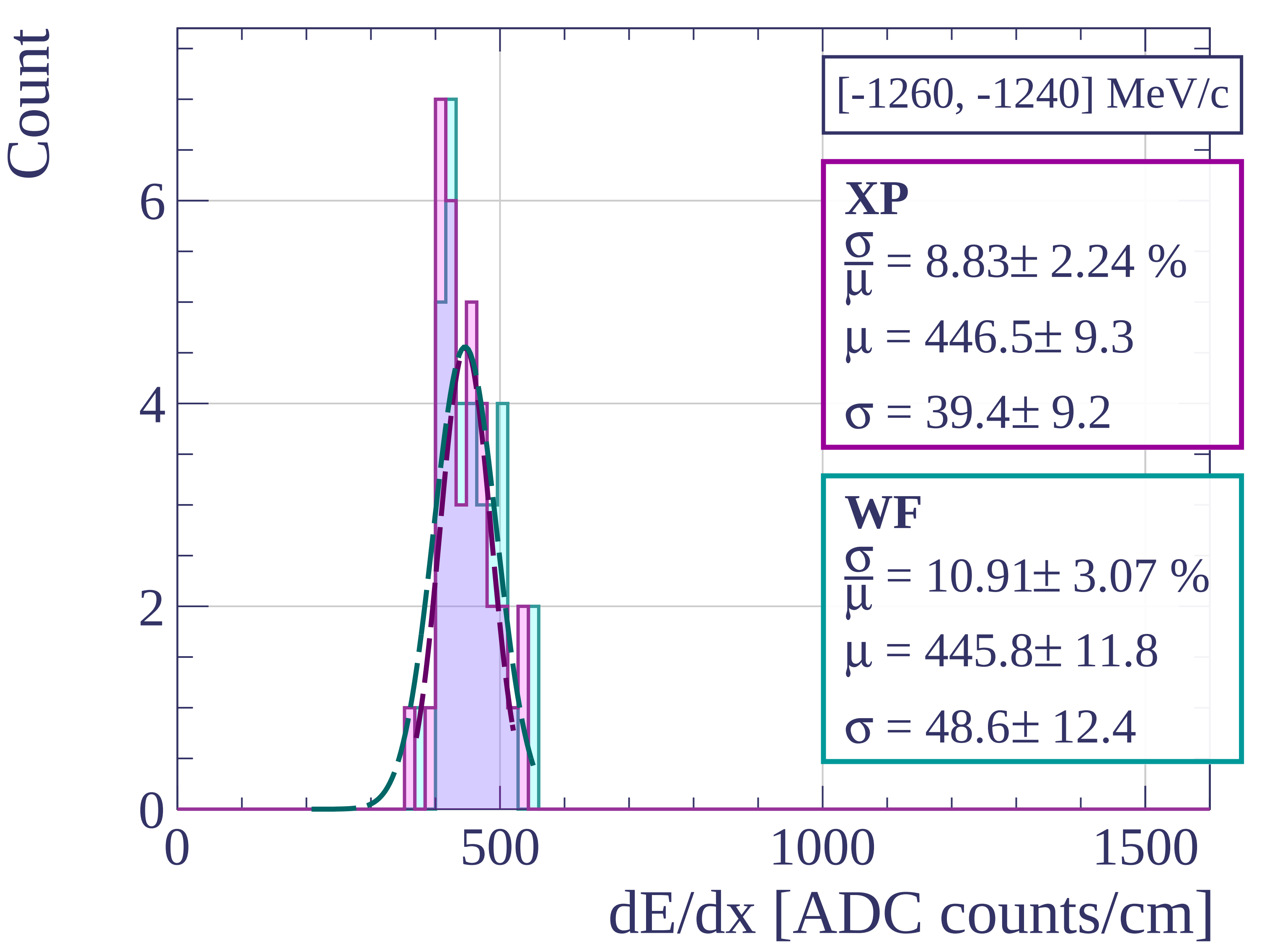
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
σ/μ = 11.80± 3.12 %
μ = 444.8± 12.0
σ = 52.5± 12.4
WF
σ/μ = 10.22± 4.00 %
μ = 445.8± 11.8
σ = 45.6± 16.6

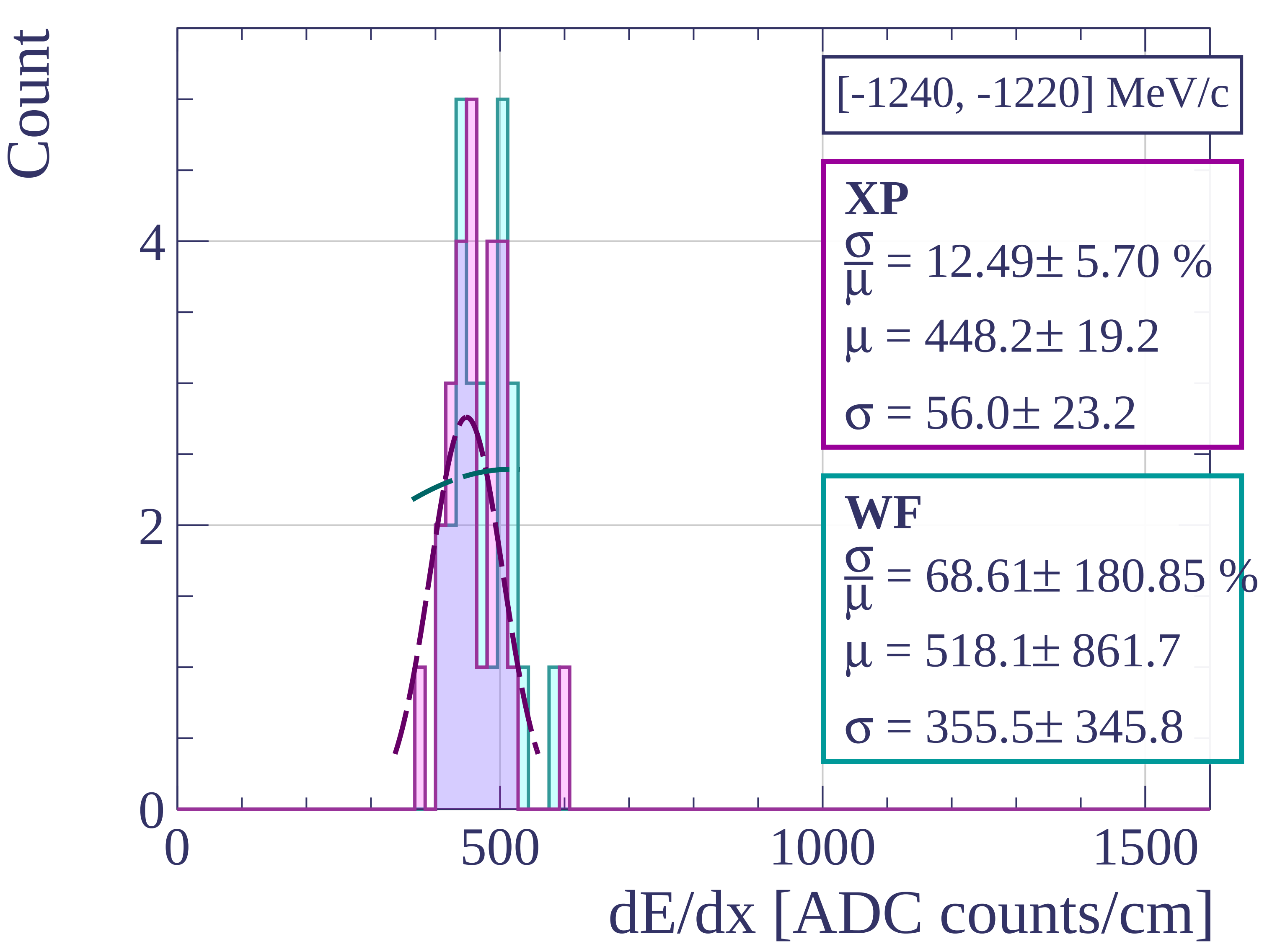


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1300, -1280] MeV/c
XP
σ/μ = 11.28± 3.69 %
μ = 464.4± 16.0
σ = 52.4± 15.3
WF
σ/μ = 4.02± 1.45 %
μ = 449.5± 5.8
σ = 18.1± 6.3

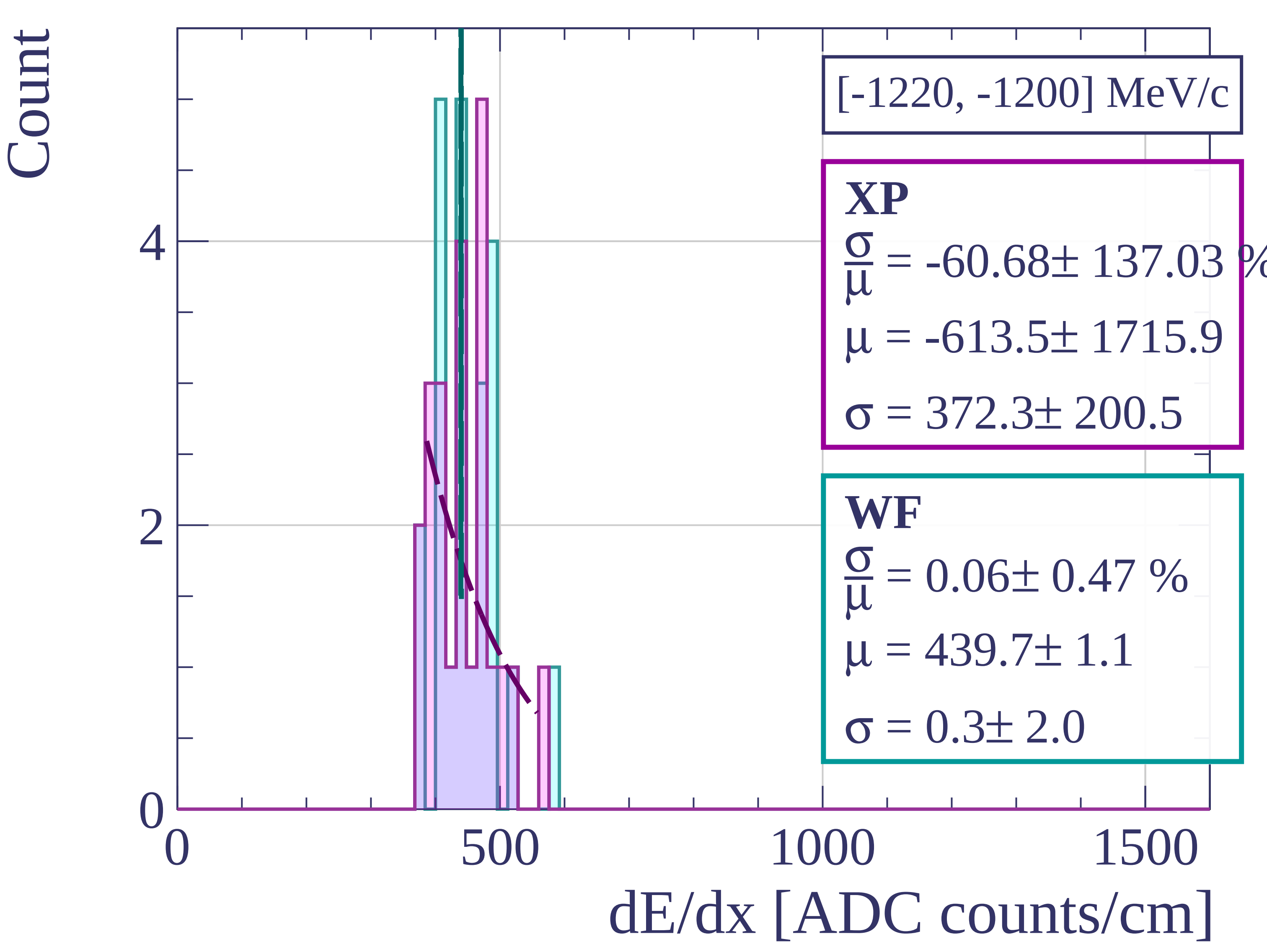


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1280, -1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.00± 1.32 %
μ = 451.2± 5.6
σ = 18.1± 5.7
WF
$\frac{\sigma}{\mu}$ = 7.31± 1.92 %
μ = 450.3± 7.5
σ = 32.9± 8.1

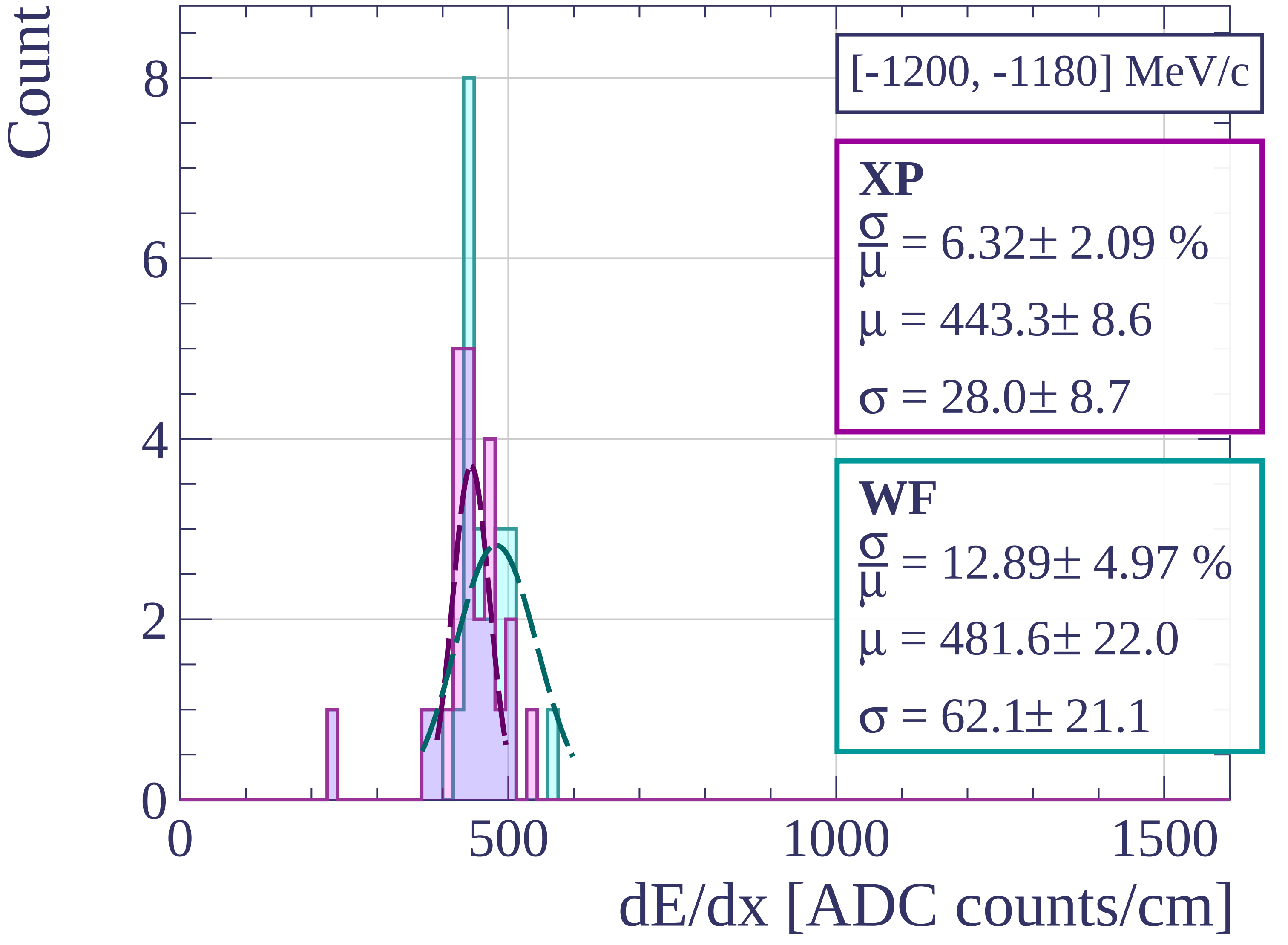

[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.83± 2.24 %
μ = 446.5± 9.3
σ = 39.4± 9.2
WF
$\frac{\sigma}{\mu}$ = 10.91± 3.07 %
μ = 445.8± 11.8
σ = 48.6± 12.4
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500



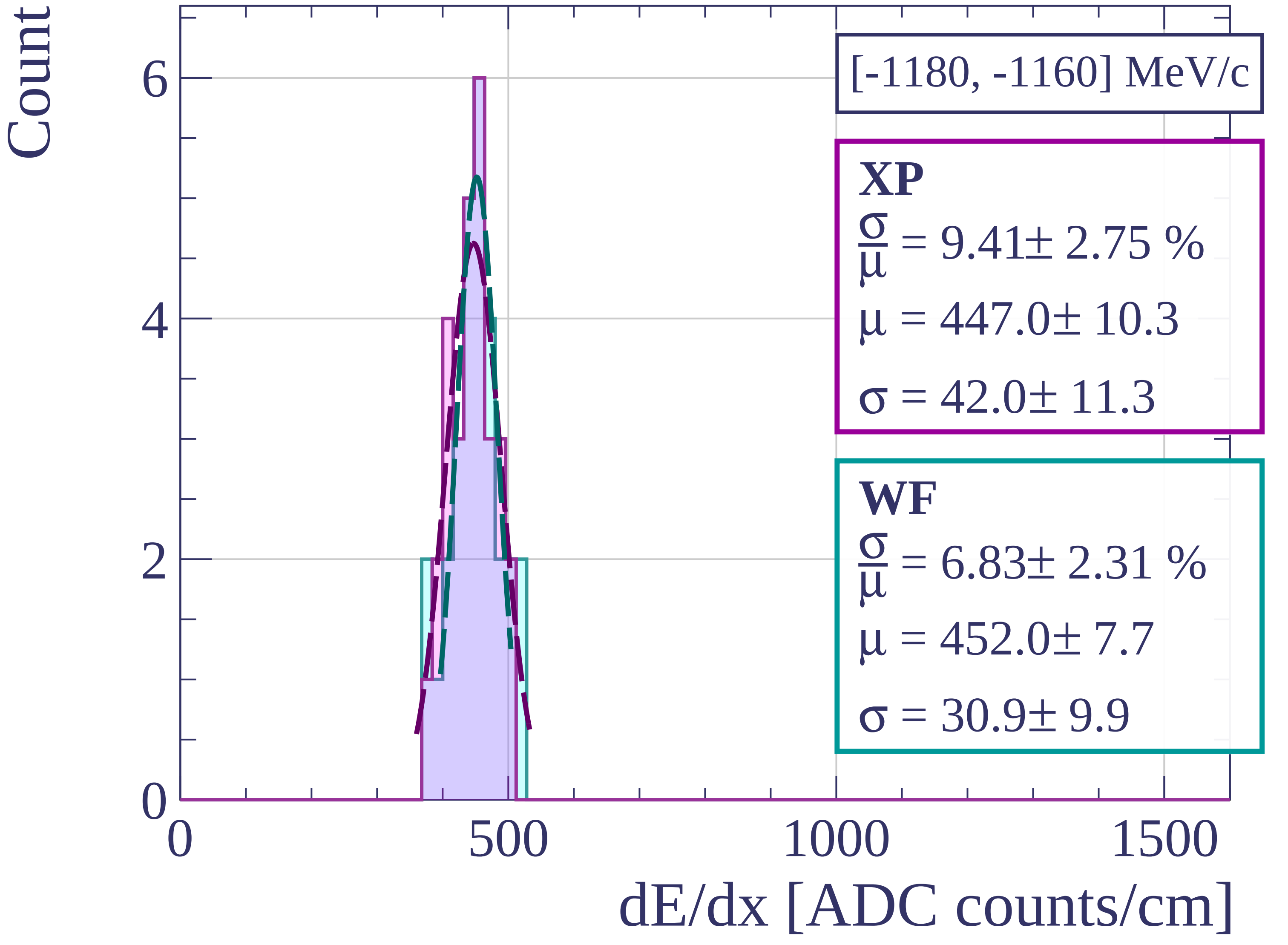
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
σ/μ = 12.49± 5.70 %
μ = 448.2± 19.2
σ = 56.0± 23.2
WF
σ/μ = 68.61± 180.85 %
μ = 518.1± 861.7
σ = 355.5± 345.8



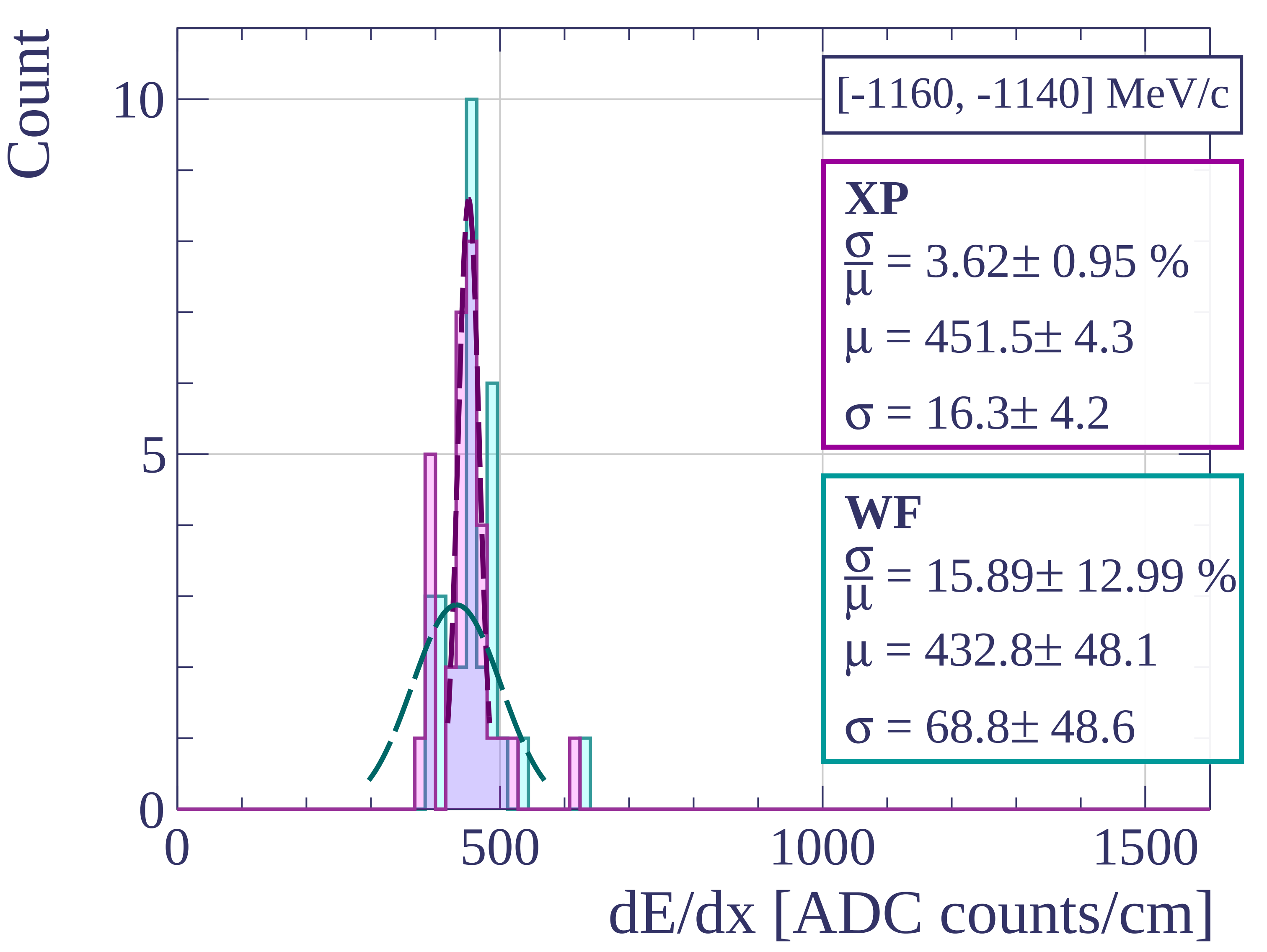
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1220, -1200] MeV/c
XP
σ/μ = -60.68± 137.03 %
μ = -613.5± 1715.9
σ = 372.3± 200.5
WF
σ/μ = 0.06± 0.47 %
μ = 439.7± 1.1
σ = 0.3± 2.0



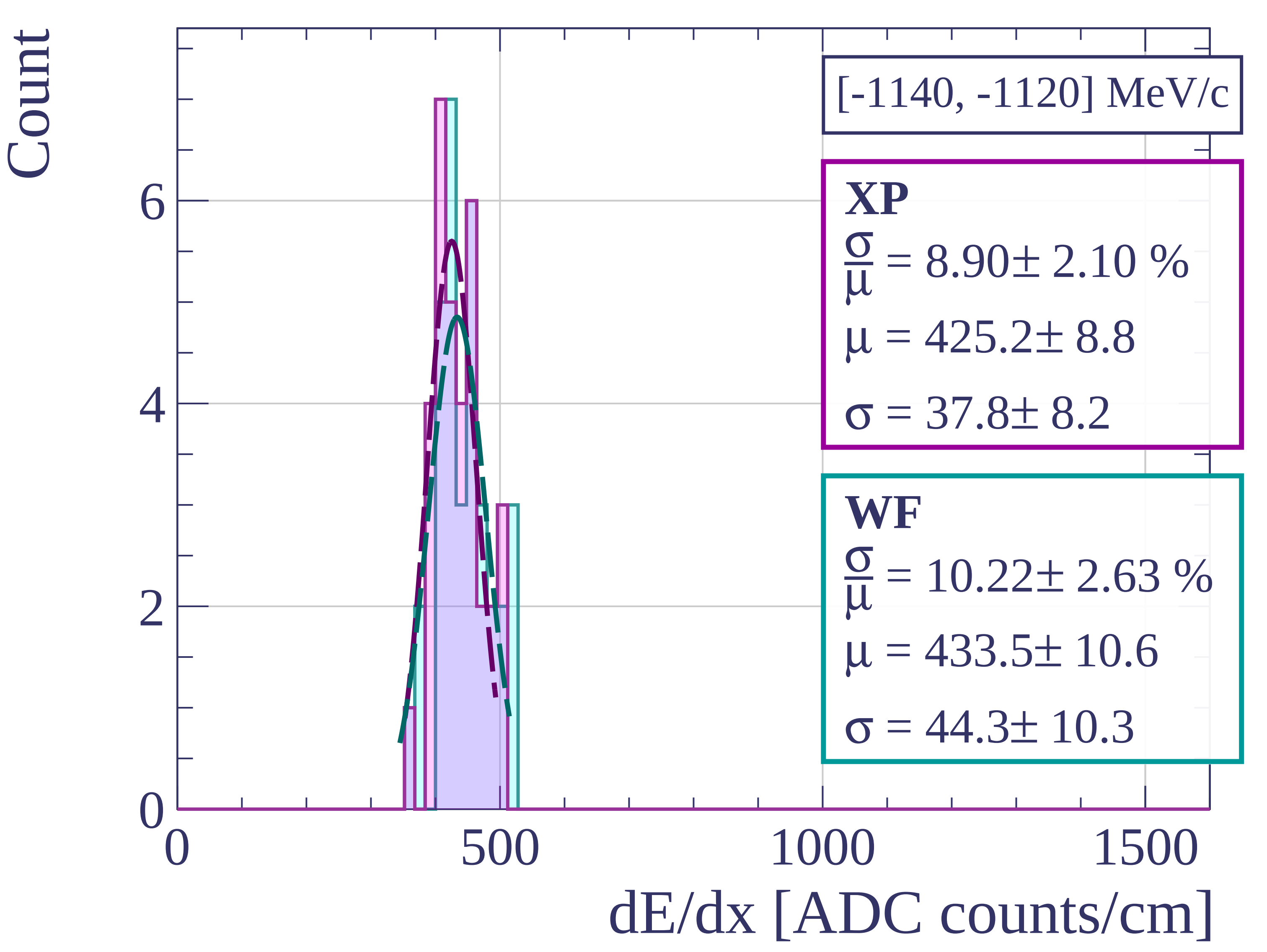
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.32± 2.09 %
μ = 443.3± 8.6
σ = 28.0± 8.7
WF
$\frac{\sigma}{\mu}$ = 12.89± 4.97 %
μ = 481.6± 22.0
σ = 62.1± 21.1



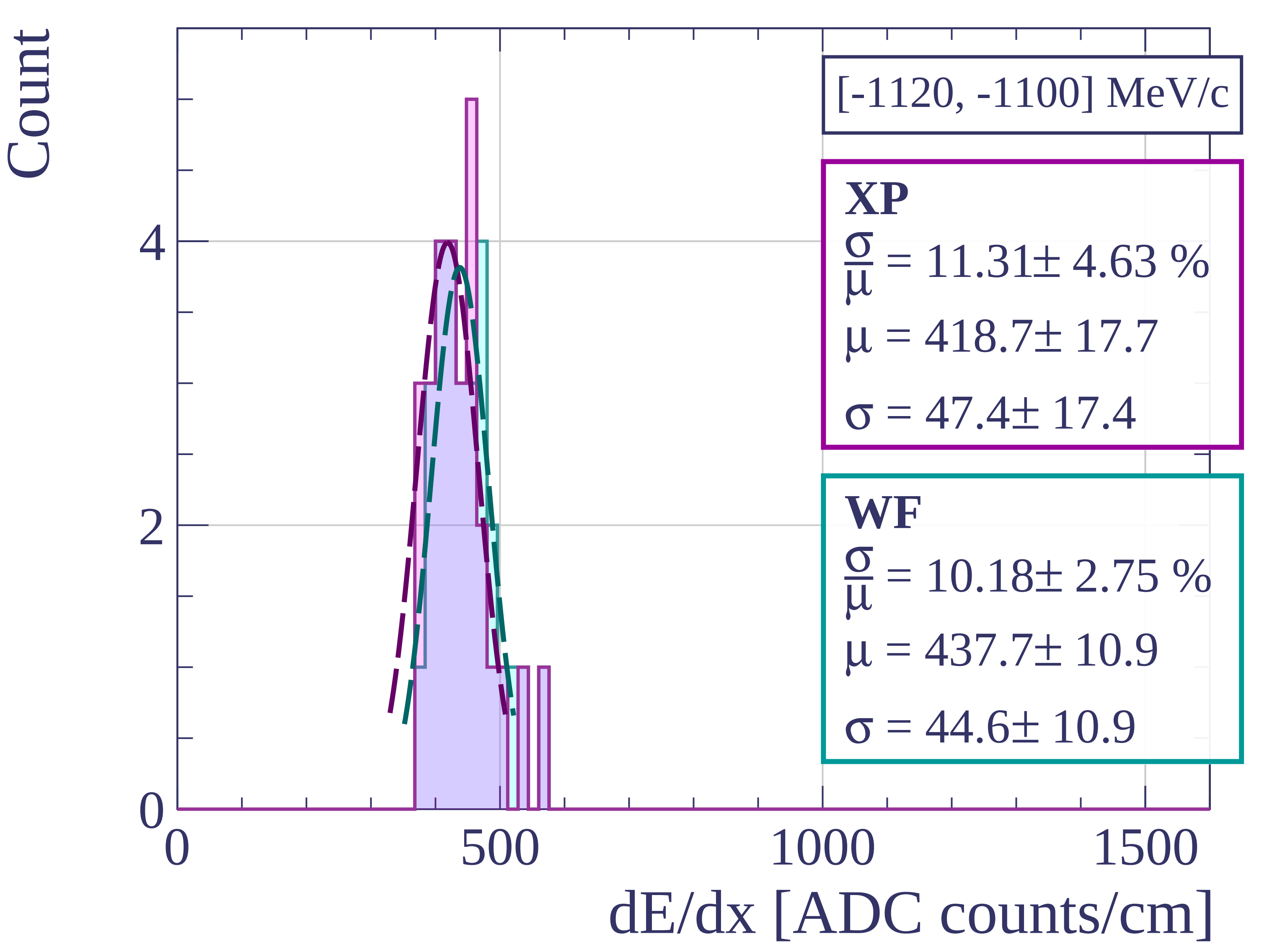
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1180, -1160] MeV/c
XP
σ/μ = 9.41± 2.75 %
μ = 447.0± 10.3
σ = 42.0± 11.3
WF
σ/μ = 6.83± 2.31 %
μ = 452.0± 7.7
σ = 30.9± 9.9



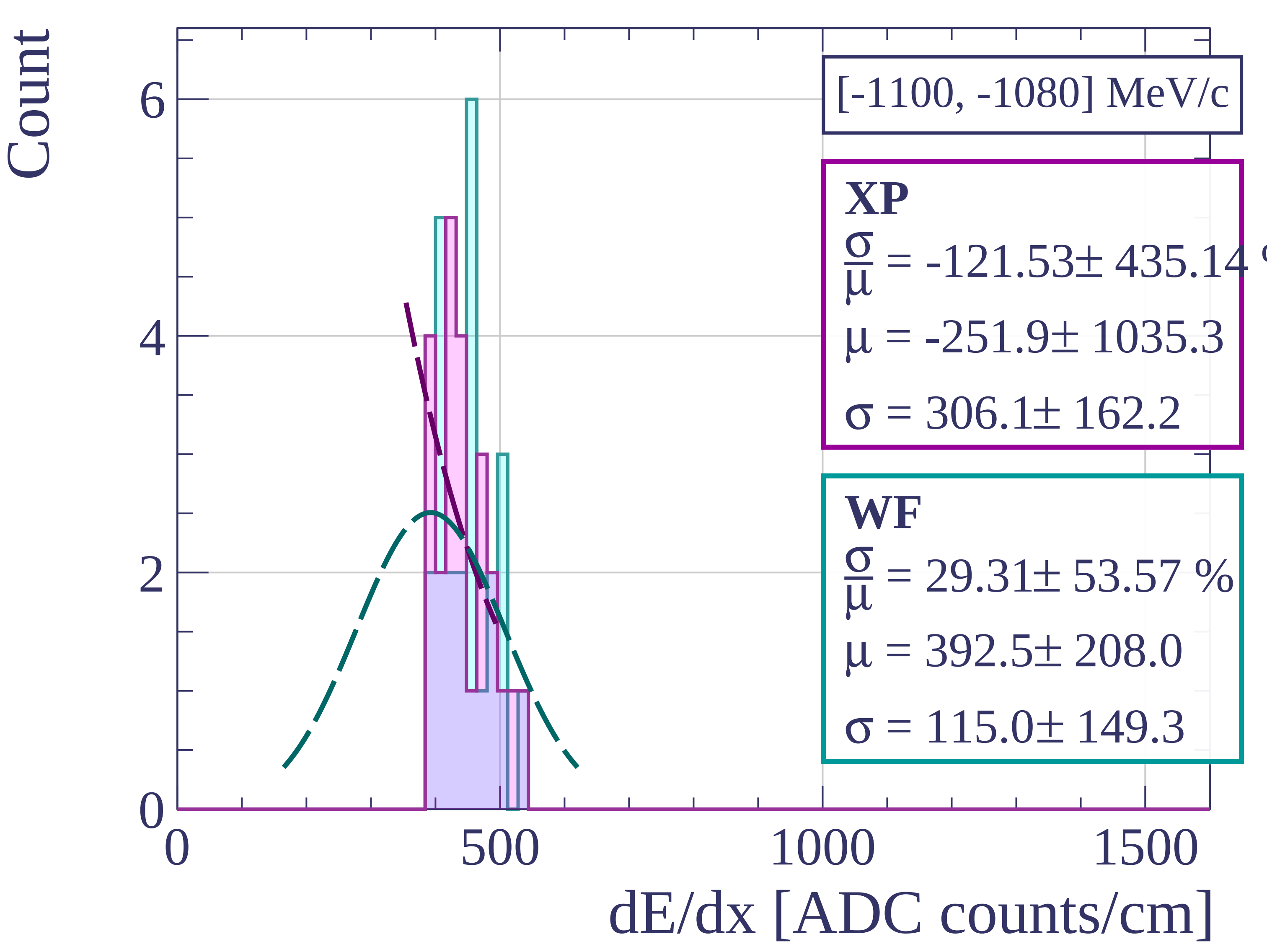
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.62± 0.95 %
μ = 451.5± 4.3
σ = 16.3± 4.2
WF
$\frac{\sigma}{\mu}$ = 15.89± 12.99 %
μ = 432.8± 48.1
σ = 68.8± 48.6



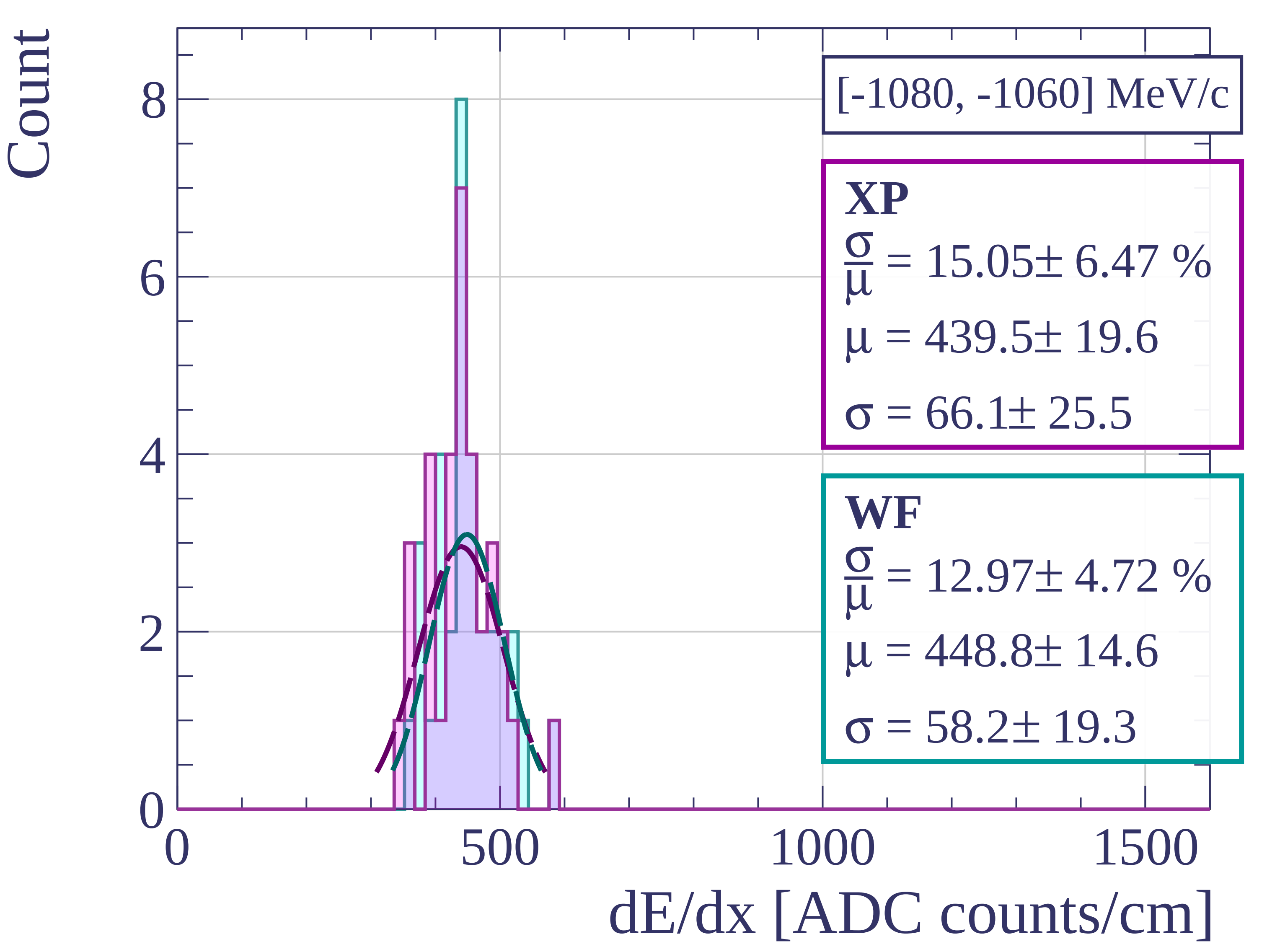
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1140, -1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.90± 2.10 %
μ = 425.2± 8.8
σ = 37.8± 8.2
WF
$\frac{\sigma}{\mu}$ = 10.22± 2.63 %
μ = 433.5± 10.6
σ = 44.3± 10.3



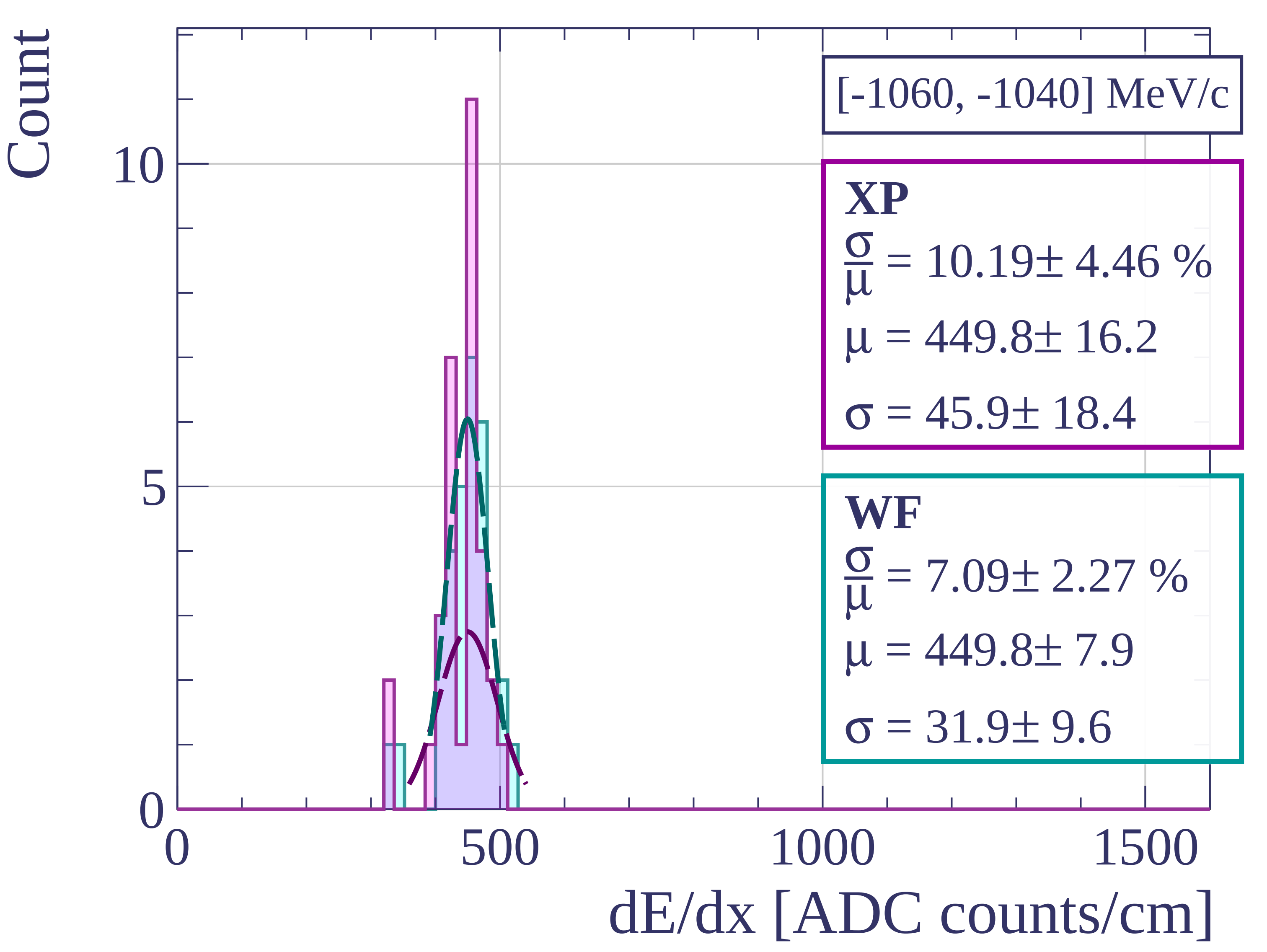
Count
[-1120, -1100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.31± 4.63 %
μ = 418.7± 17.7
σ = 47.4± 17.4
WF
$\frac{\sigma}{\mu}$ = 10.18± 2.75 %
μ = 437.7± 10.9
σ = 44.6± 10.9
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]



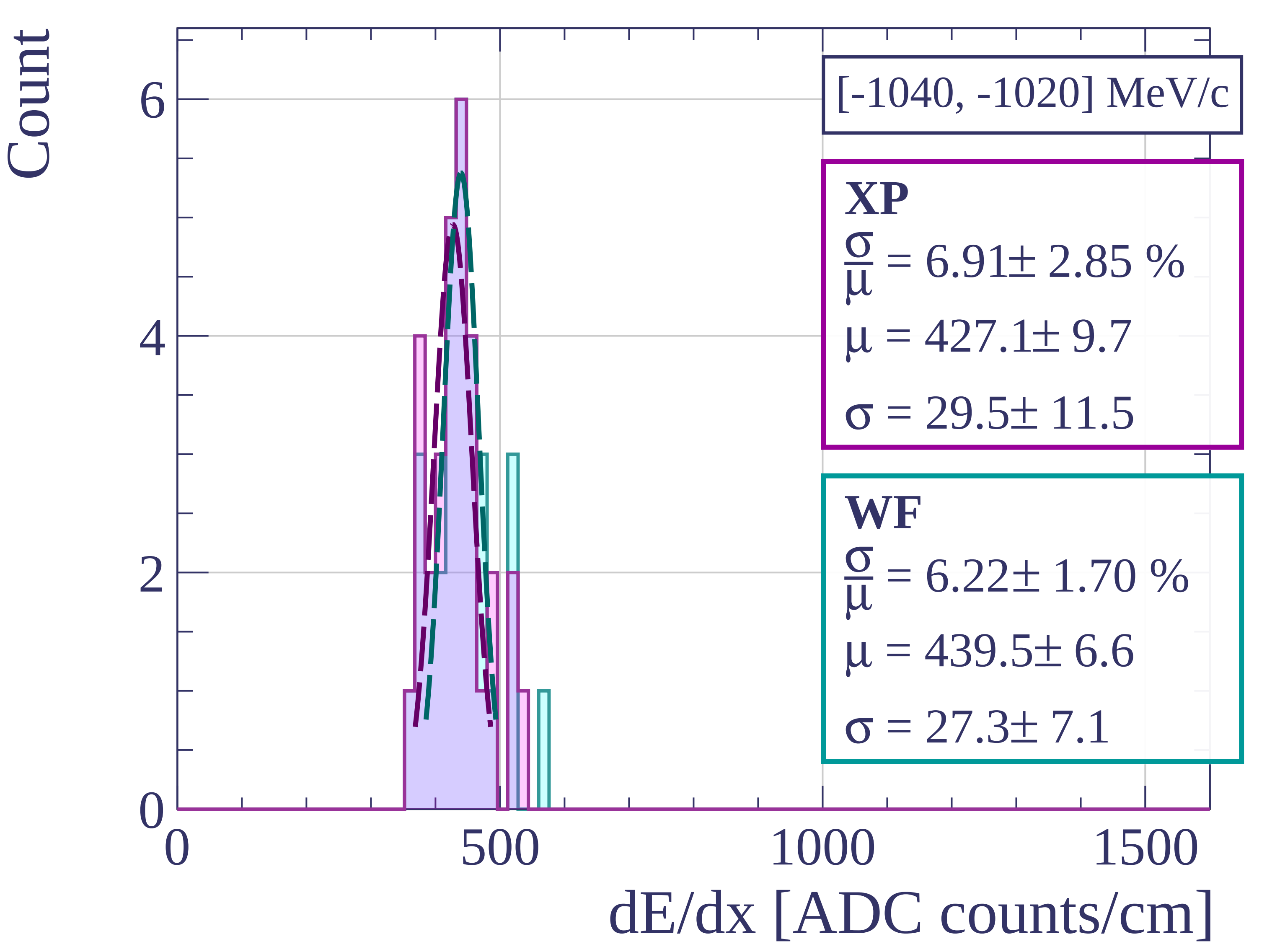
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
σ/μ = -121.53± 435.14 %
μ = -251.9± 1035.3
σ = 306.1± 162.2
WF
σ/μ = 29.31± 53.57 %
μ = 392.5± 208.0
σ = 115.0± 149.3



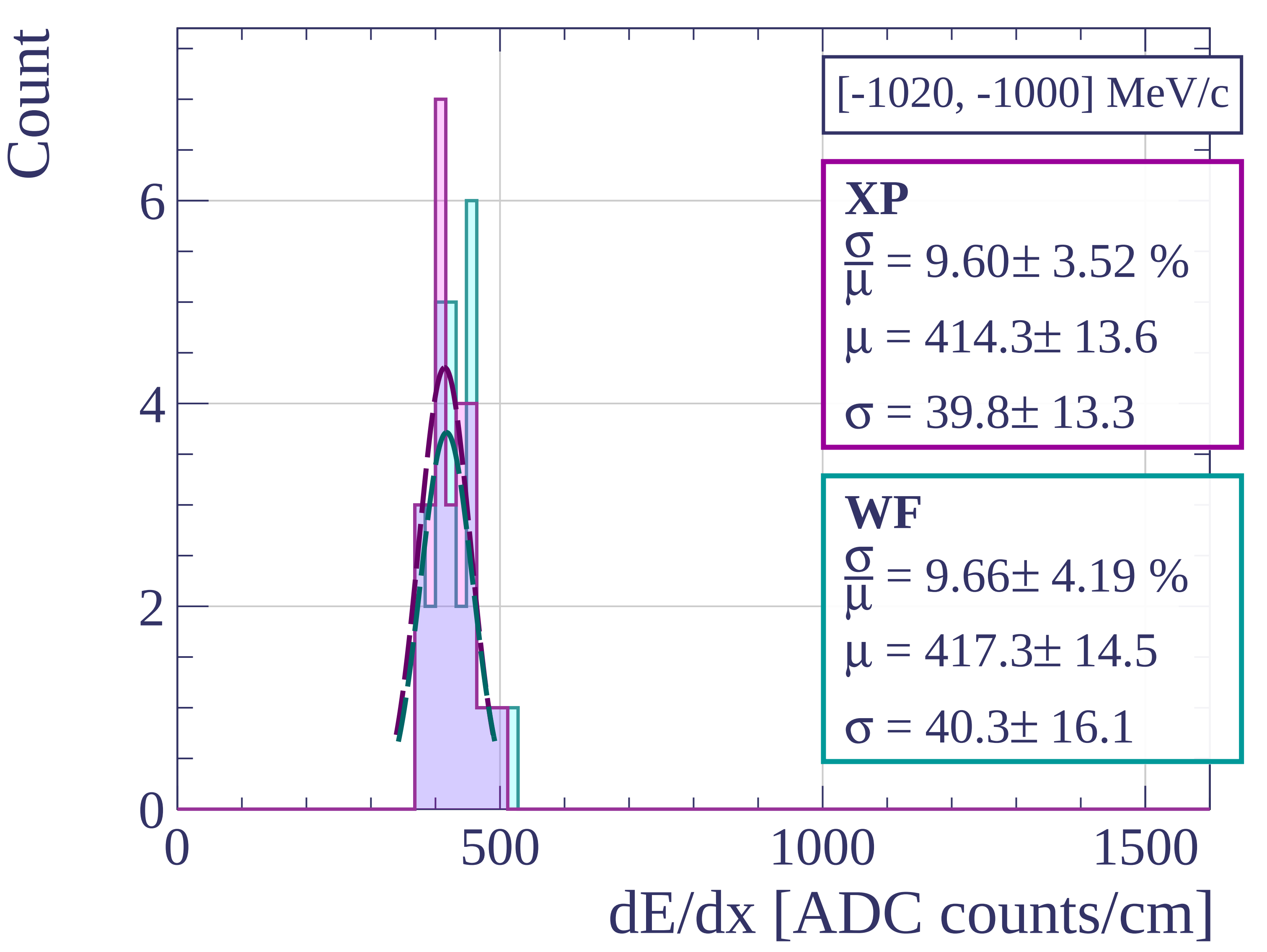
Count
dE/dx [ADC counts/cm]
[-1080, -1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 15.05± 6.47 %
μ = 439.5± 19.6
σ = 66.1± 25.5
WF
$\frac{\sigma}{\mu}$ = 12.97± 4.72 %
μ = 448.8± 14.6
σ = 58.2± 19.3

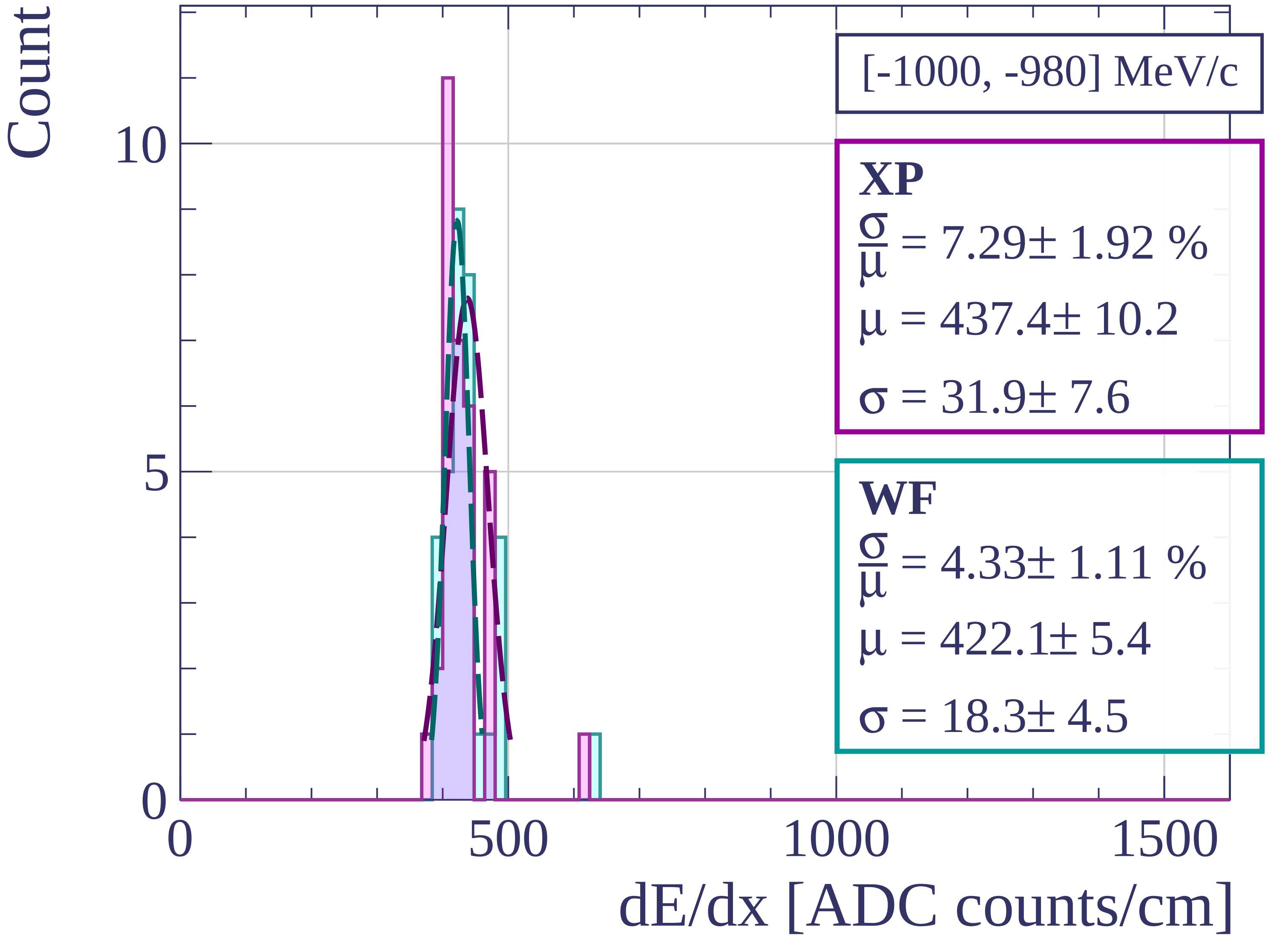


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1060, -1040] MeV/c
XP
σ/μ = 10.19± 4.46 %
μ = 449.8± 16.2
σ = 45.9± 18.4
WF
σ/μ = 7.09± 2.27 %
μ = 449.8± 7.9
σ = 31.9± 9.6

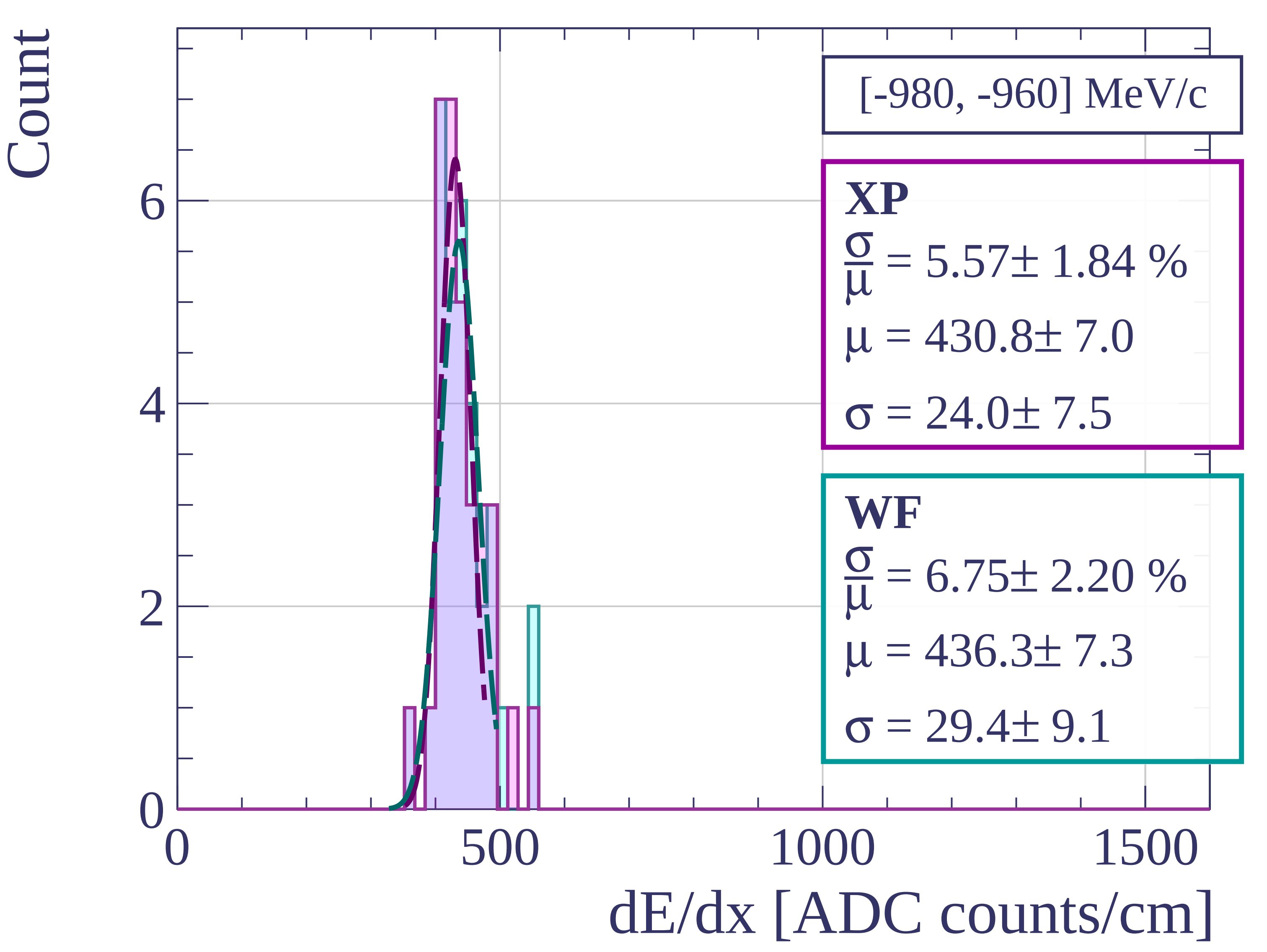


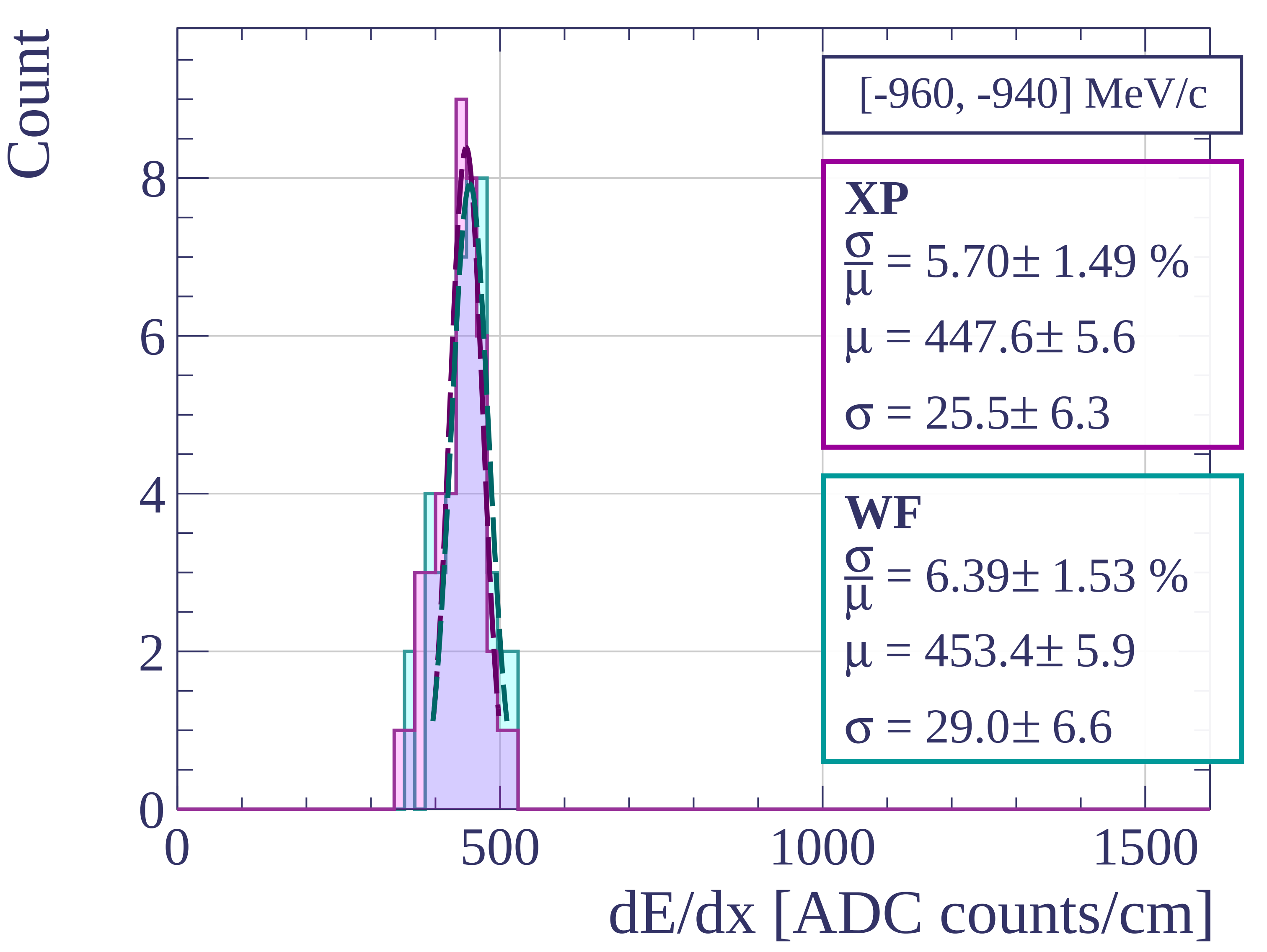
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-1040, -1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.91± 2.85 %
μ = 427.1± 9.7
σ = 29.5± 11.5
WF
$\frac{\sigma}{\mu}$ = 6.22± 1.70 %
μ = 439.5± 6.6
σ = 27.3± 7.1

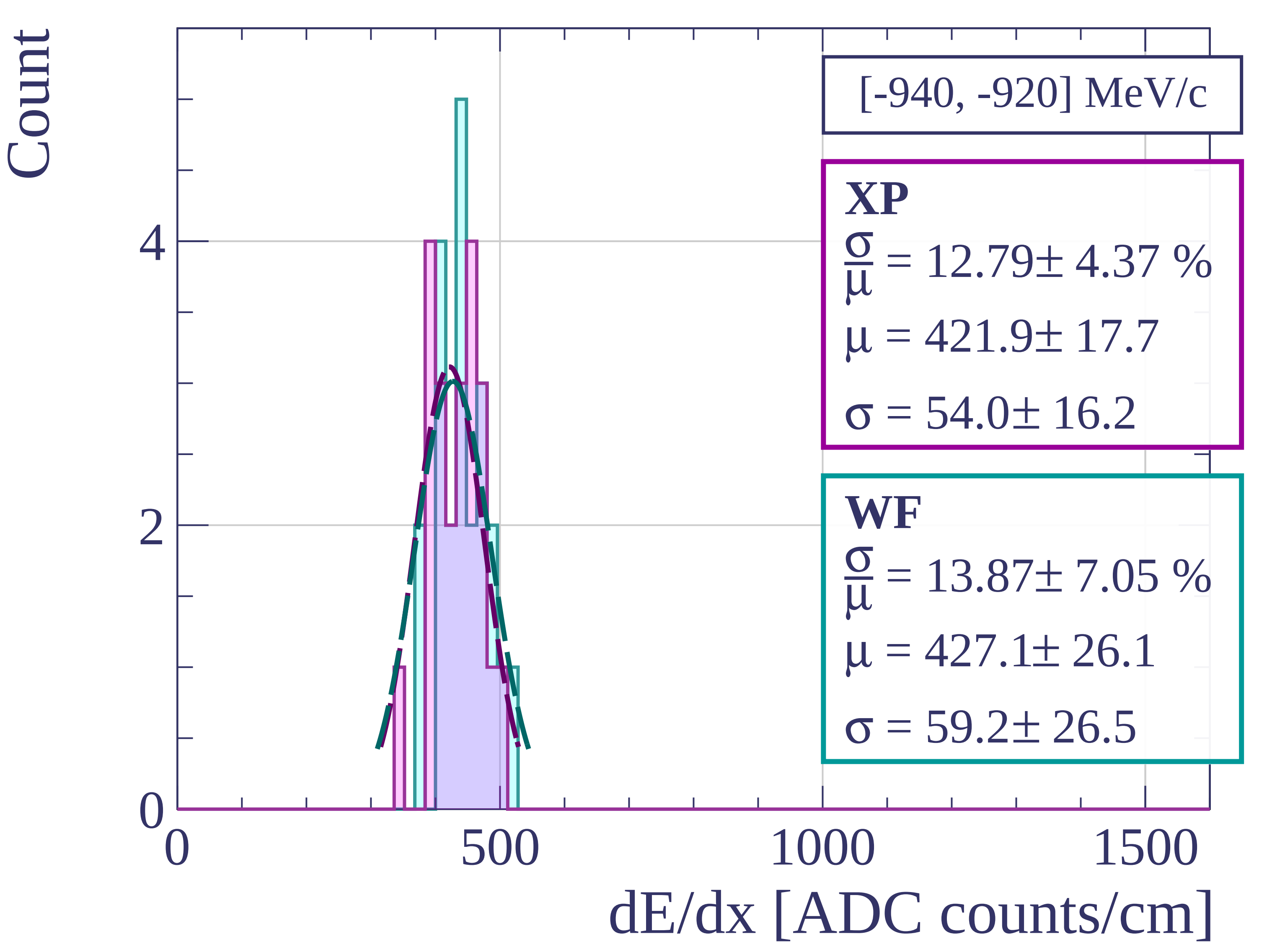

Count
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.60± 3.52 %
μ = 414.3± 13.6
σ = 39.8± 13.3
WF
$\frac{\sigma}{\mu}$ = 9.66± 4.19 %
μ = 417.3± 14.5
σ = 40.3± 16.1
0
2
4
6
0
500
1000
1500



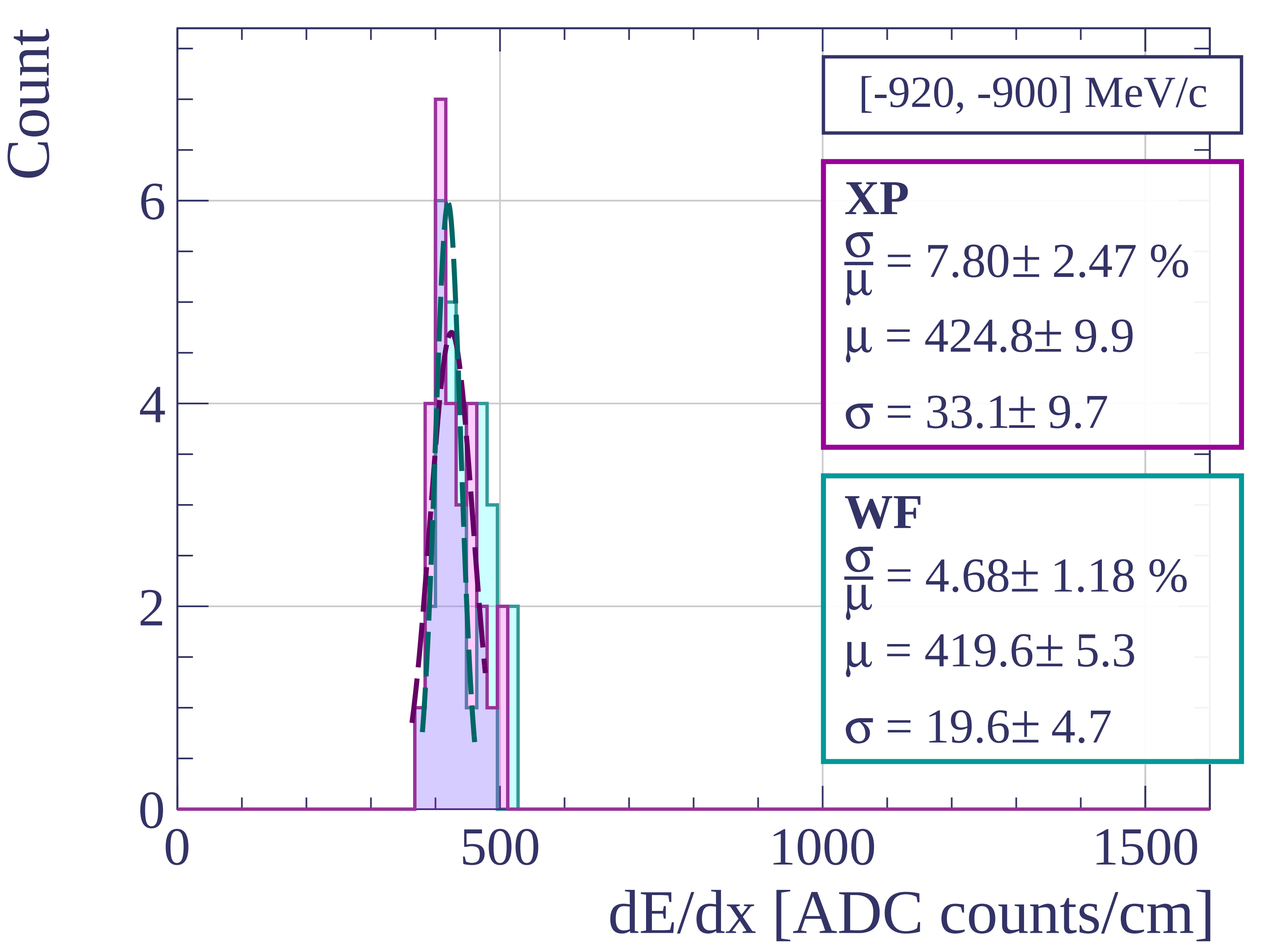
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1000, -980] MeV/c
XP
σ/μ = 7.29± 1.92 %
μ = 437.4± 10.2
σ = 31.9± 7.6
WF
σ/μ = 4.33± 1.11 %
μ = 422.1± 5.4
σ = 18.3± 4.5


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[-980, -960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.57± 1.84 %
μ = 430.8± 7.0
σ = 24.0± 7.5
WF
$\frac{\sigma}{\mu}$ = 6.75± 2.20 %
μ = 436.3± 7.3
σ = 29.4± 9.1

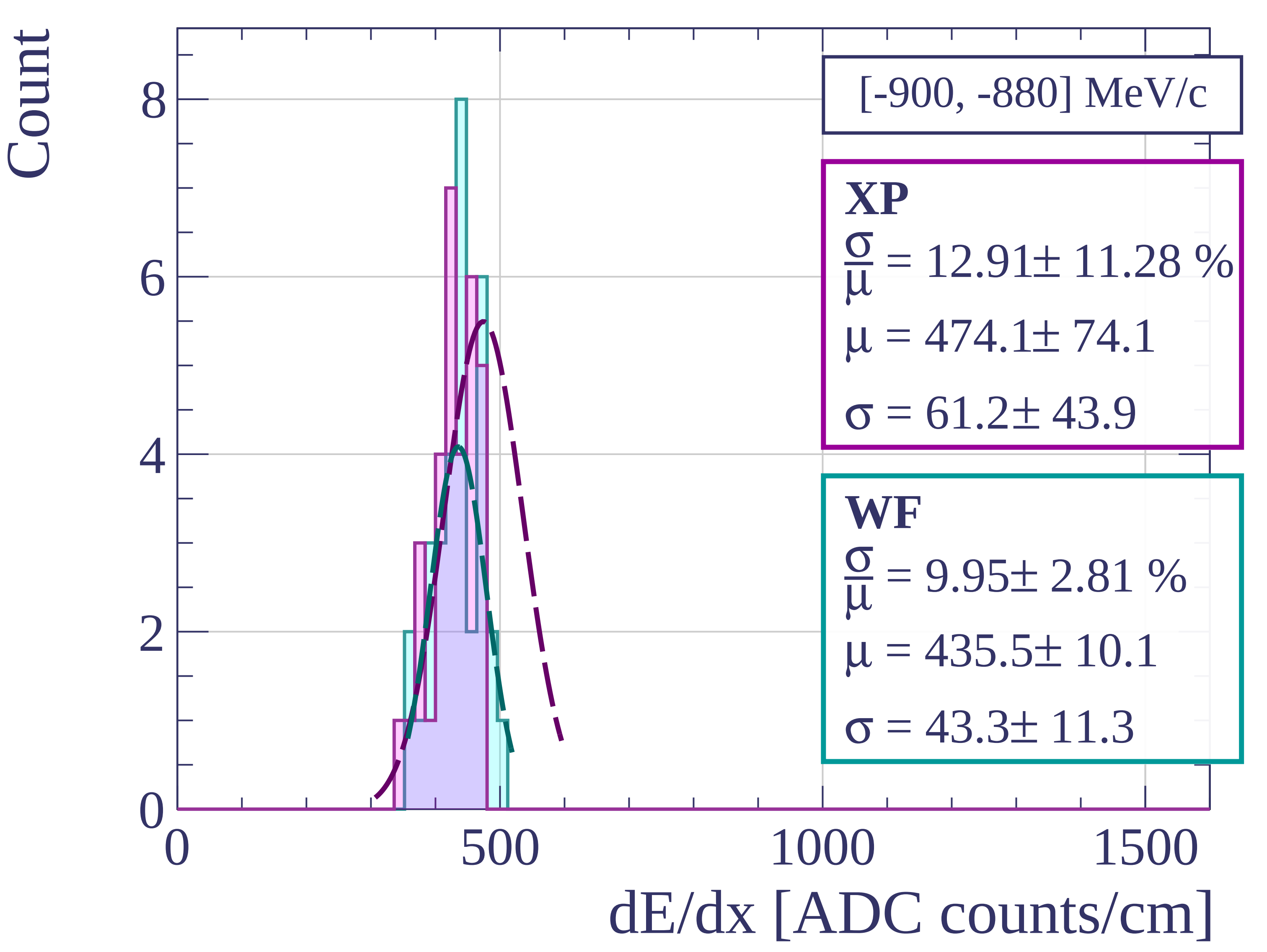

[-960, -940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.70± 1.49 %
μ = 447.6± 5.6
σ = 25.5± 6.3
WF
$\frac{\sigma}{\mu}$ = 6.39± 1.53 %
μ = 453.4± 5.9
σ = 29.0± 6.6
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500



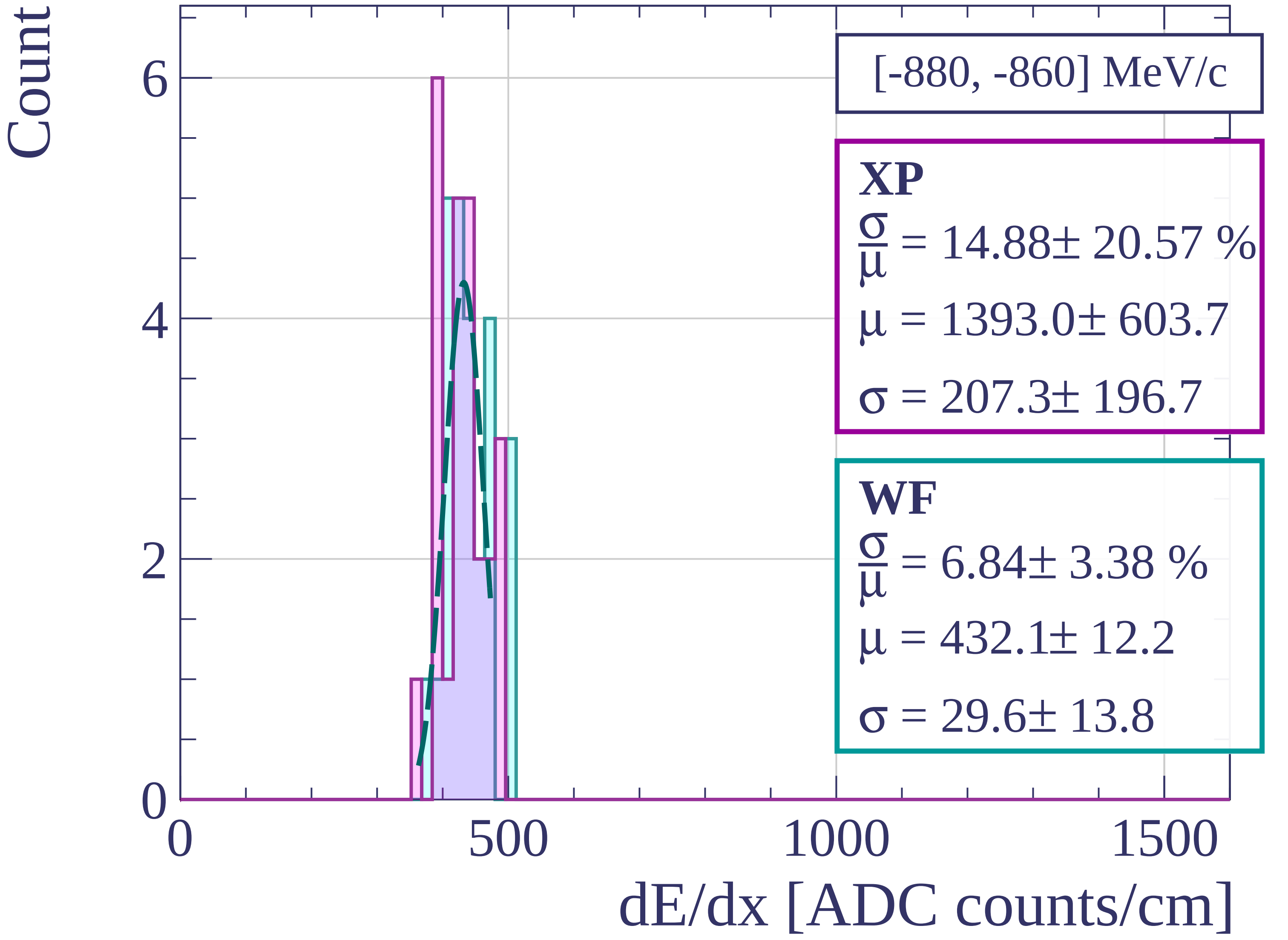
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-940, -920] MeV/c
XP
σ/μ = 12.79± 4.37 %
μ = 421.9± 17.7
σ = 54.0± 16.2
WF
σ/μ = 13.87± 7.05 %
μ = 427.1± 26.1
σ = 59.2± 26.5



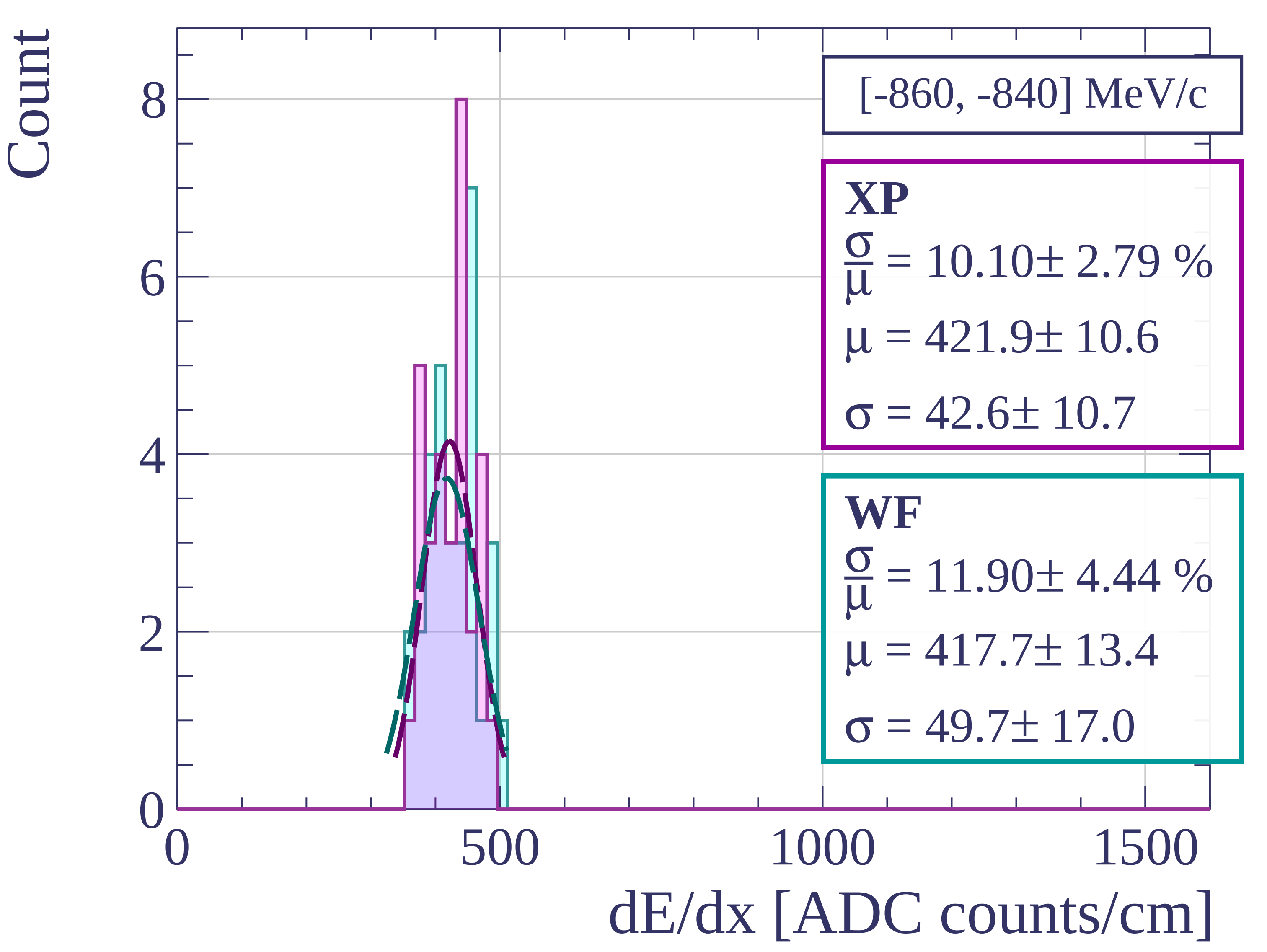
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-920, -900] MeV/c
XP
σ/μ = 7.80± 2.47 %
μ = 424.8± 9.9
σ = 33.1± 9.7
WF
σ/μ = 4.68± 1.18 %
μ = 419.6± 5.3
σ = 19.6± 4.7



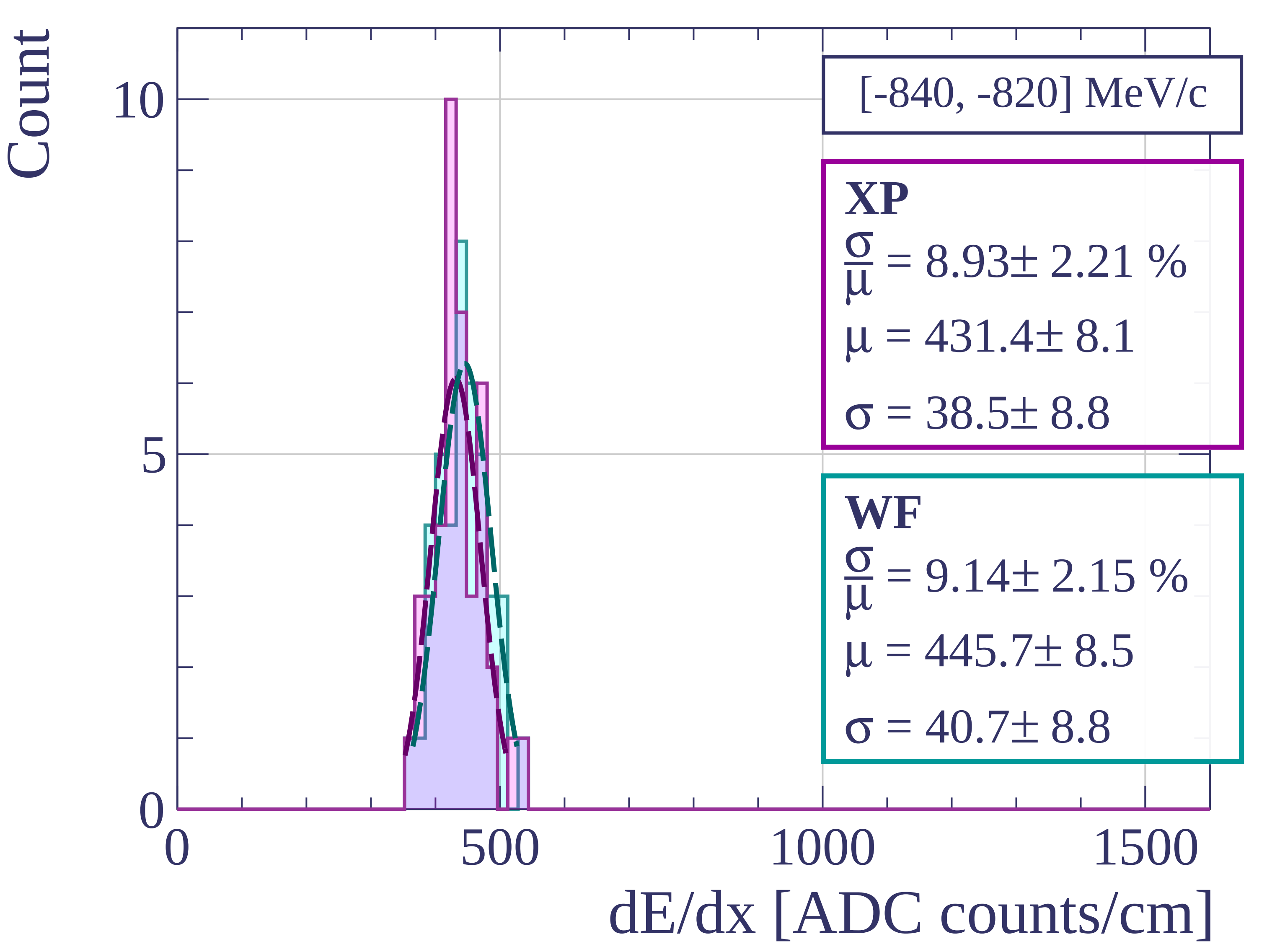
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-900, -880] MeV/c
XP
σ/μ = 12.91± 11.28 %
μ = 474.1± 74.1
σ = 61.2± 43.9
WF
σ/μ = 9.95± 2.81 %
μ = 435.5± 10.1
σ = 43.3± 11.3



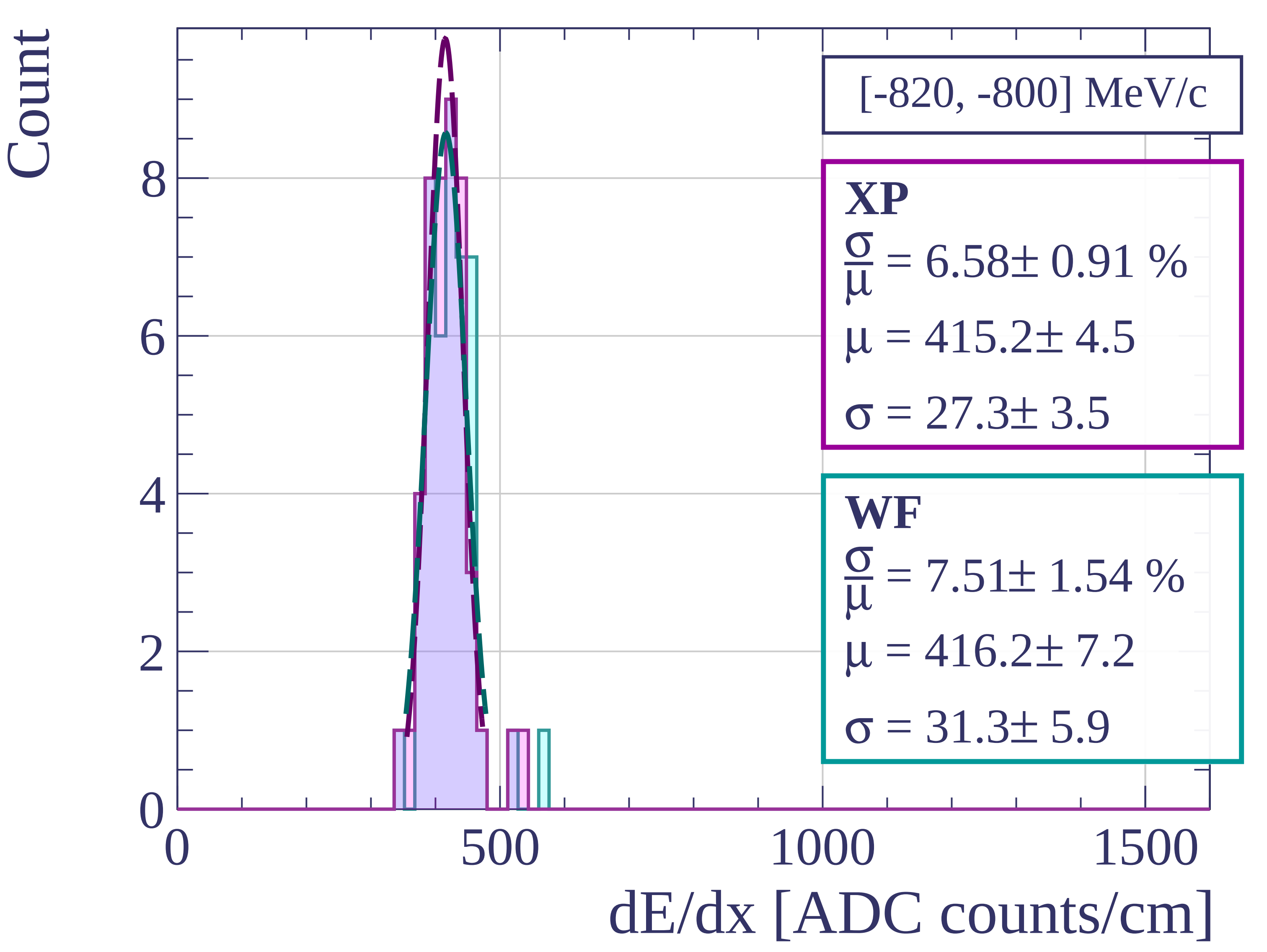
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
σ/μ = 14.88± 20.57 %
μ = 1393.0± 603.7
σ = 207.3± 196.7
WF
σ/μ = 6.84± 3.38 %
μ = 432.1± 12.2
σ = 29.6± 13.8



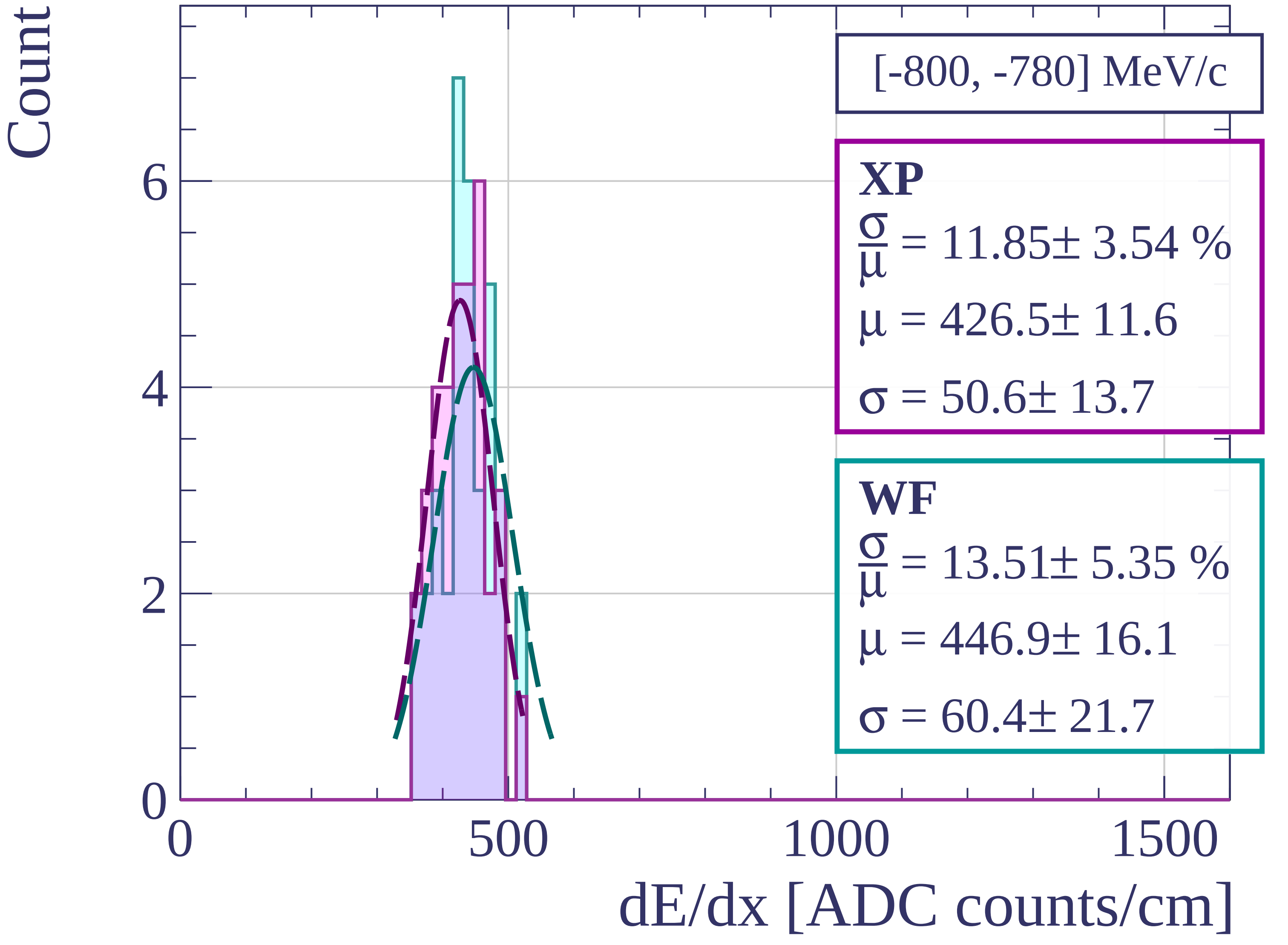
[-860, -840] MeV/c
XP
σ/μ = 10.10 ± 2.79 %
μ = 421.9 ± 10.6
σ = 42.6 ± 10.7
WF
σ/μ = 11.90 ± 4.44 %
μ = 417.7 ± 13.4
σ = 49.7 ± 17.0
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500



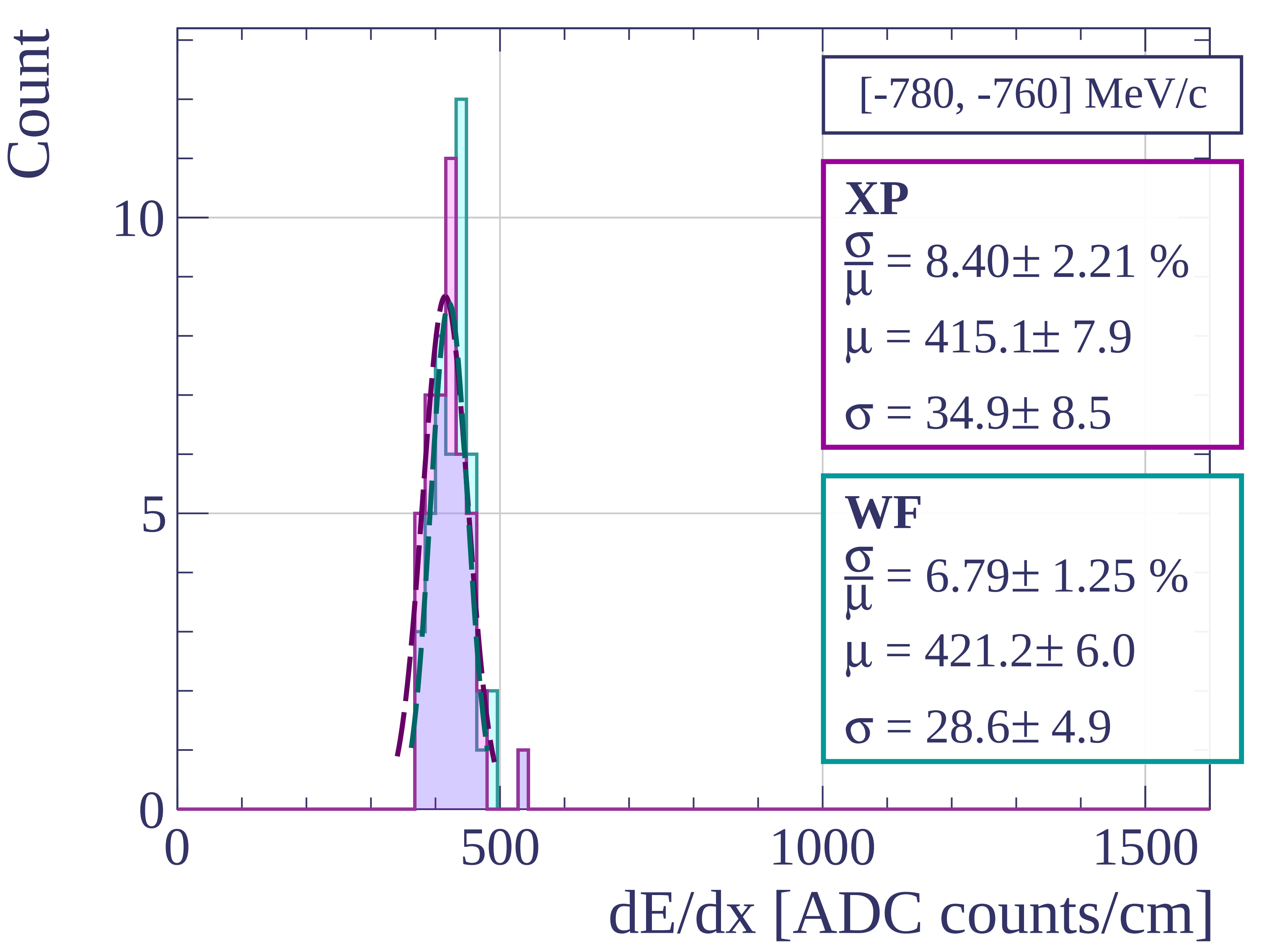
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-840, -820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.93± 2.21 %
μ = 431.4± 8.1
σ = 38.5± 8.8
WF
$\frac{\sigma}{\mu}$ = 9.14± 2.15 %
μ = 445.7± 8.5
σ = 40.7± 8.8



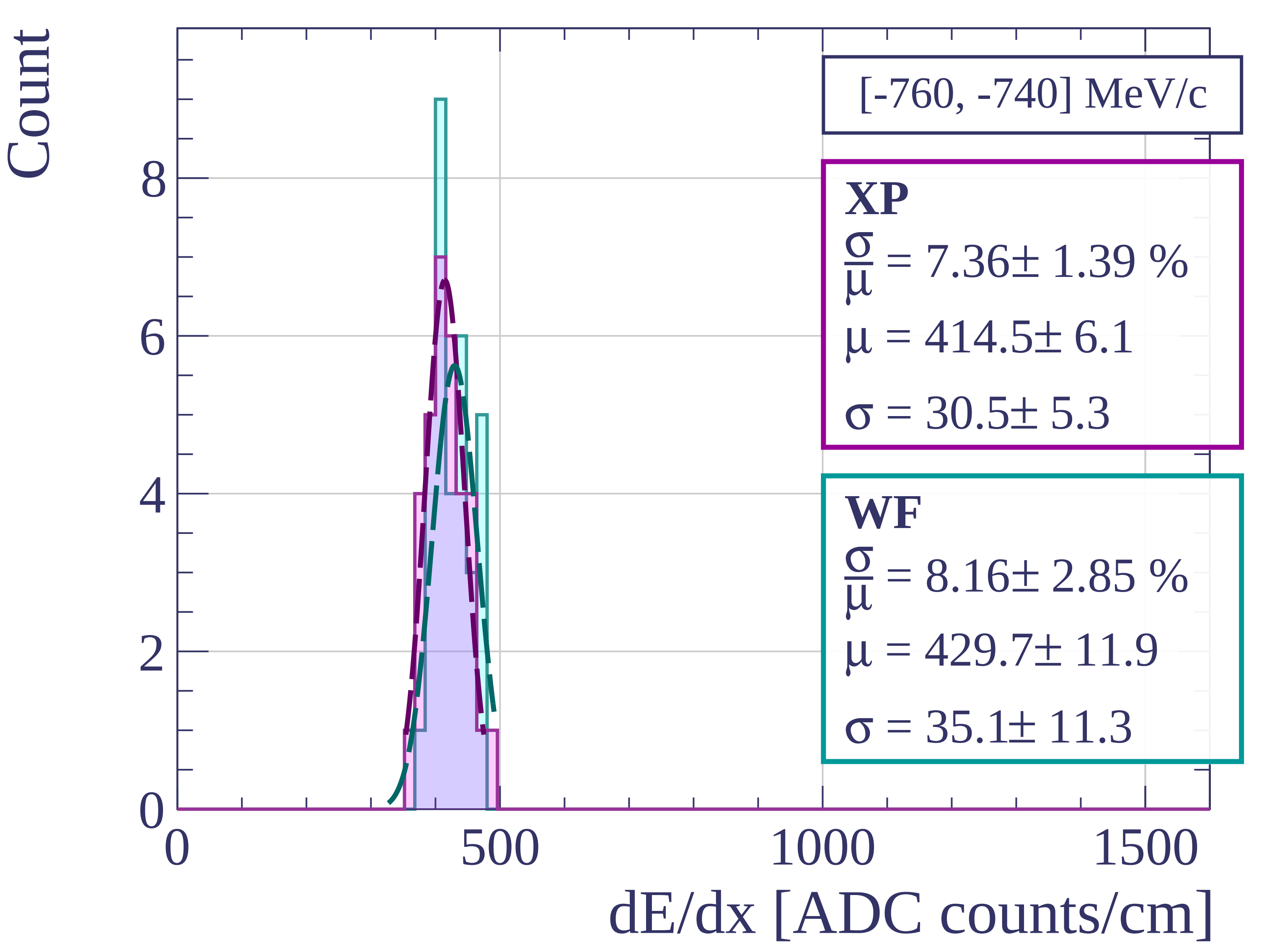
Count
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.58± 0.91 %
μ = 415.2± 4.5
σ = 27.3± 3.5
WF
$\frac{\sigma}{\mu}$ = 7.51± 1.54 %
μ = 416.2± 7.2
σ = 31.3± 5.9
0
2
4
6
8
0
500
1000
1500



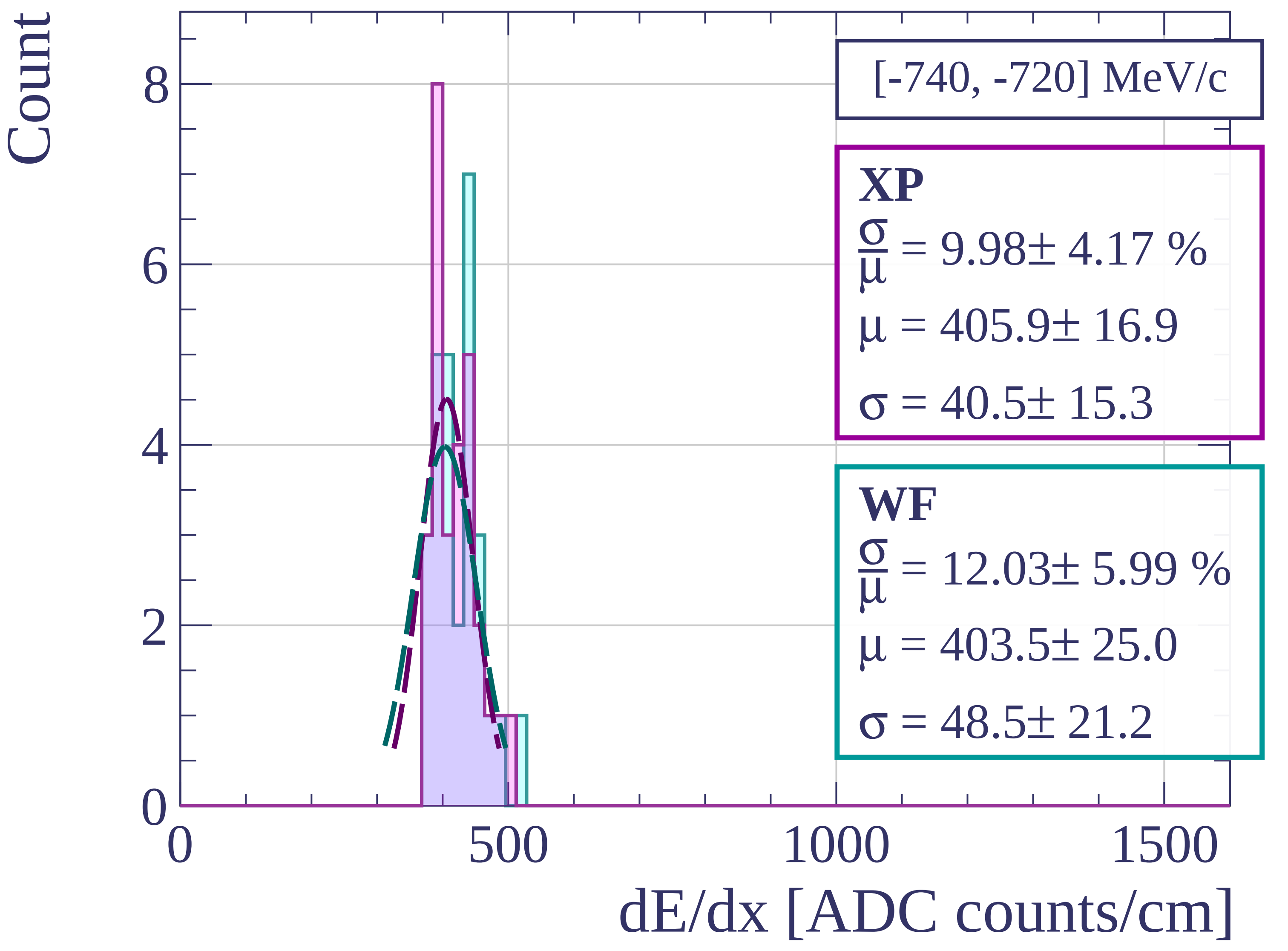
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-800, -780] MeV/c
XP
σ/μ = 11.85± 3.54 %
μ = 426.5± 11.6
σ = 50.6± 13.7
WF
σ/μ = 13.51± 5.35 %
μ = 446.9± 16.1
σ = 60.4± 21.7



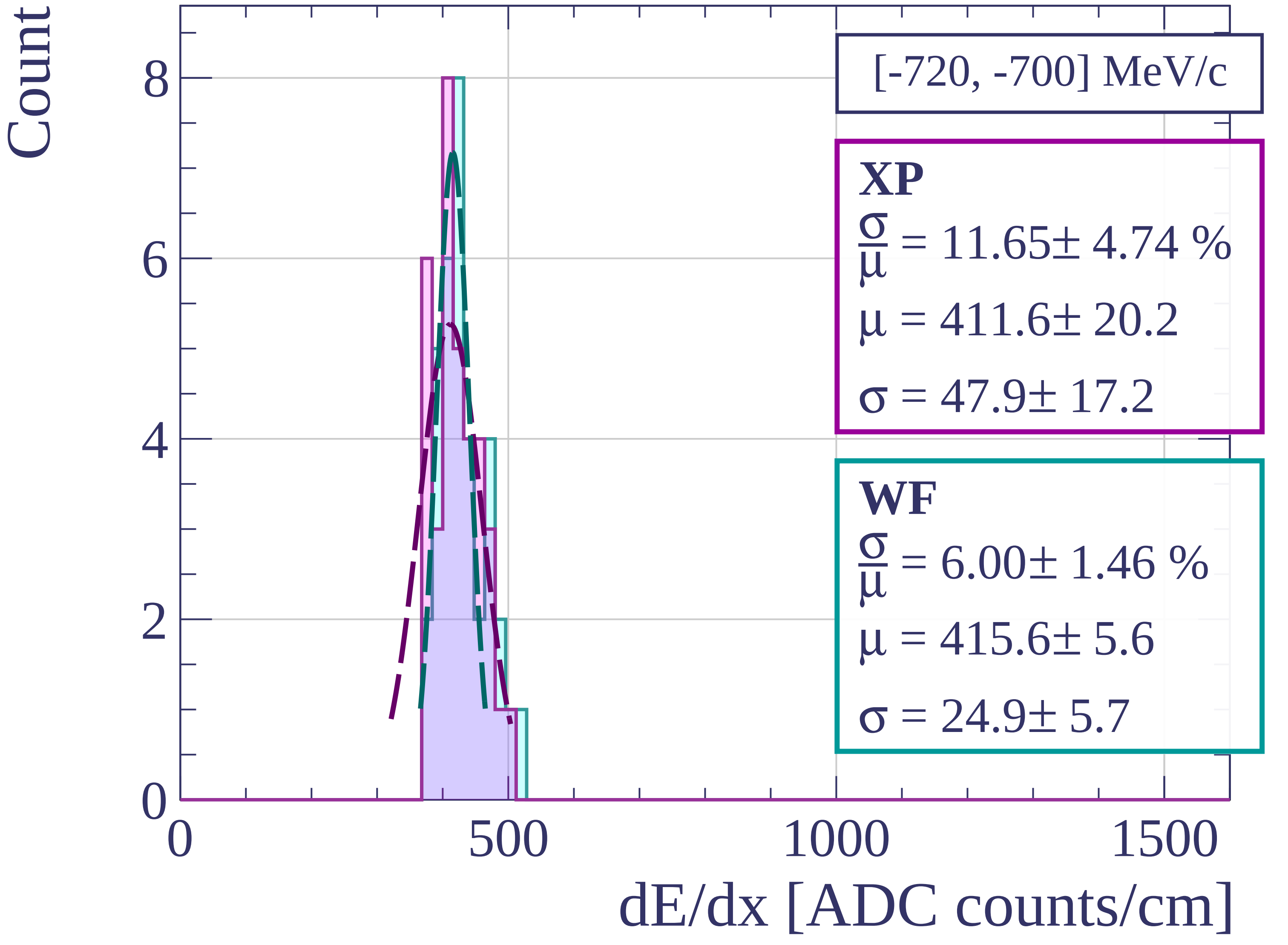
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-780, -760] MeV/c
XP
σ/μ = 8.40± 2.21 %
μ = 415.1± 7.9
σ = 34.9± 8.5
WF
σ/μ = 6.79± 1.25 %
μ = 421.2± 6.0
σ = 28.6± 4.9

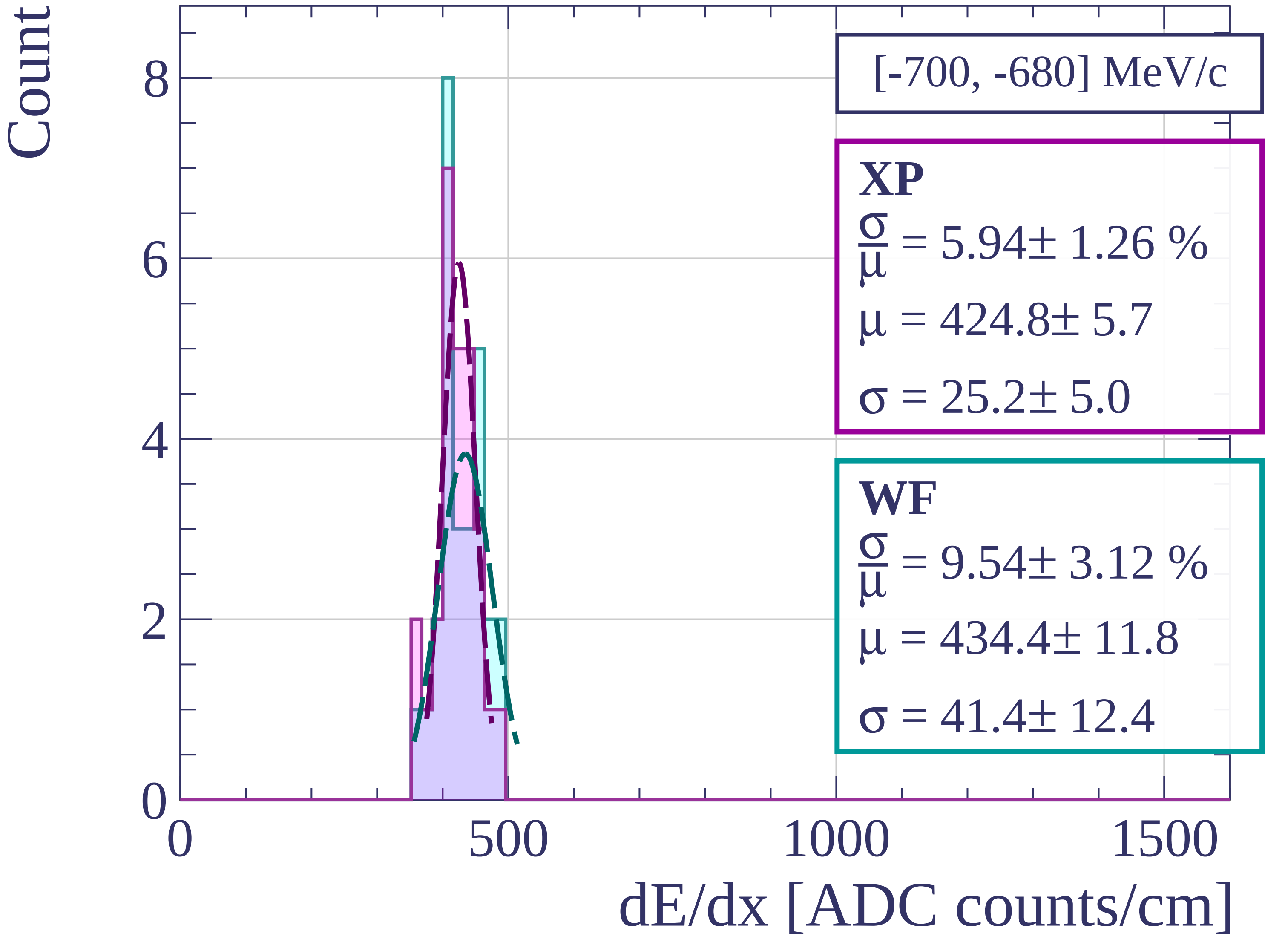


Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.36± 1.39 %
μ = 414.5± 6.1
σ = 30.5± 5.3
WF
$\frac{\sigma}{\mu}$ = 8.16± 2.85 %
μ = 429.7± 11.9
σ = 35.1± 11.3

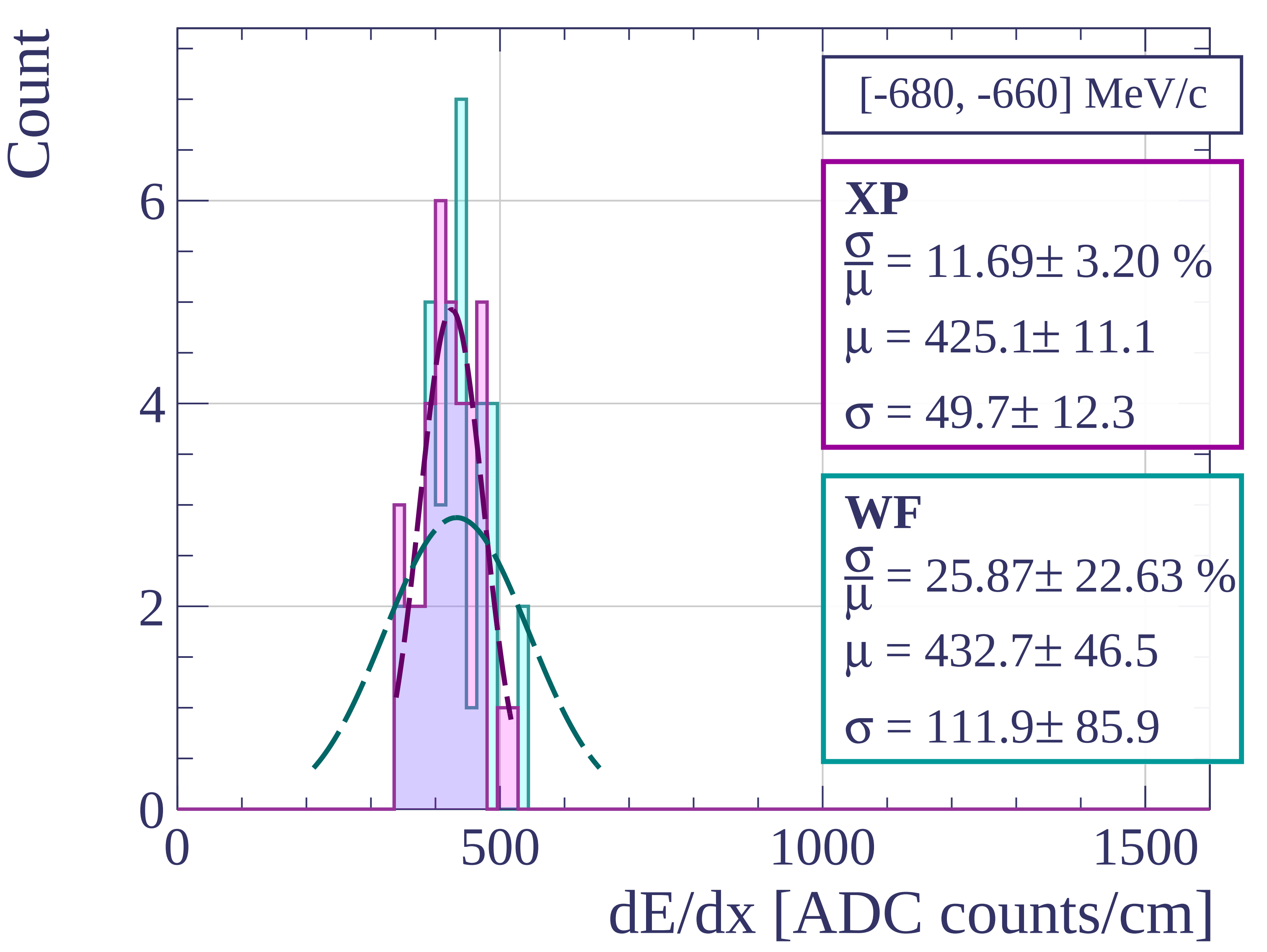


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.98± 4.17 %
μ = 405.9± 16.9
σ = 40.5± 15.3
WF
$\frac{\sigma}{\mu}$ = 12.03± 5.99 %
μ = 403.5± 25.0
σ = 48.5± 21.2

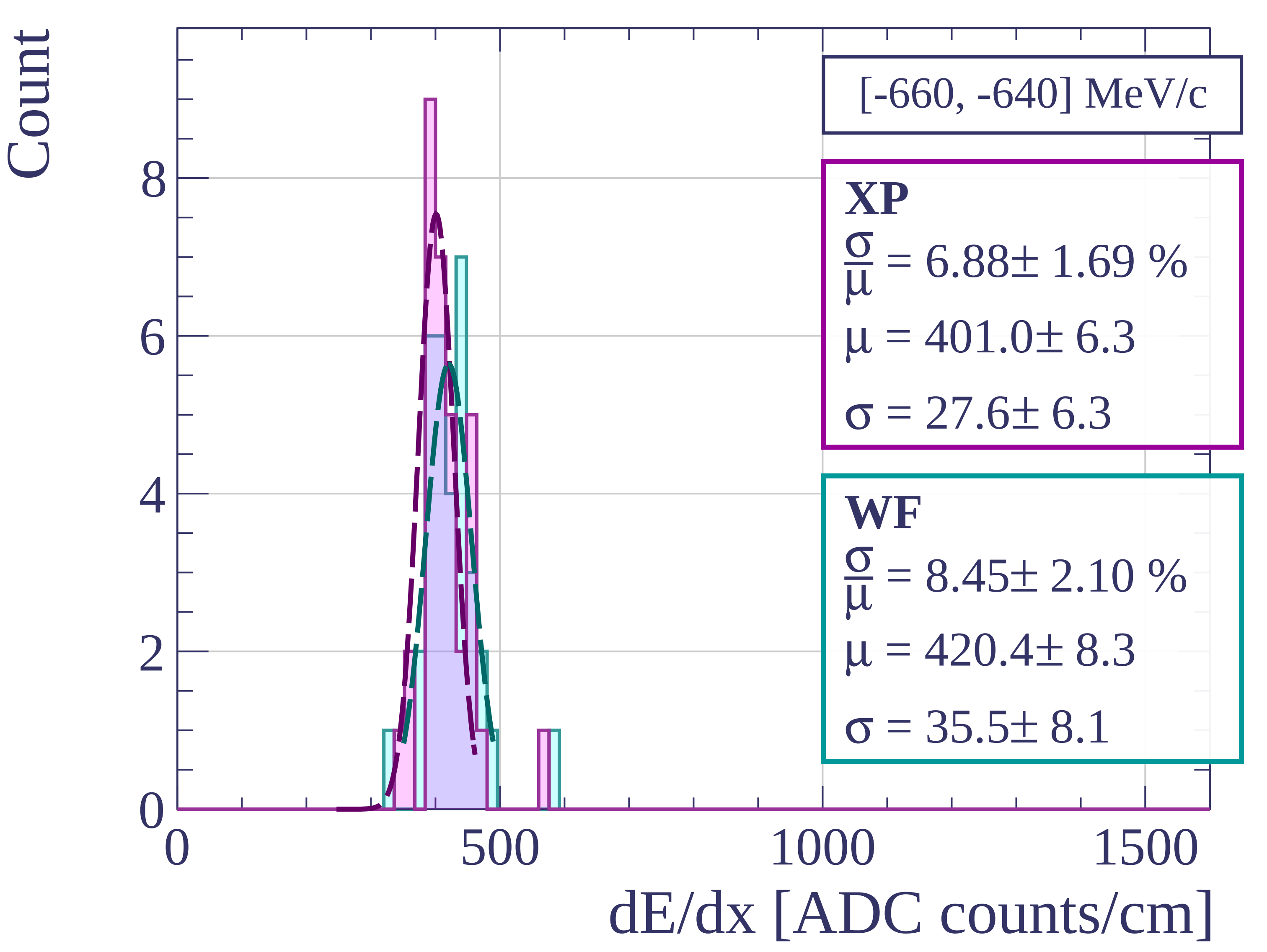

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-720, -700] MeV/c
XP
σ/μ = 11.65± 4.74 %
μ = 411.6± 20.2
σ = 47.9± 17.2
WF
σ/μ = 6.00± 1.46 %
μ = 415.6± 5.6
σ = 24.9± 5.7

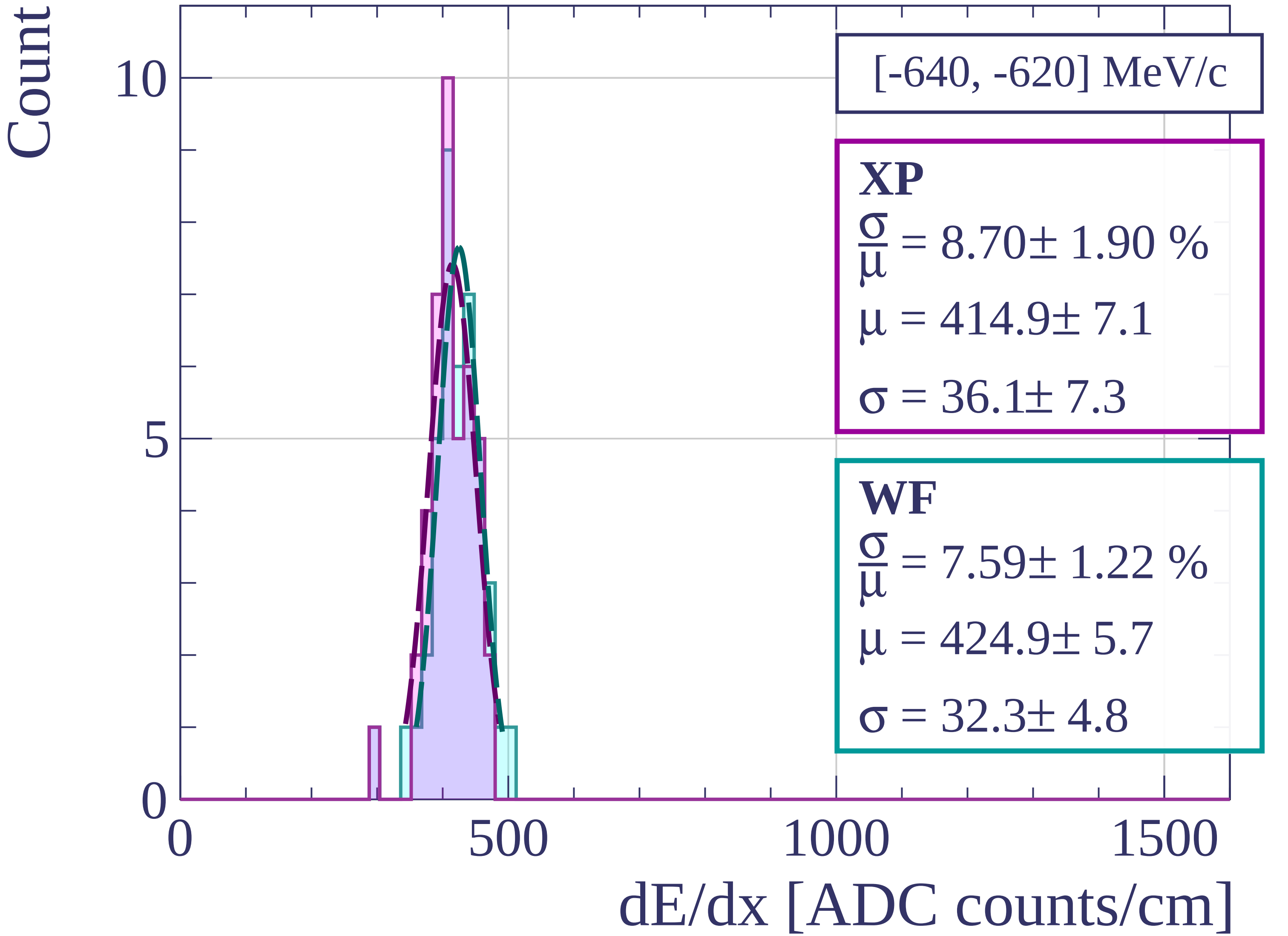


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-700, -680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.94± 1.26 %
μ = 424.8± 5.7
σ = 25.2± 5.0
WF
$\frac{\sigma}{\mu}$ = 9.54± 3.12 %
μ = 434.4± 11.8
σ = 41.4± 12.4

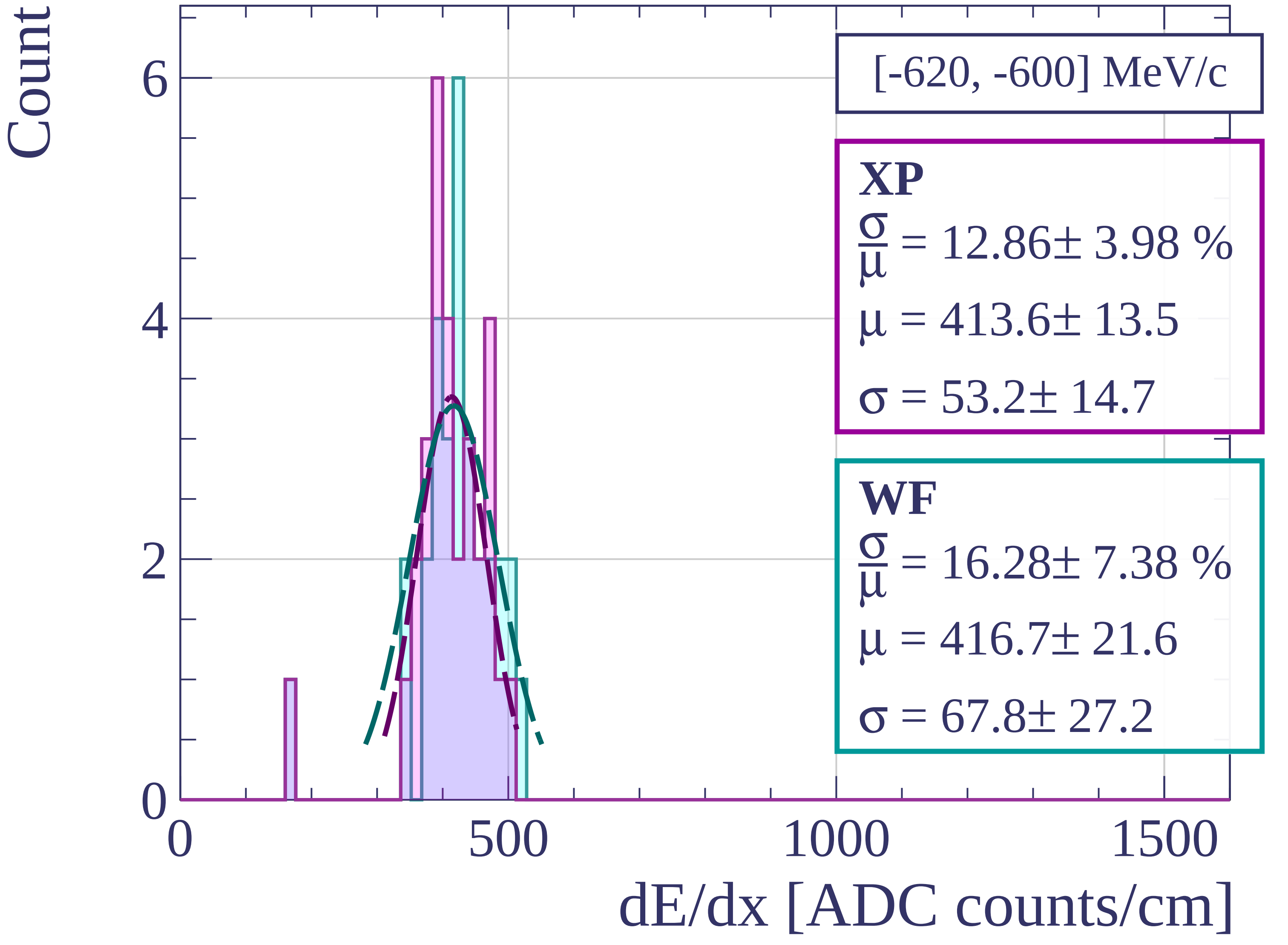


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[-680, -660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.69± 3.20 %
μ = 425.1± 11.1
σ = 49.7± 12.3
WF
$\frac{\sigma}{\mu}$ = 25.87± 22.63 %
μ = 432.7± 46.5
σ = 111.9± 85.9

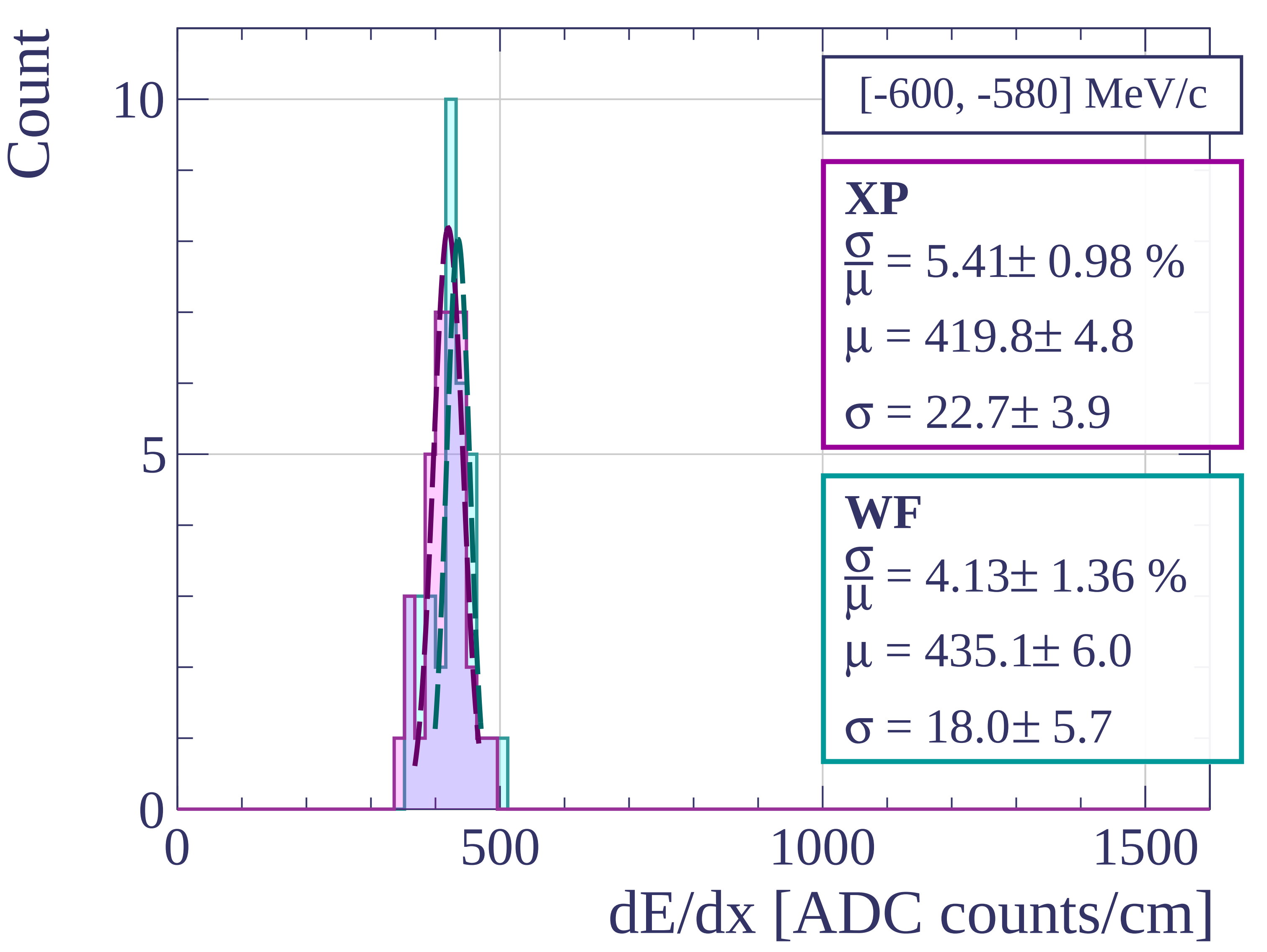

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.88± 1.69 %
μ = 401.0± 6.3
σ = 27.6± 6.3
WF
$\frac{\sigma}{\mu}$ = 8.45± 2.10 %
μ = 420.4± 8.3
σ = 35.5± 8.1



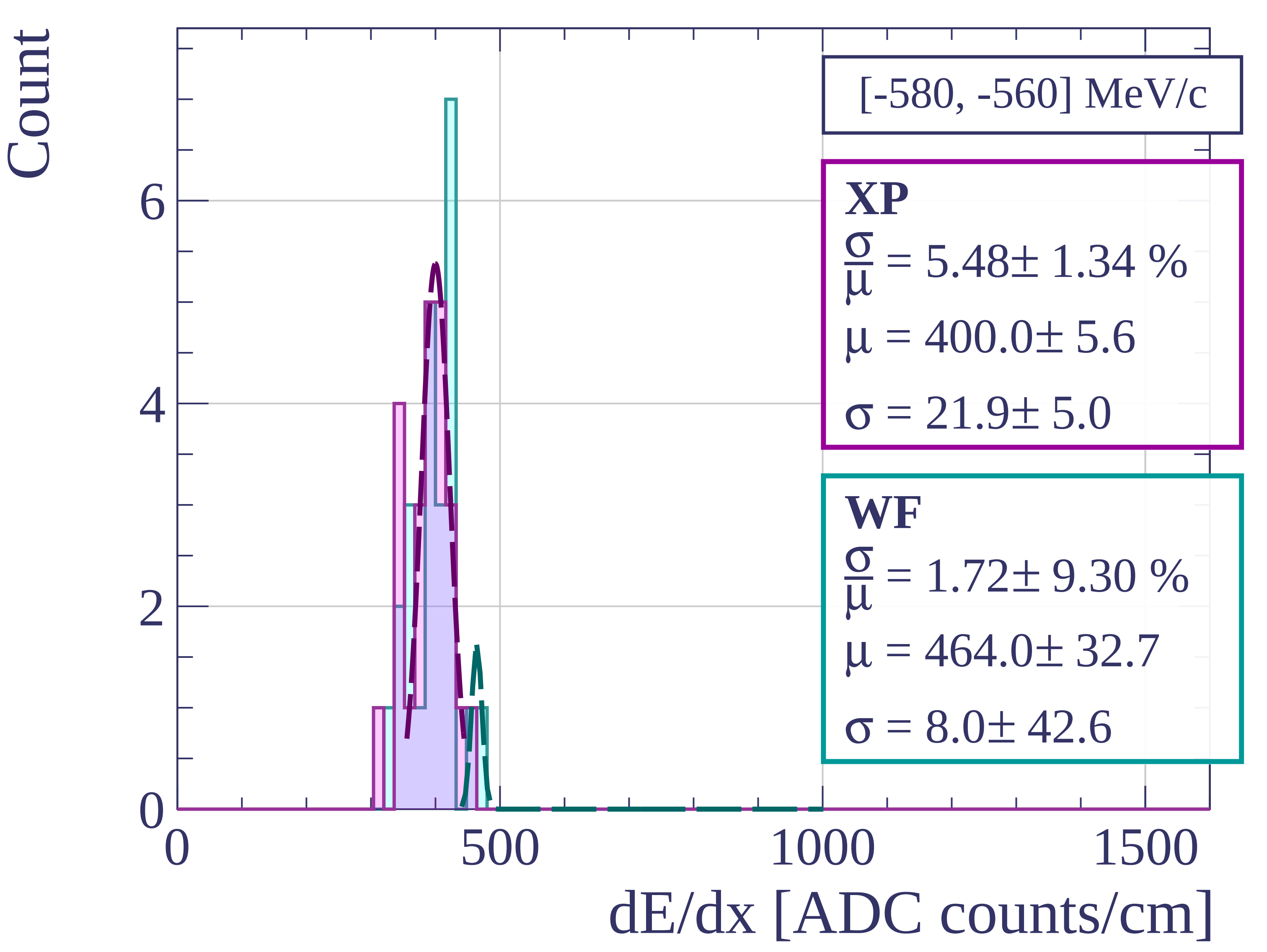
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.70± 1.90 %
μ = 414.9± 7.1
σ = 36.1± 7.3
WF
$\frac{\sigma}{\mu}$ = 7.59± 1.22 %
μ = 424.9± 5.7
σ = 32.3± 4.8



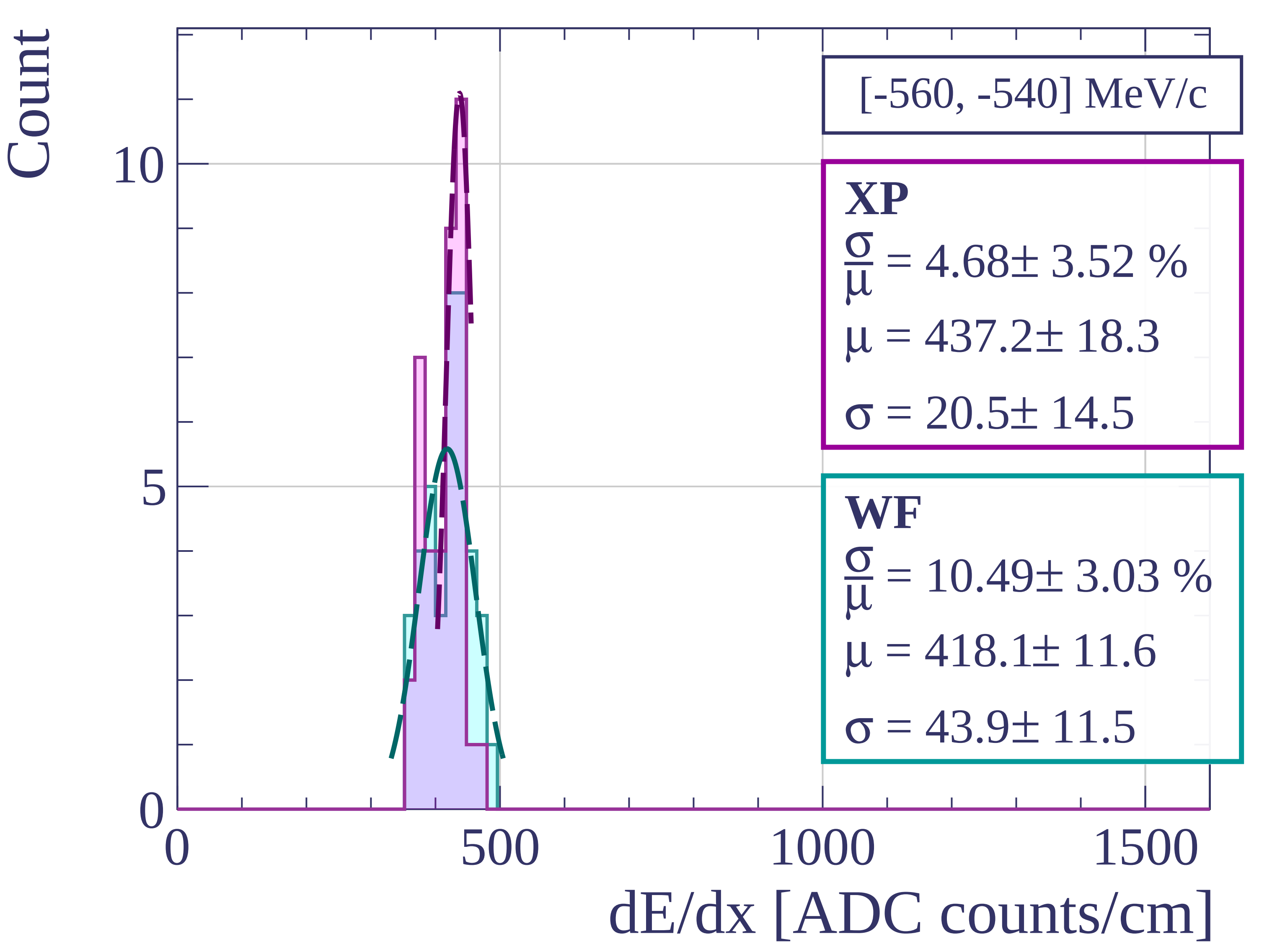
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-620, -600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.86± 3.98 %
μ = 413.6± 13.5
σ = 53.2± 14.7
WF
$\frac{\sigma}{\mu}$ = 16.28± 7.38 %
μ = 416.7± 21.6
σ = 67.8± 27.2



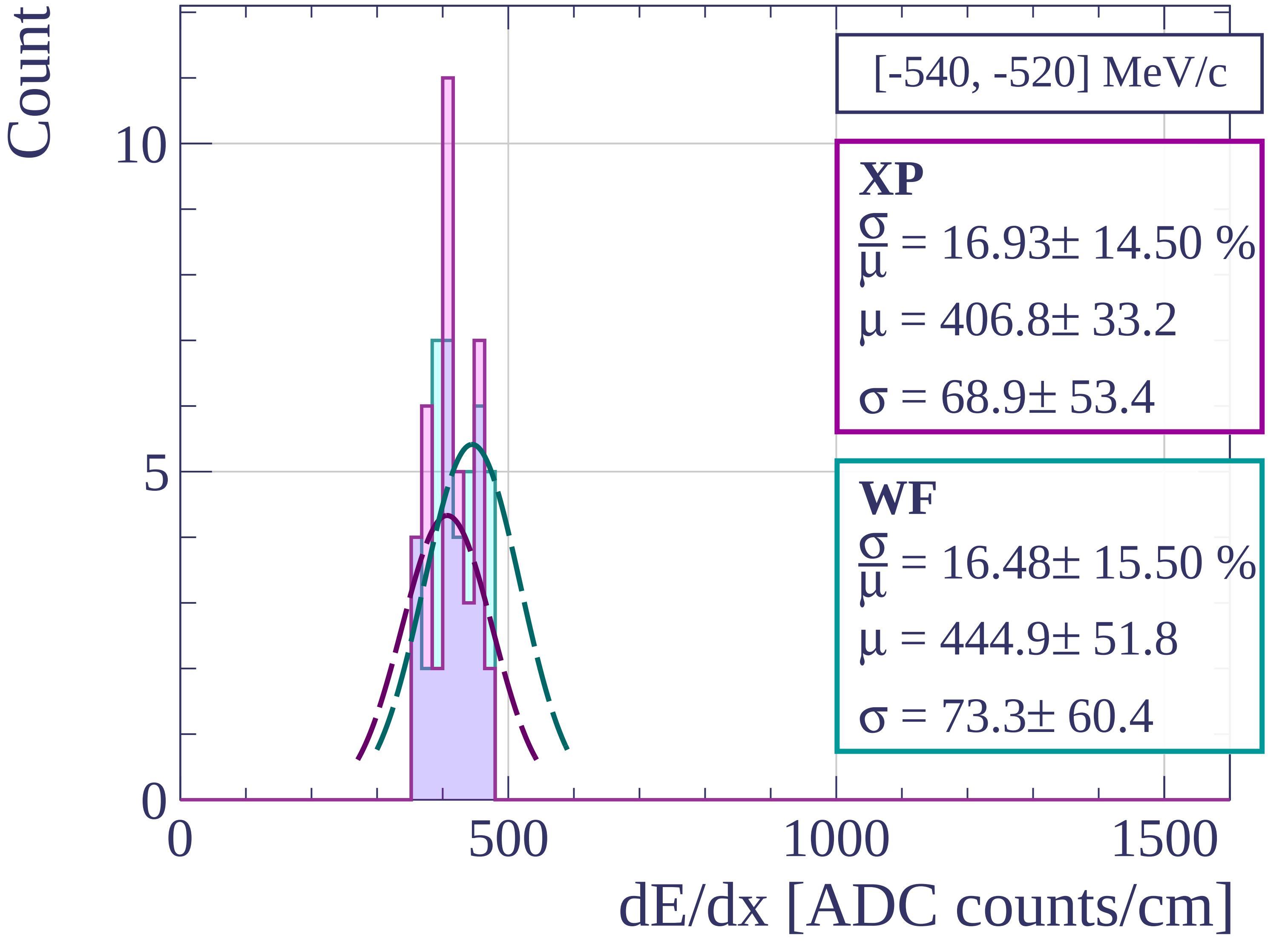
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.41± 0.98 %
μ = 419.8± 4.8
σ = 22.7± 3.9
WF
$\frac{\sigma}{\mu}$ = 4.13± 1.36 %
μ = 435.1± 6.0
σ = 18.0± 5.7

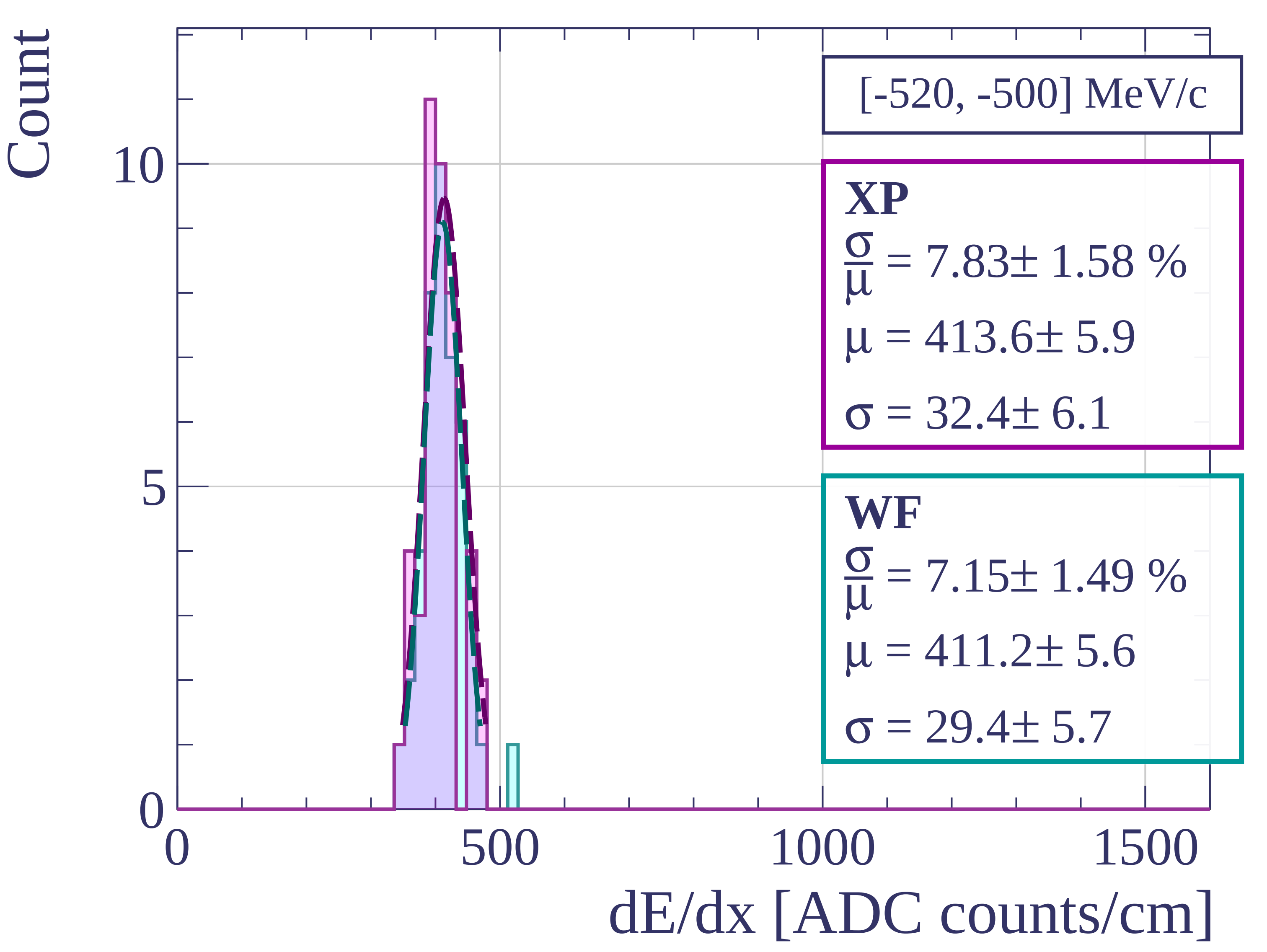


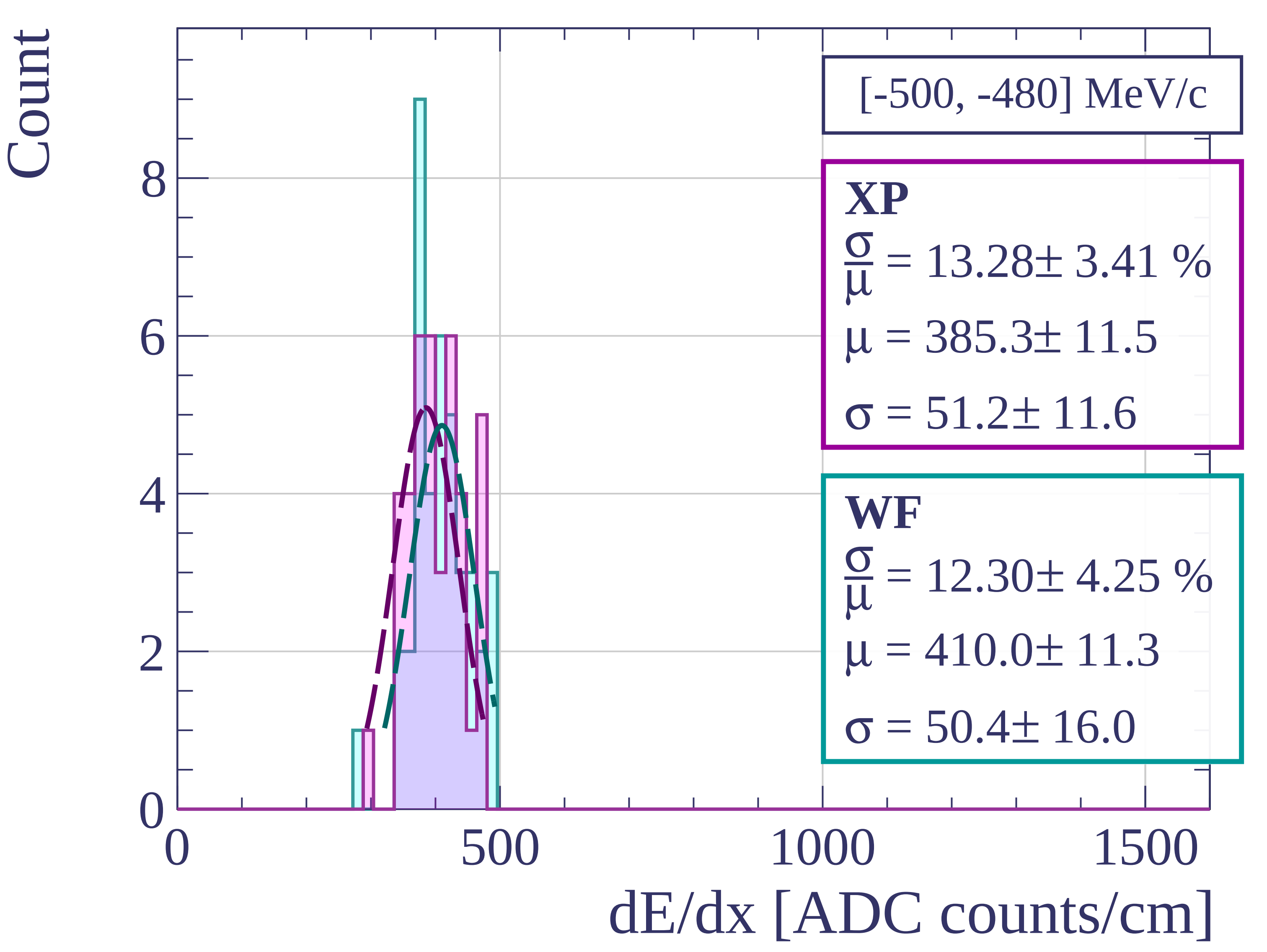
Count
dE/dx [ADC counts/cm]
[-580, -560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.48± 1.34 %
μ = 400.0± 5.6
σ = 21.9± 5.0
WF
$\frac{\sigma}{\mu}$ = 1.72± 9.30 %
μ = 464.0± 32.7
σ = 8.0± 42.6



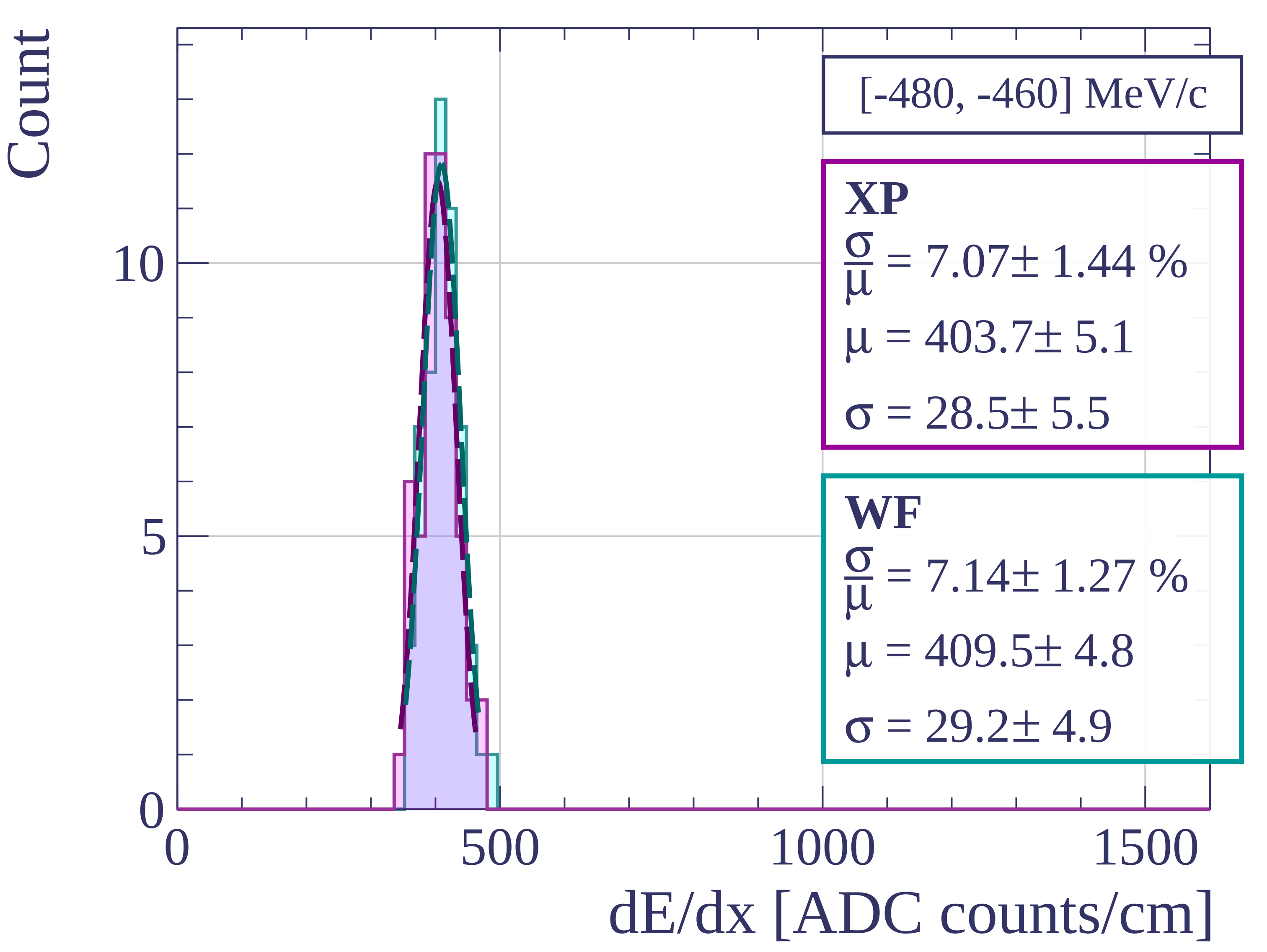
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-560, -540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.68± 3.52 %
μ = 437.2± 18.3
σ = 20.5± 14.5
WF
$\frac{\sigma}{\mu}$ = 10.49± 3.03 %
μ = 418.1± 11.6
σ = 43.9± 11.5


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
σ/μ = 16.93± 14.50 %
μ = 406.8± 33.2
σ = 68.9± 53.4
WF
σ/μ = 16.48± 15.50 %
μ = 444.9± 51.8
σ = 73.3± 60.4

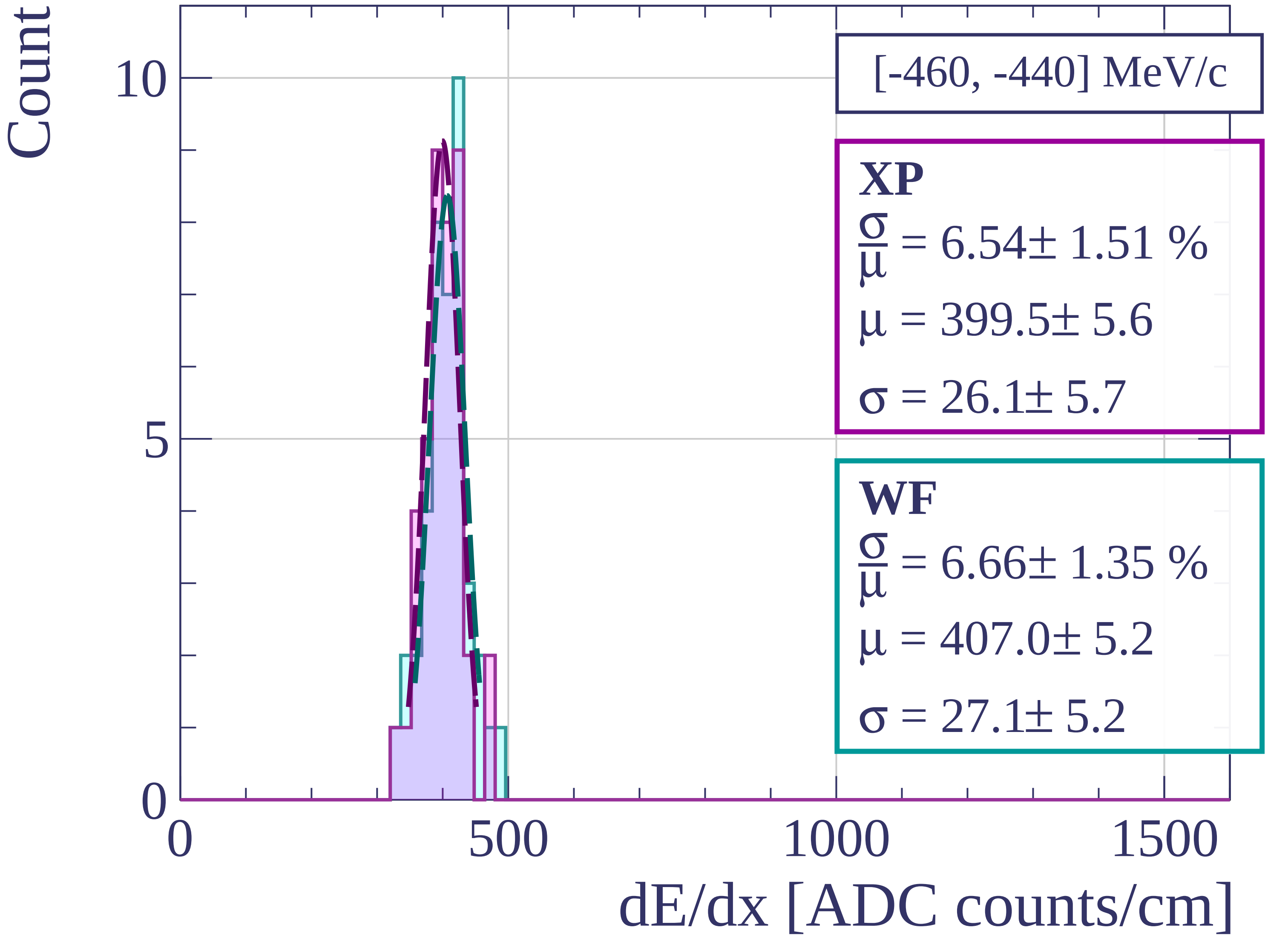

Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[-520, -500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.83± 1.58 %
μ = 413.6± 5.9
σ = 32.4± 6.1
WF
$\frac{\sigma}{\mu}$ = 7.15± 1.49 %
μ = 411.2± 5.6
σ = 29.4± 5.7



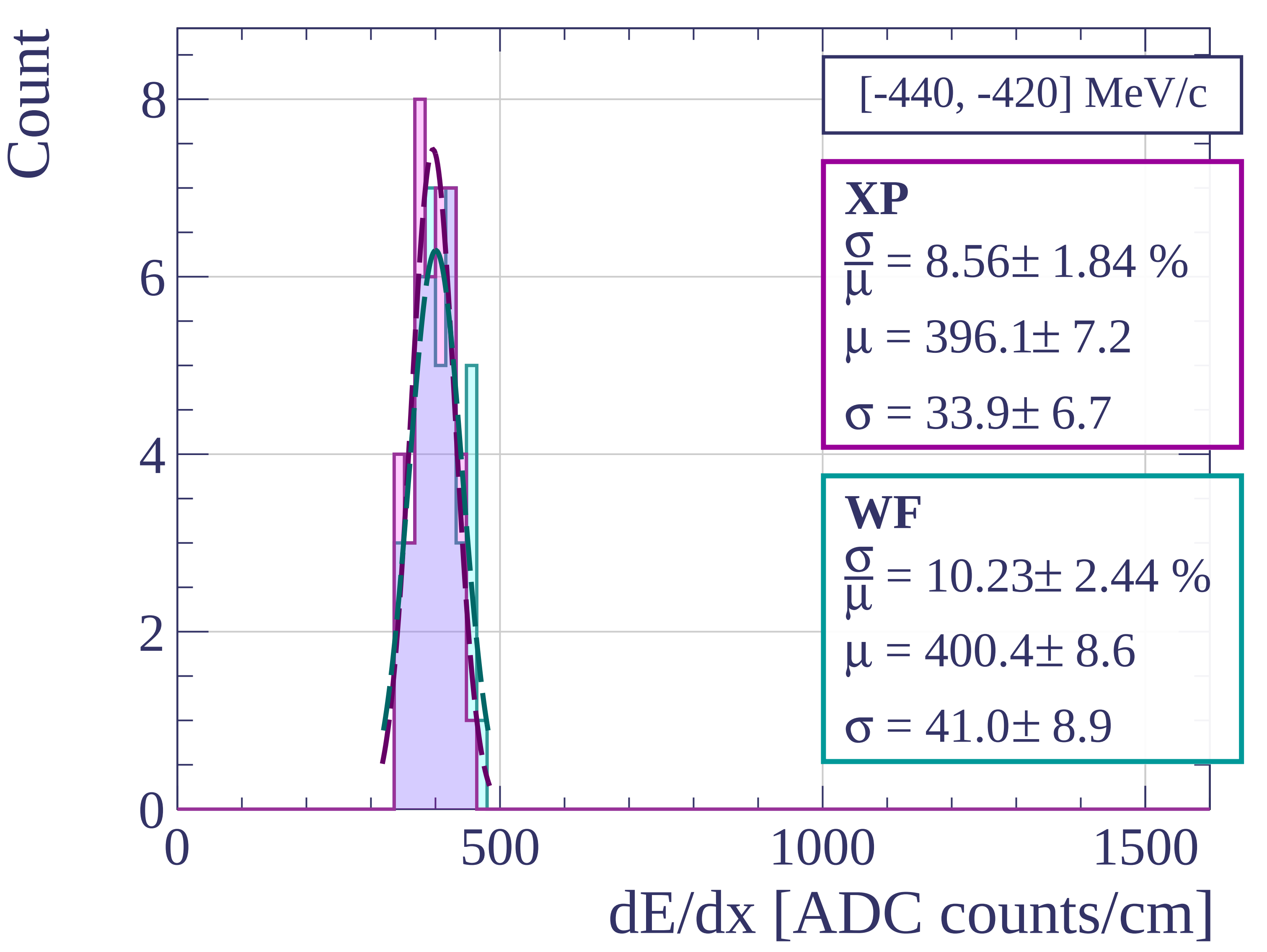
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-500, -480] MeV/c
XP
σ/μ = 13.28± 3.41 %
μ = 385.3± 11.5
σ = 51.2± 11.6
WF
σ/μ = 12.30± 4.25 %
μ = 410.0± 11.3
σ = 50.4± 16.0



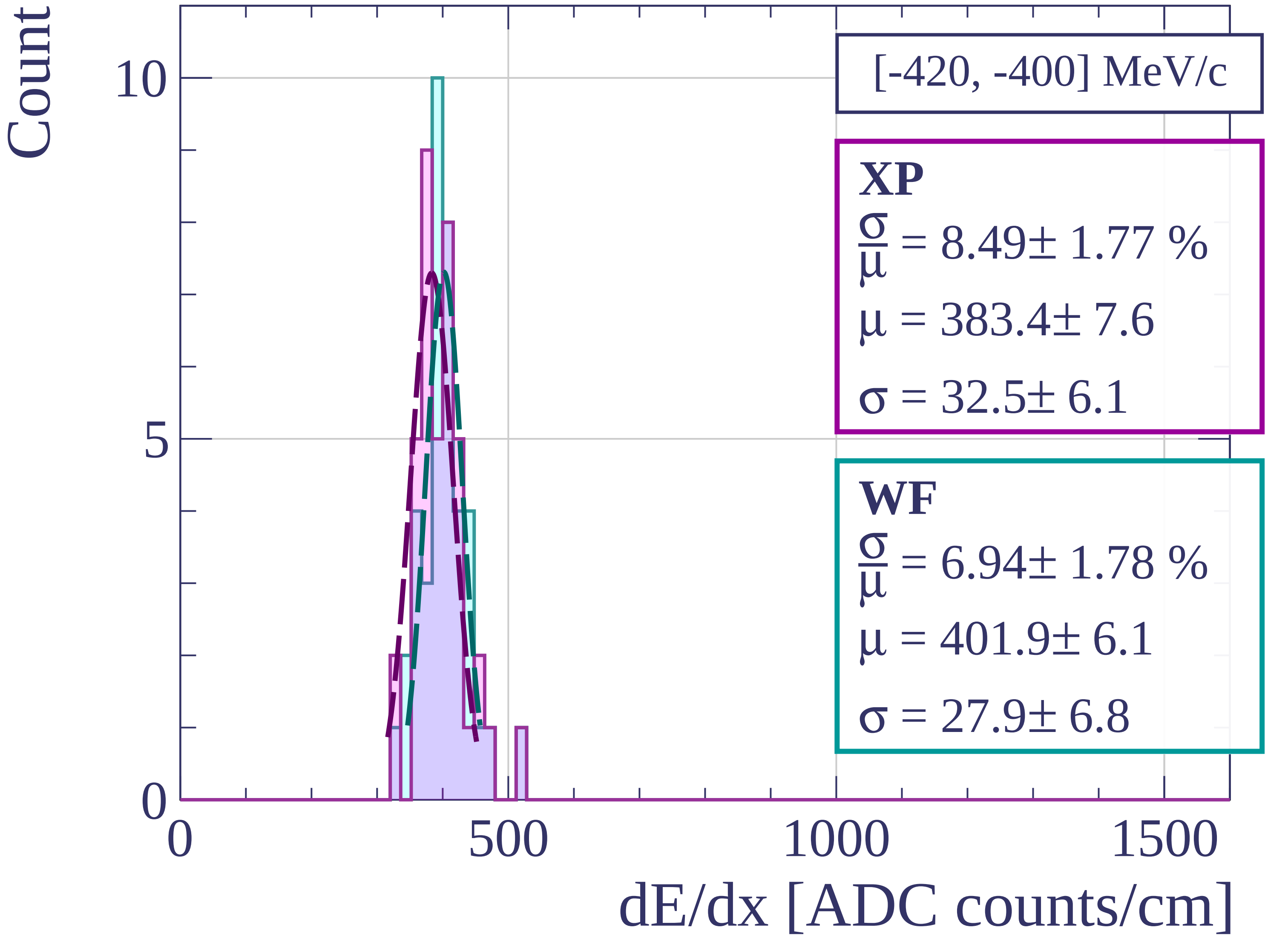
[-480, -460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.07± 1.44 %
μ = 403.7± 5.1
σ = 28.5± 5.5
WF
$\frac{\sigma}{\mu}$ = 7.14± 1.27 %
μ = 409.5± 4.8
σ = 29.2± 4.9
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500



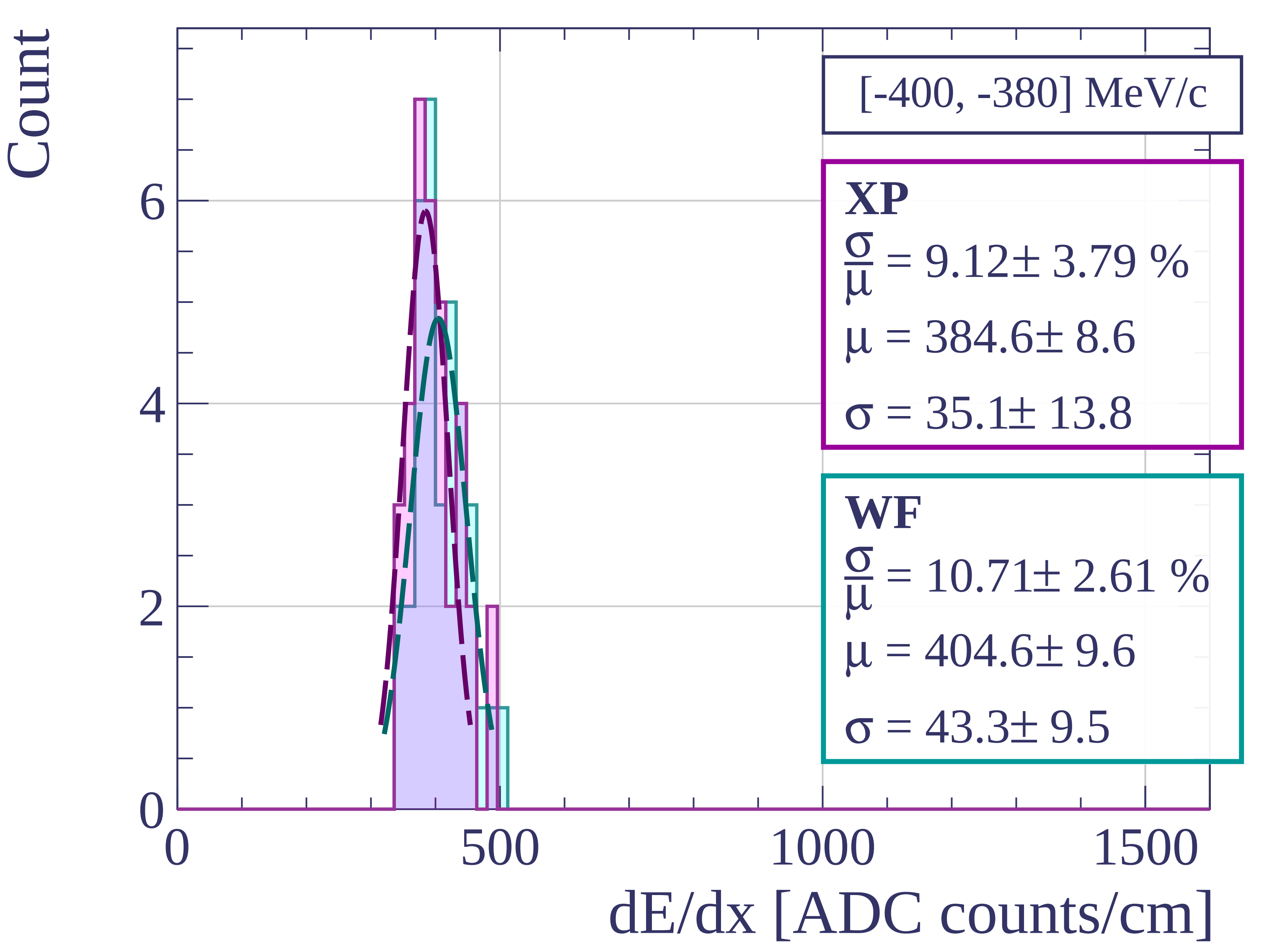
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
σ/μ = 6.54± 1.51 %
μ = 399.5± 5.6
σ = 26.1± 5.7
WF
σ/μ = 6.66± 1.35 %
μ = 407.0± 5.2
σ = 27.1± 5.2



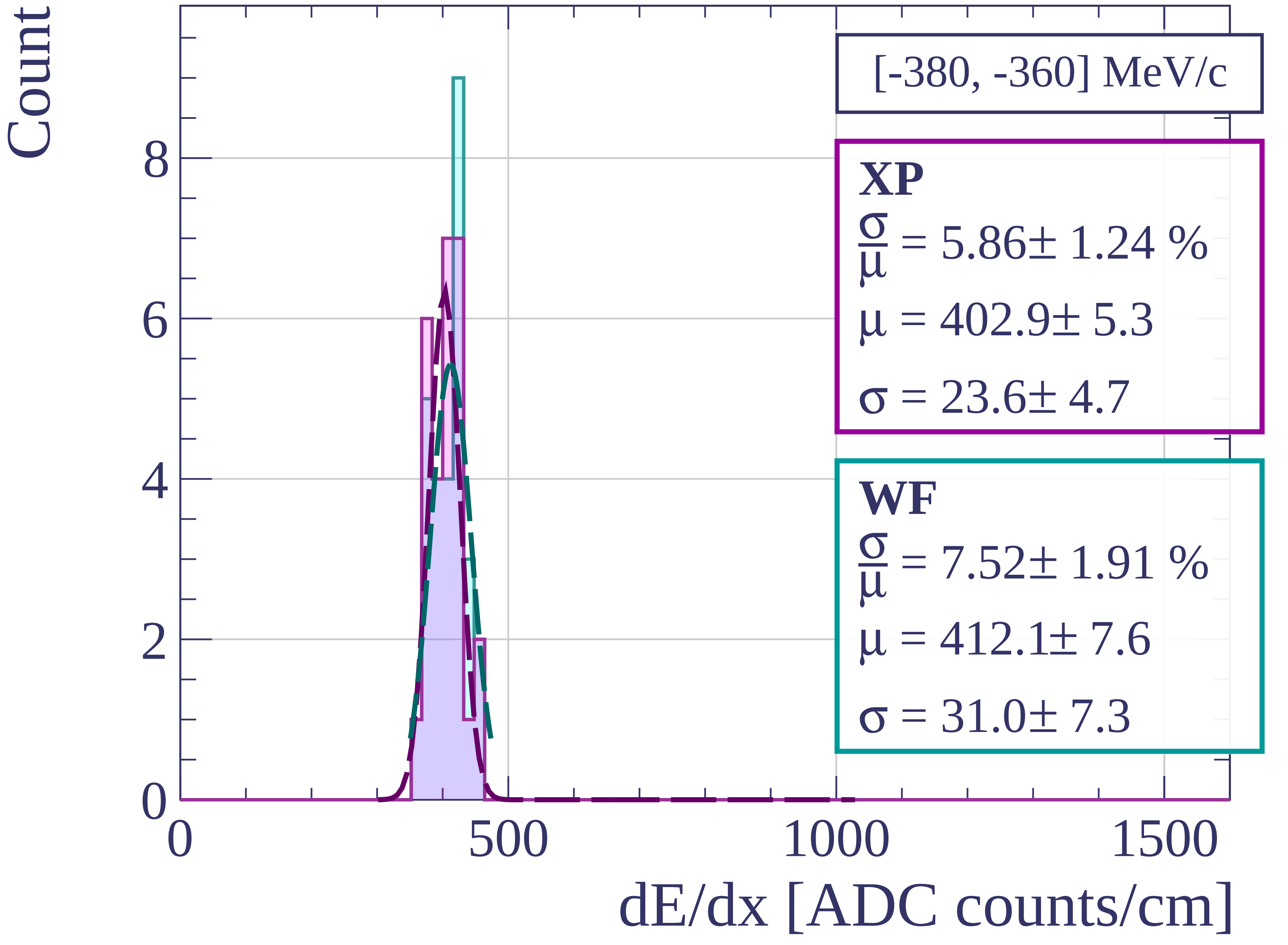
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-440, -420] MeV/c
XP
σ/μ = 8.56± 1.84 %
μ = 396.1± 7.2
σ = 33.9± 6.7
WF
σ/μ = 10.23± 2.44 %
μ = 400.4± 8.6
σ = 41.0± 8.9



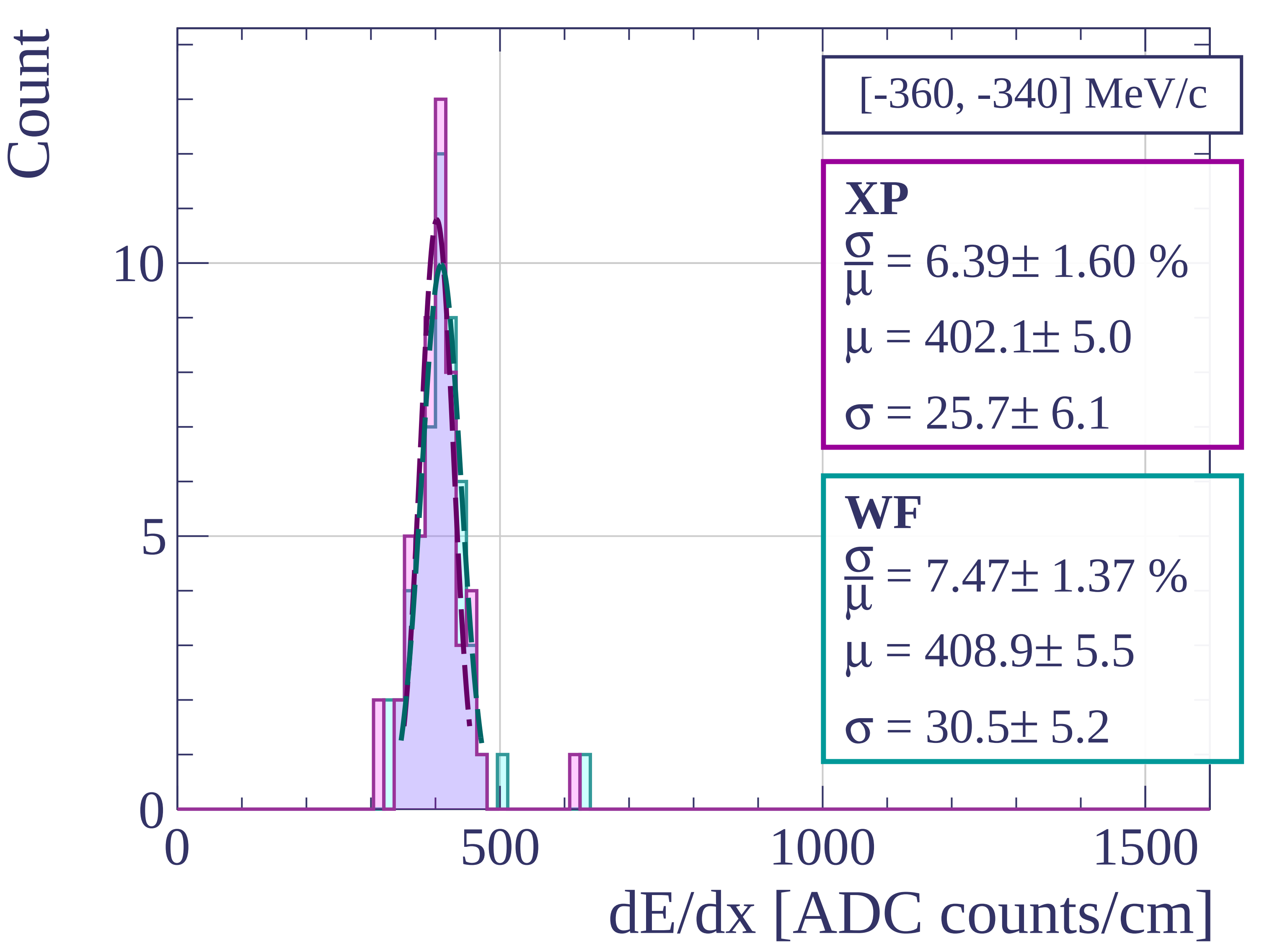
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.49± 1.77 %
μ = 383.4± 7.6
σ = 32.5± 6.1
WF
$\frac{\sigma}{\mu}$ = 6.94± 1.78 %
μ = 401.9± 6.1
σ = 27.9± 6.8



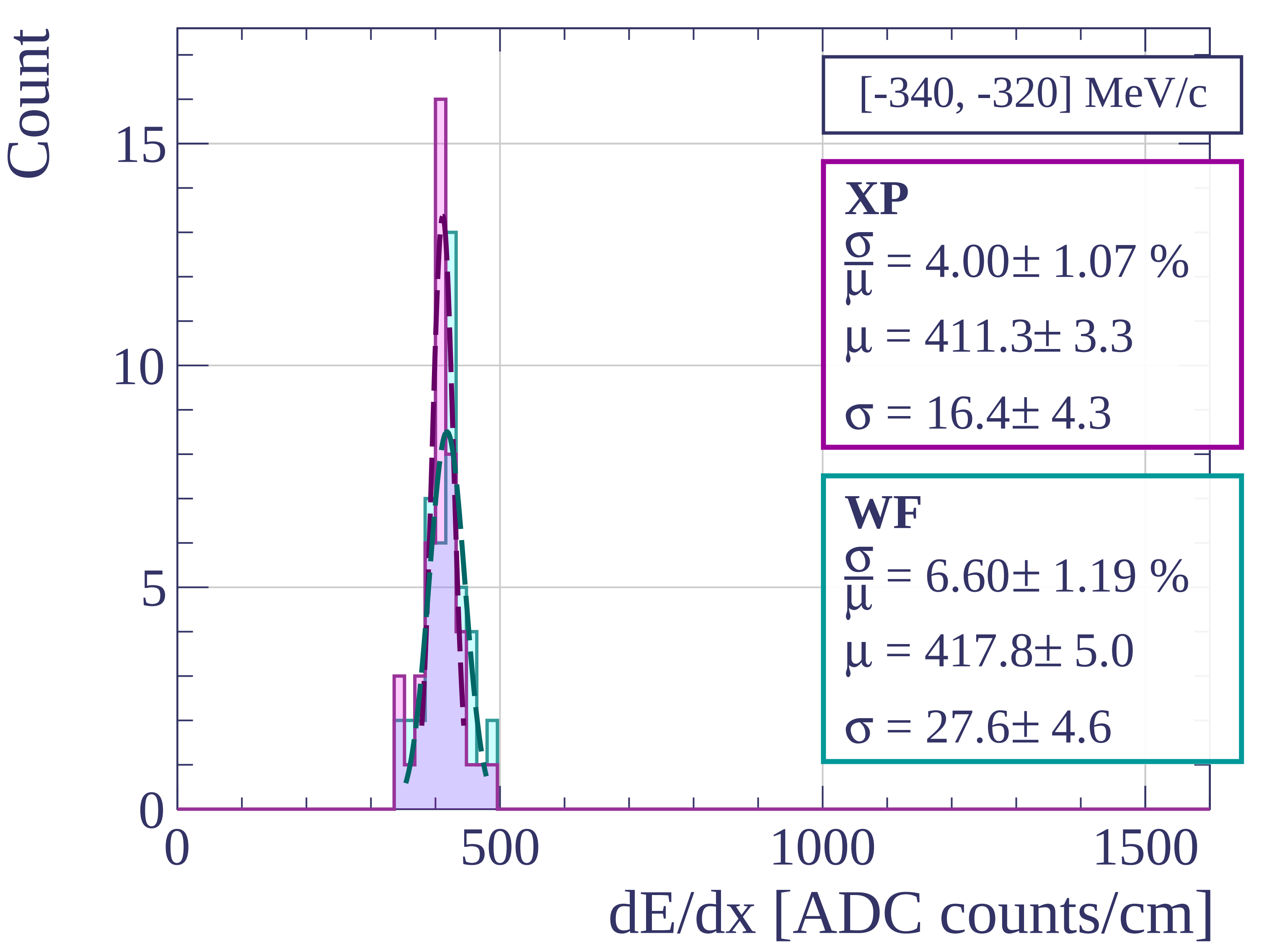
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.12± 3.79 %
μ = 384.6± 8.6
σ = 35.1± 13.8
WF
$\frac{\sigma}{\mu}$ = 10.71± 2.61 %
μ = 404.6± 9.6
σ = 43.3± 9.5



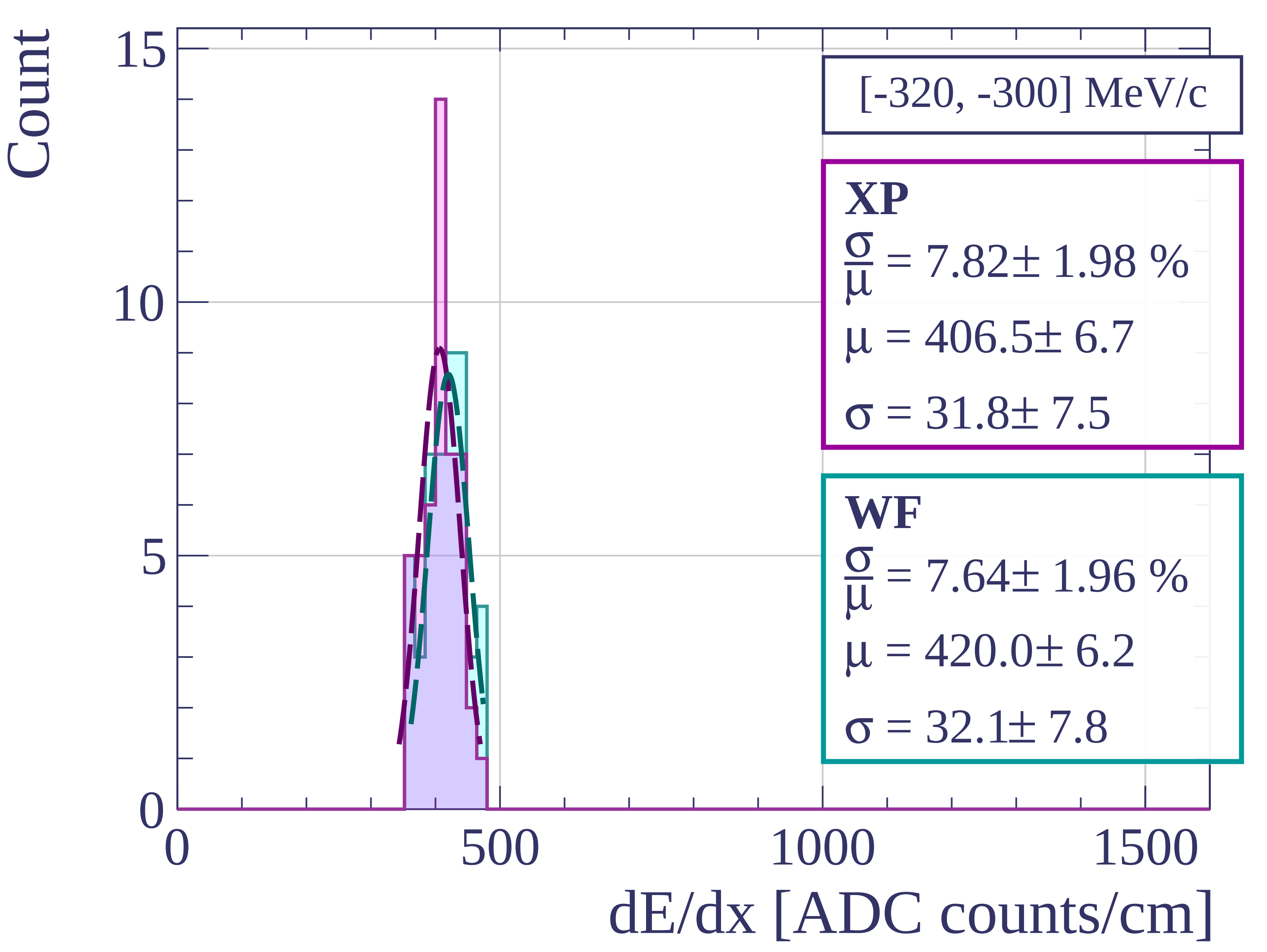
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[-380, -360] MeV/c
XP
σ/μ = 5.86± 1.24 %
μ = 402.9± 5.3
σ = 23.6± 4.7
WF
σ/μ = 7.52± 1.91 %
μ = 412.1± 7.6
σ = 31.0± 7.3



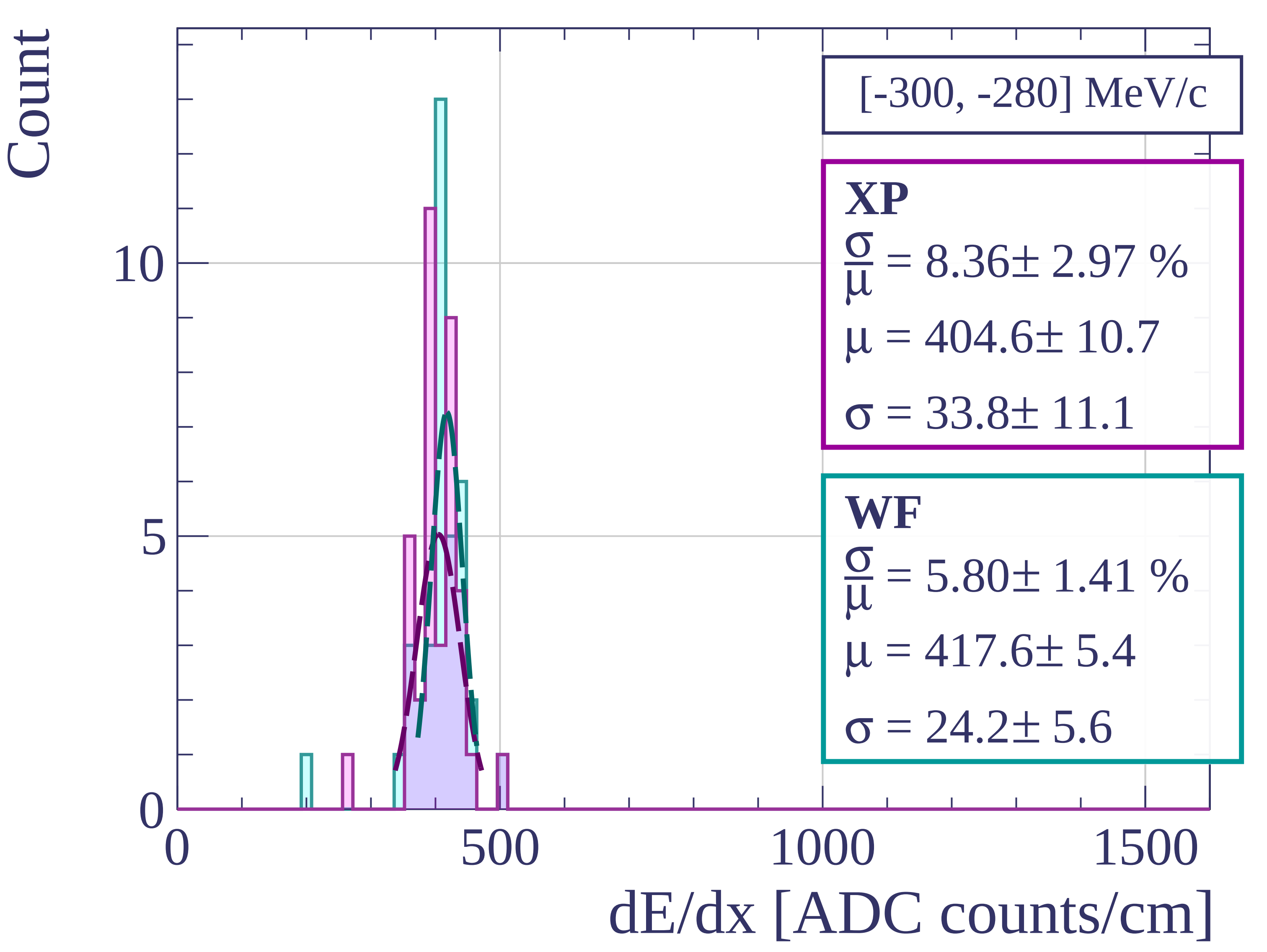
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
5
10
[-360, -340] MeV/c
XP
σ/μ = 6.39± 1.60 %
μ = 402.1± 5.0
σ = 25.7± 6.1
WF
σ/μ = 7.47± 1.37 %
μ = 408.9± 5.5
σ = 30.5± 5.2



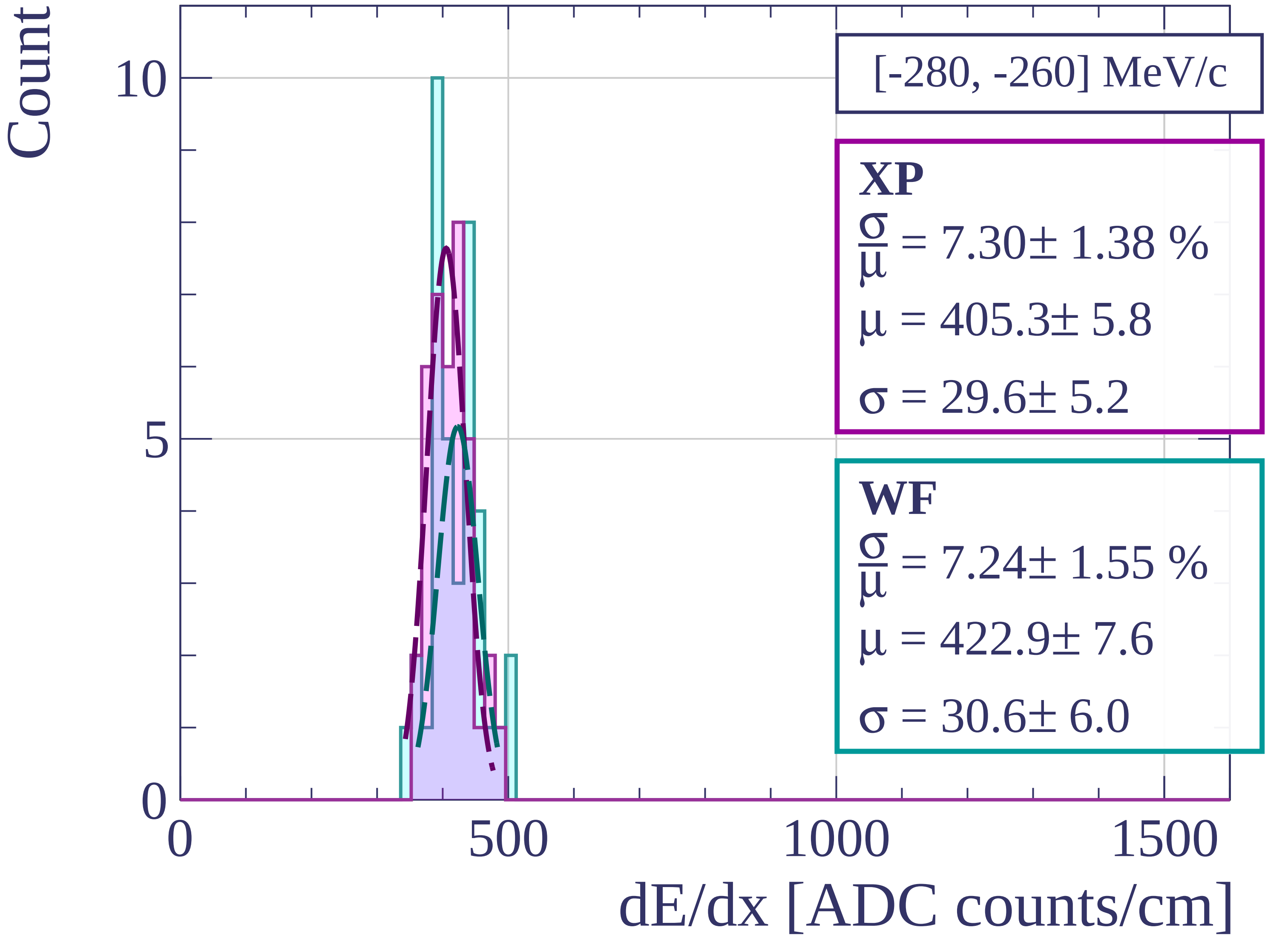
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
[-340, -320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.00± 1.07 %
μ = 411.3± 3.3
σ = 16.4± 4.3
WF
$\frac{\sigma}{\mu}$ = 6.60± 1.19 %
μ = 417.8± 5.0
σ = 27.6± 4.6

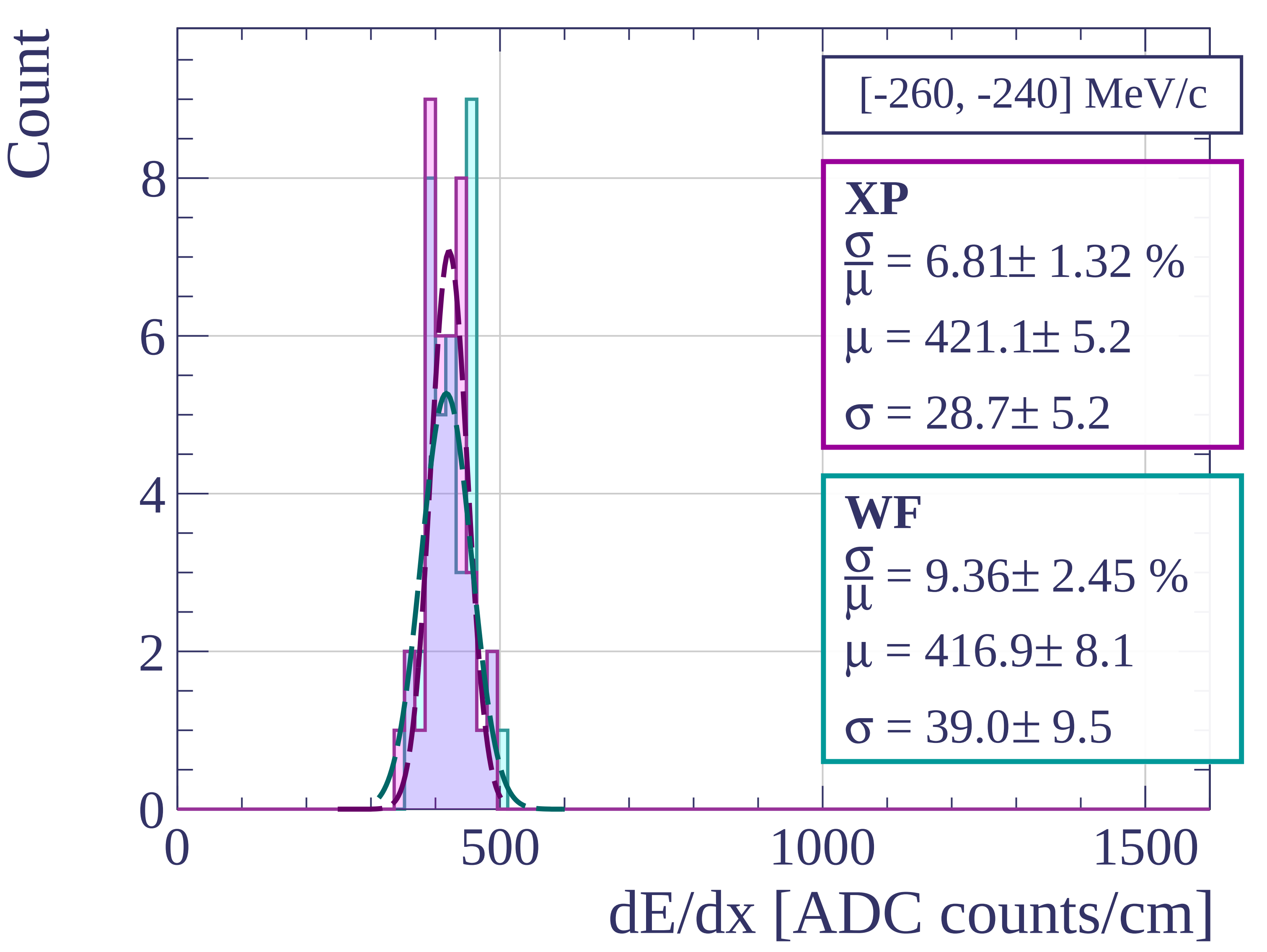


Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-320, -300] MeV/c
XP
σ/μ = 7.82± 1.98 %
μ = 406.5± 6.7
σ = 31.8± 7.5
WF
σ/μ = 7.64± 1.96 %
μ = 420.0± 6.2
σ = 32.1± 7.8

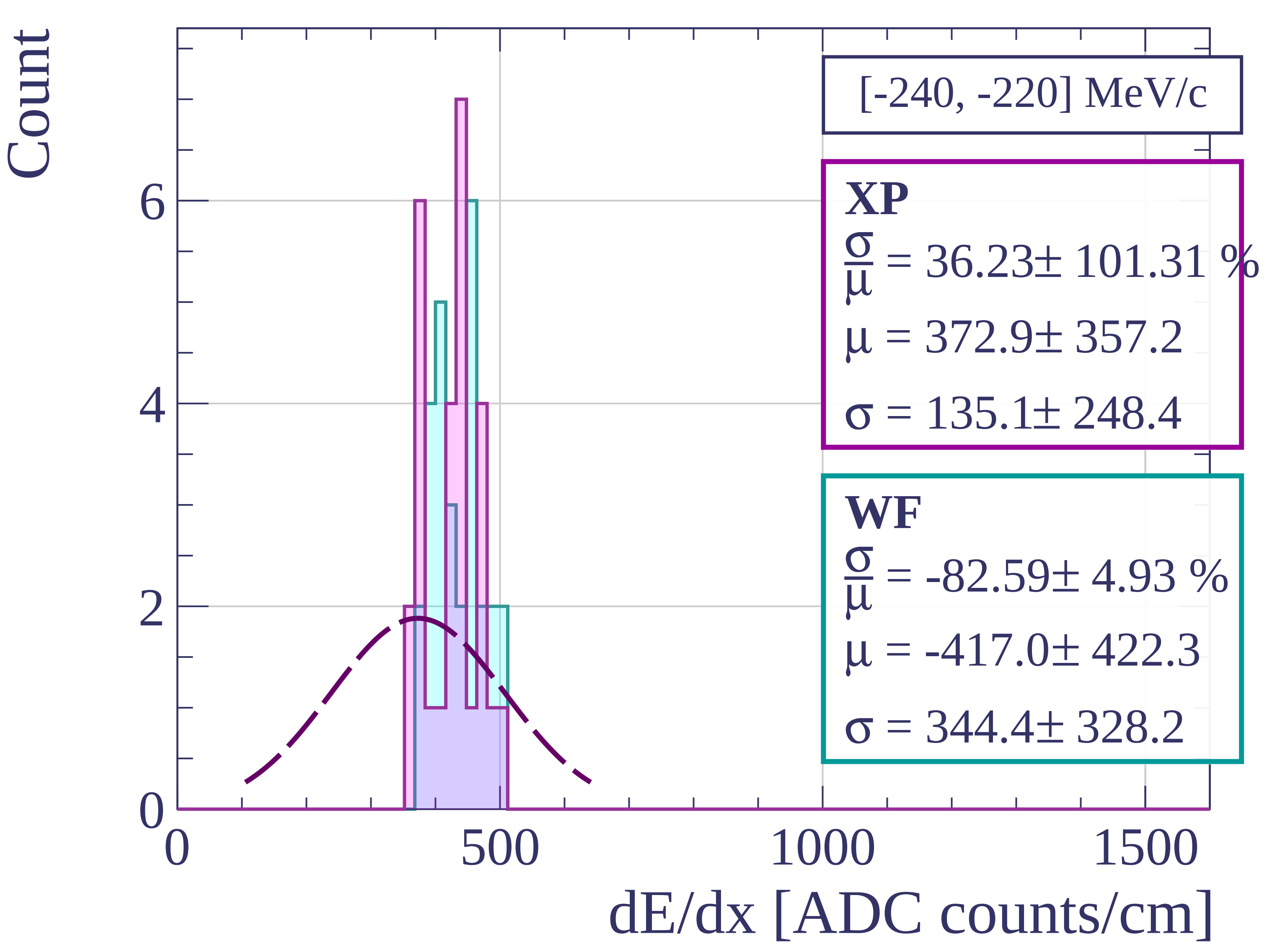


Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[-300, -280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.36± 2.97 %
μ = 404.6± 10.7
σ = 33.8± 11.1
WF
$\frac{\sigma}{\mu}$ = 5.80± 1.41 %
μ = 417.6± 5.4
σ = 24.2± 5.6

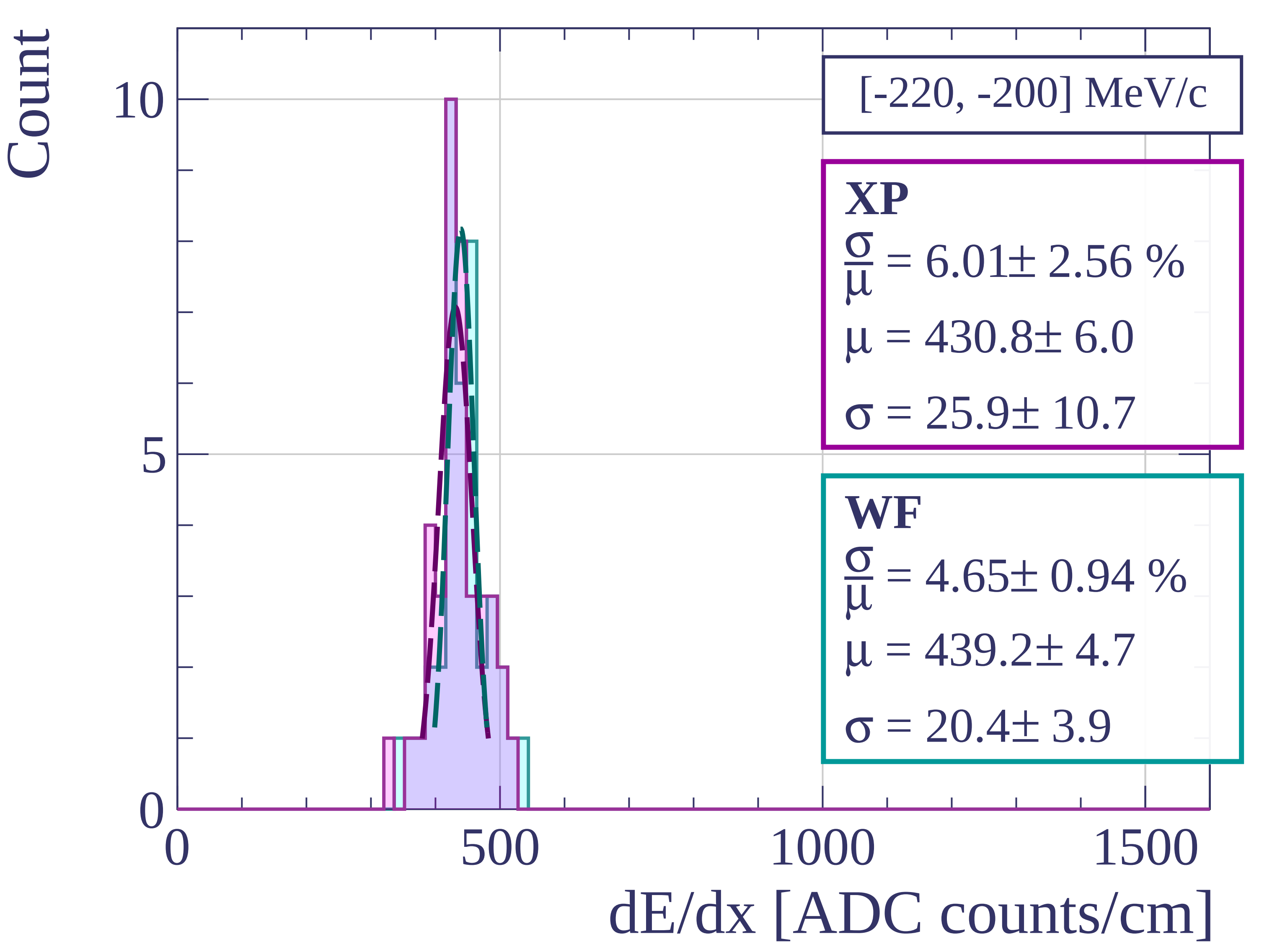

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-280, -260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.30± 1.38 %
μ = 405.3± 5.8
σ = 29.6± 5.2
WF
$\frac{\sigma}{\mu}$ = 7.24± 1.55 %
μ = 422.9± 7.6
σ = 30.6± 6.0

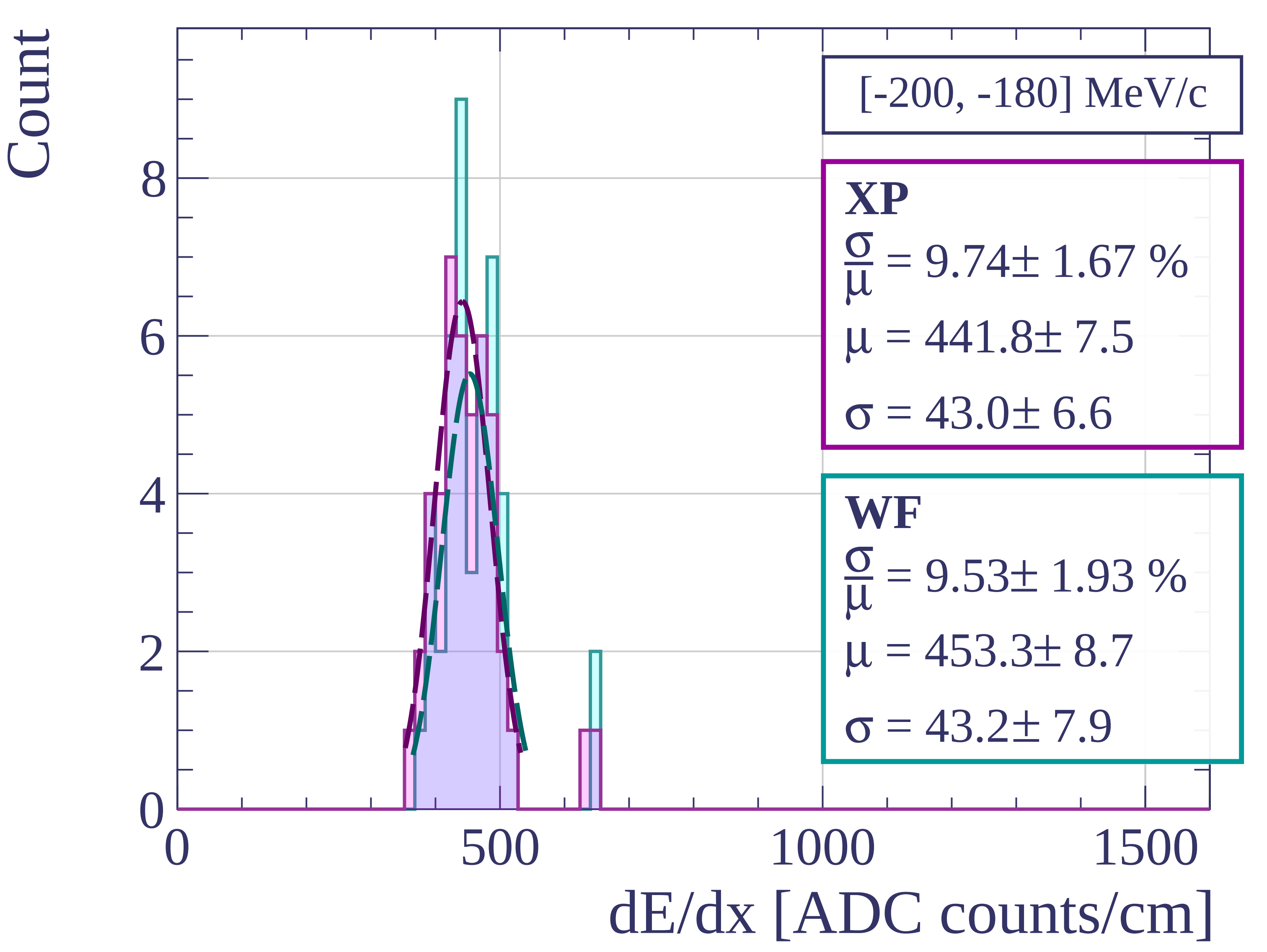


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-260, -240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.81± 1.32 %
μ = 421.1± 5.2
σ = 28.7± 5.2
WF
$\frac{\sigma}{\mu}$ = 9.36± 2.45 %
μ = 416.9± 8.1
σ = 39.0± 9.5

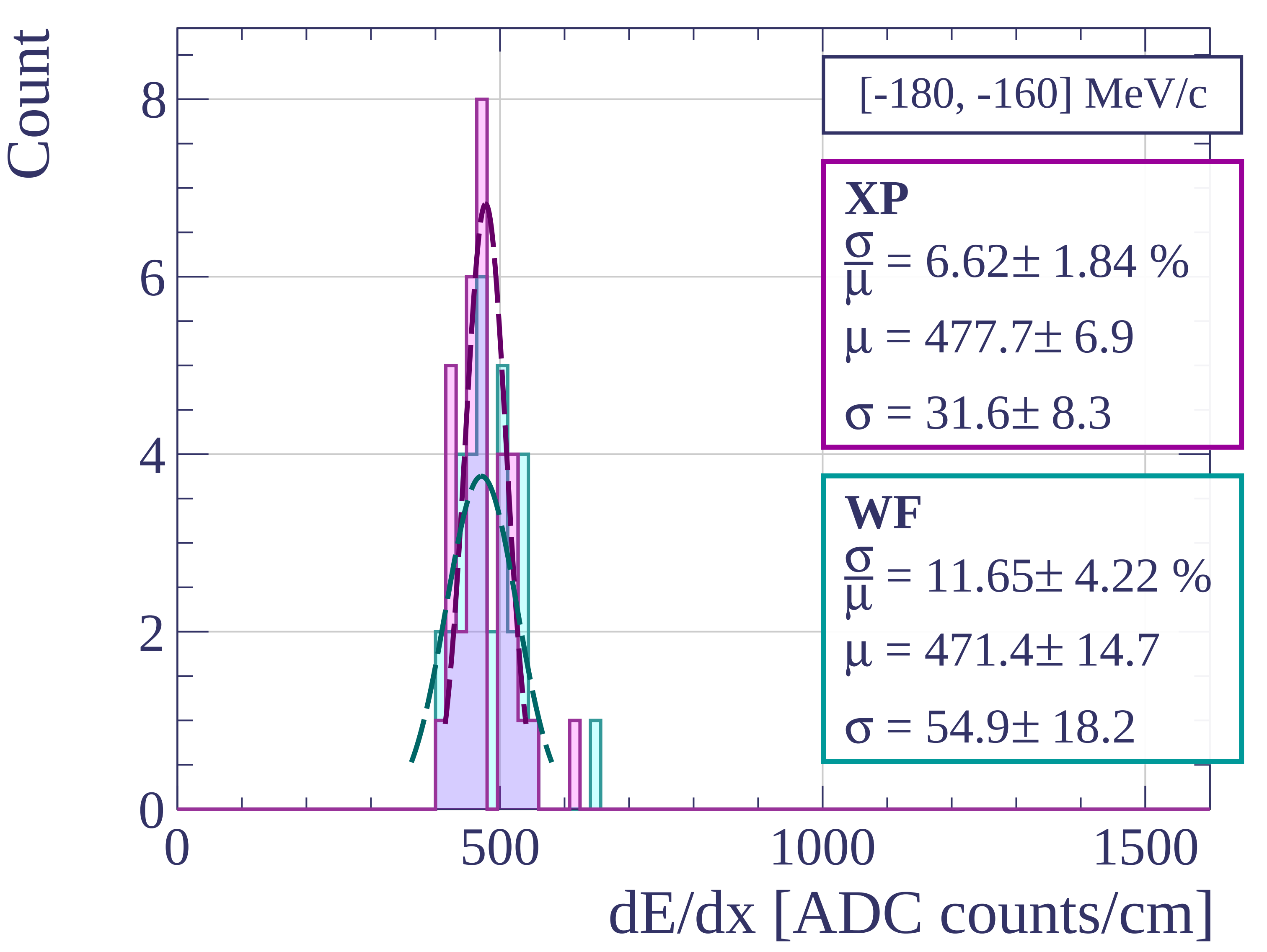


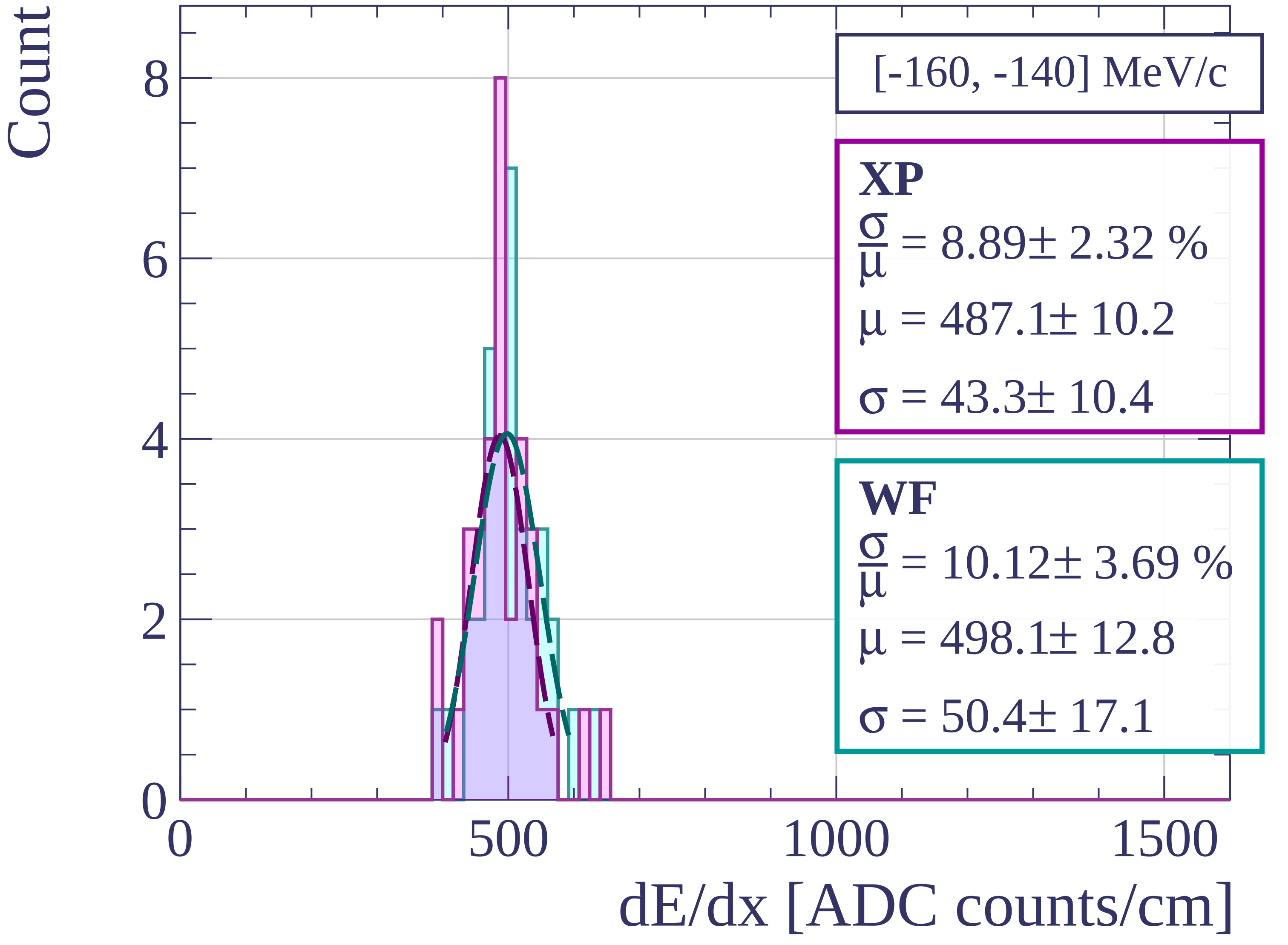
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-240, -220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 36.23± 101.31 %
μ = 372.9± 357.2
σ = 135.1± 248.4
WF
$\frac{\sigma}{\mu}$ = -82.59± 4.93 %
μ = -417.0± 422.3
σ = 344.4± 328.2


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
σ/μ = 6.01± 2.56 %
μ = 430.8± 6.0
σ = 25.9± 10.7
WF
σ/μ = 4.65± 0.94 %
μ = 439.2± 4.7
σ = 20.4± 3.9

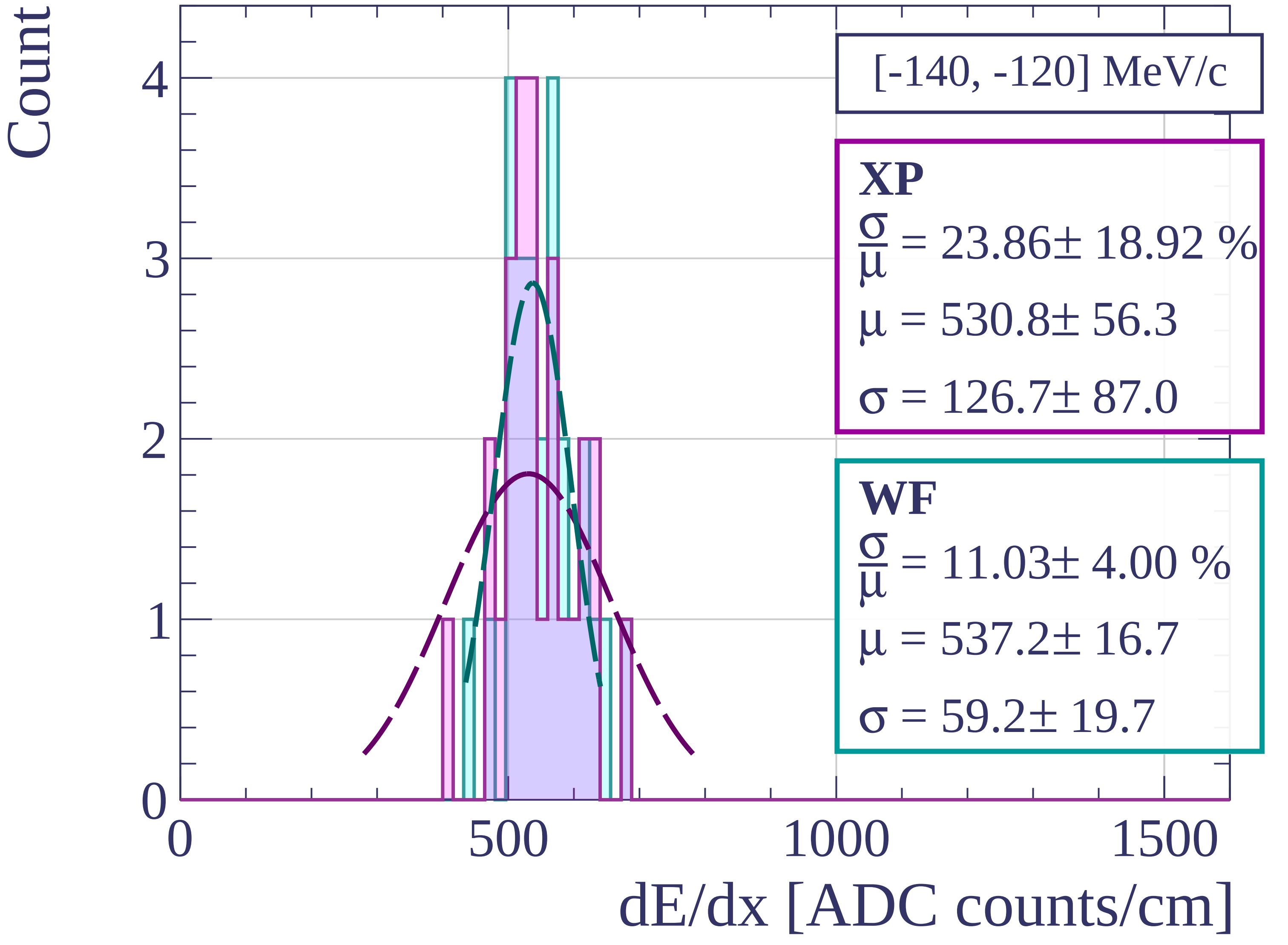


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-200, -180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.74± 1.67 %
μ = 441.8± 7.5
σ = 43.0± 6.6
WF
$\frac{\sigma}{\mu}$ = 9.53± 1.93 %
μ = 453.3± 8.7
σ = 43.2± 7.9

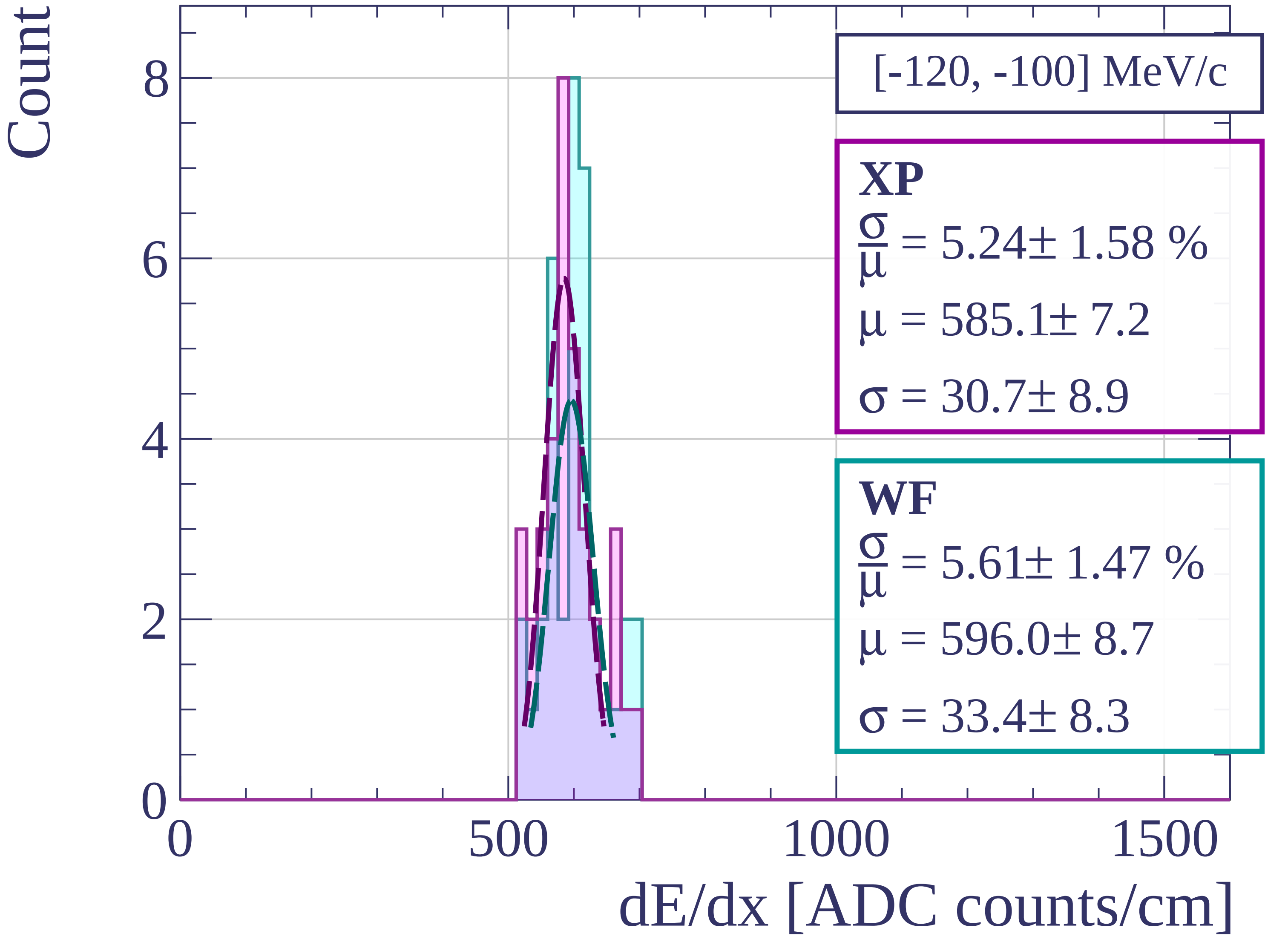

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.62± 1.84 %
μ = 477.7± 6.9
σ = 31.6± 8.3
WF
$\frac{\sigma}{\mu}$ = 11.65± 4.22 %
μ = 471.4± 14.7
σ = 54.9± 18.2



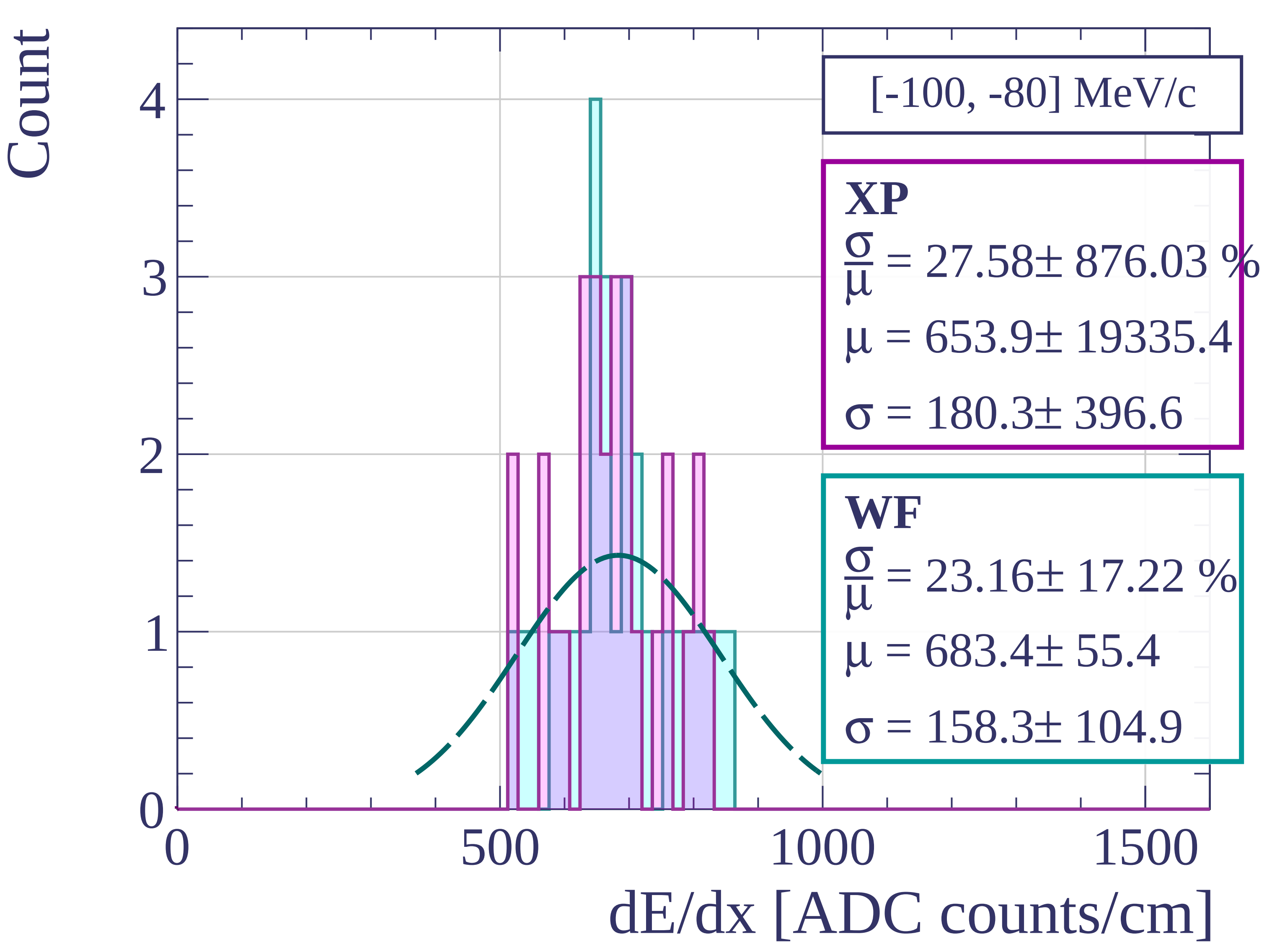
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.89± 2.32 %
μ = 487.1± 10.2
σ = 43.3± 10.4
WF
$\frac{\sigma}{\mu}$ = 10.12± 3.69 %
μ = 498.1± 12.8
σ = 50.4± 17.1

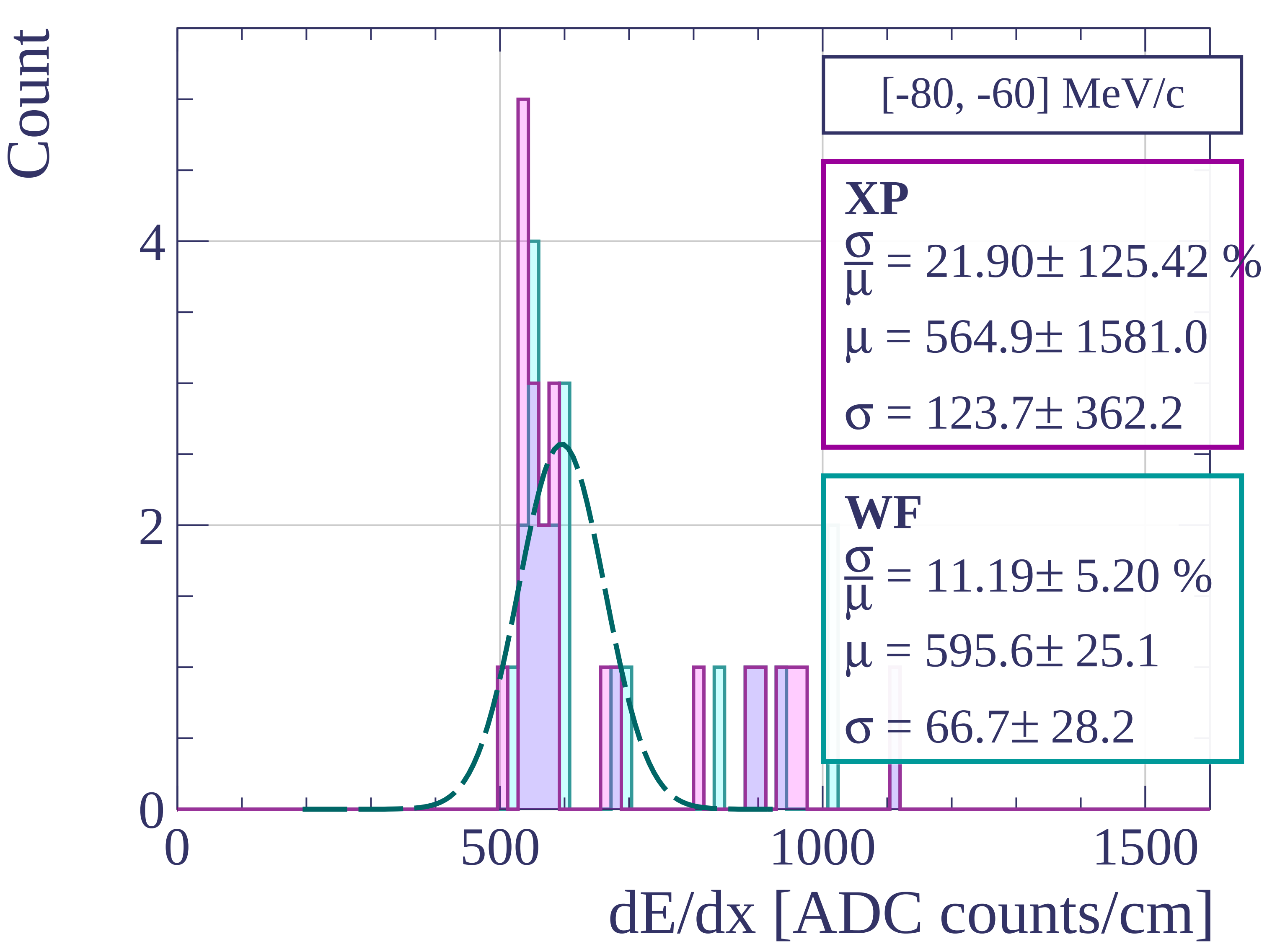


Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.86± 18.92 %
μ = 530.8± 56.3
σ = 126.7± 87.0
WF
$\frac{\sigma}{\mu}$ = 11.03± 4.00 %
μ = 537.2± 16.7
σ = 59.2± 19.7

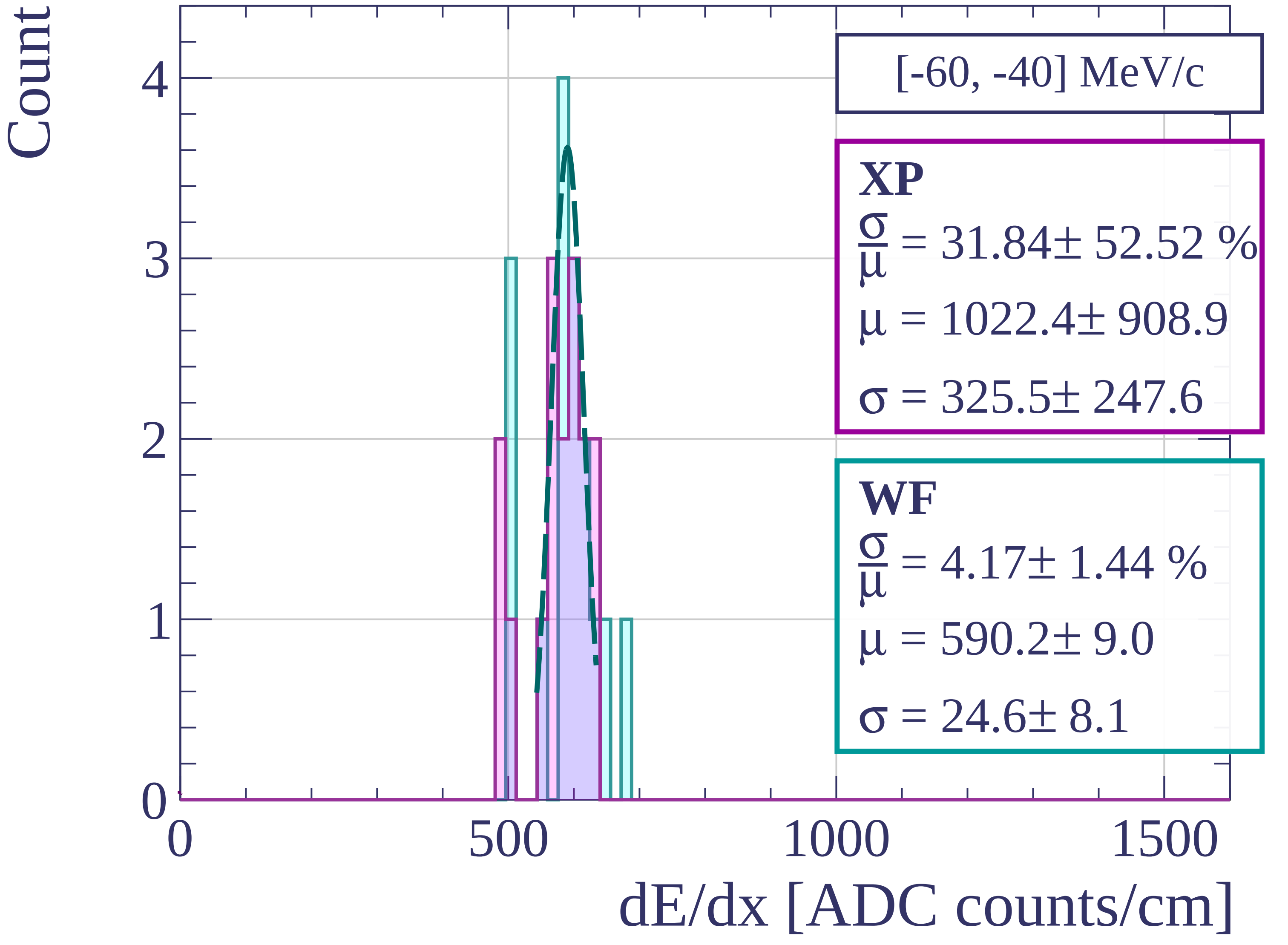


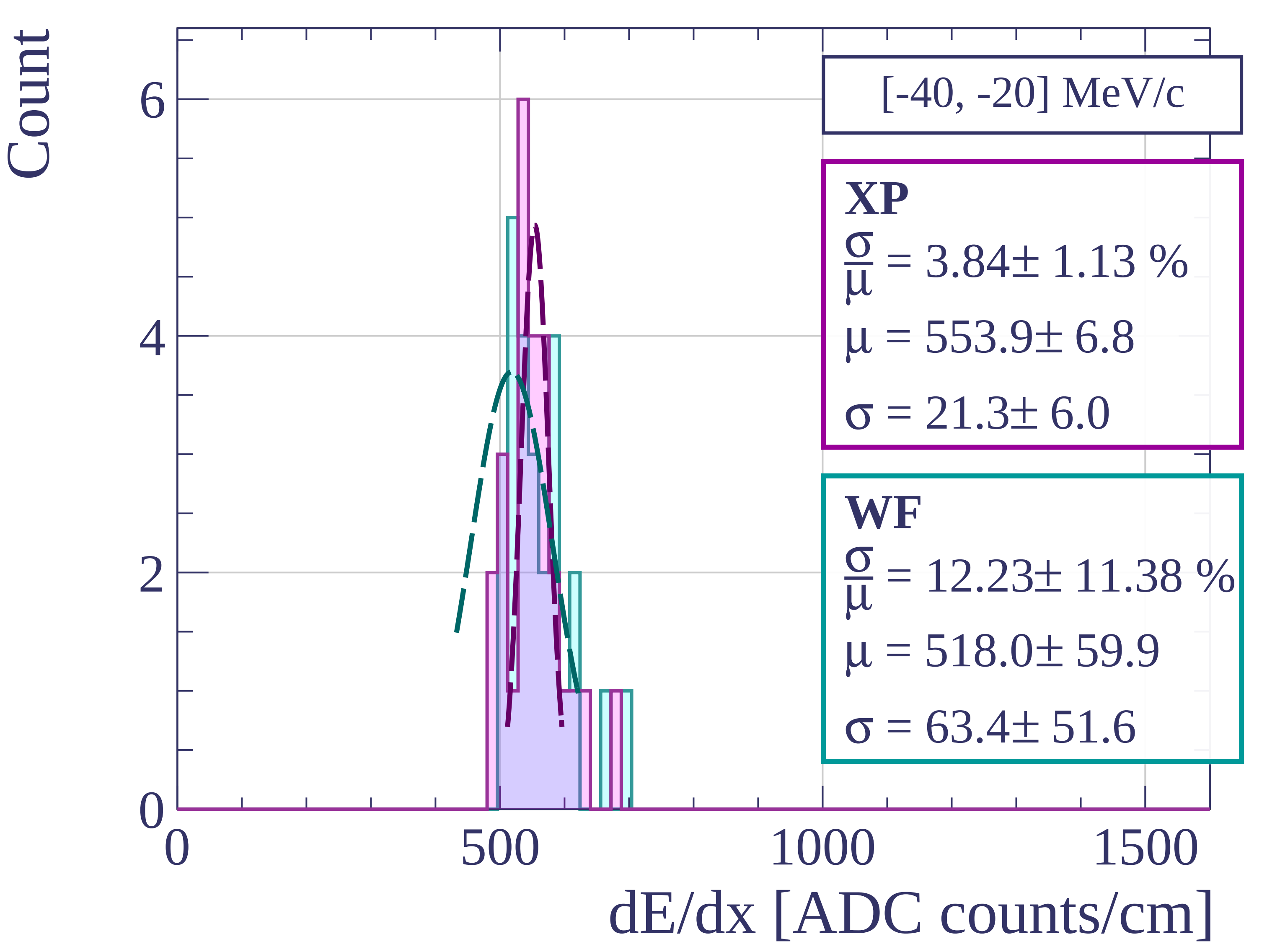
Count
dE/dx [ADC counts/cm]
[-120, -100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.24± 1.58 %
μ = 585.1± 7.2
σ = 30.7± 8.9
WF
$\frac{\sigma}{\mu}$ = 5.61± 1.47 %
μ = 596.0± 8.7
σ = 33.4± 8.3
0
2
4
6
8
0
500
1000
1500


Count
dE/dx [ADC counts/cm]
[-100, -80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 27.58± 876.03 %
μ = 653.9± 19335.4
σ = 180.3± 396.6
WF
$\frac{\sigma}{\mu}$ = 23.16± 17.22 %
μ = 683.4± 55.4
σ = 158.3± 104.9

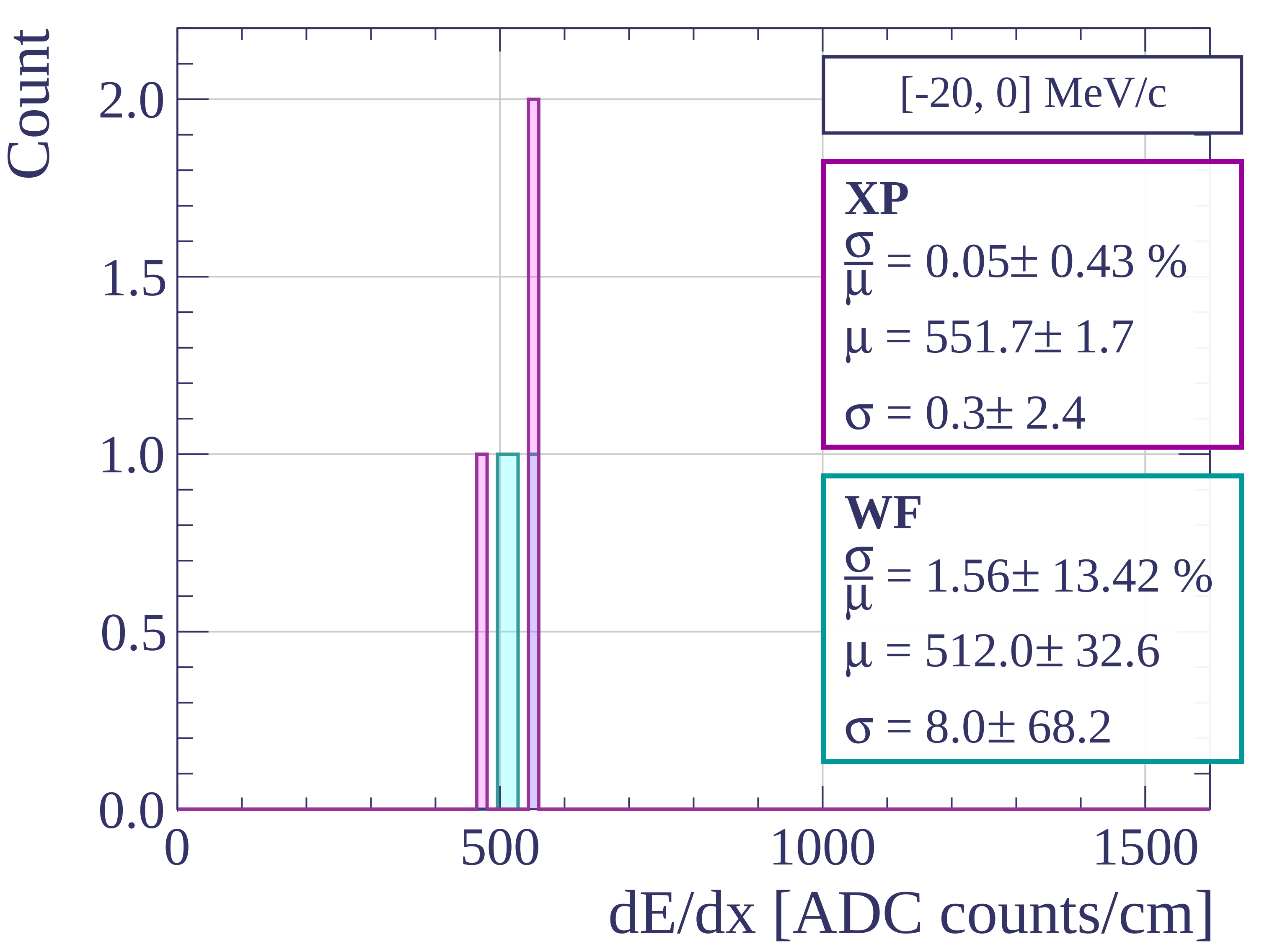


Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[-80, -60] MeV/c
XP
σ/μ = 21.90± 125.42 %
μ = 564.9± 1581.0
σ = 123.7± 362.2
WF
σ/μ = 11.19± 5.20 %
μ = 595.6± 25.1
σ = 66.7± 28.2

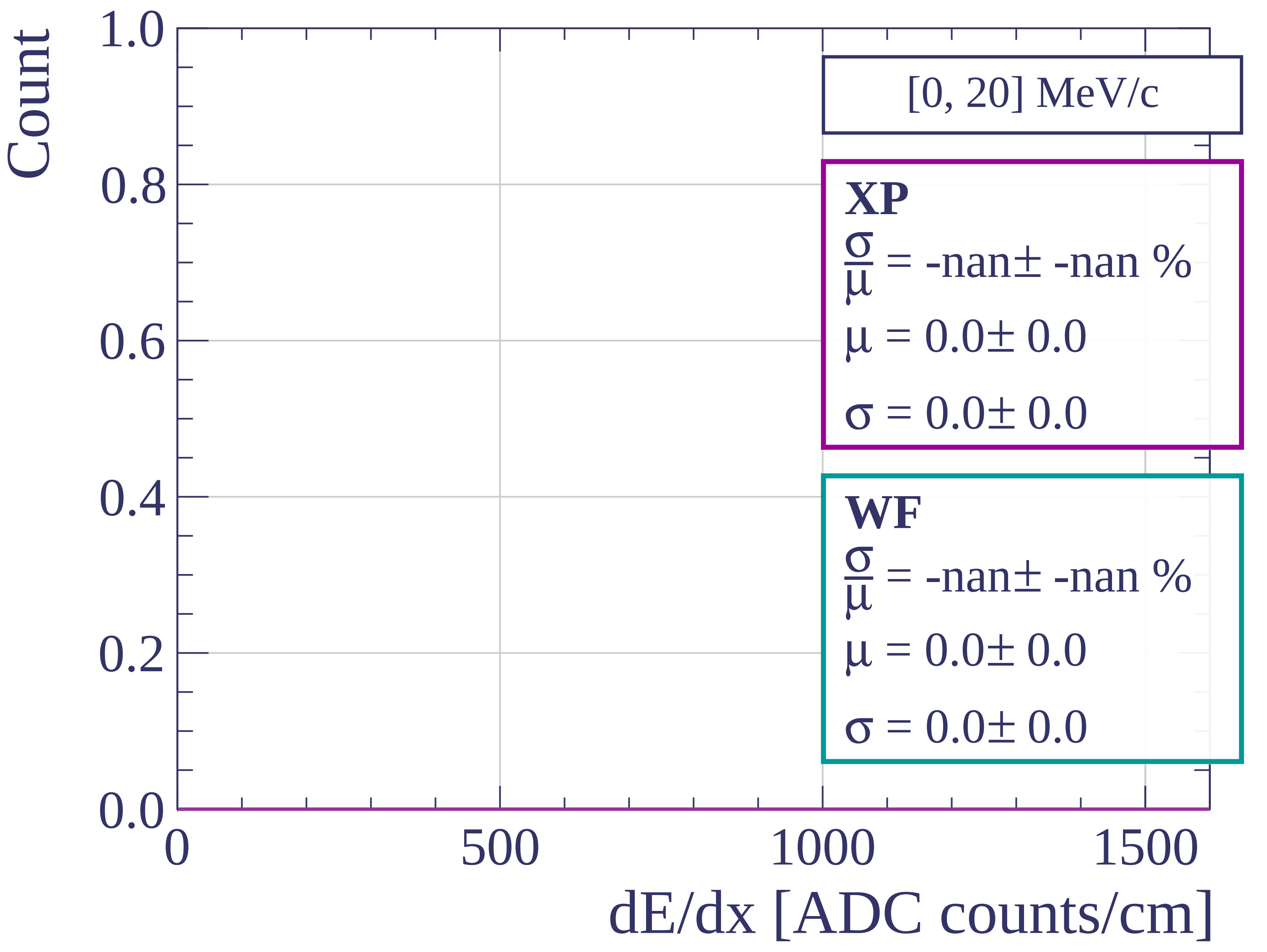

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1
2
3
4
[-60, -40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 31.84± 52.52 %
μ = 1022.4± 908.9
σ = 325.5± 247.6
WF
$\frac{\sigma}{\mu}$ = 4.17± 1.44 %
μ = 590.2± 9.0
σ = 24.6± 8.1



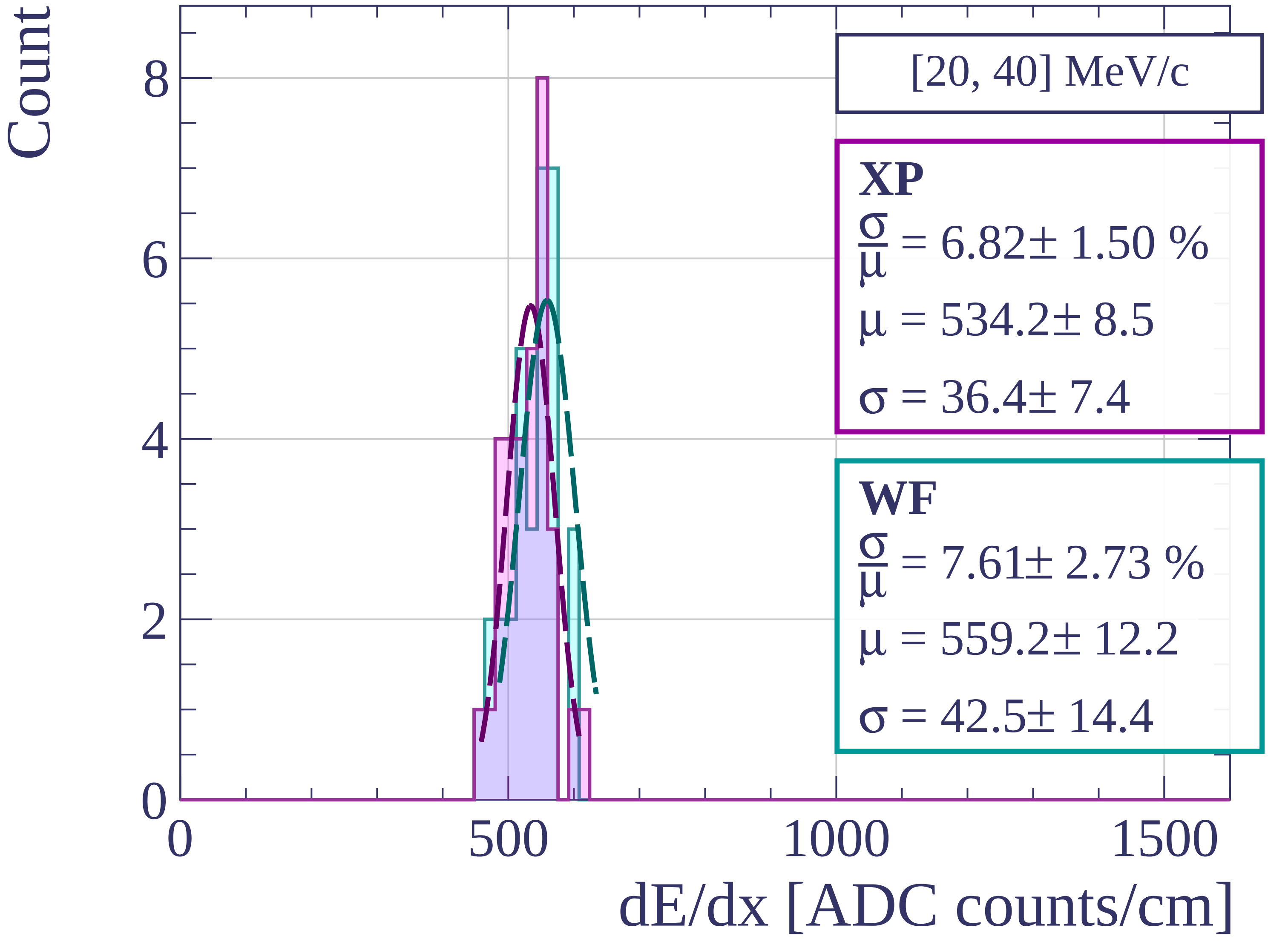
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[-40, -20] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.84± 1.13 %
μ = 553.9± 6.8
σ = 21.3± 6.0
WF
$\frac{\sigma}{\mu}$ = 12.23± 11.38 %
μ = 518.0± 59.9
σ = 63.4± 51.6



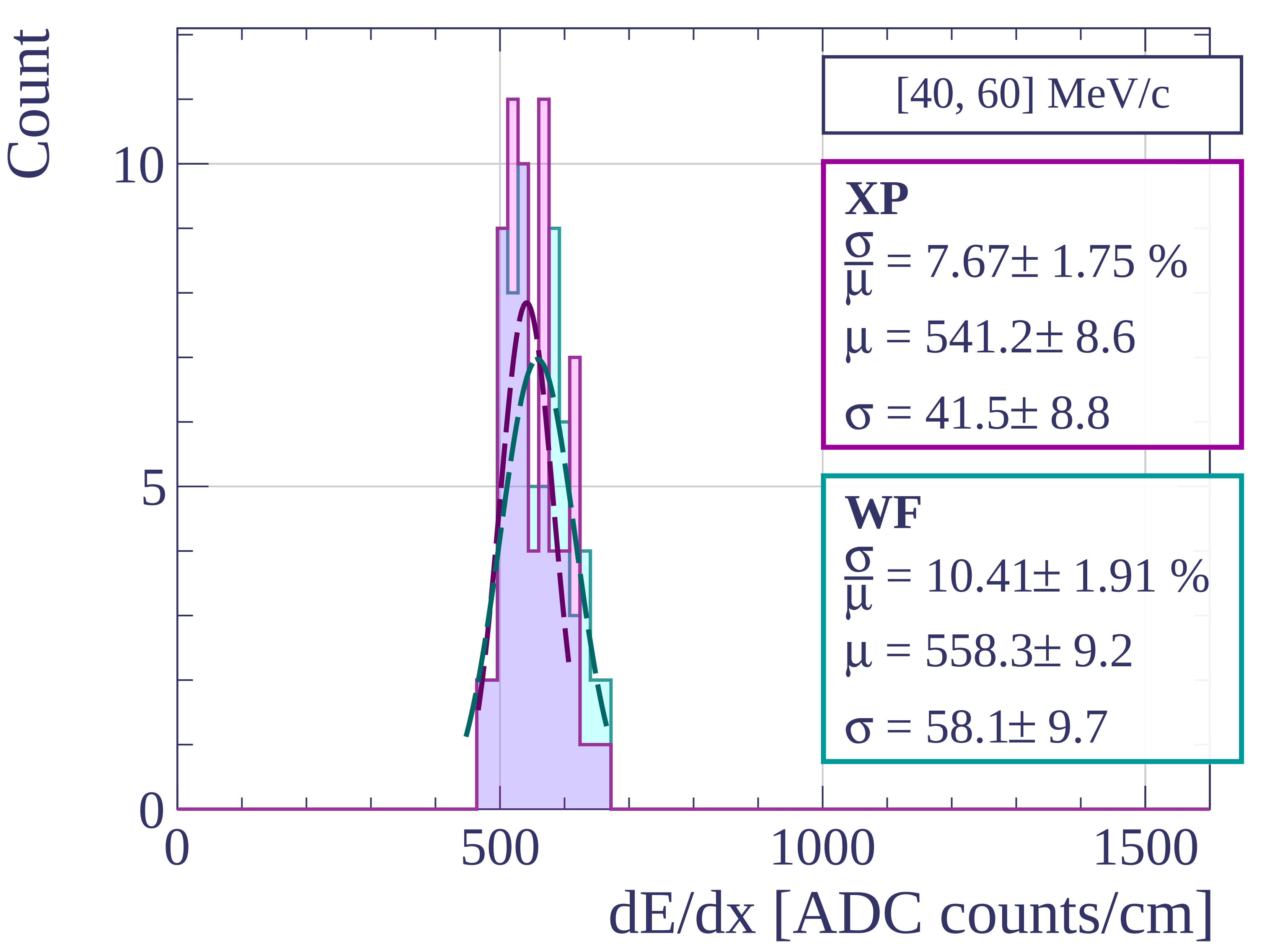
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.43 %
μ = 551.7± 1.7
σ = 0.3± 2.4
WF
$\frac{\sigma}{\mu}$ = 1.56± 13.42 %
μ = 512.0± 32.6
σ = 8.0± 68.2



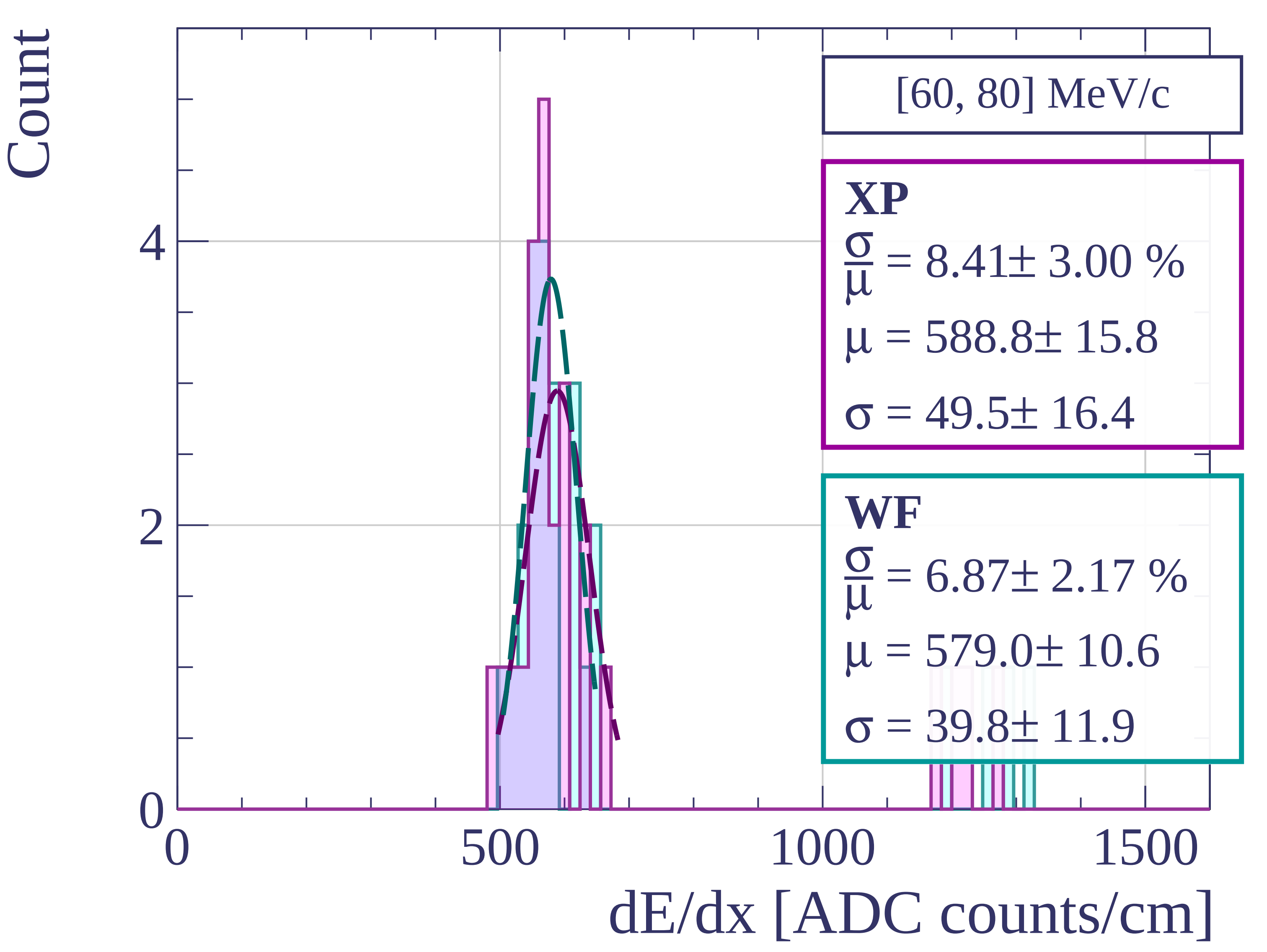
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



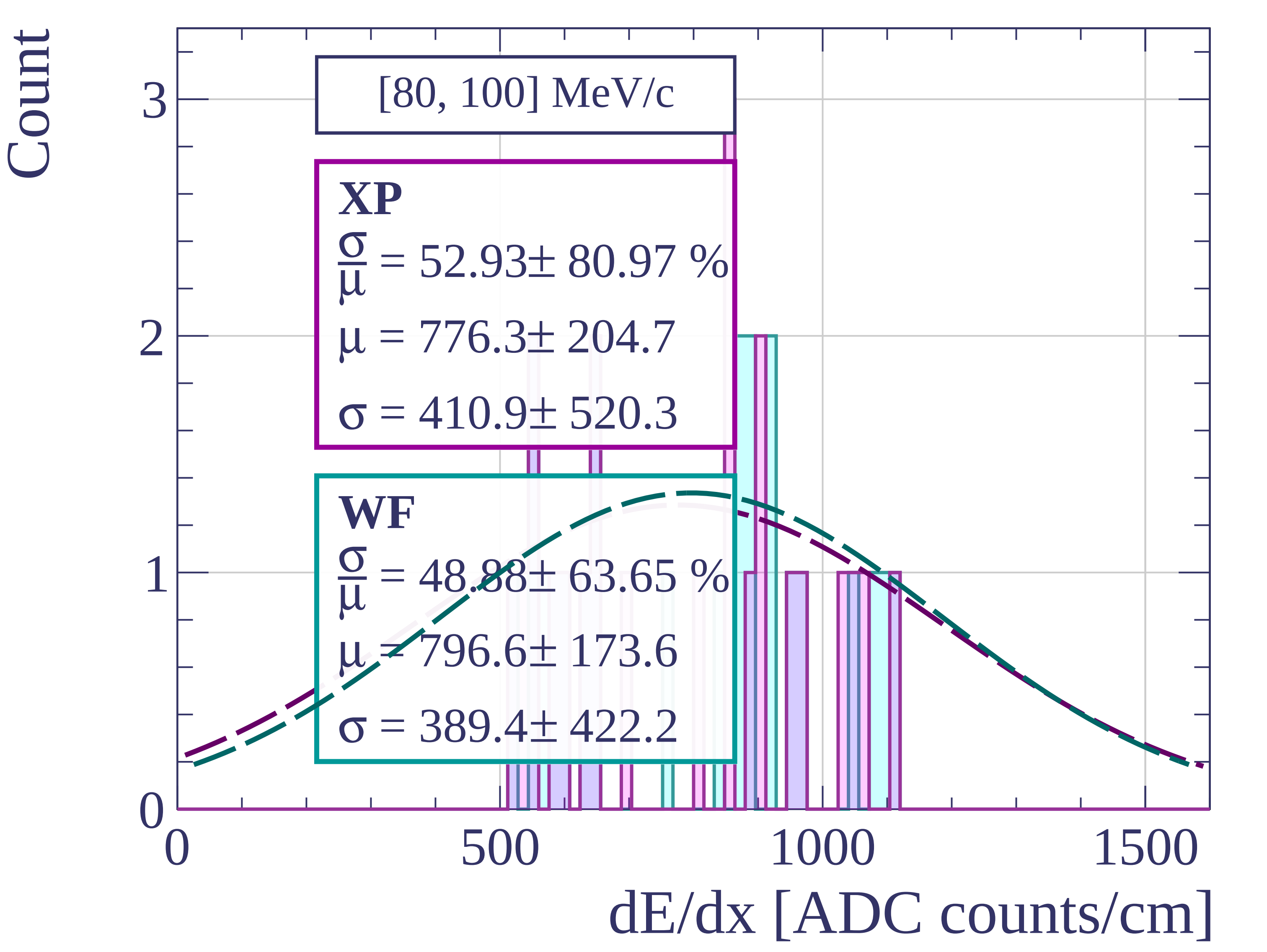
Count
[20, 40] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.82± 1.50 %
μ = 534.2± 8.5
σ = 36.4± 7.4
WF
$\frac{\sigma}{\mu}$ = 7.61± 2.73 %
μ = 559.2± 12.2
σ = 42.5± 14.4
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]



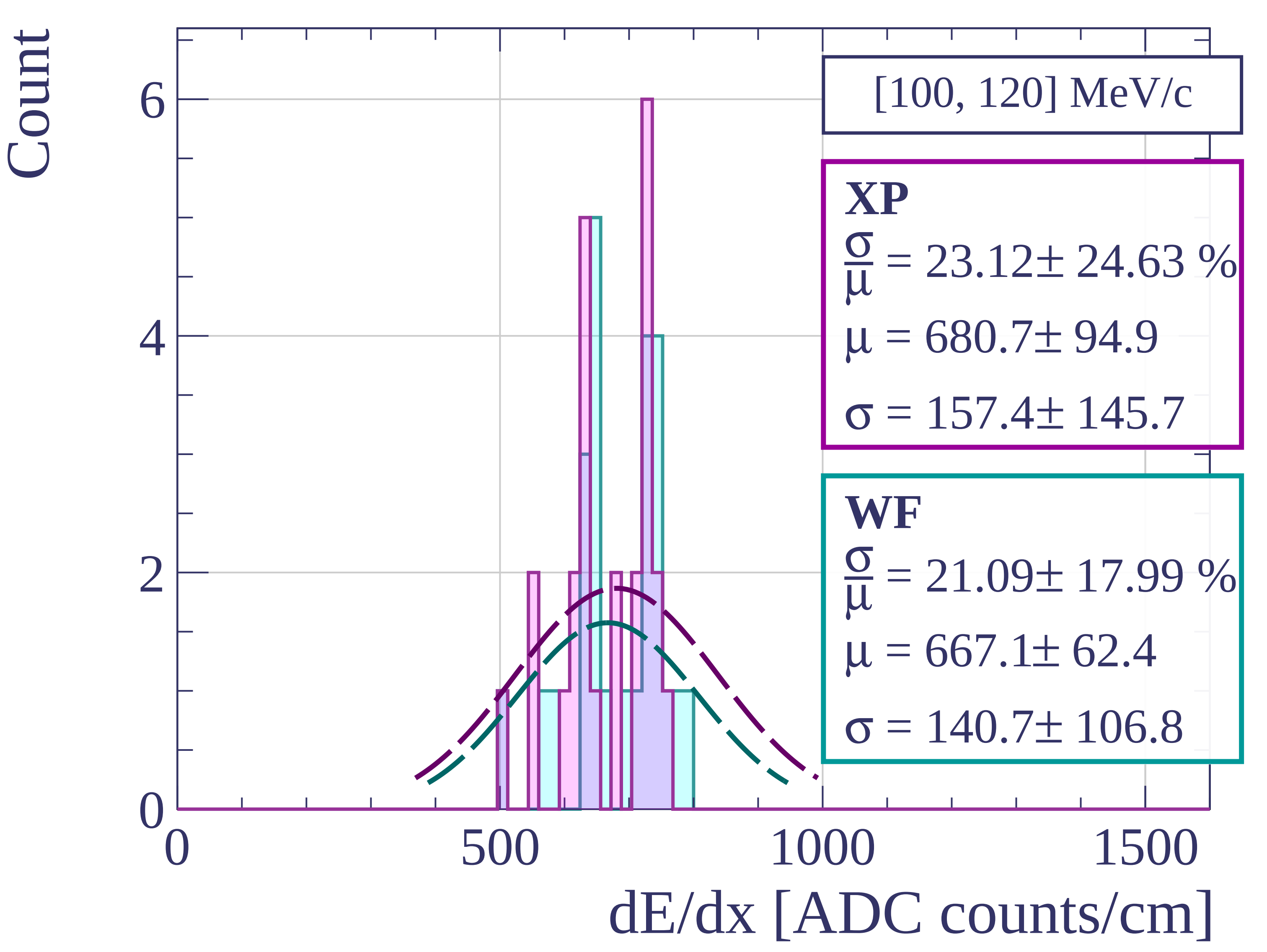
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.67± 1.75 %
μ = 541.2± 8.6
σ = 41.5± 8.8
WF
$\frac{\sigma}{\mu}$ = 10.41± 1.91 %
μ = 558.3± 9.2
σ = 58.1± 9.7



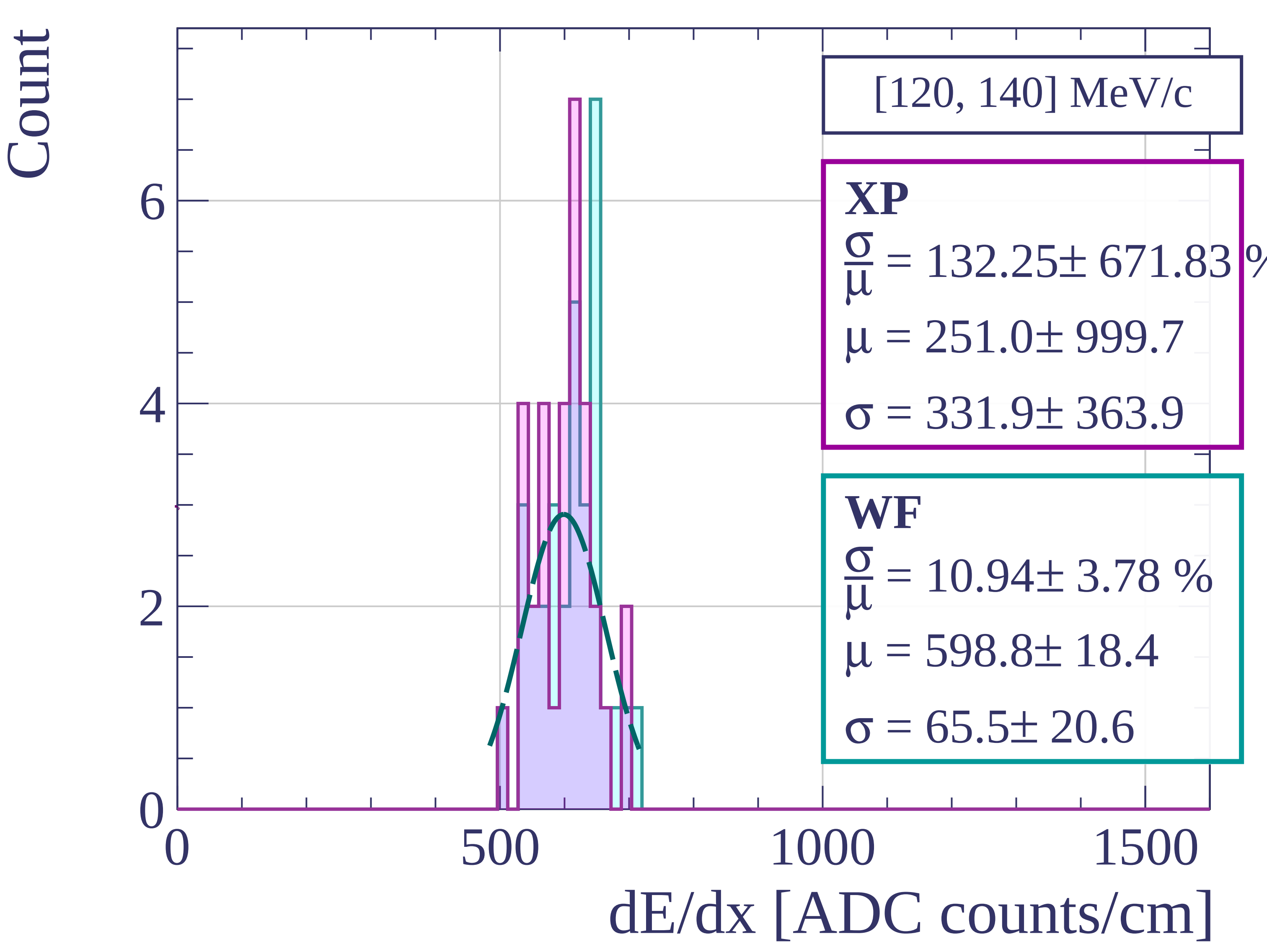
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
[60, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.41± 3.00 %
μ = 588.8± 15.8
σ = 49.5± 16.4
WF
$\frac{\sigma}{\mu}$ = 6.87± 2.17 %
μ = 579.0± 10.6
σ = 39.8± 11.9



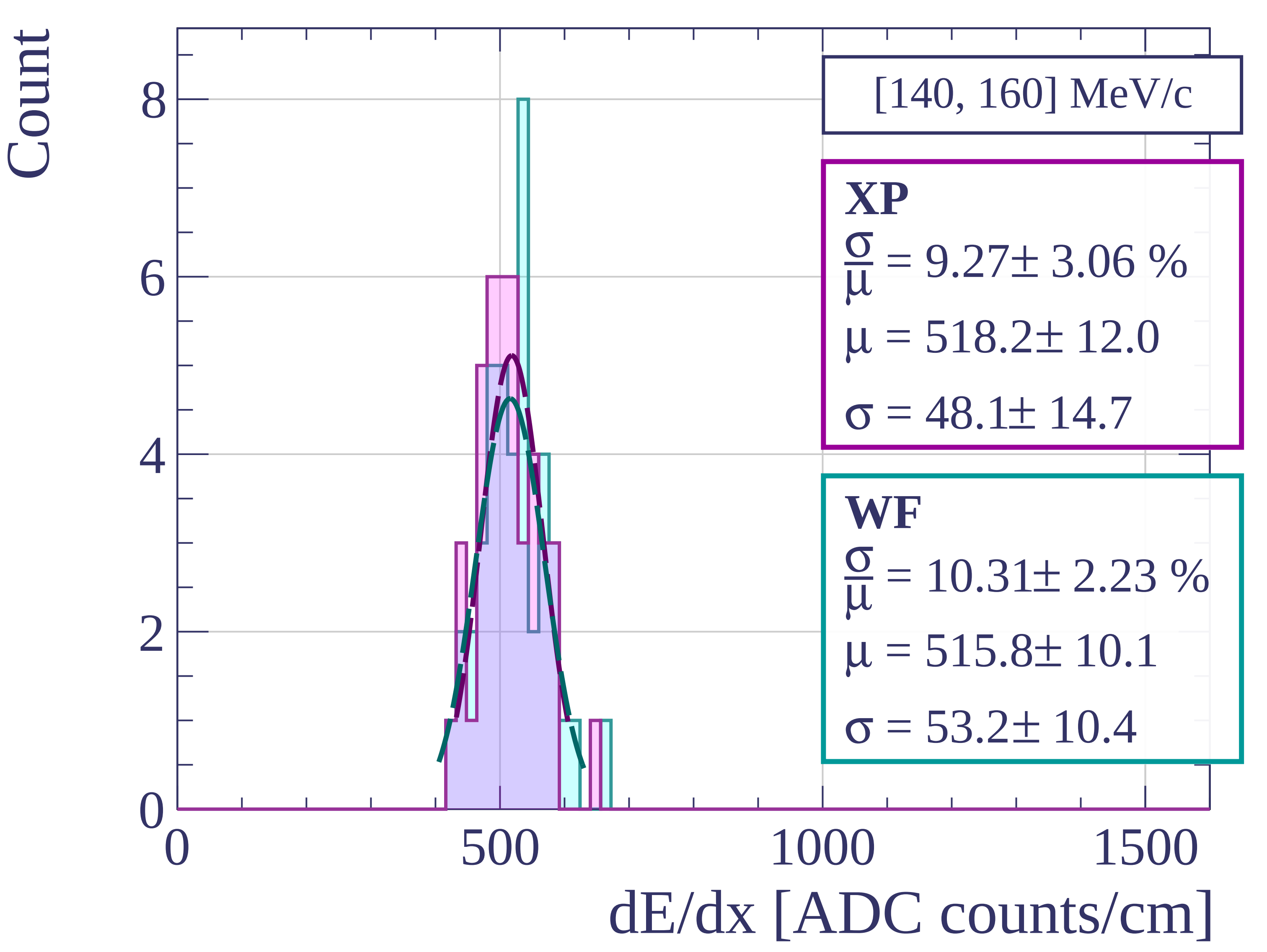
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 100] MeV/c
XP
σ/μ = 52.93± 80.97 %
μ = 776.3± 204.7
σ = 410.9± 520.3
WF
σ/μ = 48.88± 63.65 %
μ = 796.6± 173.6
σ = 389.4± 422.2



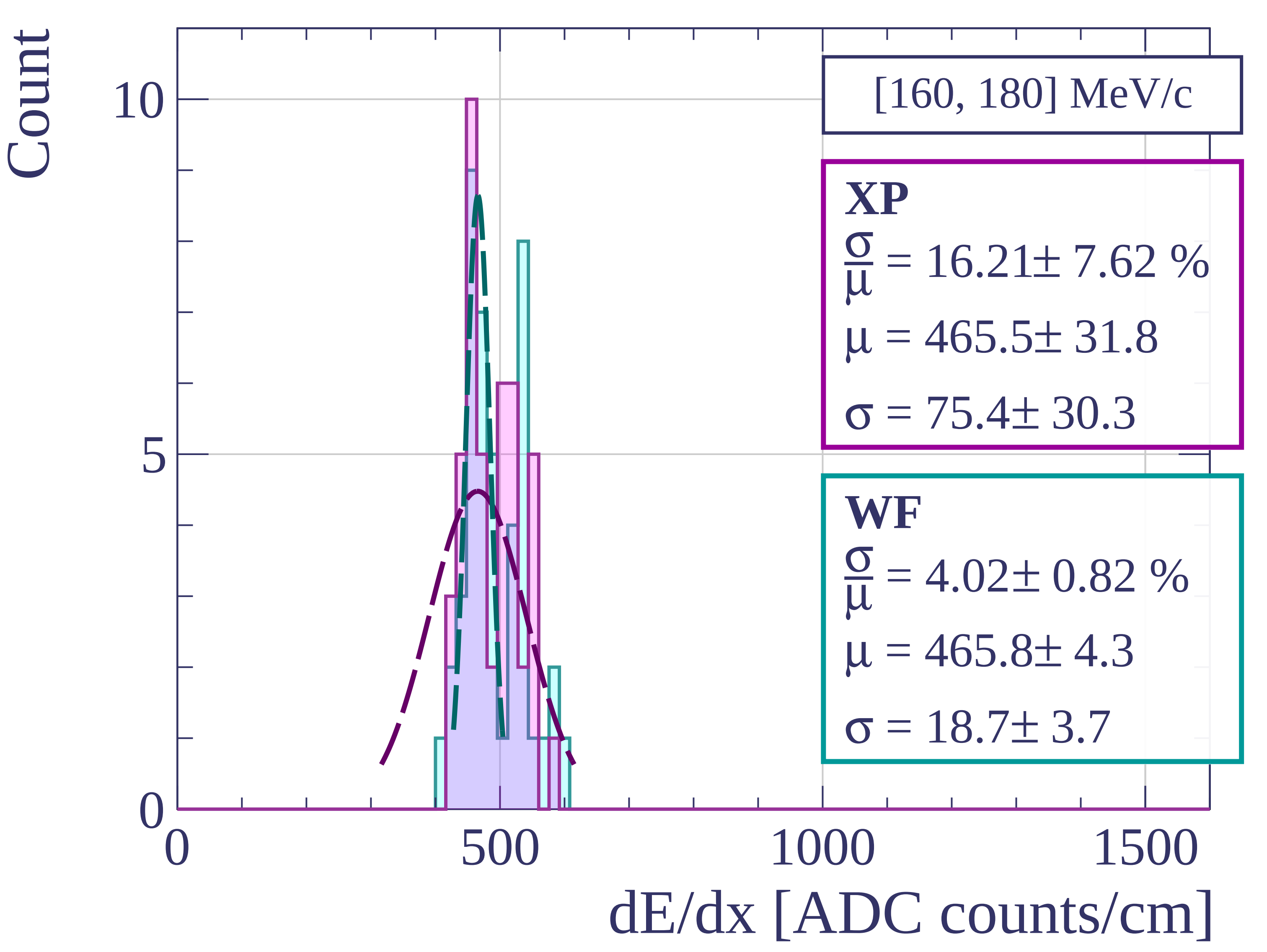
Count
[100, 120] MeV/c
XP
σ/μ = 23.12± 24.63 %
μ = 680.7± 94.9
σ = 157.4± 145.7
WF
σ/μ = 21.09± 17.99 %
μ = 667.1± 62.4
σ = 140.7± 106.8
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



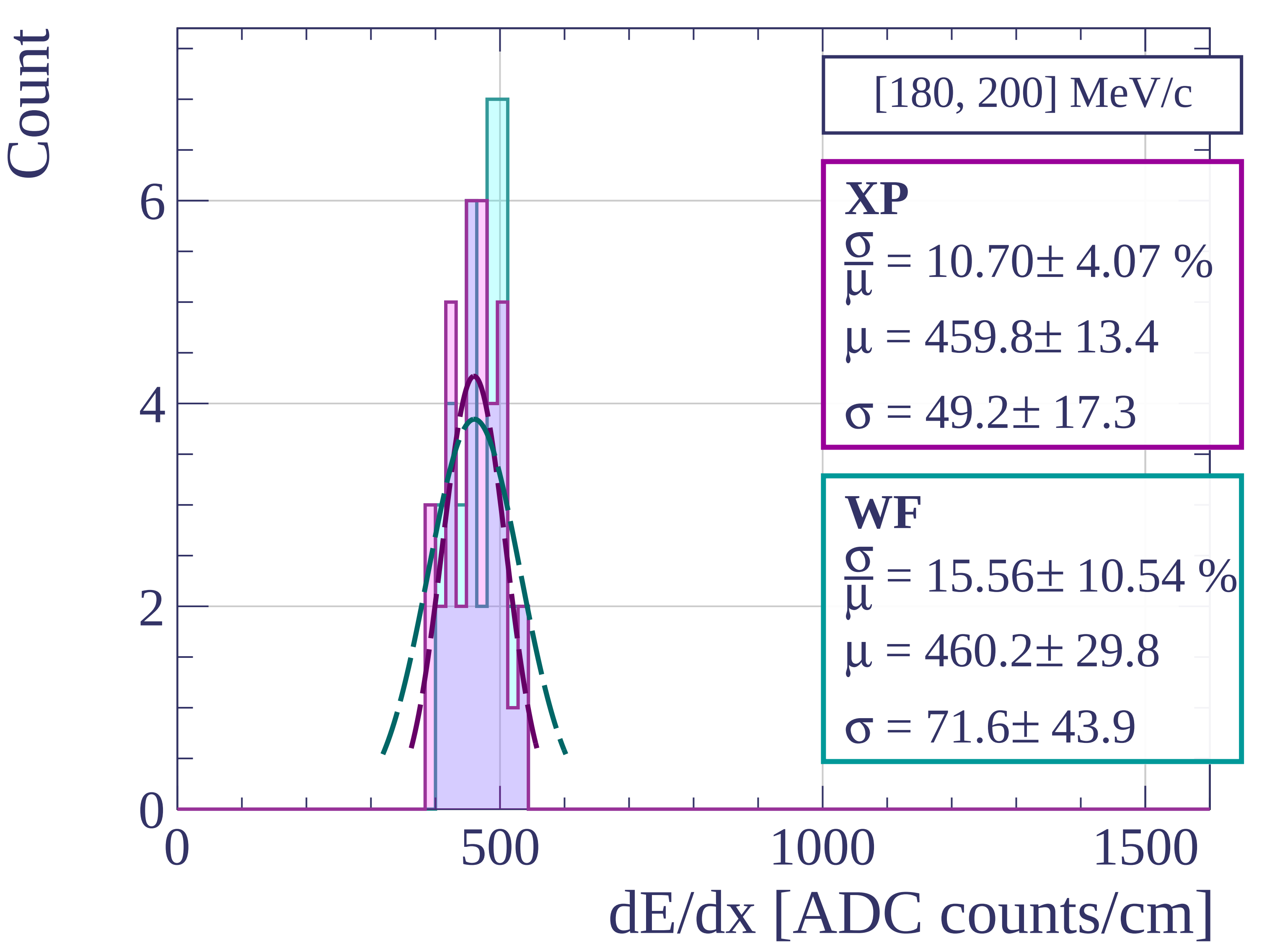
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[120, 140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 132.25± 671.83 %
μ = 251.0± 999.7
σ = 331.9± 363.9
WF
$\frac{\sigma}{\mu}$ = 10.94± 3.78 %
μ = 598.8± 18.4
σ = 65.5± 20.6

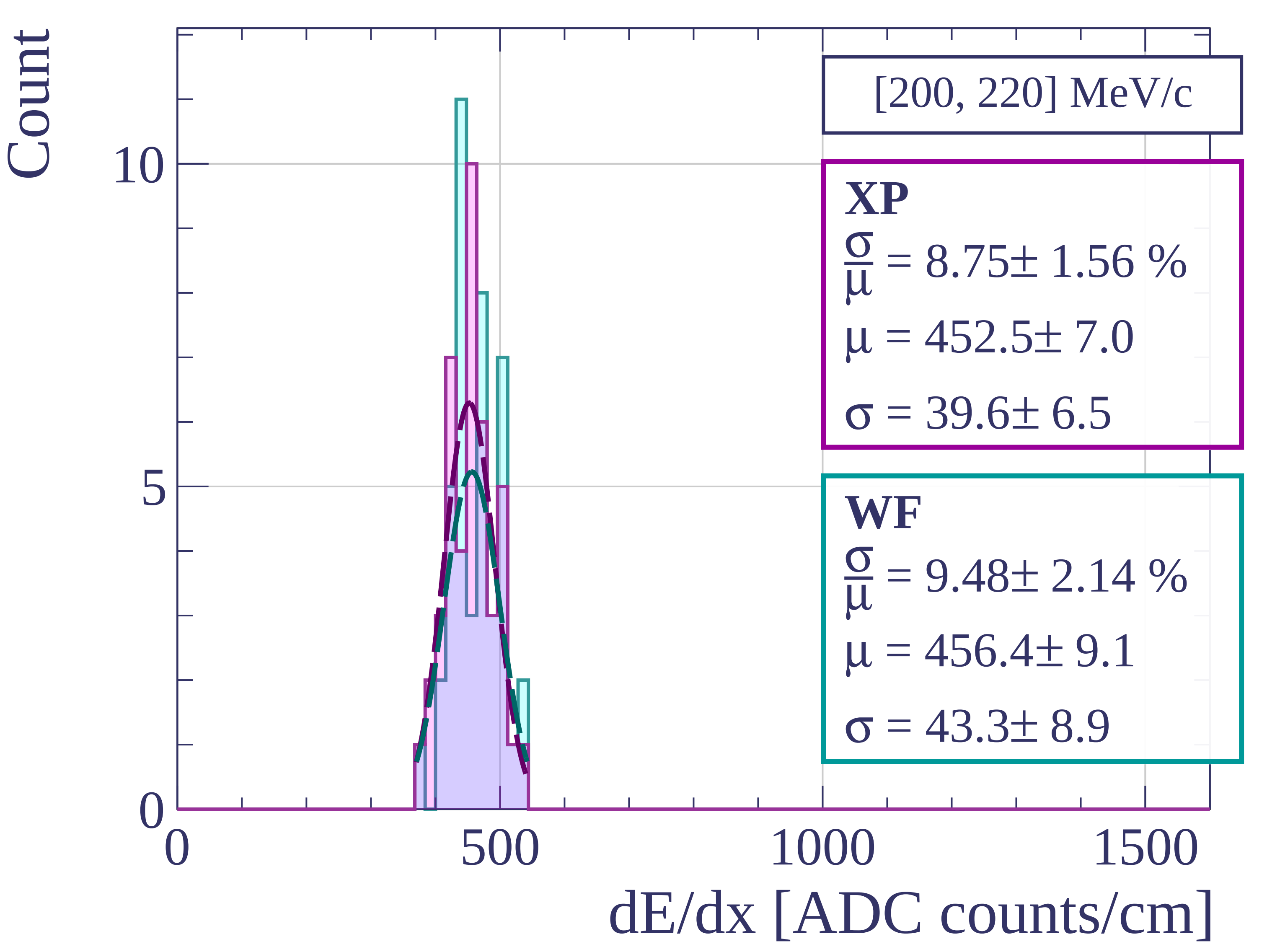


Count
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.27± 3.06 %
μ = 518.2± 12.0
σ = 48.1± 14.7
WF
$\frac{\sigma}{\mu}$ = 10.31± 2.23 %
μ = 515.8± 10.1
σ = 53.2± 10.4

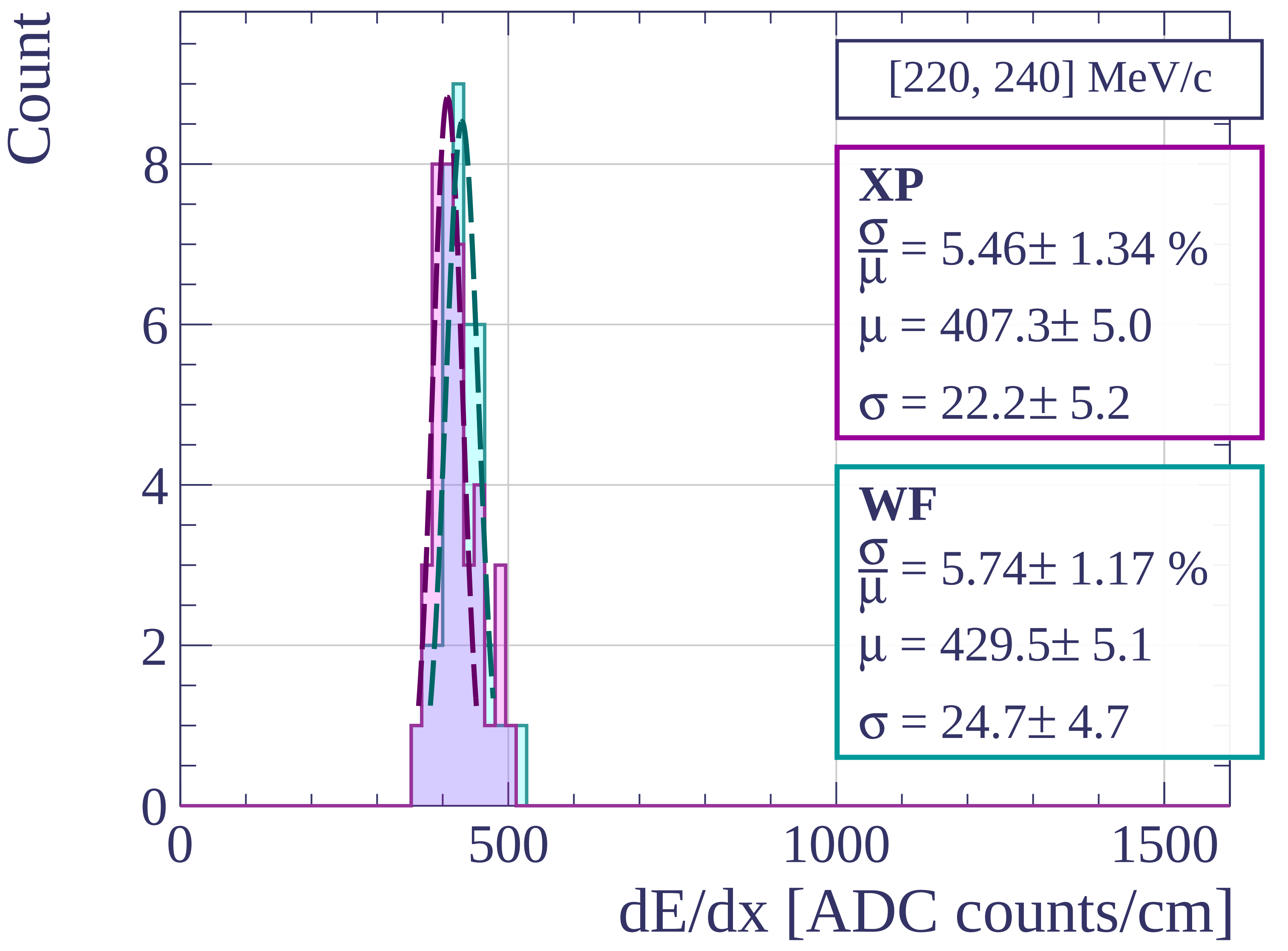


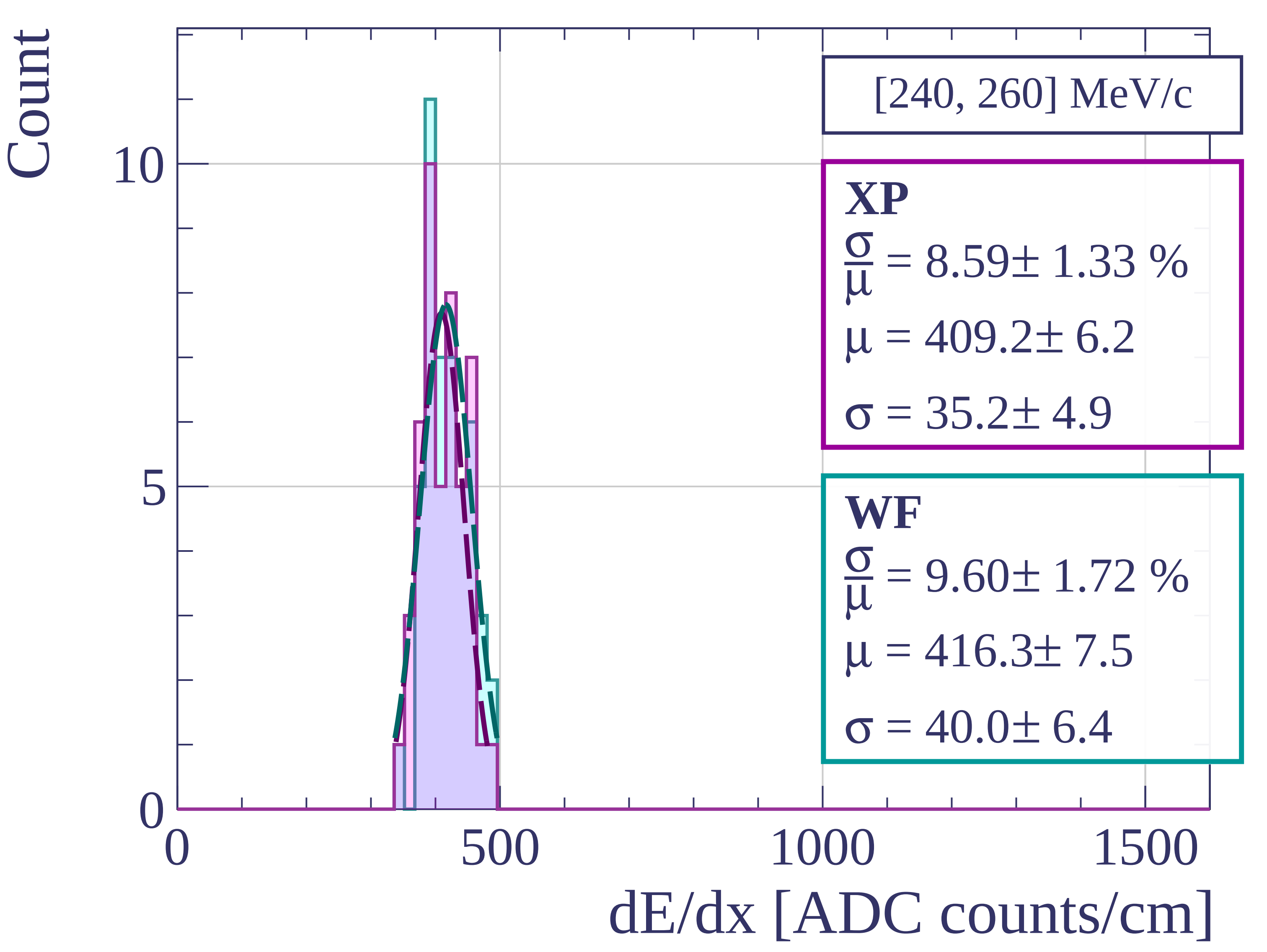
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
σ/μ = 16.21± 7.62 %
μ = 465.5± 31.8
σ = 75.4± 30.3
WF
σ/μ = 4.02± 0.82 %
μ = 465.8± 4.3
σ = 18.7± 3.7


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.70± 4.07 %
μ = 459.8± 13.4
σ = 49.2± 17.3
WF
$\frac{\sigma}{\mu}$ = 15.56± 10.54 %
μ = 460.2± 29.8
σ = 71.6± 43.9

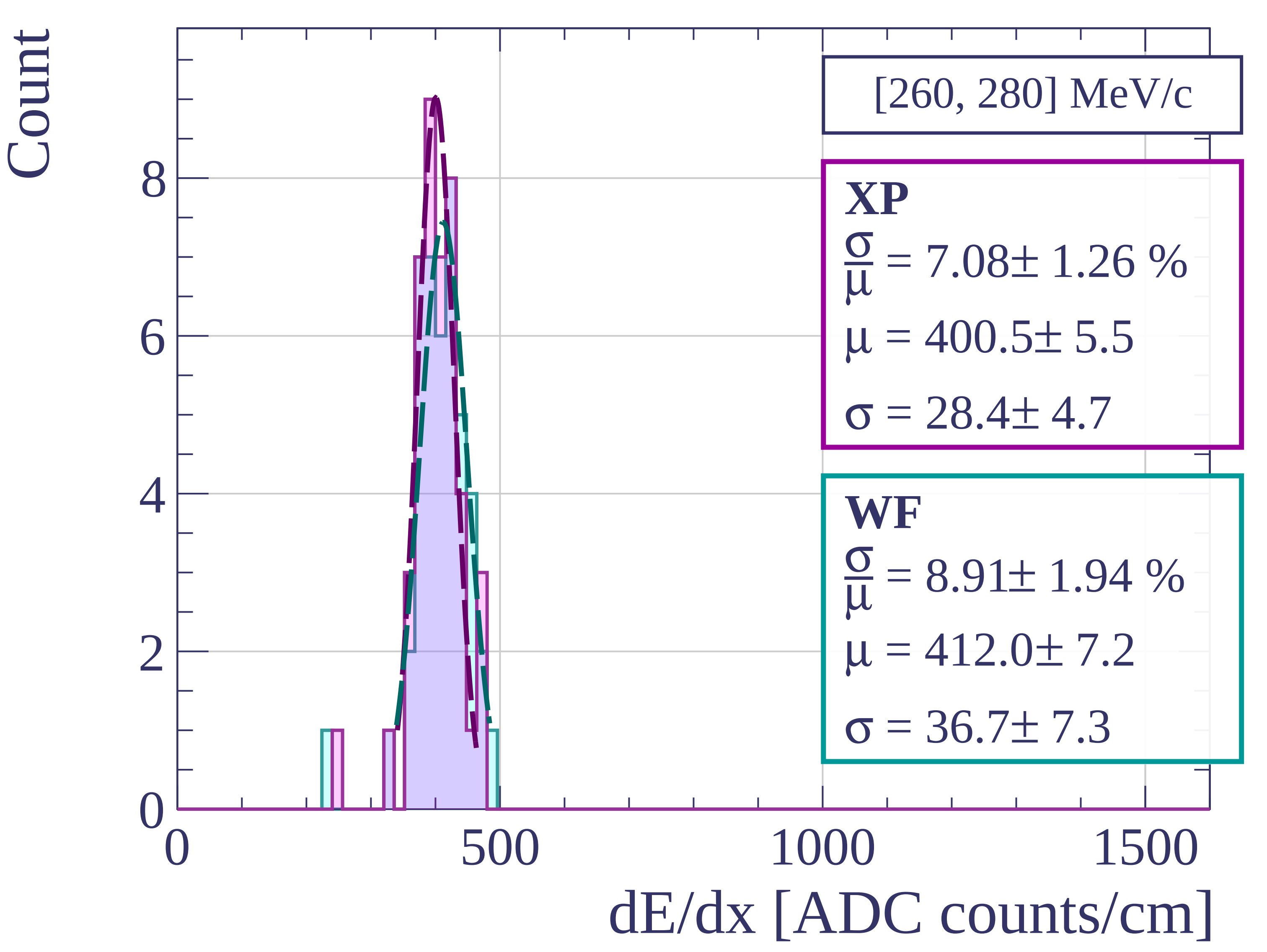


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[200, 220] MeV/c
XP
σ/μ = 8.75± 1.56 %
μ = 452.5± 7.0
σ = 39.6± 6.5
WF
σ/μ = 9.48± 2.14 %
μ = 456.4± 9.1
σ = 43.3± 8.9

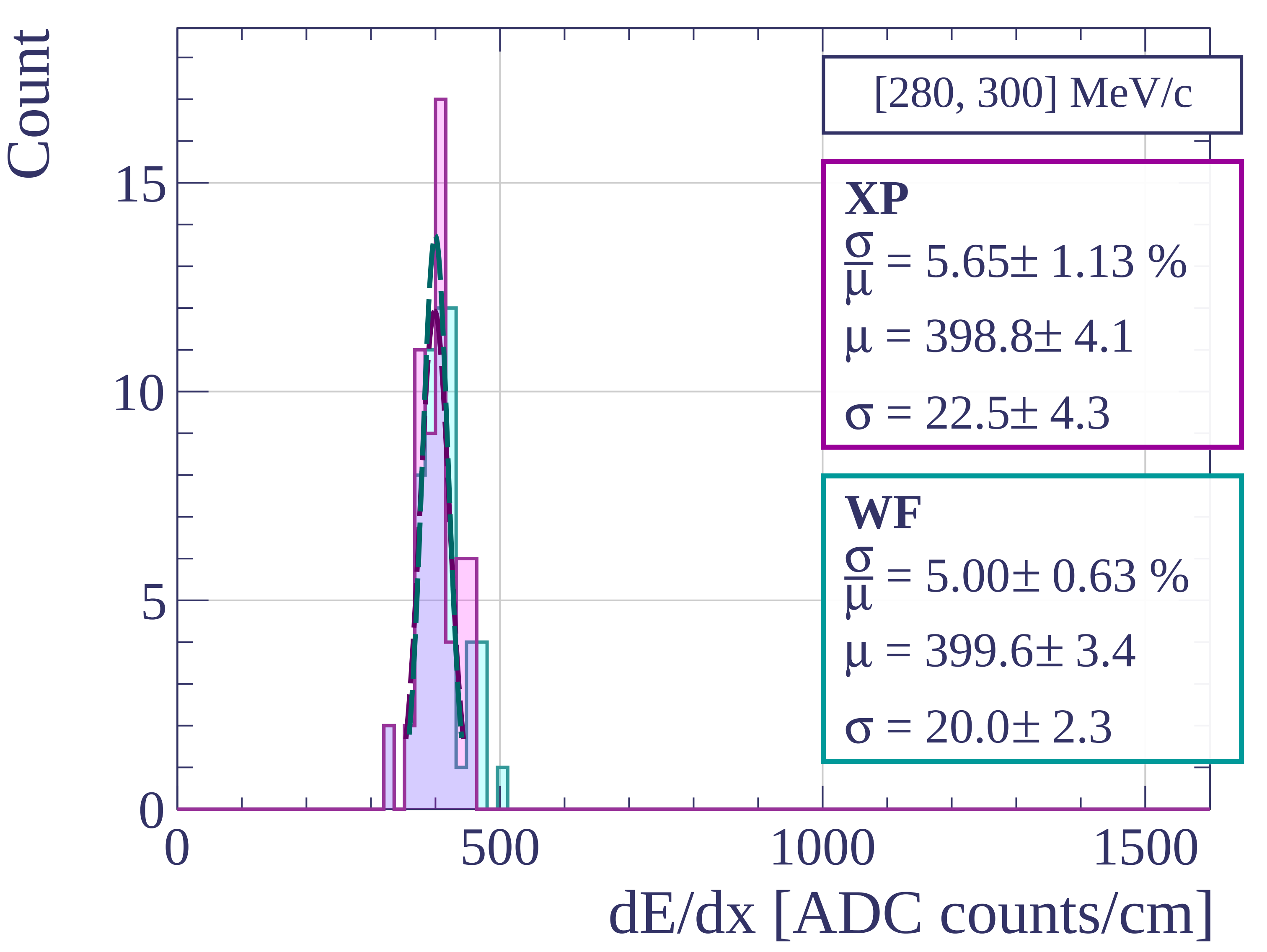

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[220, 240] MeV/c
XP
σ/μ = 5.46± 1.34 %
μ = 407.3± 5.0
σ = 22.2± 5.2
WF
σ/μ = 5.74± 1.17 %
μ = 429.5± 5.1
σ = 24.7± 4.7



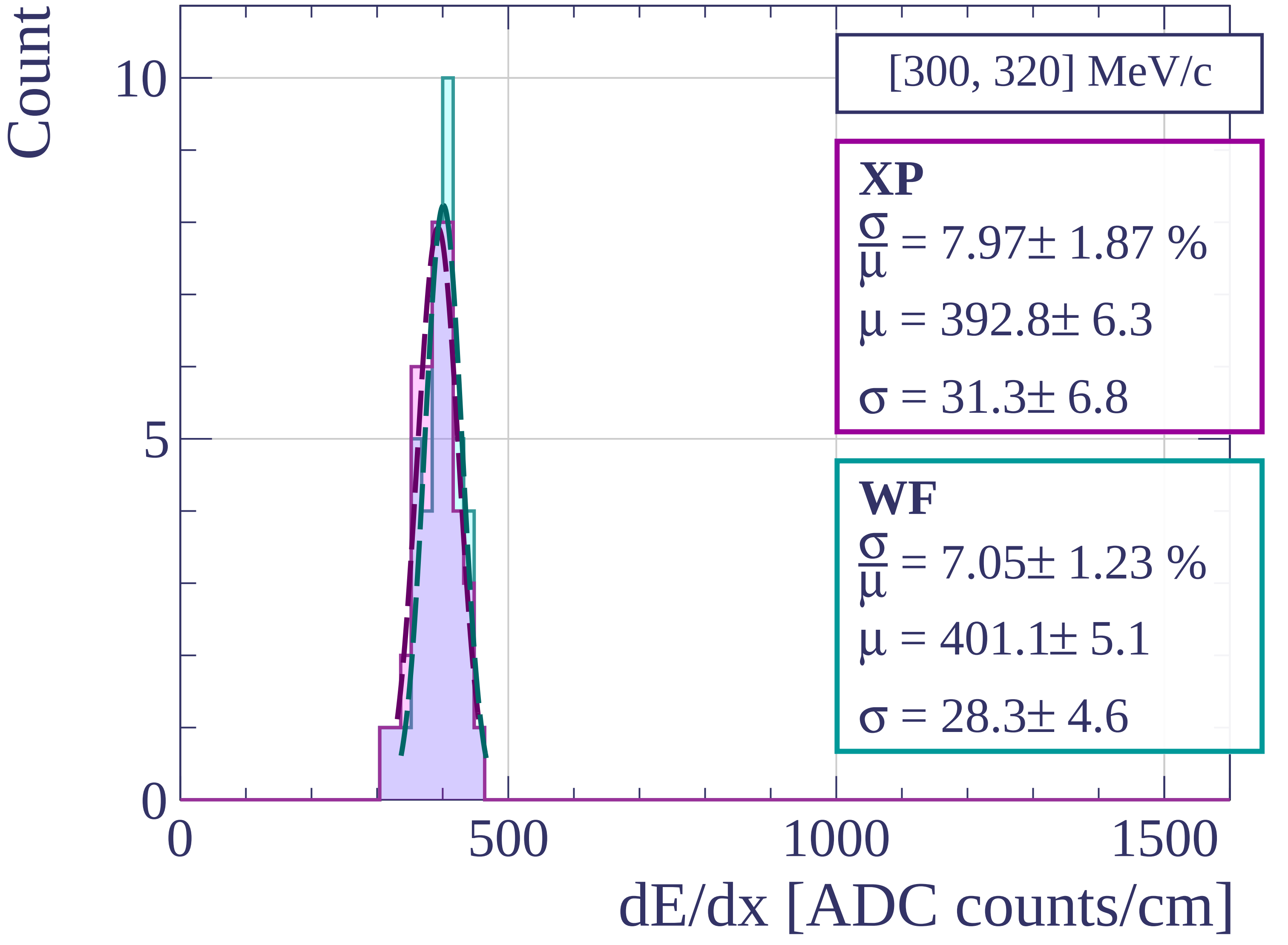
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.59± 1.33 %
μ = 409.2± 6.2
σ = 35.2± 4.9
WF
$\frac{\sigma}{\mu}$ = 9.60± 1.72 %
μ = 416.3± 7.5
σ = 40.0± 6.4



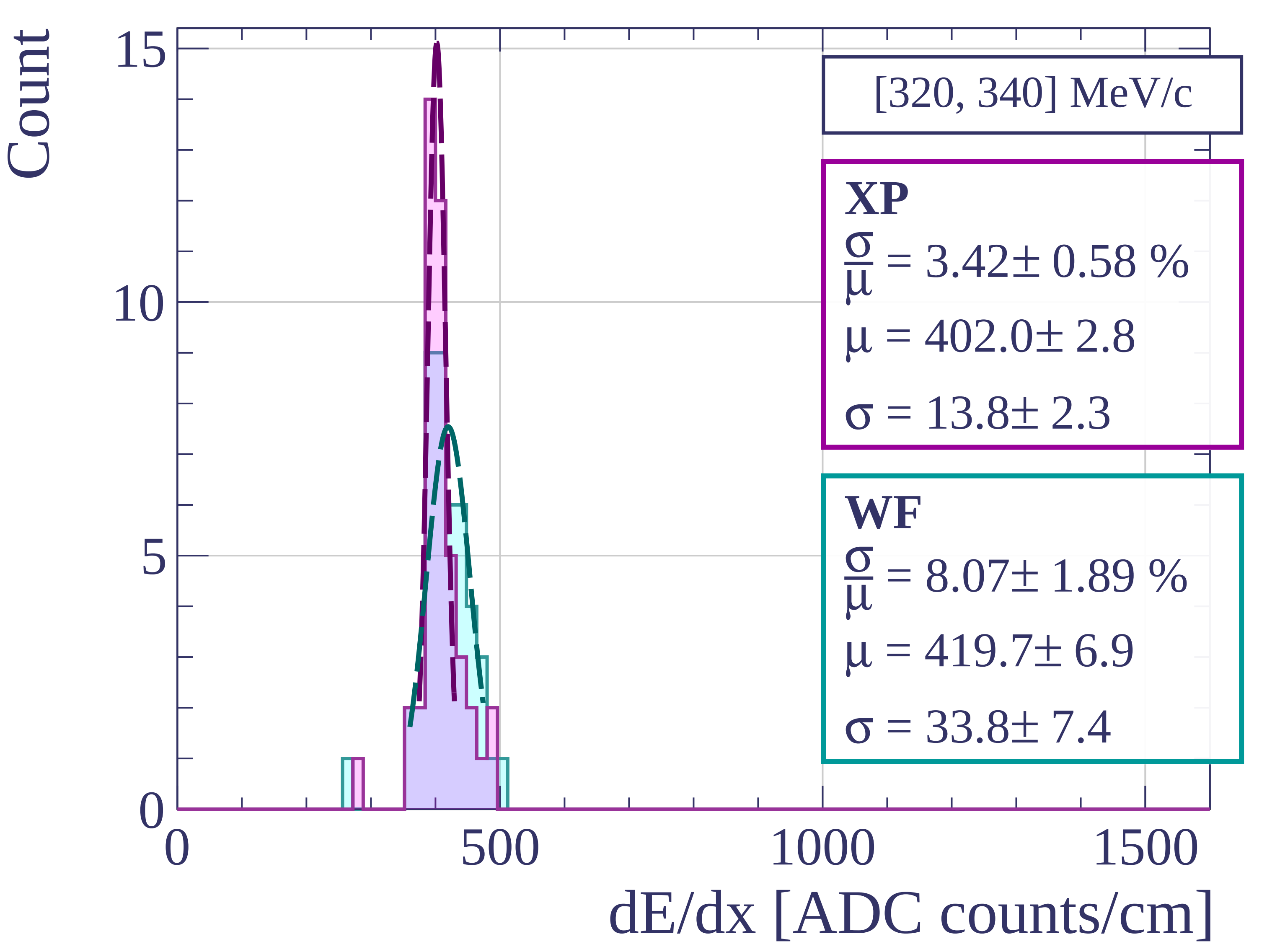
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[260, 280] MeV/c
XP
σ/μ = 7.08± 1.26 %
μ = 400.5± 5.5
σ = 28.4± 4.7
WF
σ/μ = 8.91± 1.94 %
μ = 412.0± 7.2
σ = 36.7± 7.3



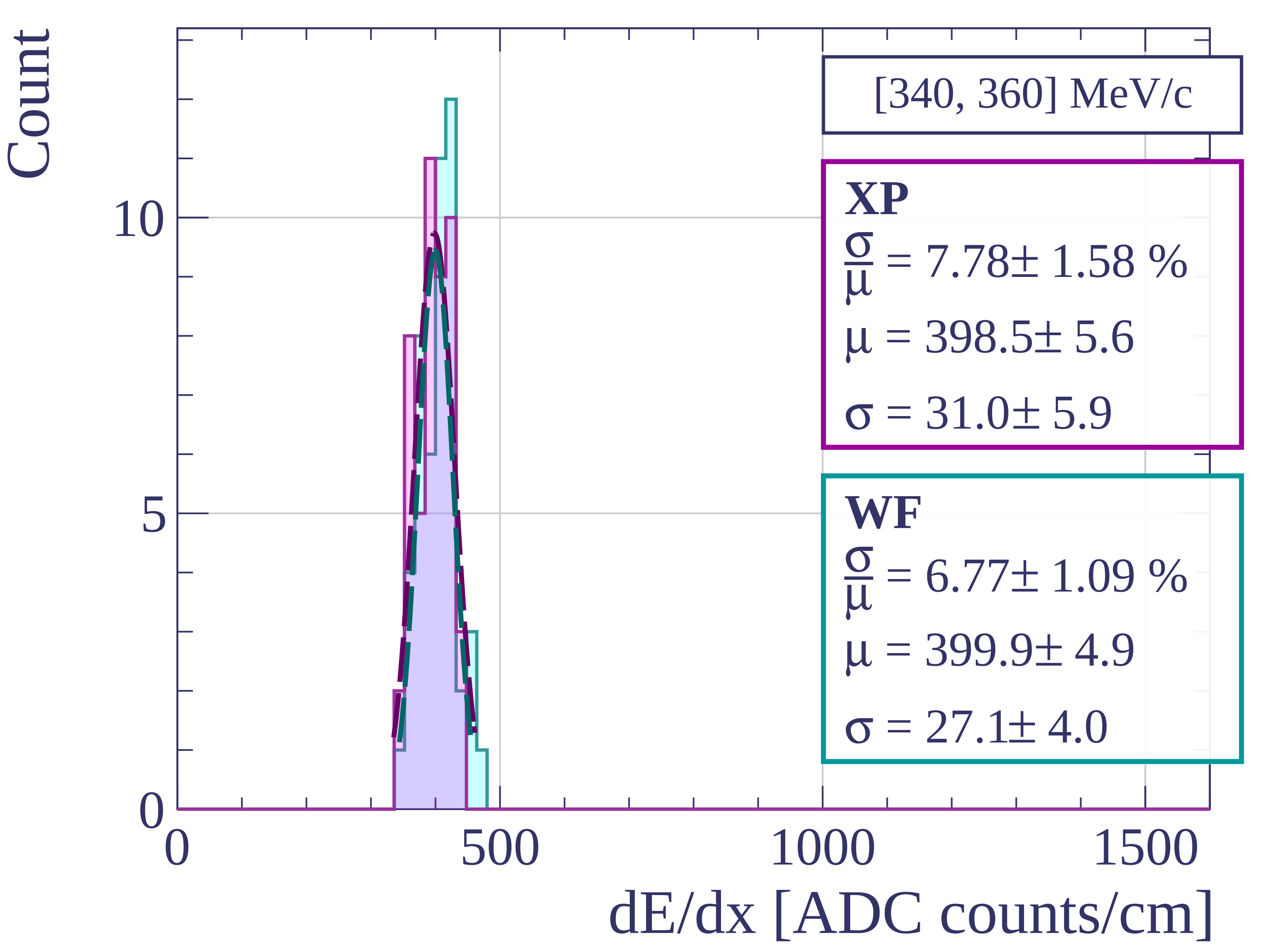
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.65± 1.13 %
μ = 398.8± 4.1
σ = 22.5± 4.3
WF
$\frac{\sigma}{\mu}$ = 5.00± 0.63 %
μ = 399.6± 3.4
σ = 20.0± 2.3



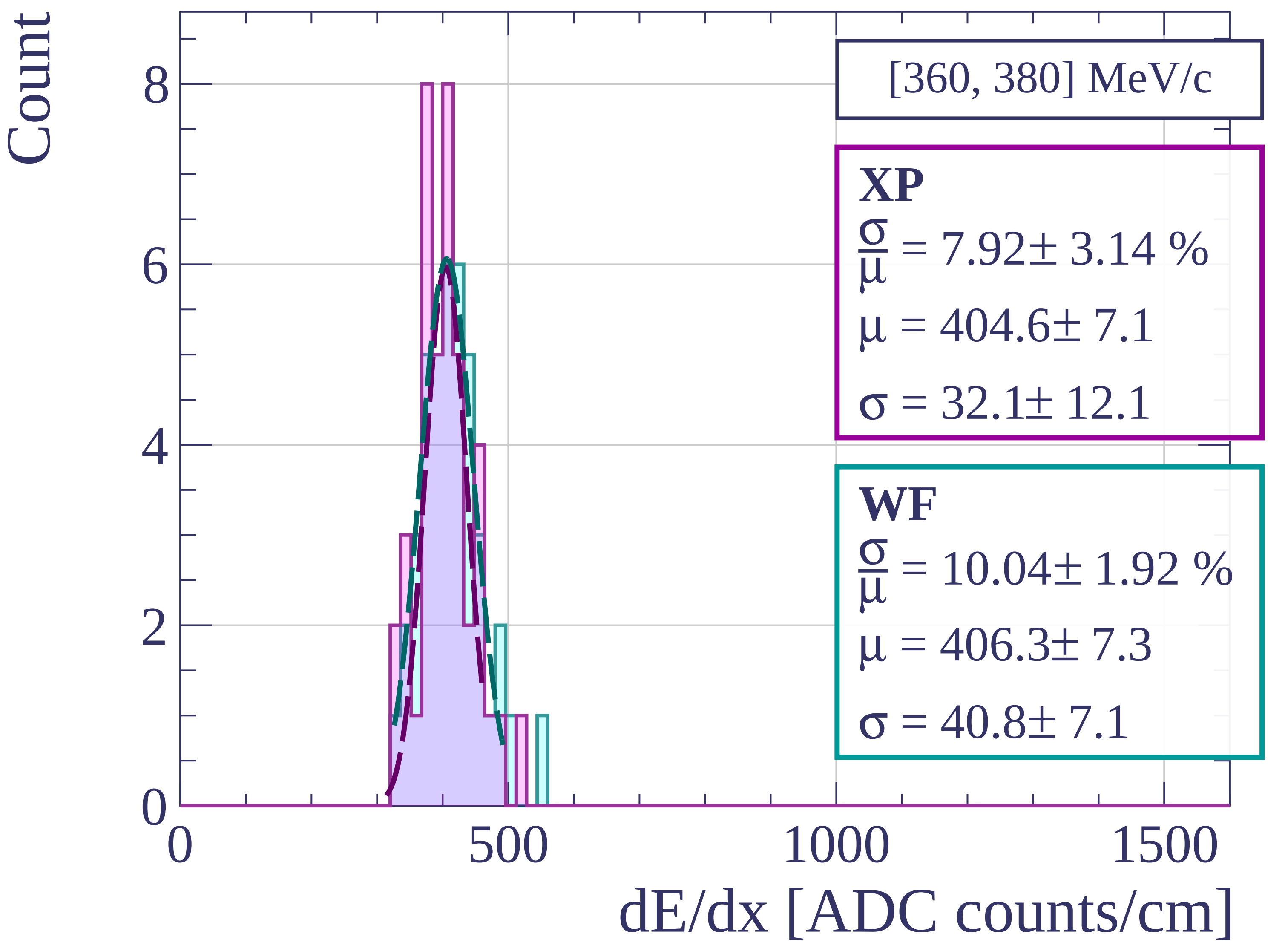
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[300, 320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.97± 1.87 %
μ = 392.8± 6.3
σ = 31.3± 6.8
WF
$\frac{\sigma}{\mu}$ = 7.05± 1.23 %
μ = 401.1± 5.1
σ = 28.3± 4.6



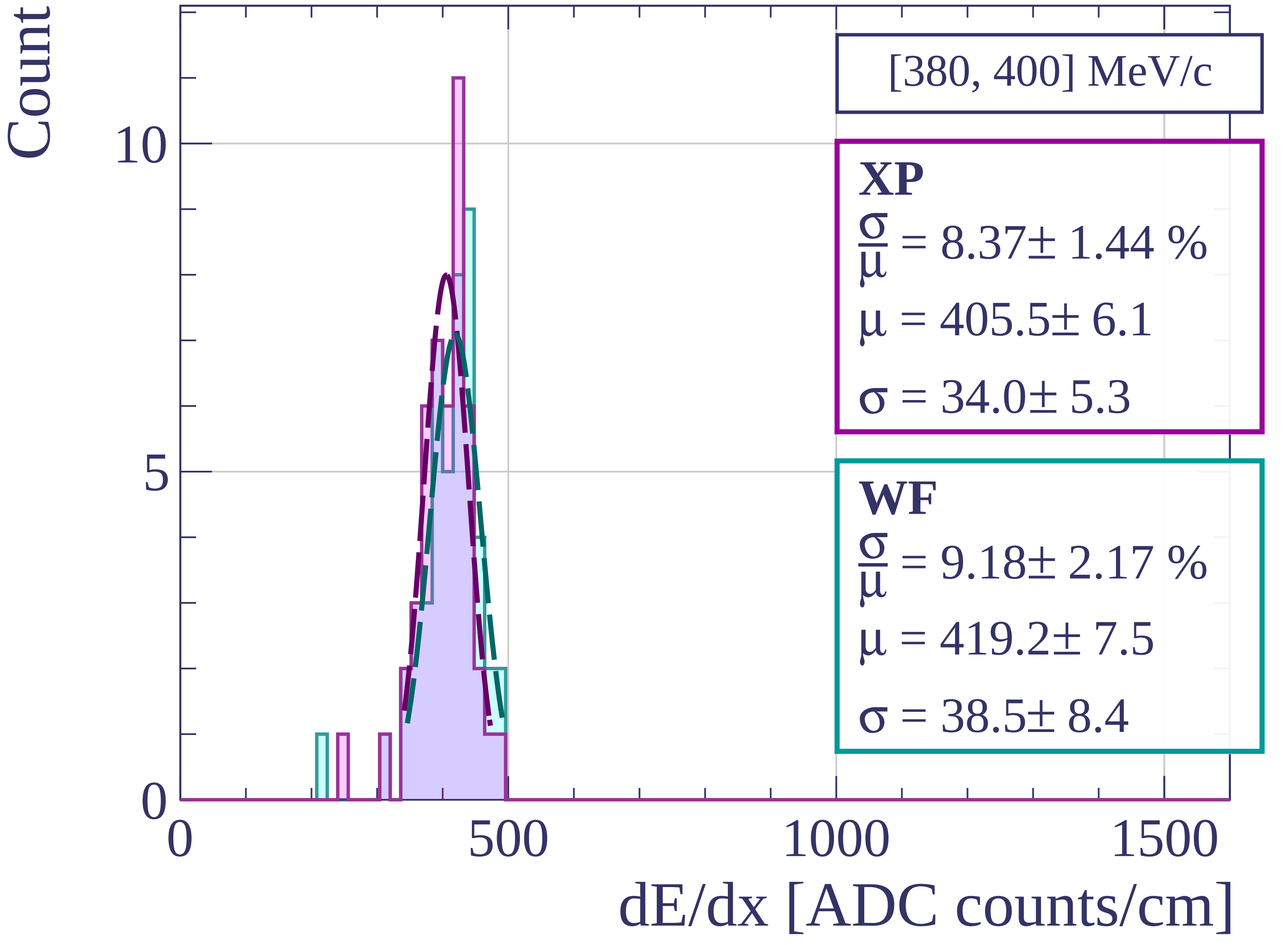
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 340] MeV/c
XP
σ/μ = 3.42± 0.58 %
μ = 402.0± 2.8
σ = 13.8± 2.3
WF
σ/μ = 8.07± 1.89 %
μ = 419.7± 6.9
σ = 33.8± 7.4



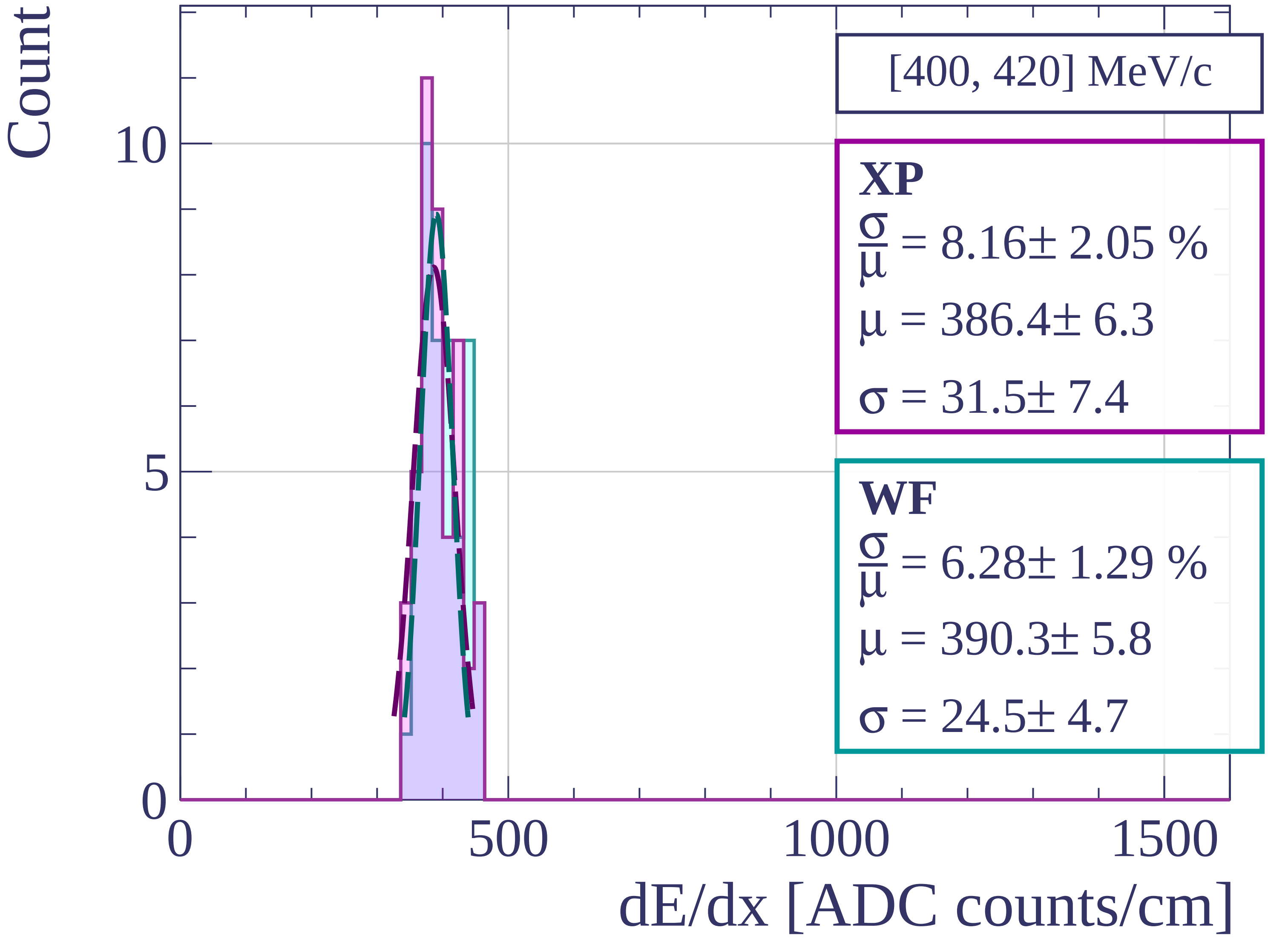
Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
σ/μ = 7.78± 1.58 %
μ = 398.5± 5.6
σ = 31.0± 5.9
WF
σ/μ = 6.77± 1.09 %
μ = 399.9± 4.9
σ = 27.1± 4.0

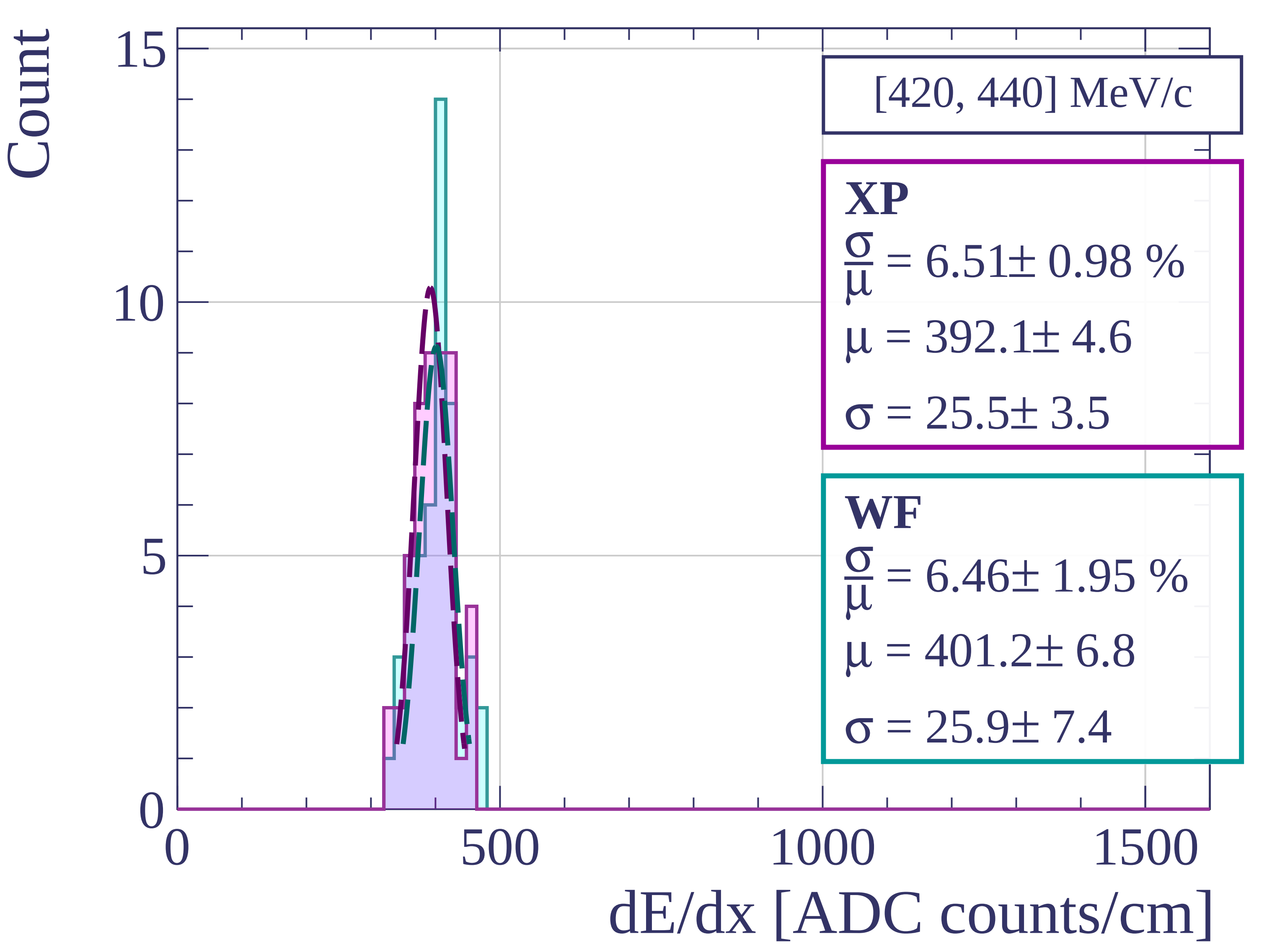


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
σ/μ = 7.92± 3.14 %
μ = 404.6± 7.1
σ = 32.1± 12.1
WF
σ/μ = 10.04± 1.92 %
μ = 406.3± 7.3
σ = 40.8± 7.1

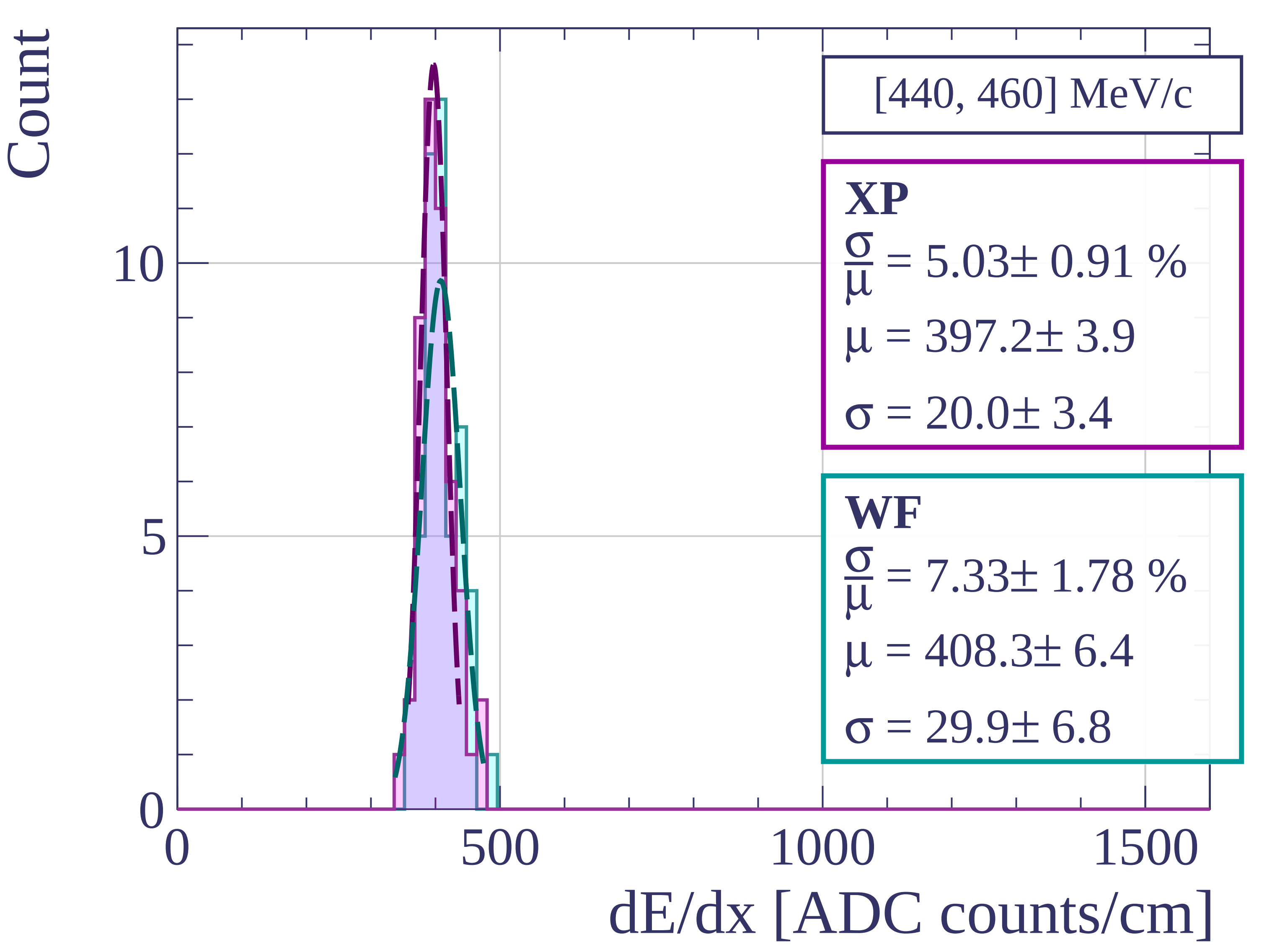


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[380, 400] MeV/c
XP
σ/μ = 8.37± 1.44 %
μ = 405.5± 6.1
σ = 34.0± 5.3
WF
σ/μ = 9.18± 2.17 %
μ = 419.2± 7.5
σ = 38.5± 8.4

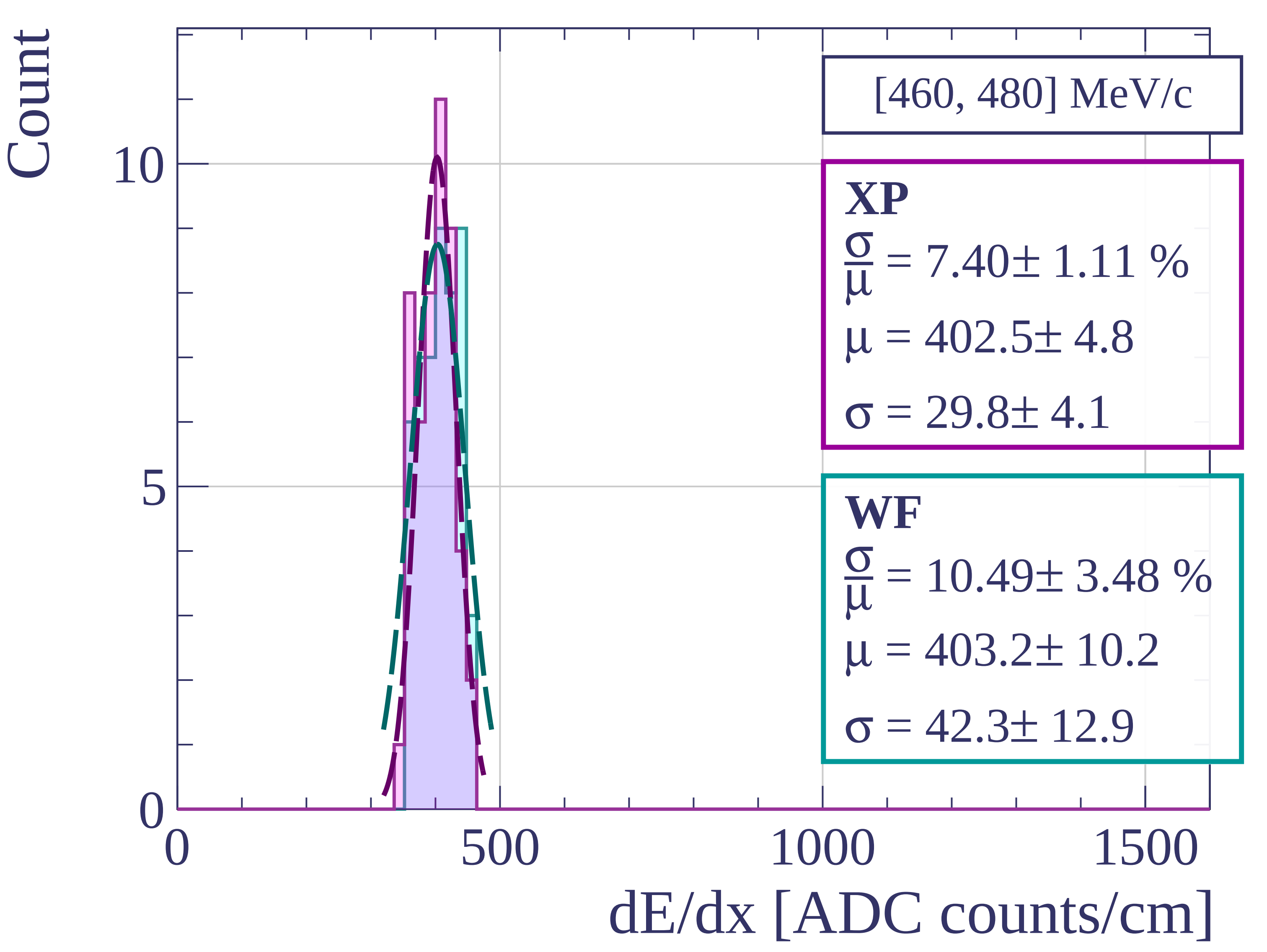

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.16± 2.05 %
μ = 386.4± 6.3
σ = 31.5± 7.4
WF
$\frac{\sigma}{\mu}$ = 6.28± 1.29 %
μ = 390.3± 5.8
σ = 24.5± 4.7



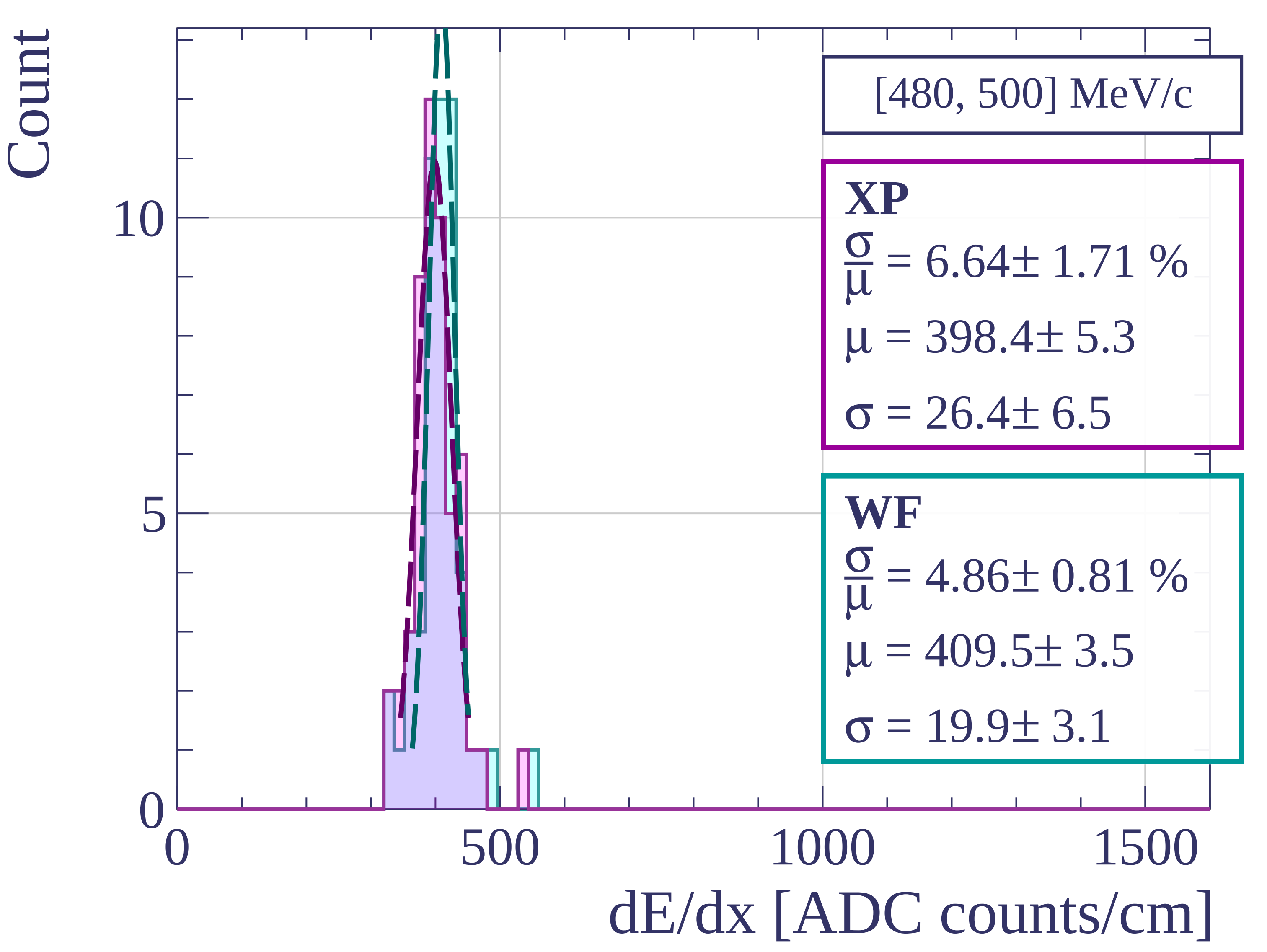
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
[420, 440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.51± 0.98 %
μ = 392.1± 4.6
σ = 25.5± 3.5
WF
$\frac{\sigma}{\mu}$ = 6.46± 1.95 %
μ = 401.2± 6.8
σ = 25.9± 7.4

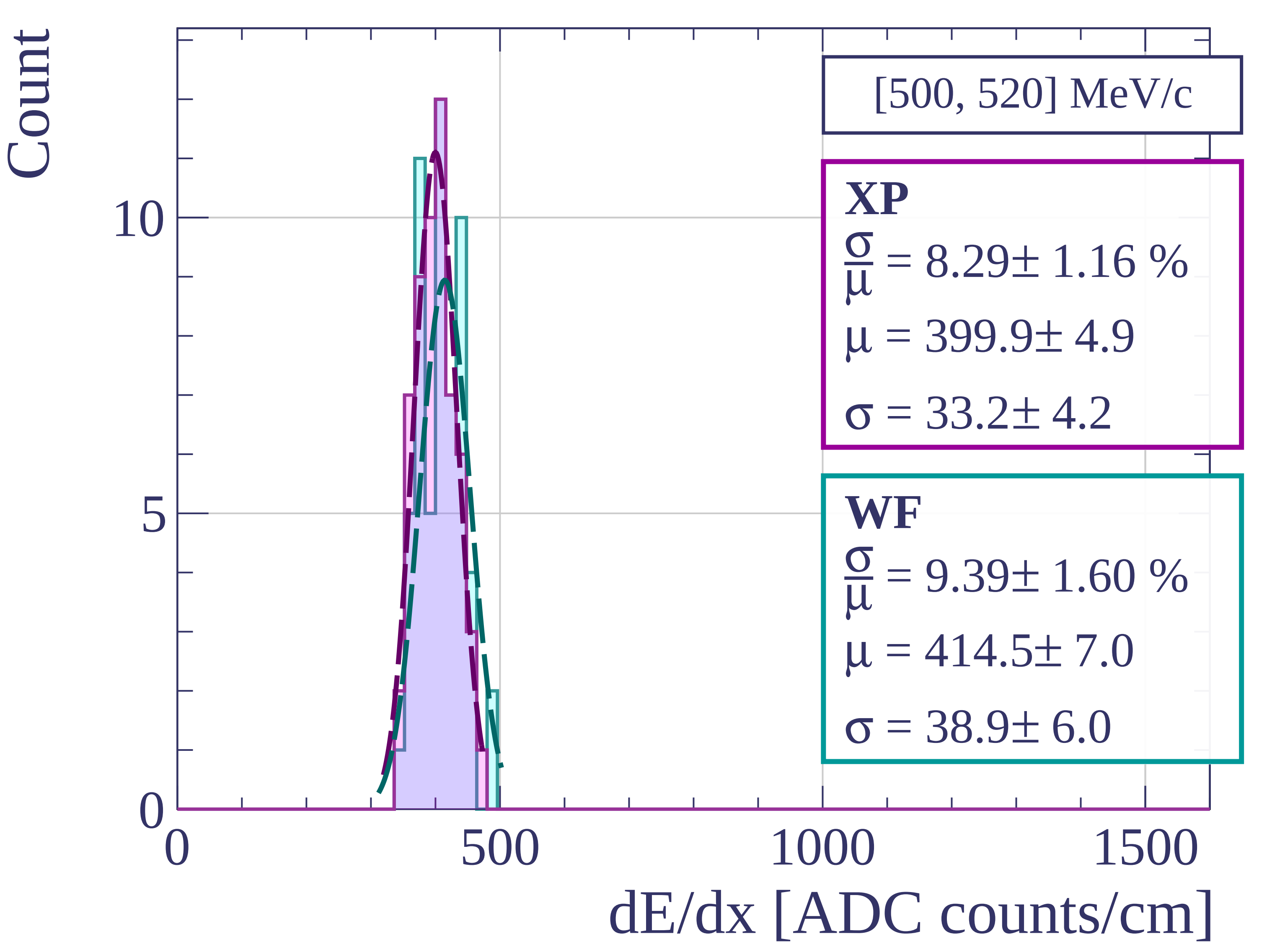


Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[440, 460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.03± 0.91 %
μ = 397.2± 3.9
σ = 20.0± 3.4
WF
$\frac{\sigma}{\mu}$ = 7.33± 1.78 %
μ = 408.3± 6.4
σ = 29.9± 6.8

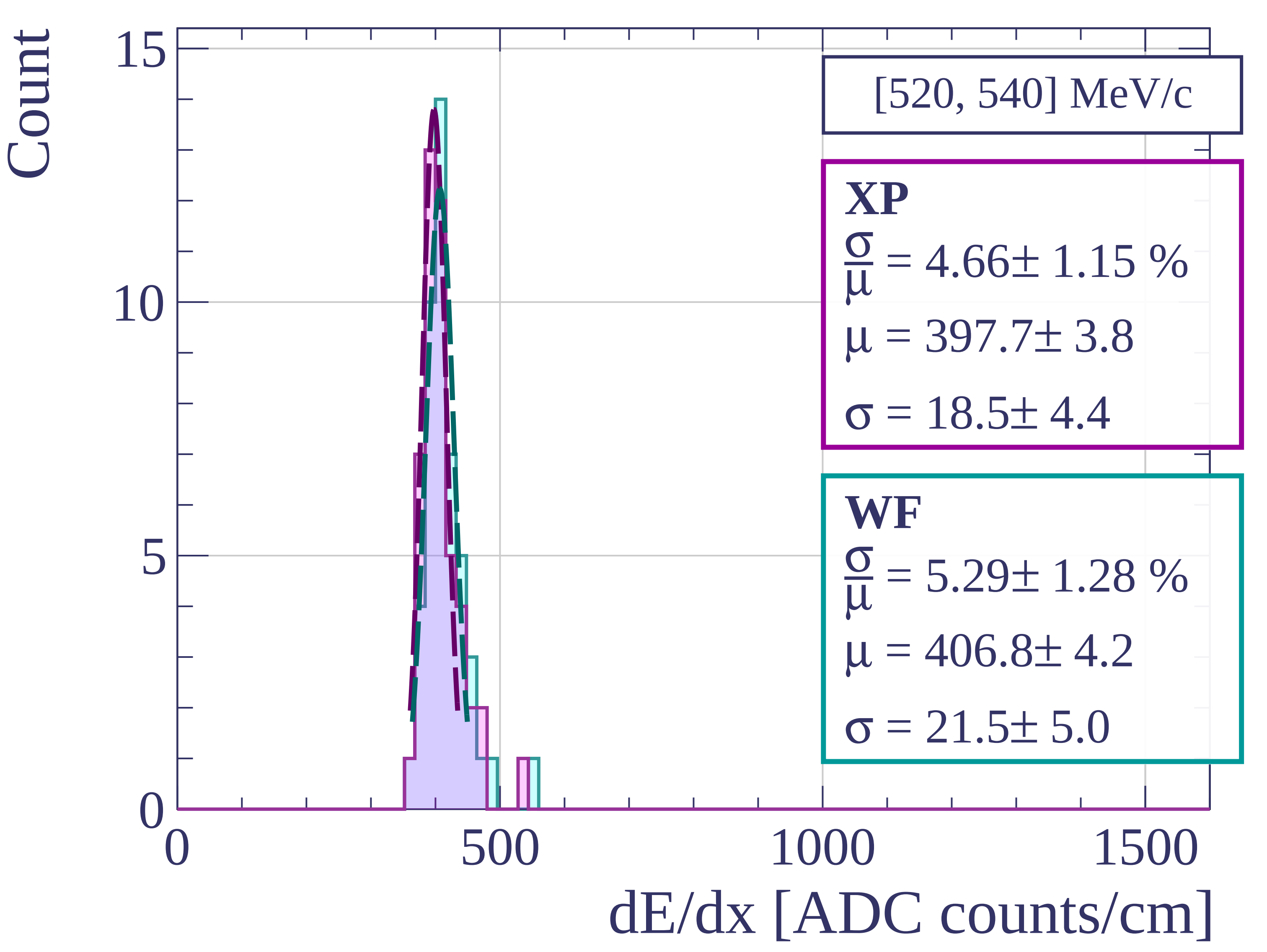


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[460, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.40± 1.11 %
μ = 402.5± 4.8
σ = 29.8± 4.1
WF
$\frac{\sigma}{\mu}$ = 10.49± 3.48 %
μ = 403.2± 10.2
σ = 42.3± 12.9

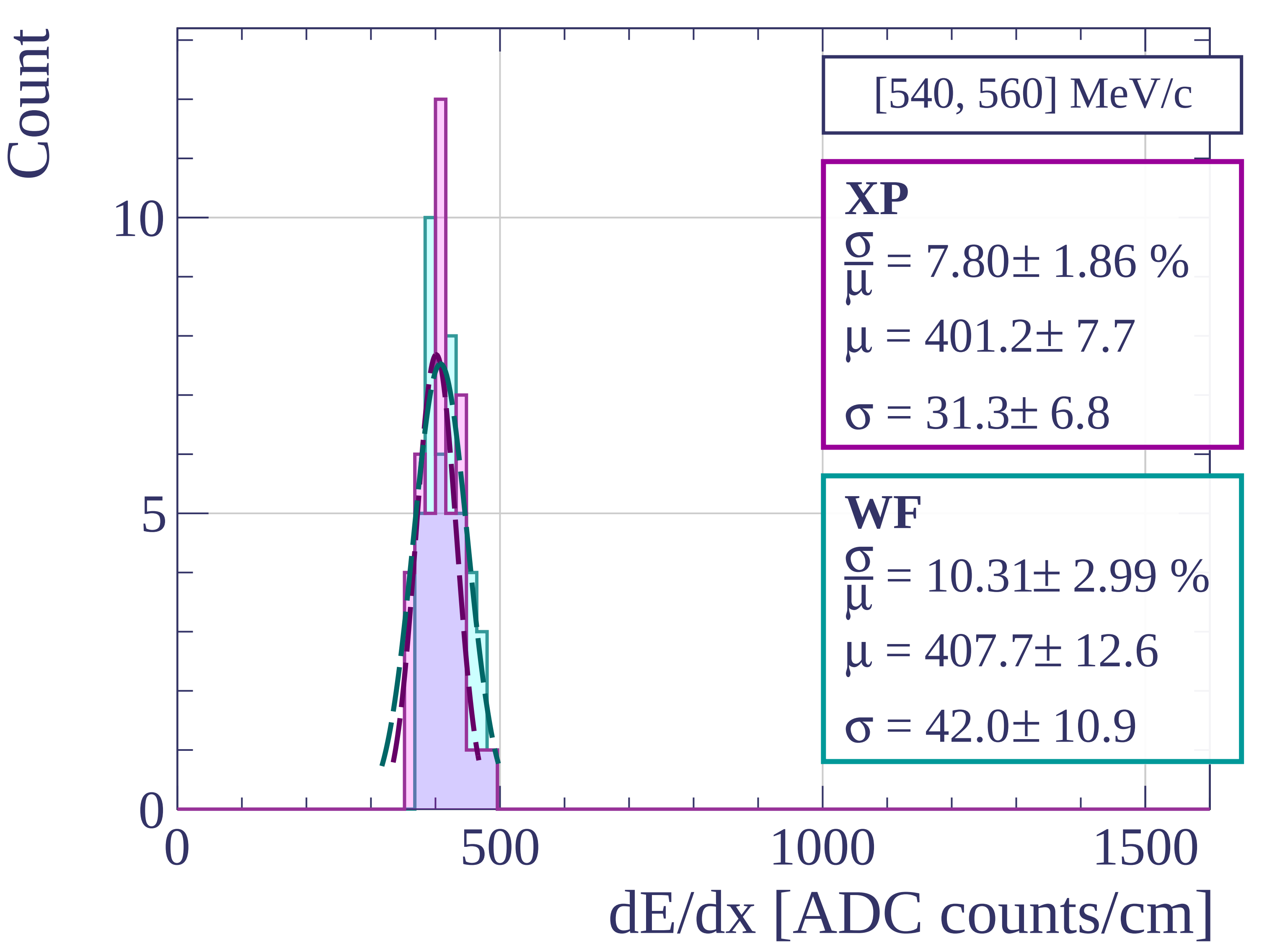

[480, 500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.64± 1.71 %
μ = 398.4± 5.3
σ = 26.4± 6.5
WF
$\frac{\sigma}{\mu}$ = 4.86± 0.81 %
μ = 409.5± 3.5
σ = 19.9± 3.1
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]



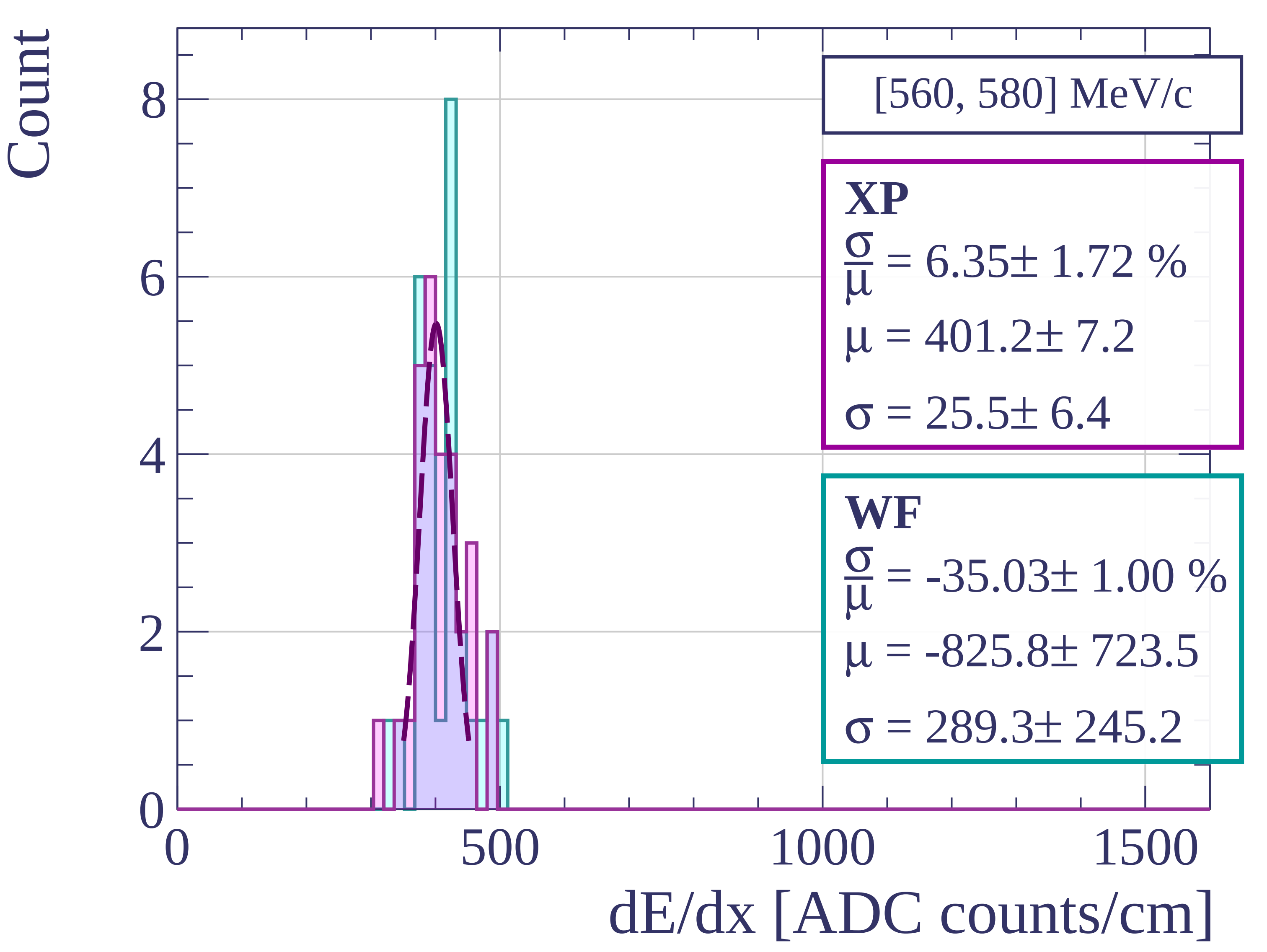
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[500, 520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.29± 1.16 %
μ = 399.9± 4.9
σ = 33.2± 4.2
WF
$\frac{\sigma}{\mu}$ = 9.39± 1.60 %
μ = 414.5± 7.0
σ = 38.9± 6.0



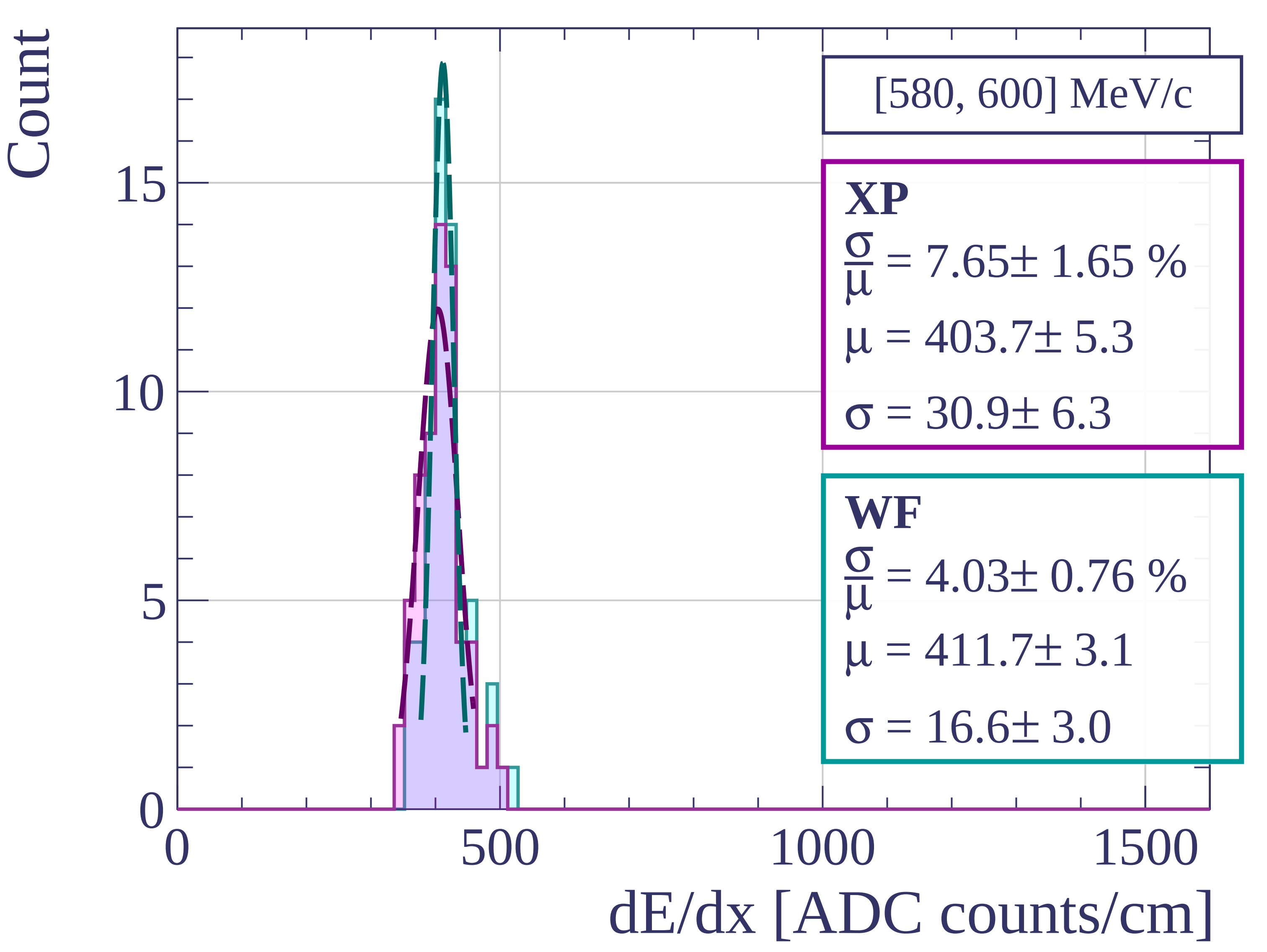
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.66± 1.15 %
μ = 397.7± 3.8
σ = 18.5± 4.4
WF
$\frac{\sigma}{\mu}$ = 5.29± 1.28 %
μ = 406.8± 4.2
σ = 21.5± 5.0



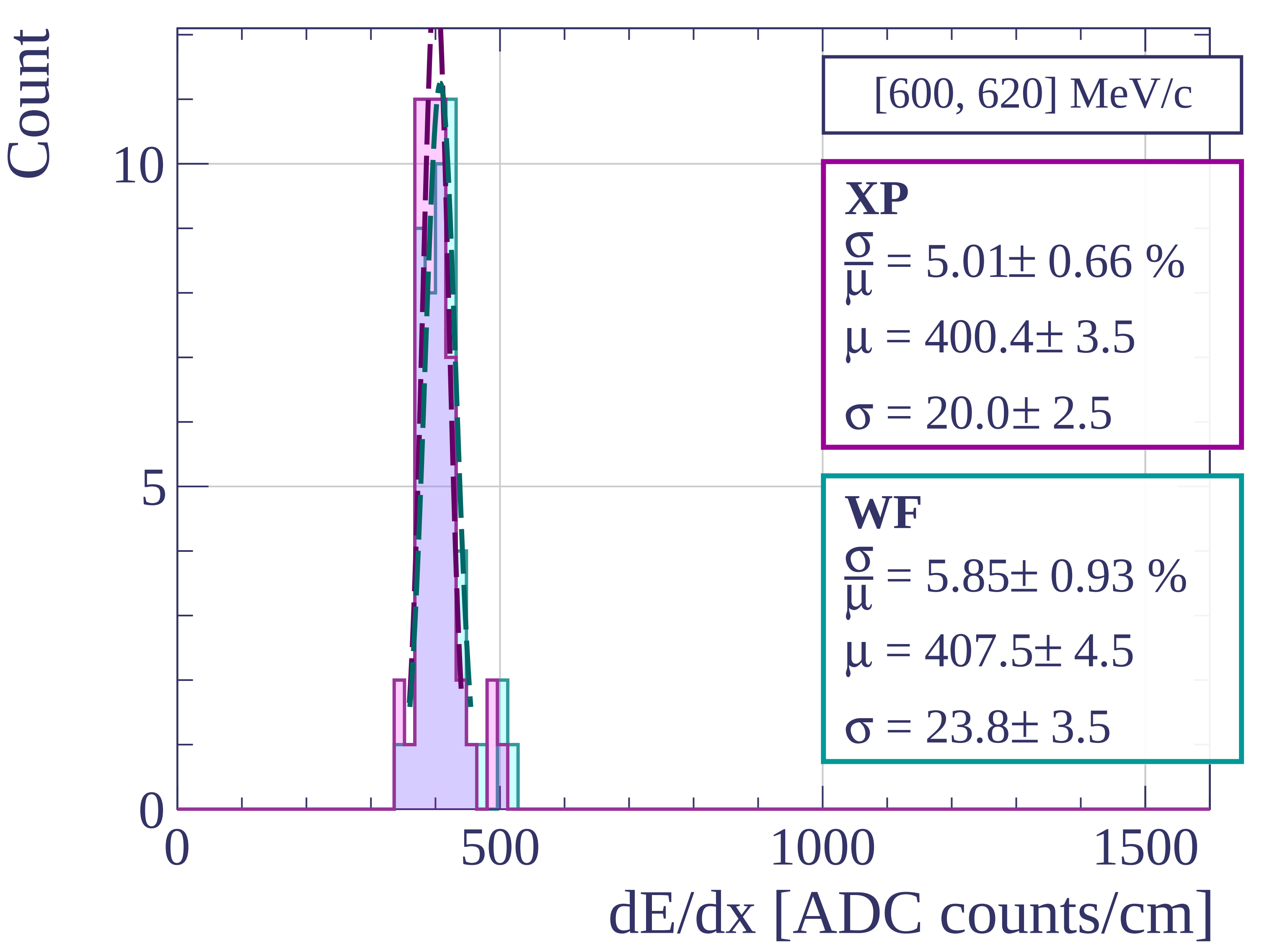
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[540, 560] MeV/c
XP
σ/μ = 7.80± 1.86 %
μ = 401.2± 7.7
σ = 31.3± 6.8
WF
σ/μ = 10.31± 2.99 %
μ = 407.7± 12.6
σ = 42.0± 10.9



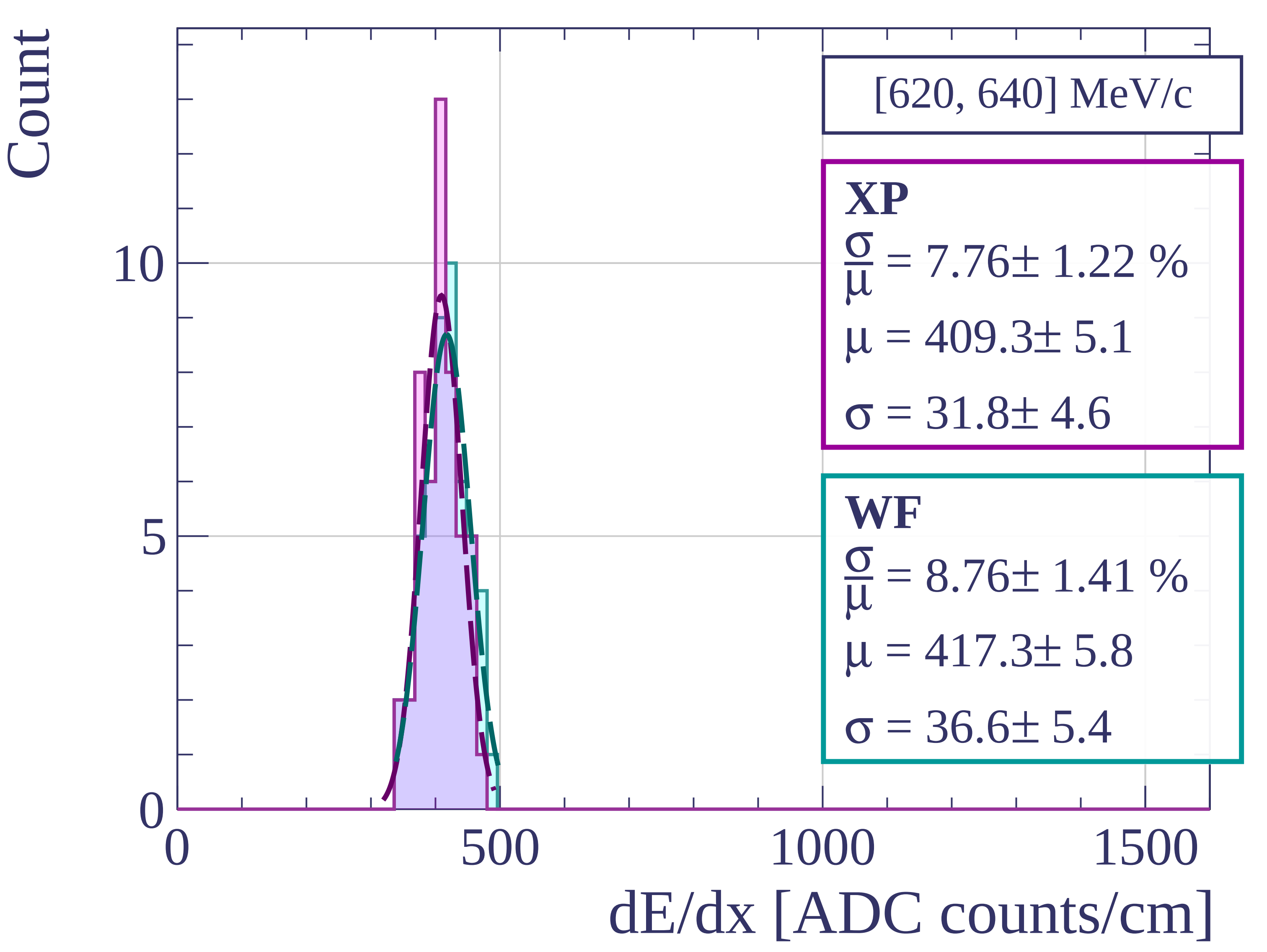
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 580] MeV/c
XP
σ/μ = 6.35± 1.72 %
μ = 401.2± 7.2
σ = 25.5± 6.4
WF
σ/μ = -35.03± 1.00 %
μ = -825.8± 723.5
σ = 289.3± 245.2



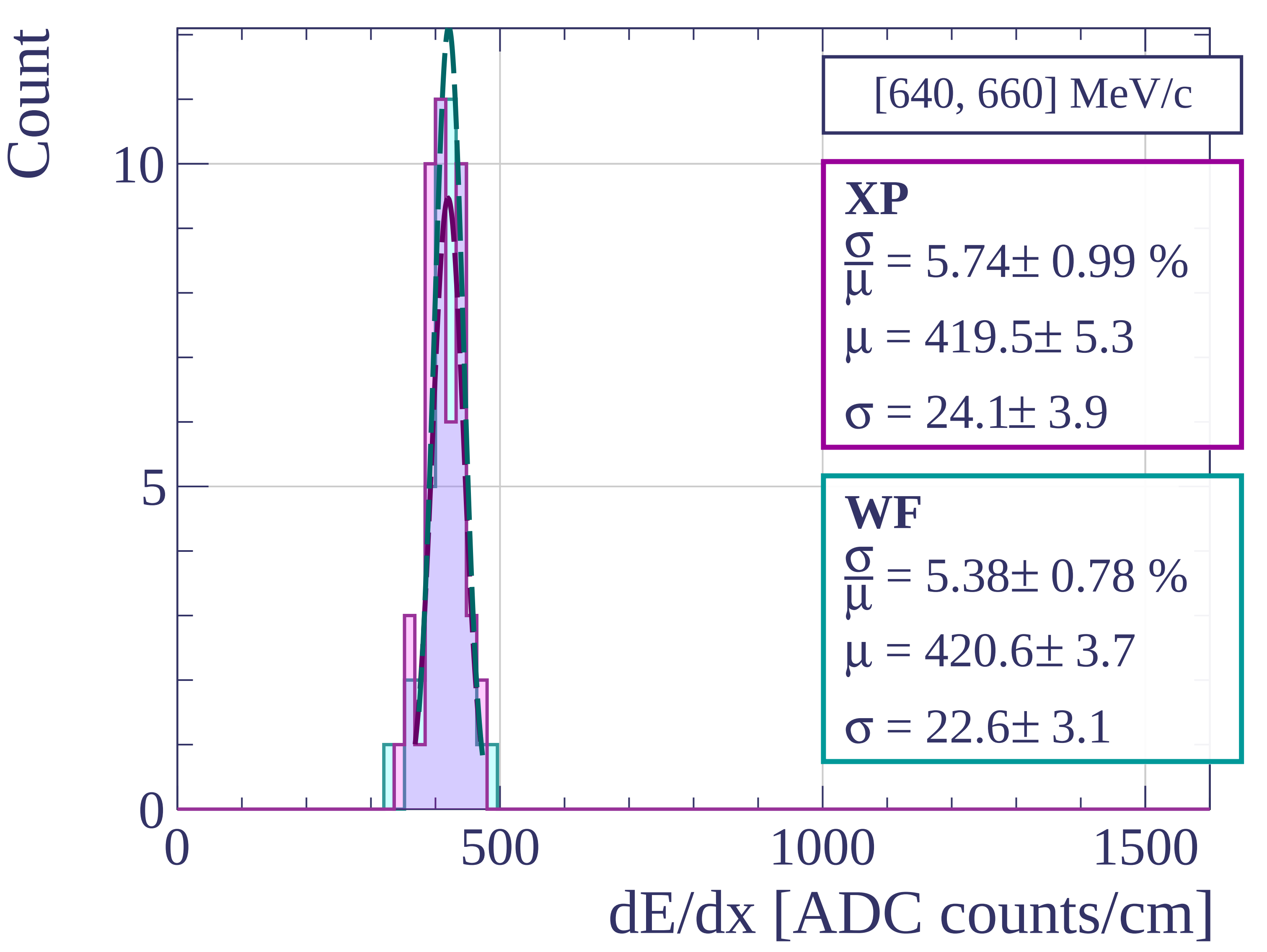
Count
[580, 600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.65± 1.65 %
μ = 403.7± 5.3
σ = 30.9± 6.3
WF
$\frac{\sigma}{\mu}$ = 4.03± 0.76 %
μ = 411.7± 3.1
σ = 16.6± 3.0
0
5
10
15
0
500
1000
1500
dE/dx [ADC counts/cm]



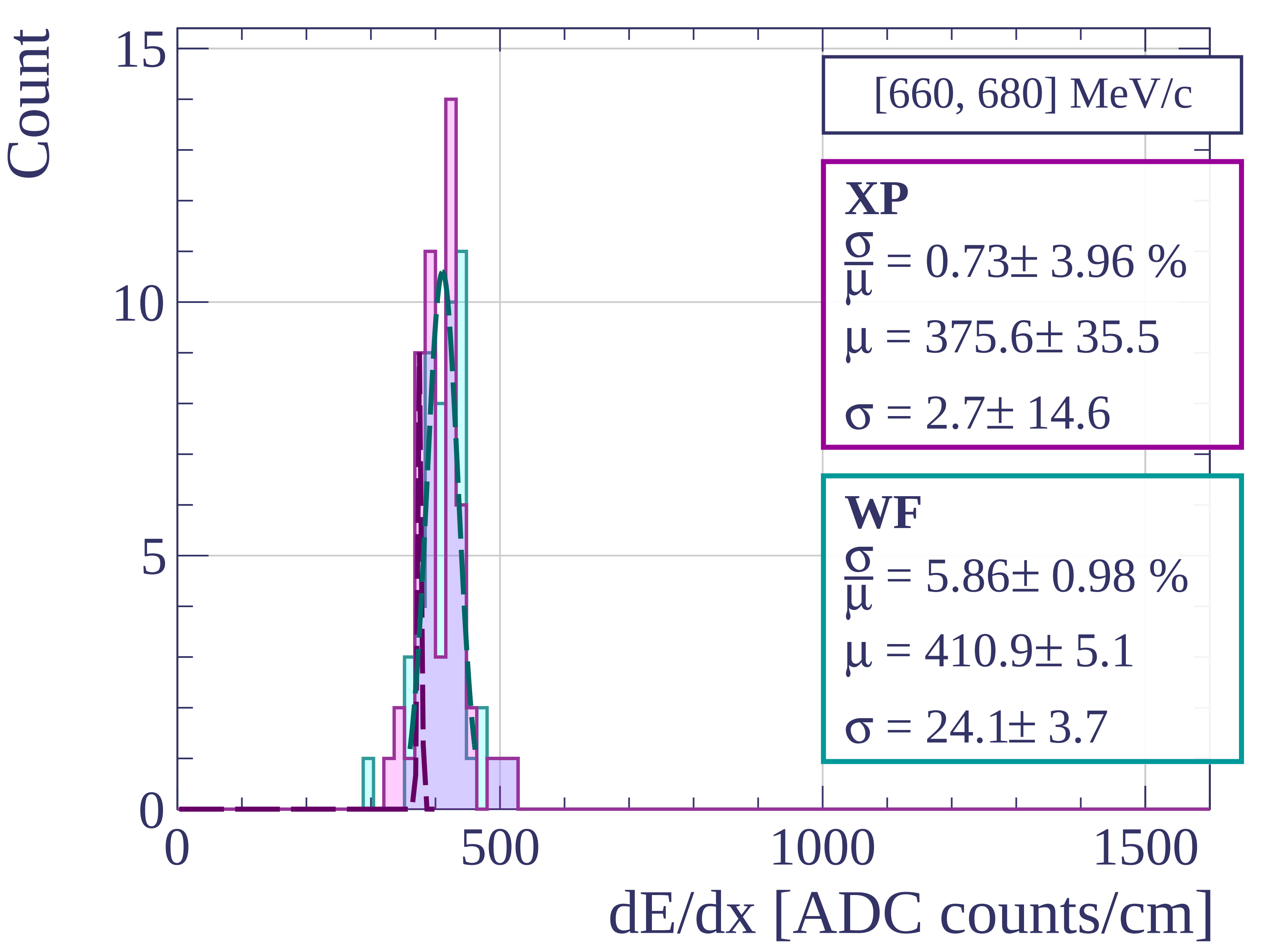
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.01± 0.66 %
μ = 400.4± 3.5
σ = 20.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 5.85± 0.93 %
μ = 407.5± 4.5
σ = 23.8± 3.5



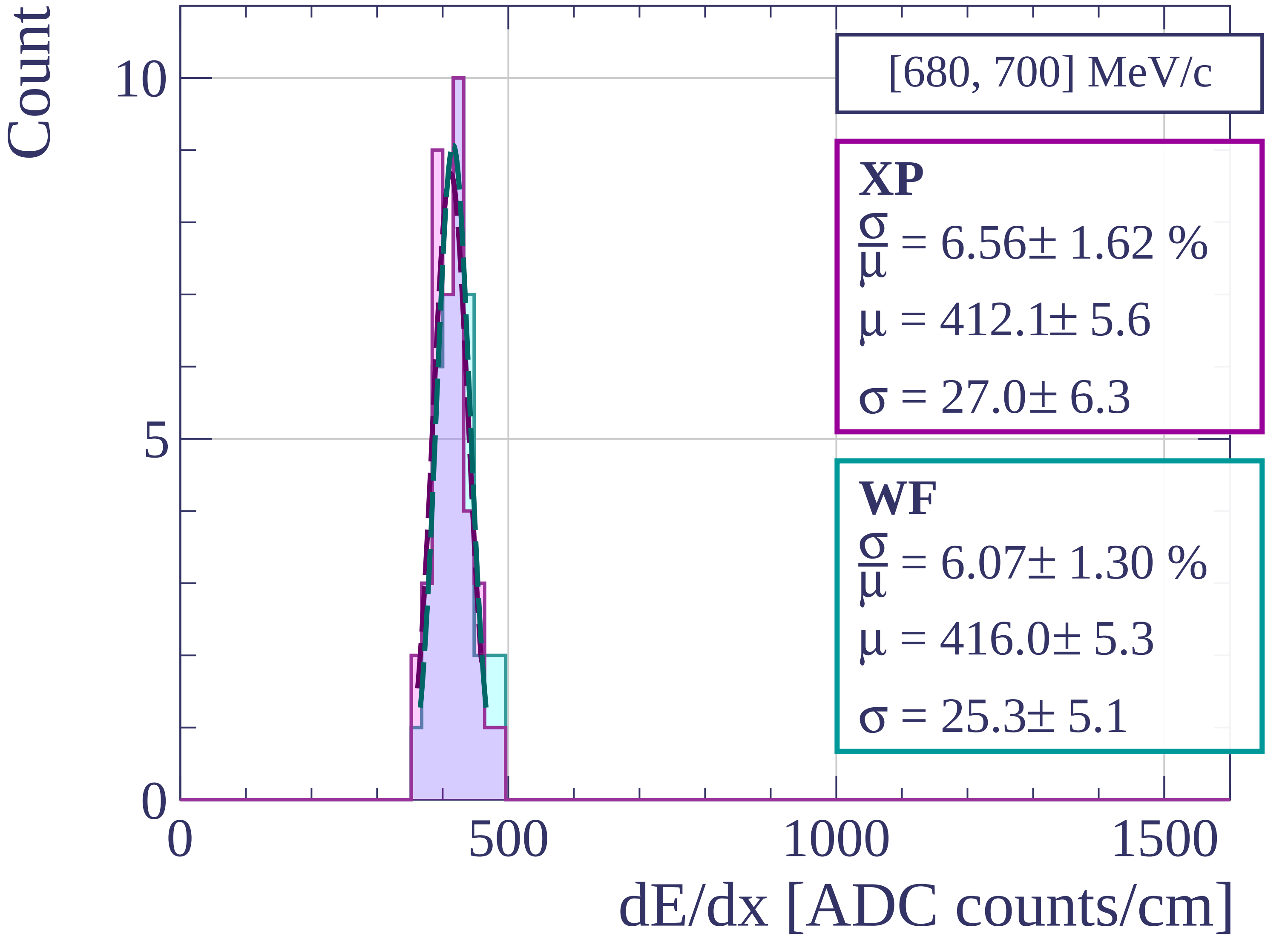
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[620, 640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.76± 1.22 %
μ = 409.3± 5.1
σ = 31.8± 4.6
WF
$\frac{\sigma}{\mu}$ = 8.76± 1.41 %
μ = 417.3± 5.8
σ = 36.6± 5.4



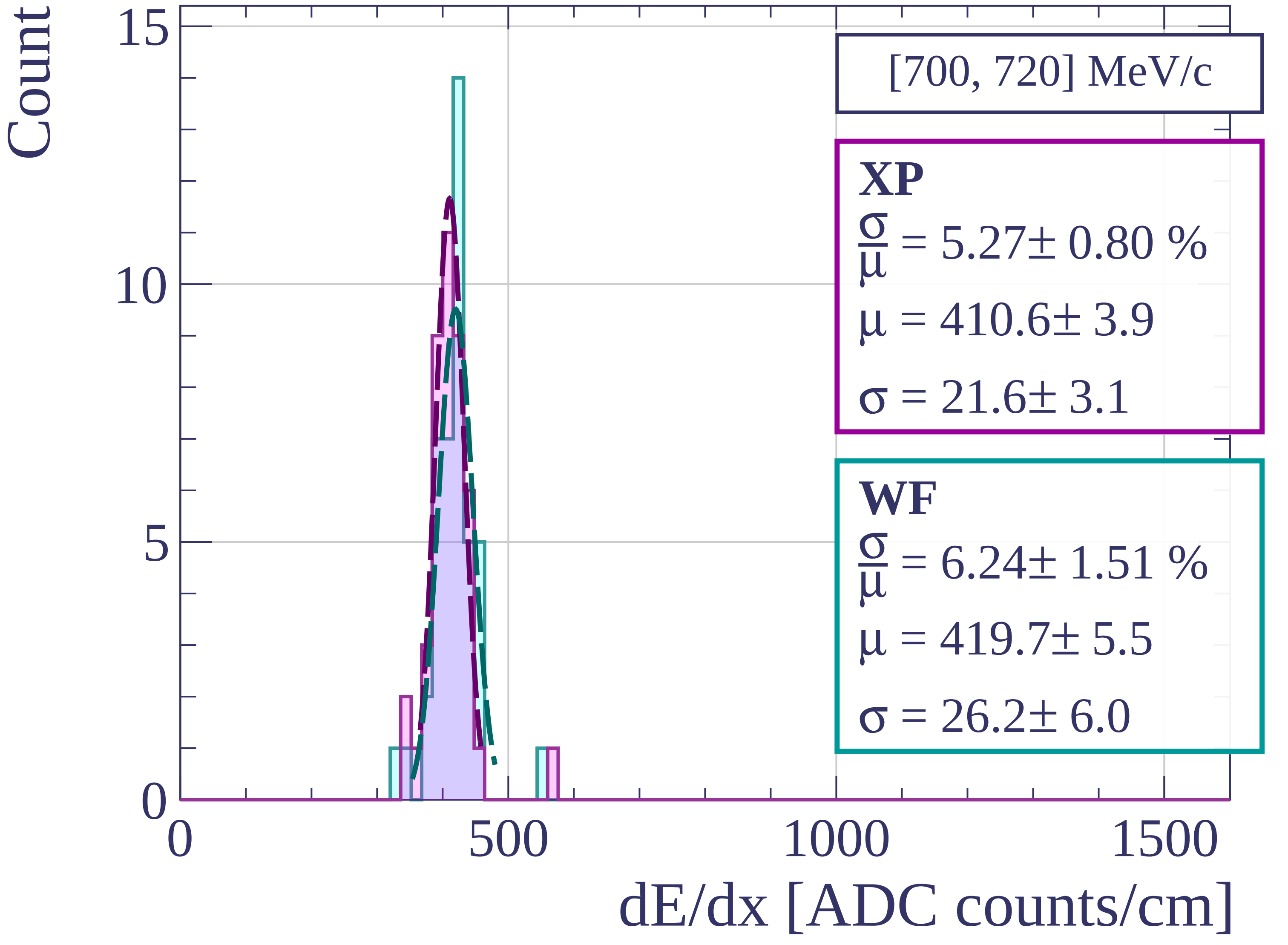
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 660] MeV/c
XP
σ/μ = 5.74± 0.99 %
μ = 419.5± 5.3
σ = 24.1± 3.9
WF
σ/μ = 5.38± 0.78 %
μ = 420.6± 3.7
σ = 22.6± 3.1



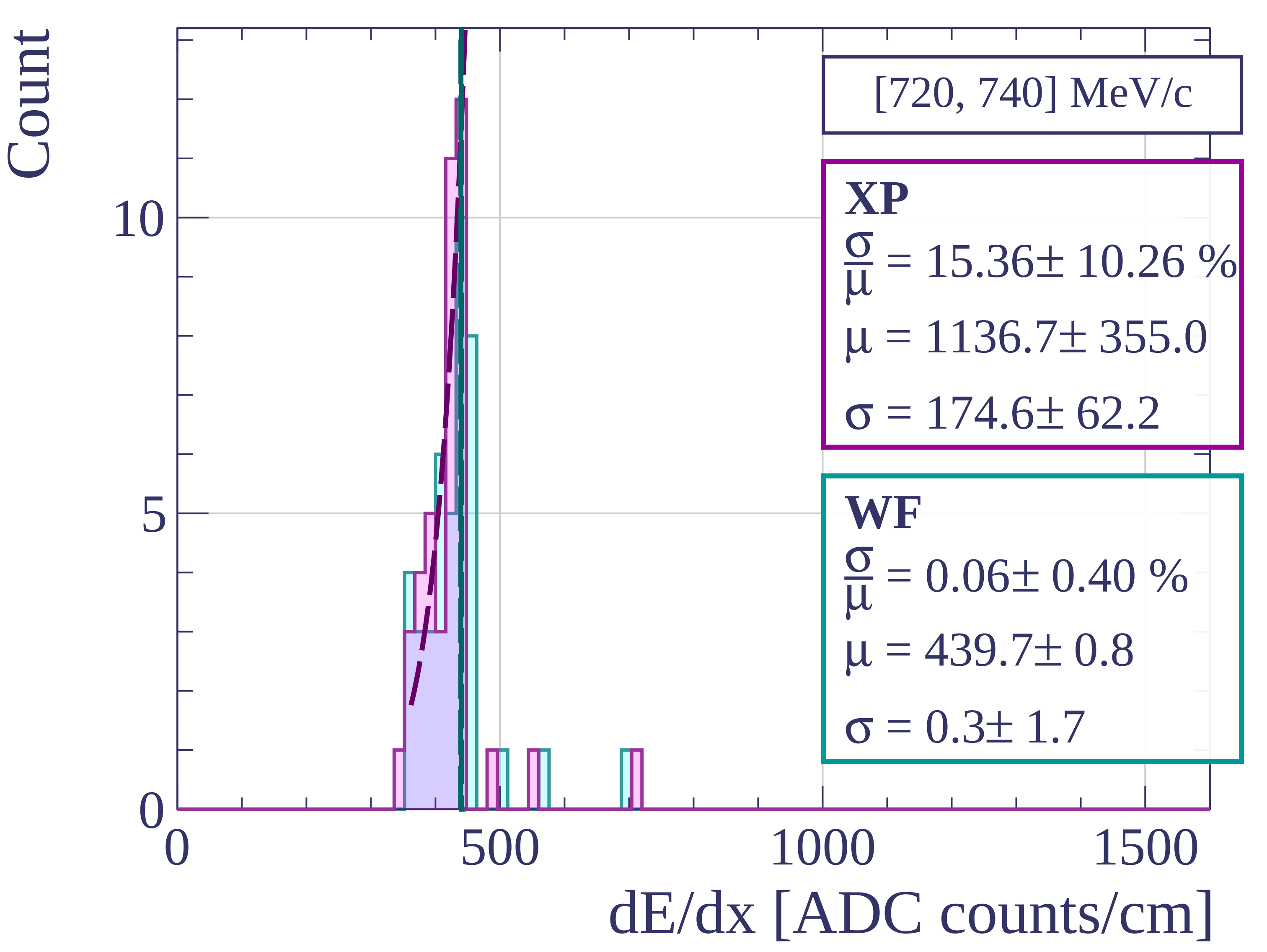
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 0.73± 3.96 %
μ = 375.6± 35.5
σ = 2.7± 14.6
WF
$\frac{\sigma}{\mu}$ = 5.86± 0.98 %
μ = 410.9± 5.1
σ = 24.1± 3.7



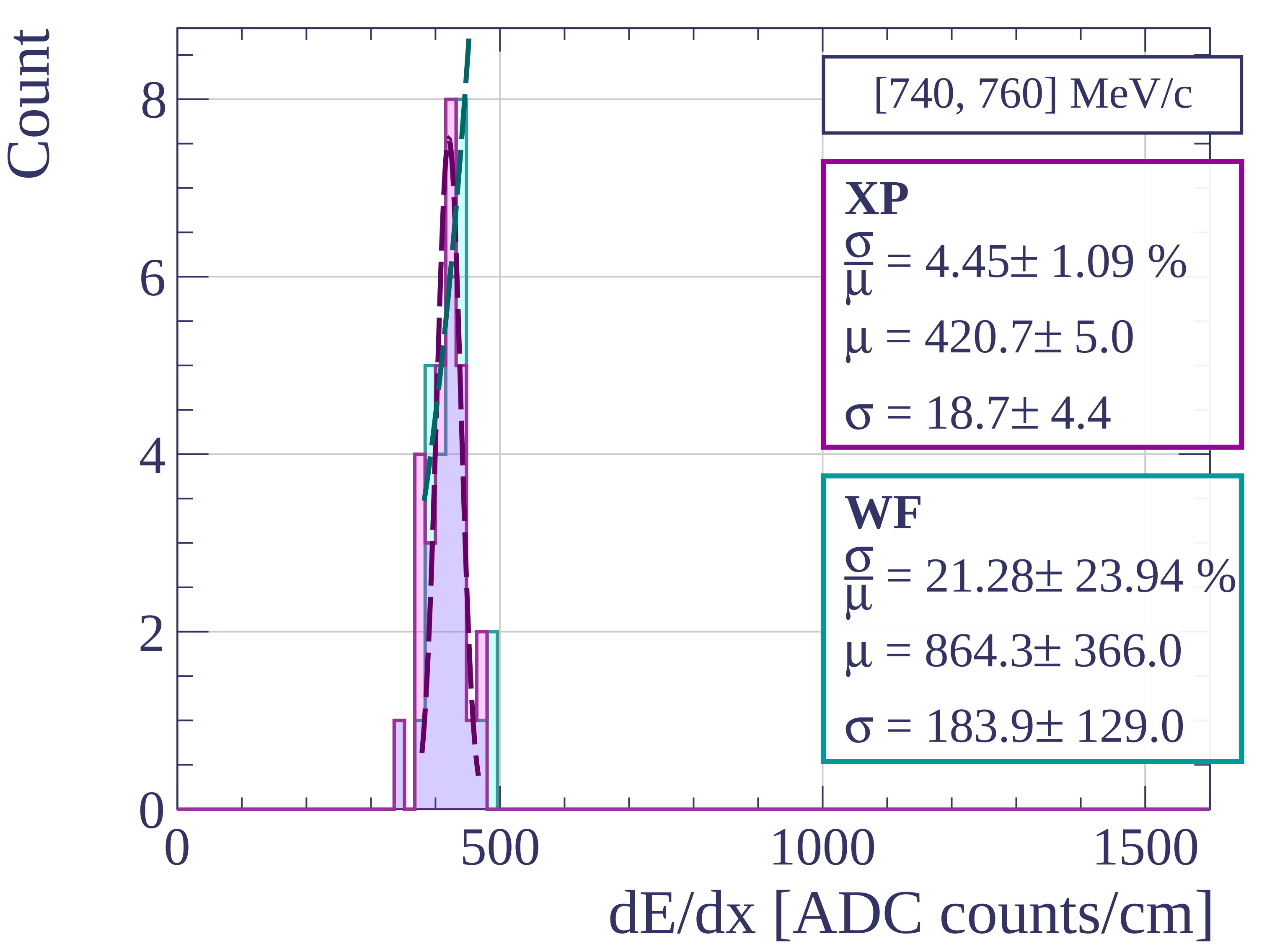
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[680, 700] MeV/c
XP
σ/μ = 6.56± 1.62 %
μ = 412.1± 5.6
σ = 27.0± 6.3
WF
σ/μ = 6.07± 1.30 %
μ = 416.0± 5.3
σ = 25.3± 5.1



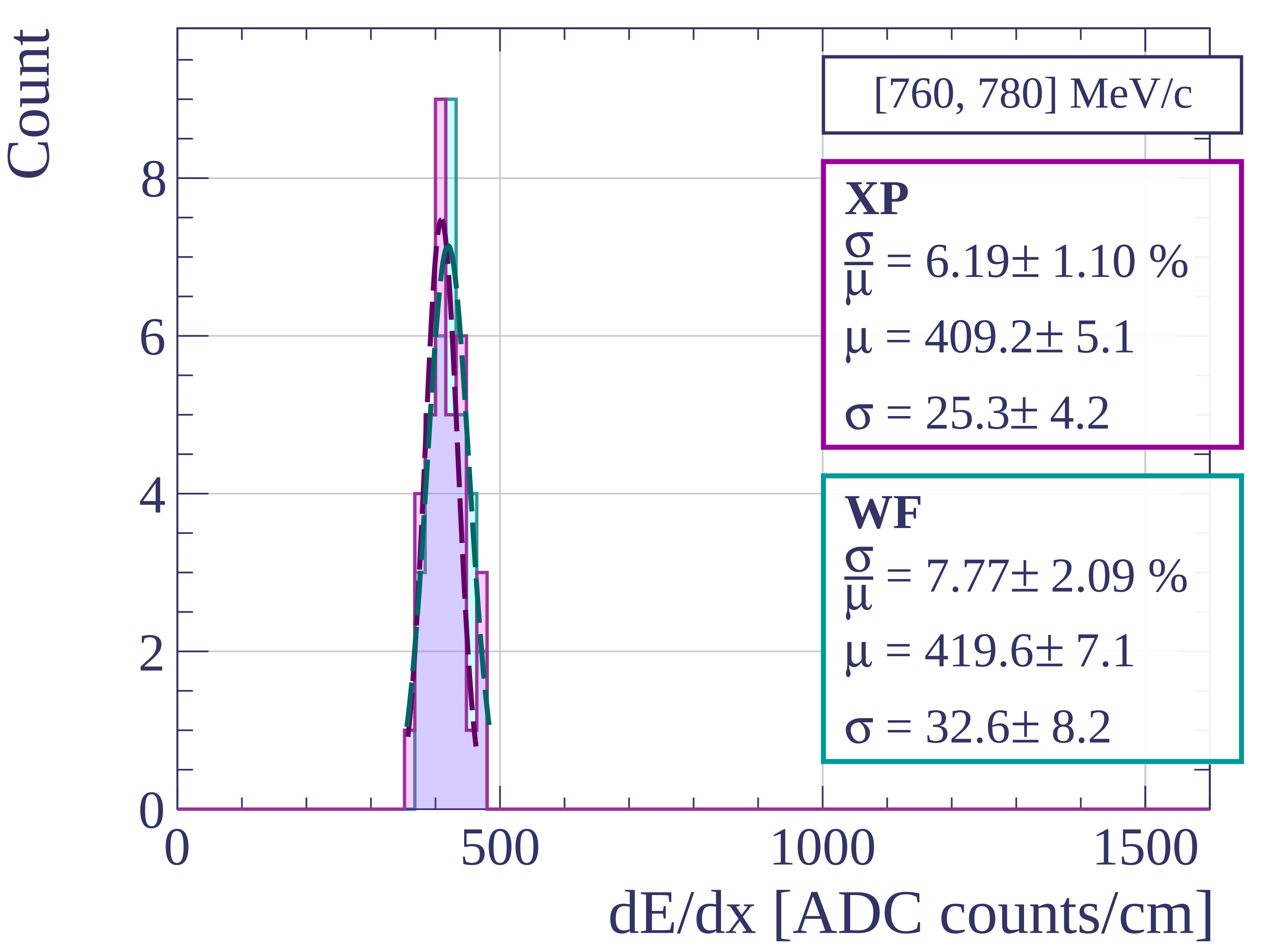
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.27± 0.80 %
μ = 410.6± 3.9
σ = 21.6± 3.1
WF
$\frac{\sigma}{\mu}$ = 6.24± 1.51 %
μ = 419.7± 5.5
σ = 26.2± 6.0



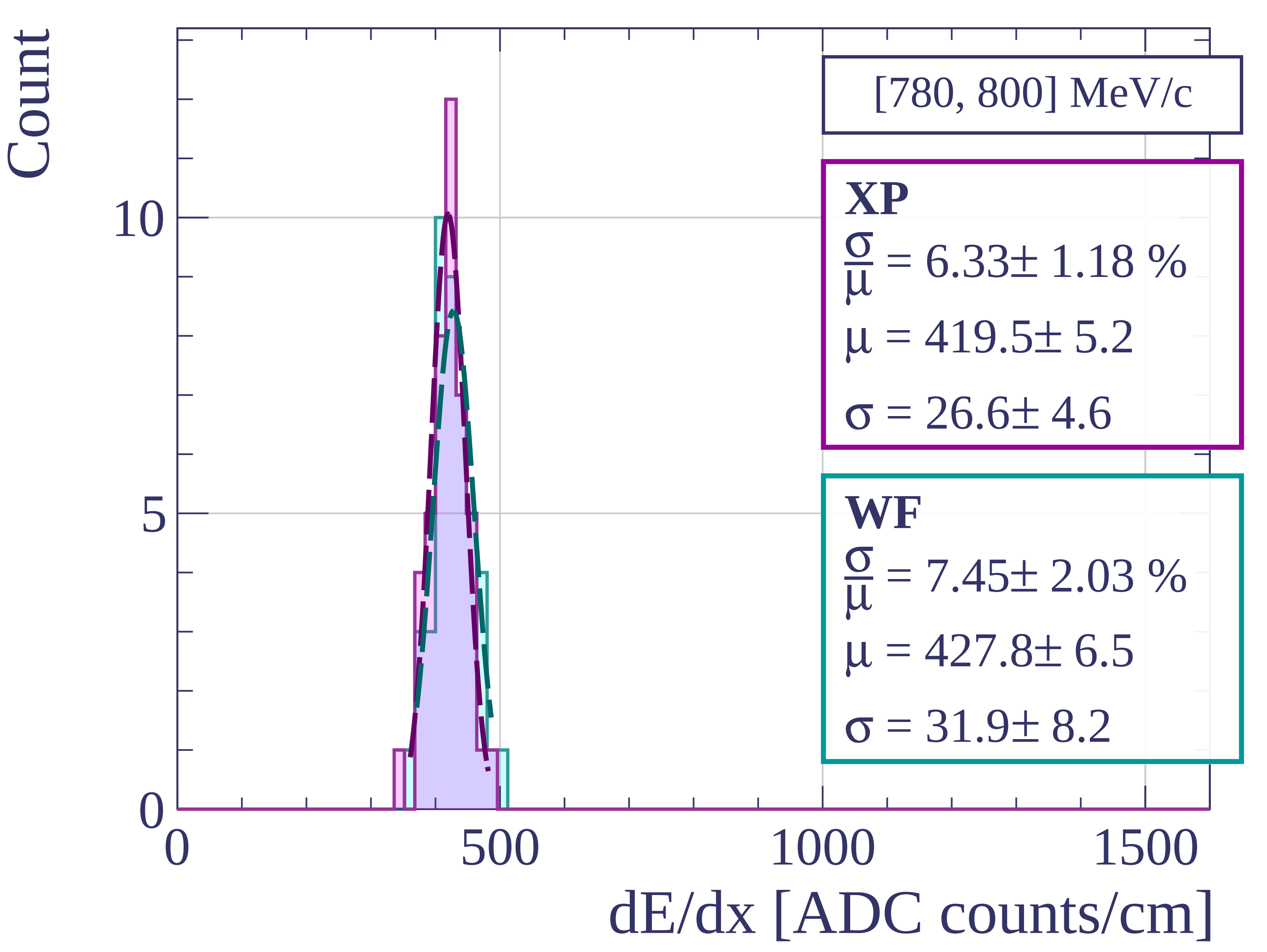
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[720, 740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 15.36± 10.26 %
μ = 1136.7± 355.0
σ = 174.6± 62.2
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.40 %
μ = 439.7± 0.8
σ = 0.3± 1.7



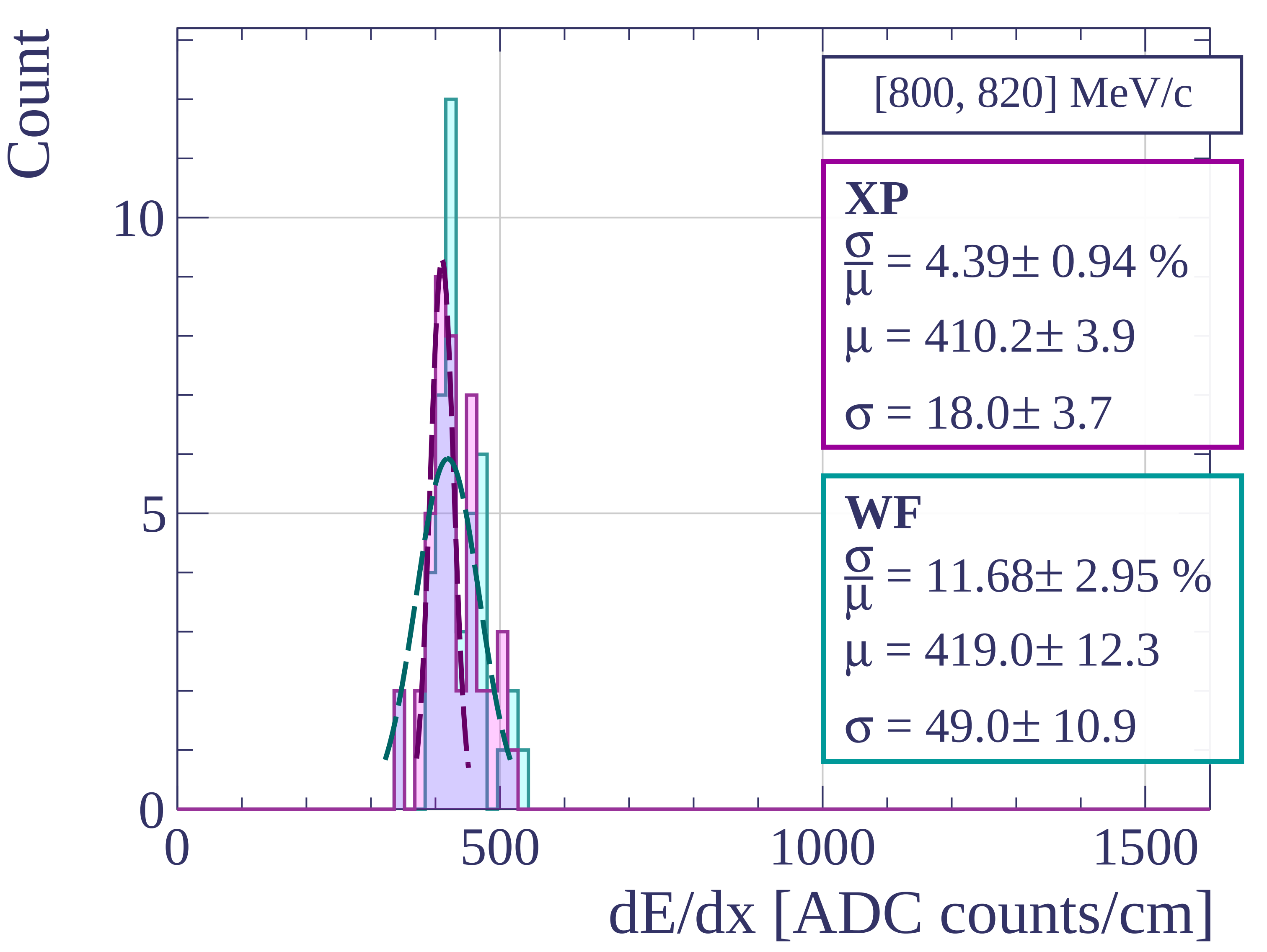
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[740, 760] MeV/c
XP
σ/μ = 4.45± 1.09 %
μ = 420.7± 5.0
σ = 18.7± 4.4
WF
σ/μ = 21.28± 23.94 %
μ = 864.3± 366.0
σ = 183.9± 129.0



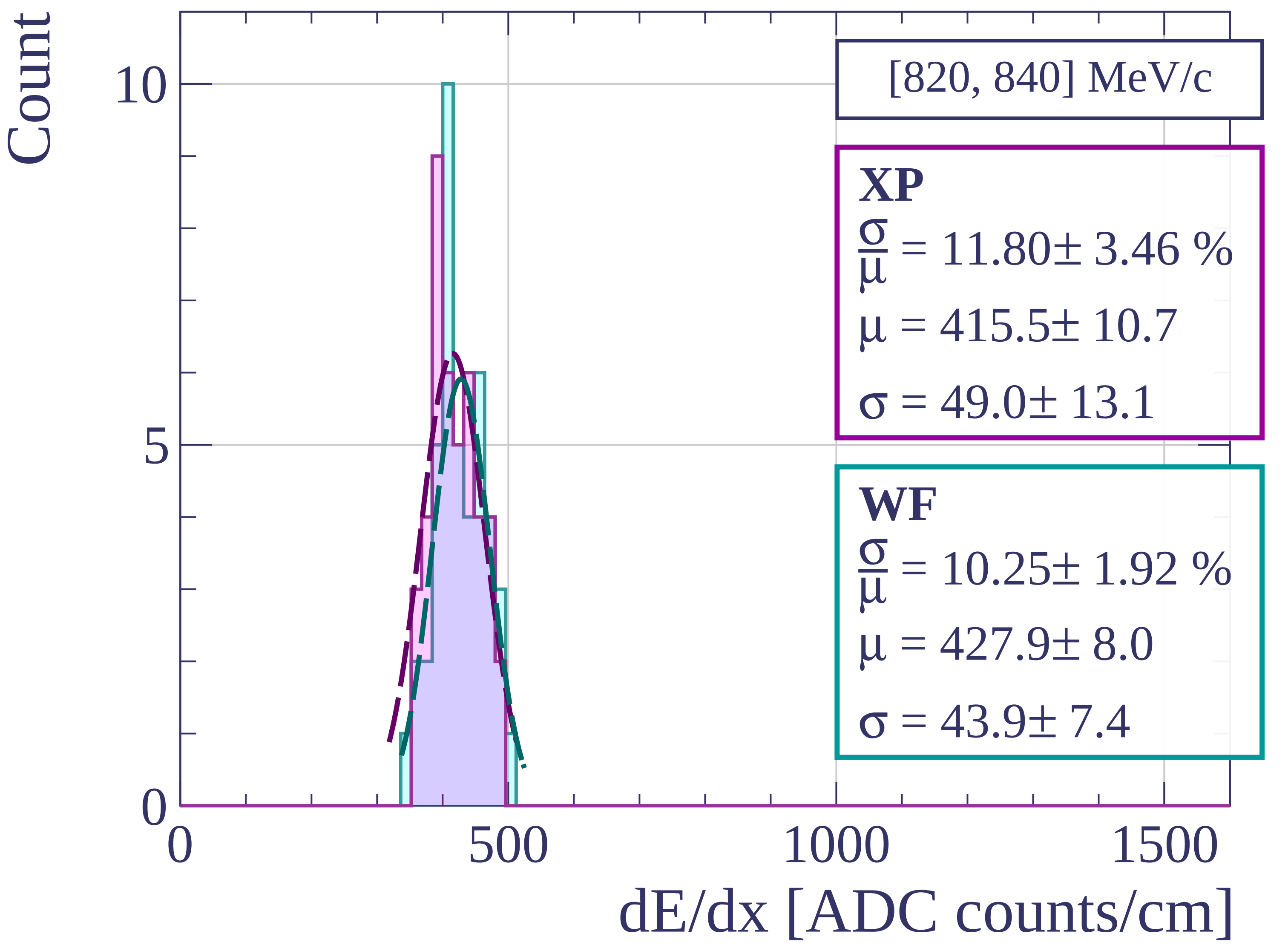
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[760, 780] MeV/c
XP
σ/μ = 6.19± 1.10 %
μ = 409.2± 5.1
σ = 25.3± 4.2
WF
σ/μ = 7.77± 2.09 %
μ = 419.6± 7.1
σ = 32.6± 8.2



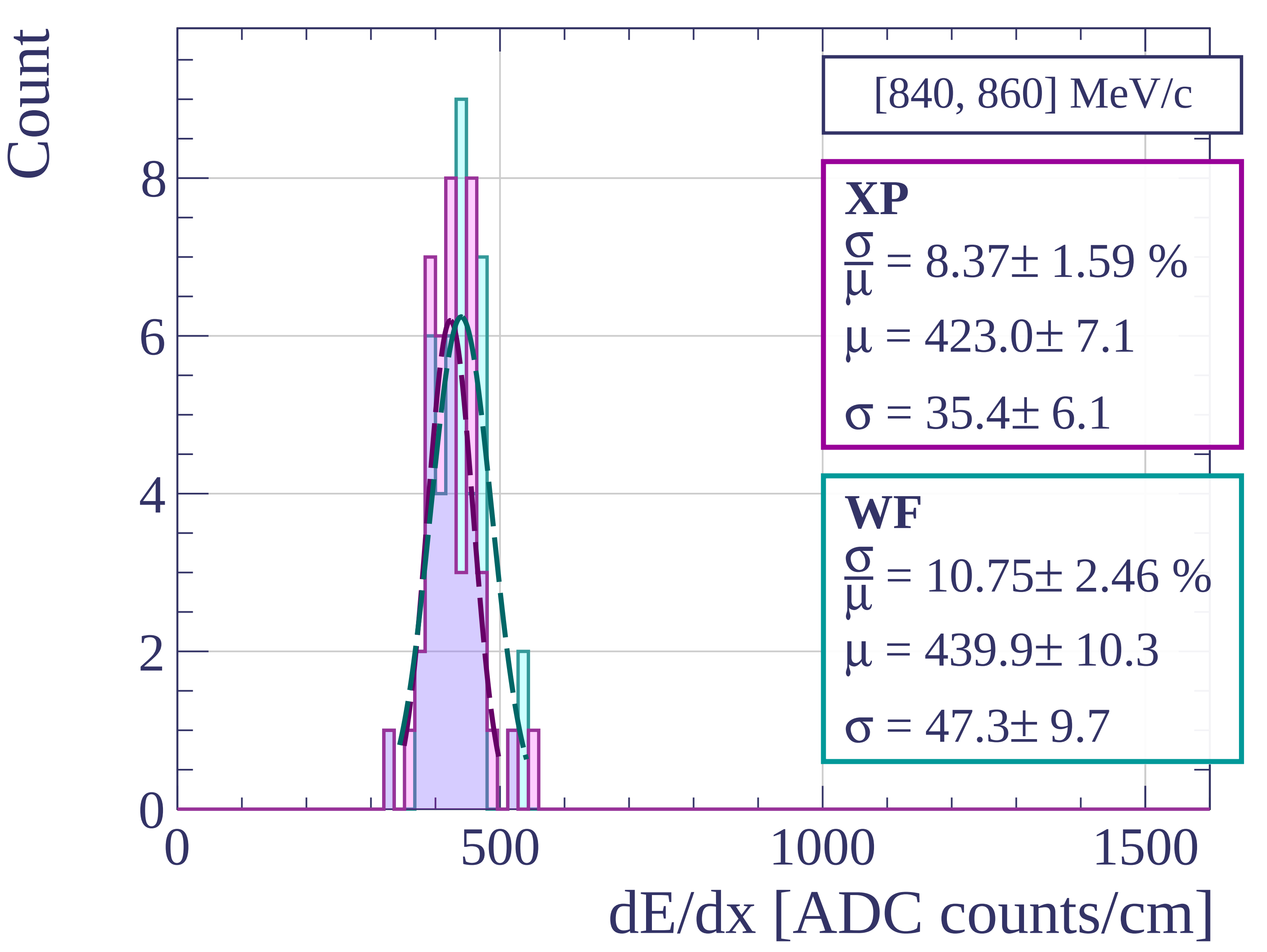
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[780, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.33± 1.18 %
μ = 419.5± 5.2
σ = 26.6± 4.6
WF
$\frac{\sigma}{\mu}$ = 7.45± 2.03 %
μ = 427.8± 6.5
σ = 31.9± 8.2



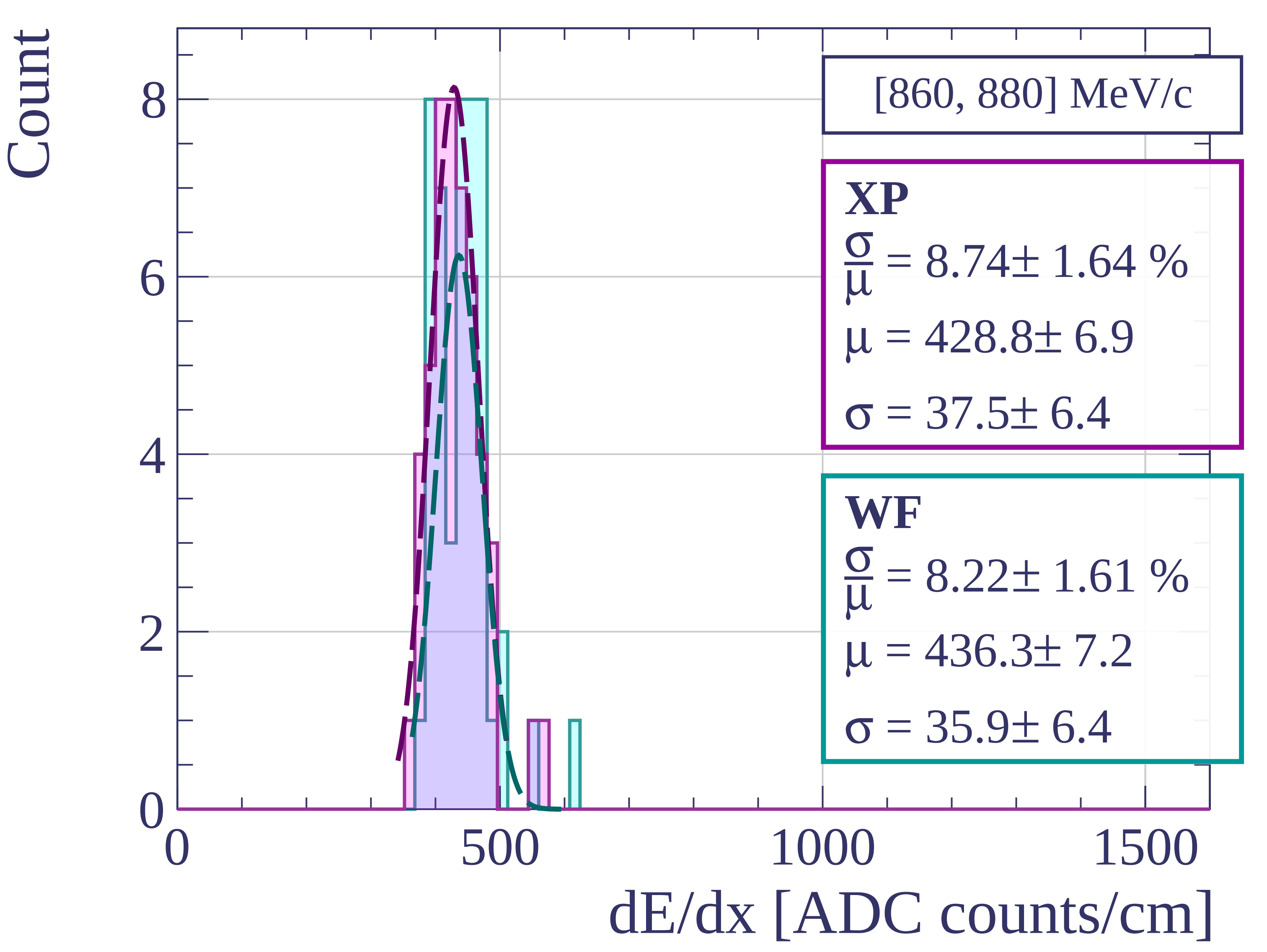
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.39± 0.94 %
μ = 410.2± 3.9
σ = 18.0± 3.7
WF
$\frac{\sigma}{\mu}$ = 11.68± 2.95 %
μ = 419.0± 12.3
σ = 49.0± 10.9



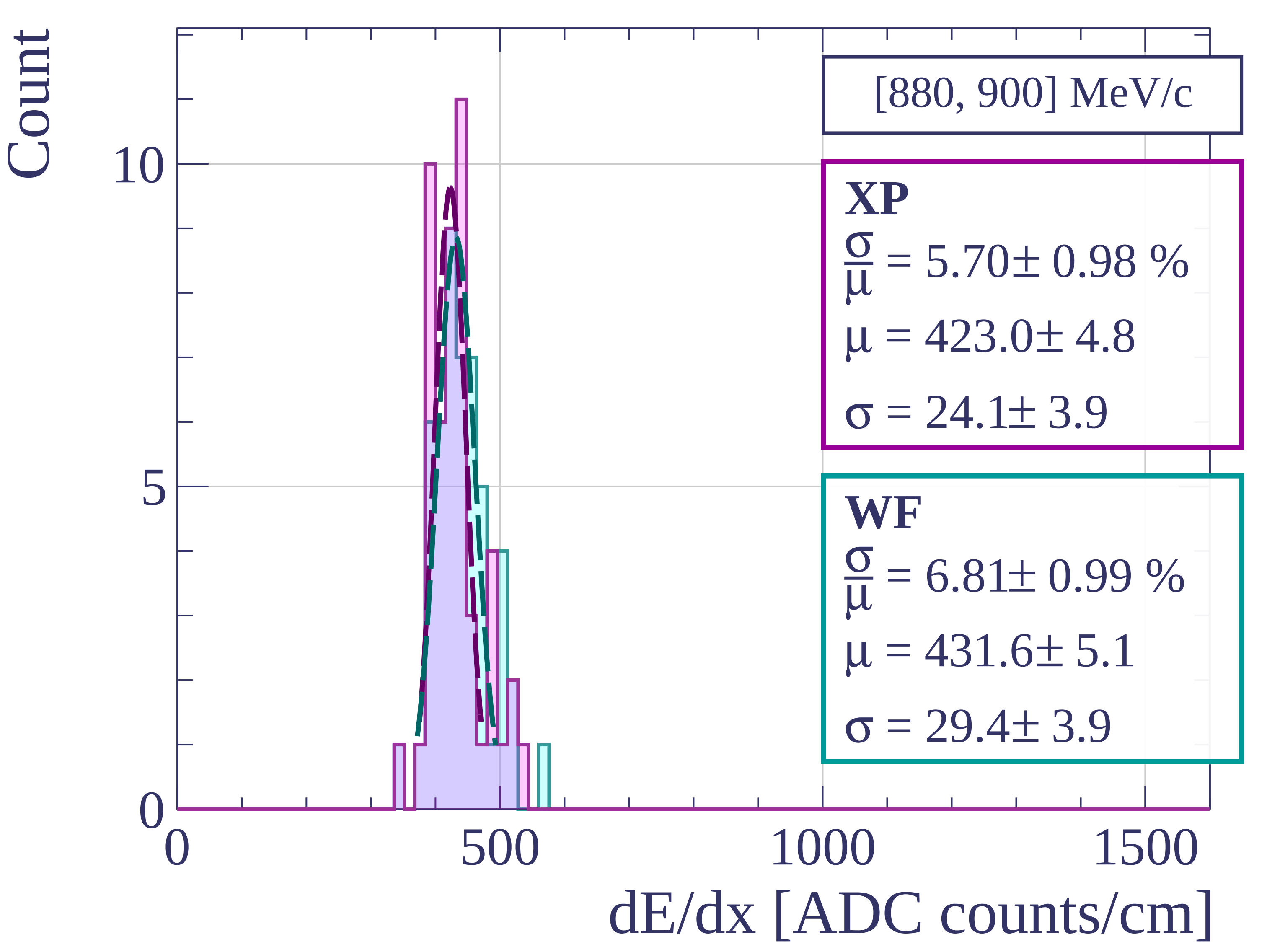
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[820, 840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.80± 3.46 %
μ = 415.5± 10.7
σ = 49.0± 13.1
WF
$\frac{\sigma}{\mu}$ = 10.25± 1.92 %
μ = 427.9± 8.0
σ = 43.9± 7.4



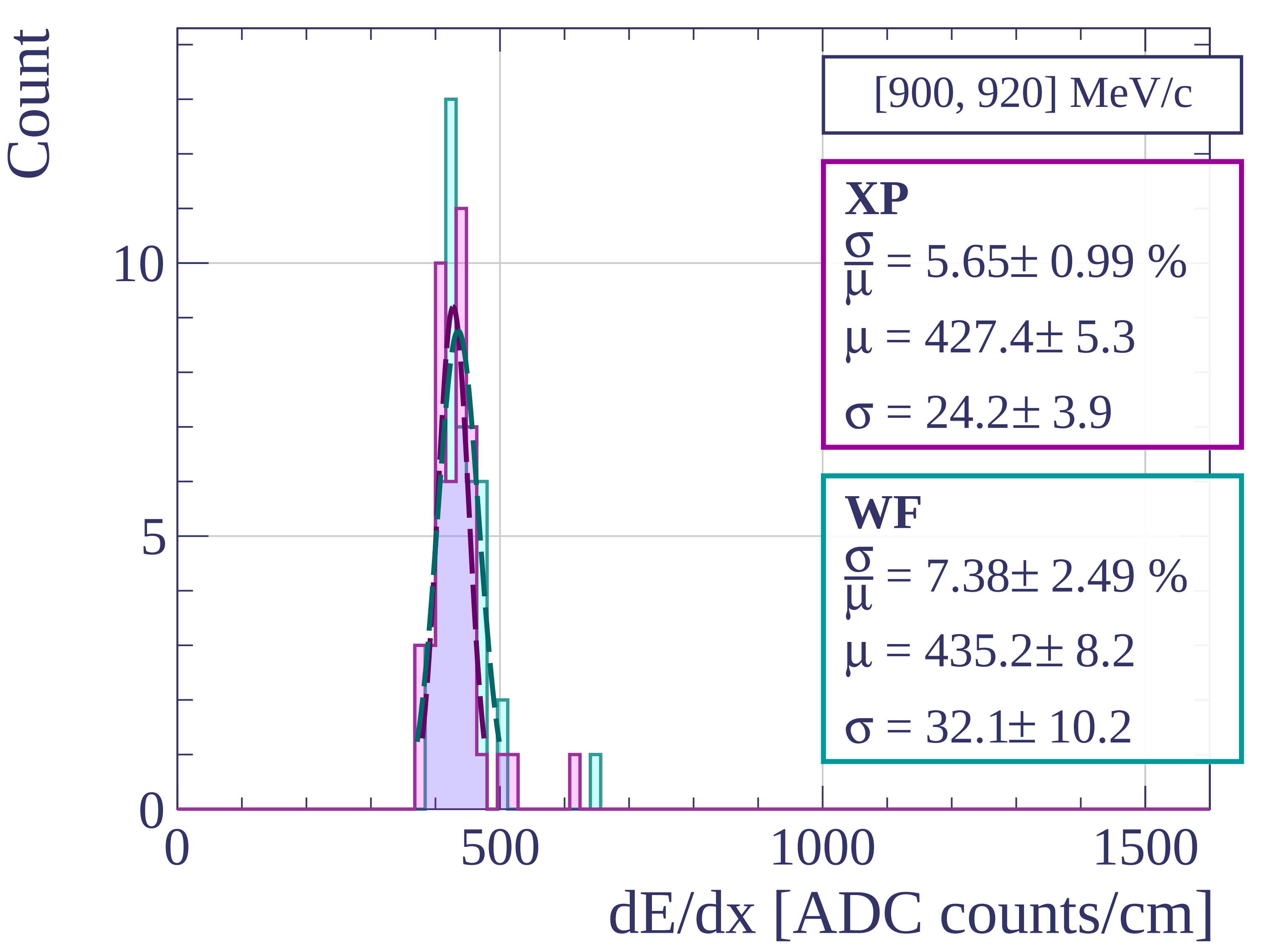
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[840, 860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.37± 1.59 %
μ = 423.0± 7.1
σ = 35.4± 6.1
WF
$\frac{\sigma}{\mu}$ = 10.75± 2.46 %
μ = 439.9± 10.3
σ = 47.3± 9.7



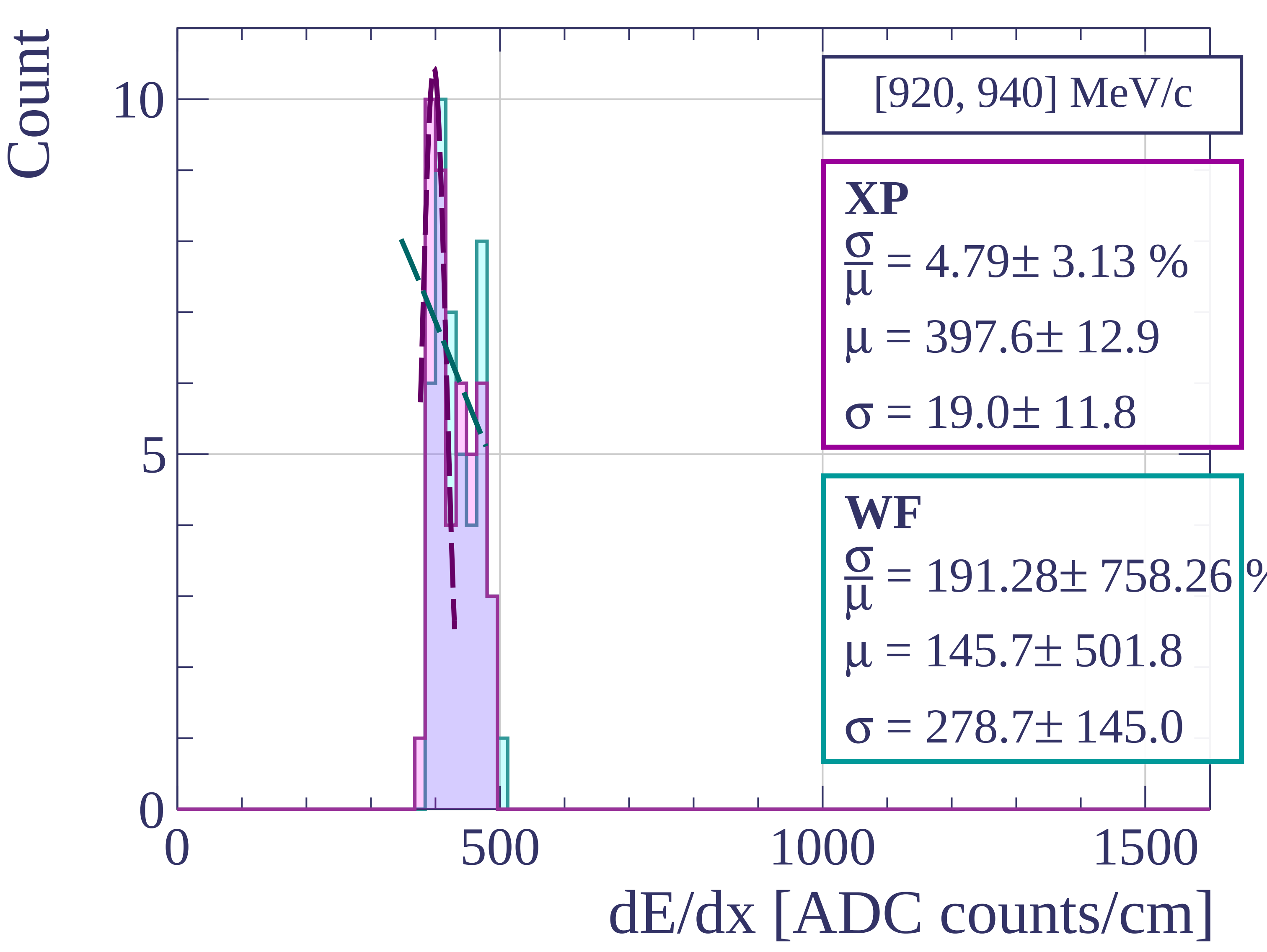
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[860, 880] MeV/c
XP
σ/μ = 8.74± 1.64 %
μ = 428.8± 6.9
σ = 37.5± 6.4
WF
σ/μ = 8.22± 1.61 %
μ = 436.3± 7.2
σ = 35.9± 6.4



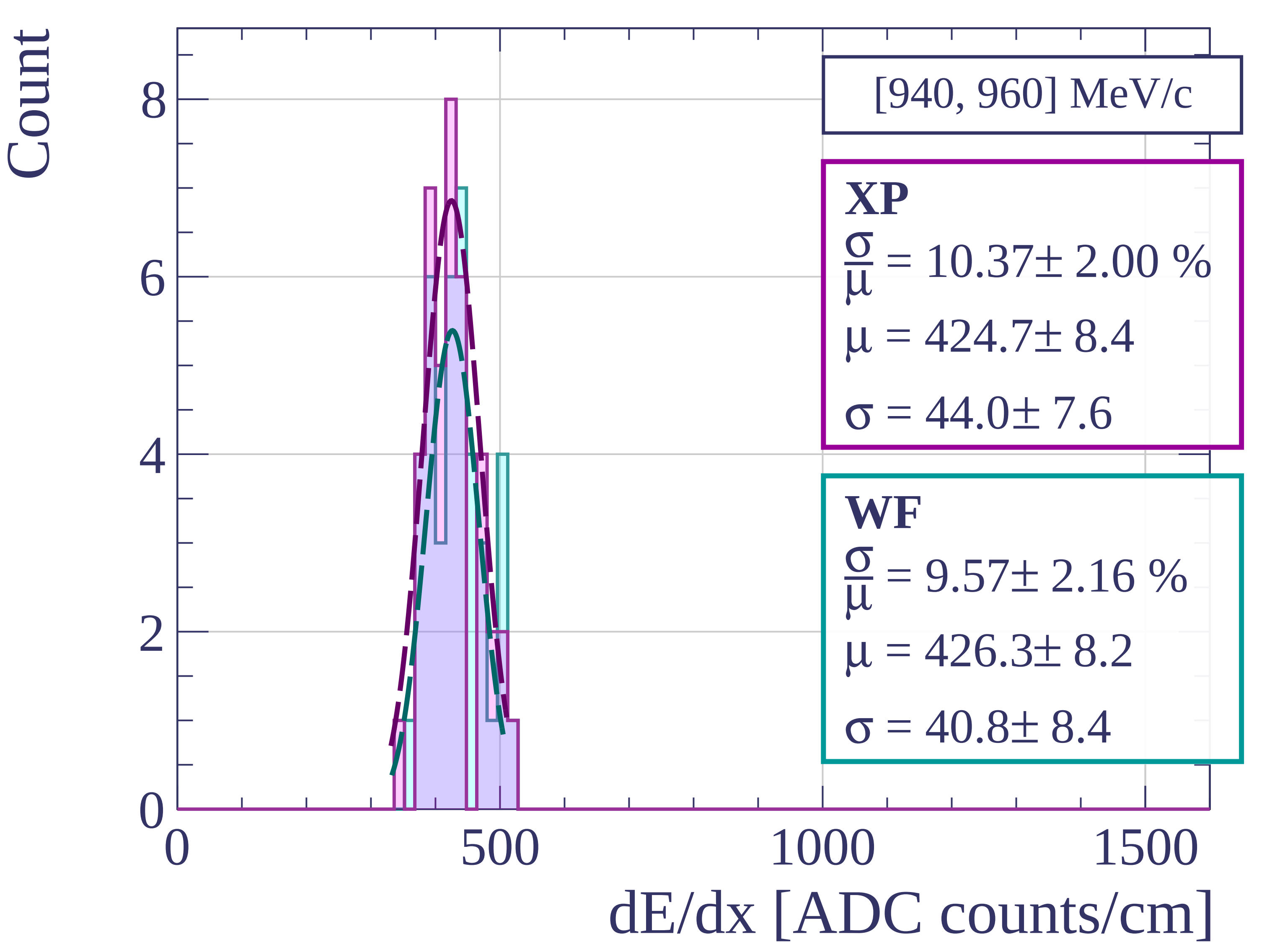
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.70± 0.98 %
μ = 423.0± 4.8
σ = 24.1± 3.9
WF
$\frac{\sigma}{\mu}$ = 6.81± 0.99 %
μ = 431.6± 5.1
σ = 29.4± 3.9



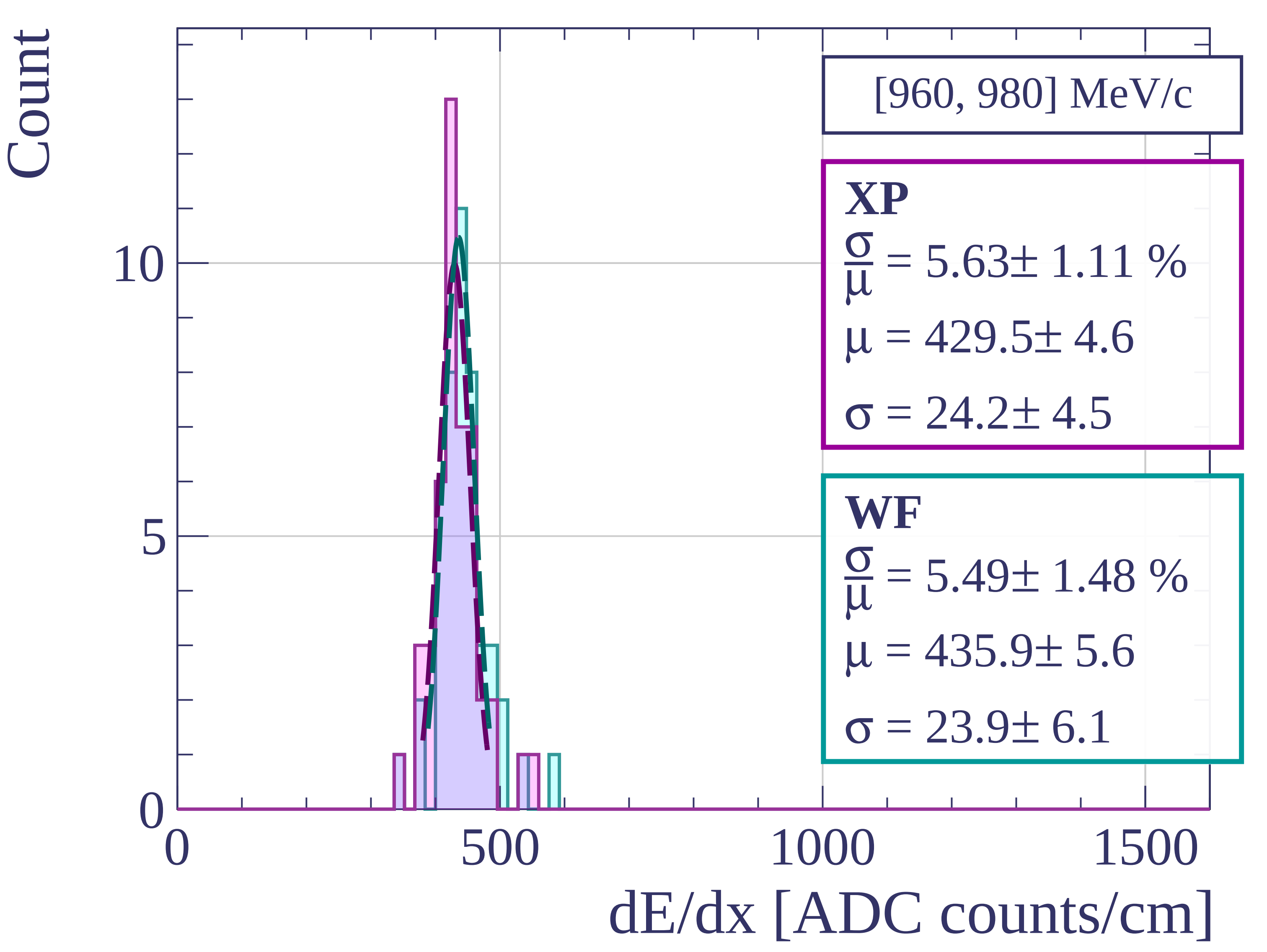
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
[900, 920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.65± 0.99 %
μ = 427.4± 5.3
σ = 24.2± 3.9
WF
$\frac{\sigma}{\mu}$ = 7.38± 2.49 %
μ = 435.2± 8.2
σ = 32.1± 10.2



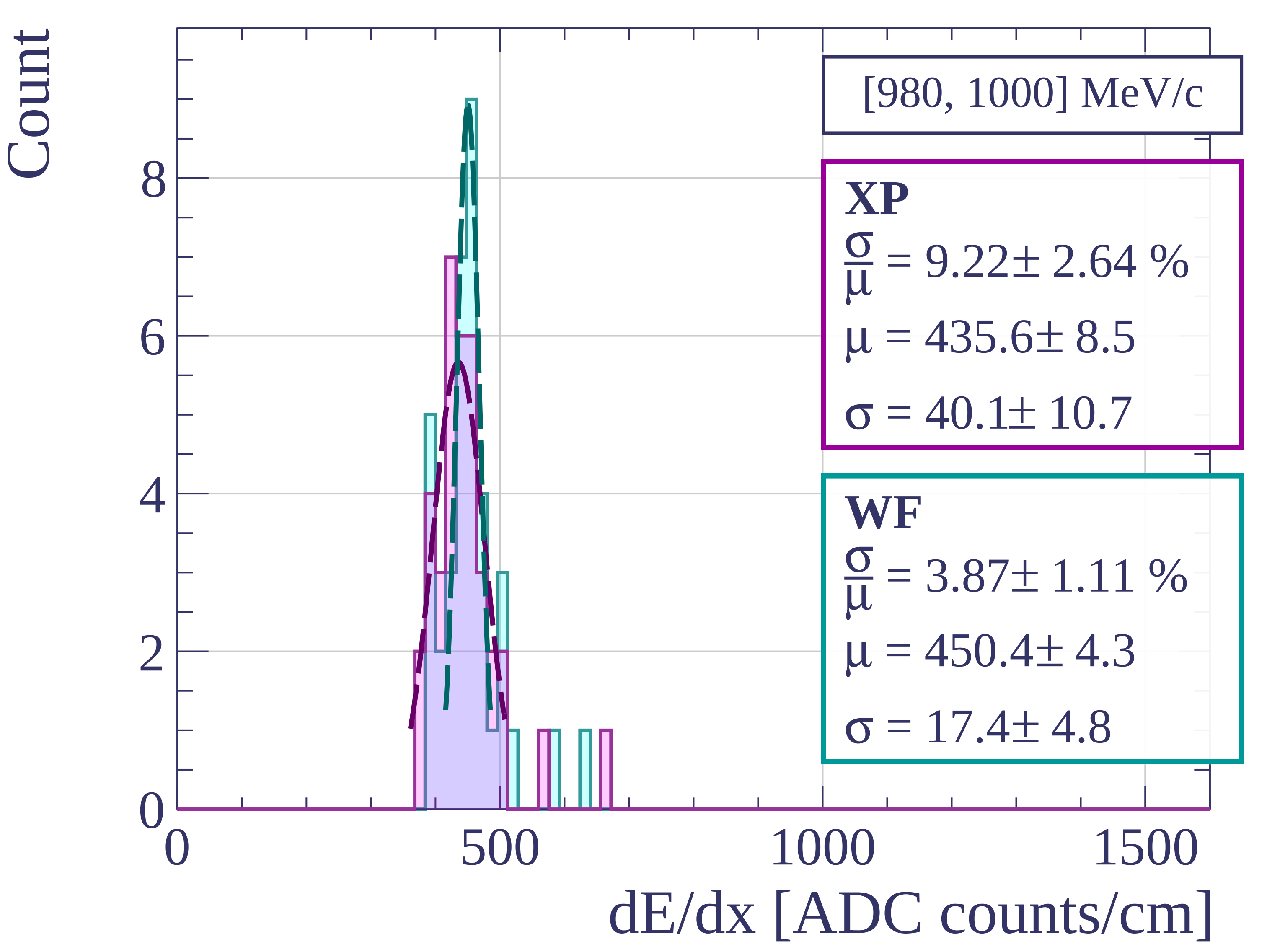
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[920, 940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 4.79± 3.13 %
μ = 397.6± 12.9
σ = 19.0± 11.8
WF
$\frac{\sigma}{\mu}$ = 191.28± 758.26 %
μ = 145.7± 501.8
σ = 278.7± 145.0



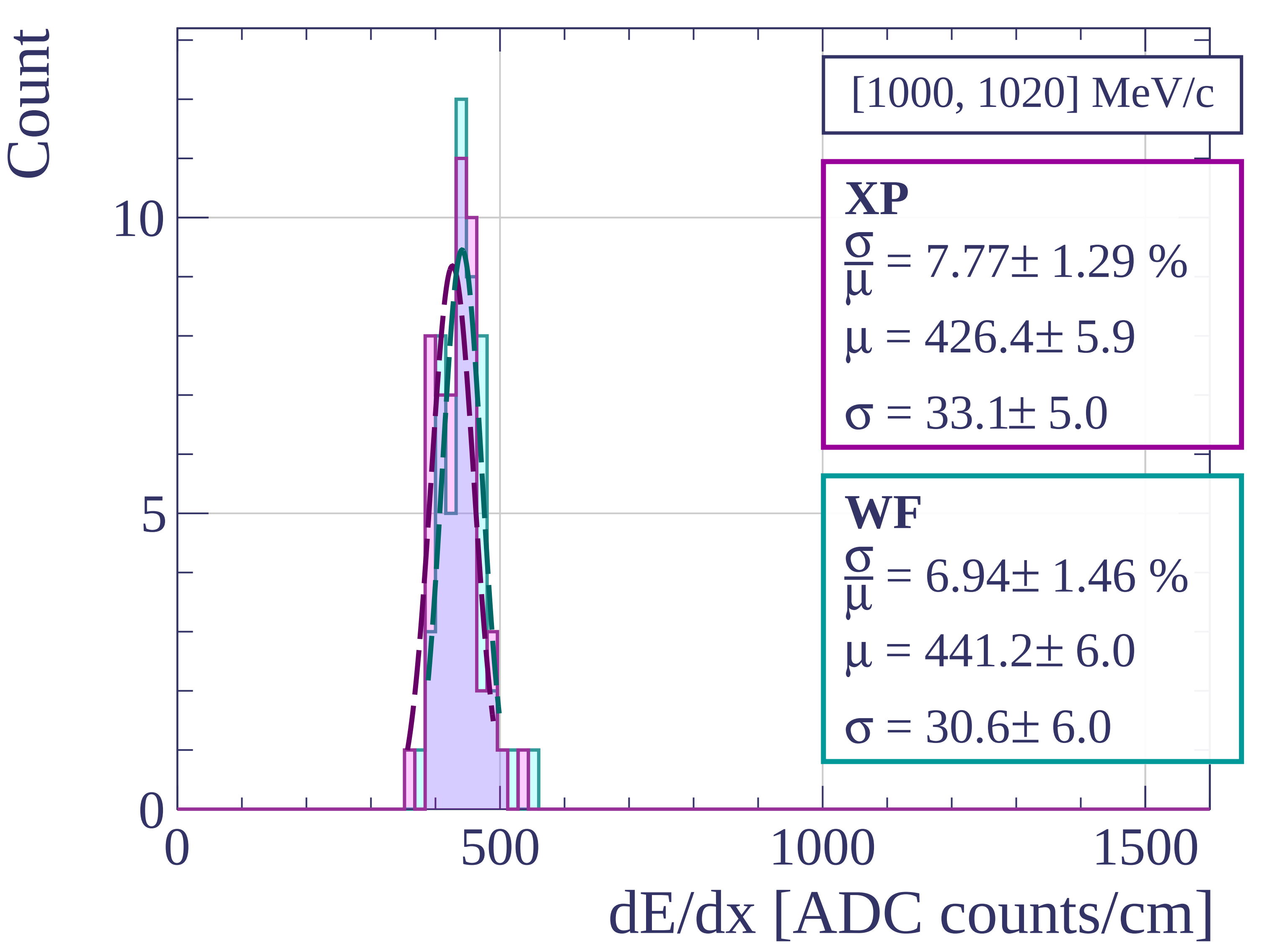
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
σ/μ = 10.37± 2.00 %
μ = 424.7± 8.4
σ = 44.0± 7.6
WF
σ/μ = 9.57± 2.16 %
μ = 426.3± 8.2
σ = 40.8± 8.4

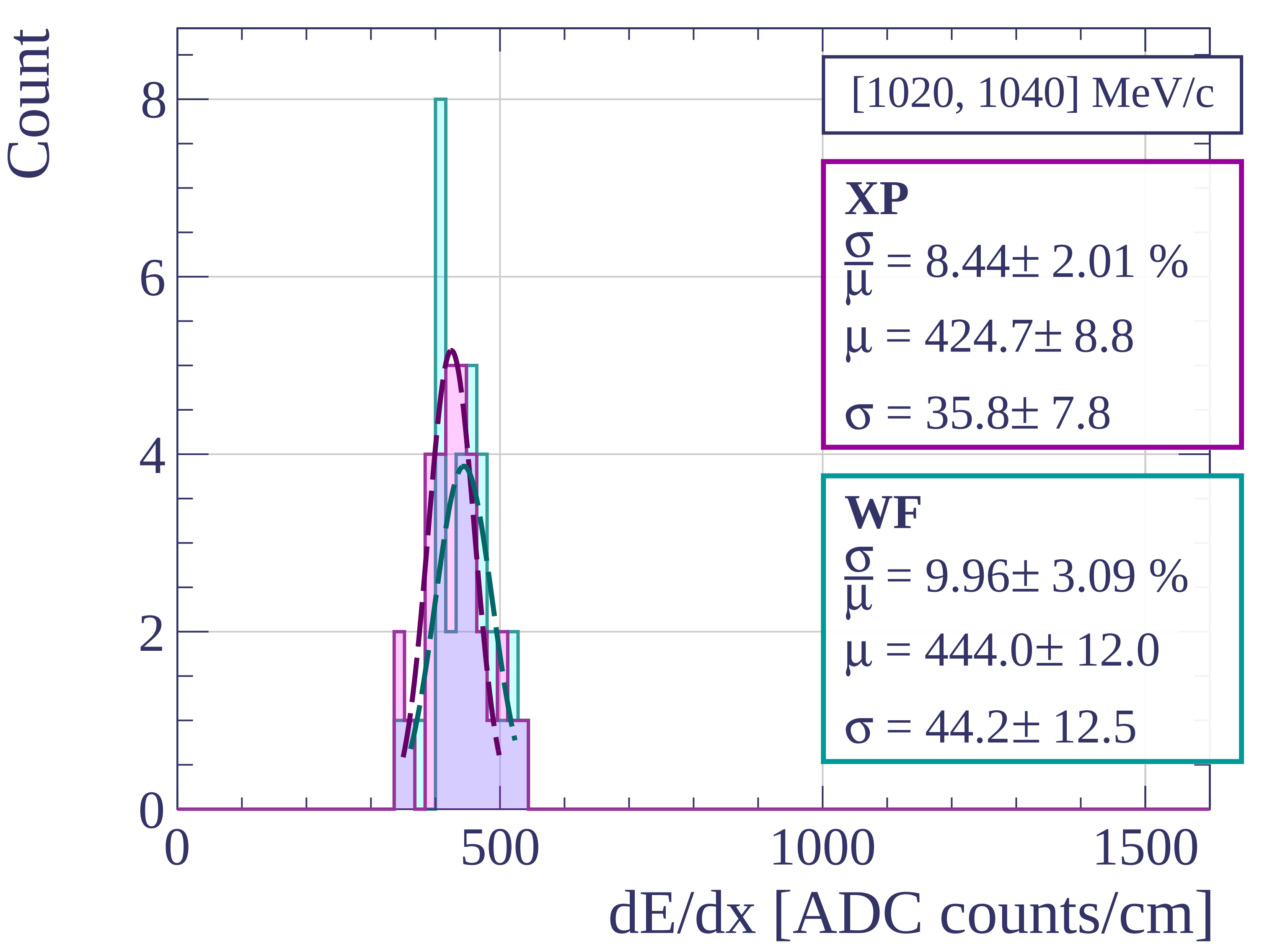


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 980] MeV/c
XP
σ/μ = 5.63± 1.11 %
μ = 429.5± 4.6
σ = 24.2± 4.5
WF
σ/μ = 5.49± 1.48 %
μ = 435.9± 5.6
σ = 23.9± 6.1

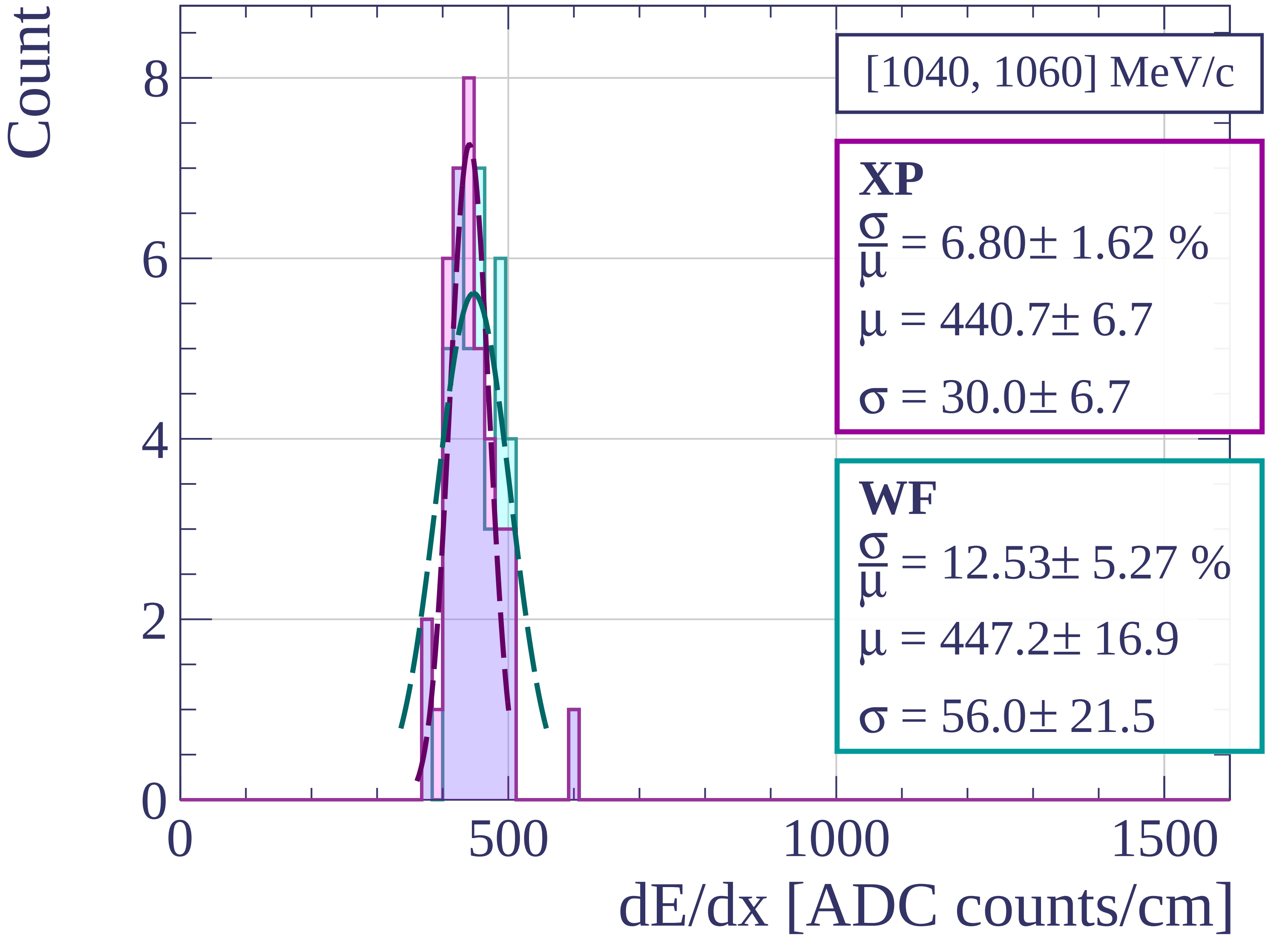


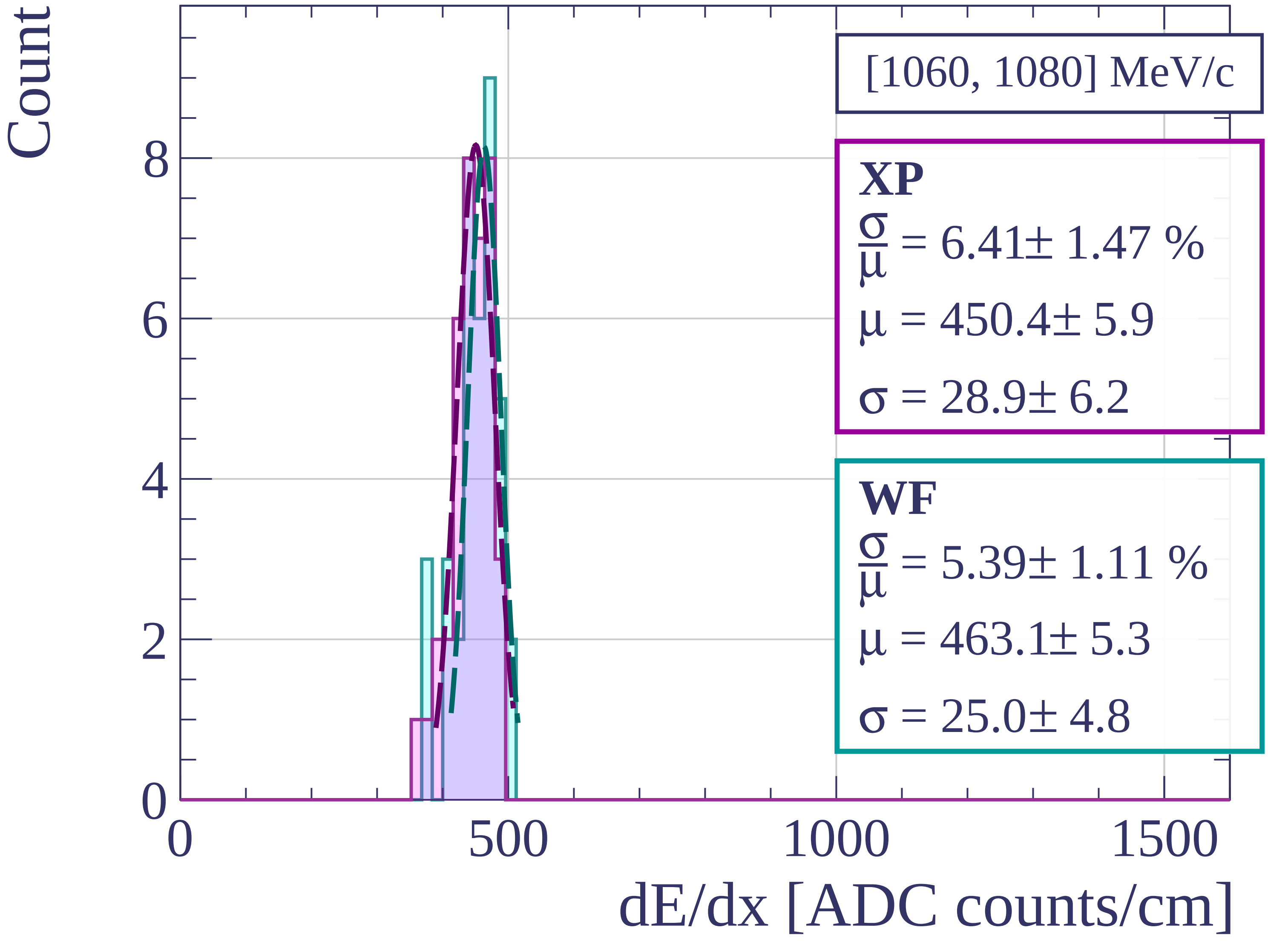
Count
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
σ/μ = 9.22± 2.64 %
μ = 435.6± 8.5
σ = 40.1± 10.7
WF
σ/μ = 3.87± 1.11 %
μ = 450.4± 4.3
σ = 17.4± 4.8
0
2
4
6
8
0
500
1000
1500


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1000, 1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.77± 1.29 %
μ = 426.4± 5.9
σ = 33.1± 5.0
WF
$\frac{\sigma}{\mu}$ = 6.94± 1.46 %
μ = 441.2± 6.0
σ = 30.6± 6.0

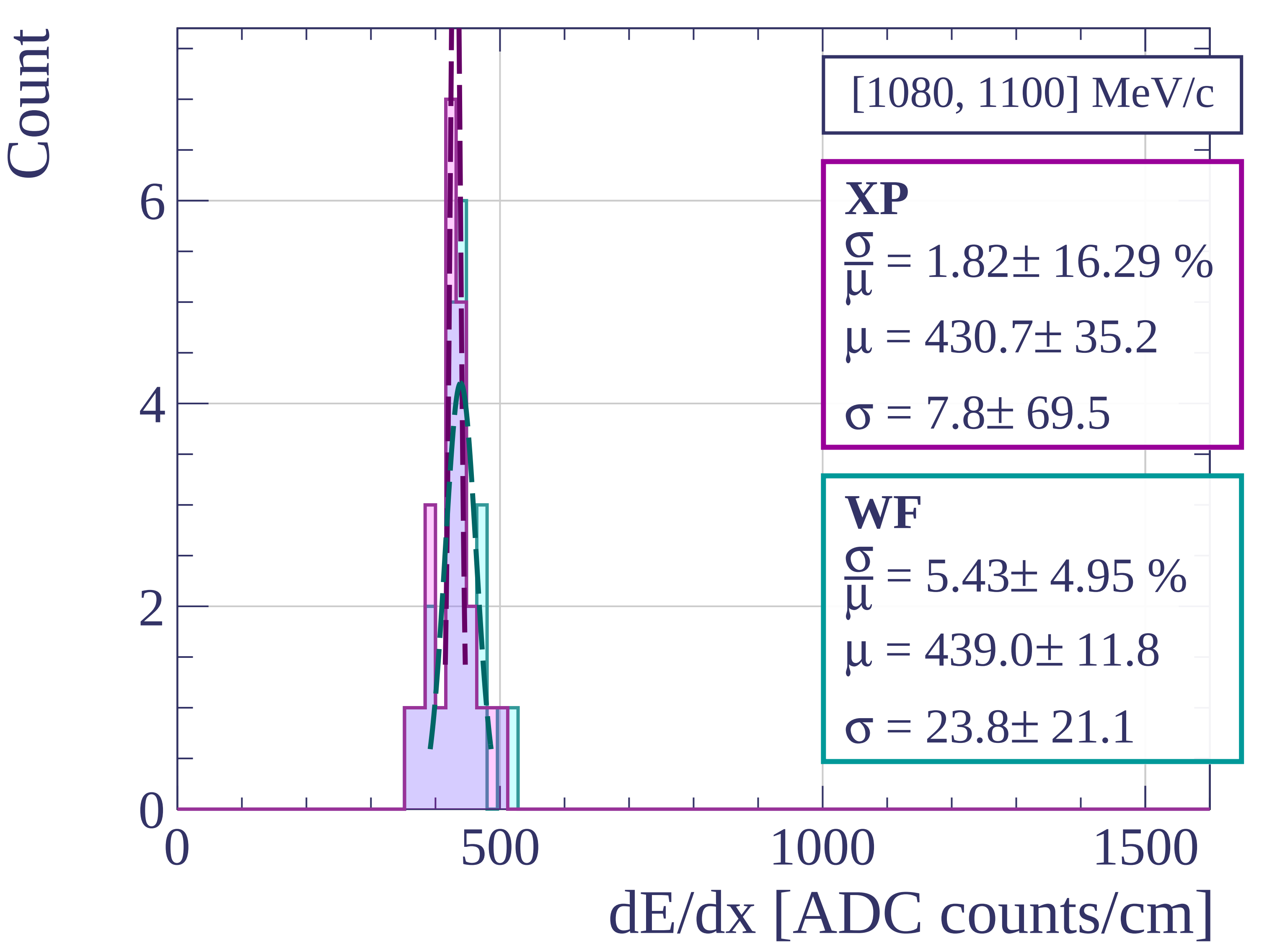


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
σ/μ = 8.44± 2.01 %
μ = 424.7± 8.8
σ = 35.8± 7.8
WF
σ/μ = 9.96± 3.09 %
μ = 444.0± 12.0
σ = 44.2± 12.5

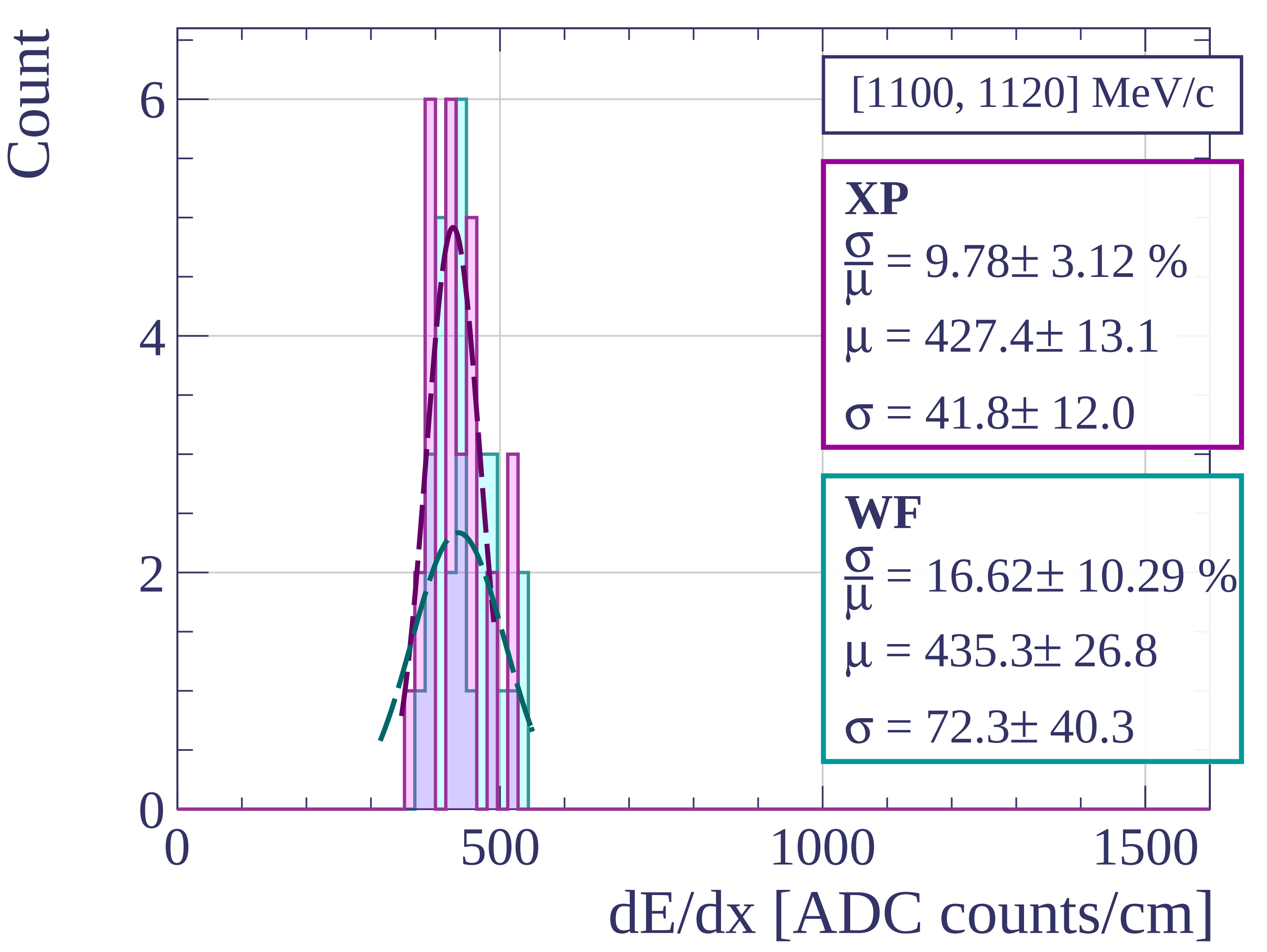

Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
σ/μ = 6.80± 1.62 %
μ = 440.7± 6.7
σ = 30.0± 6.7
WF
σ/μ = 12.53± 5.27 %
μ = 447.2± 16.9
σ = 56.0± 21.5

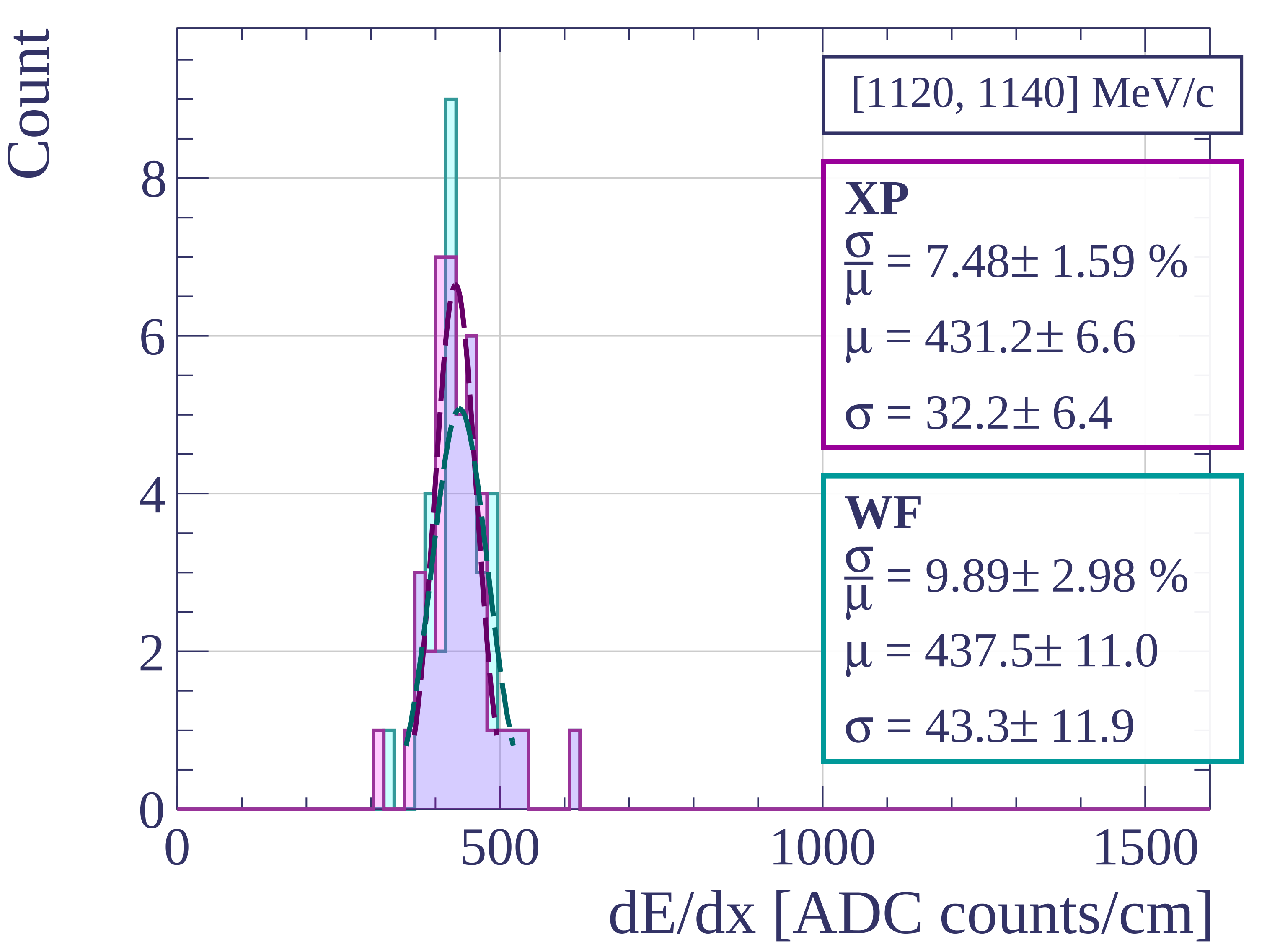


Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.41± 1.47 %
μ = 450.4± 5.9
σ = 28.9± 6.2
WF
$\frac{\sigma}{\mu}$ = 5.39± 1.11 %
μ = 463.1± 5.3
σ = 25.0± 4.8

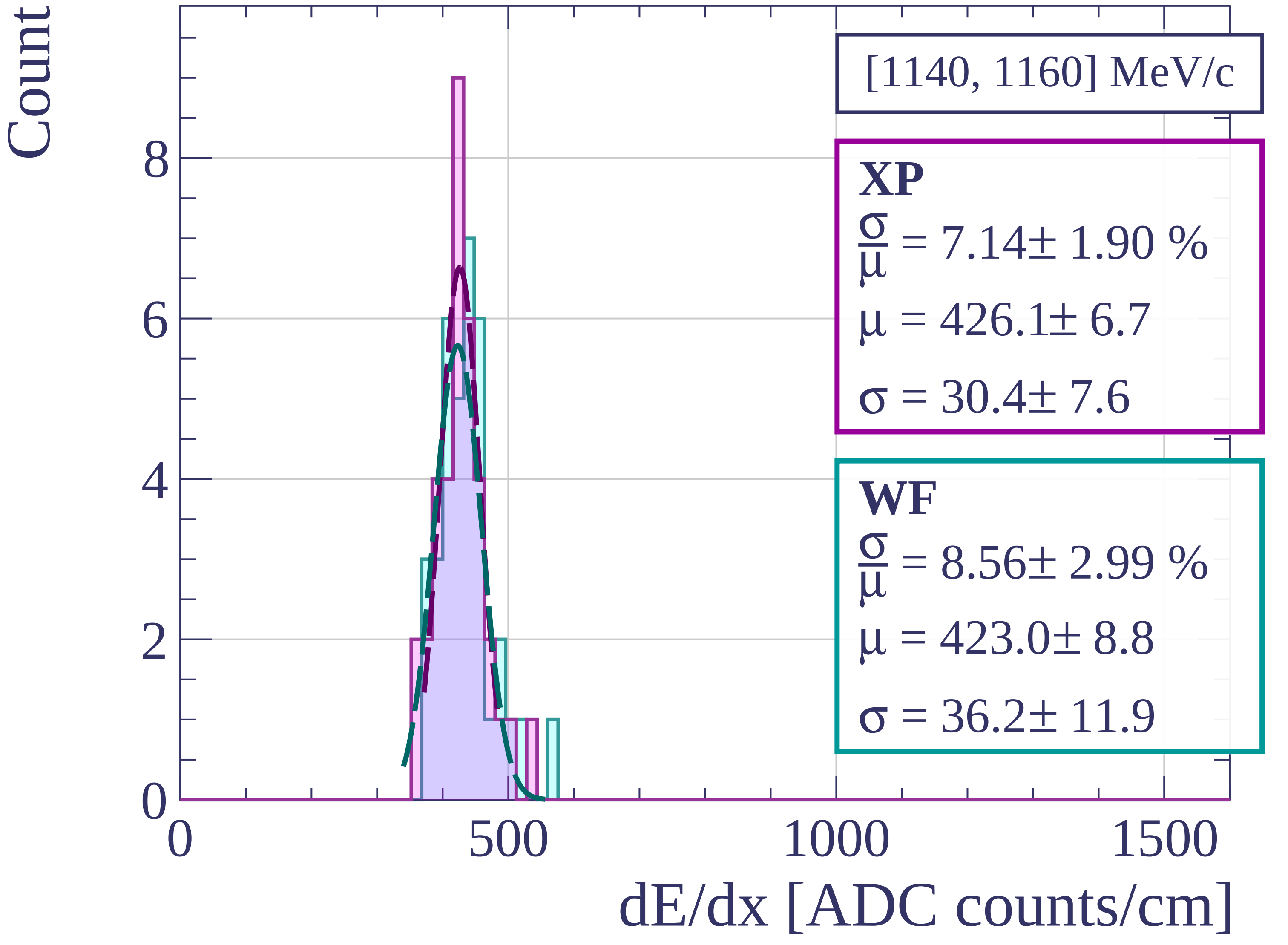


Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1080, 1100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 1.82± 16.29 %
μ = 430.7± 35.2
σ = 7.8± 69.5
WF
$\frac{\sigma}{\mu}$ = 5.43± 4.95 %
μ = 439.0± 11.8
σ = 23.8± 21.1

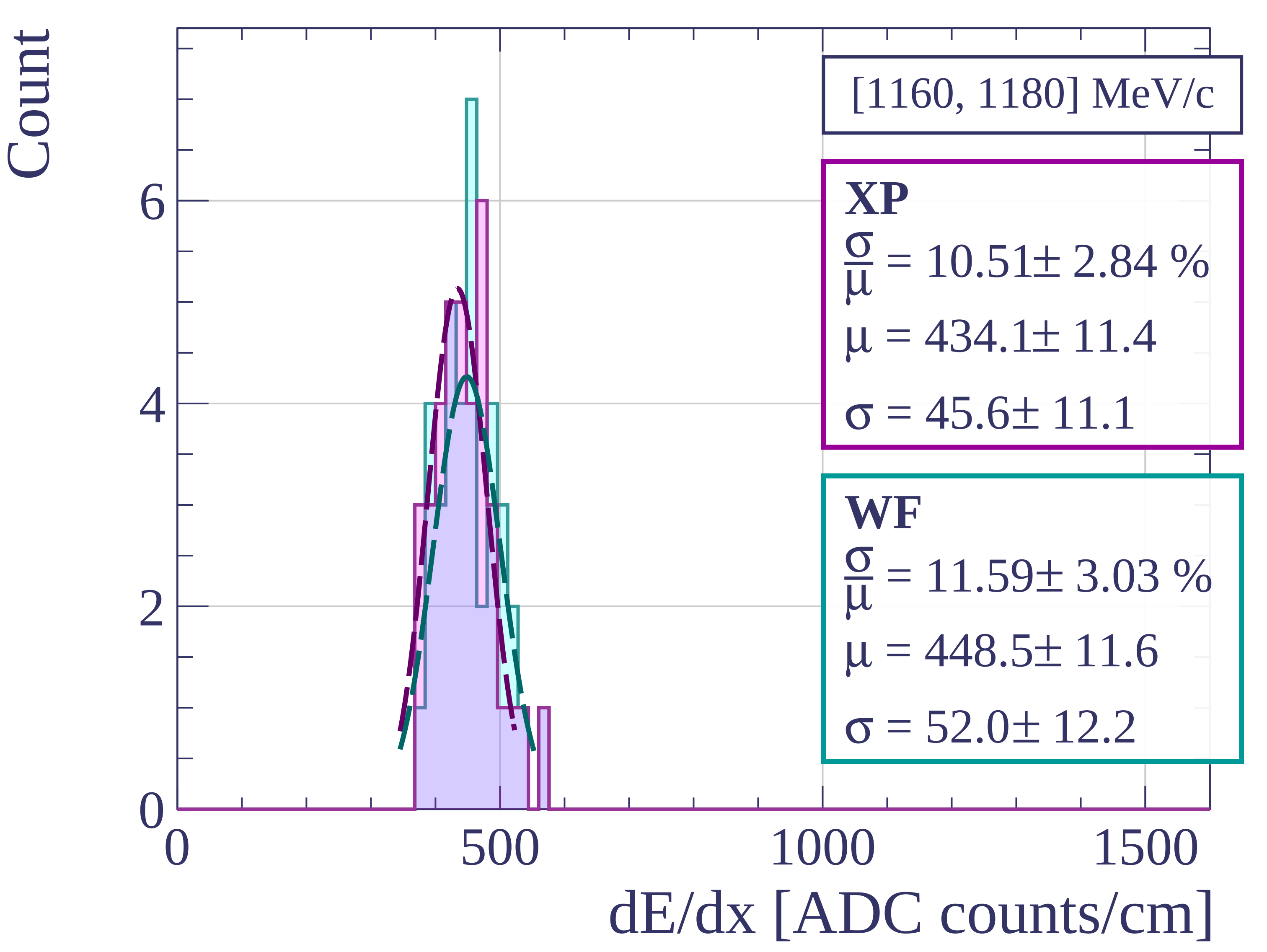

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
σ/μ = 9.78± 3.12 %
μ = 427.4± 13.1
σ = 41.8± 12.0
WF
σ/μ = 16.62± 10.29 %
μ = 435.3± 26.8
σ = 72.3± 40.3



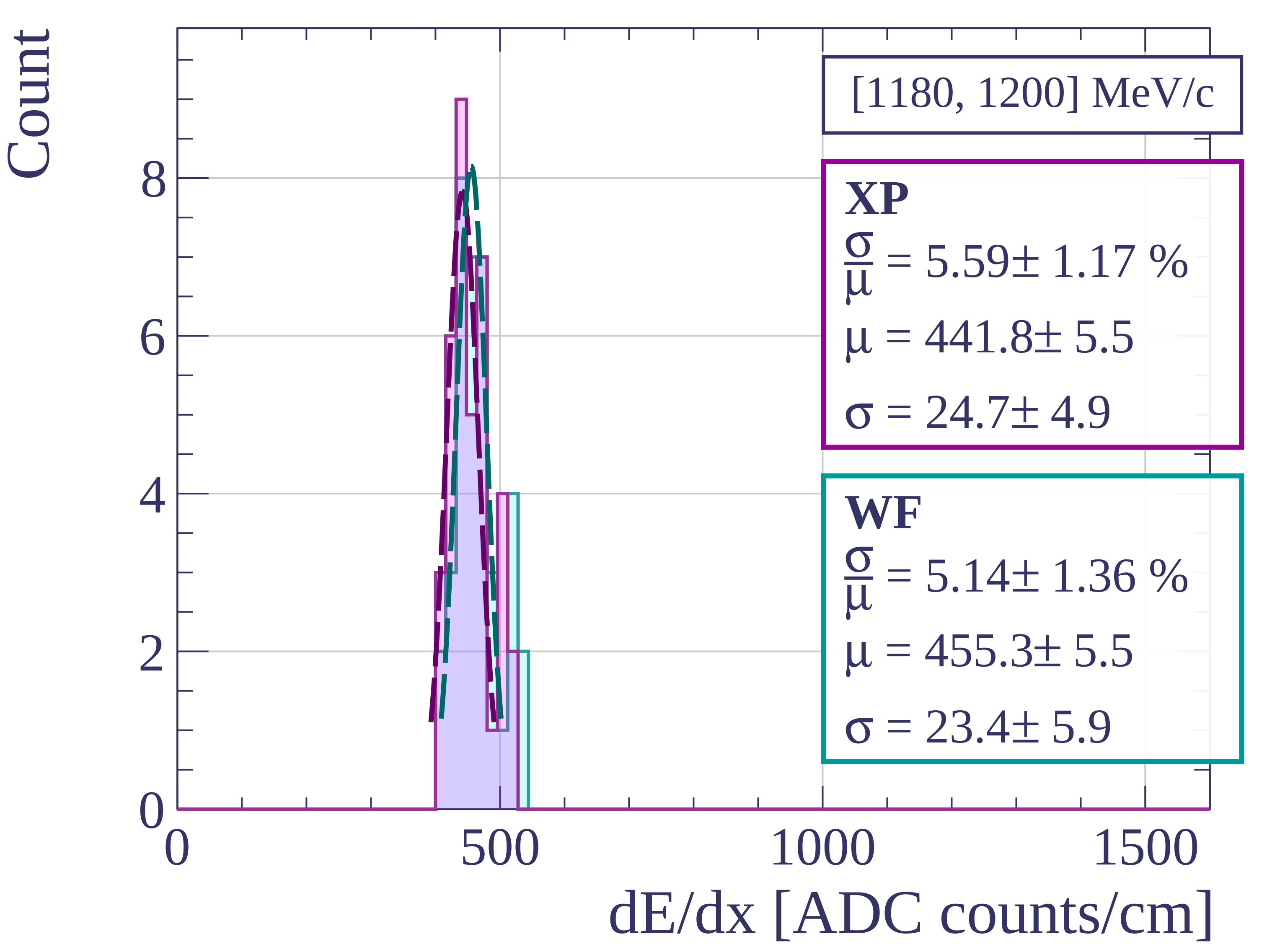
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.48± 1.59 %
μ = 431.2± 6.6
σ = 32.2± 6.4
WF
$\frac{\sigma}{\mu}$ = 9.89± 2.98 %
μ = 437.5± 11.0
σ = 43.3± 11.9



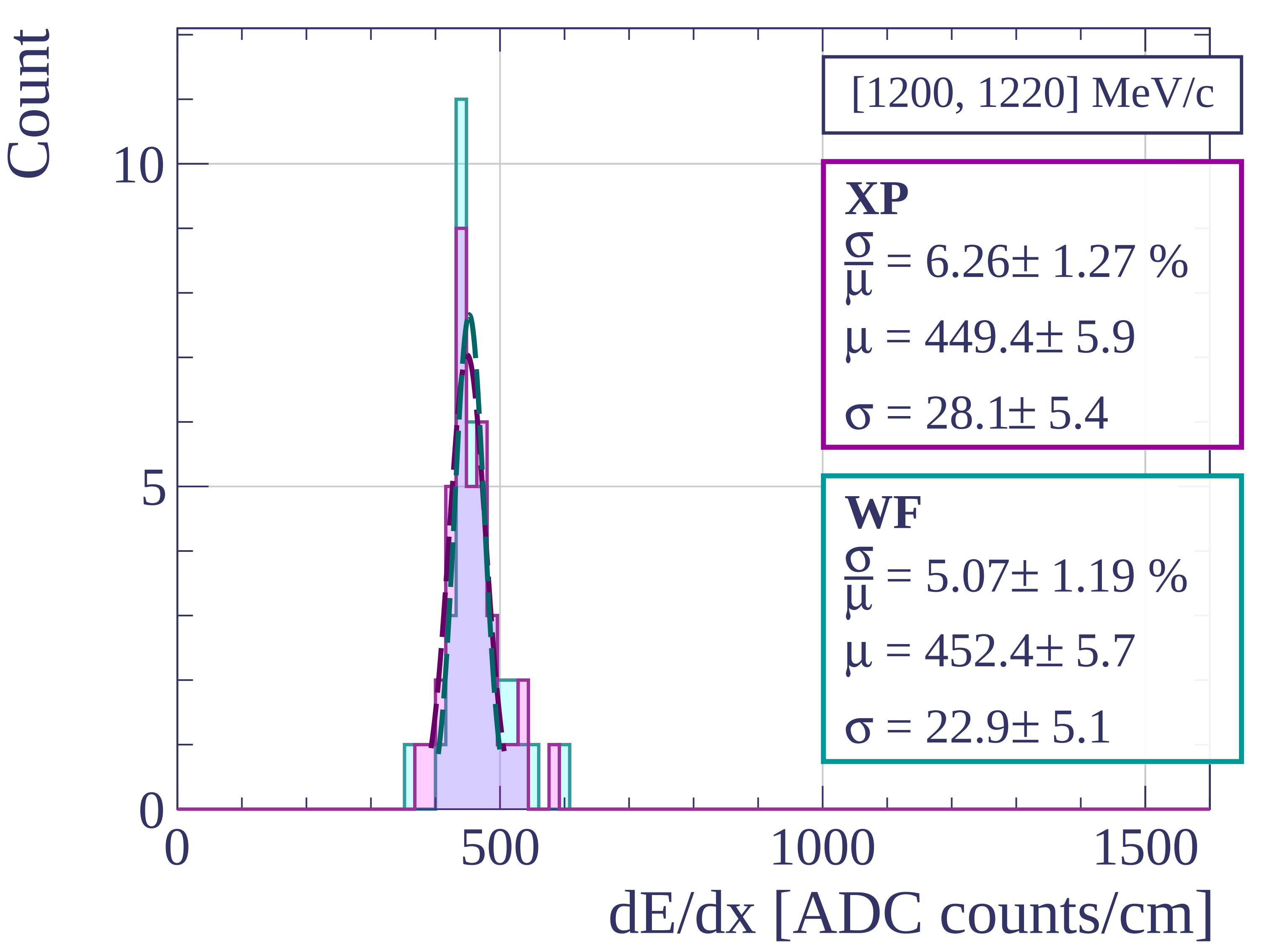
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1140, 1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.14± 1.90 %
μ = 426.1± 6.7
σ = 30.4± 7.6
WF
$\frac{\sigma}{\mu}$ = 8.56± 2.99 %
μ = 423.0± 8.8
σ = 36.2± 11.9

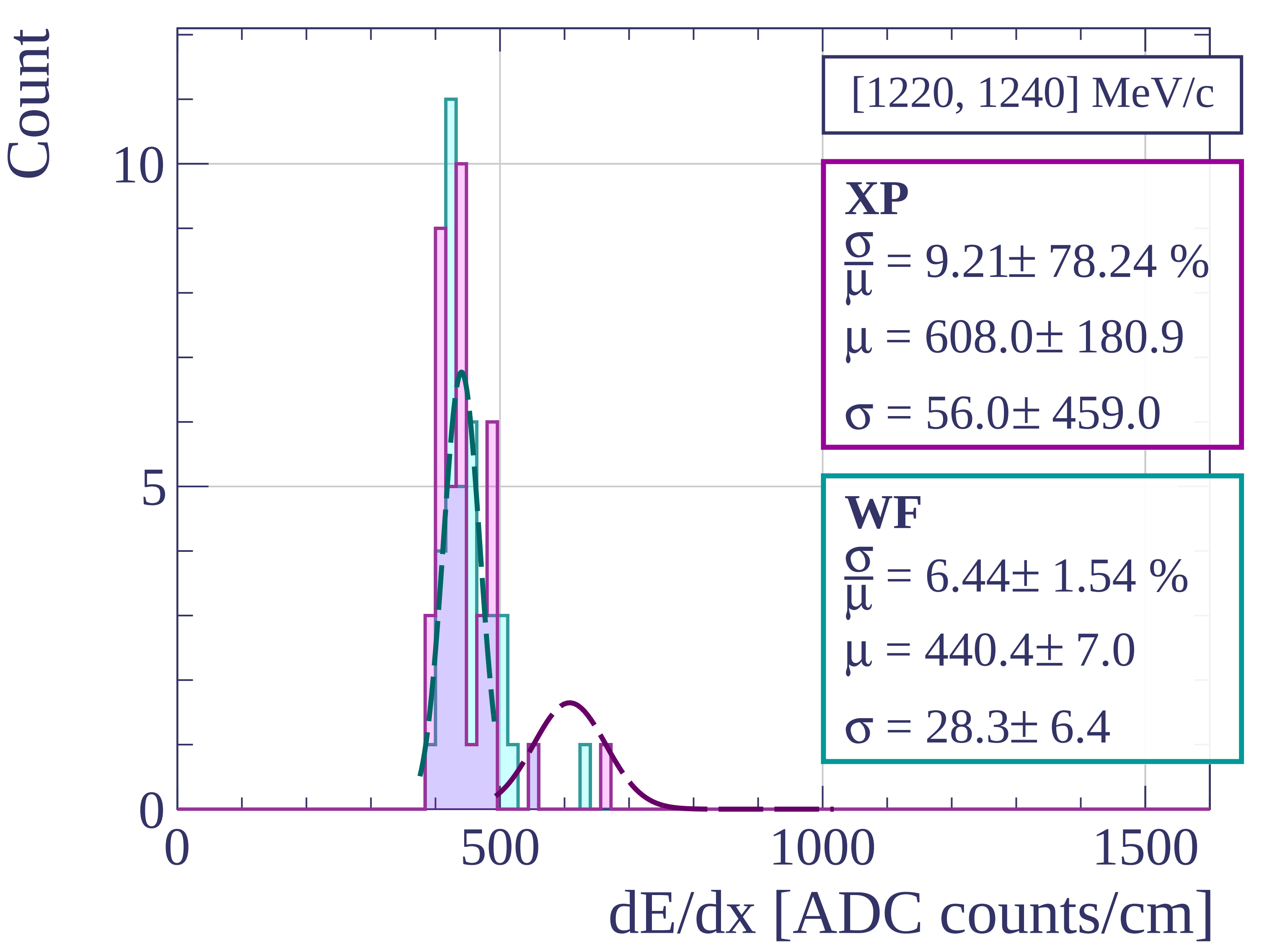


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1160, 1180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.51± 2.84 %
μ = 434.1± 11.4
σ = 45.6± 11.1
WF
$\frac{\sigma}{\mu}$ = 11.59± 3.03 %
μ = 448.5± 11.6
σ = 52.0± 12.2

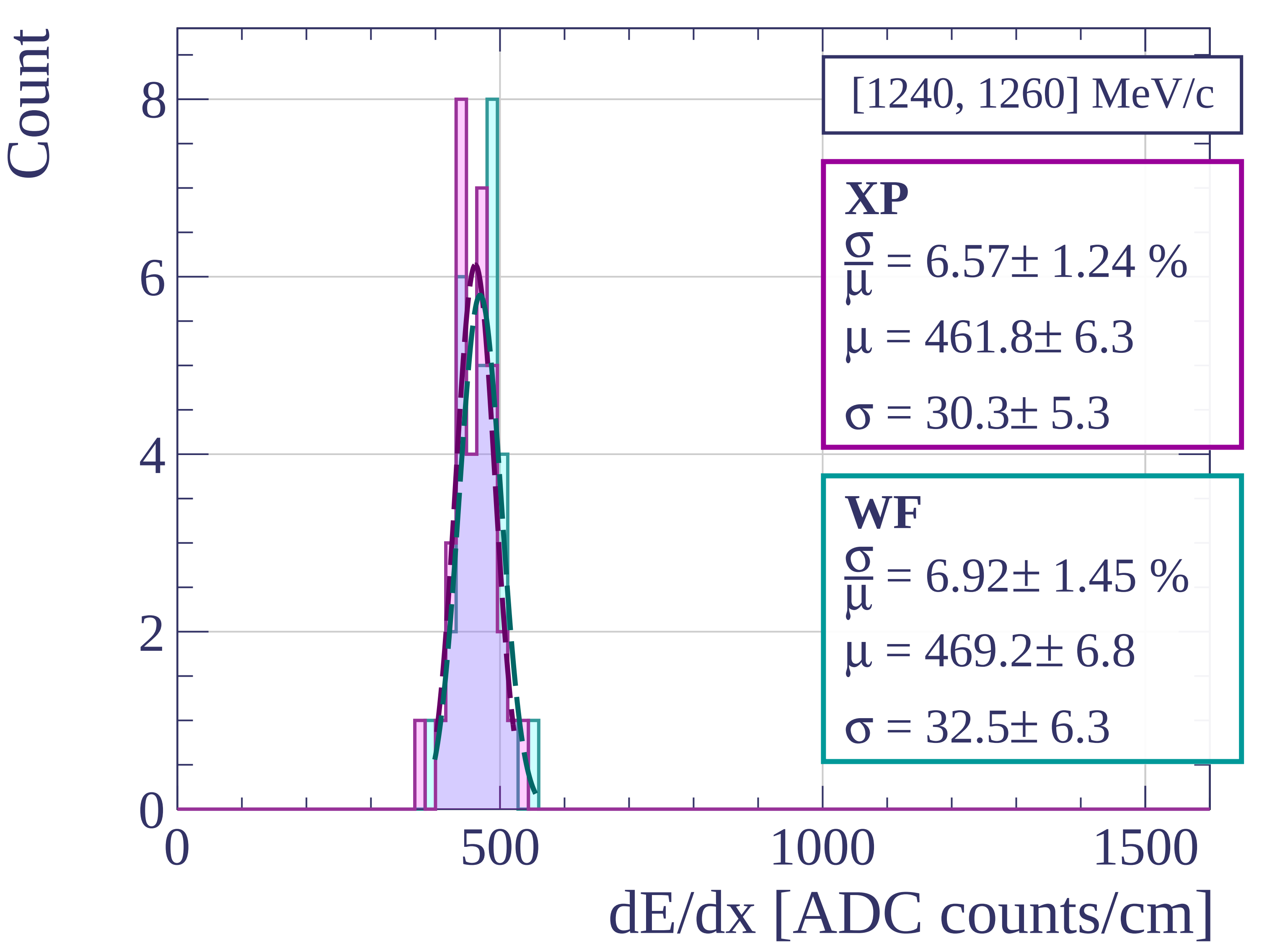


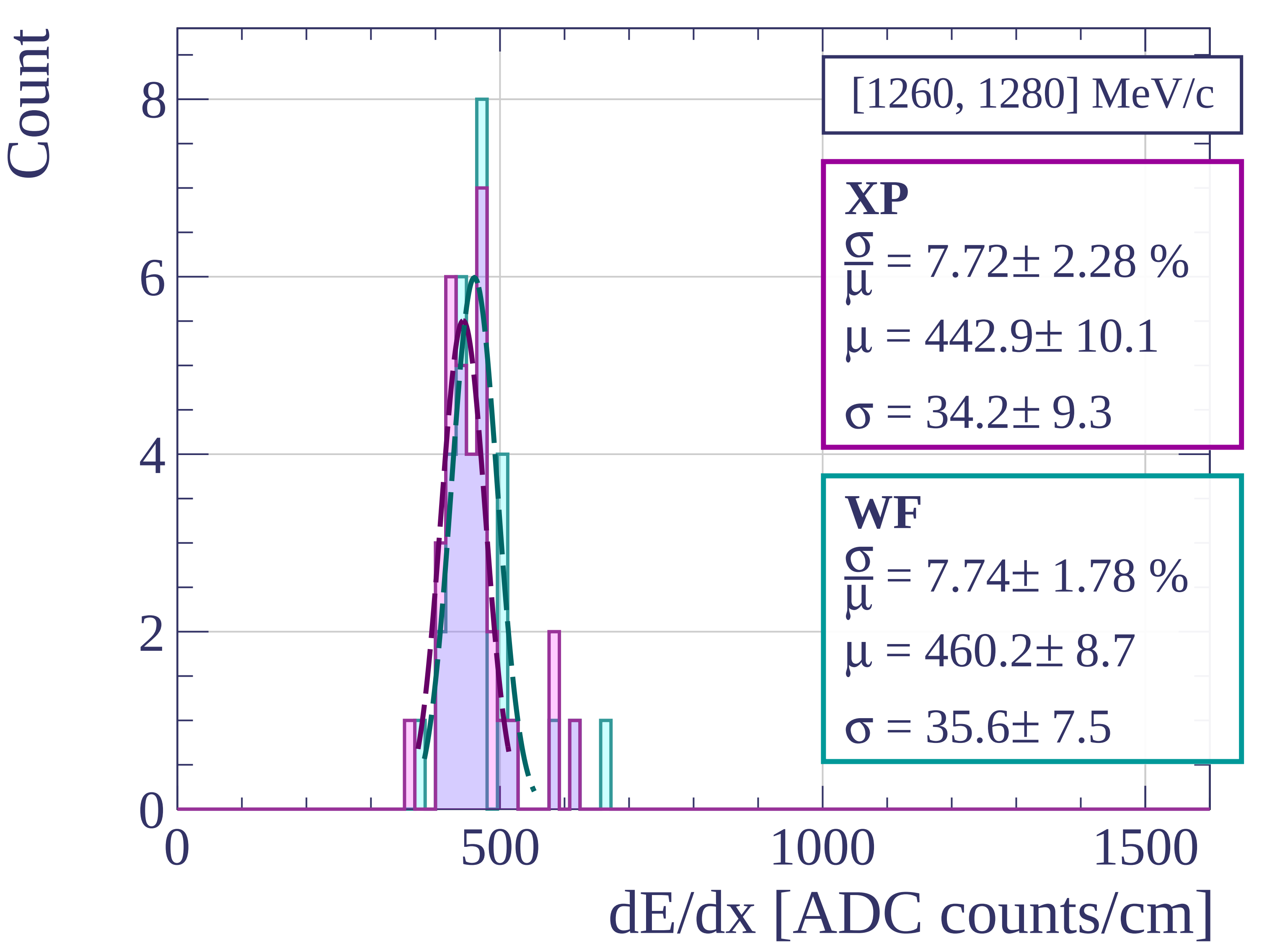
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
[1180, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.59± 1.17 %
μ = 441.8± 5.5
σ = 24.7± 4.9
WF
$\frac{\sigma}{\mu}$ = 5.14± 1.36 %
μ = 455.3± 5.5
σ = 23.4± 5.9


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.26± 1.27 %
μ = 449.4± 5.9
σ = 28.1± 5.4
WF
$\frac{\sigma}{\mu}$ = 5.07± 1.19 %
μ = 452.4± 5.7
σ = 22.9± 5.1

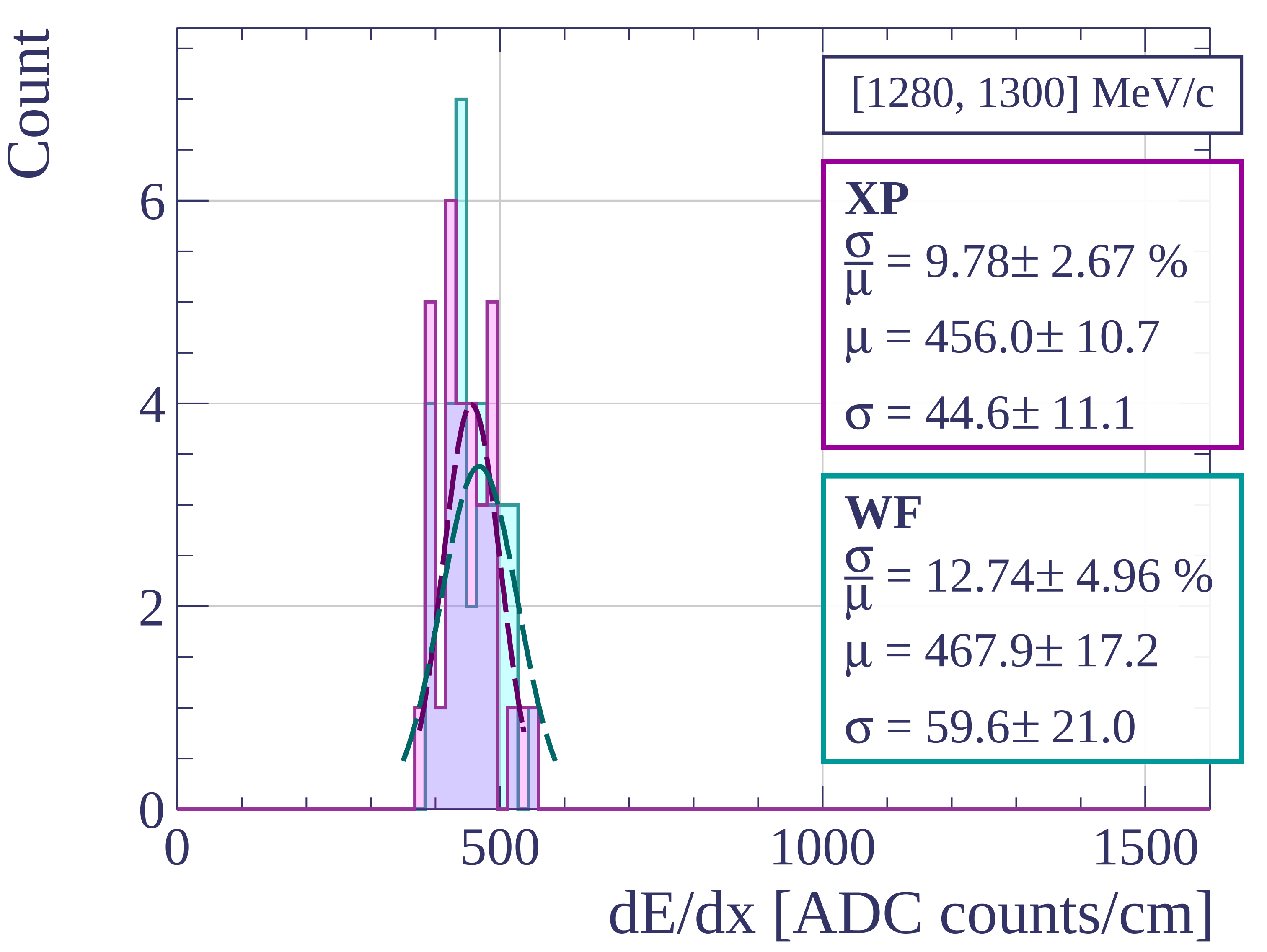


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1220, 1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.21± 78.24 %
μ = 608.0± 180.9
σ = 56.0± 459.0
WF
$\frac{\sigma}{\mu}$ = 6.44± 1.54 %
μ = 440.4± 7.0
σ = 28.3± 6.4

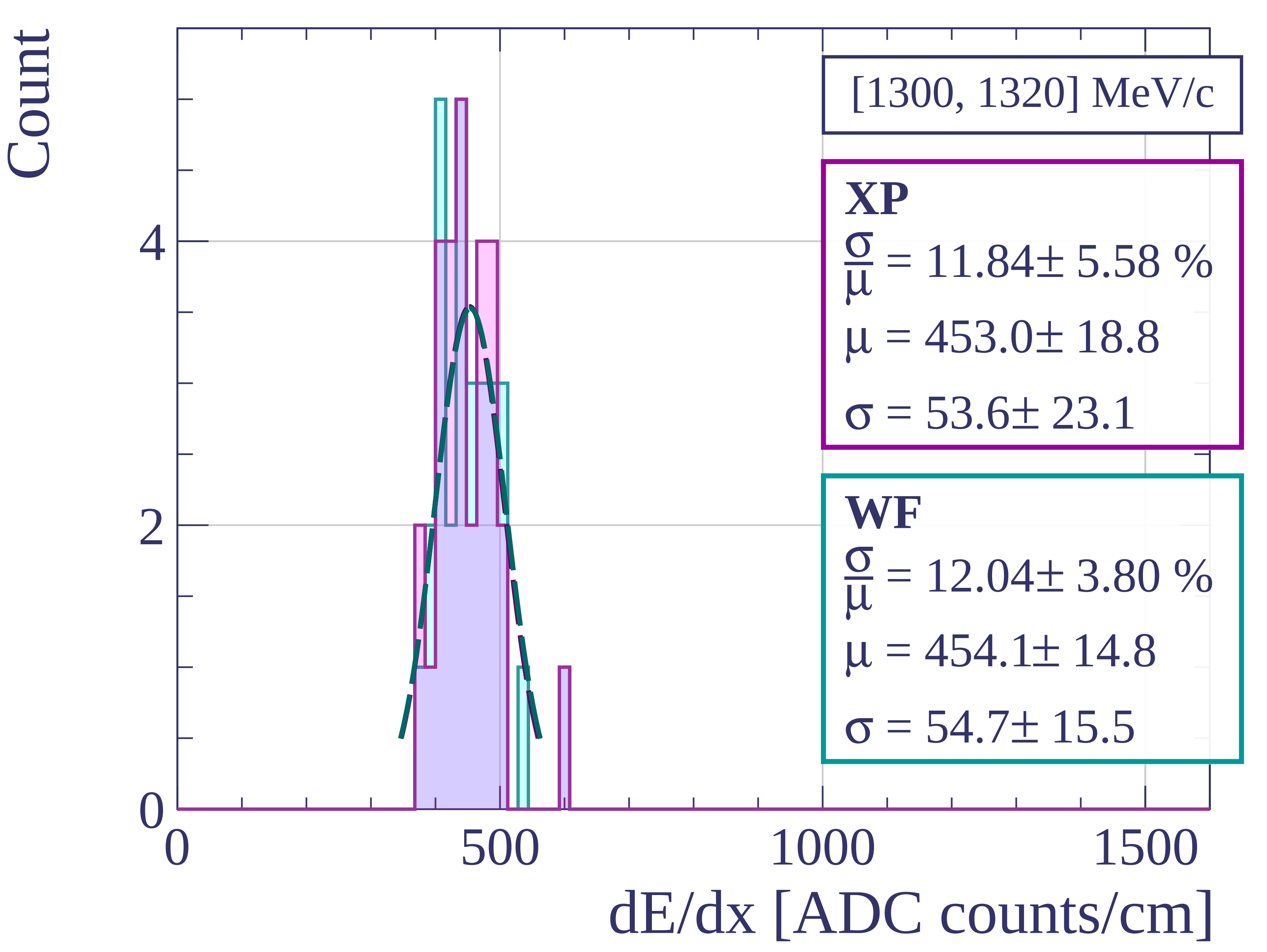

Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
[1240, 1260] MeV/c
XP
σ/μ = 6.57± 1.24 %
μ = 461.8± 6.3
σ = 30.3± 5.3
WF
σ/μ = 6.92± 1.45 %
μ = 469.2± 6.8
σ = 32.5± 6.3



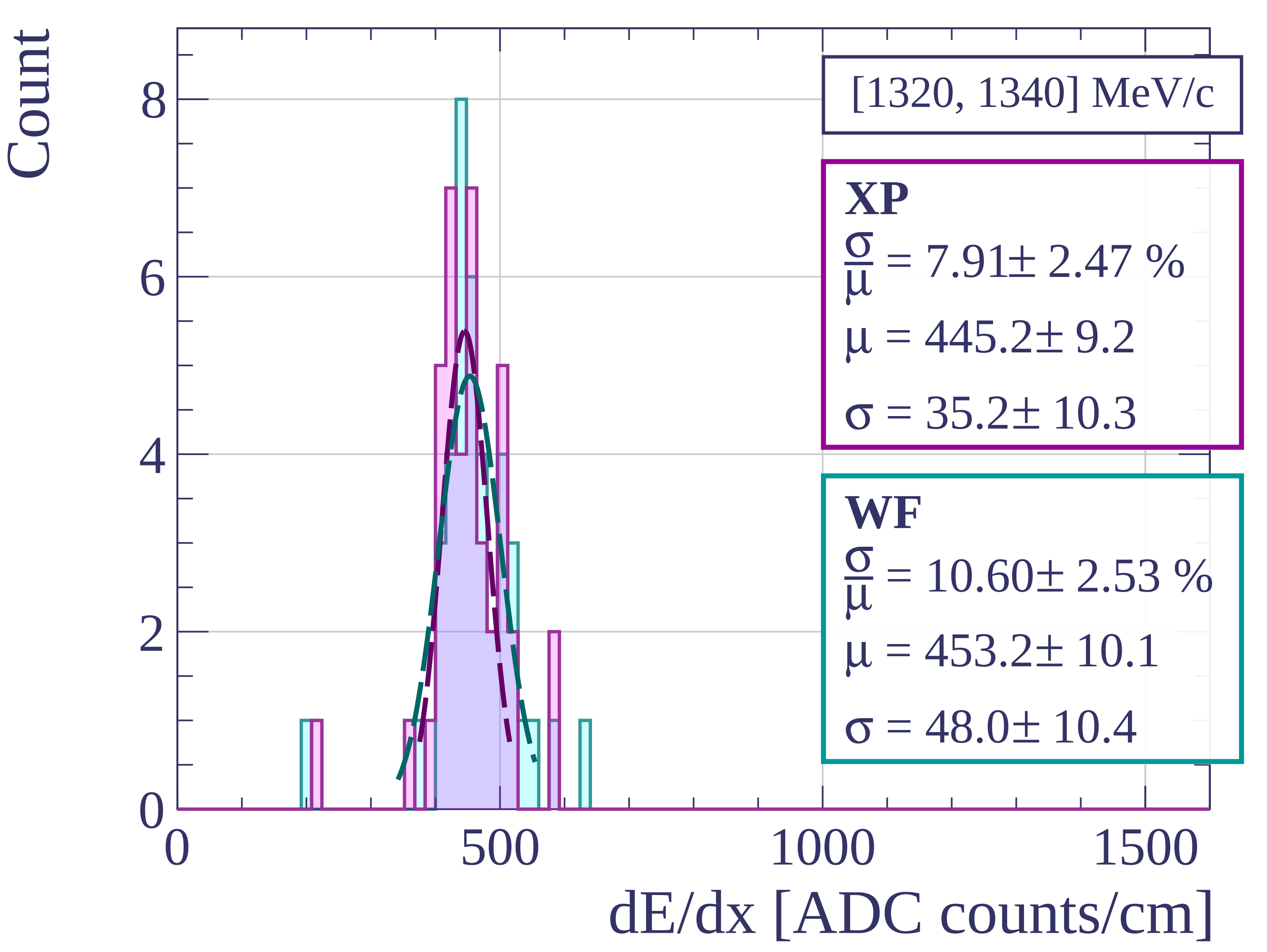
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1260, 1280] MeV/c
XP
σ/μ = 7.72± 2.28 %
μ = 442.9± 10.1
σ = 34.2± 9.3
WF
σ/μ = 7.74± 1.78 %
μ = 460.2± 8.7
σ = 35.6± 7.5



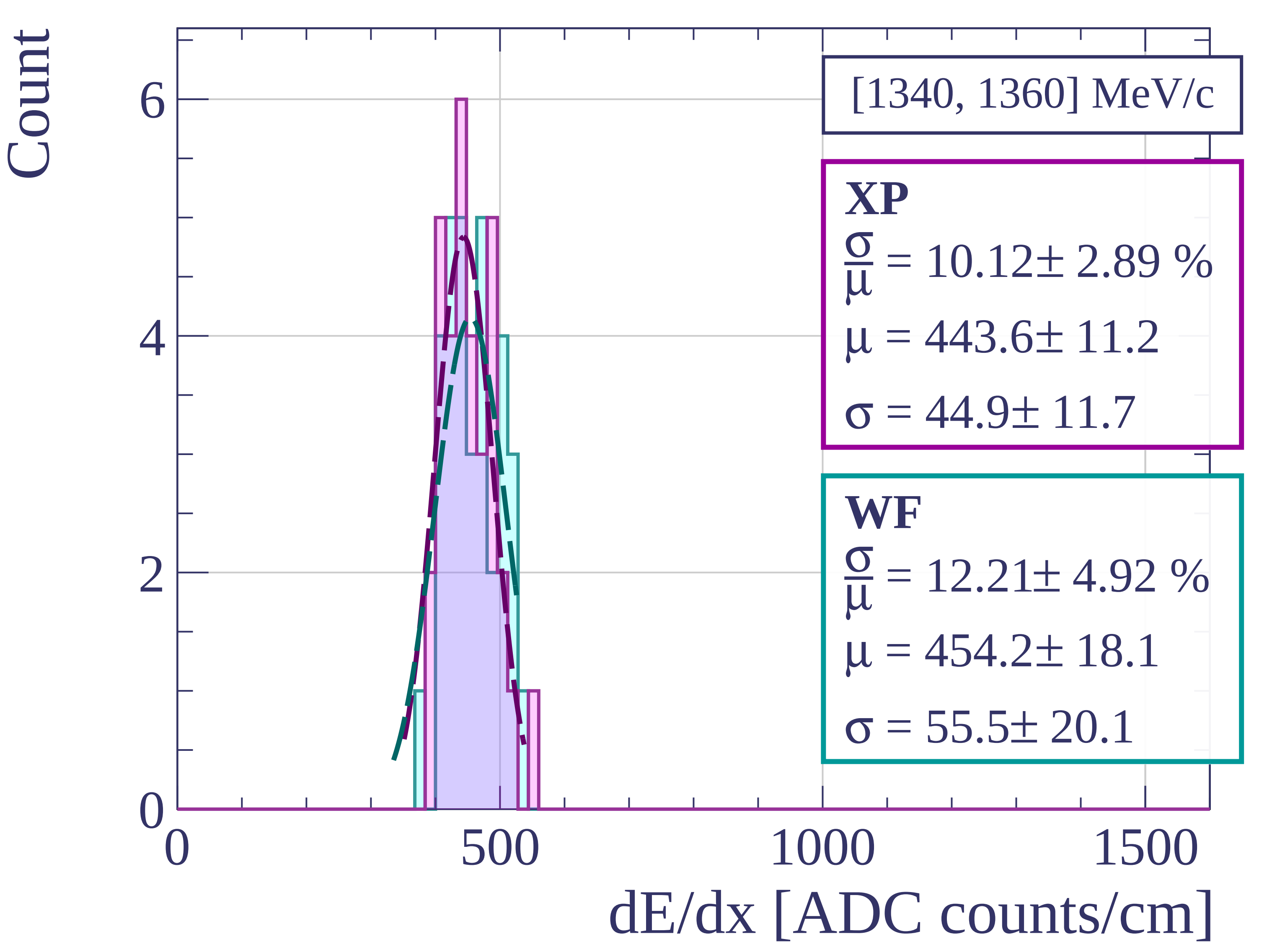
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.78± 2.67 %
μ = 456.0± 10.7
σ = 44.6± 11.1
WF
$\frac{\sigma}{\mu}$ = 12.74± 4.96 %
μ = 467.9± 17.2
σ = 59.6± 21.0

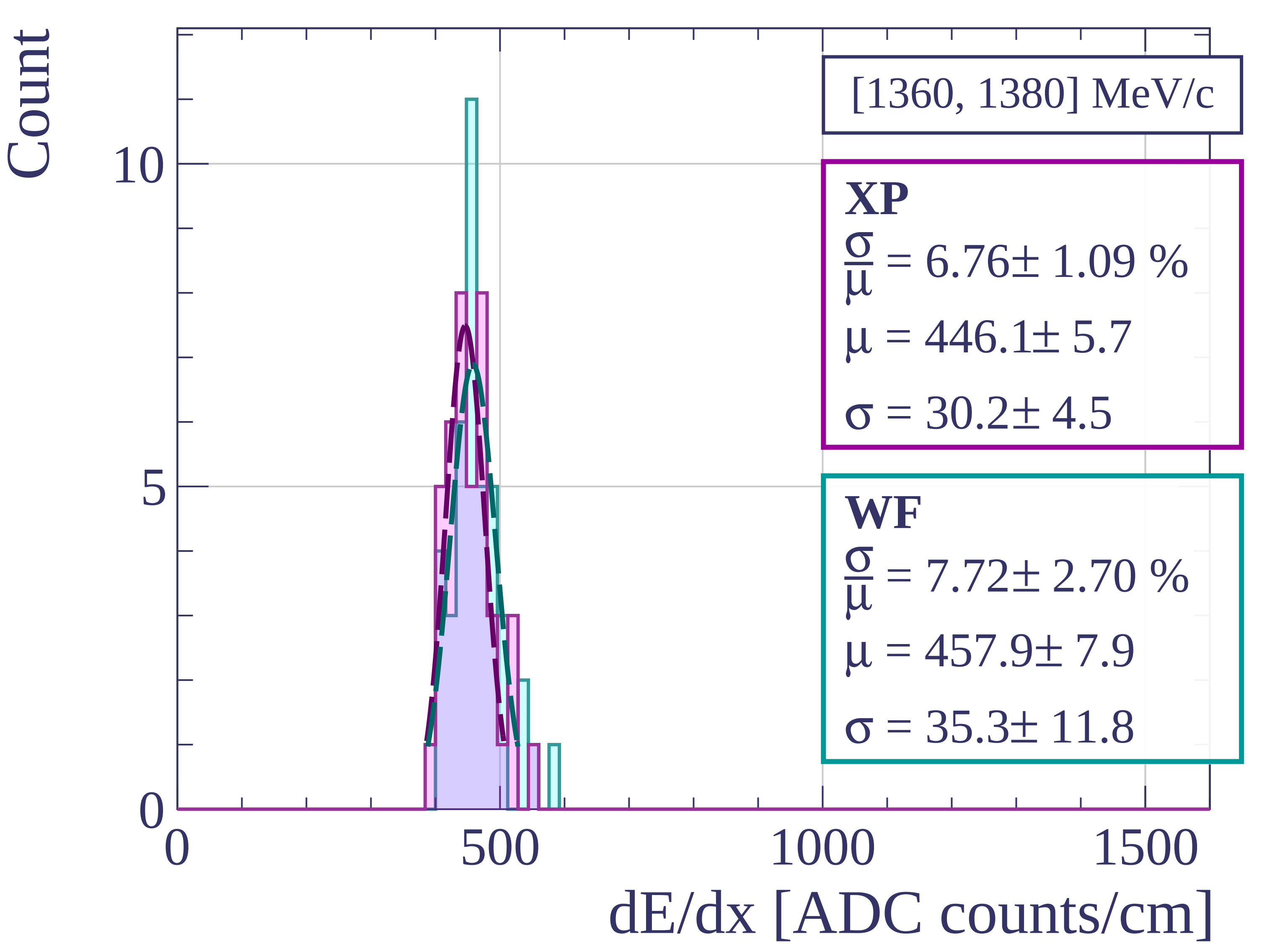


Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1300, 1320] MeV/c
XP
σ/μ = 11.84± 5.58 %
μ = 453.0± 18.8
σ = 53.6± 23.1
WF
σ/μ = 12.04± 3.80 %
μ = 454.1± 14.8
σ = 54.7± 15.5

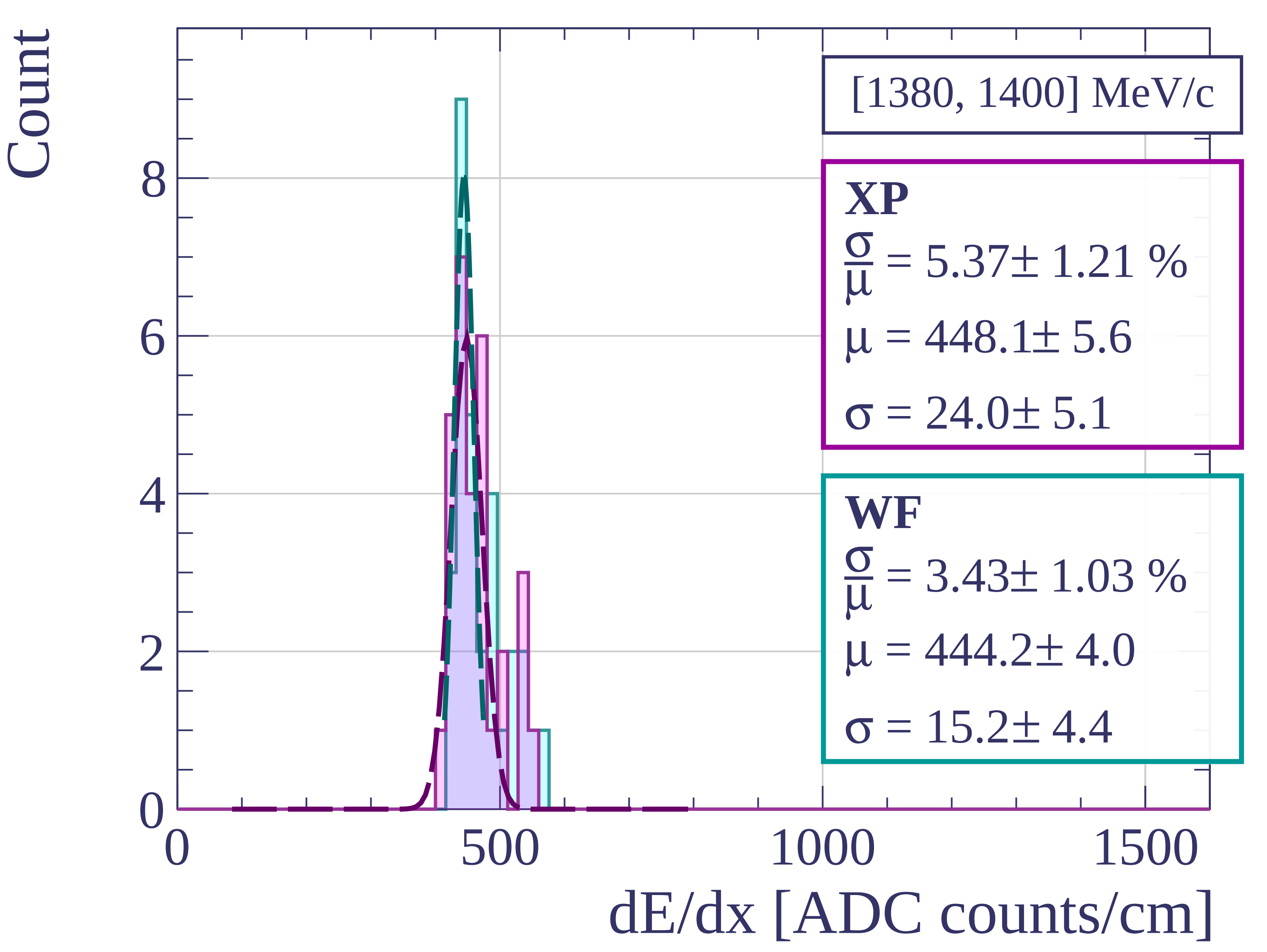


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1320, 1340] MeV/c
XP
σ/μ = 7.91± 2.47 %
μ = 445.2± 9.2
σ = 35.2± 10.3
WF
σ/μ = 10.60± 2.53 %
μ = 453.2± 10.1
σ = 48.0± 10.4

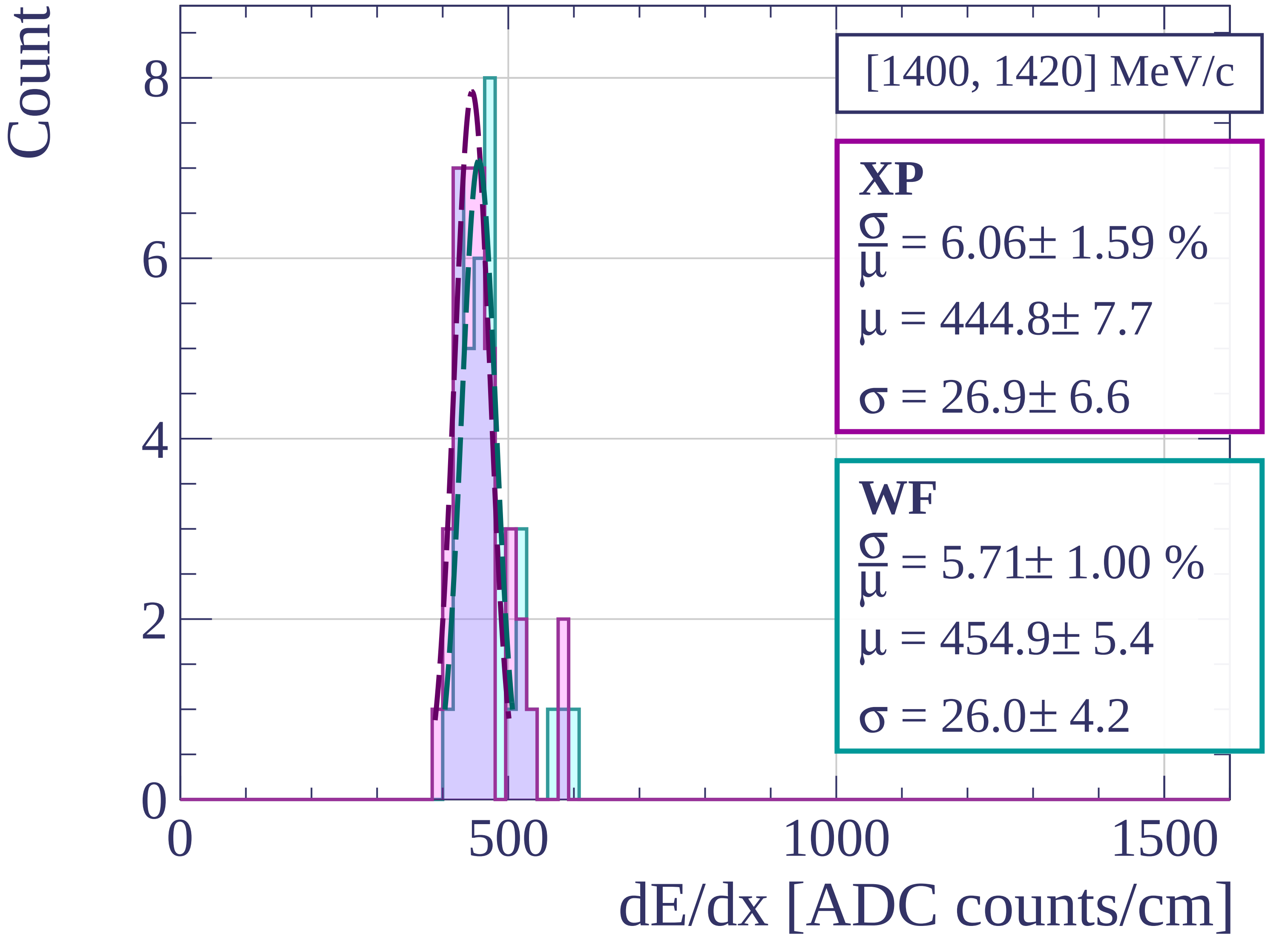

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[1340, 1360] MeV/c
XP
σ/μ = 10.12± 2.89 %
μ = 443.6± 11.2
σ = 44.9± 11.7
WF
σ/μ = 12.21± 4.92 %
μ = 454.2± 18.1
σ = 55.5± 20.1

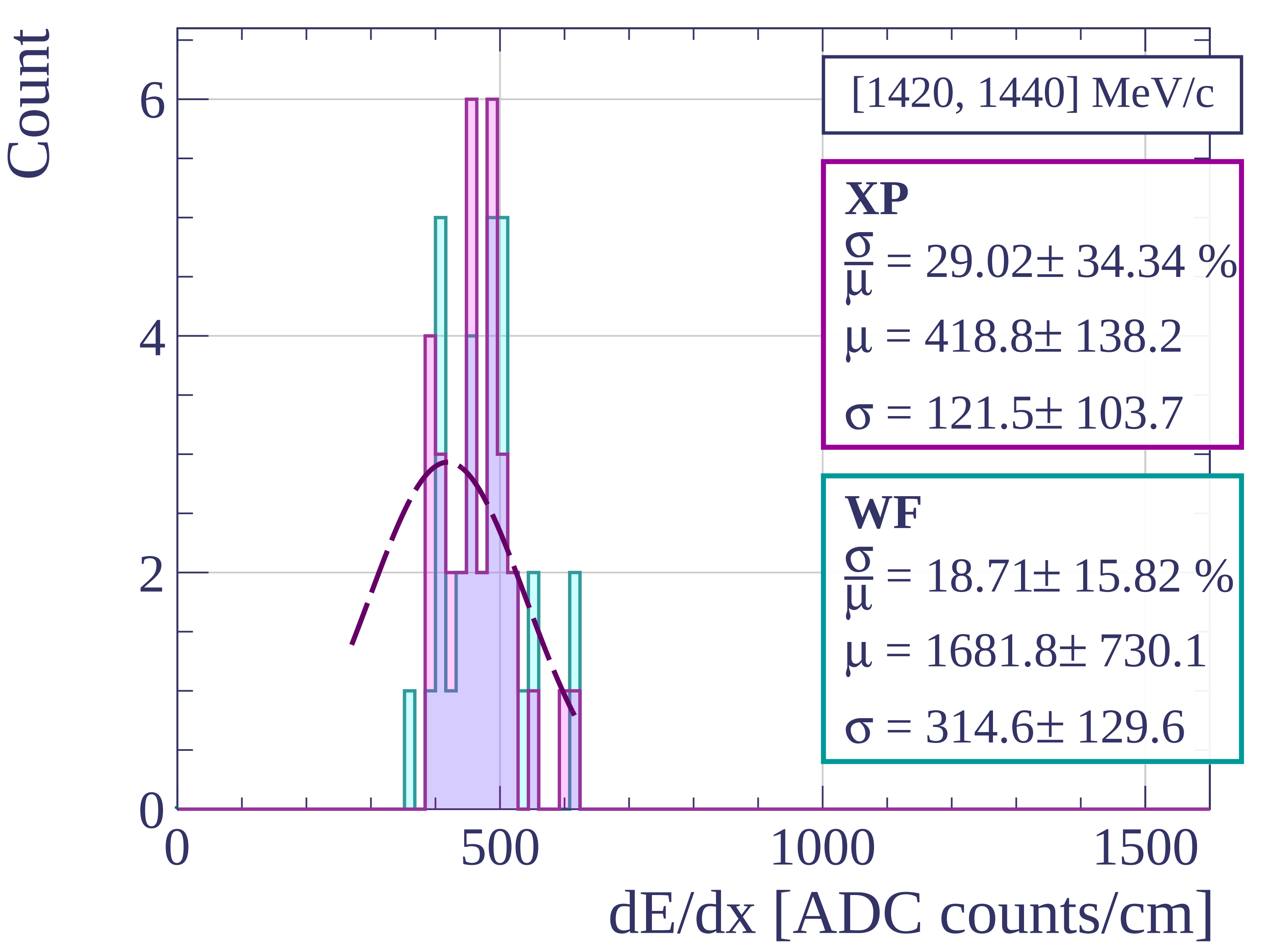


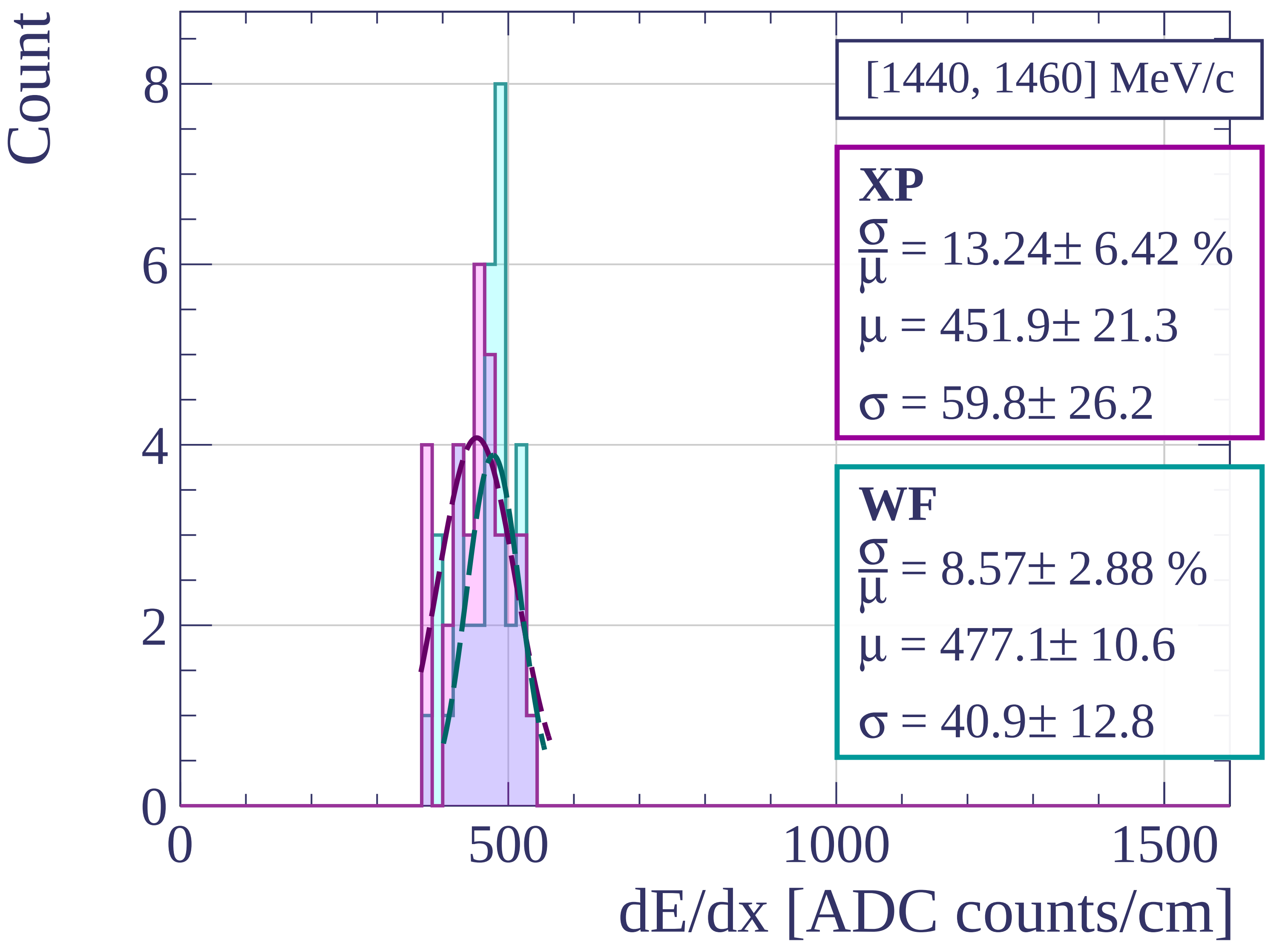
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.76± 1.09 %
μ = 446.1± 5.7
σ = 30.2± 4.5
WF
$\frac{\sigma}{\mu}$ = 7.72± 2.70 %
μ = 457.9± 7.9
σ = 35.3± 11.8



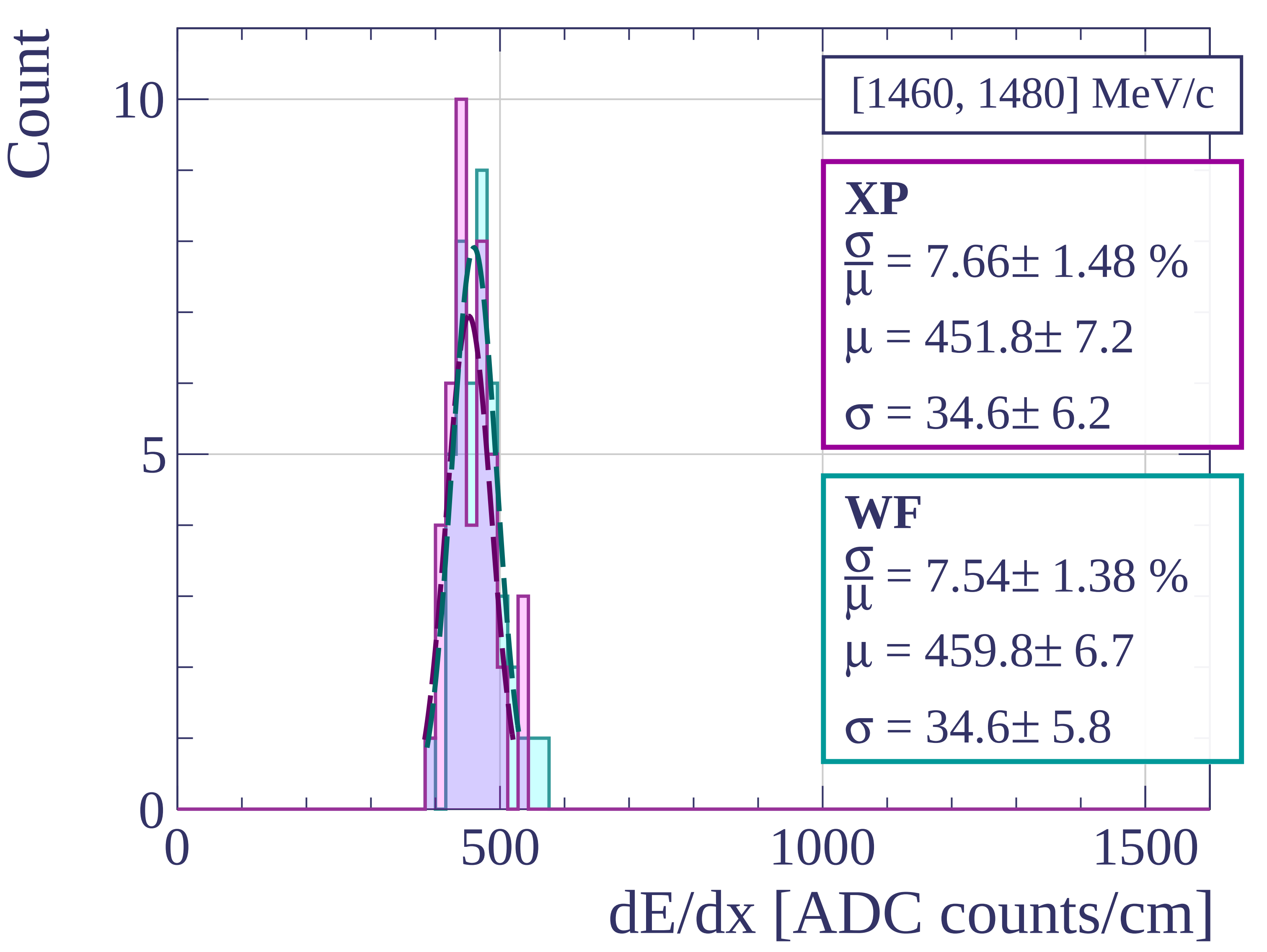
Count
[1380, 1400] MeV/c
XP
σ/μ = 5.37± 1.21 %
μ = 448.1± 5.6
σ = 24.0± 5.1
WF
σ/μ = 3.43± 1.03 %
μ = 444.2± 4.0
σ = 15.2± 4.4
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1400, 1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.06± 1.59 %
μ = 444.8± 7.7
σ = 26.9± 6.6
WF
$\frac{\sigma}{\mu}$ = 5.71± 1.00 %
μ = 454.9± 5.4
σ = 26.0± 4.2

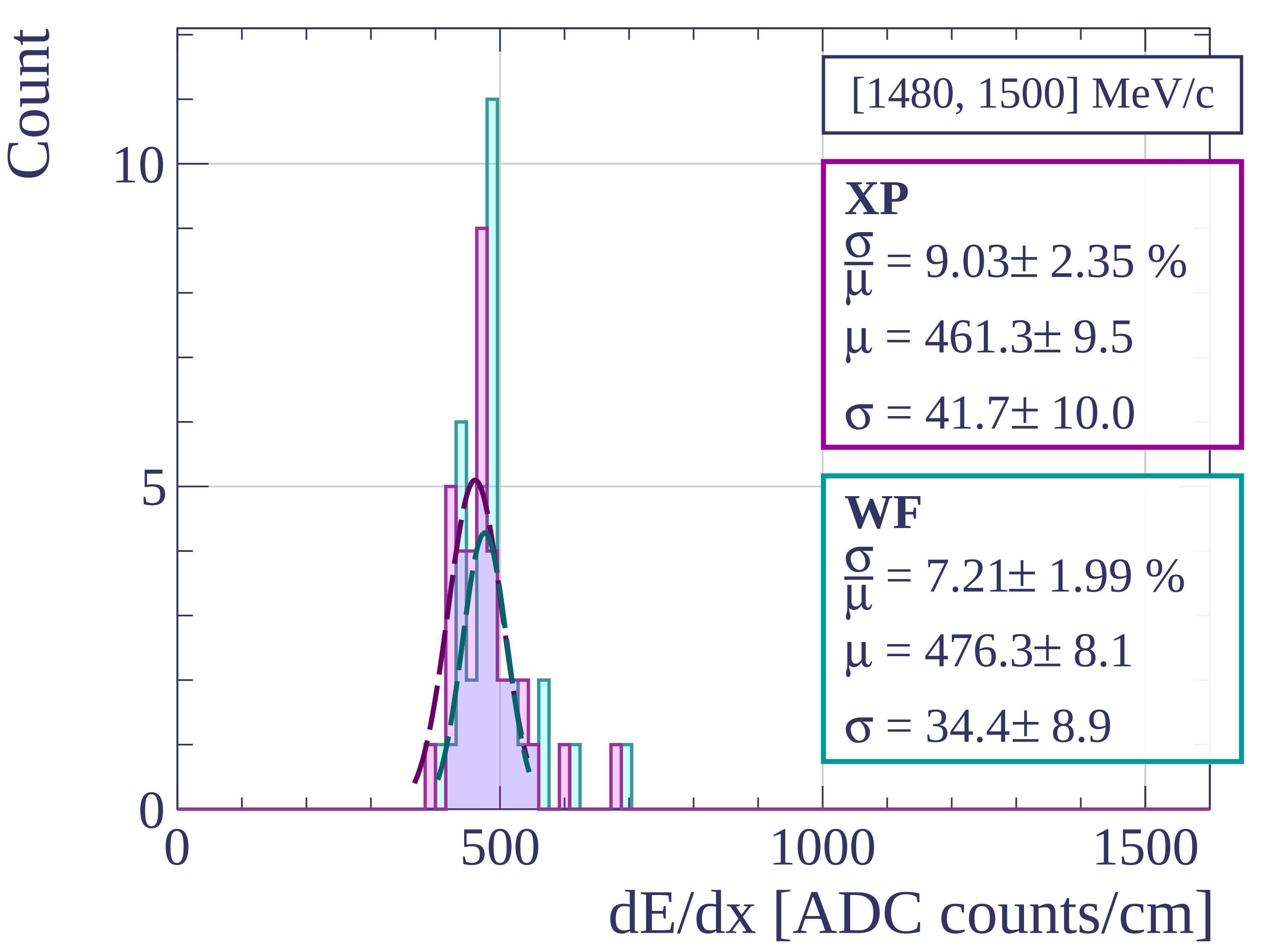

Count
[1420, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 29.02± 34.34 %
μ = 418.8± 138.2
σ = 121.5± 103.7
WF
$\frac{\sigma}{\mu}$ = 18.71± 15.82 %
μ = 1681.8± 730.1
σ = 314.6± 129.6
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]



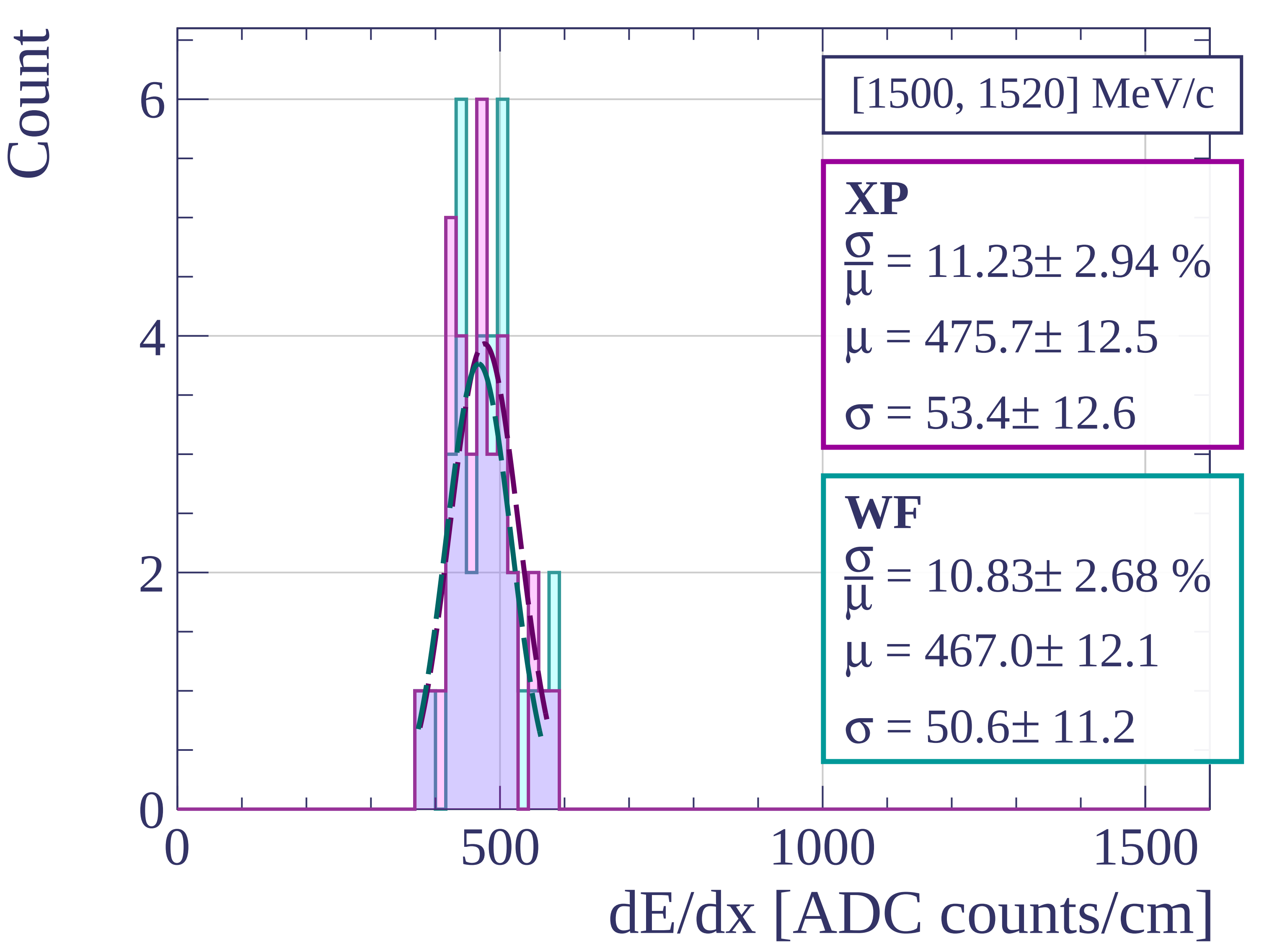
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.24± 6.42 %
μ = 451.9± 21.3
σ = 59.8± 26.2
WF
$\frac{\sigma}{\mu}$ = 8.57± 2.88 %
μ = 477.1± 10.6
σ = 40.9± 12.8



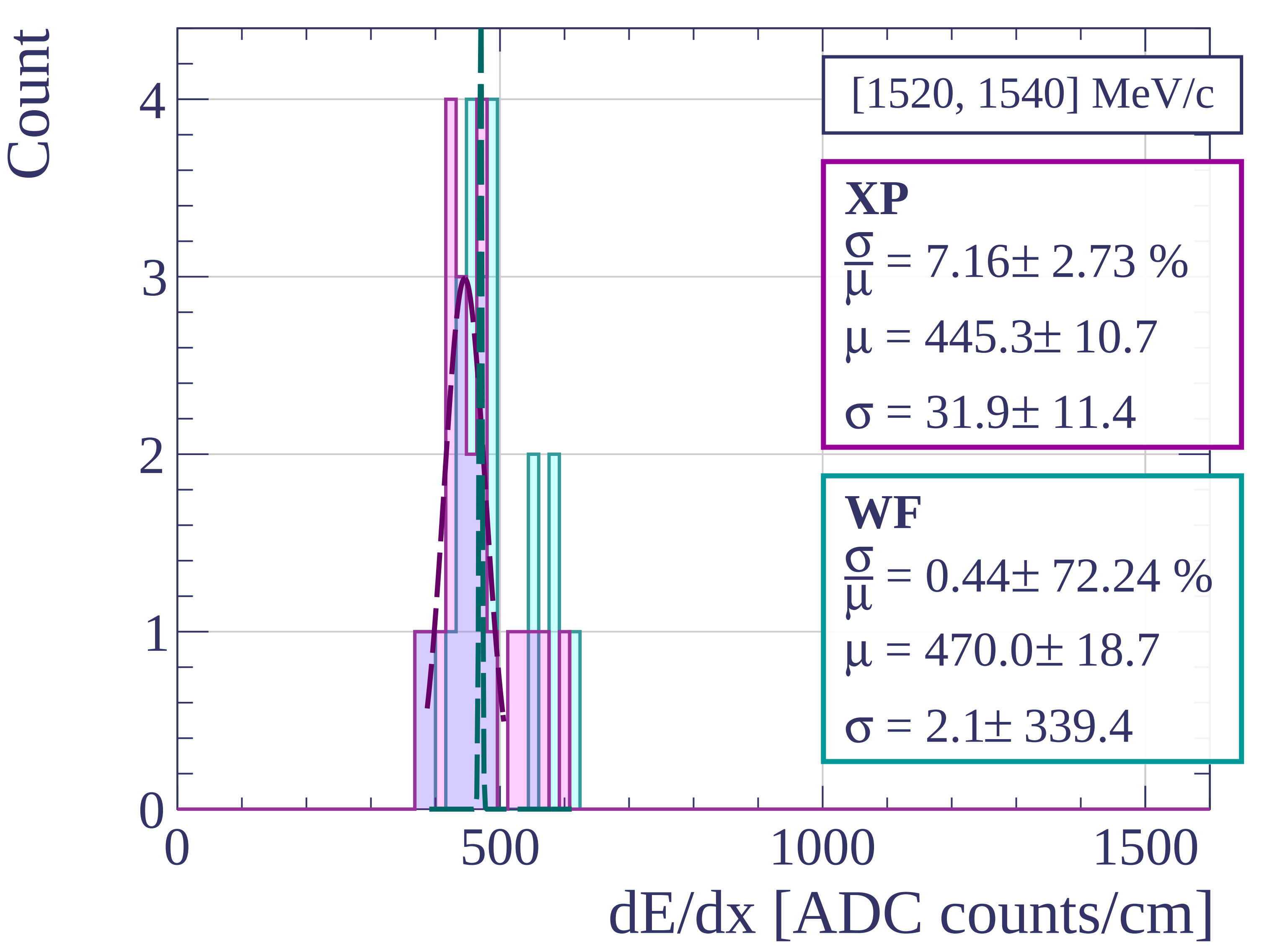
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
σ/μ = 7.66± 1.48 %
μ = 451.8± 7.2
σ = 34.6± 6.2
WF
σ/μ = 7.54± 1.38 %
μ = 459.8± 6.7
σ = 34.6± 5.8



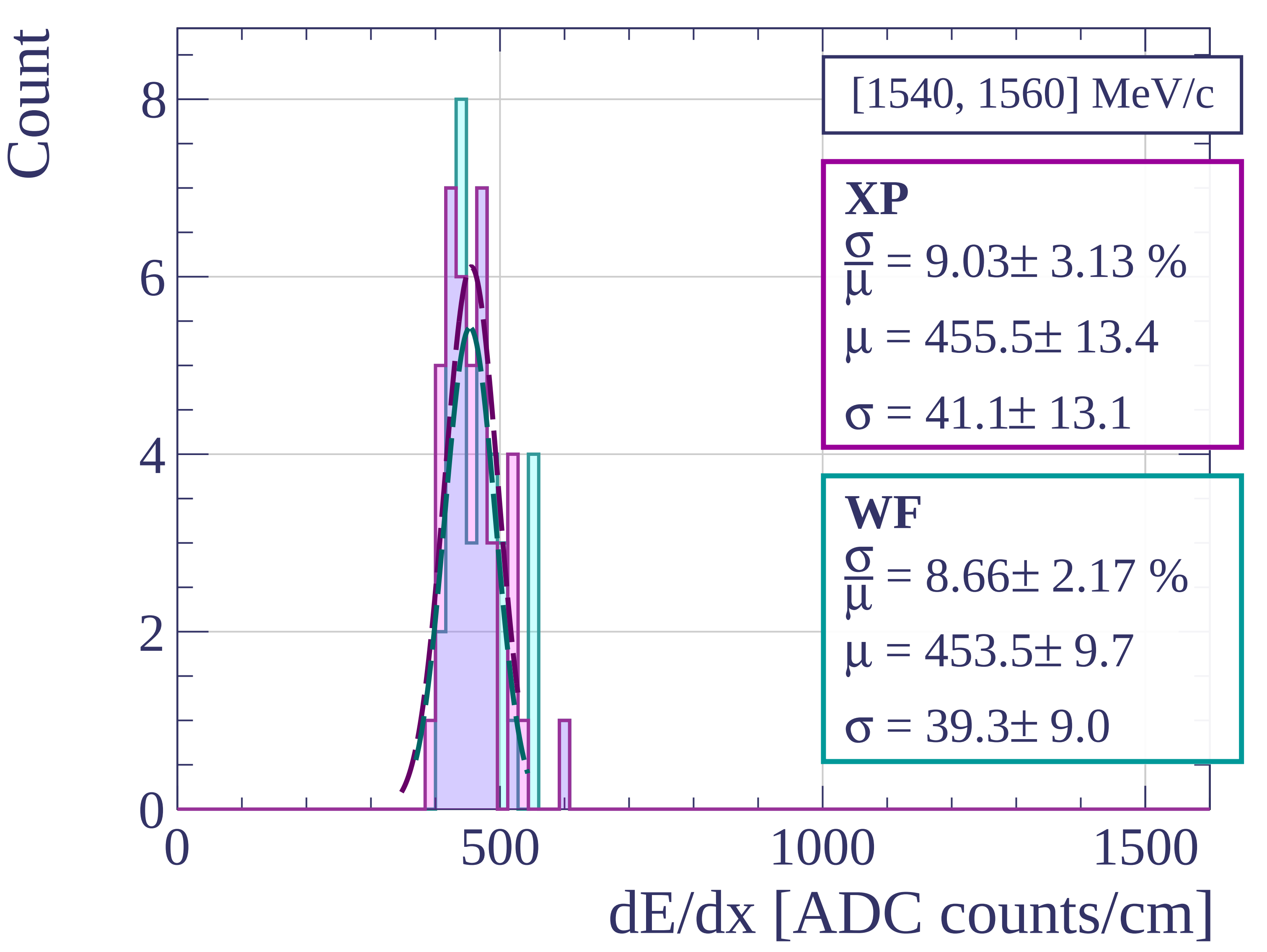
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
σ/μ = 9.03± 2.35 %
μ = 461.3± 9.5
σ = 41.7± 10.0
WF
σ/μ = 7.21± 1.99 %
μ = 476.3± 8.1
σ = 34.4± 8.9



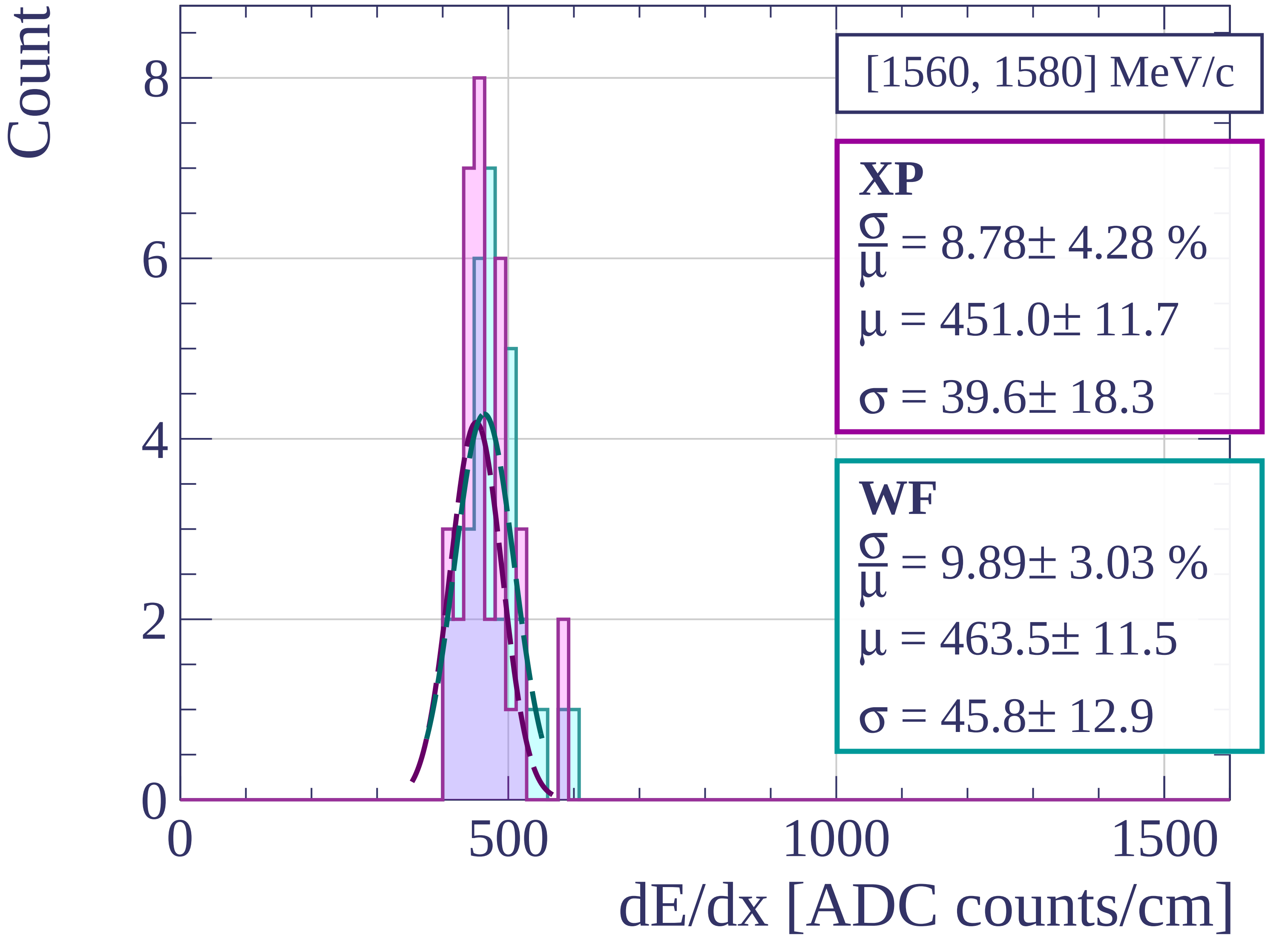
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1500, 1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.23± 2.94 %
μ = 475.7± 12.5
σ = 53.4± 12.6
WF
$\frac{\sigma}{\mu}$ = 10.83± 2.68 %
μ = 467.0± 12.1
σ = 50.6± 11.2



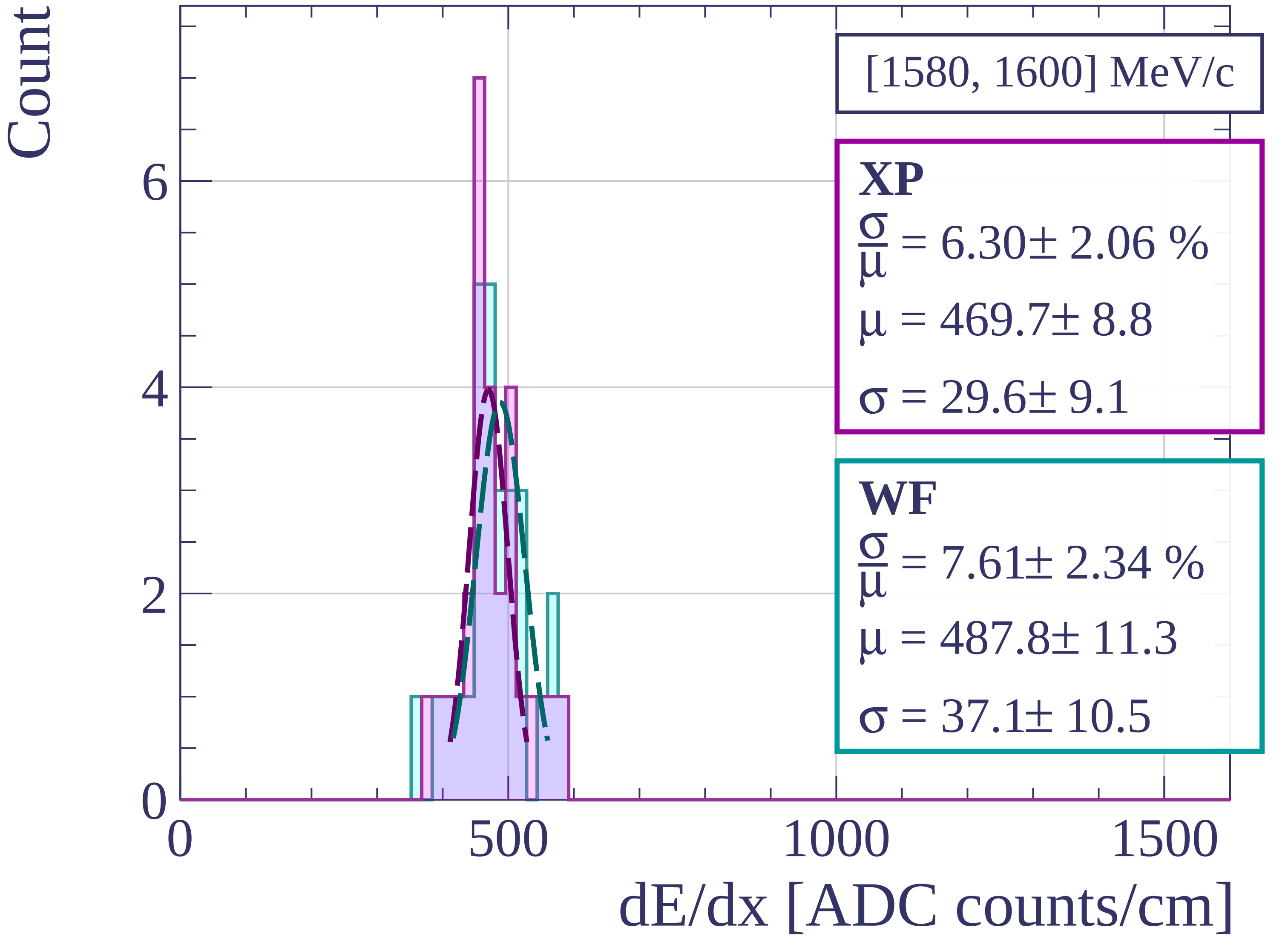
Count
dE/dx [ADC counts/cm]
[1520, 1540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.16± 2.73 %
μ = 445.3± 10.7
σ = 31.9± 11.4
WF
$\frac{\sigma}{\mu}$ = 0.44± 72.24 %
μ = 470.0± 18.7
σ = 2.1± 339.4
0
500
1000
1500
0
1
2
3
4



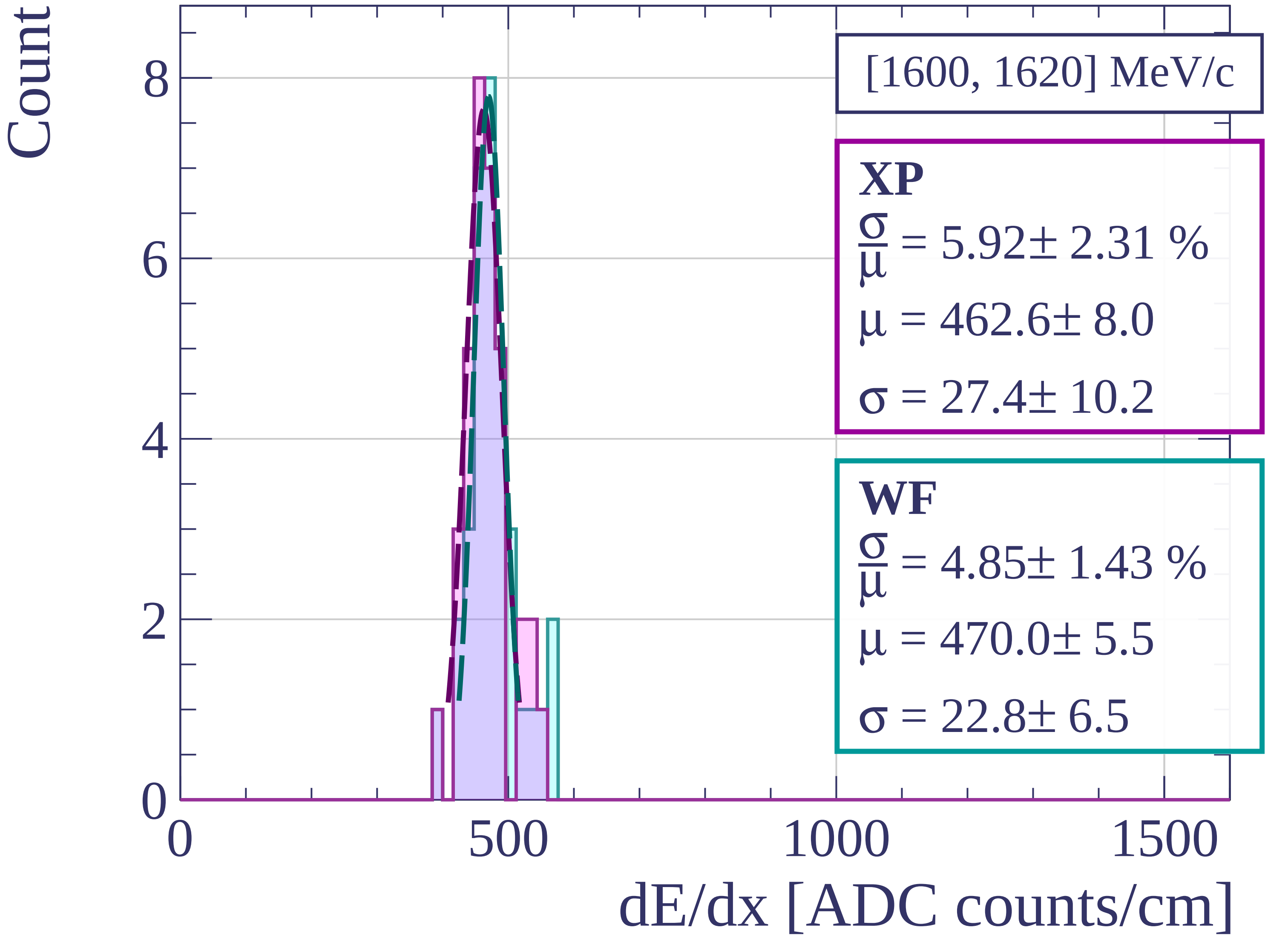
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1540, 1560] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.03± 3.13 %
μ = 455.5± 13.4
σ = 41.1± 13.1
WF
$\frac{\sigma}{\mu}$ = 8.66± 2.17 %
μ = 453.5± 9.7
σ = 39.3± 9.0



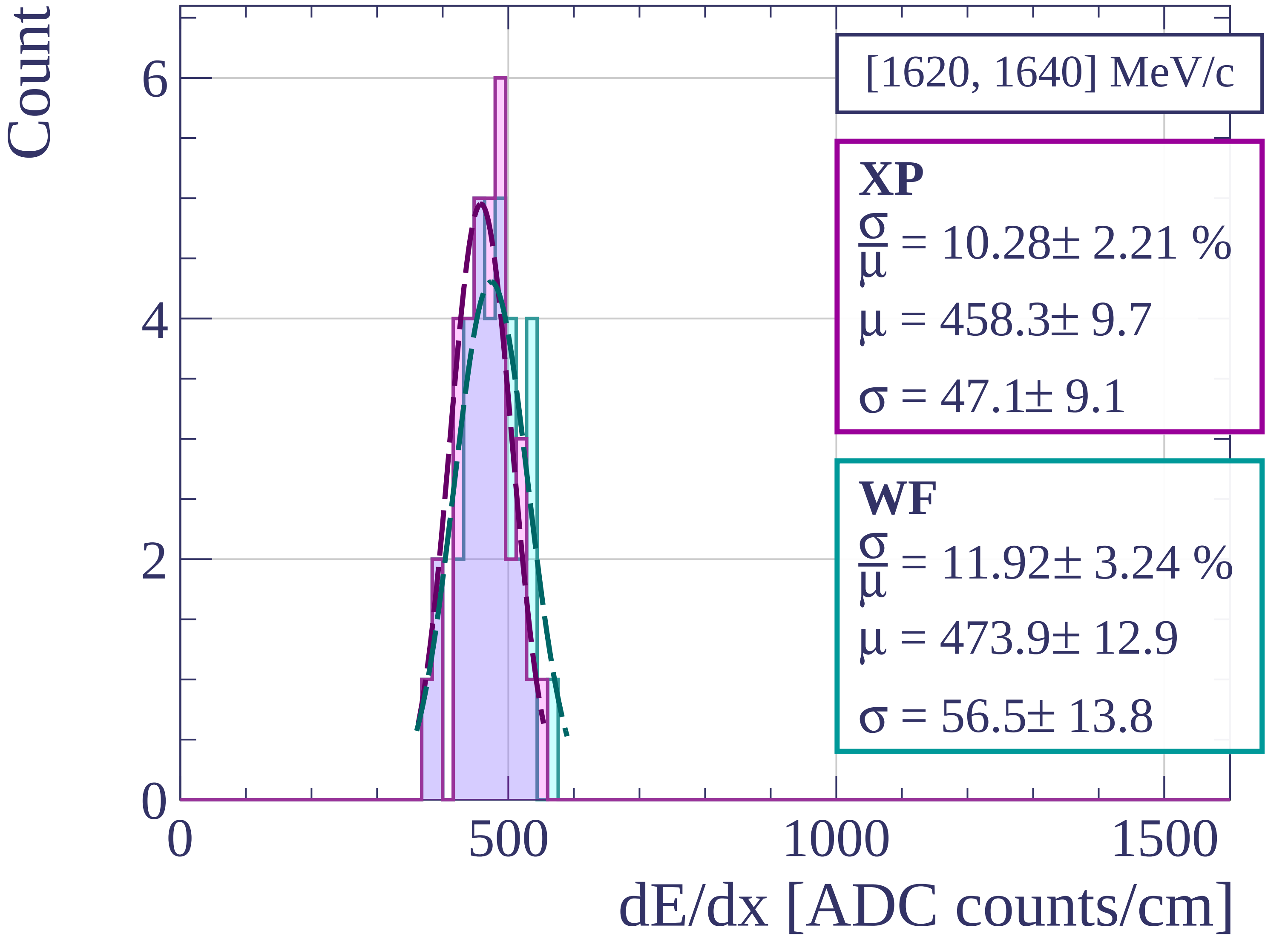
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1560, 1580] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.78± 4.28 %
μ = 451.0± 11.7
σ = 39.6± 18.3
WF
$\frac{\sigma}{\mu}$ = 9.89± 3.03 %
μ = 463.5± 11.5
σ = 45.8± 12.9



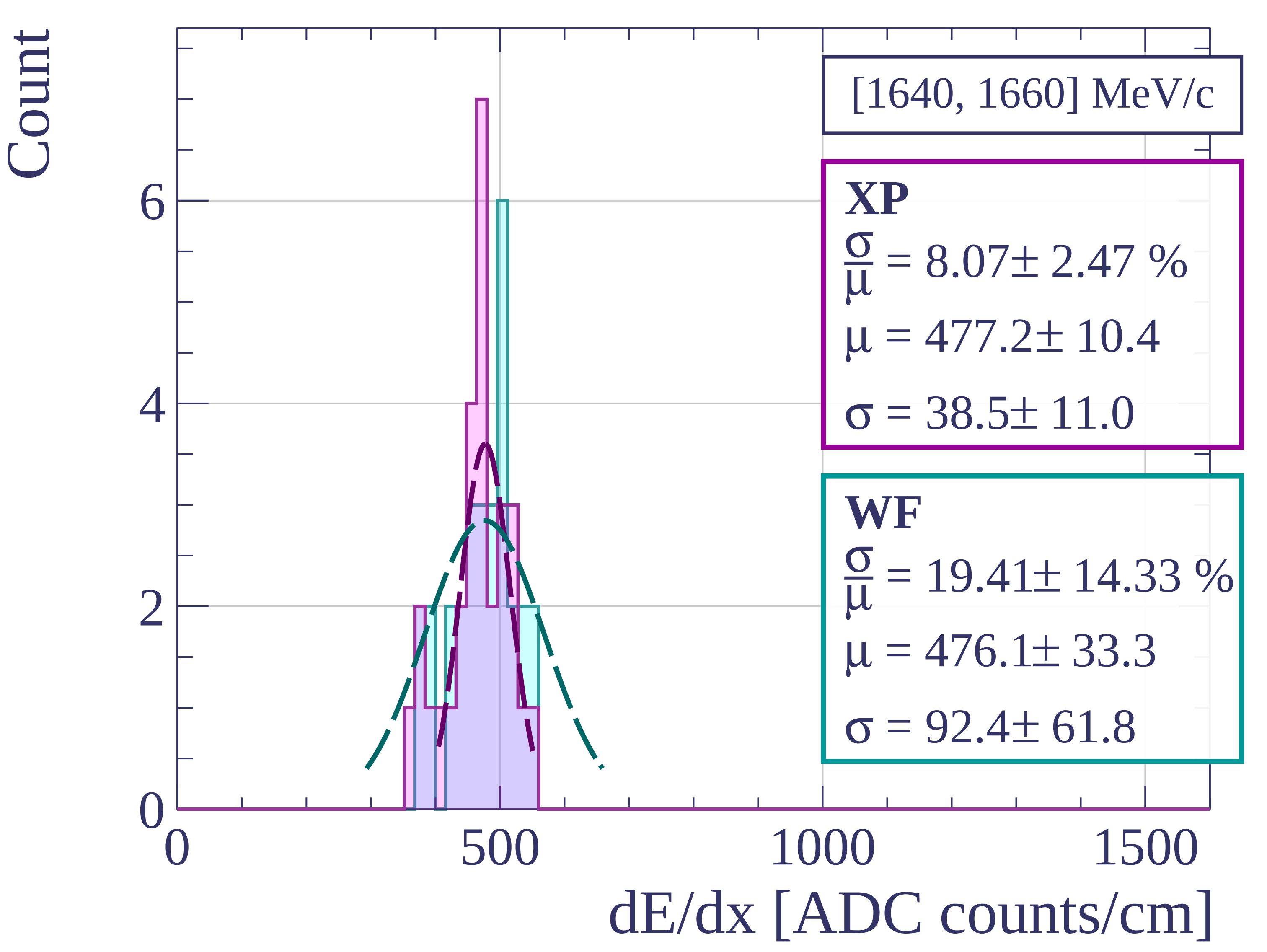
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1580, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 6.30± 2.06 %
μ = 469.7± 8.8
σ = 29.6± 9.1
WF
$\frac{\sigma}{\mu}$ = 7.61± 2.34 %
μ = 487.8± 11.3
σ = 37.1± 10.5

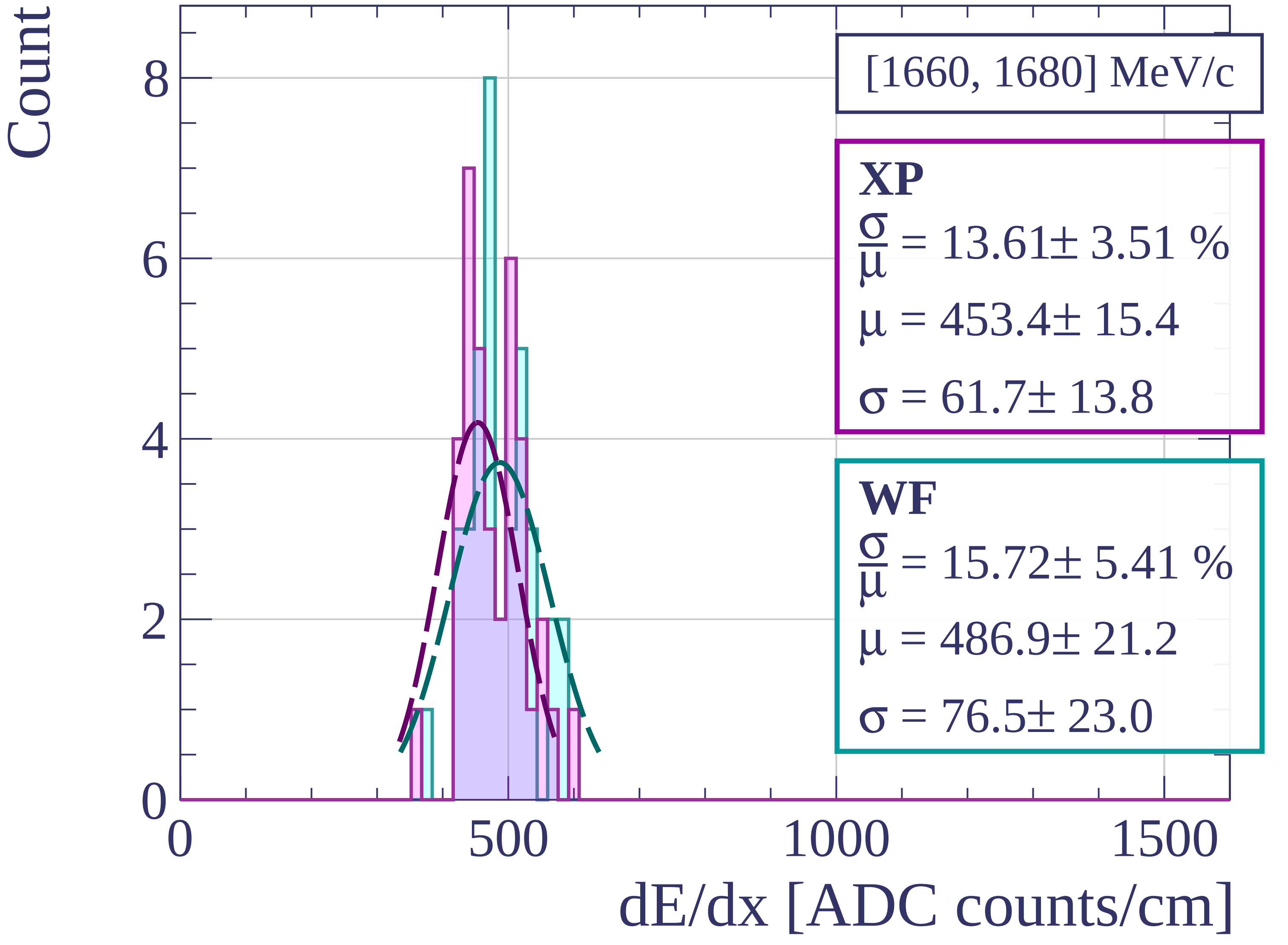


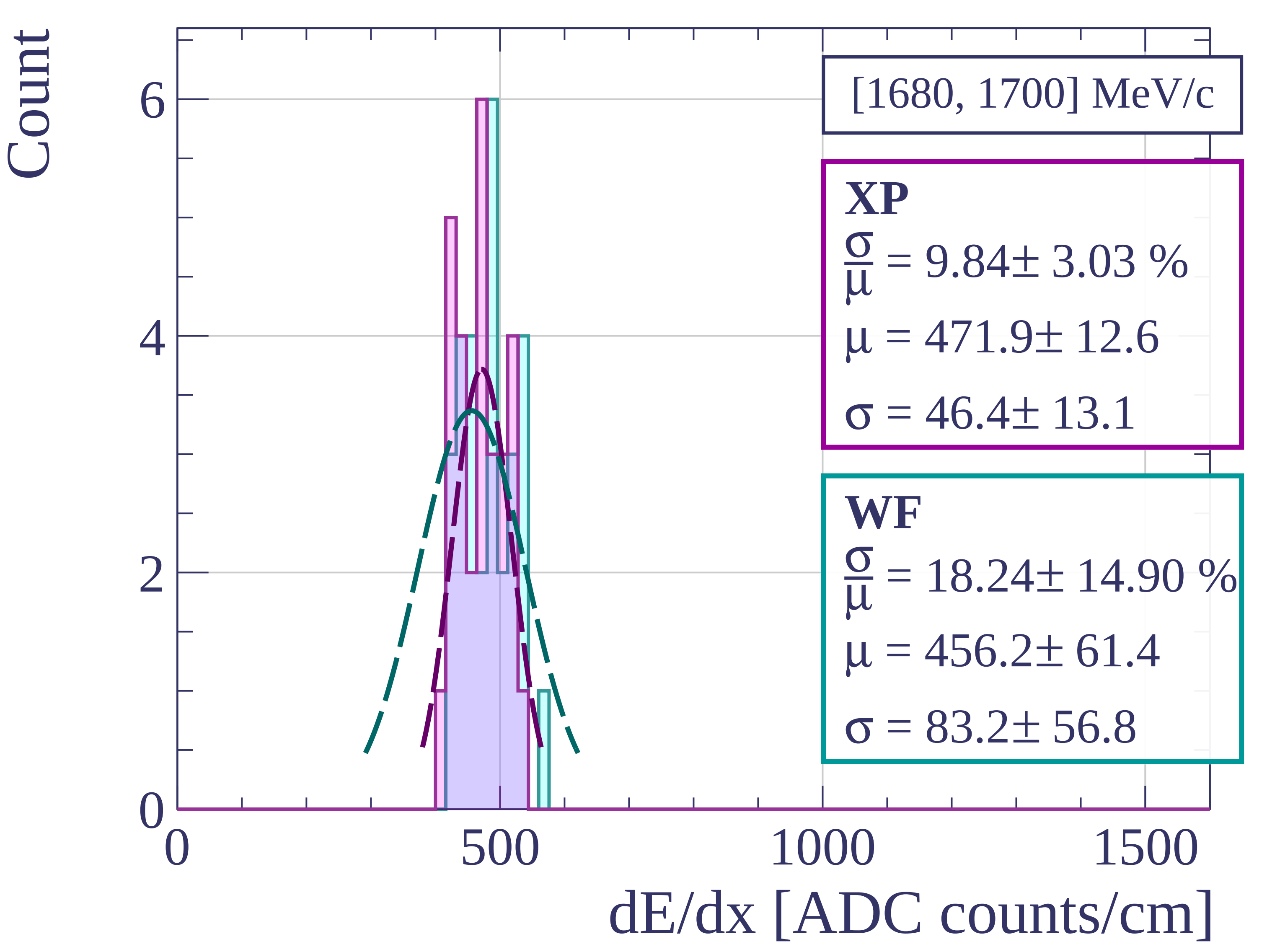
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.92± 2.31 %
μ = 462.6± 8.0
σ = 27.4± 10.2
WF
$\frac{\sigma}{\mu}$ = 4.85± 1.43 %
μ = 470.0± 5.5
σ = 22.8± 6.5



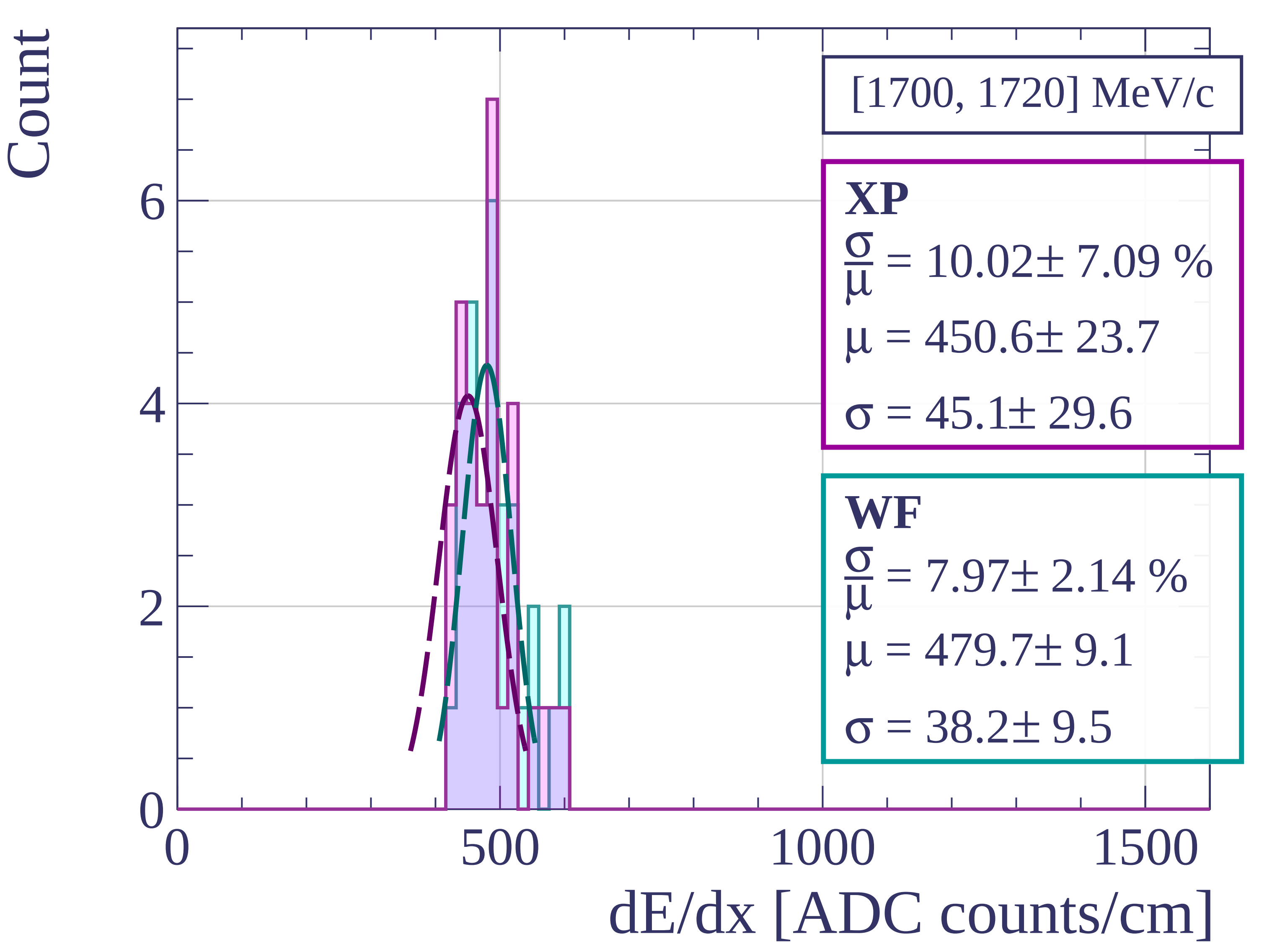
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[1620, 1640] MeV/c
XP
σ/μ = 10.28± 2.21 %
μ = 458.3± 9.7
σ = 47.1± 9.1
WF
σ/μ = 11.92± 3.24 %
μ = 473.9± 12.9
σ = 56.5± 13.8


Count
[1640, 1660] MeV/c
XP
σ/μ = 8.07± 2.47 %
μ = 477.2± 10.4
σ = 38.5± 11.0
WF
σ/μ = 19.41± 14.33 %
μ = 476.1± 33.3
σ = 92.4± 61.8
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]

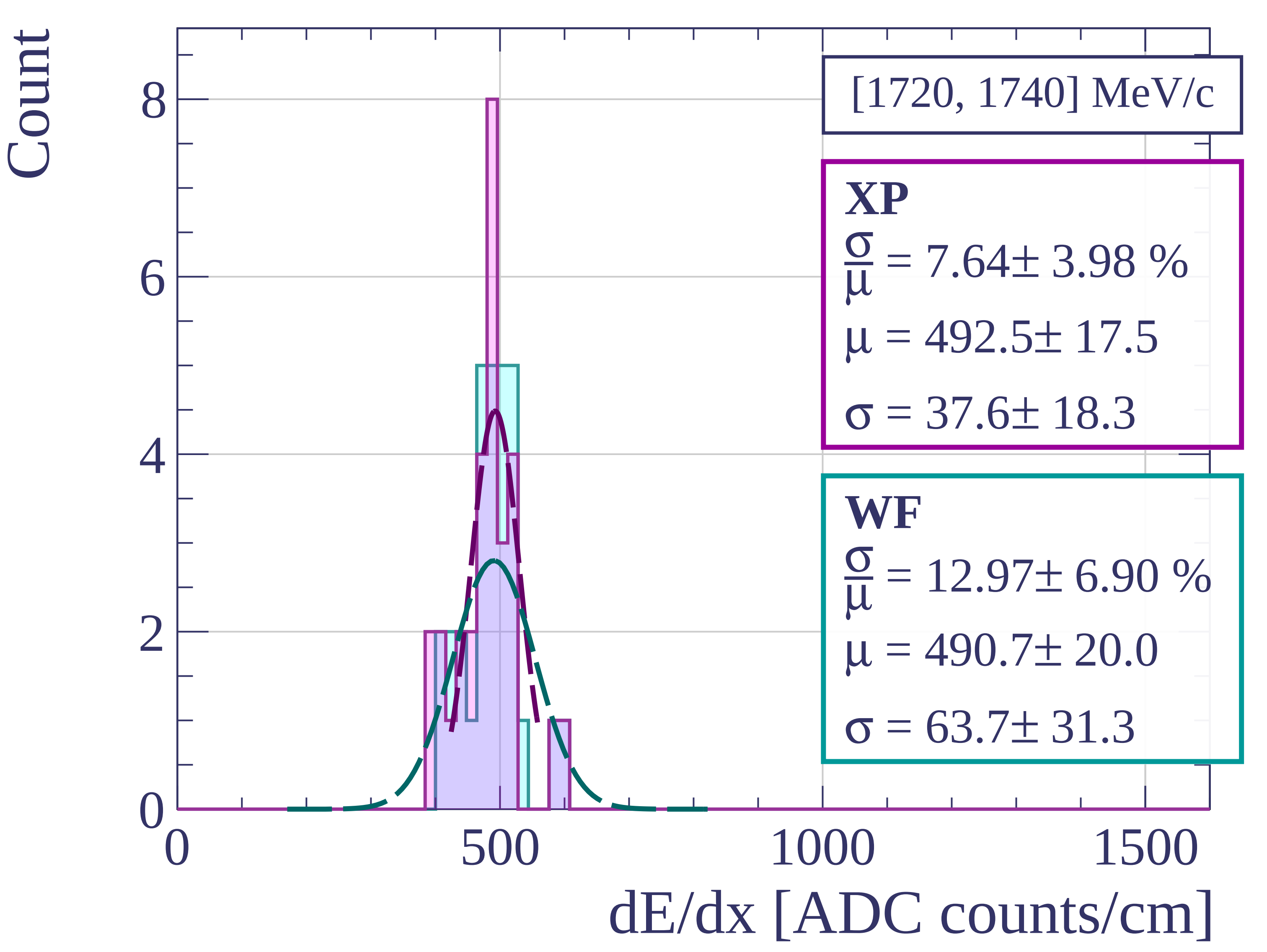

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
8
[1660, 1680] MeV/c
XP
σ/μ = 13.61± 3.51 %
μ = 453.4± 15.4
σ = 61.7± 13.8
WF
σ/μ = 15.72± 5.41 %
μ = 486.9± 21.2
σ = 76.5± 23.0



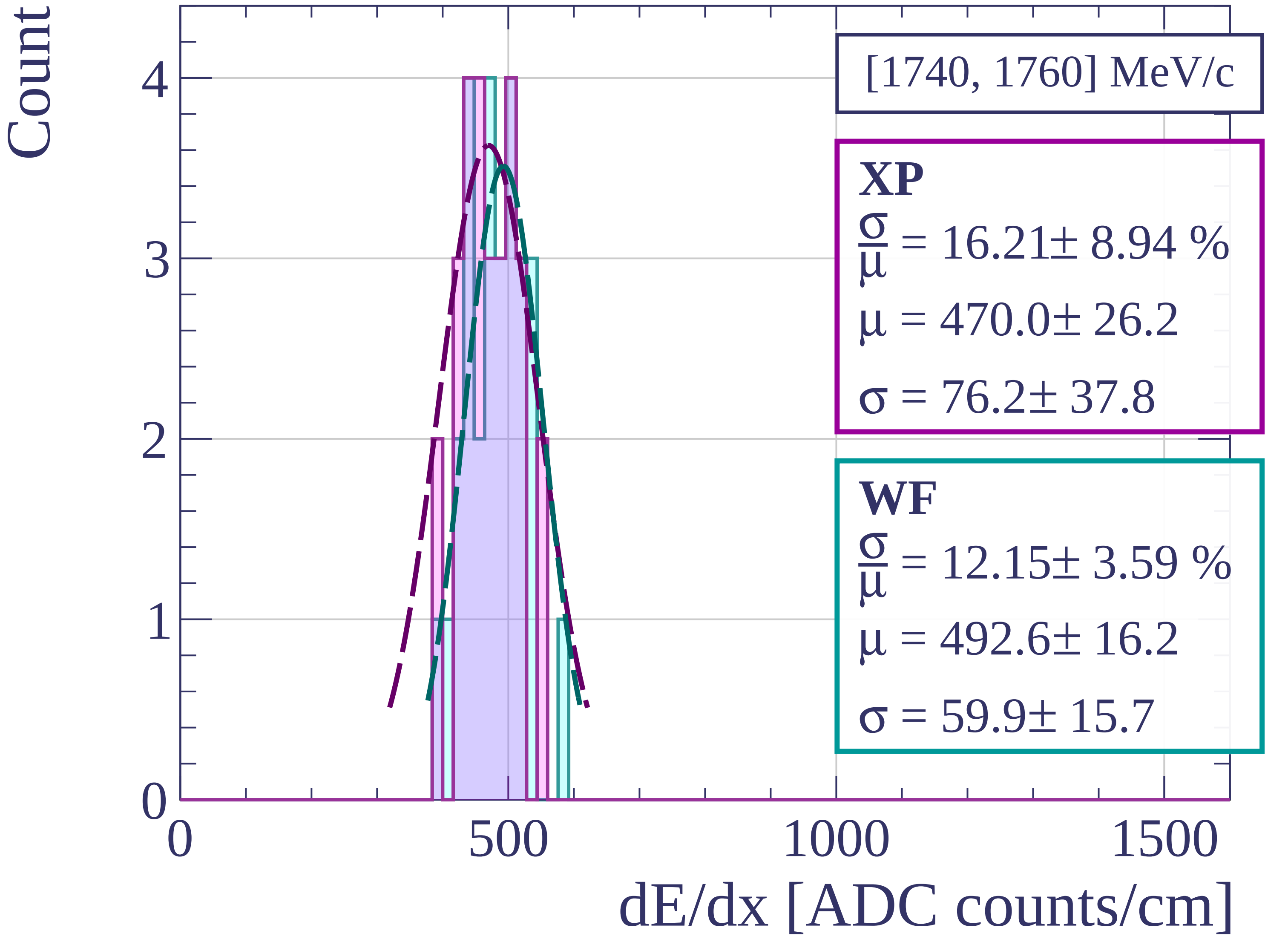
Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.84± 3.03 %
μ = 471.9± 12.6
σ = 46.4± 13.1
WF
$\frac{\sigma}{\mu}$ = 18.24± 14.90 %
μ = 456.2± 61.4
σ = 83.2± 56.8



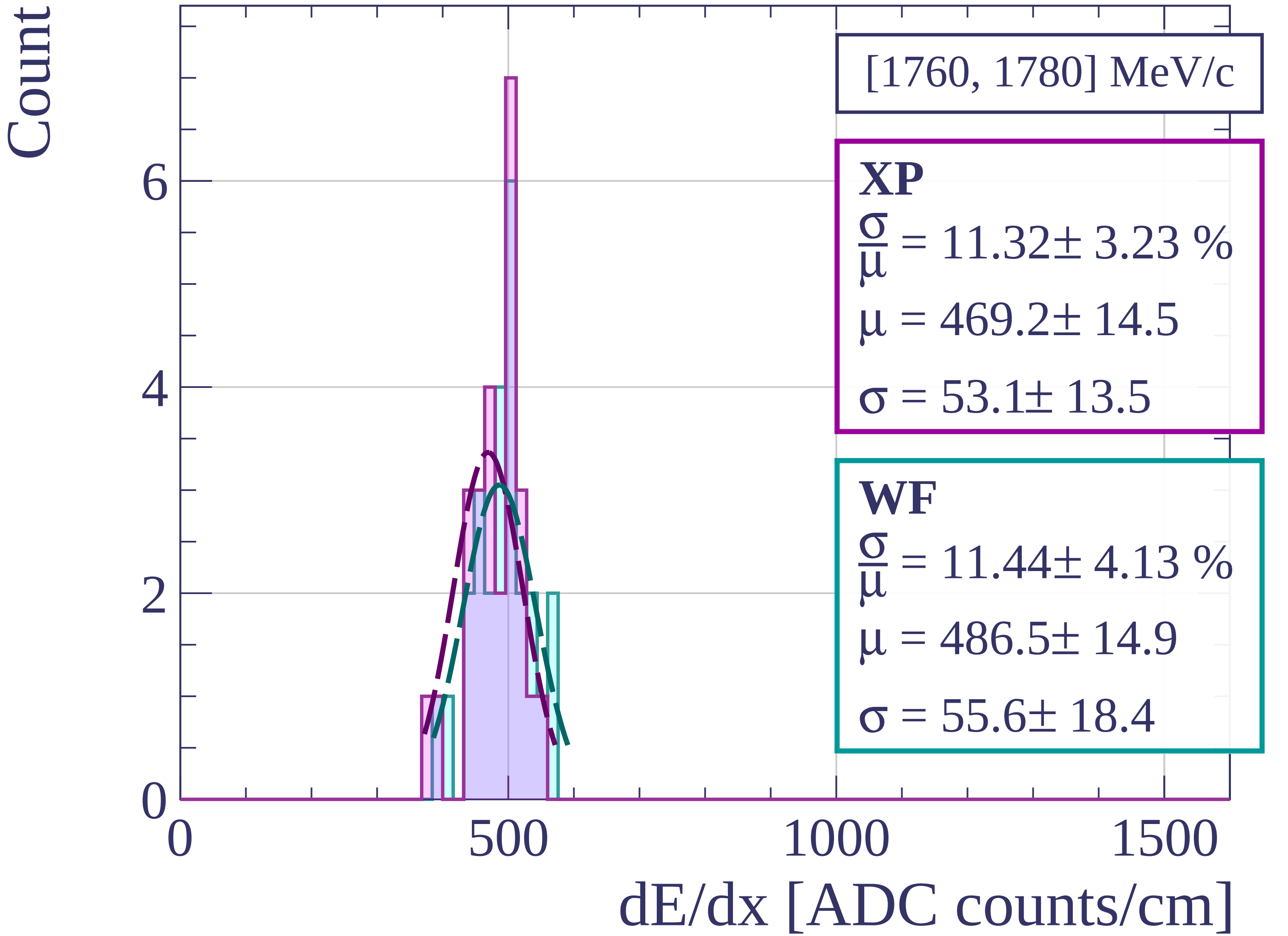
Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1700, 1720] MeV/c
XP
σ/μ = 10.02± 7.09 %
μ = 450.6± 23.7
σ = 45.1± 29.6
WF
σ/μ = 7.97± 2.14 %
μ = 479.7± 9.1
σ = 38.2± 9.5



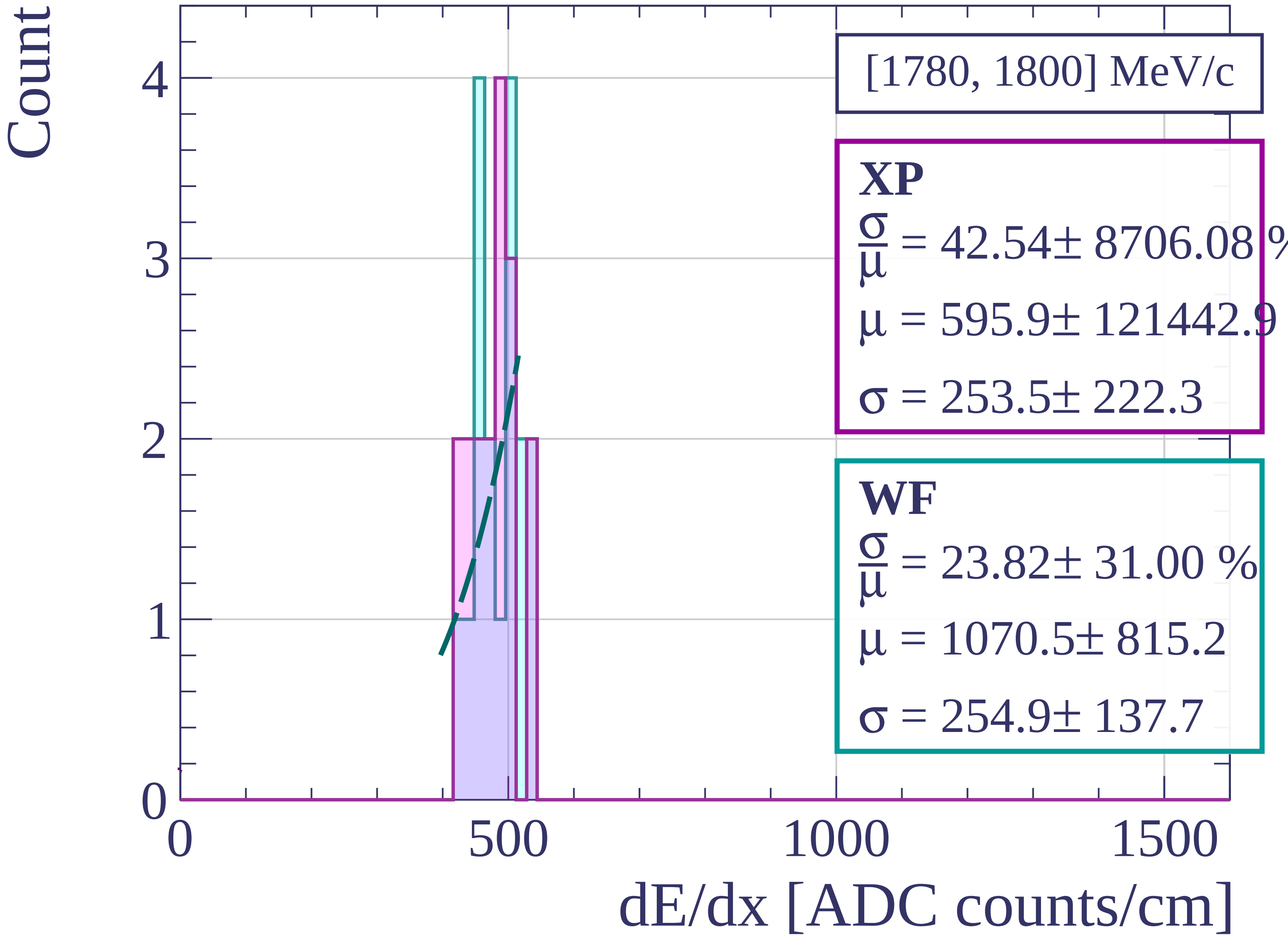
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1720, 1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 7.64± 3.98 %
μ = 492.5± 17.5
σ = 37.6± 18.3
WF
$\frac{\sigma}{\mu}$ = 12.97± 6.90 %
μ = 490.7± 20.0
σ = 63.7± 31.3

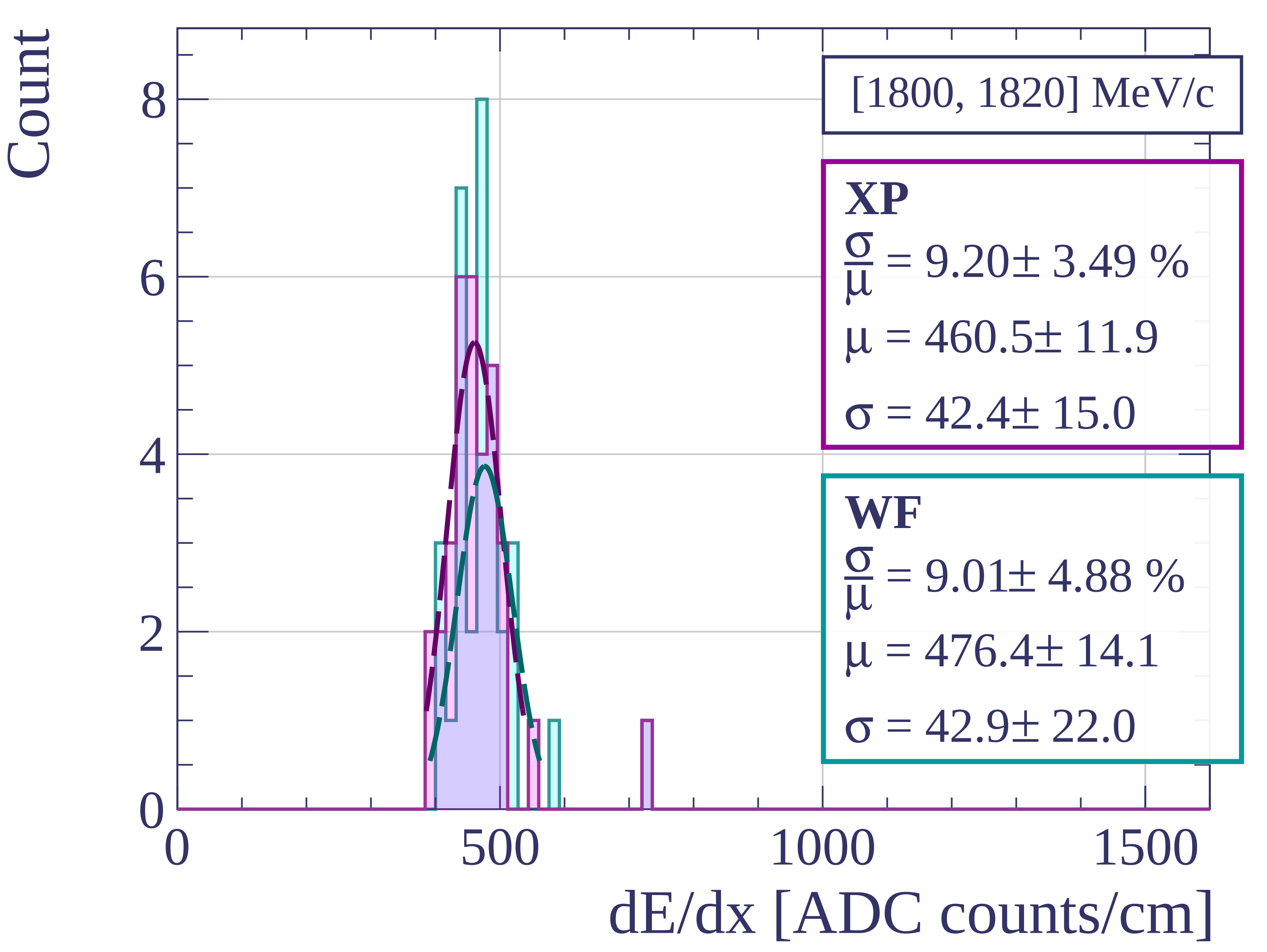


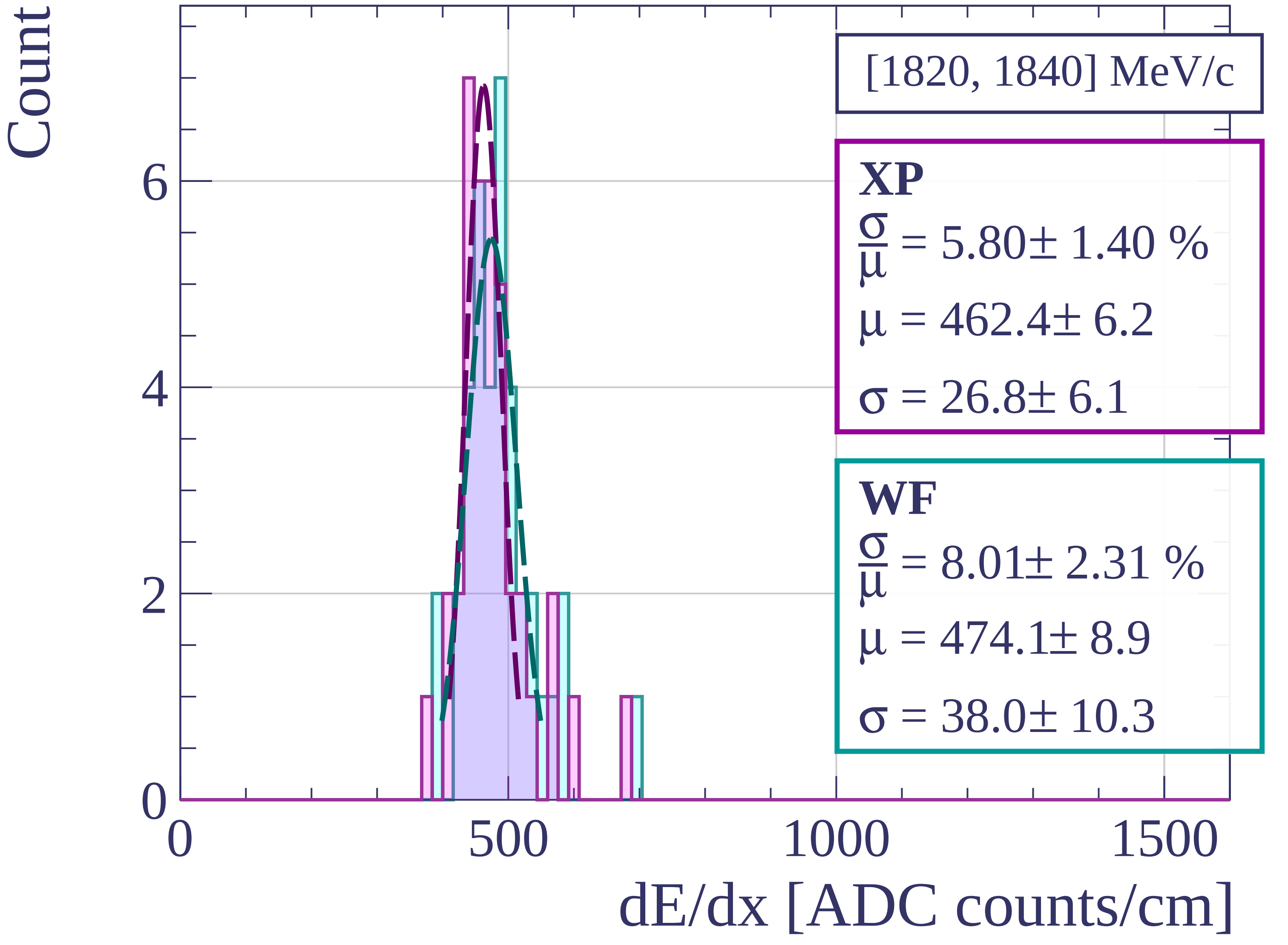
Count
0
1
2
3
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1740, 1760] MeV/c
XP
σ/μ = 16.21± 8.94 %
μ = 470.0± 26.2
σ = 76.2± 37.8
WF
σ/μ = 12.15± 3.59 %
μ = 492.6± 16.2
σ = 59.9± 15.7



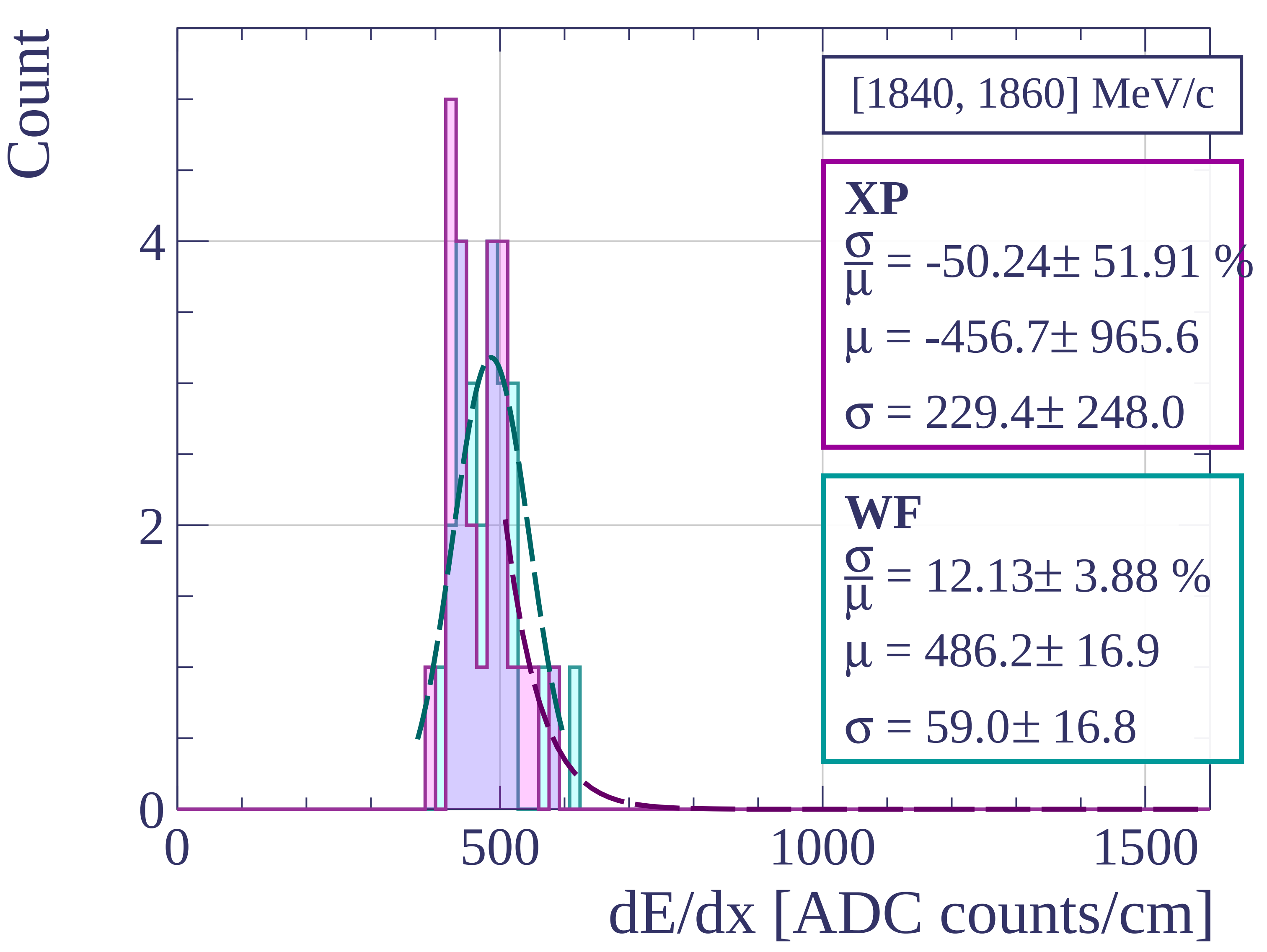
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.32± 3.23 %
μ = 469.2± 14.5
σ = 53.1± 13.5
WF
$\frac{\sigma}{\mu}$ = 11.44± 4.13 %
μ = 486.5± 14.9
σ = 55.6± 18.4

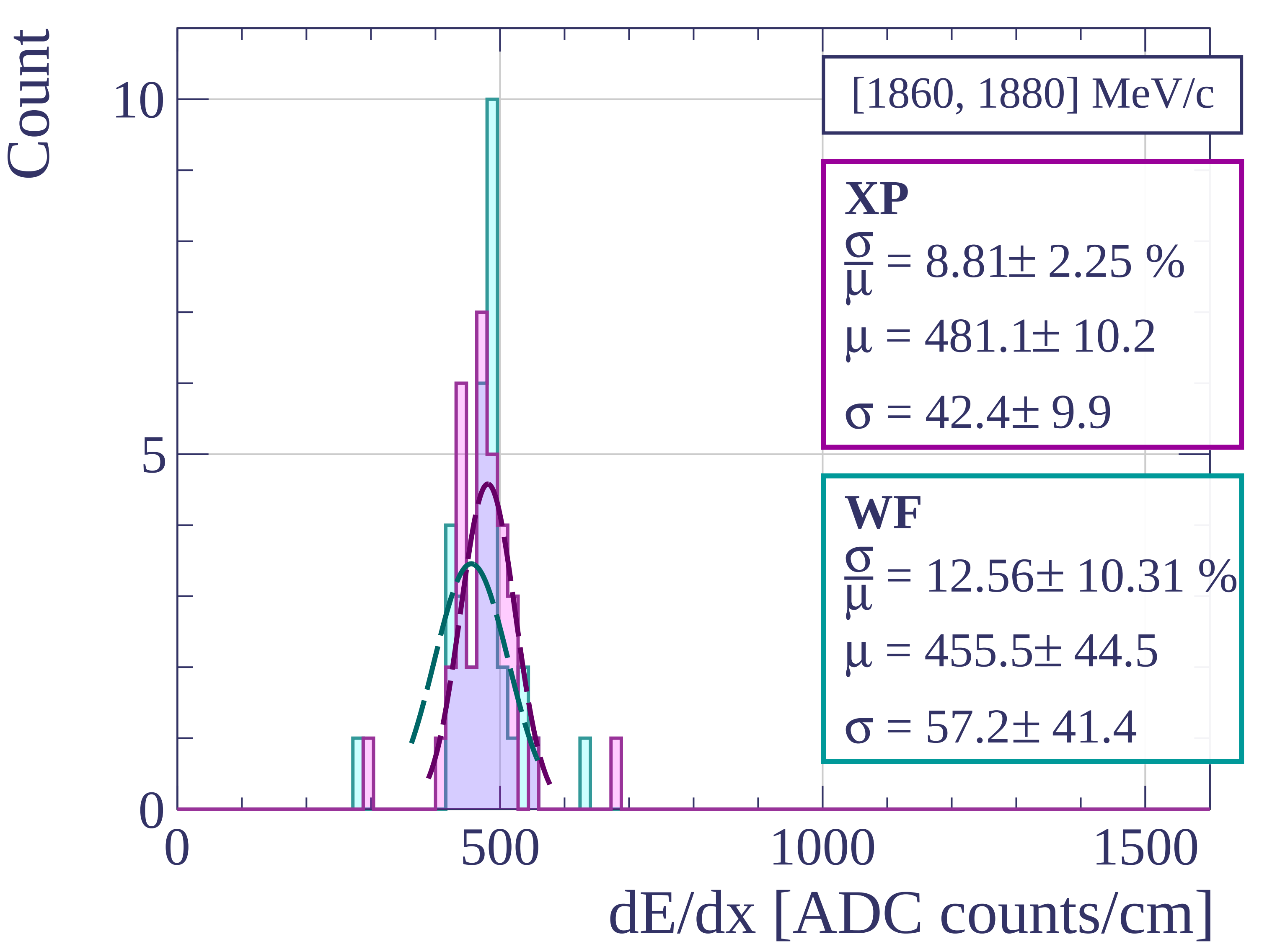

[1780, 1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 42.54± 8706.08 %
μ = 595.9± 121442.9
σ = 253.5± 222.3
WF
$\frac{\sigma}{\mu}$ = 23.82± 31.00 %
μ = 1070.5± 815.2
σ = 254.9± 137.7
Count
dE/dx [ADC counts/cm]
0
1
2
3
4
0
500
1000
1500


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1800, 1820] MeV/c
XP
σ/μ = 9.20± 3.49 %
μ = 460.5± 11.9
σ = 42.4± 15.0
WF
σ/μ = 9.01± 4.88 %
μ = 476.4± 14.1
σ = 42.9± 22.0

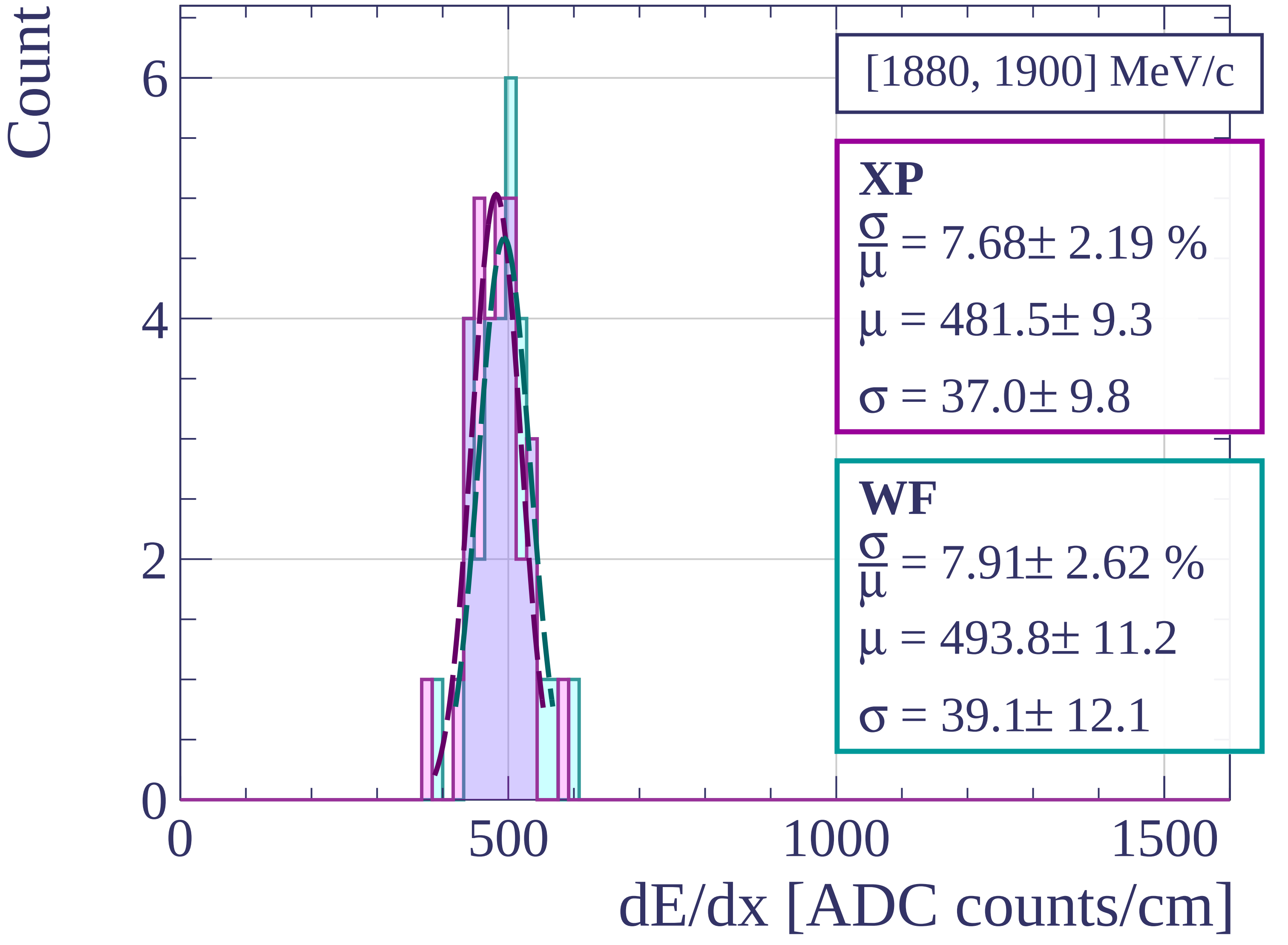


Count
[1820, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 5.80± 1.40 %
μ = 462.4± 6.2
σ = 26.8± 6.1
WF
$\frac{\sigma}{\mu}$ = 8.01± 2.31 %
μ = 474.1± 8.9
σ = 38.0± 10.3
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]

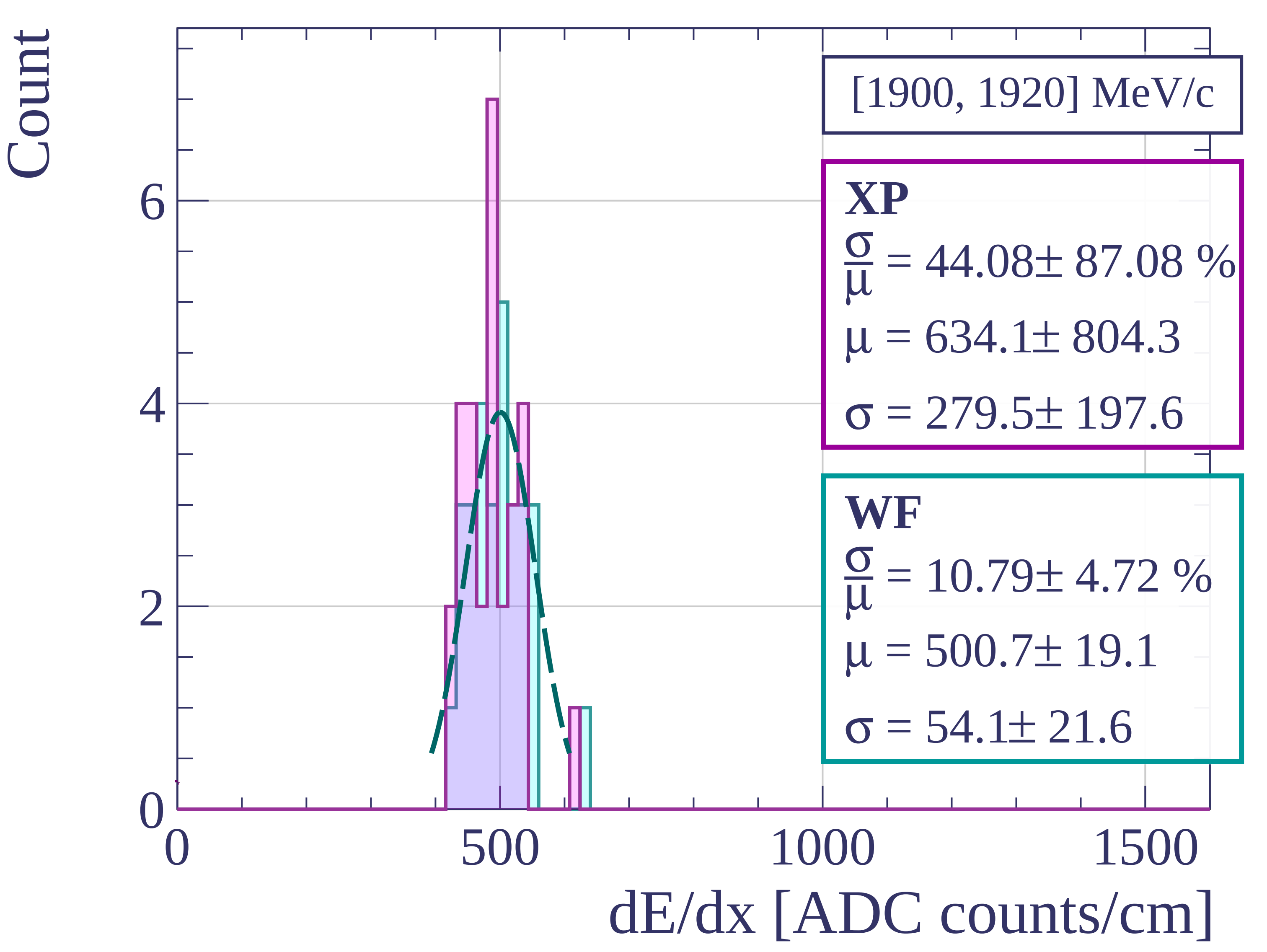

Count
0
2
4
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1860] MeV/c
XP
σ/μ = -50.24± 51.91 %
μ = -456.7± 965.6
σ = 229.4± 248.0
WF
σ/μ = 12.13± 3.88 %
μ = 486.2± 16.9
σ = 59.0± 16.8

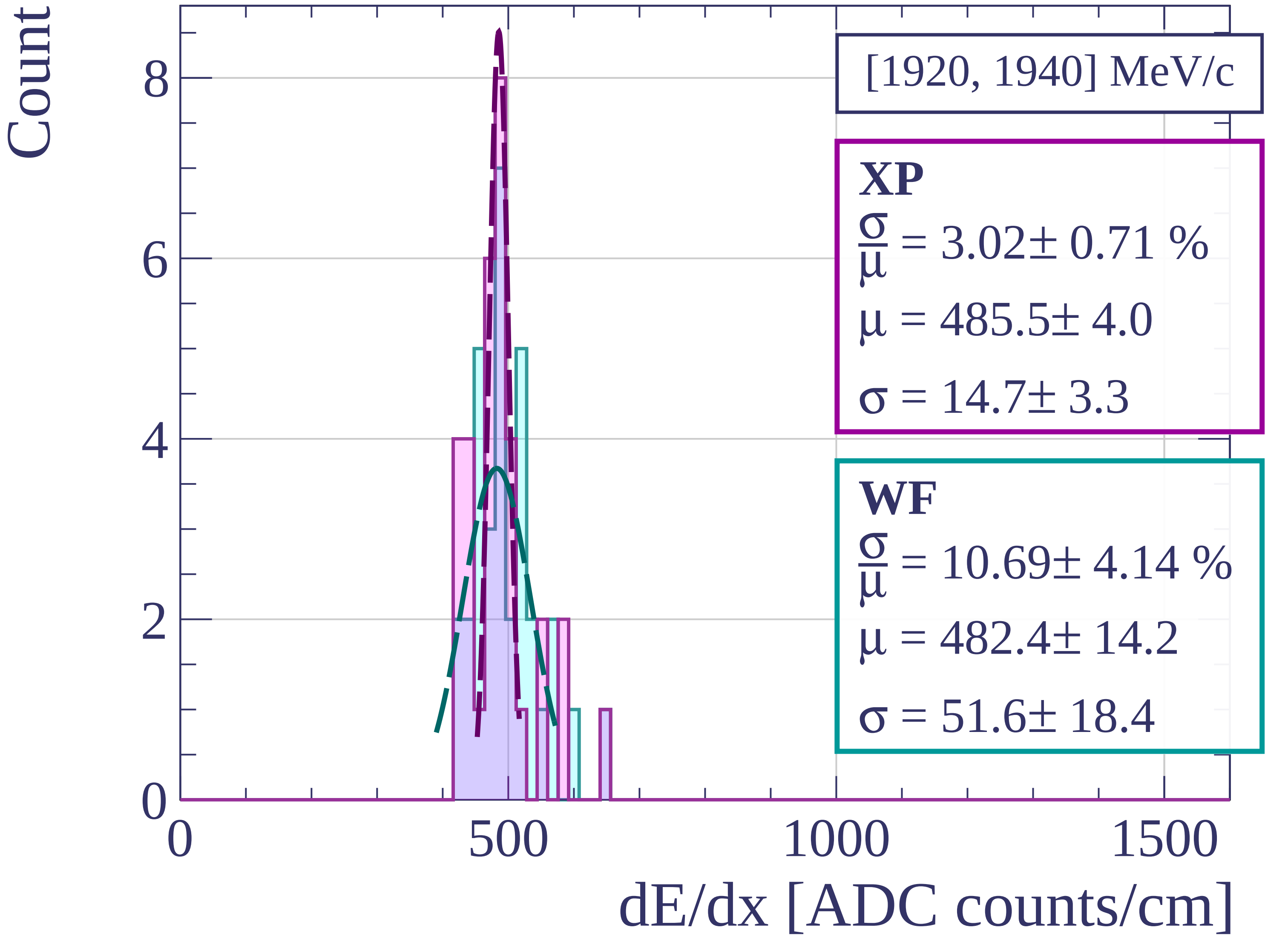


Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 8.81± 2.25 %
μ = 481.1± 10.2
σ = 42.4± 9.9
WF
$\frac{\sigma}{\mu}$ = 12.56± 10.31 %
μ = 455.5± 44.5
σ = 57.2± 41.4

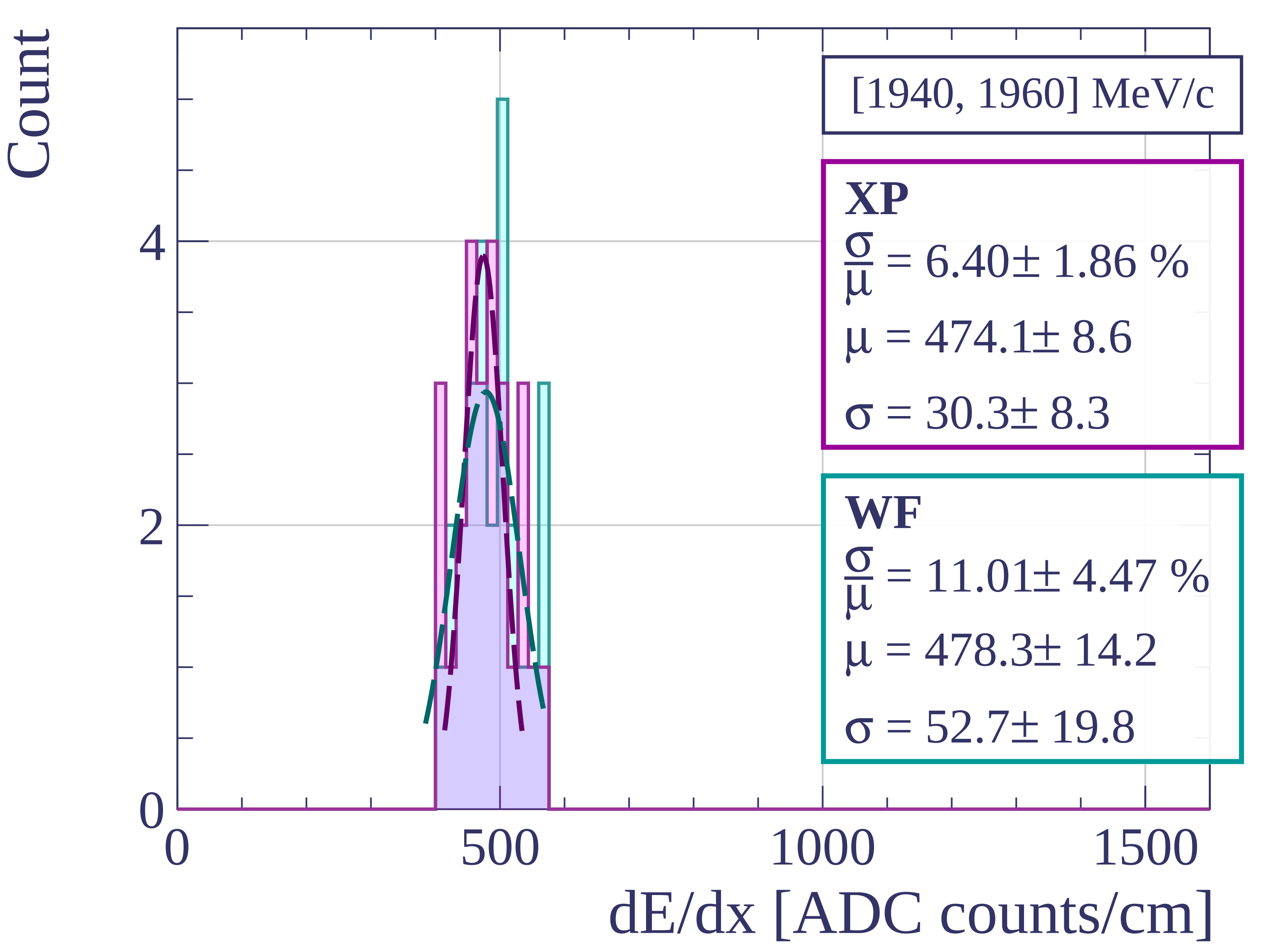


Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1880, 1900] MeV/c
XP
σ/μ = 7.68± 2.19 %
μ = 481.5± 9.3
σ = 37.0± 9.8
WF
σ/μ = 7.91± 2.62 %
μ = 493.8± 11.2
σ = 39.1± 12.1

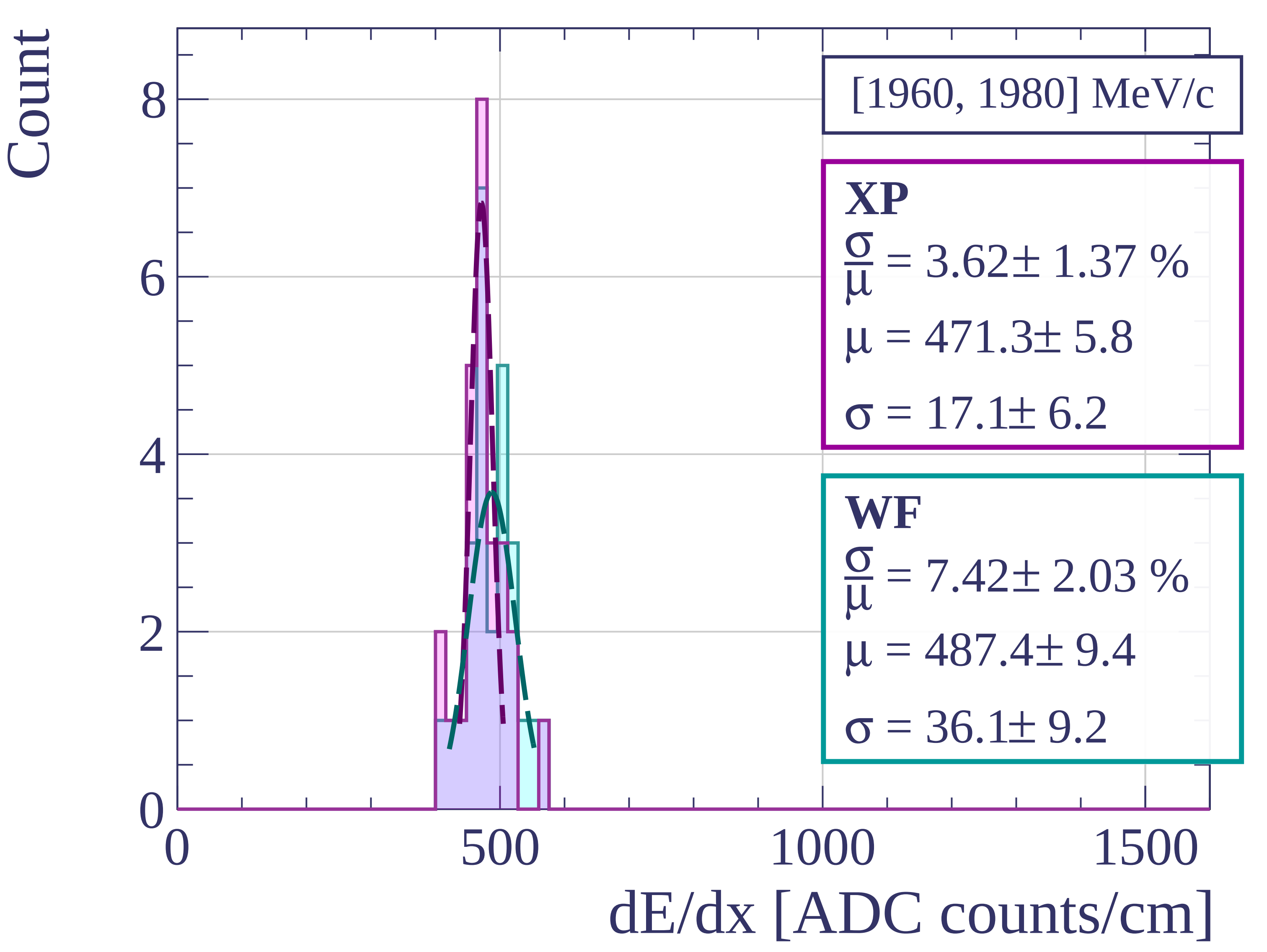

Count
dE/dx [ADC counts/cm]
0
2
4
6
0
500
1000
1500
[1900, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 44.08± 87.08 %
μ = 634.1± 804.3
σ = 279.5± 197.6
WF
$\frac{\sigma}{\mu}$ = 10.79± 4.72 %
μ = 500.7± 19.1
σ = 54.1± 21.6

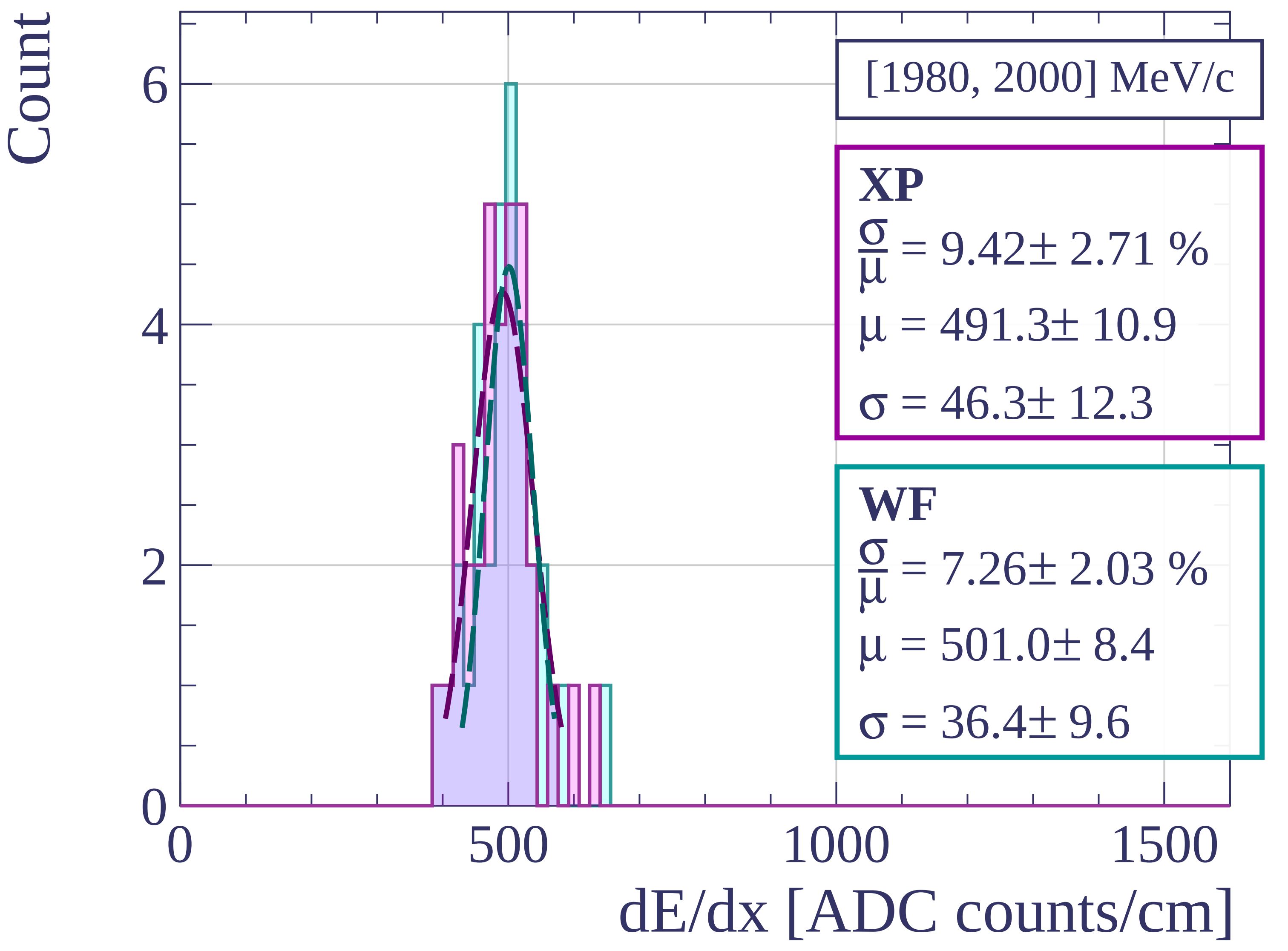


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.02± 0.71 %
μ = 485.5± 4.0
σ = 14.7± 3.3
WF
$\frac{\sigma}{\mu}$ = 10.69± 4.14 %
μ = 482.4± 14.2
σ = 51.6± 18.4

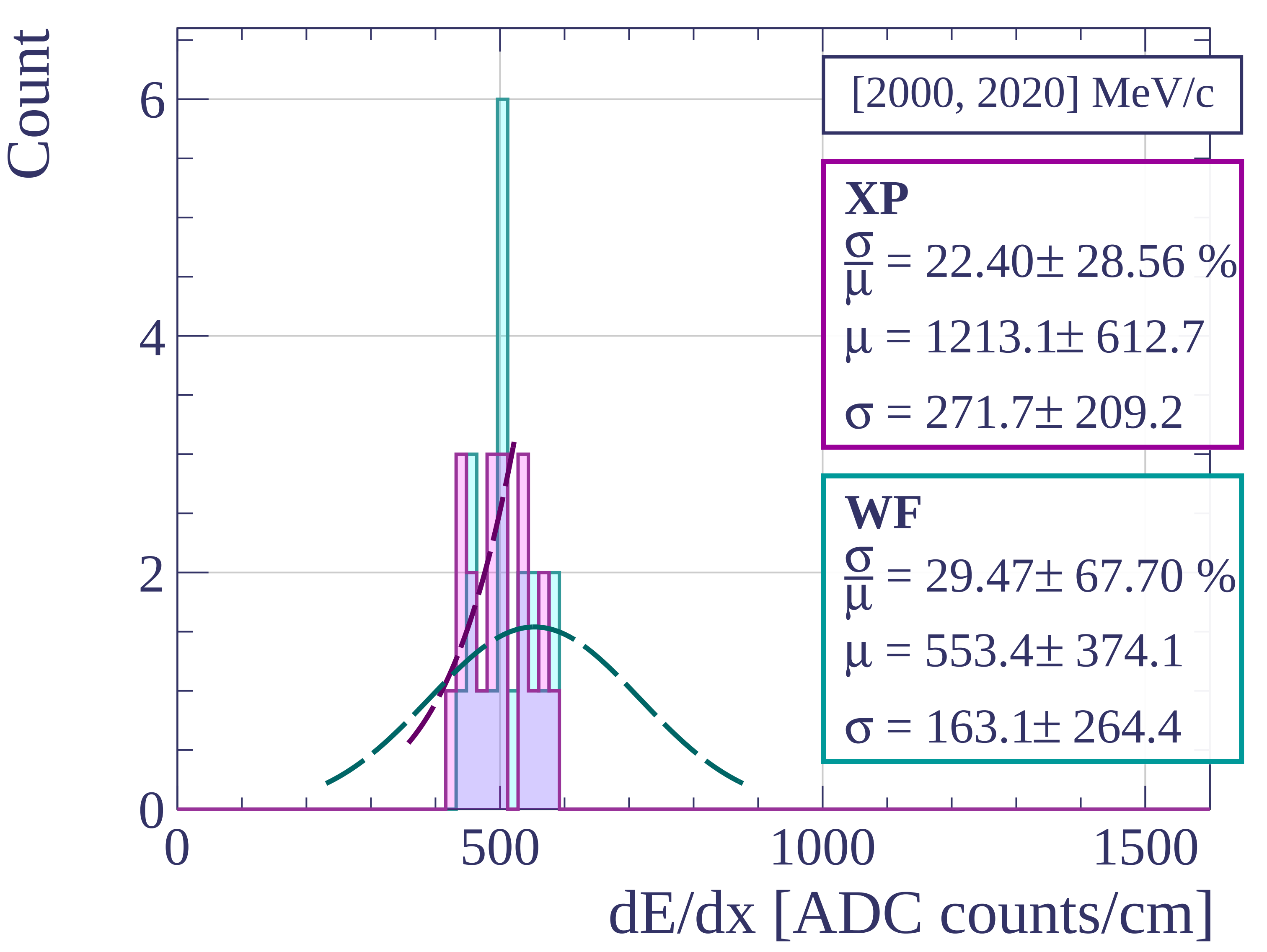


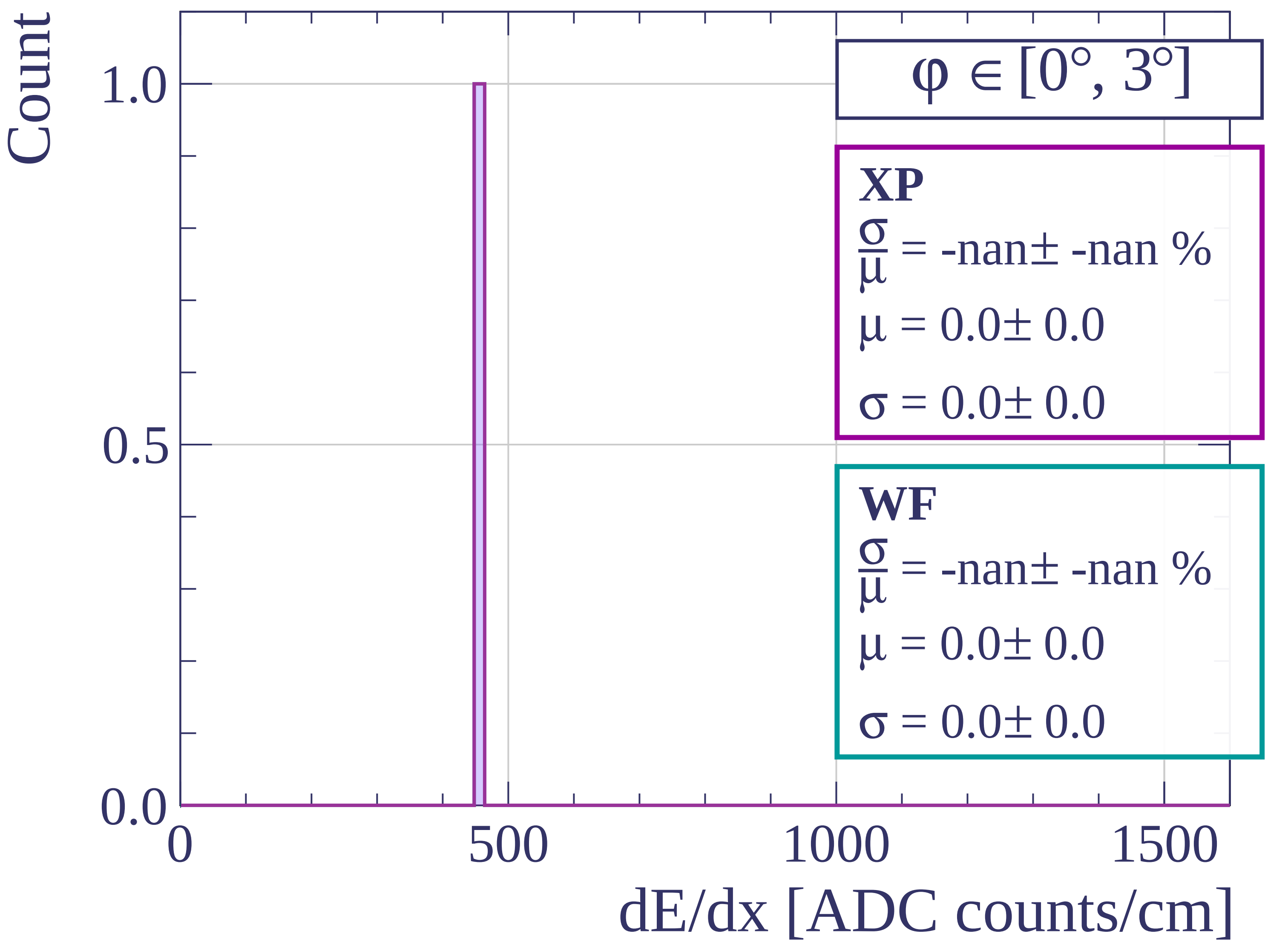
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
[1940, 1960] MeV/c
XP
σ/μ = 6.40± 1.86 %
μ = 474.1± 8.6
σ = 30.3± 8.3
WF
σ/μ = 11.01± 4.47 %
μ = 478.3± 14.2
σ = 52.7± 19.8


Count
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 3.62± 1.37 %
μ = 471.3± 5.8
σ = 17.1± 6.2
WF
$\frac{\sigma}{\mu}$ = 7.42± 2.03 %
μ = 487.4± 9.4
σ = 36.1± 9.2
0
2
4
6
8
0
500
1000
1500

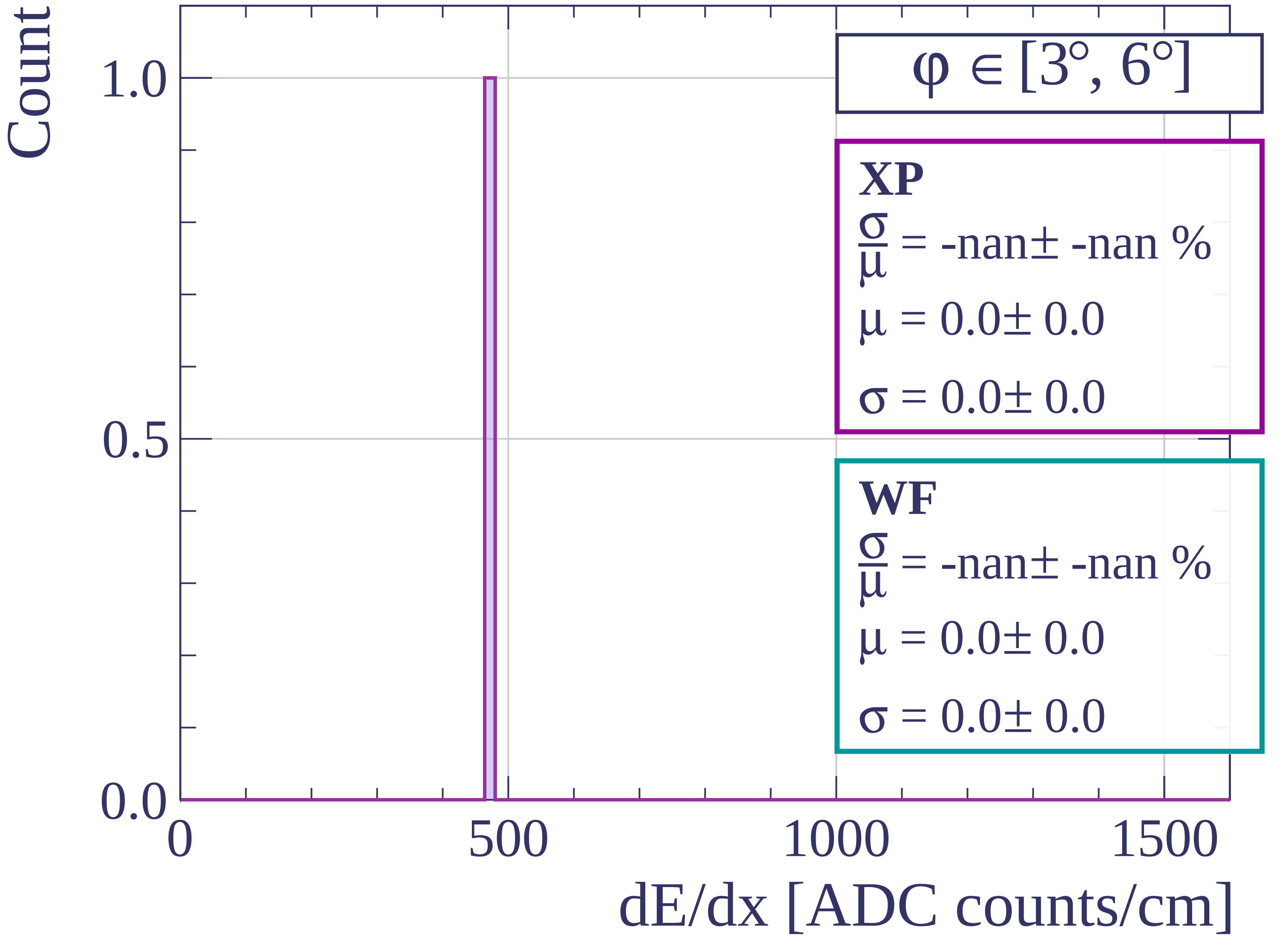


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
[1980, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 9.42± 2.71 %
μ = 491.3± 10.9
σ = 46.3± 12.3
WF
$\frac{\sigma}{\mu}$ = 7.26± 2.03 %
μ = 501.0± 8.4
σ = 36.4± 9.6

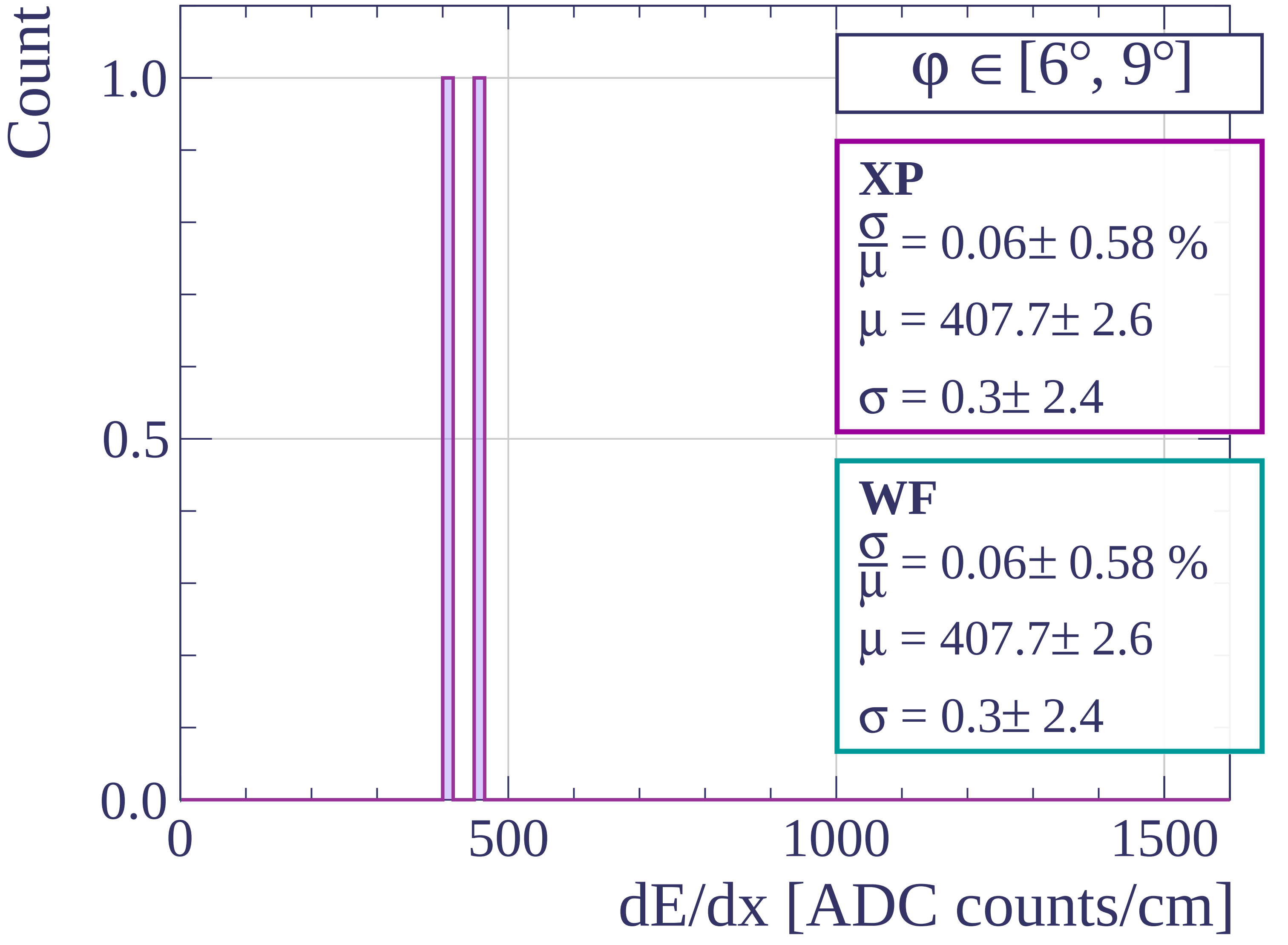

Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
2
4
6
[2000, 2020] MeV/c
XP
σ/μ = 22.40± 28.56 %
μ = 1213.1± 612.7
σ = 271.7± 209.2
WF
σ/μ = 29.47± 67.70 %
μ = 553.4± 374.1
σ = 163.1± 264.4



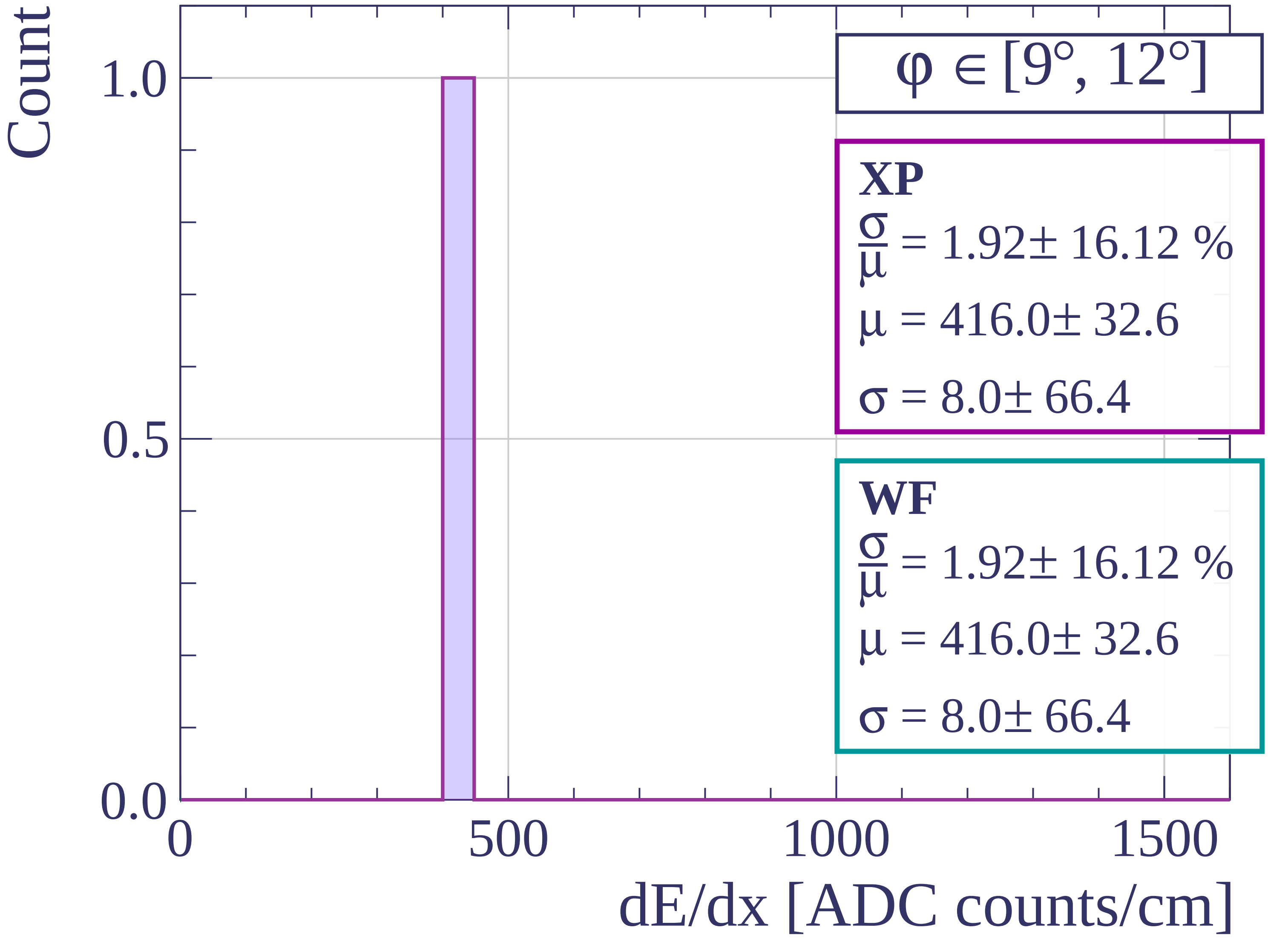
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [0°, 3°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



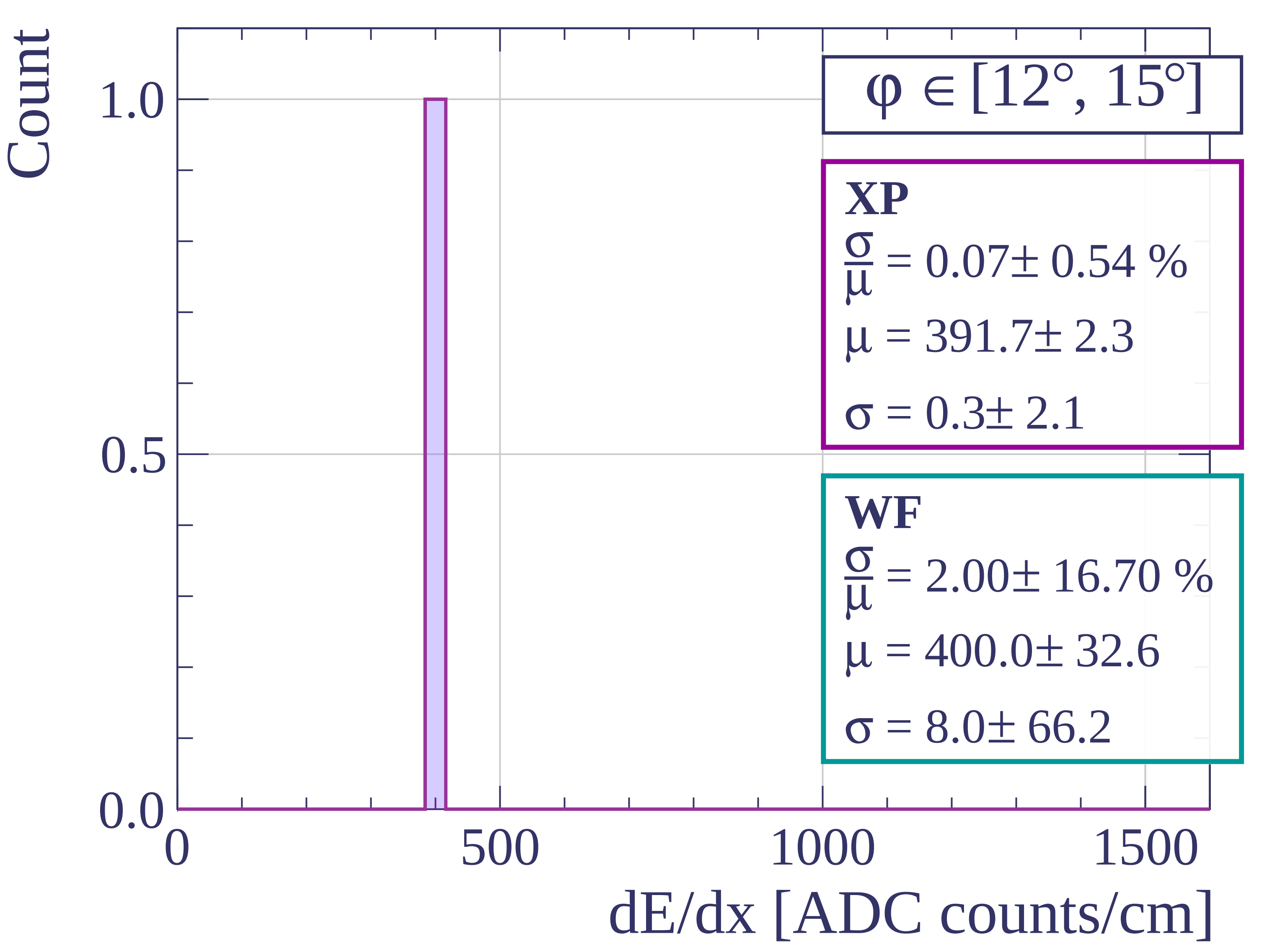
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [3°, 6°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



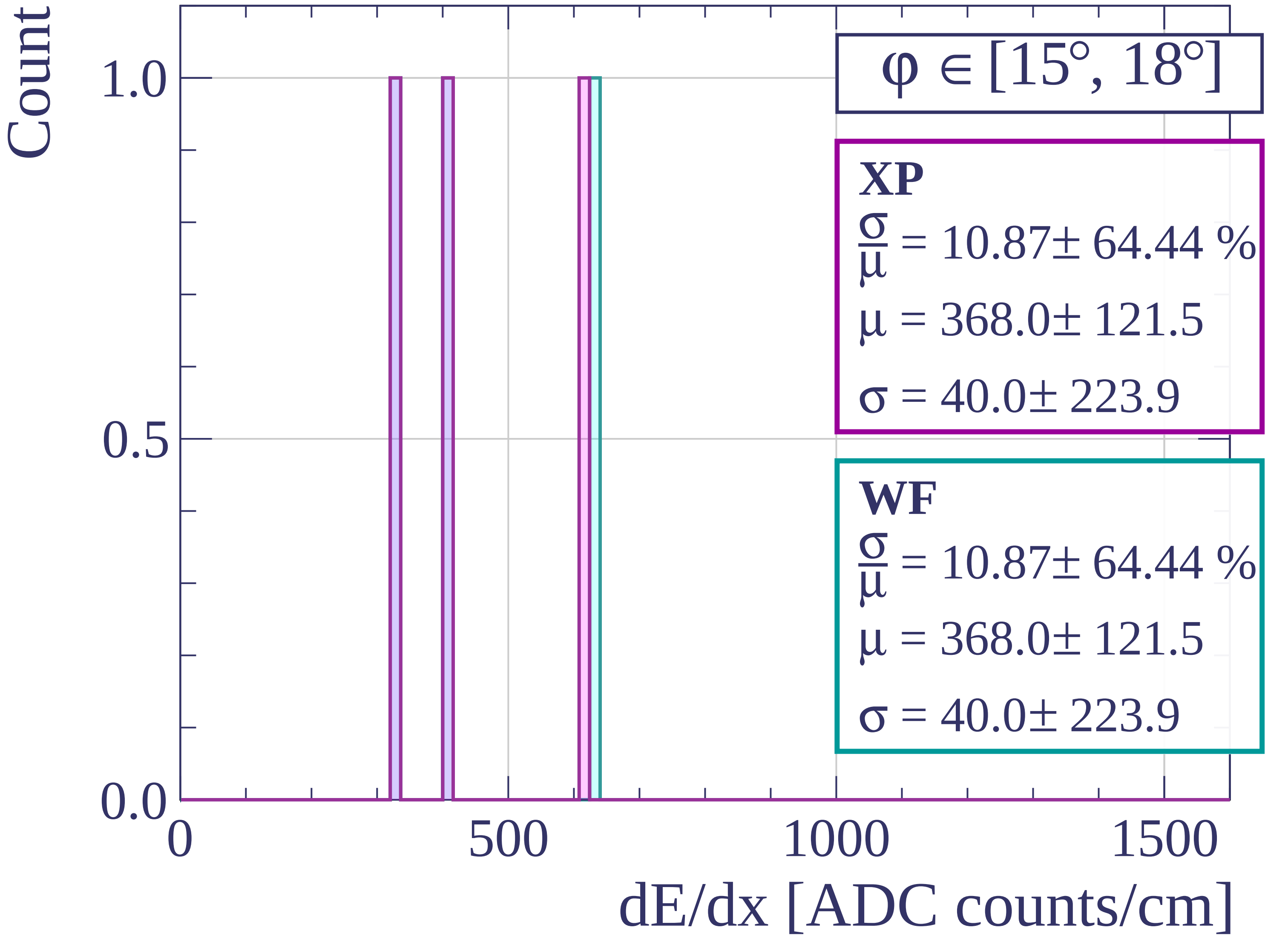
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = 0.06± 0.58 %
μ = 407.7± 2.6
σ = 0.3± 2.4
WF
σ/μ = 0.06± 0.58 %
μ = 407.7± 2.6
σ = 0.3± 2.4



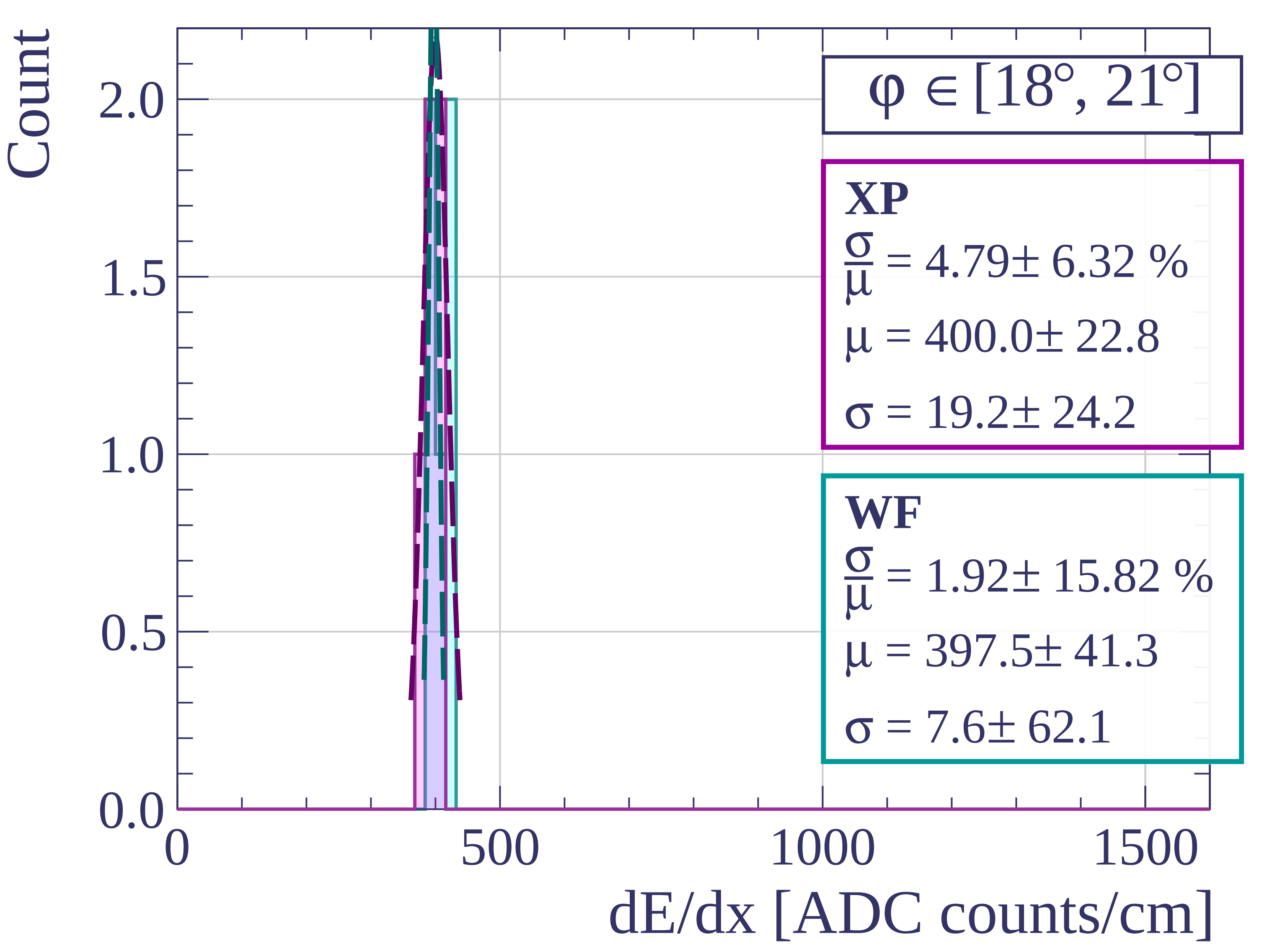
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = 1.92± 16.12 %
μ = 416.0± 32.6
σ = 8.0± 66.4
WF
σ/μ = 1.92± 16.12 %
μ = 416.0± 32.6
σ = 8.0± 66.4

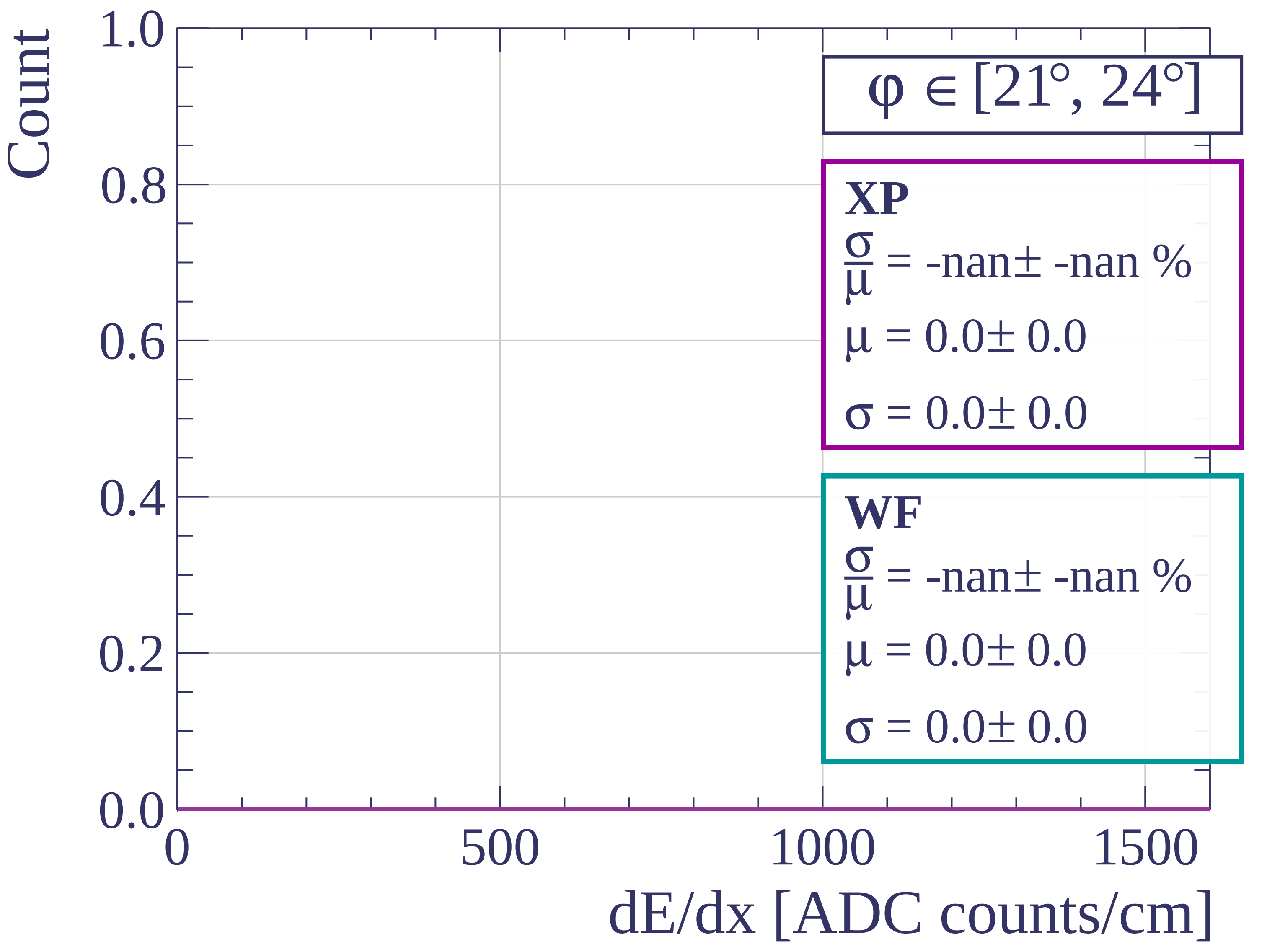


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 0.07± 0.54 %
μ = 391.7± 2.3
σ = 0.3± 2.1
WF
σ/μ = 2.00± 16.70 %
μ = 400.0± 32.6
σ = 8.0± 66.2

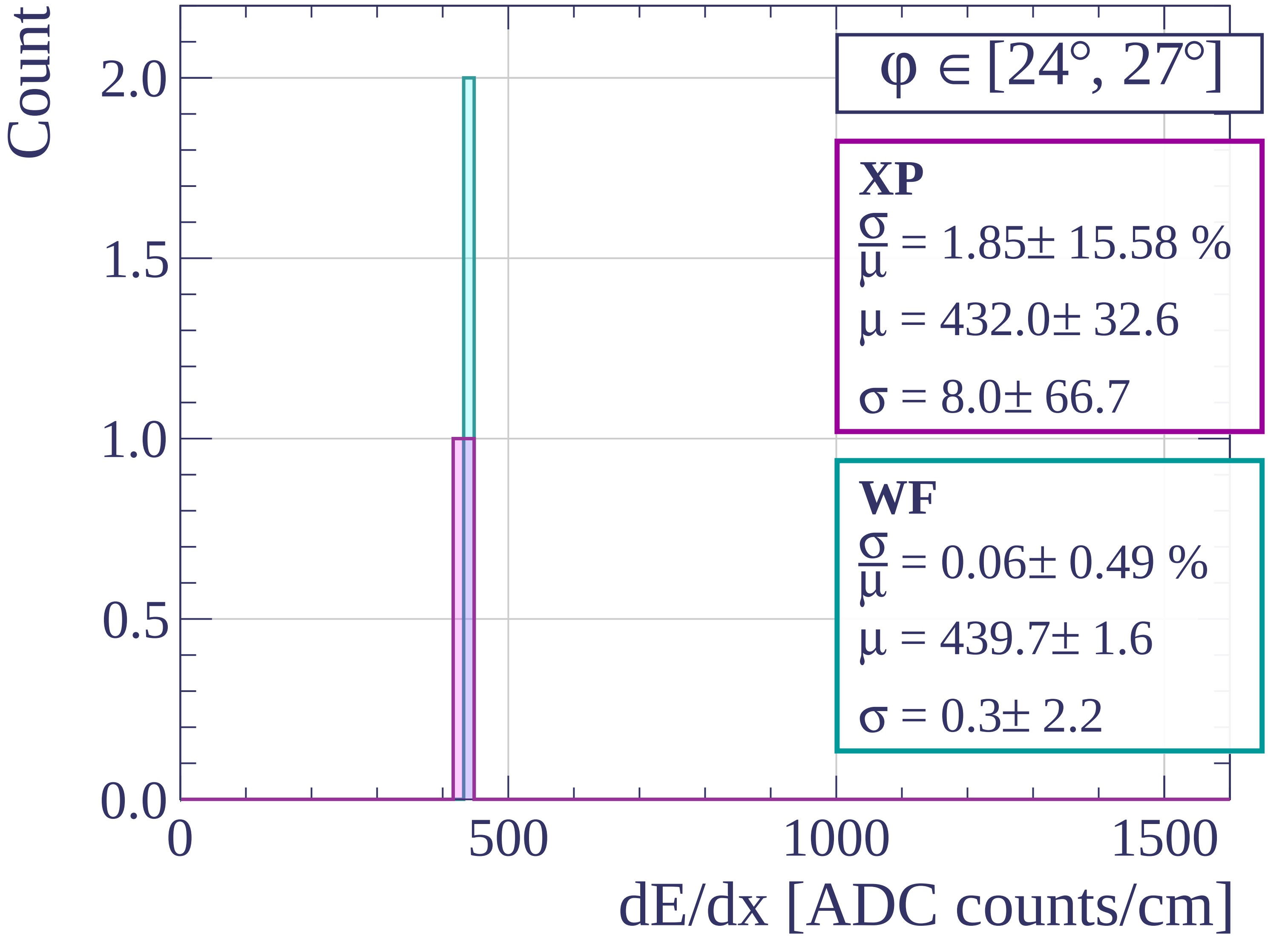


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 10.87± 64.44 %
μ = 368.0± 121.5
σ = 40.0± 223.9
WF
σ/μ = 10.87± 64.44 %
μ = 368.0± 121.5
σ = 40.0± 223.9

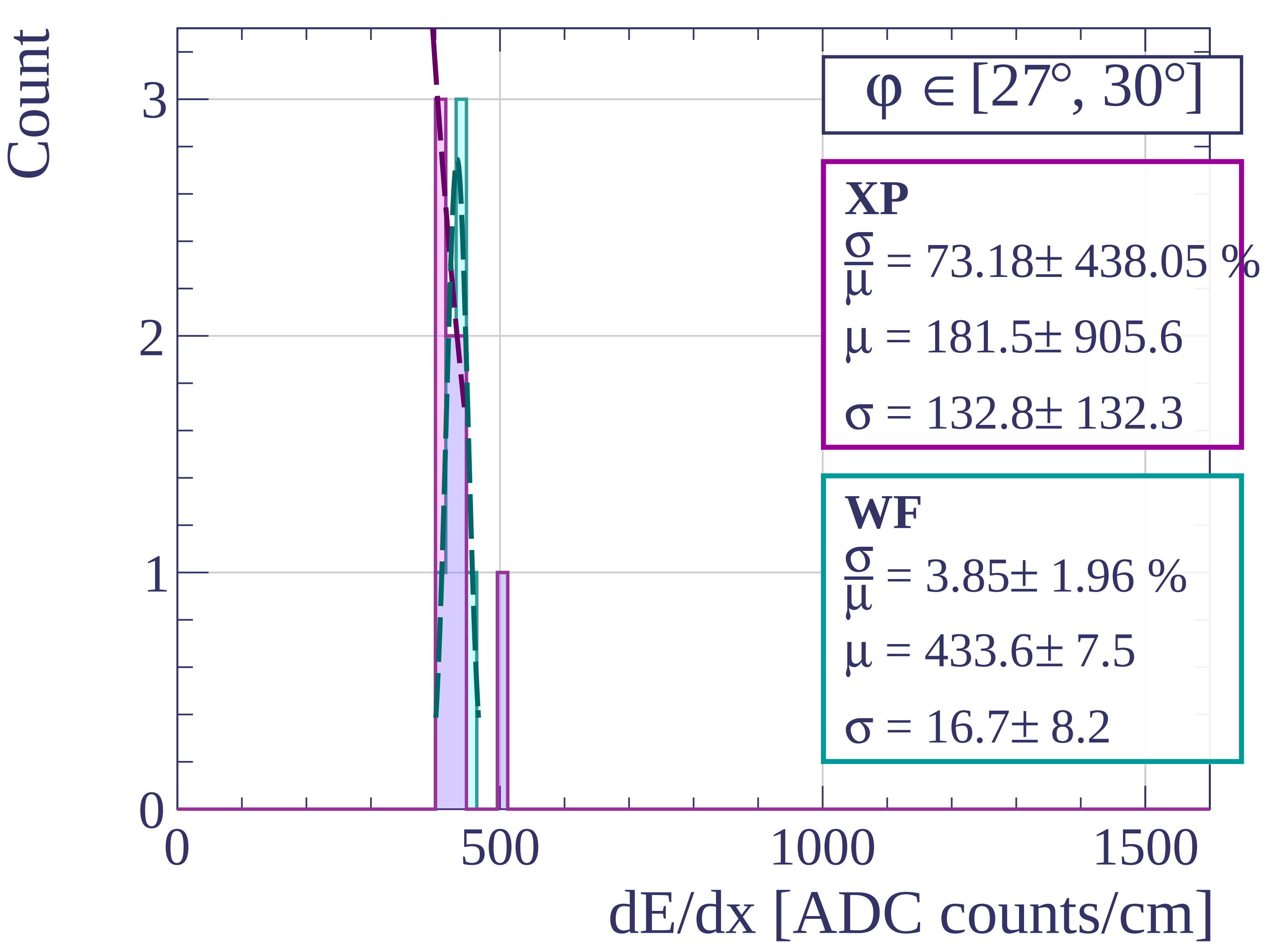

Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [18°, 21°]
XP
σ/μ = 4.79± 6.32 %
μ = 400.0± 22.8
σ = 19.2± 24.2
WF
σ/μ = 1.92± 15.82 %
μ = 397.5± 41.3
σ = 7.6± 62.1



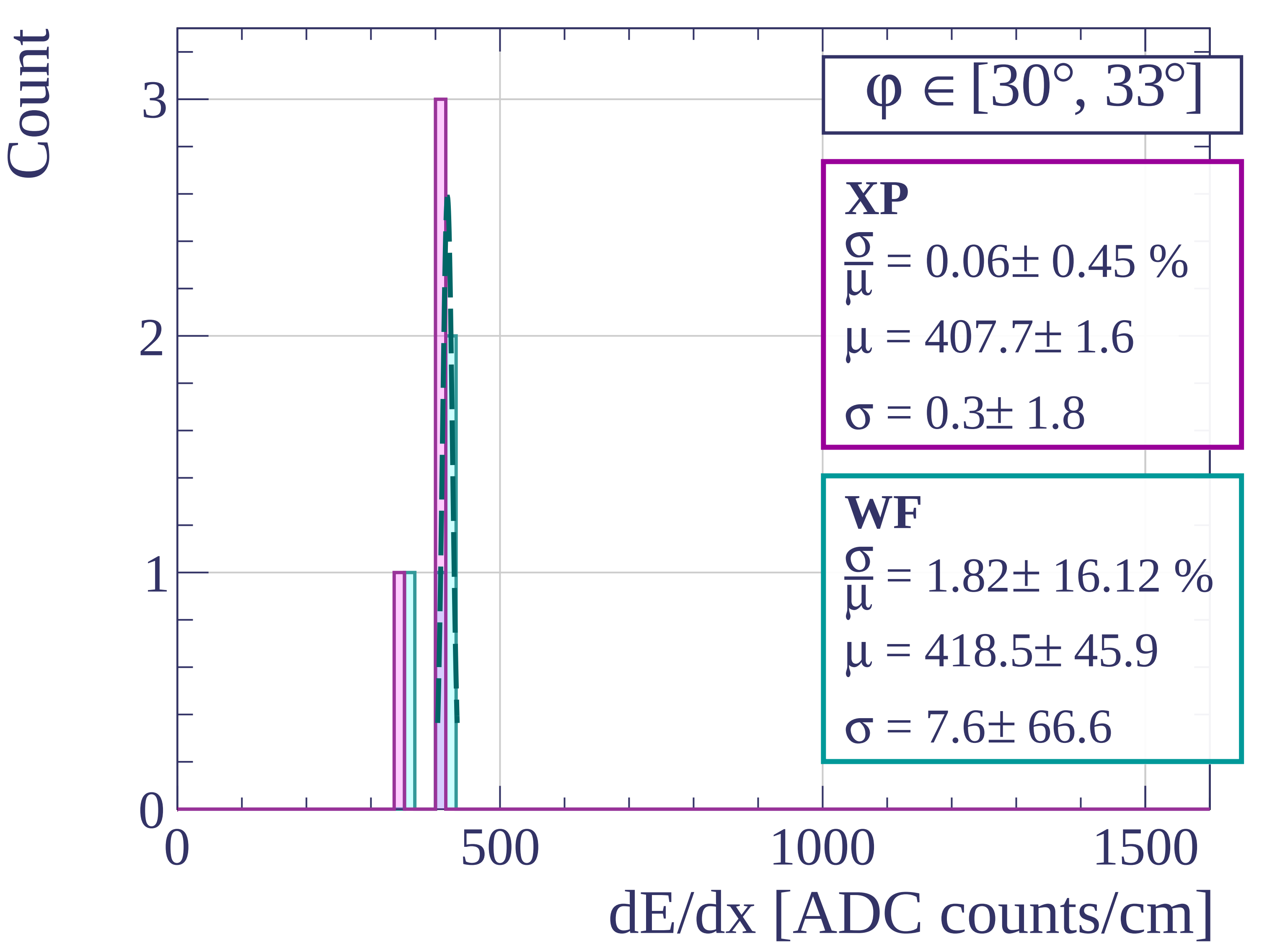
Count
dE/dx [ADC counts/cm]
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
$\varphi \in [21°, 24°]$
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



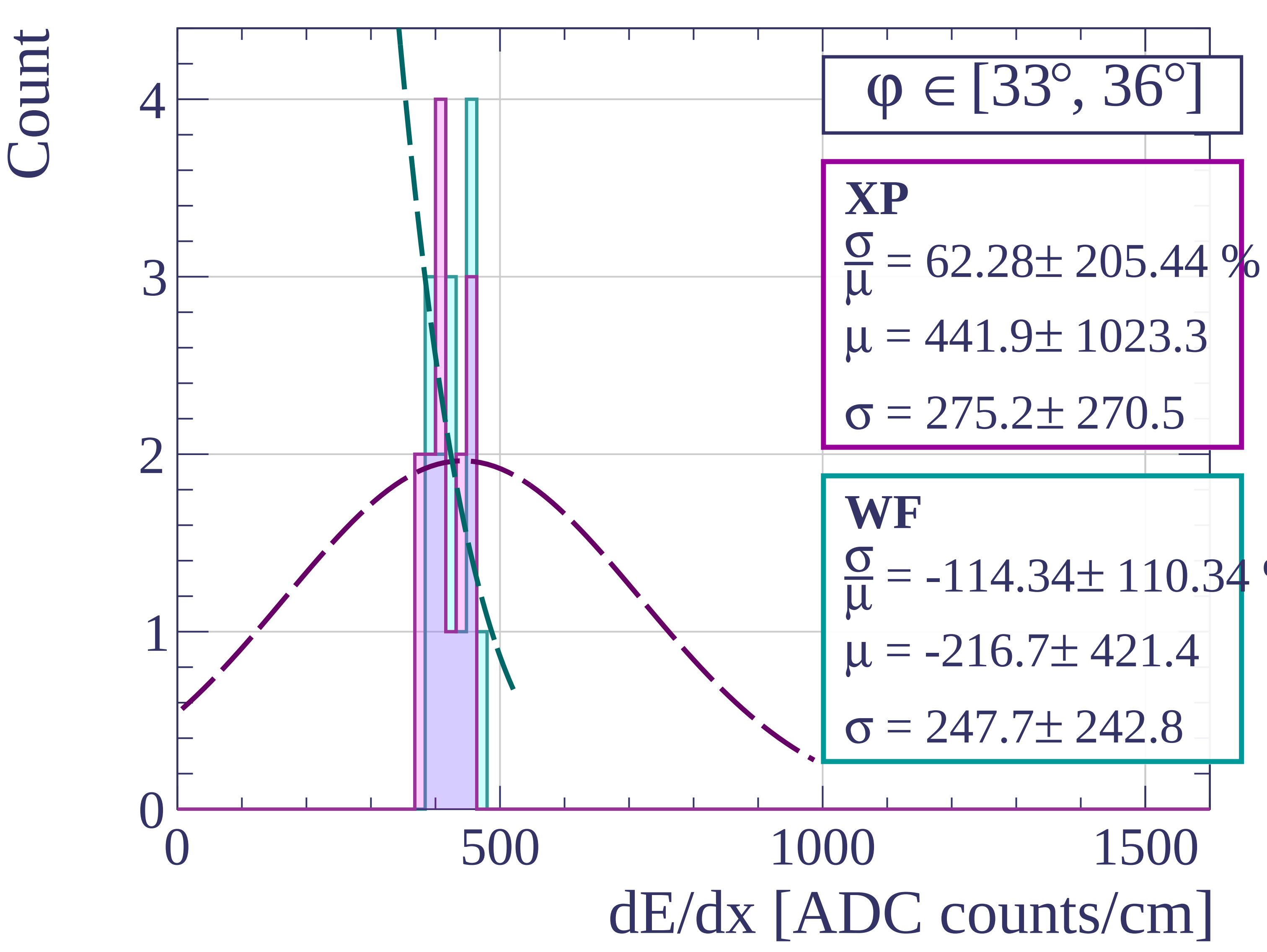
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 1.85± 15.58 %
μ = 432.0± 32.6
σ = 8.0± 66.7
WF
σ/μ = 0.06± 0.49 %
μ = 439.7± 1.6
σ = 0.3± 2.2



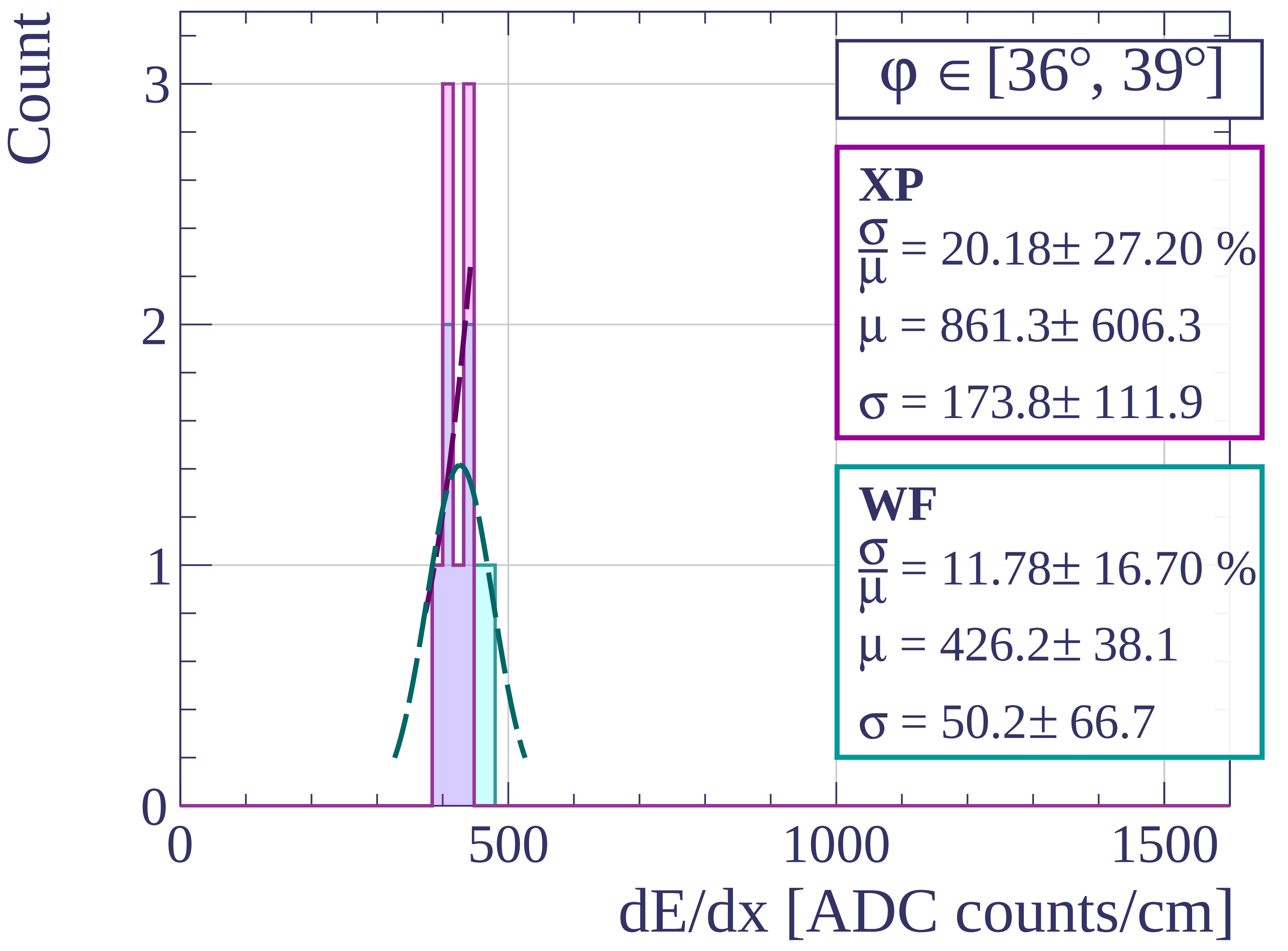
Count
dE/dx [ADC counts/cm]
0
1
2
3
0
500
1000
1500
φ ∈ [27°, 30°]
XP
σ/μ = 73.18± 438.05 %
μ = 181.5± 905.6
σ = 132.8± 132.3
WF
σ/μ = 3.85± 1.96 %
μ = 433.6± 7.5
σ = 16.7± 8.2



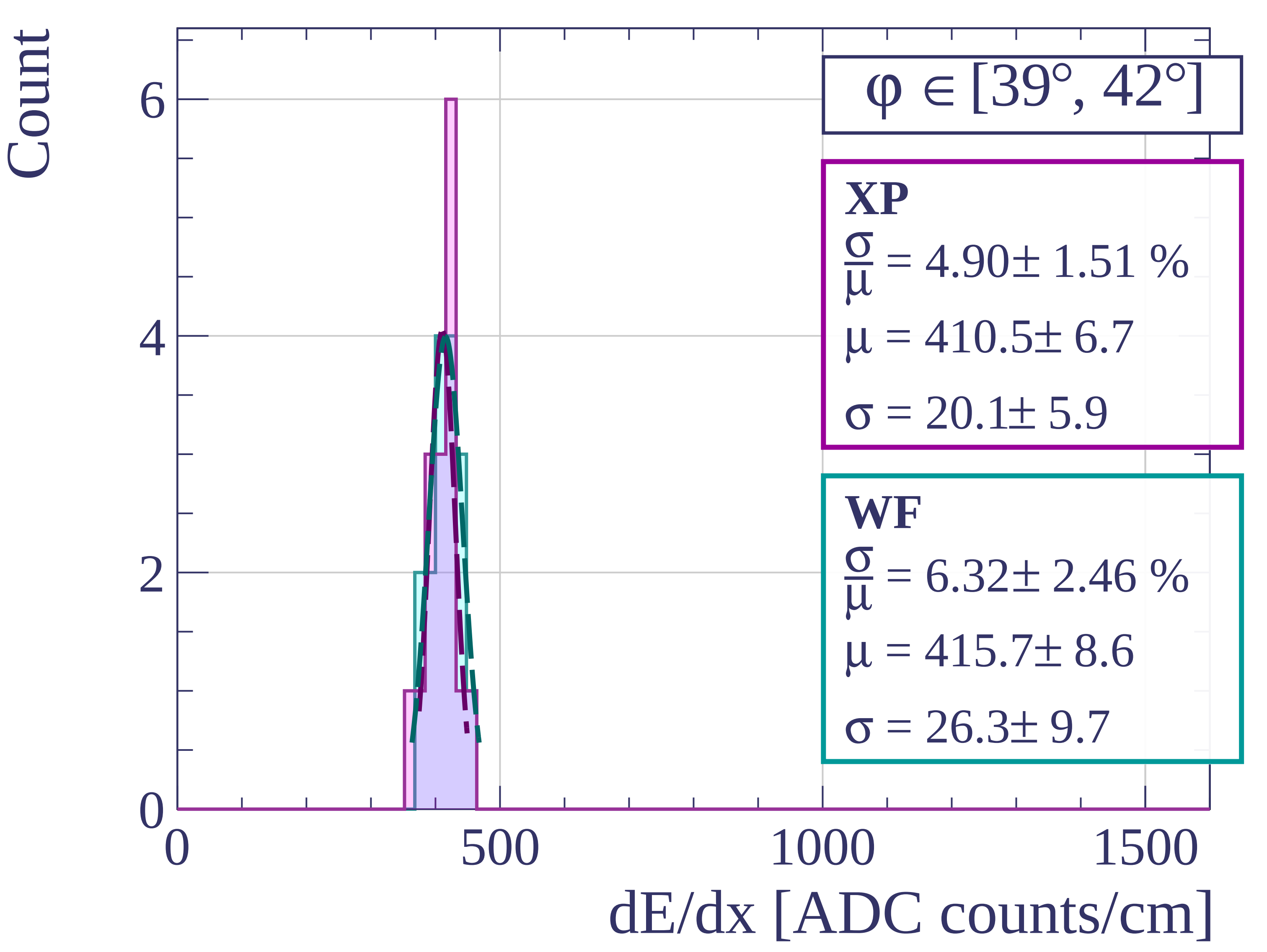
Count
dE/dx [ADC counts/cm]
φ ∈ [30°, 33°]
XP
σ/μ = 0.06± 0.45 %
μ = 407.7± 1.6
σ = 0.3± 1.8
WF
σ/μ = 1.82± 16.12 %
μ = 418.5± 45.9
σ = 7.6± 66.6
0
1
2
3
0
500
1000
1500

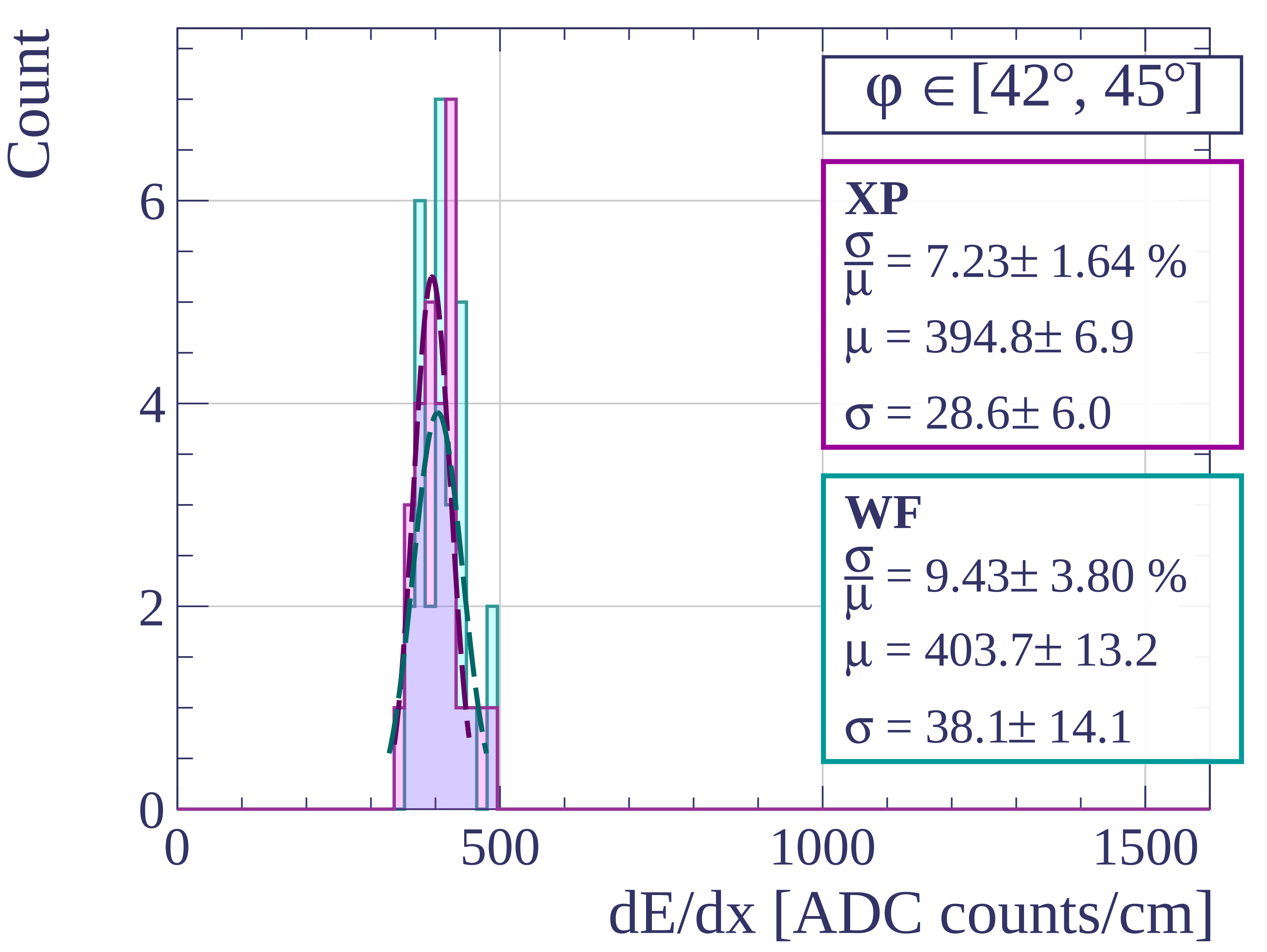


Count
dE/dx [ADC counts/cm]
0
1
2
3
4
0
500
1000
1500
φ ∈ [33°, 36°]
XP
σ/μ = 62.28± 205.44 %
μ = 441.9± 1023.3
σ = 275.2± 270.5
WF
σ/μ = -114.34± 110.34
μ = -216.7± 421.4
σ = 247.7± 242.8

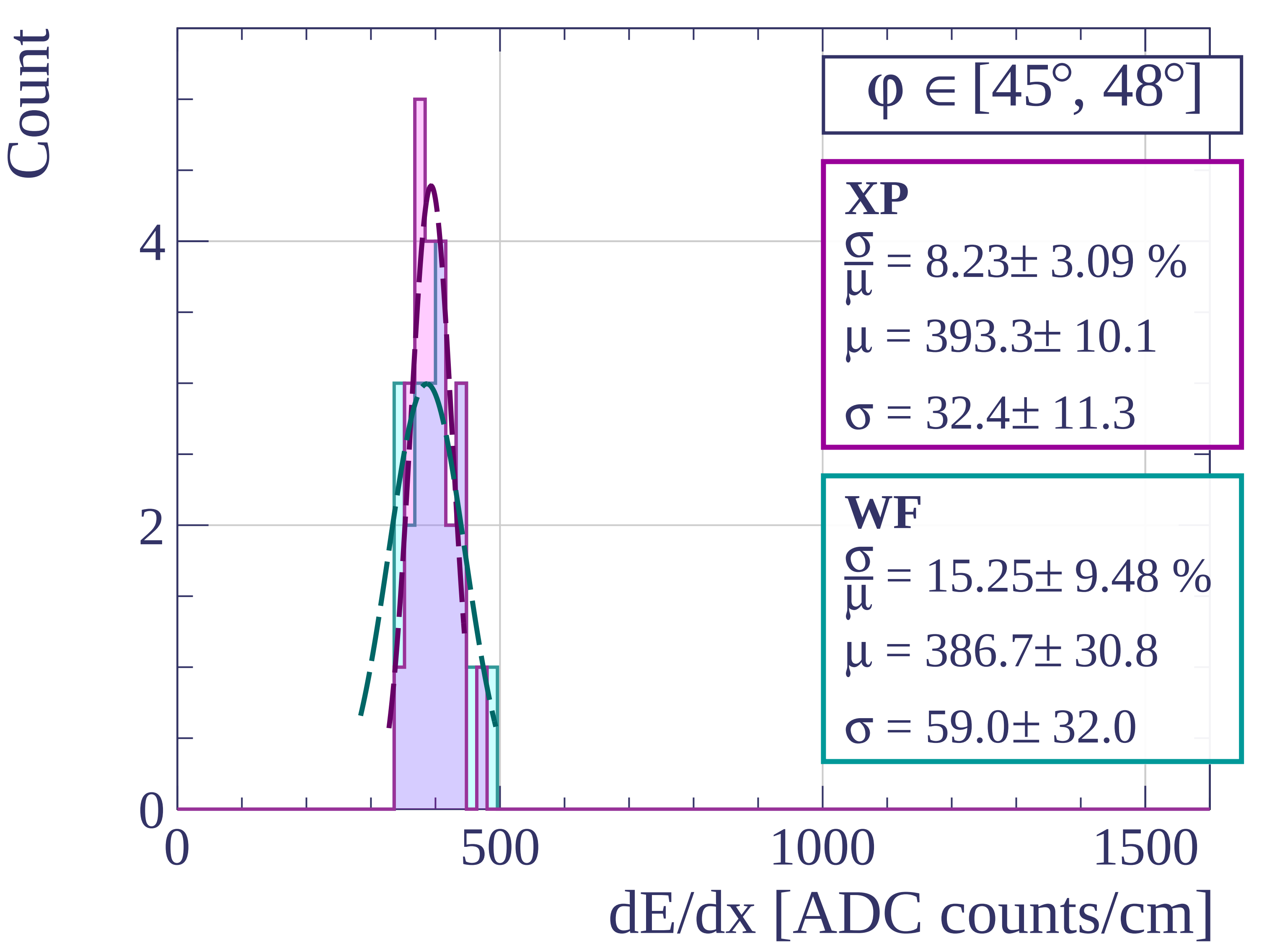


Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 20.18± 27.20 %
μ = 861.3± 606.3
σ = 173.8± 111.9
WF
σ/μ = 11.78± 16.70 %
μ = 426.2± 38.1
σ = 50.2± 66.7

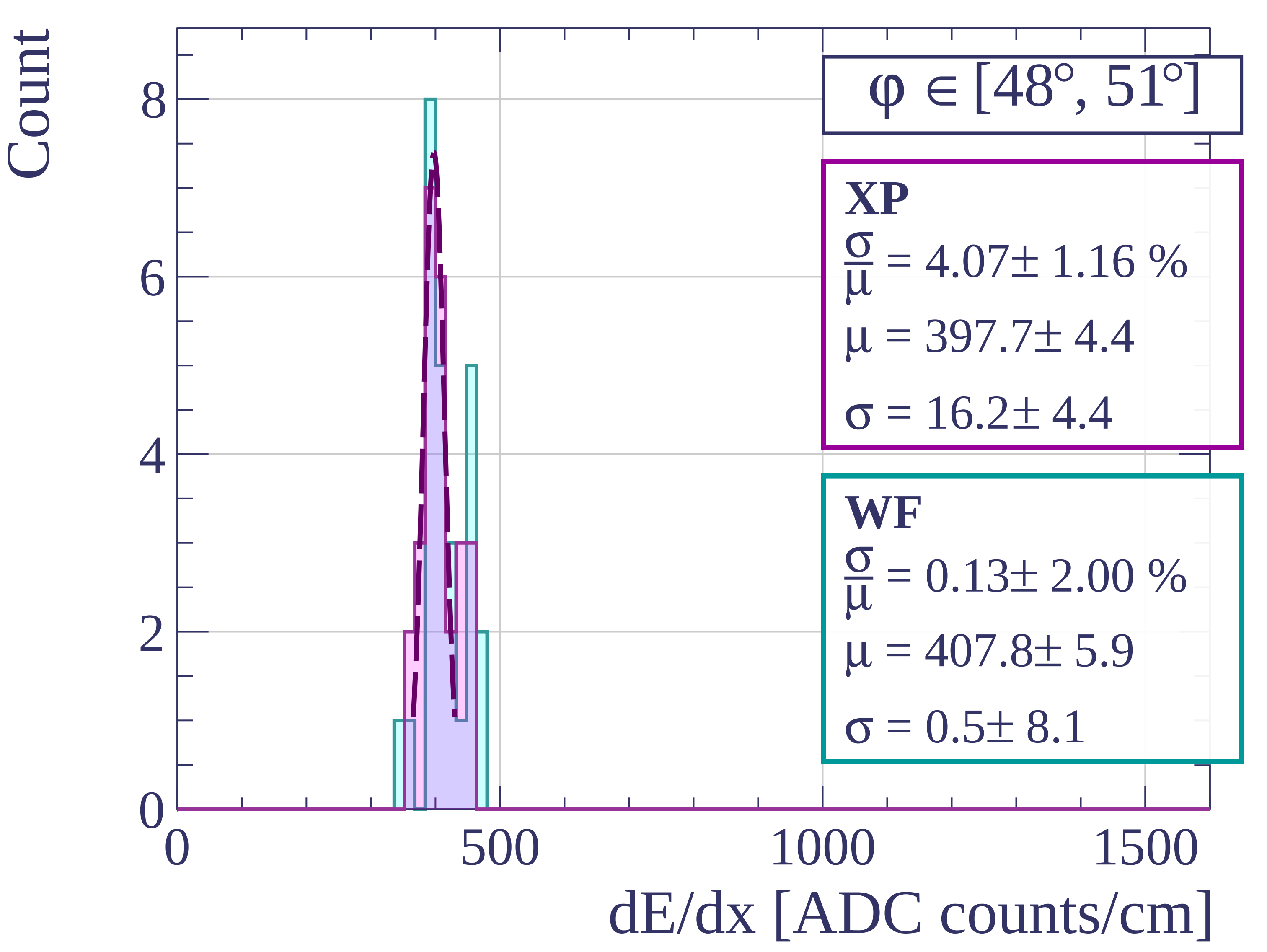

Count
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
σ/μ = 4.90± 1.51 %
μ = 410.5± 6.7
σ = 20.1± 5.9
WF
σ/μ = 6.32± 2.46 %
μ = 415.7± 8.6
σ = 26.3± 9.7



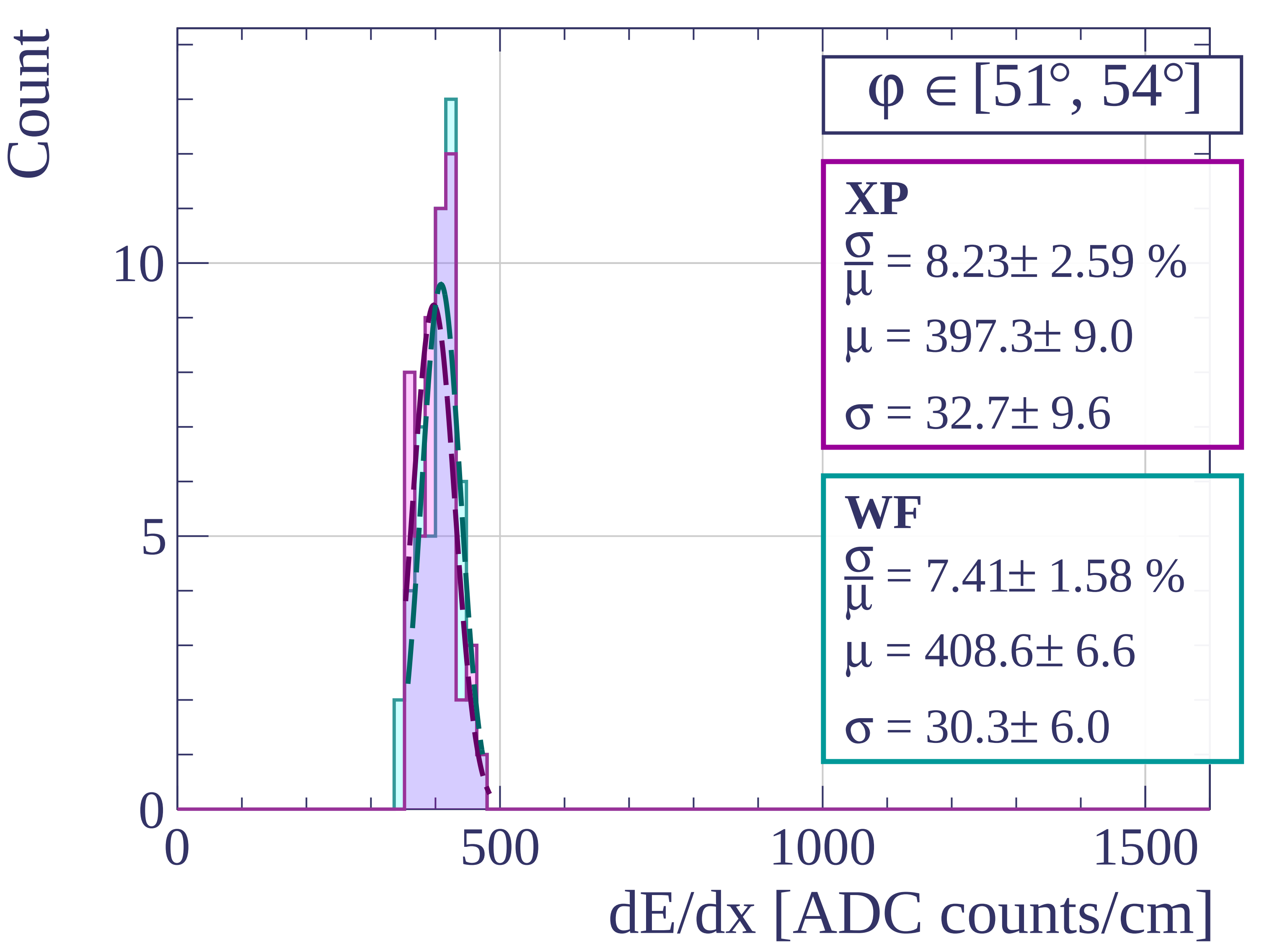
Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
σ/μ = 7.23± 1.64 %
μ = 394.8± 6.9
σ = 28.6± 6.0
WF
σ/μ = 9.43± 3.80 %
μ = 403.7± 13.2
σ = 38.1± 14.1



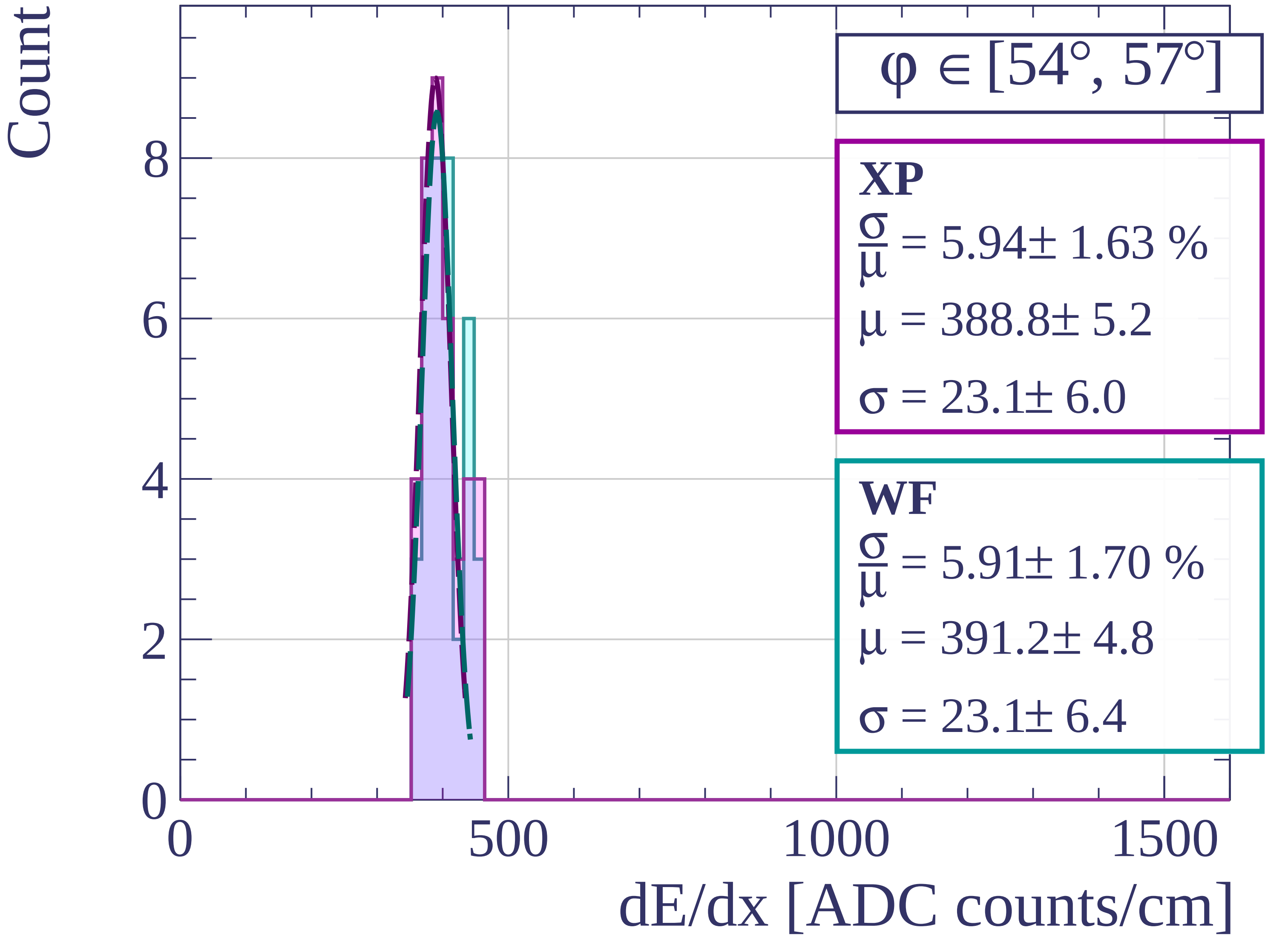
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
φ ∈ [45°, 48°]
XP
σ/μ = 8.23± 3.09 %
μ = 393.3± 10.1
σ = 32.4± 11.3
WF
σ/μ = 15.25± 9.48 %
μ = 386.7± 30.8
σ = 59.0± 32.0



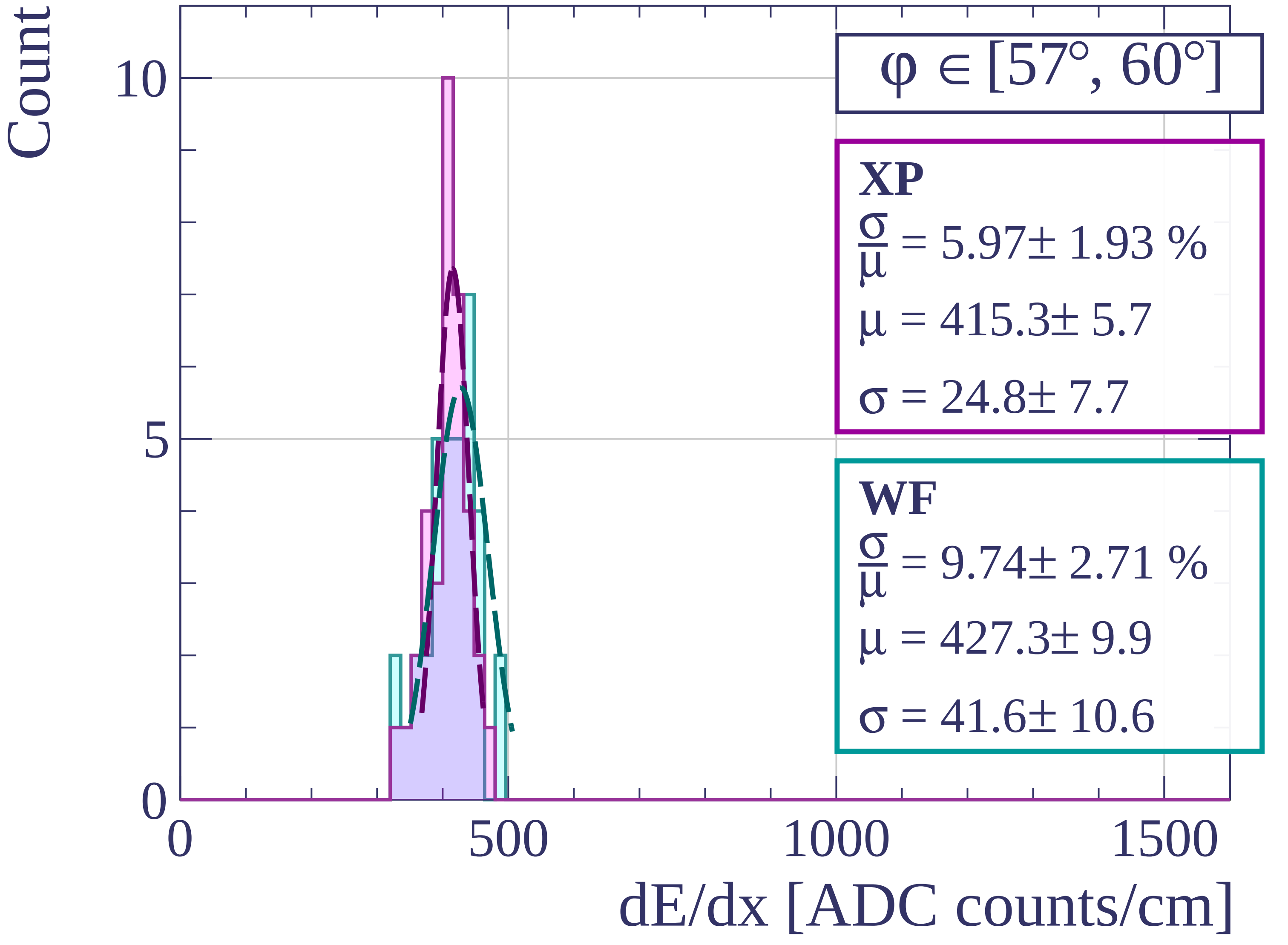
Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [48°, 51°]
XP
σ/μ = 4.07± 1.16 %
μ = 397.7± 4.4
σ = 16.2± 4.4
WF
σ/μ = 0.13± 2.00 %
μ = 407.8± 5.9
σ = 0.5± 8.1



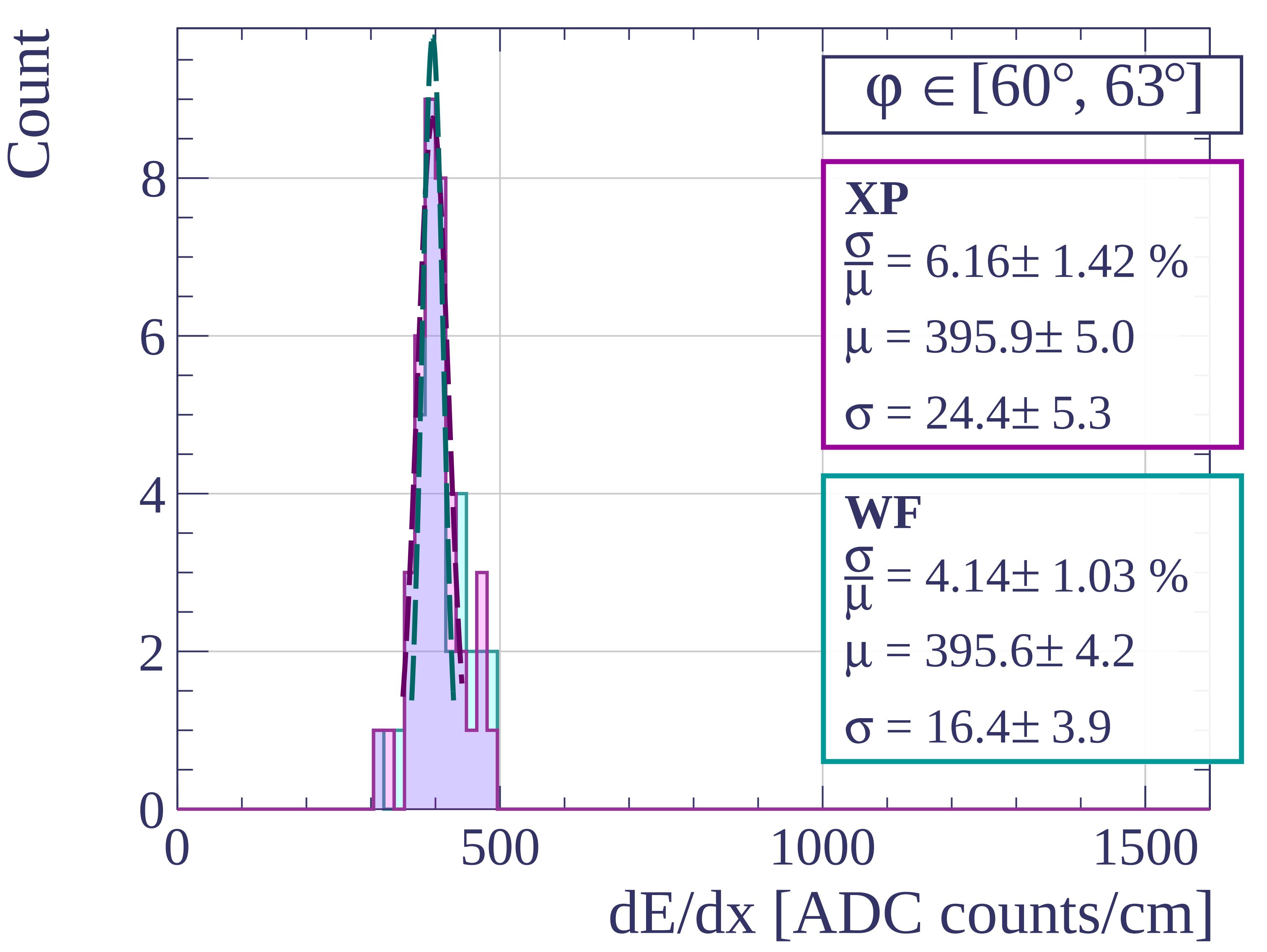
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
φ ∈ [51°, 54°]
XP
σ/μ = 8.23± 2.59 %
μ = 397.3± 9.0
σ = 32.7± 9.6
WF
σ/μ = 7.41± 1.58 %
μ = 408.6± 6.6
σ = 30.3± 6.0



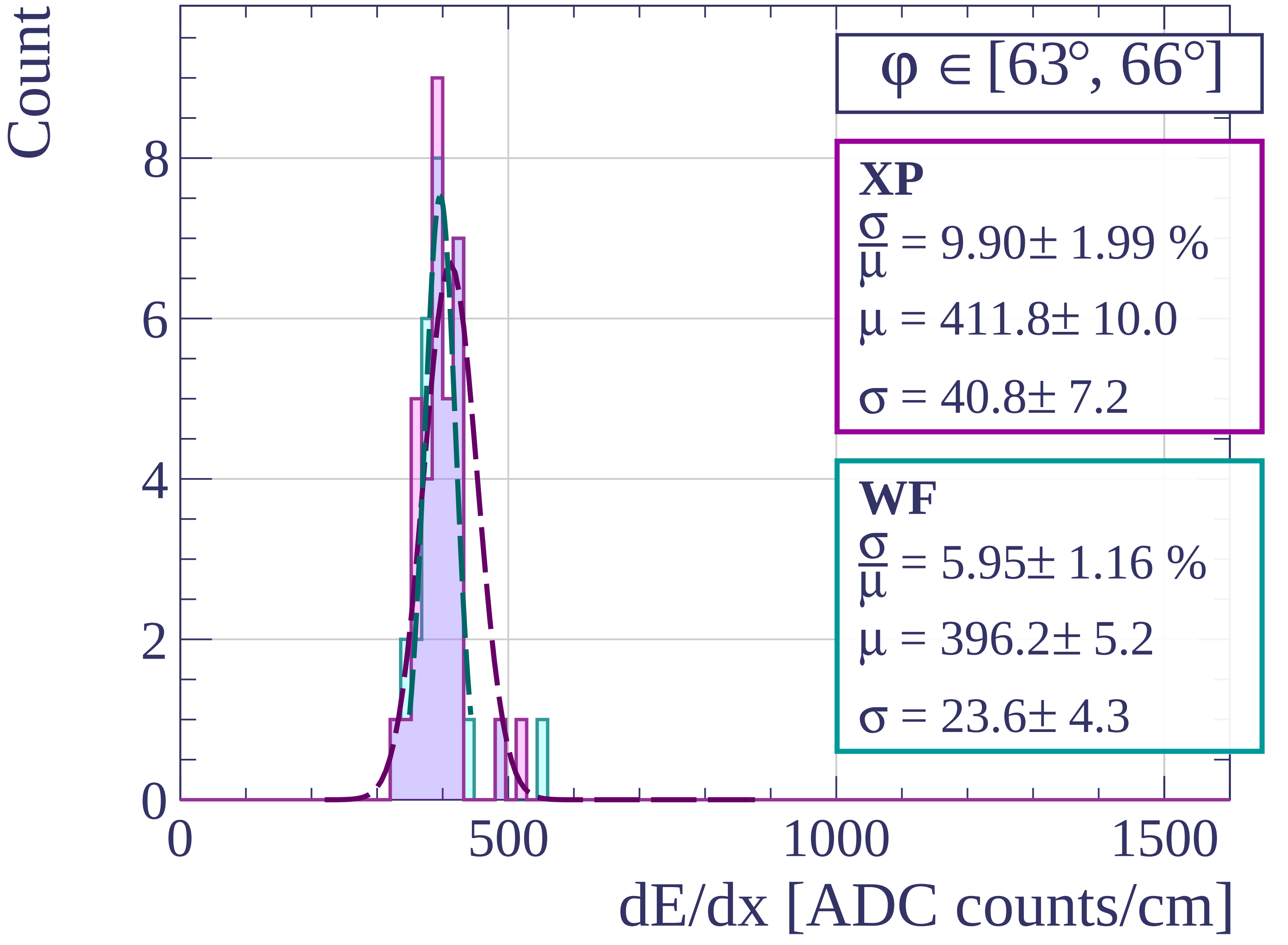
Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
φ ∈ [54°, 57°]
XP
σ/μ = 5.94± 1.63 %
μ = 388.8± 5.2
σ = 23.1± 6.0
WF
σ/μ = 5.91± 1.70 %
μ = 391.2± 4.8
σ = 23.1± 6.4

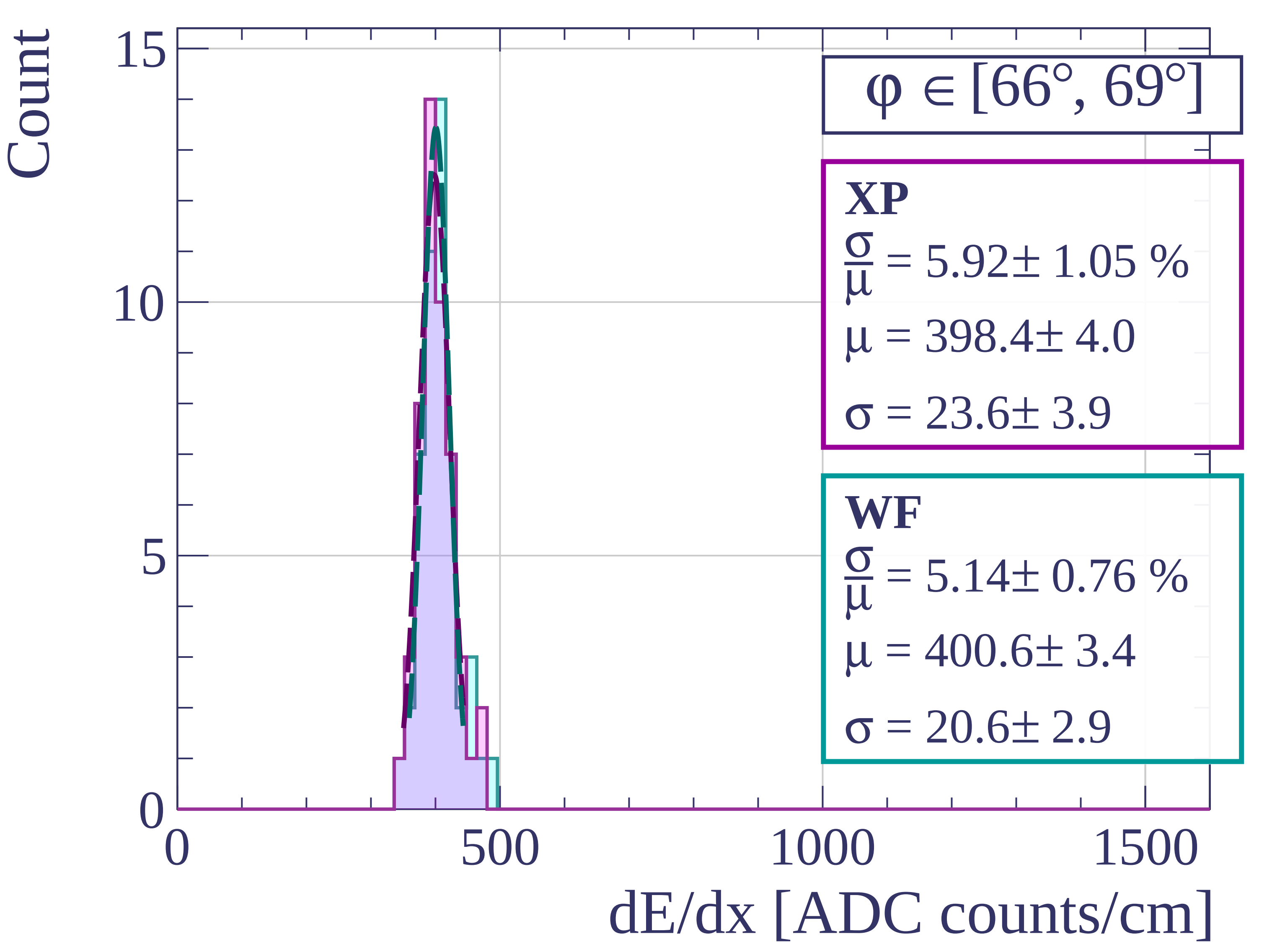


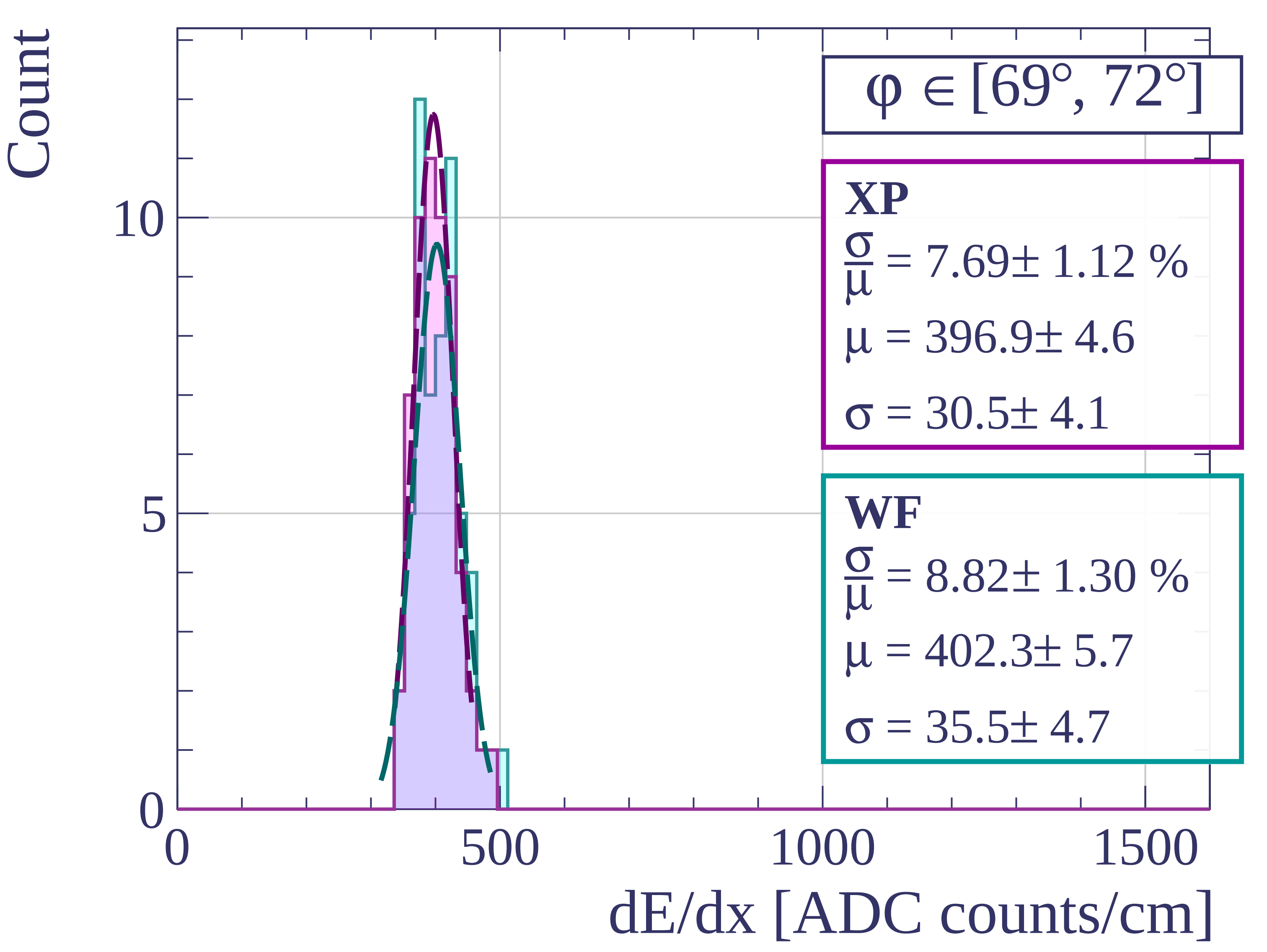
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [57°, 60°]
XP
σ/μ = 5.97± 1.93 %
μ = 415.3± 5.7
σ = 24.8± 7.7
WF
σ/μ = 9.74± 2.71 %
μ = 427.3± 9.9
σ = 41.6± 10.6



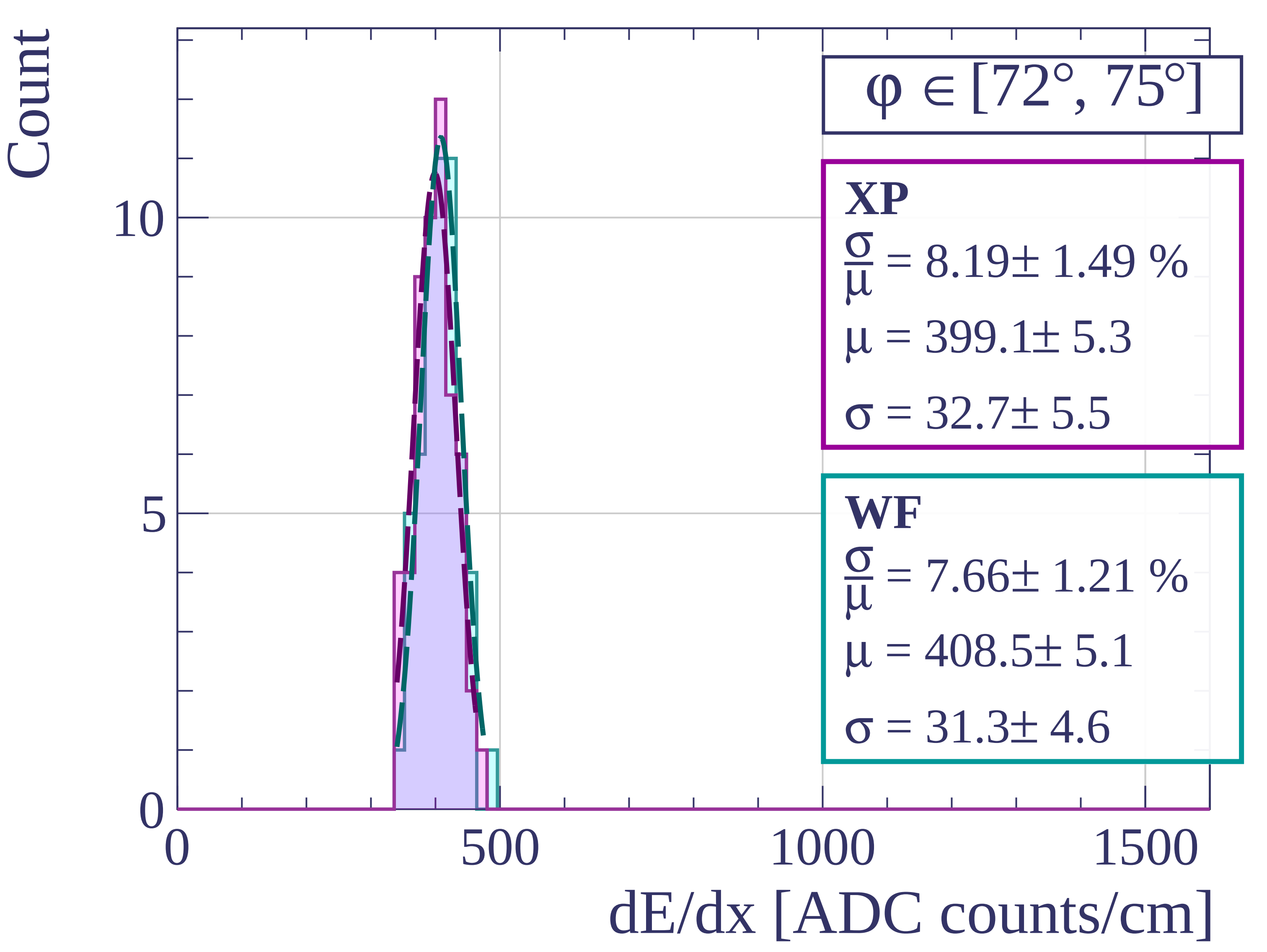
Count
0
2
4
6
8
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
σ/μ = 6.16± 1.42 %
μ = 395.9± 5.0
σ = 24.4± 5.3
WF
σ/μ = 4.14± 1.03 %
μ = 395.6± 4.2
σ = 16.4± 3.9

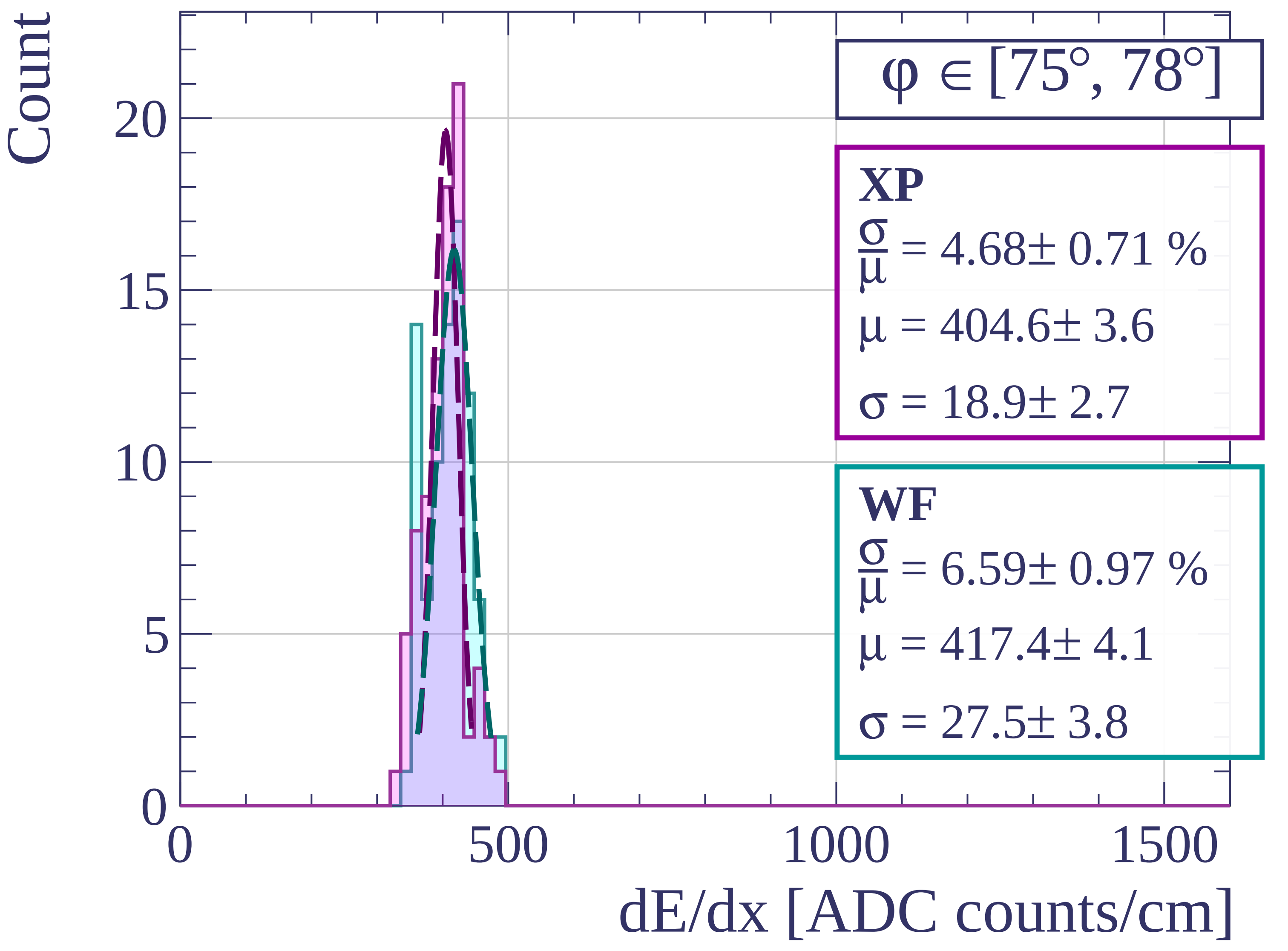

Count
dE/dx [ADC counts/cm]
0
2
4
6
8
0
500
1000
1500
φ ∈ [63°, 66°]
XP
σ/μ = 9.90± 1.99 %
μ = 411.8± 10.0
σ = 40.8± 7.2
WF
σ/μ = 5.95± 1.16 %
μ = 396.2± 5.2
σ = 23.6± 4.3


Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [66°, 69°]
XP
σ/μ = 5.92± 1.05 %
μ = 398.4± 4.0
σ = 23.6± 3.9
WF
σ/μ = 5.14± 0.76 %
μ = 400.6± 3.4
σ = 20.6± 2.9

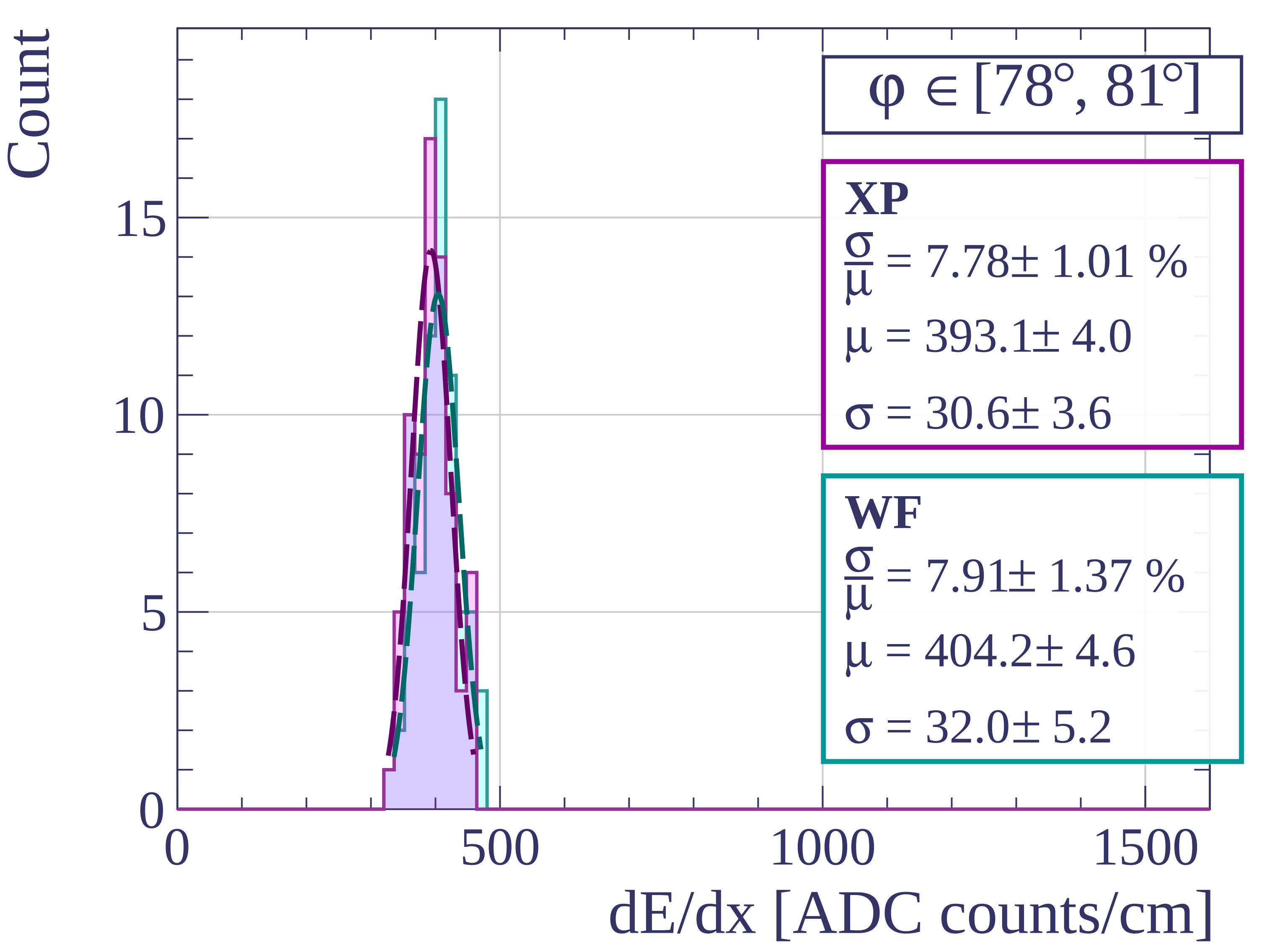


Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
φ ∈ [69°, 72°]
XP
σ/μ = 7.69± 1.12 %
μ = 396.9± 4.6
σ = 30.5± 4.1
WF
σ/μ = 8.82± 1.30 %
μ = 402.3± 5.7
σ = 35.5± 4.7

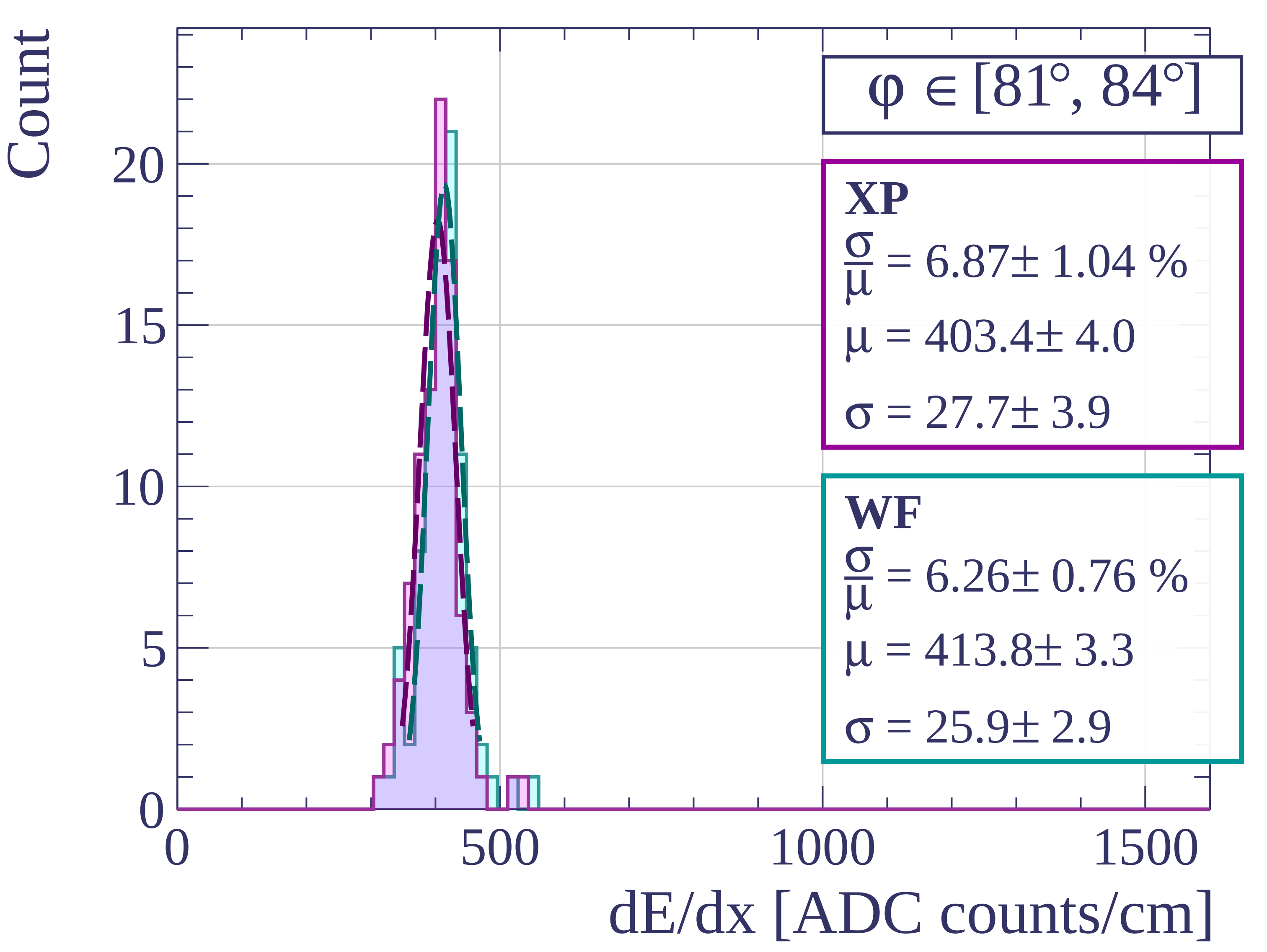

Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [72°, 75°]
XP
$\frac{\sigma}{\mu}$ = 8.19± 1.49 %
μ = 399.1± 5.3
σ = 32.7± 5.5
WF
$\frac{\sigma}{\mu}$ = 7.66± 1.21 %
μ = 408.5± 5.1
σ = 31.3± 4.6



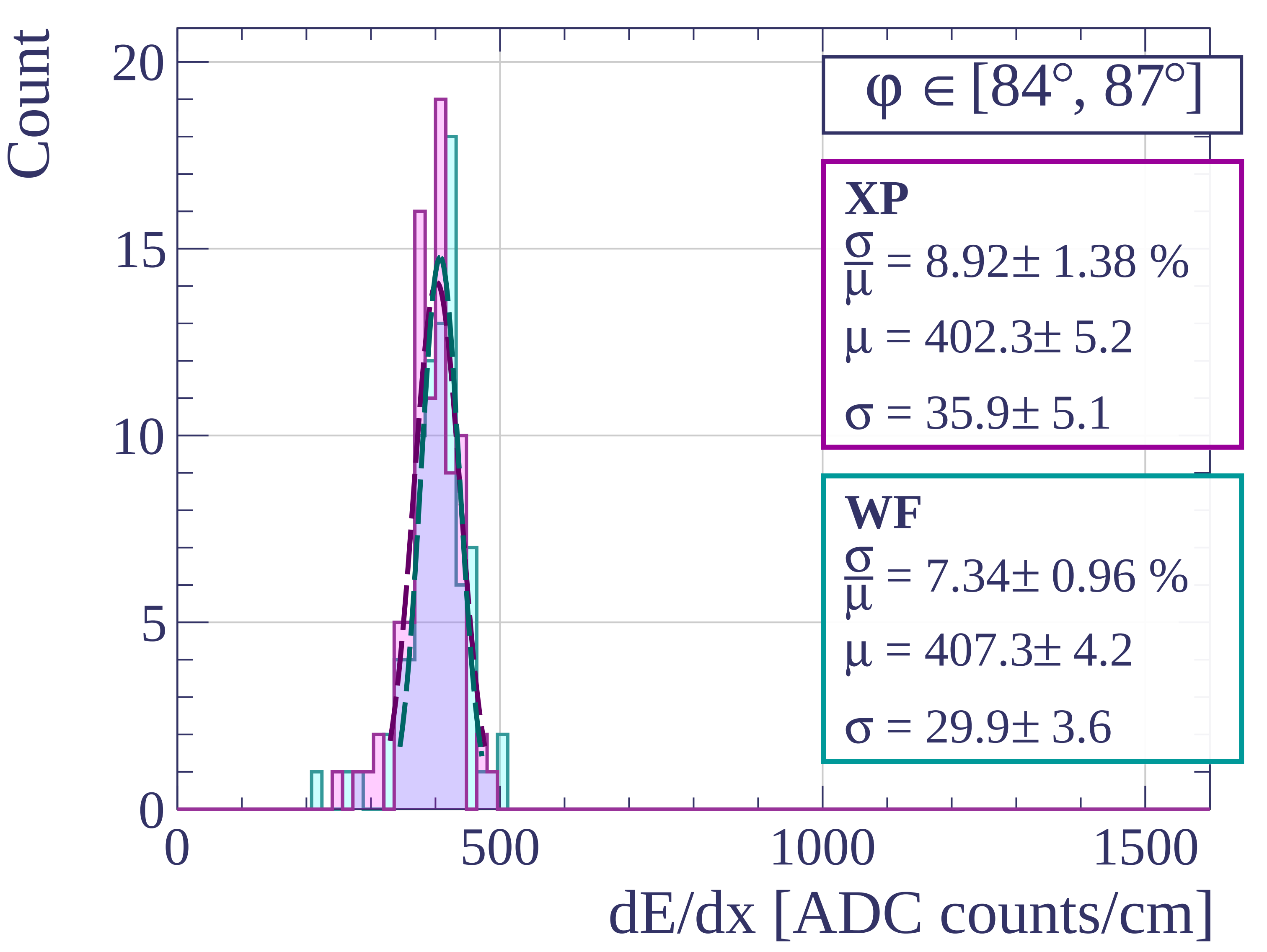
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
5
10
15
20
φ ∈ [75°, 78°]
XP
σ/μ = 4.68± 0.71 %
μ = 404.6± 3.6
σ = 18.9± 2.7
WF
σ/μ = 6.59± 0.97 %
μ = 417.4± 4.1
σ = 27.5± 3.8



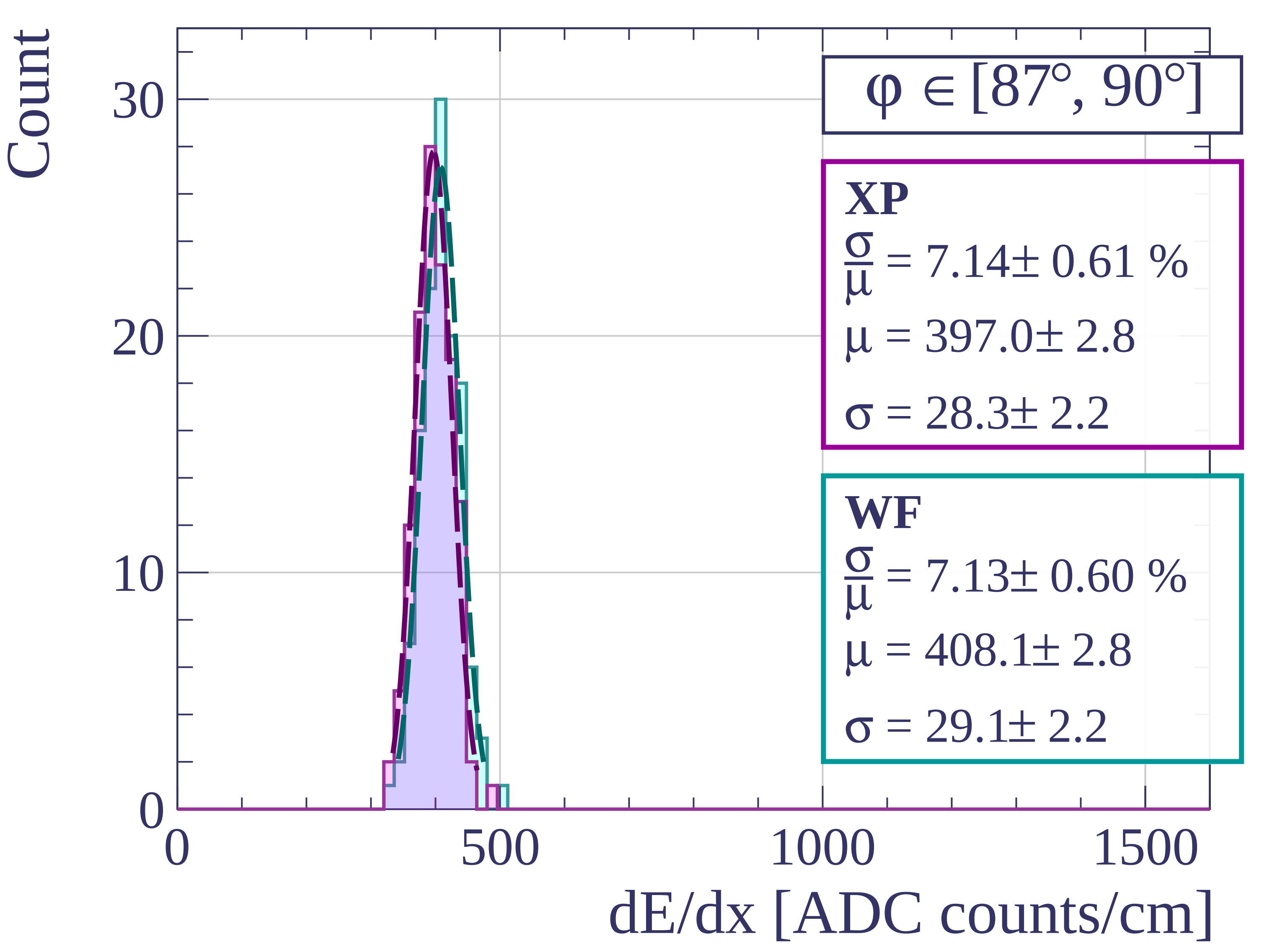
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
φ ∈ [78°, 81°]
XP
σ/μ = 7.78± 1.01 %
μ = 393.1± 4.0
σ = 30.6± 3.6
WF
σ/μ = 7.91± 1.37 %
μ = 404.2± 4.6
σ = 32.0± 5.2



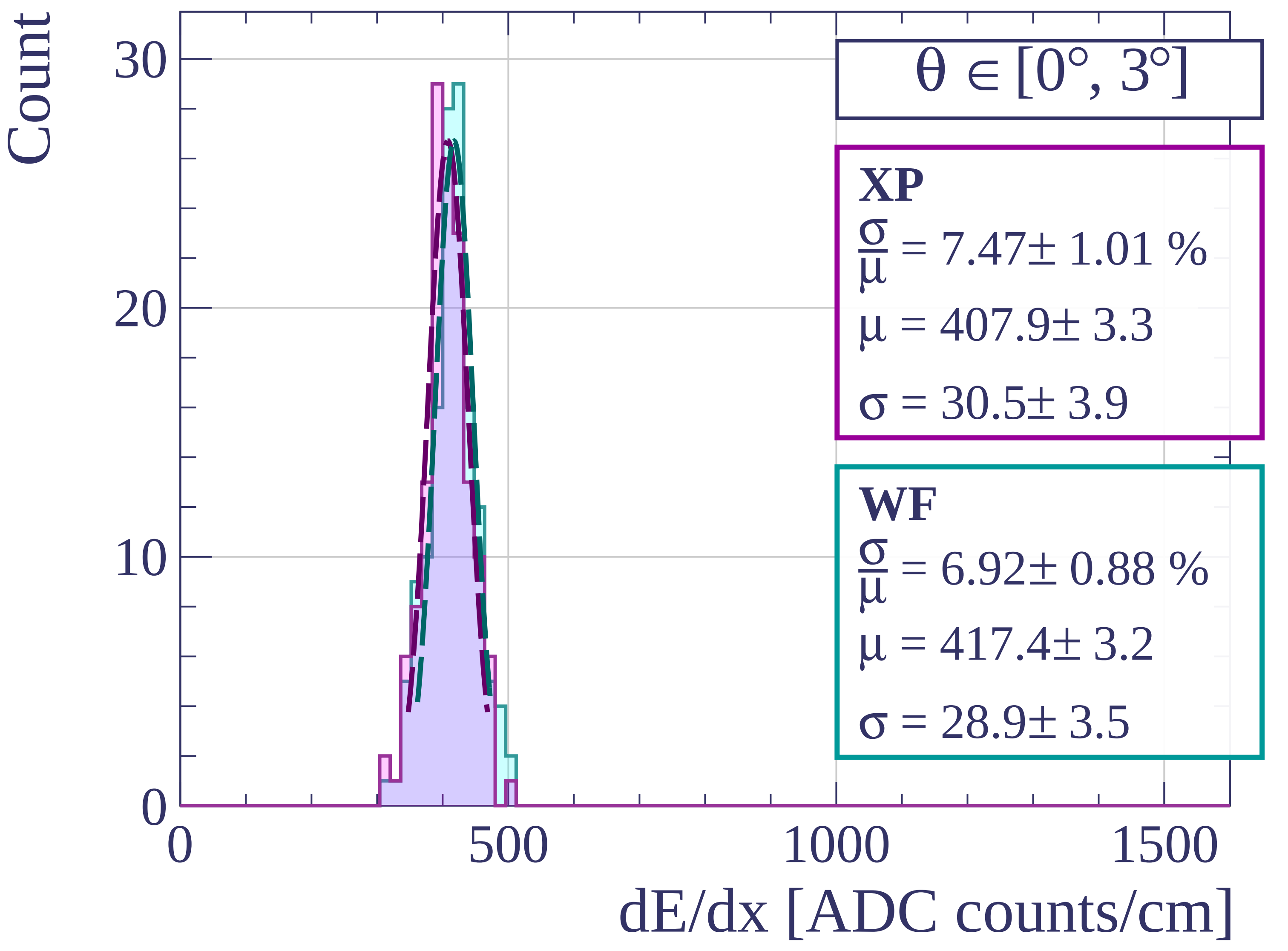
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
20
φ ∈ [81°, 84°]
XP
σ/μ = 6.87± 1.04 %
μ = 403.4± 4.0
σ = 27.7± 3.9
WF
σ/μ = 6.26± 0.76 %
μ = 413.8± 3.3
σ = 25.9± 2.9



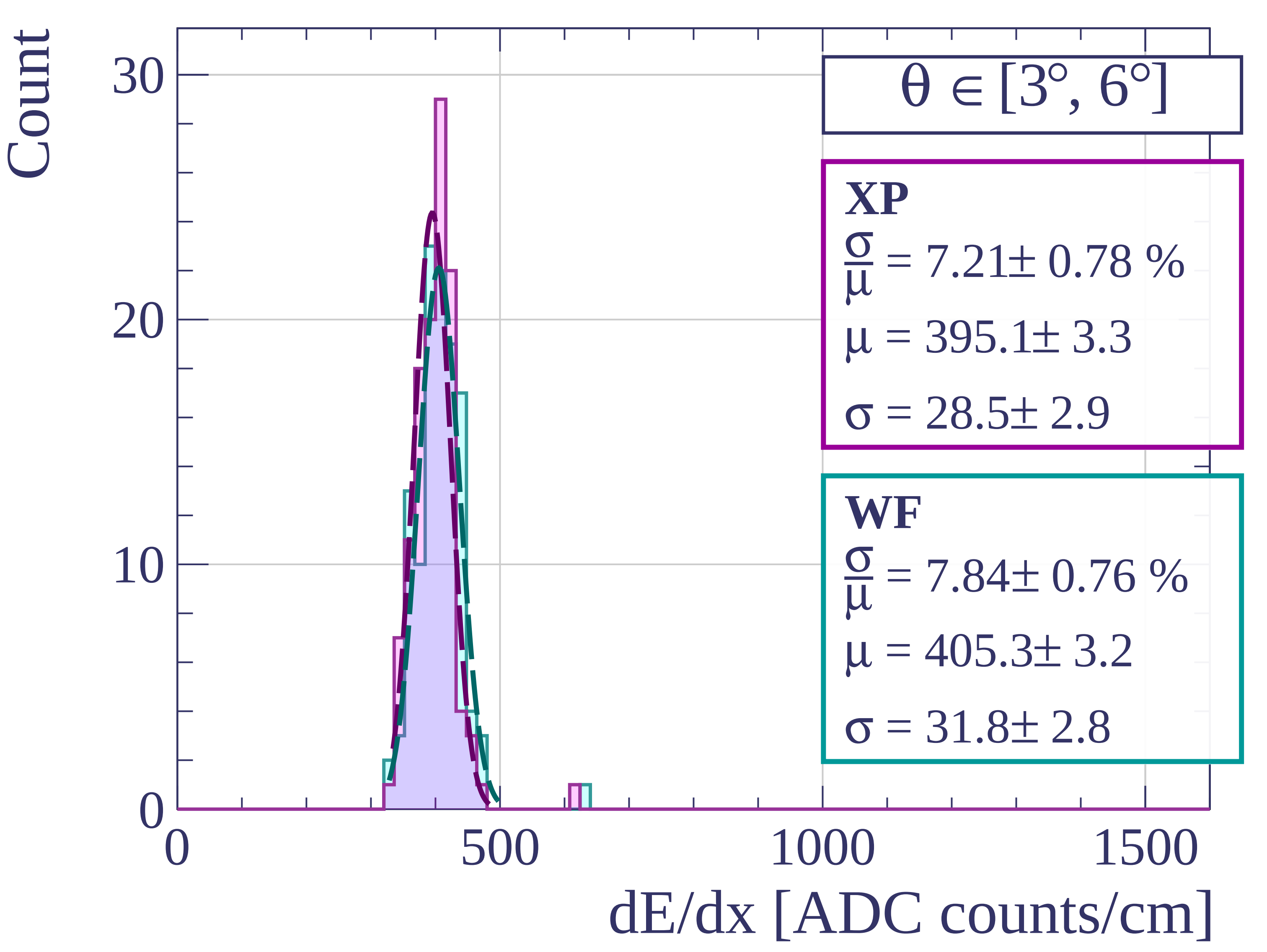
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [84°, 87°]
XP
σ/μ = 8.92± 1.38 %
μ = 402.3± 5.2
σ = 35.9± 5.1
WF
σ/μ = 7.34± 0.96 %
μ = 407.3± 4.2
σ = 29.9± 3.6



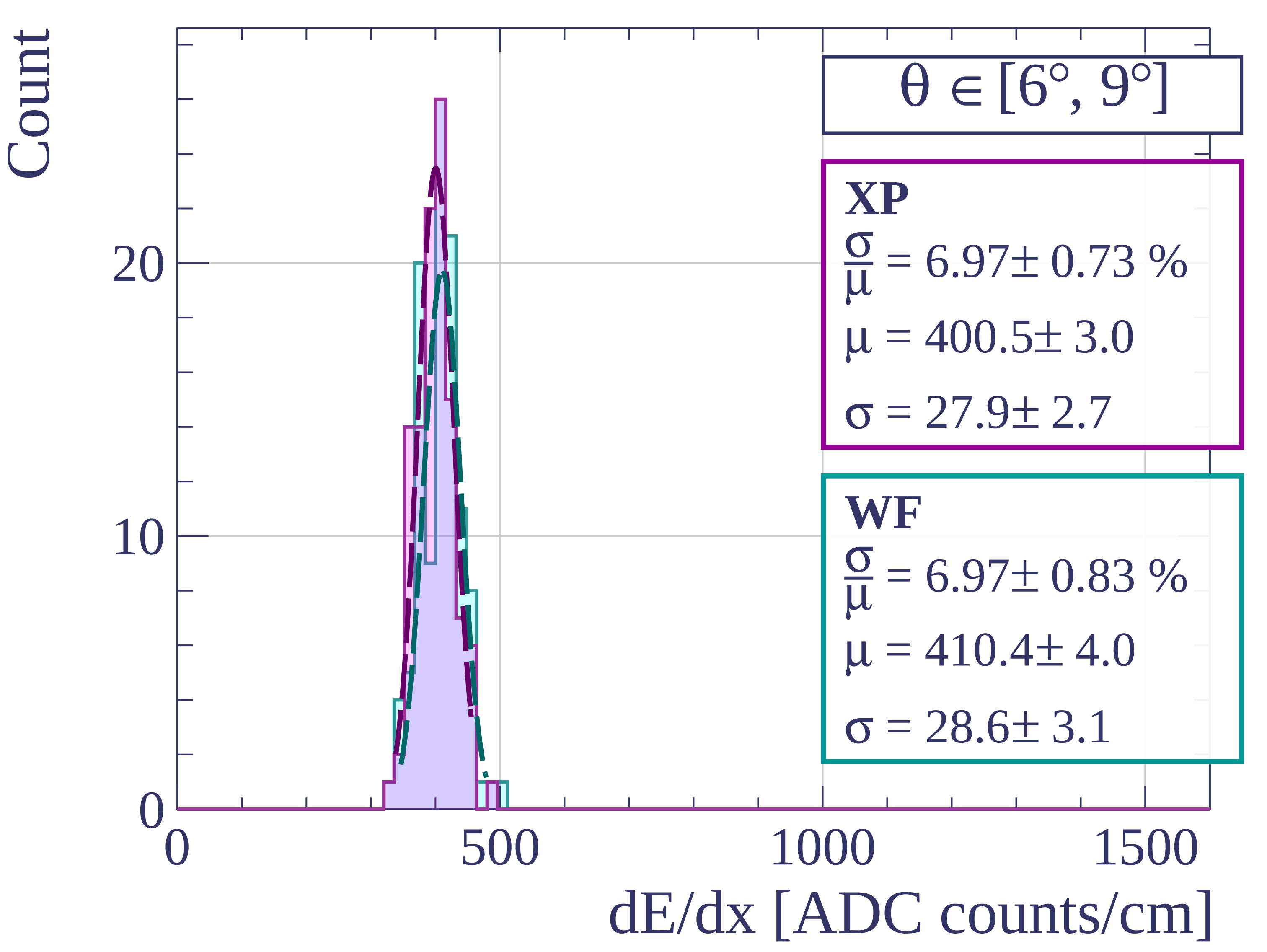
Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [87°, 90°]
XP
σ/μ = 7.14± 0.61 %
μ = 397.0± 2.8
σ = 28.3± 2.2
WF
σ/μ = 7.13± 0.60 %
μ = 408.1± 2.8
σ = 29.1± 2.2



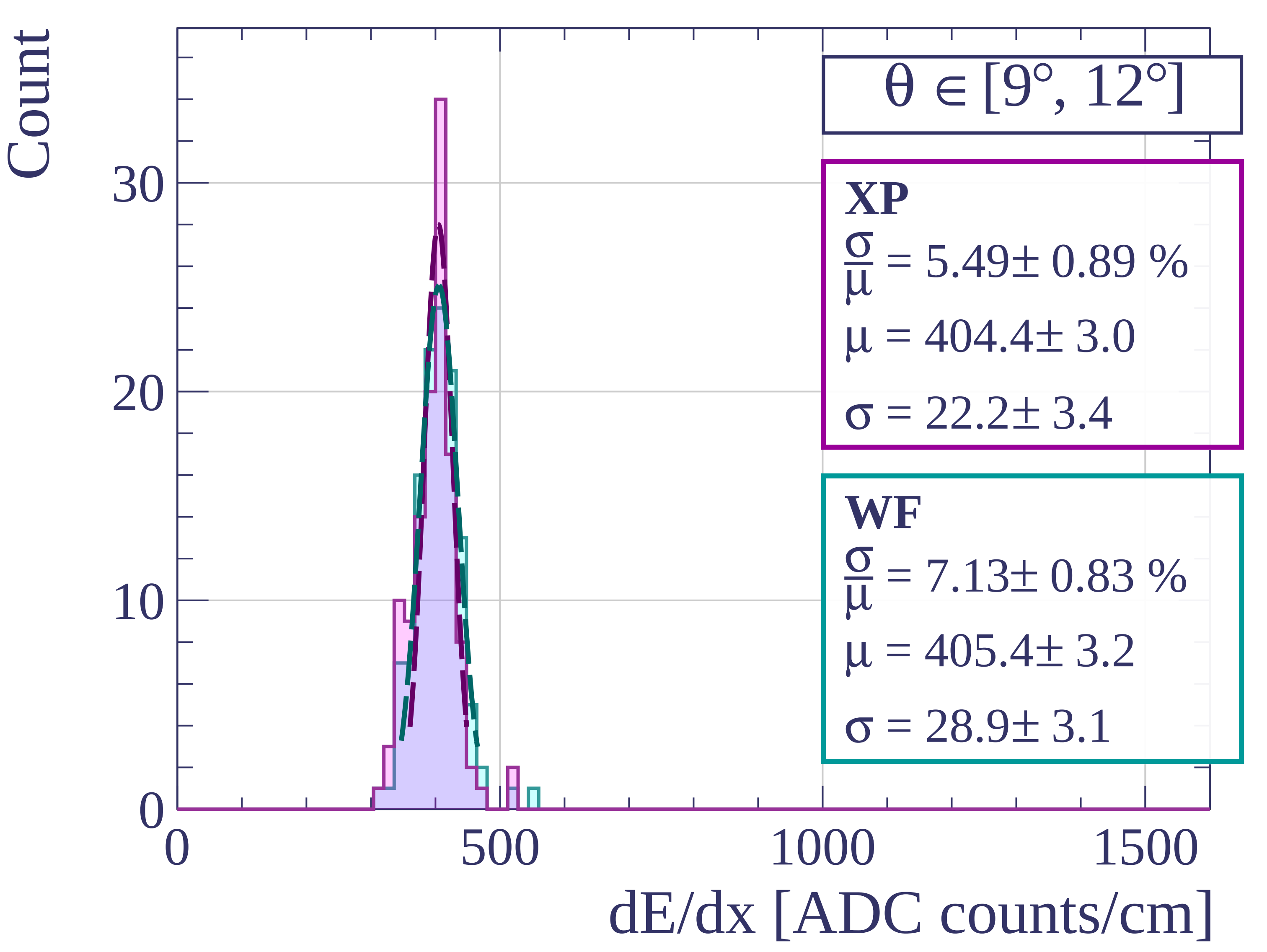
Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [0°, 3°]
XP
σ/μ = 7.47± 1.01 %
μ = 407.9± 3.3
σ = 30.5± 3.9
WF
σ/μ = 6.92± 0.88 %
μ = 417.4± 3.2
σ = 28.9± 3.5

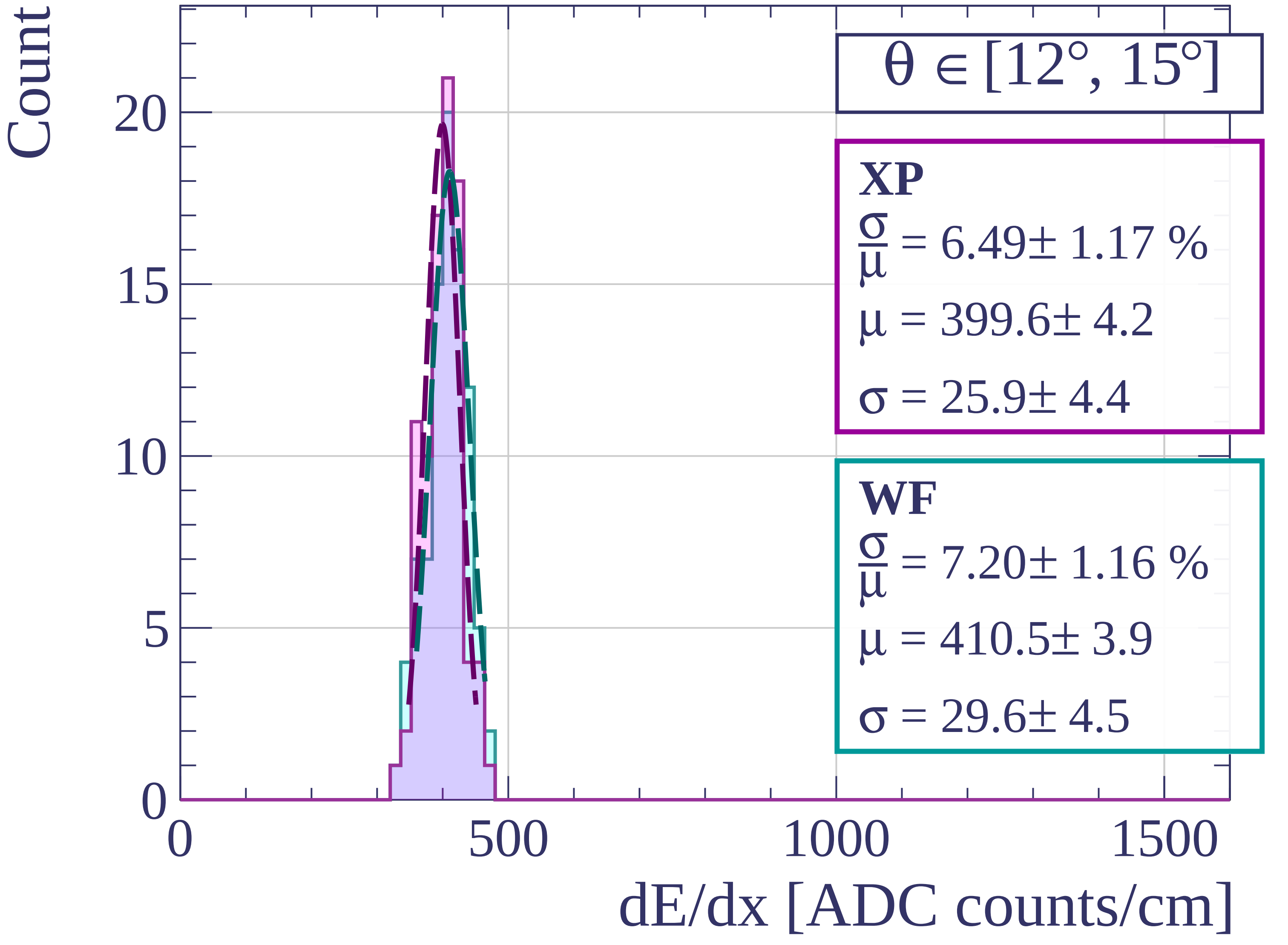


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [3°, 6°]
XP
σ/μ = 7.21± 0.78 %
μ = 395.1± 3.3
σ = 28.5± 2.9
WF
σ/μ = 7.84± 0.76 %
μ = 405.3± 3.2
σ = 31.8± 2.8

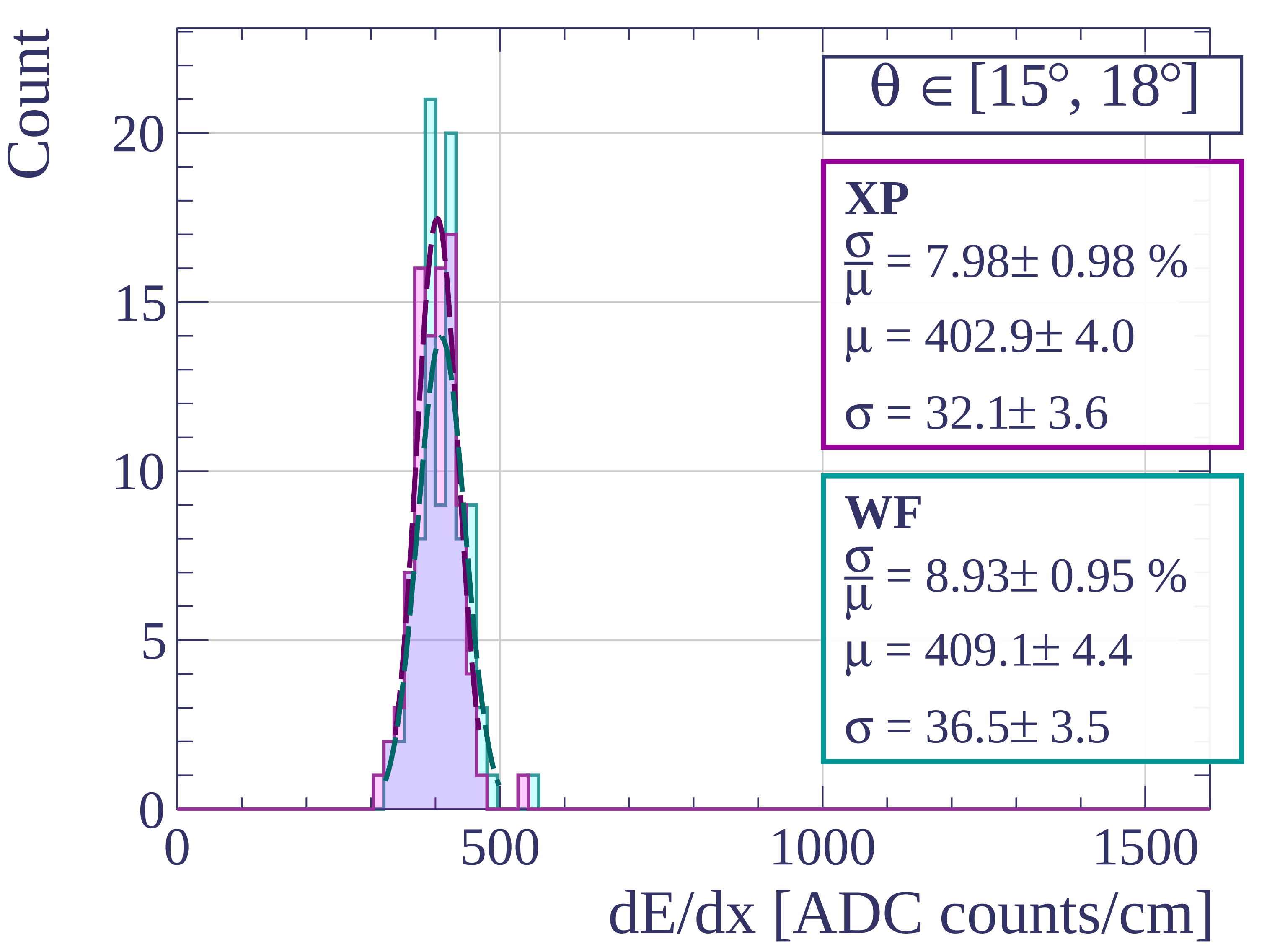


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
10
20
θ ∈ [6°, 9°]
XP
σ/μ = 6.97± 0.73 %
μ = 400.5± 3.0
σ = 27.9± 2.7
WF
σ/μ = 6.97± 0.83 %
μ = 410.4± 4.0
σ = 28.6± 3.1

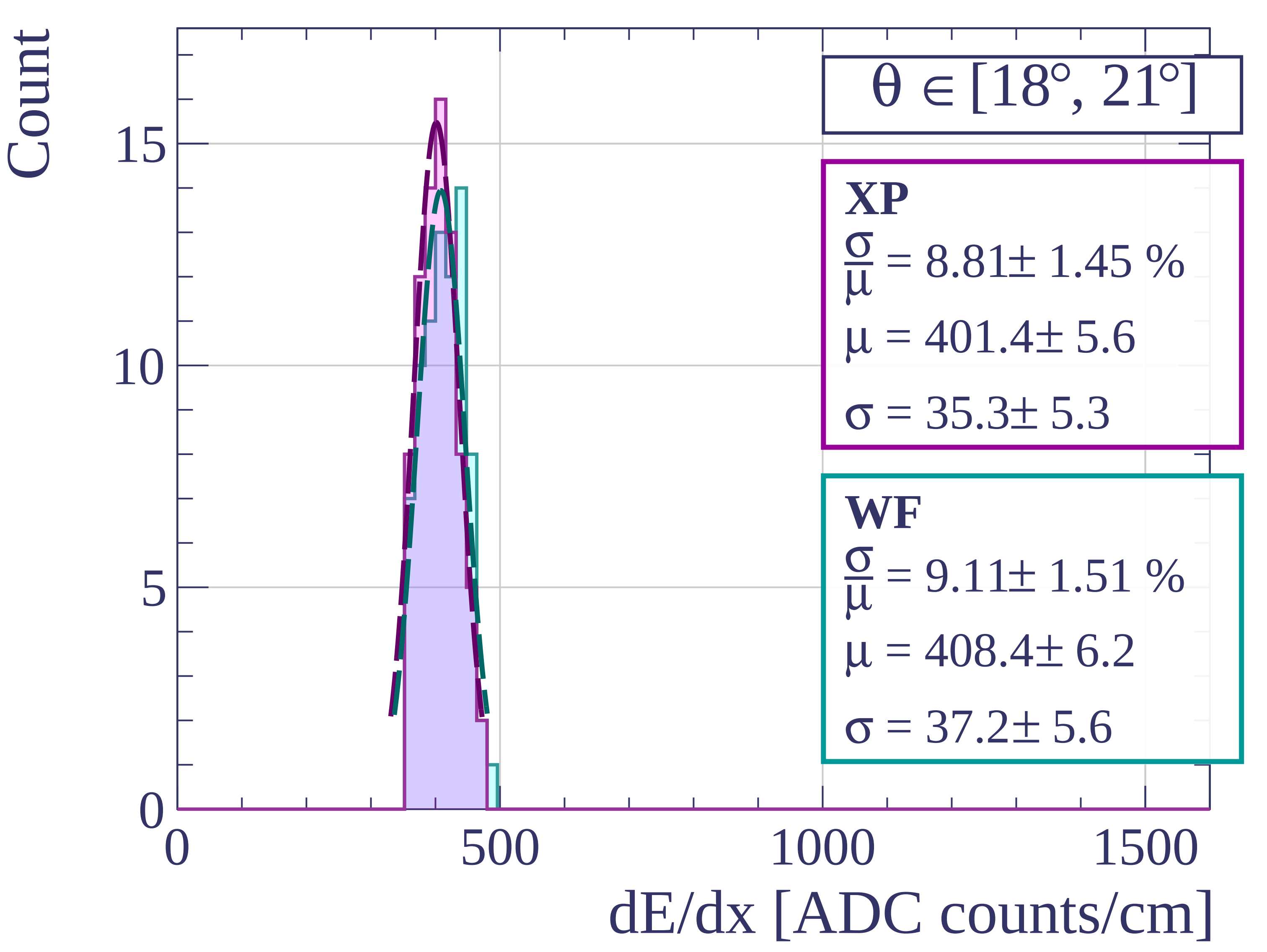

Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 5.49± 0.89 %
μ = 404.4± 3.0
σ = 22.2± 3.4
WF
σ/μ = 7.13± 0.83 %
μ = 405.4± 3.2
σ = 28.9± 3.1



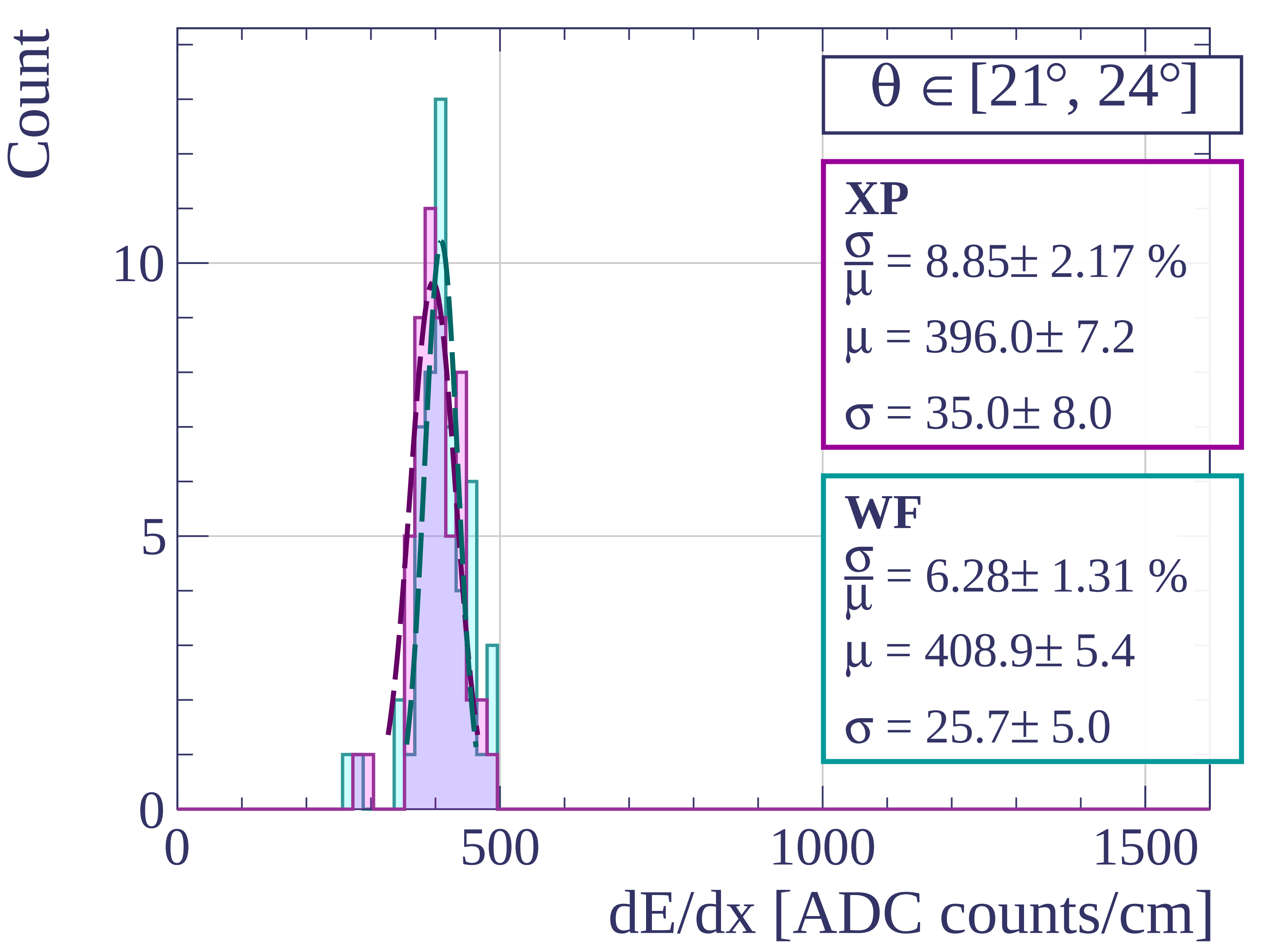
Count
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 6.49± 1.17 %
μ = 399.6± 4.2
σ = 25.9± 4.4
WF
σ/μ = 7.20± 1.16 %
μ = 410.5± 3.9
σ = 29.6± 4.5
0
5
10
15
20
0
500
1000
1500



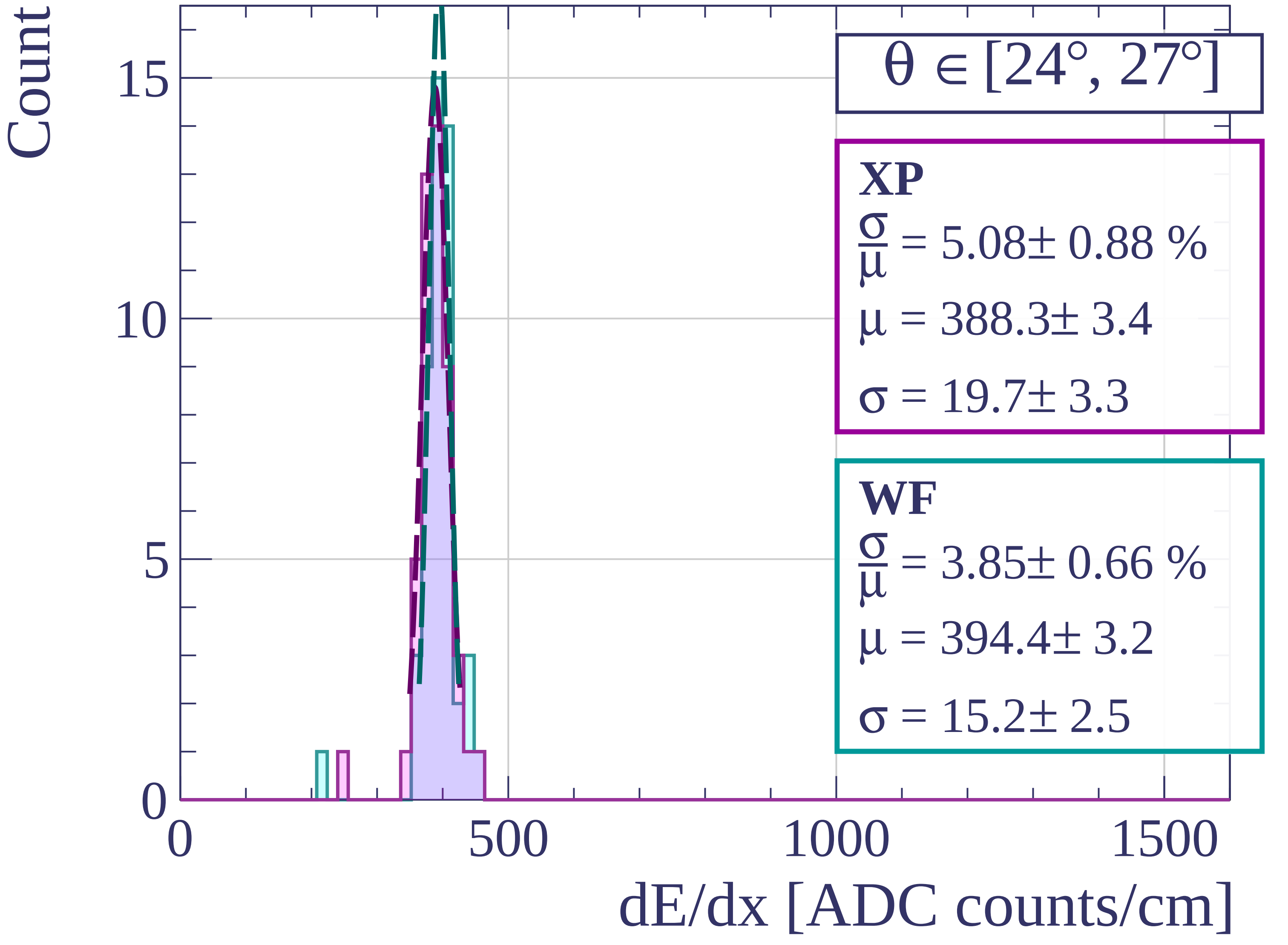
θ ∈ [15°, 18°]
XP
σ/μ = 7.98± 0.98 %
μ = 402.9± 4.0
σ = 32.1± 3.6
WF
σ/μ = 8.93± 0.95 %
μ = 409.1± 4.4
σ = 36.5± 3.5
Count
0
5
10
15
20
0
500
1000
1500
dE/dx [ADC counts/cm]



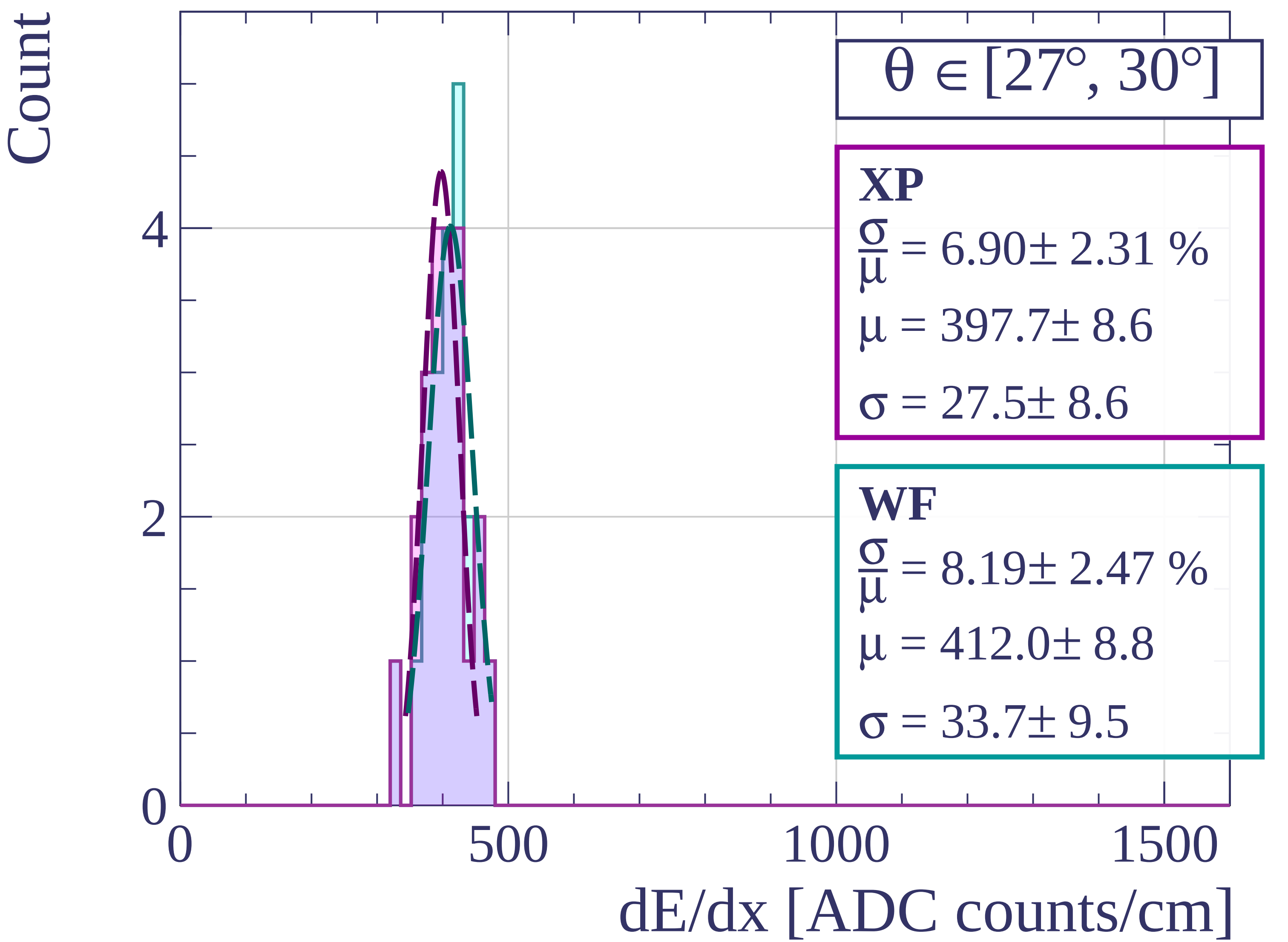
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 8.81± 1.45 %
μ = 401.4± 5.6
σ = 35.3± 5.3
WF
σ/μ = 9.11± 1.51 %
μ = 408.4± 6.2
σ = 37.2± 5.6



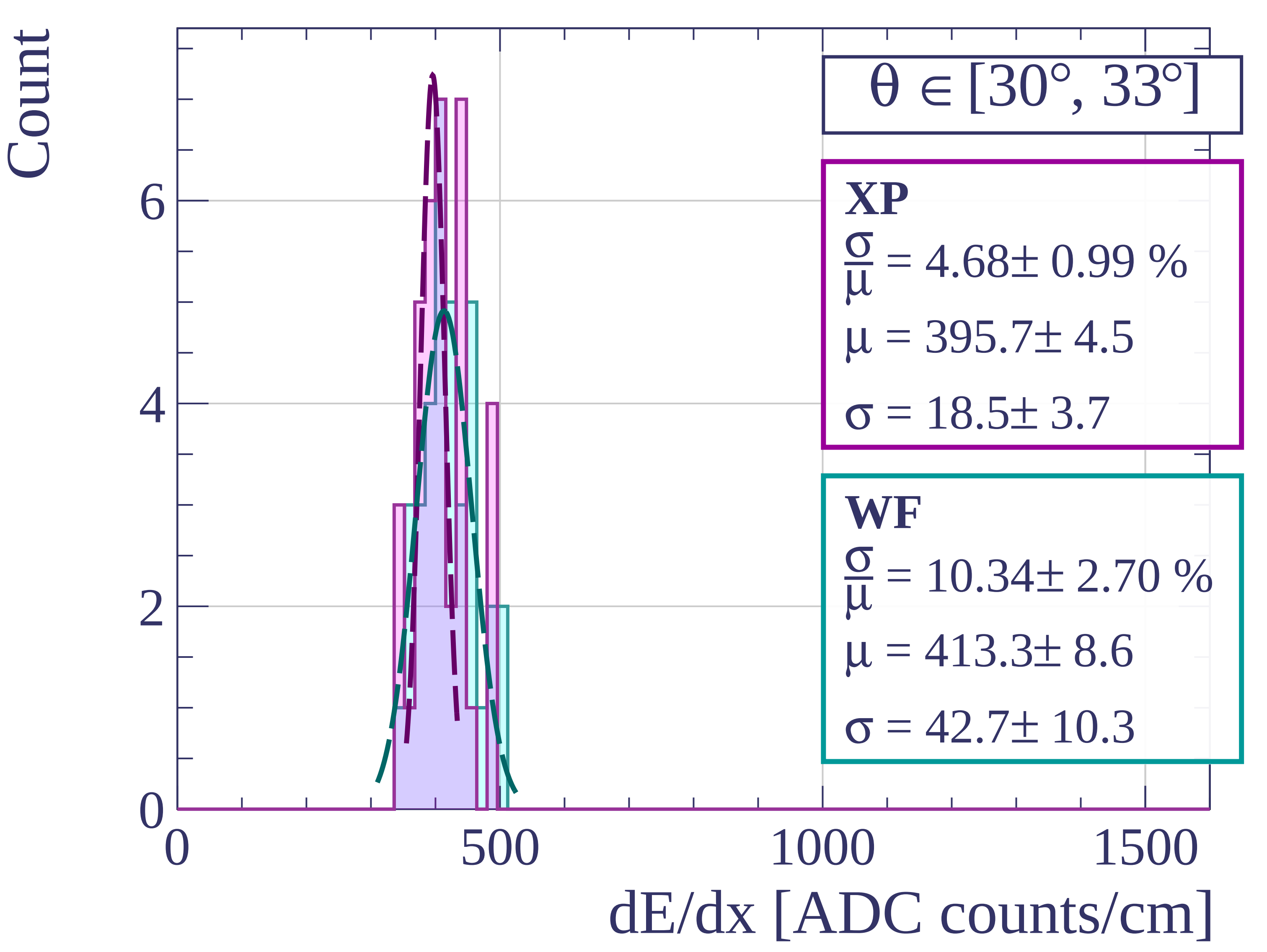
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [21°, 24°]
XP
σ/μ = 8.85± 2.17 %
μ = 396.0± 7.2
σ = 35.0± 8.0
WF
σ/μ = 6.28± 1.31 %
μ = 408.9± 5.4
σ = 25.7± 5.0



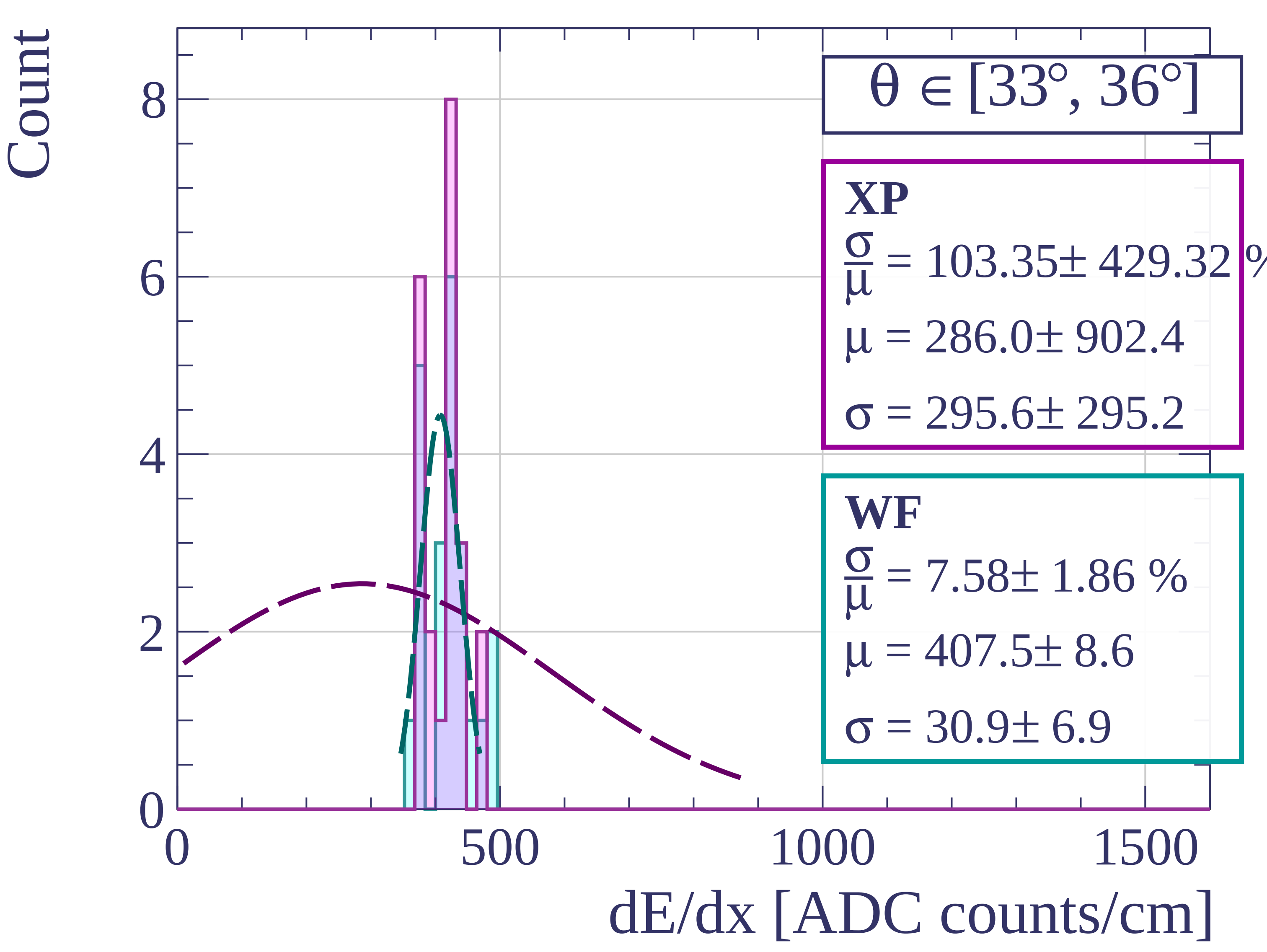
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [24°, 27°]
XP
σ/μ = 5.08± 0.88 %
μ = 388.3± 3.4
σ = 19.7± 3.3
WF
σ/μ = 3.85± 0.66 %
μ = 394.4± 3.2
σ = 15.2± 2.5



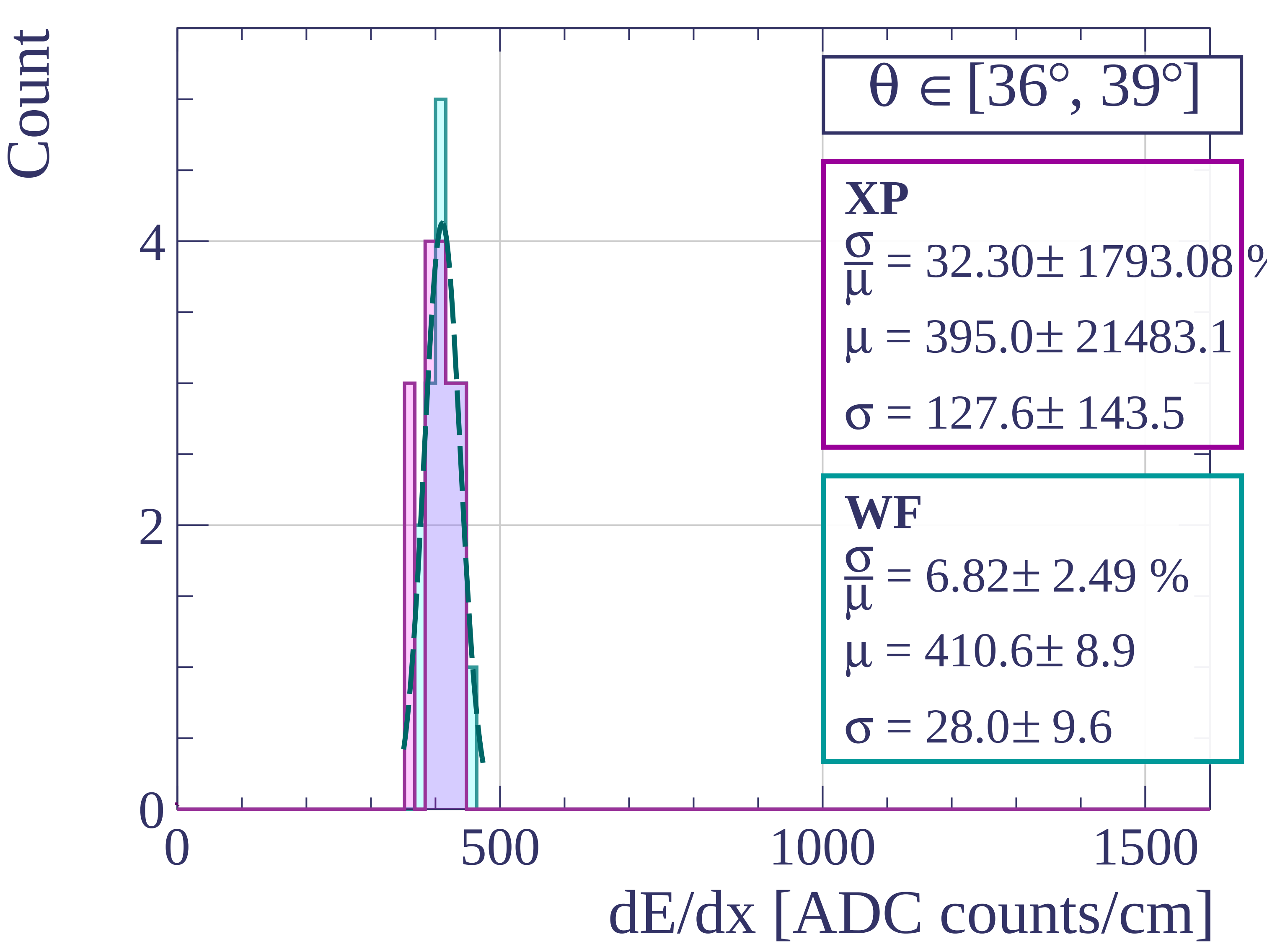
Count
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [27°, 30°]
XP
σ/μ = 6.90± 2.31 %
μ = 397.7± 8.6
σ = 27.5± 8.6
WF
σ/μ = 8.19± 2.47 %
μ = 412.0± 8.8
σ = 33.7± 9.5

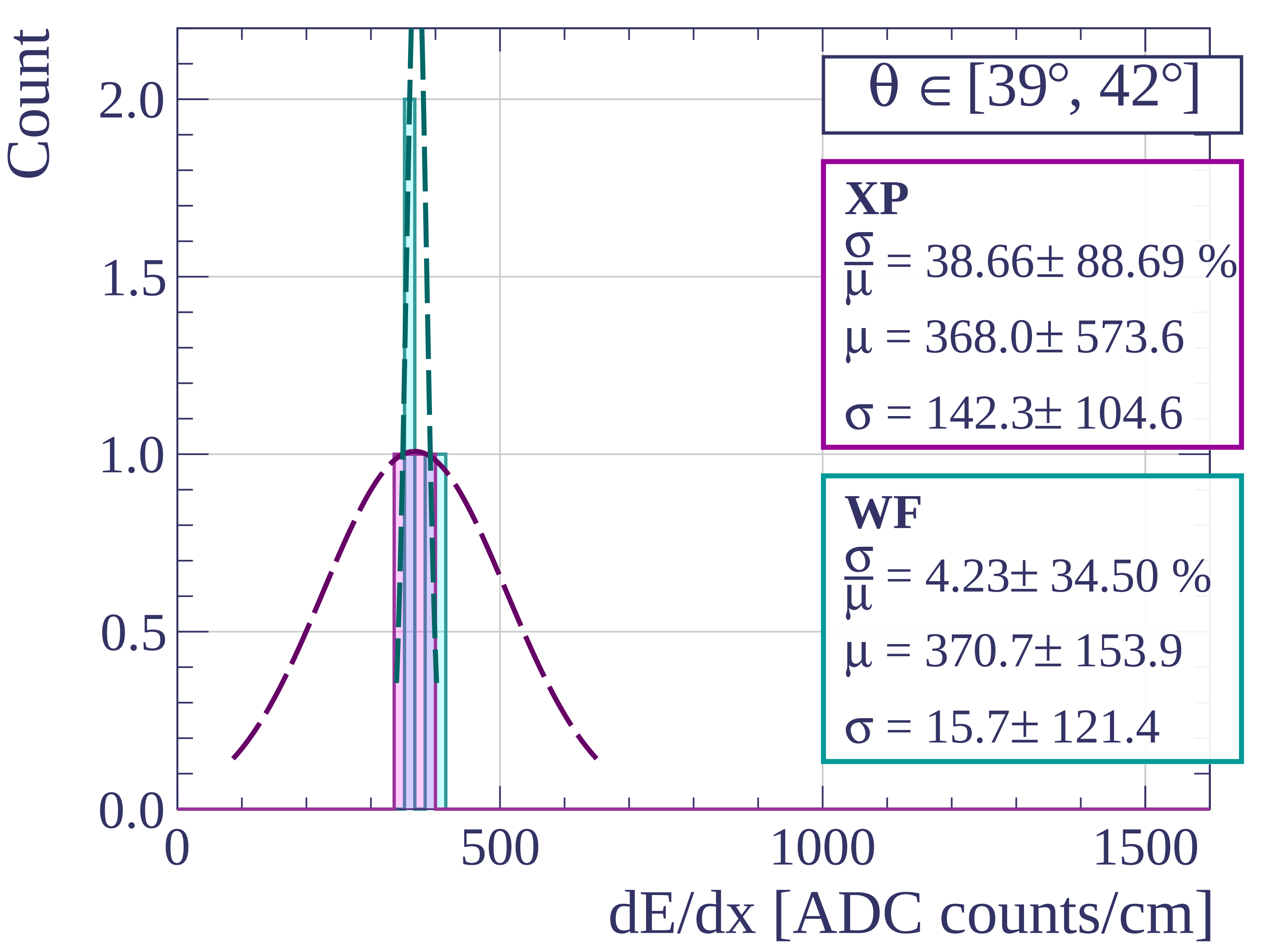


Count
0
2
4
6
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [30°, 33°]
XP
σ/μ = 4.68± 0.99 %
μ = 395.7± 4.5
σ = 18.5± 3.7
WF
σ/μ = 10.34± 2.70 %
μ = 413.3± 8.6
σ = 42.7± 10.3

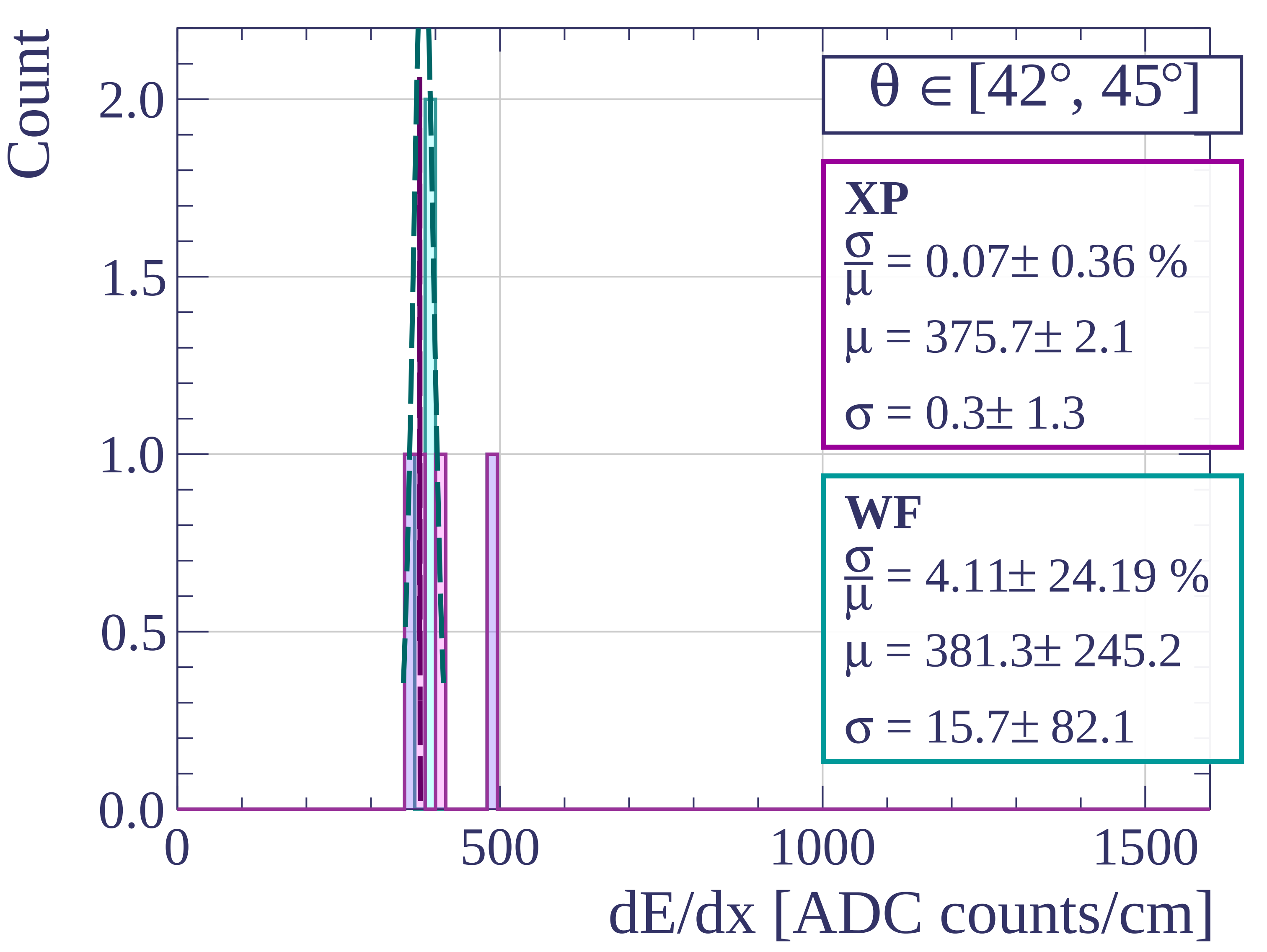


Count
8
6
4
2
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 103.35± 429.32 %
μ = 286.0± 902.4
σ = 295.6± 295.2
WF
σ/μ = 7.58± 1.86 %
μ = 407.5± 8.6
σ = 30.9± 6.9

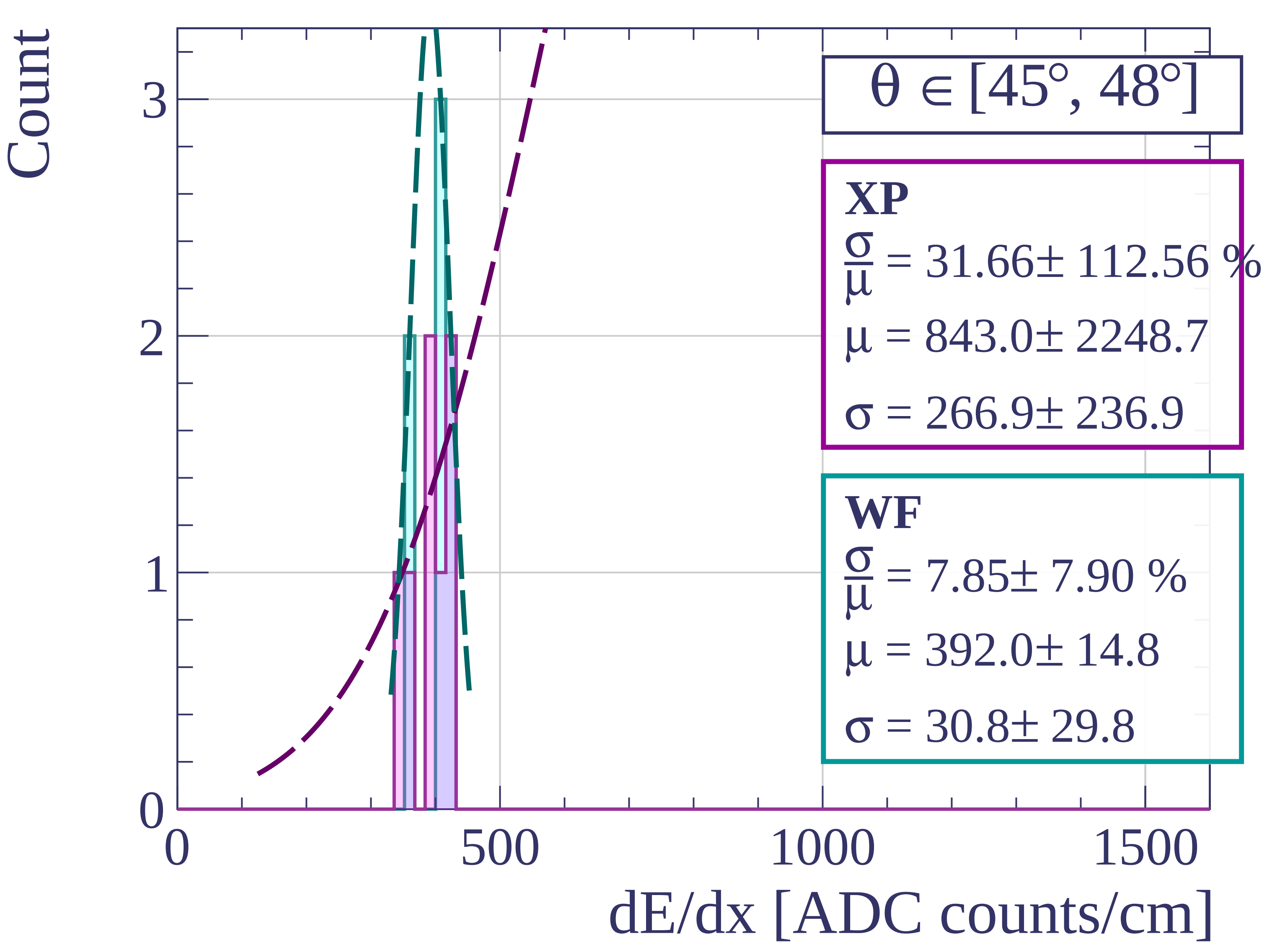

Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
θ ∈ [36°, 39°]
XP
σ/μ = 32.30± 1793.08 %
μ = 395.0± 21483.1
σ = 127.6± 143.5
WF
σ/μ = 6.82± 2.49 %
μ = 410.6± 8.9
σ = 28.0± 9.6



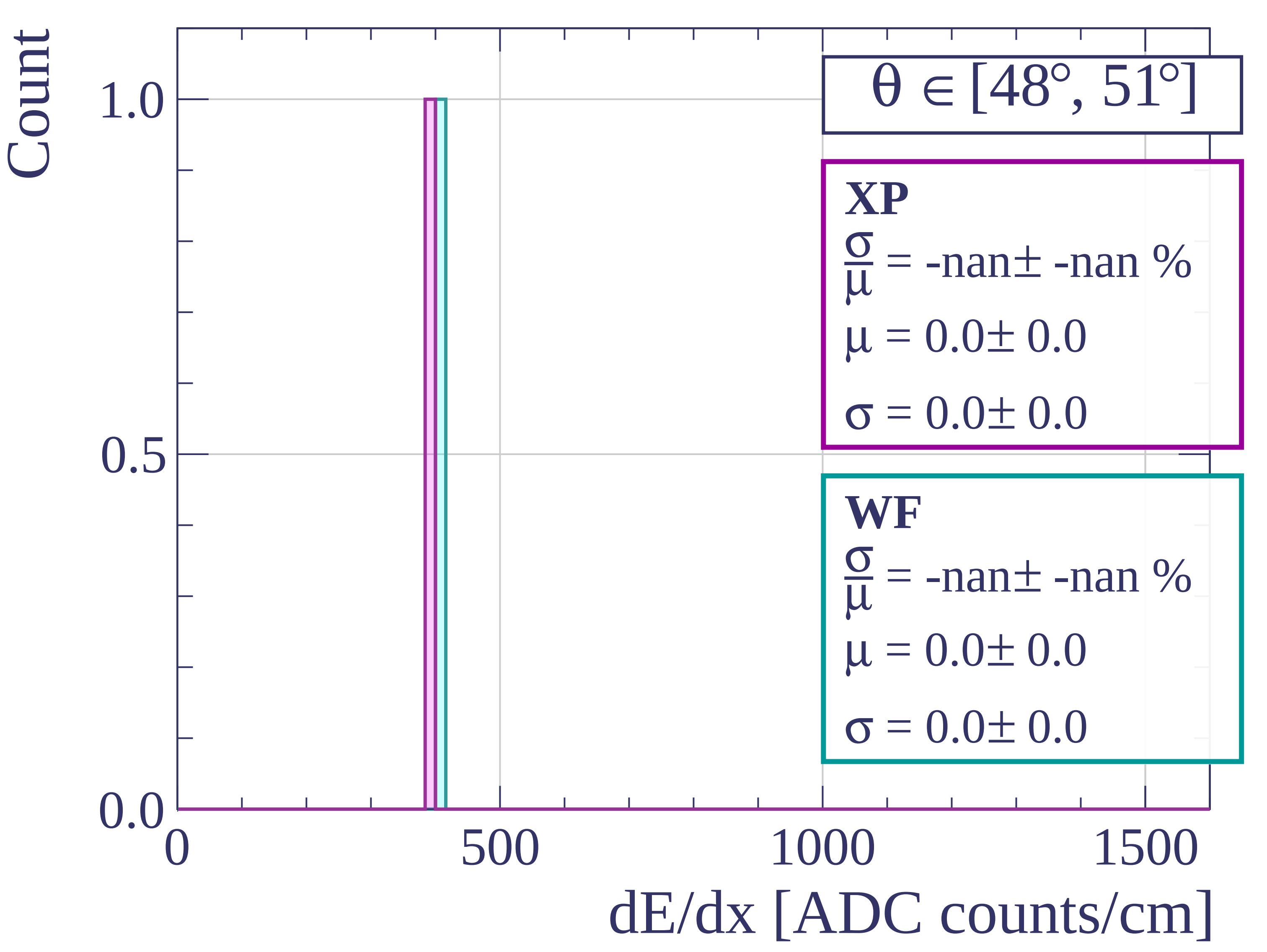
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [39°, 42°]
XP
σ/μ = 38.66± 88.69 %
μ = 368.0± 573.6
σ = 142.3± 104.6
WF
σ/μ = 4.23± 34.50 %
μ = 370.7± 153.9
σ = 15.7± 121.4

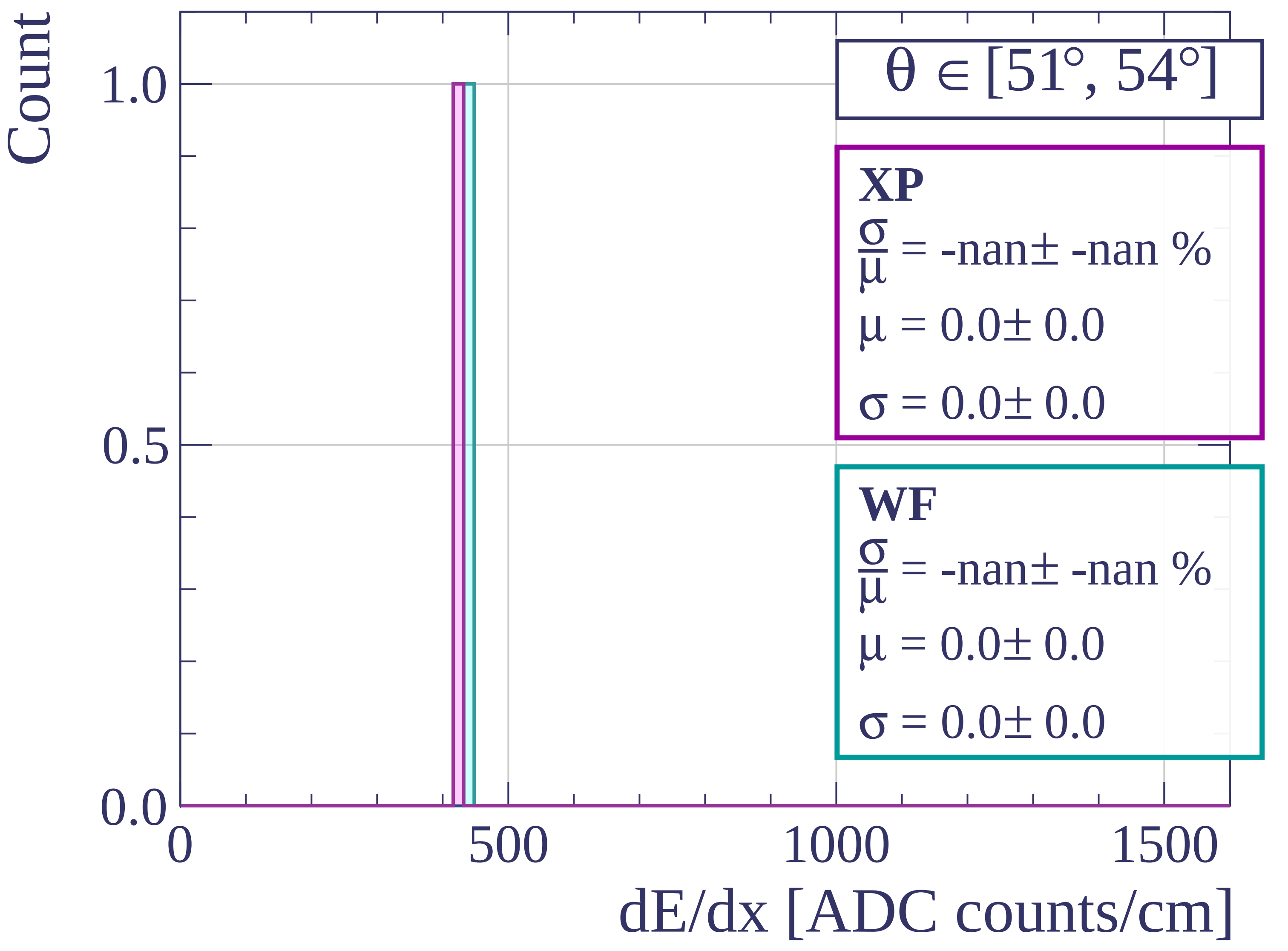


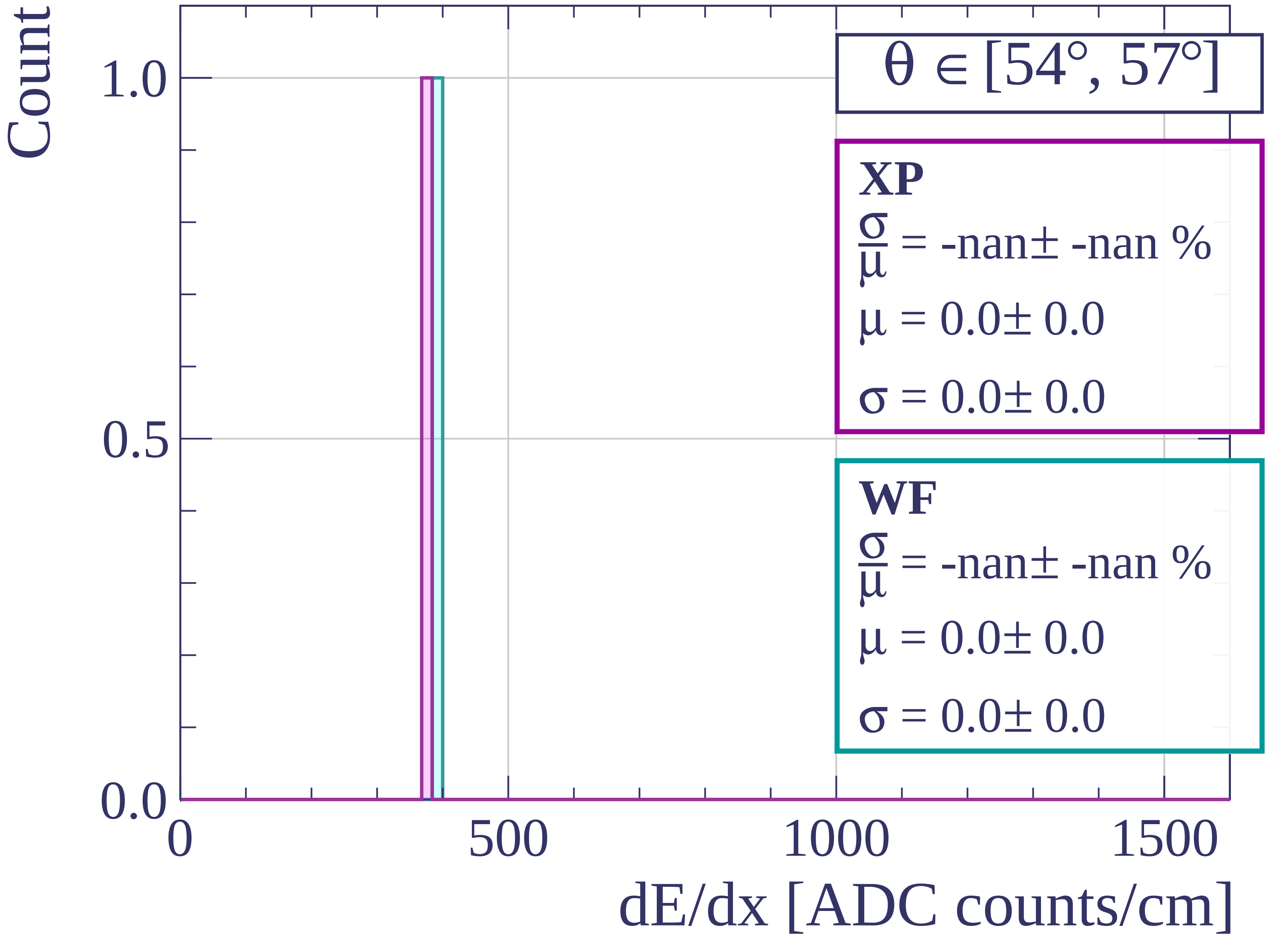
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [42°, 45°]
XP
σ/μ = 0.07± 0.36 %
μ = 375.7± 2.1
σ = 0.3± 1.3
WF
σ/μ = 4.11± 24.19 %
μ = 381.3± 245.2
σ = 15.7± 82.1



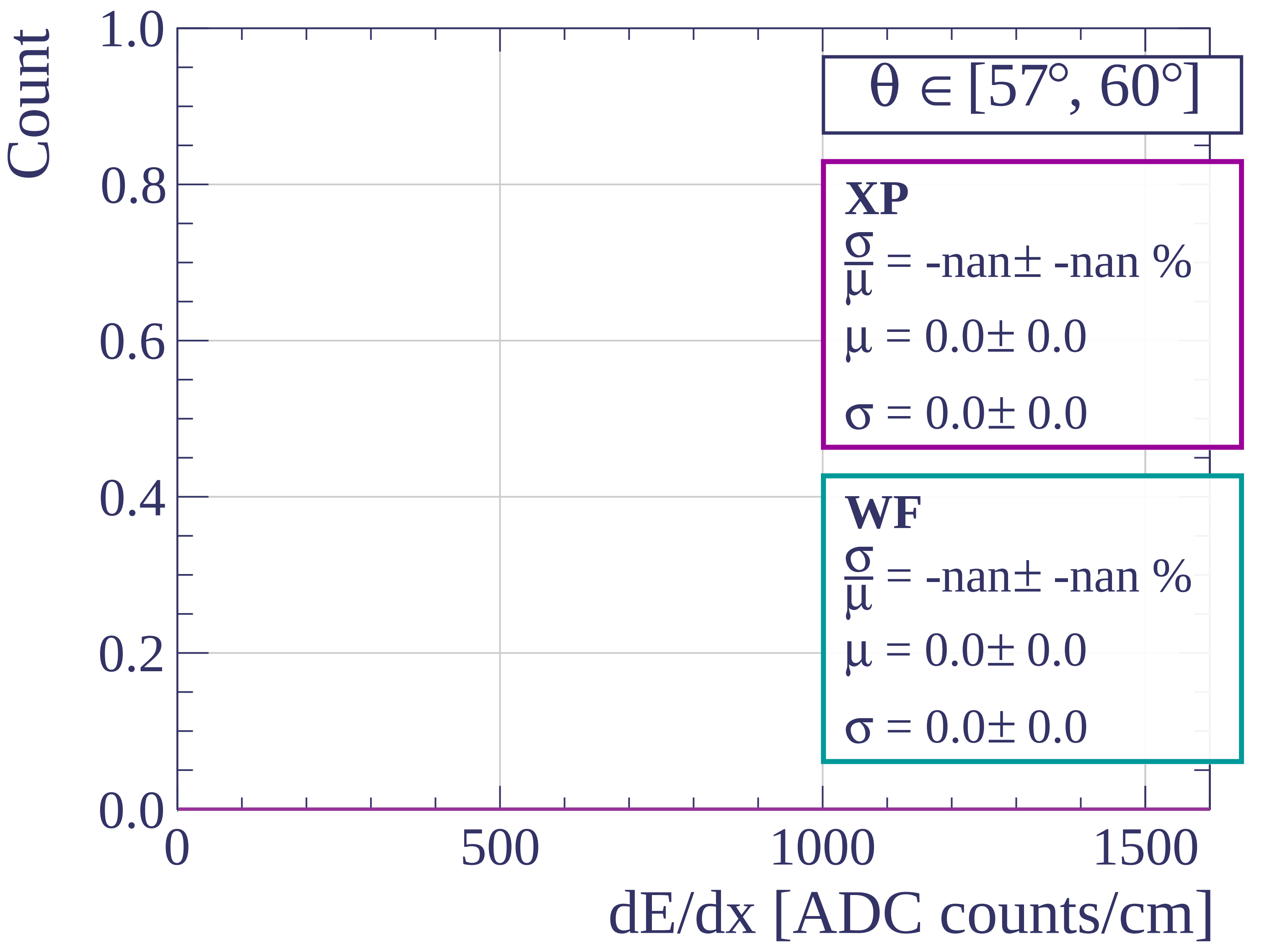
Count
3
2
1
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [45°, 48°]
XP
σ/μ = 31.66± 112.56 %
μ = 843.0± 2248.7
σ = 266.9± 236.9
WF
σ/μ = 7.85± 7.90 %
μ = 392.0± 14.8
σ = 30.8± 29.8

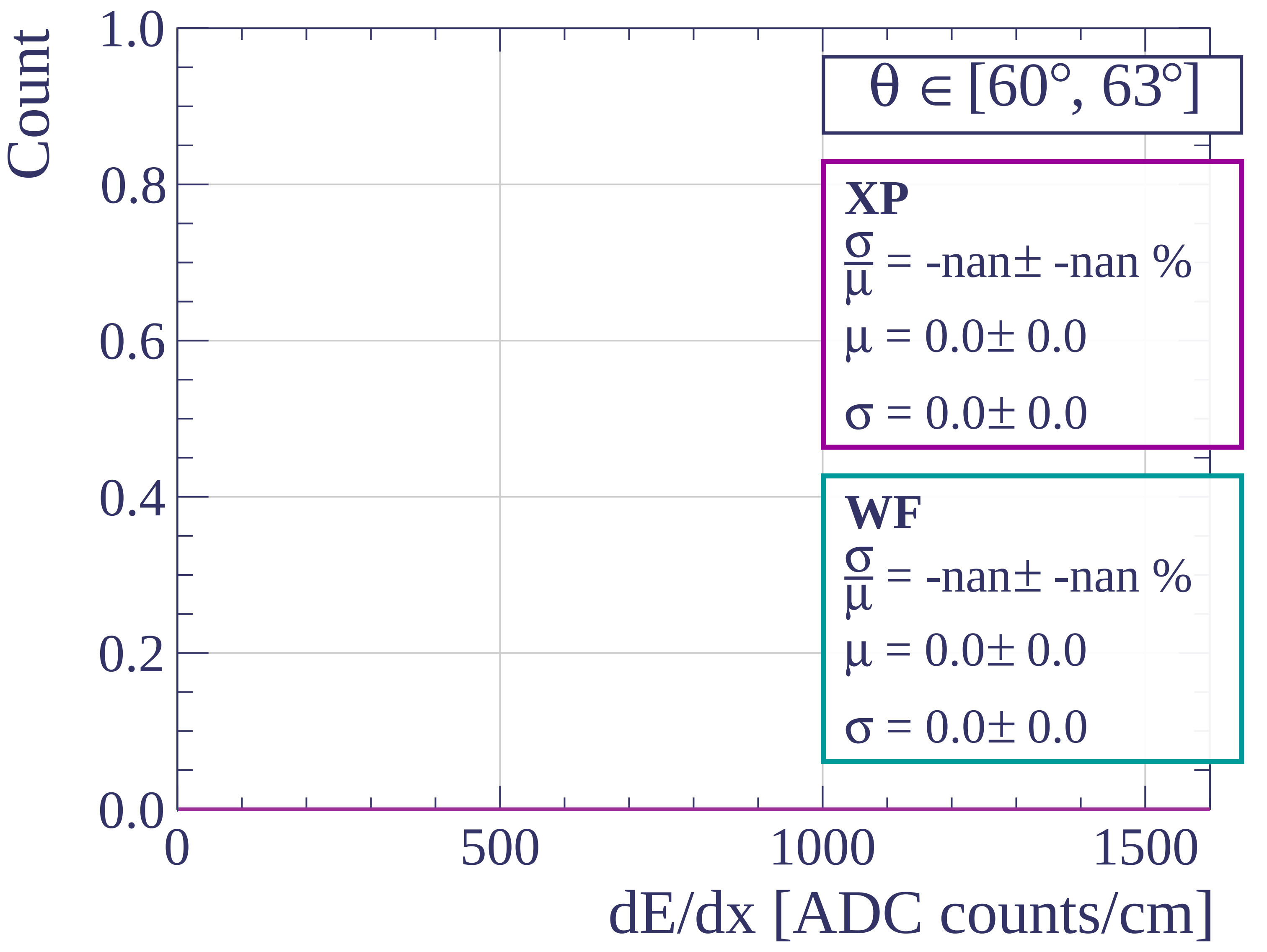

Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

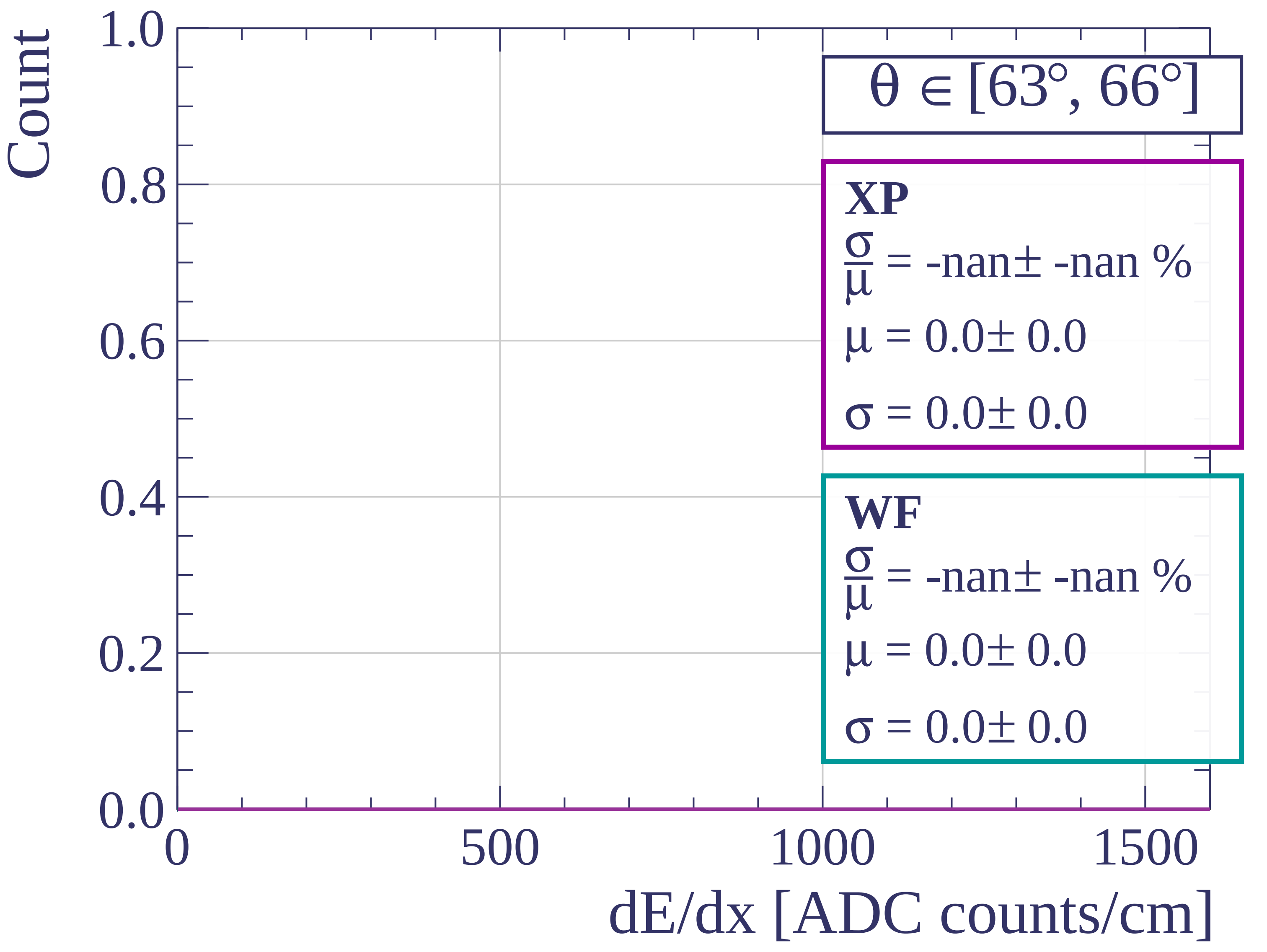


Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [54°, 57°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

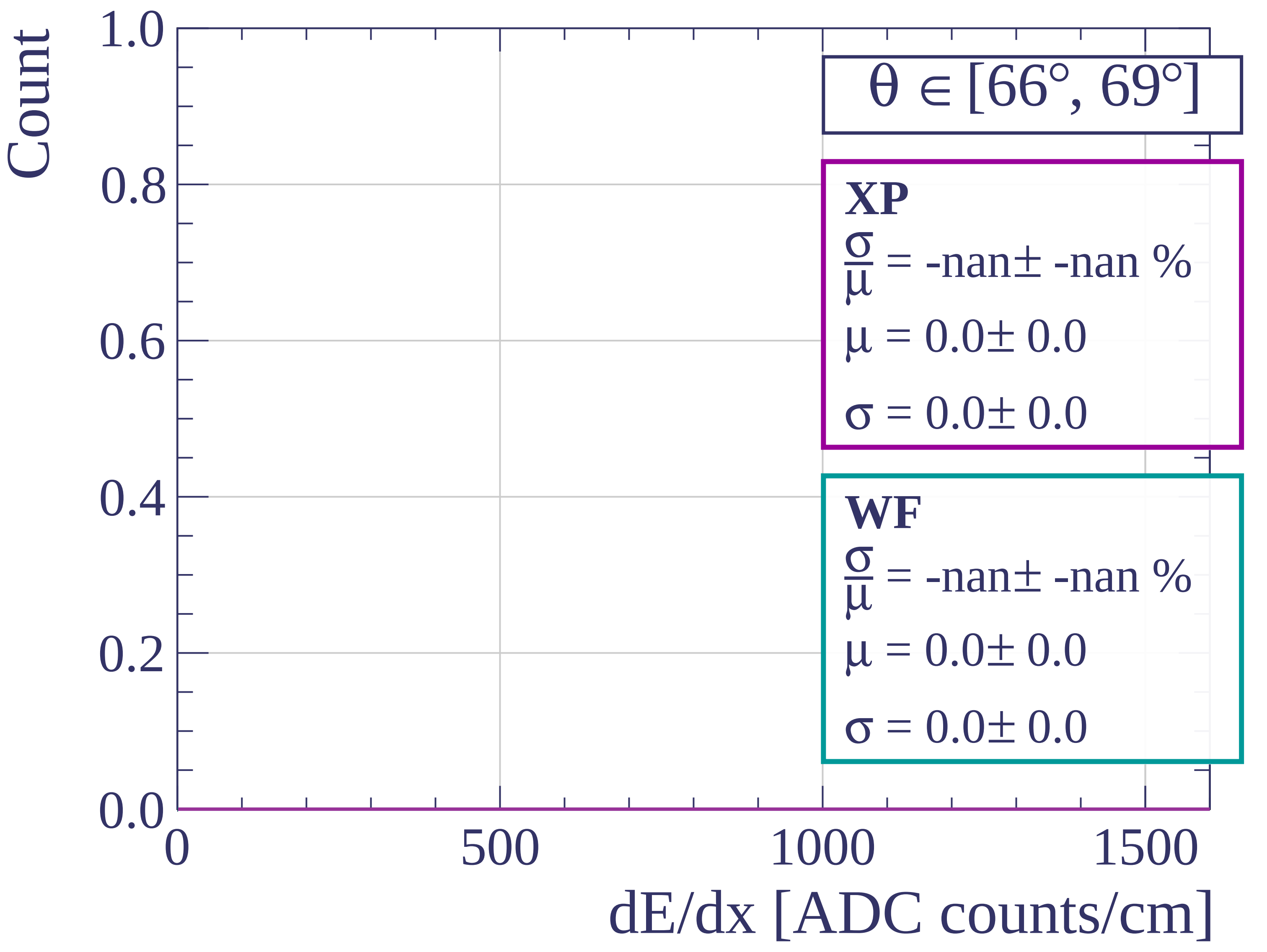

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



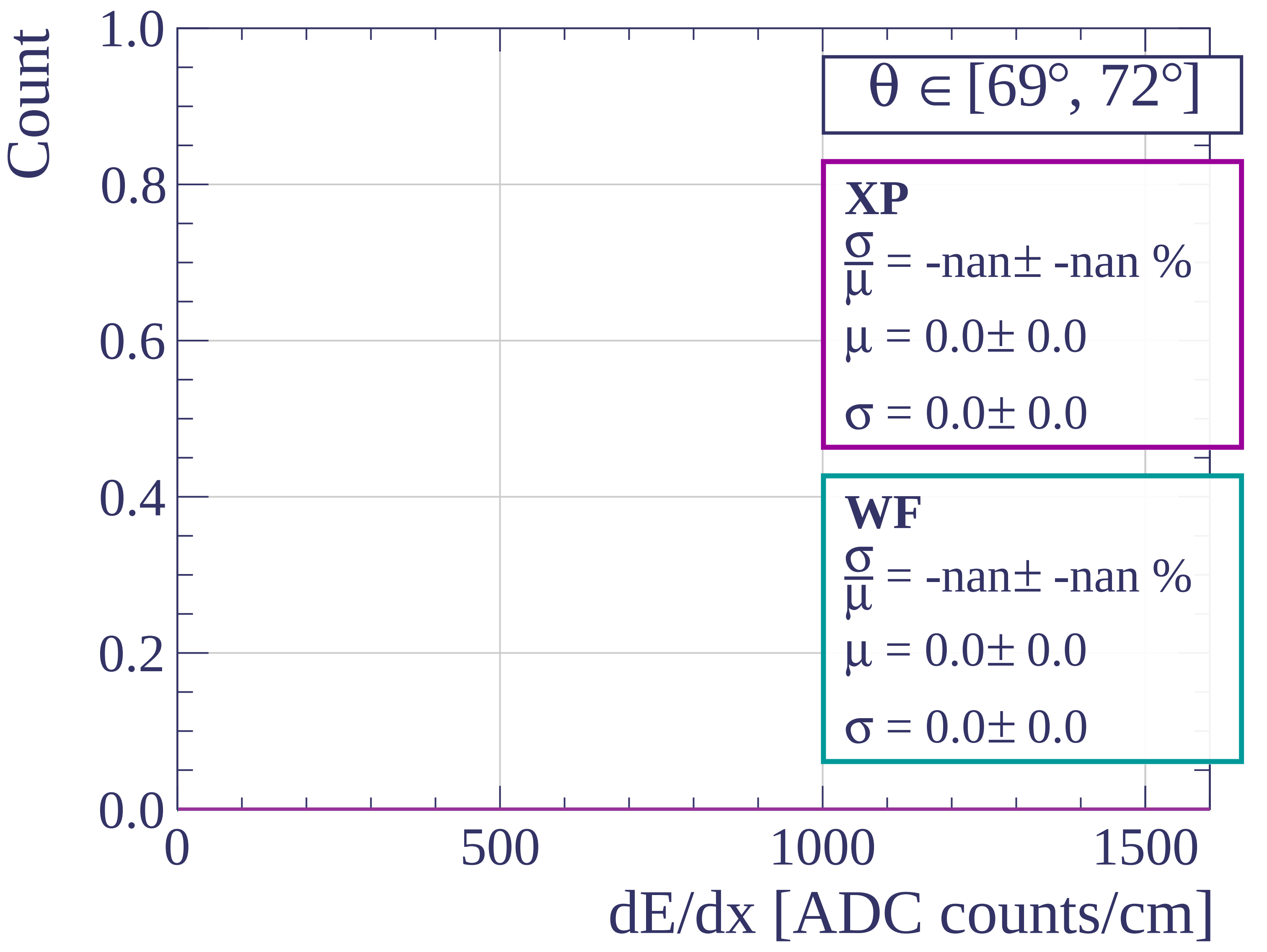
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

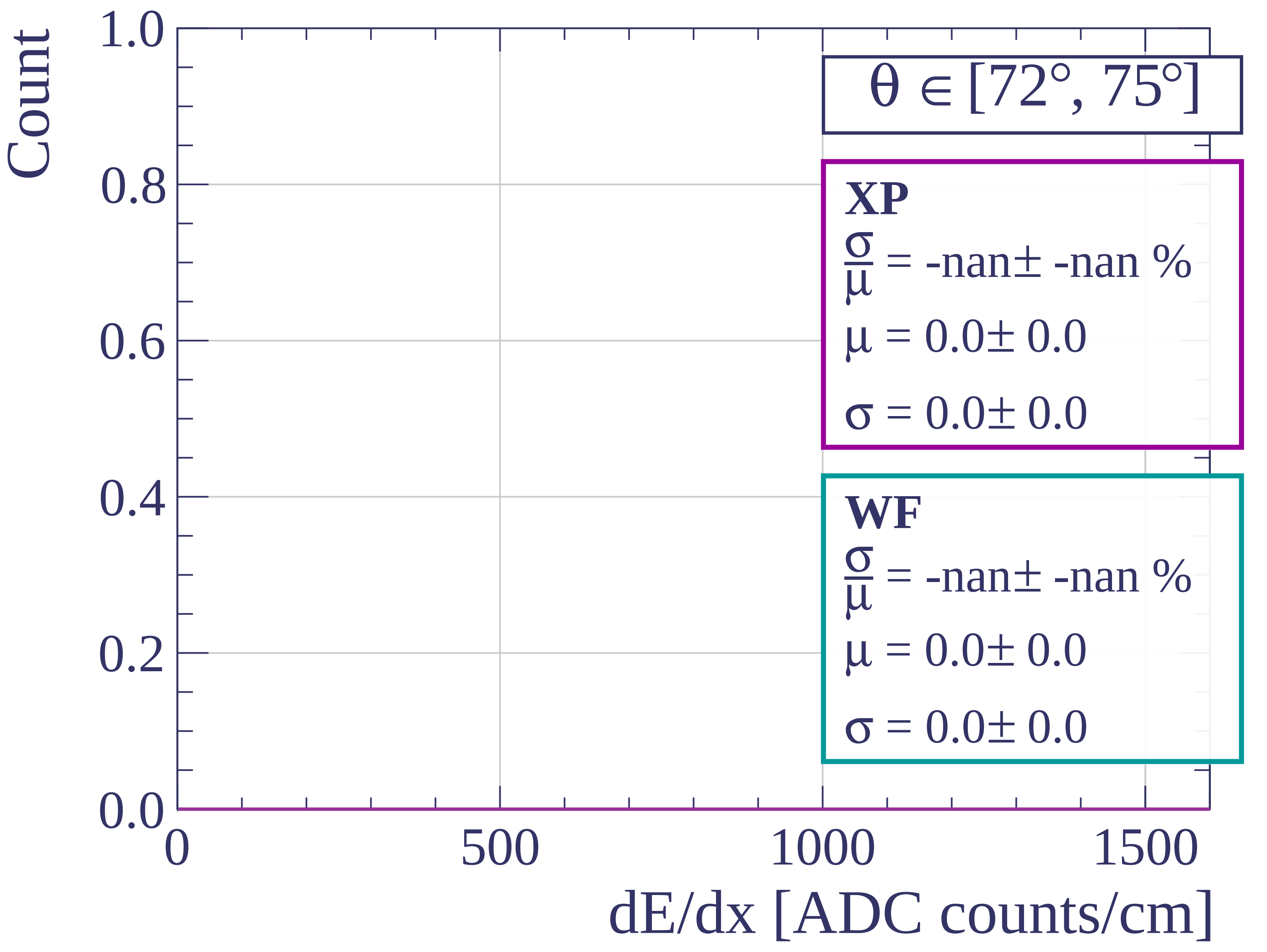


θ ∈ [63°, 66°]
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
Count
dE/dx [ADC counts/cm]
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500

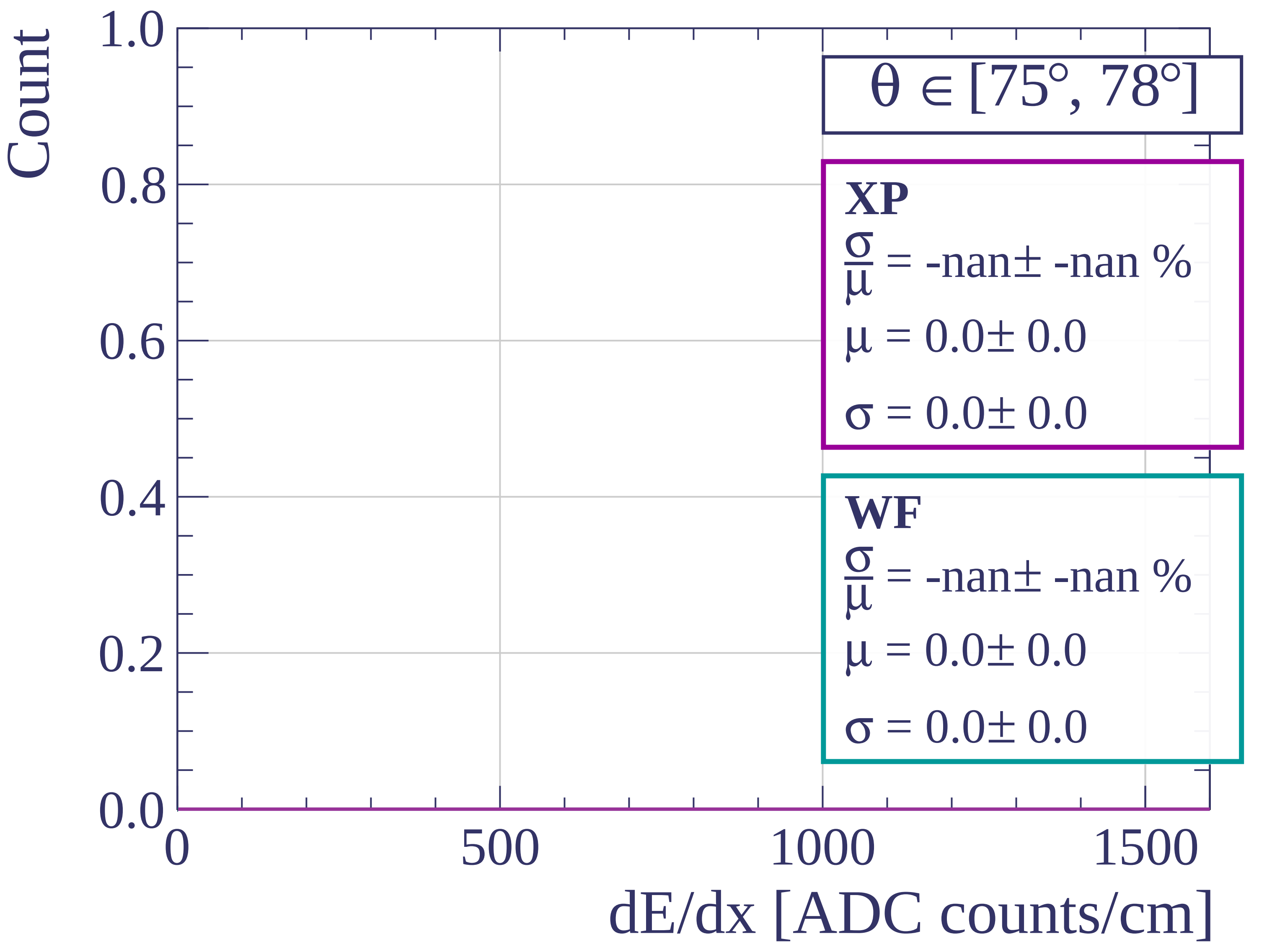


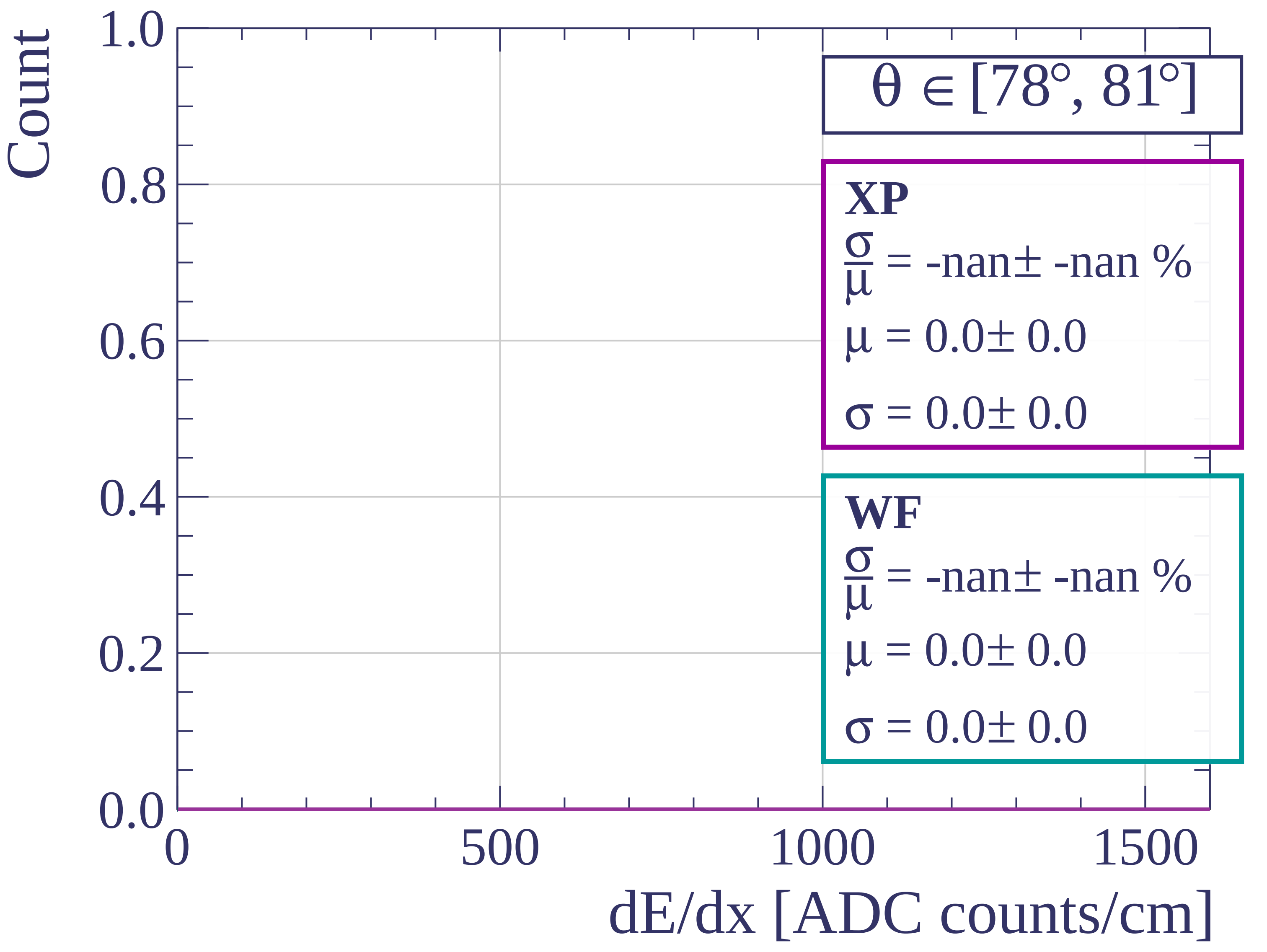
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [66°, 69°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

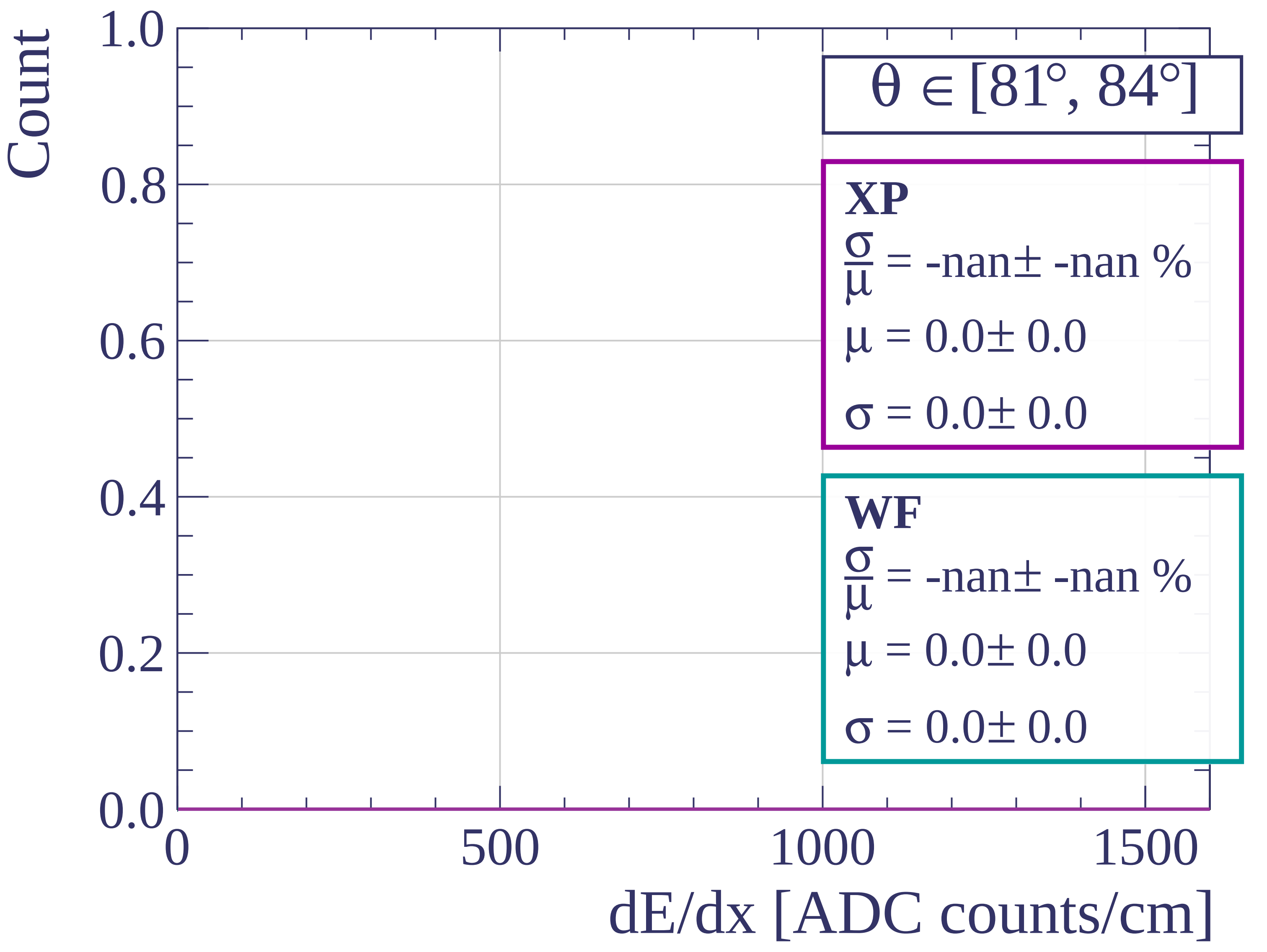

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [69°, 72°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



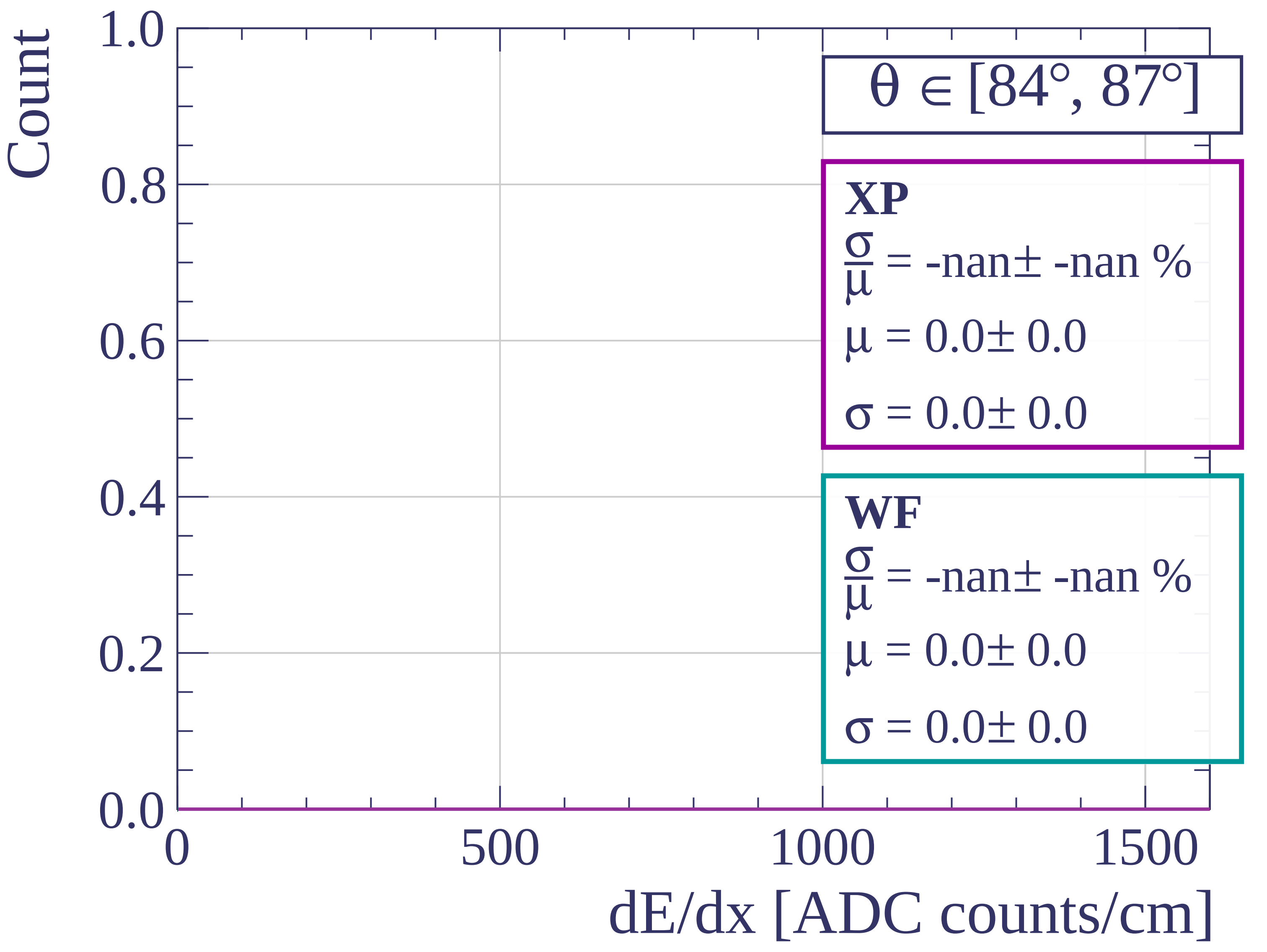
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500

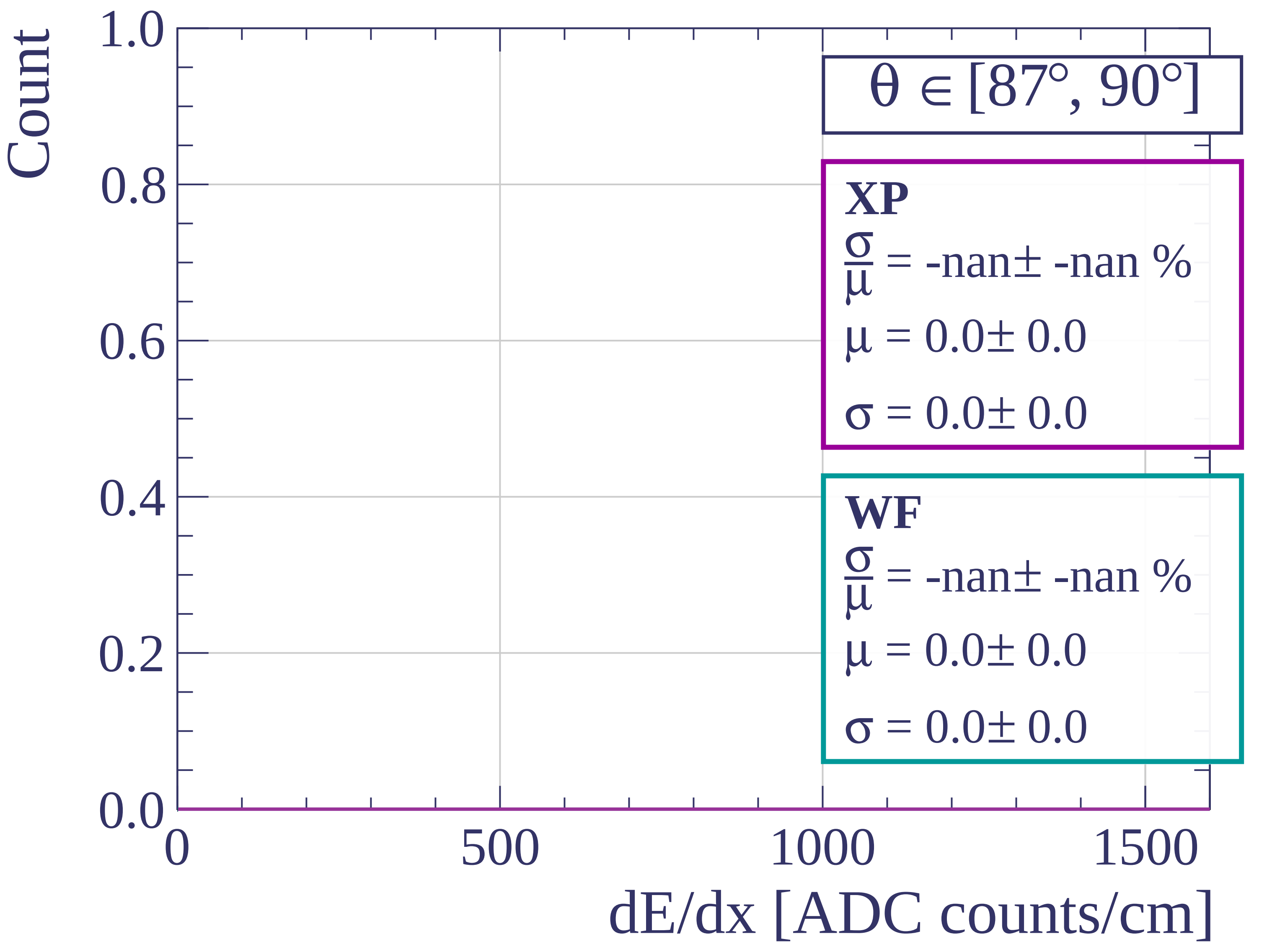

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



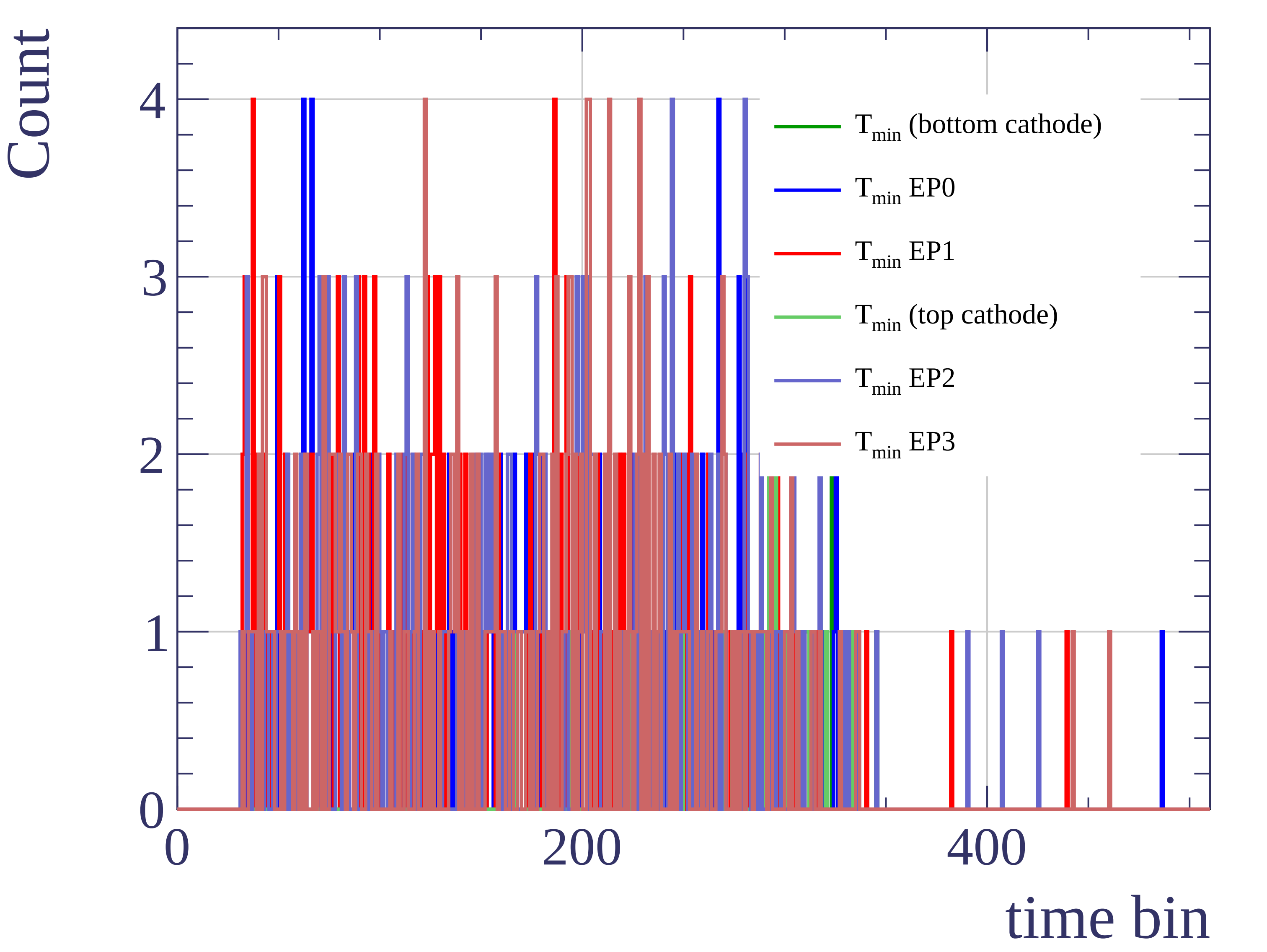
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



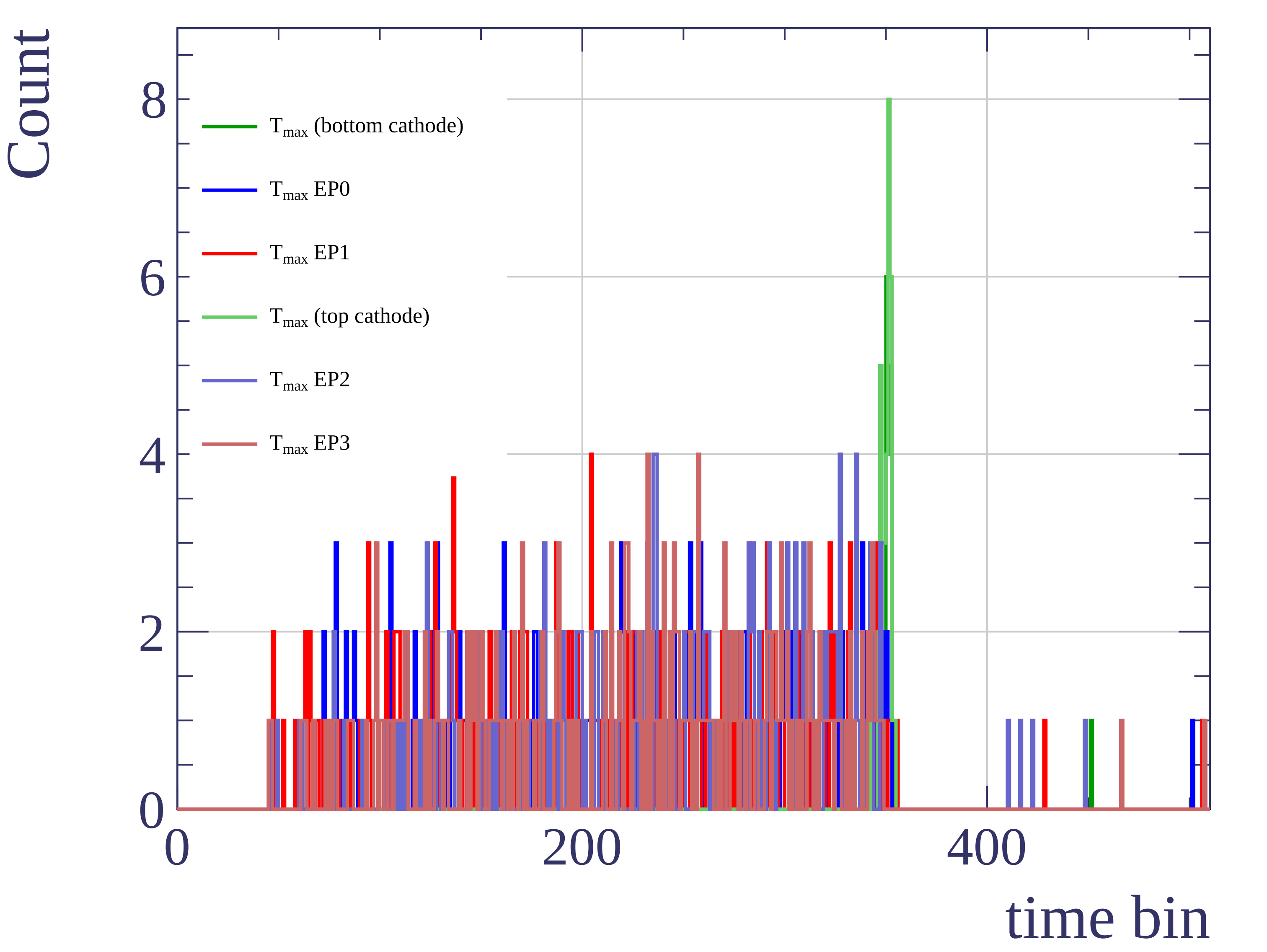
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
4
3
2
1
0
0
200
400
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
8
6
4
2
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3